



The 3PL Language Version 11.7.0M

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2023-01-13 15:00:30 +1100

Open report - CSIRO Mineral Resources

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Chapter 1 - Introduction

The use of Field Programmable Gate Arrays (FPGAs) as real-time data processors is attracting growing interest. The Xilinx Virtex FPGA series is aimed directly at the DSP market and promises spectacular performance in comparison to a conventional CPU.

However, the programming of FPGAs still presents considerable difficulties, and digital circuit design is still the predominant method. Parallel languages for FPGAs have emerged in recent years, but they have arisen largely from computer science departments where the focus has been on a wide range of general parallel algorithms. The environment of parallel CPUs which prompted early work, the very generality of these parallel languages, and their evolution from serial languages has resulted in compilers that arguably do not use FPGAs very efficiently. In particular these parallel programming languages do not support pipelining very well.

Pipeline processing seems a natural way to implement machine vision (and other) algorithms which work on real-time data streams. Even where data is stored in memory, in many cases it can be traversed to generate a data stream. Even where an algorithm is sequential, pipelining results in some parallel execution - the parallelism is effectively the length of the pipeline.

1.1 The 3PL language

This document describes a *dataflow*¹ parallel programming language for FPGAs which has been explicitly designed to support pipelining. The name **3PL** stands for "Parallel Pipeline Programming Language" (the name was suggested by Alen Yokhanis).

In principle we would like our pipeline to have the following characteristics -

1. Each pipeline module should process input data when -
 - (a) its internal state is appropriate
 - (b) required input data are available
 - (c) required output can take place
2. Movement of data through the pipeline should depend only on the behaviour of individual modules and their interaction.

Data will flow through the pipeline as rapidly as possible, retarded only when modules or groups of modules are not in a state to accept them.

Data cannot be assumed to be contiguous in a pipeline.

3PL supports pipelined data, and modules are inherently parallel. While the pipeline/module model works well for real-time data streams, there are some sequential algorithms for which pipeline processing

¹Dataflow was a name given to data-driven computing architectures explored at Manchester University by Ian Watson and John Gurd in the late 1970s.

is clumsy and laborious. While non-pipelined sequential operations are inefficient in an FPGA (a modest conventional CPU will be significantly faster), occasions arise where the sequential code is but a small part of a larger, predominantly parallel, algorithm. For this reason **3PL** supports the usual conventional sequential constructs as well.

Sequential constructs can also be used in code which is executed at compile time, rather than in the target architecture. This *immediate* code is used to control target code generation, for example to generate a network topology from an algorithm or to systematically initialise a table.

The 3PL compiler is not specific to any particular FPGA, but the code generator supports only Xilinx FPGAs and CPLDs at the moment. Currently these platforms are Spartan2, Spartan3, Spartan6, Virtex, Virtex-E, Virtex2, Virtex2-Pro, Virtex4, Virtex5, Virtex6 and series 7 FPGAs (Artix, Kintex, Virtex7 and Zynq). It also supports Coolrunner and XC95 CPLDs though only a much restricted range of 3PL features can be used for these devices.

1.2 Current version

The current version of 3PL is **11.7**. It requires Java 1.6 or later. A list of compiler versions appears as an appendix.

1.3 Known deficiencies

1.3.1 enumerated types where ordinal 0 is not used

When a queue variable whose type is non-primitive is assigned, any words in the type that are not assigned in that clock cycle are by default assigned a value corresponding to all bits low. For numeric types that is zero and for logical that is false. This is a useful convention that can be employed to simplify source code. For an enum that default will be the value which has ordinal 0. If an enum is created using inbuilt procedure `enumv()` then ordinal 0 may not be used. In that case the above scenario would result in an invalid enum value. This problem can be easily avoided by either using ordinal 0 or else always explicitly assigning that field, so it is unlikely that this will be fixed.

1.3.2 ++ and – operators limited

The increment and decrement operators work on integer primitives, including members of arrays and structs. They do not parse if applied to an expression and hence cannot be used on an indirect reference via the `->` operator.

1.3.3 cannot have arrays of queues, clocks, memories etc

Arrays and structs are part of a type specification. Clock mode variables do not have a type and hence there cannot be an array of clocks. The two memory modes do not have types, though their address and data types must be supplied, and so again there cannot be an array of these. Priority mode variables have a type which is "log" but again there cannot be an array of priority variables. Queue mode variables can have an array type, but the entire variable is read and written as a unit. You can have a queue whose type is an array but you cannot have an array of queues.

Immediate, value, selectvalue and static mode variables can be assigned or evaluated in part or whole and hence do not present a problem in this respect.

This inconvenience is a result of the language semantics and is not simply an implementation deficiency.

Arrays of any mode of variable can be created and accessed by means of pointers. An array of pointers is created and then initialised to point to created variables. This can be done in a function so that the

created variables are hidden and the pointer array is returned as the value of the function. There is an example in library stdlib.3pl, `queuearray()`, which creates a number of queue mode variables and returns an array of pointers to the queues.

1.3.4 petrinets transitions do not wait on code blocks

When a petrinet transition occurs an associated code block may wait on queue availability. Transitions should wait for completion of associated code blocks. To implement this the transition would have to occur simultaneously with the last execution cycle of the code block. At present there is no easy way to do this. This will be further considered since this problem limits the applicability of petrinets.

1.4 Future development

1.4.1 static multiply-assignment operator

The `*<-` assignment may be implemented later.

1.4.2 extension to round function

The round function takes both immediate and target mode arguments. For the latter it rounds a fixed to int or ufixed to uint. It may be extended in the future to allow rounding of fixed, ufixed, int or uint at any bit position, producing an appropriate result type, but so far no particular need for this has been felt.

1.4.3 target code pointers

Immediate-mode pointers are available in 3PL. In immediate code, pointers to immediate or target mode variables can be created, assigned and dereferenced. There is no equivalent in target code. Pointer variables which could be assigned and dereferenced in target code could be implemented, but the resulting decoder logic would be extensive and slow except in the case of very limited usage, hence this is probably not worth considering.

1.4.4 module and procedure call syntax

Currently the syntax for calling a module is identical to that for a procedure. It would enhance readability if these calls had distinctive syntaxes. This would help emphasise the difference between a module and a procedure. In particular the fact that a called module is a completely separate (new) environment from that of the call. I am considering changing module calls to the following syntax -

```
[modname(arg, arg, ....)]
```

using the brackets to suggest a box as for inline modules. Note that an inline module containing a procedure call is distinguished by the `”;` terminating the call -

```
[procname(arg, arg, ....);]
```

The proposed change has not met with enthusiasm from other users so it may never be done.

1.4.5 debugging facilities

Six debugging tools have been envisaged -

- an immediate execution interactive debugger
- a target code simulator
- a timing constraint error report analyser which will determine which parts of the source are responsible for failure to meet place-and-route timing constraints
- a target run-time exception mechanism which can report exceptions or freeze execution when they occur
- a single step facility, at least for single clock applications
- a configuration read-back utility which would relate variable and memory contents back to the source code

The second, a target simulator, has been included as part of the compiler/interpreter. As 3PL evolved the simulator atrophied and hence it has been disabled, though the source code is still included to allow later re-enabling if required.

Provision of the other five facilities has not yet been started. The third tool listed is the most urgent because failure to meet timing constraints will be a major problem for users of the language, but it has not yet been tackled.

1.4.6 variable types

Thus far bit, integer, unsigned integer, fixed point, floating point (limited), enumerated and logical types have been implemented for target variables. The string type can be used for immediate variables but there is no plan to extend it to target variables.

1.4.7 linkage to precompiled 3PL code

No mechanism for this has been designed or implemented. Currently all 3PL libraries are in 3PL source code and are compiled along with the application program.

1.5 Acknowledgements

This work was done within what was the Machine Vision group in CSIRO Manufacturing and Infrastructure Technology, Preston laboratory, Melbourne. The group is now called High Speed Instrumentation and is in CSIRO Mineral Resources, Clayton, Melbourne.

Paul Dunn conceived the language and Paul and Andrew Bainbridge-Smith developed the original (incomplete) compiler. Andrew did all the groundwork finding java tools and set up the make file, the directory structure and the javadoc framework and wrote the original parser. Paul wrote the code generator.

The first version of the compiler was a good learning exercise but soon proved to have serious shortcomings and was never completed. Paul Dunn redesigned the language and re-implemented the compiler from scratch. This benefited greatly from the experience of the first attempt.

Andrew Tulloh has worked through both compiler versions developing the target simulator (currently disabled as it has not kept up with 3PL evolution).

Robin Kirkham has been an invaluable critic and expert user of 3PL throughout and has contributed considerable 3PL library code.

Chapter 2 - Overview

3PL is a language intended to allow parallel algorithms to be coded for implementation in an FPGA. A language which compiled directly into an FPGA configuration was considered insufficient on its own since it is often necessary to run an algorithm which in turn generates the target algorithm. For example an extreme case is a batcher sorting network where the batcher algorithm must be executed to generate the network topology. In simpler cases it is an advantage to parameterise the target algorithm which again implies some sort of preliminary processing. This could be accomplished by using a general purpose language such as C, Java or Python to pre-process the target algorithm, but it was decided that it was preferable to combine all of this in one language. Thus 3PL is actually a sequential language similar in most respects to C, but it has additional statement types that generate FPGA target code. This might have been accomplished by adding appropriate classes to Java or Python but it was felt that a new language tailored to the task would be preferable.

3PL was designed to meet several criteria -

- it must provide constructs within the FPGA such as variables, memory and combinatorial logic.
- it must provide execution threads which include sequential operations, parallel operations and looping and conditional control.
- it should provide a mechanism whereby systolic pipelines can be constructed which are data driven.
- the control of execution threads must co-exist with data-driven execution.
- it should be concise.

3PL is really two languages in one. 3PL is compiled and then executed by an interpreter. That execution, called *immediate* execution, is strictly sequential. During immediate execution target (FPGA) statements are encountered. These target statements are also subject to immediate execution during which target code is generated which will later be executed in the FPGA. For example the code

```
par {
  a <- b;
  c <- d;
}
```

looks like a parallel block in which two assignments execute simultaneously. While that is exactly what it ultimately represents the way 3PL handles this is a little more complicated. The *par* construct and the assignments are subject to immediate execution - first the *par* and then each of the nested assignment statements in turn. Immediate execution of the *par* creates a parallel block structure in the target code and then each nested statement in the block is executed in turn. If immediate execution of a nested statement generates in-line target code then that code is linked into the parallel block that has been created for subsequent parallel execution in the target. In the example each assignment statement generates in-line target code and that target code will be linked into the parallel block.

This description may seem pedantic, but once immediate and target statements are intermixed it becomes important to understand the way 3PL executes. As a more complicated example consider the following -

```

par {
  for(int(i) ; i<10 ; i=i+2)
    a[i] <- b[i];
}

```

In this example we can again follow the previous description to see what is generated. Immediate execution of the **par** creates a parallel block for the target. Immediate execution of the nested **for** does not itself create target code, but the immediate execution of the code within the loop creates a target assignment statement on each iteration. Immediate execution of each target assignment statement generates in-line target code which is passed back to the enclosing **par**. In this case the 5 iterations produce 5 target assignments which are handled by the enclosing **par**. The generated target code is equivalent to having written -

```

par {
  a[0] <- b[0];
  a[2] <- b[2];
  a[4] <- b[4];
  a[6] <- b[6];
  a[8] <- b[8];
}

```

The **3PL** language contains two distinct types of statements -

- *immediate* statements
- *target* statements

Immediate statements are executed by the compiler following parsing. They control the processing of target statements.

Target statements are to be executed in the FPGA later - they perform the actual data processing.

The **3PL** compiler parses the source code and stores a representation of the code internally.

The compiler then executes the entire **3PL** program sequentially. Immediate statements are explicitly executed. Target statements when encountered generate logic elements into a target code structure. Module, procedure and function calls are executed by the interpreter but any target code they generate is produced for each instance. Immediate variables occurring in target code are replaced by their current value.

Immediate and target statements are distinguished by the compiler by their different syntax. This was a design decision intended to make the code more readable - immediate and target constructs are easily distinguished by inspection. The alternative would be to use the same syntax for immediate and target code and distinguish them by the mode of variables - easily implemented but resulting in more difficulty in reading and maintaining 3PL code.

The philosophy of queue variables is that pipelines of processing modules can be simply strung together using queue variables to convey data from one to the next. Where this differs from the usual sort of readable/writable channel is that -

- queue variables can go to multiple destinations, all of which must read before the source will make the next datum available
- queue reads and writes occur in the source code as normal variable value references and assignments, not explicit read and write statements

Such a pipeline does not necessarily advance on every clock cycle, but rather modules process data in their own time when it is available, then pass it on. This mechanism does not incur any speed penalty as a pipeline can run at the full clock rate unless a module takes more than one cycle because of the nature of the algorithm.

In addition to the dataflow mechanism provided by queue variables 3PL fully supports simple register variables and execution control constructs such as sequential and parallel blocks, loops and conditionals. 3PL also provides a simple implementation of Petrinets which are more general than state machines. Queues and other variable modes can be compound types such as arrays or structs. These can be nested, and array members or struct fields can be accessed in parallel. Target array members can also be selected using variable subscripts.

Hardware elements explicitly implemented include 3-state buses, priority encoders, combinatorial output multiport RAMs (Xilinx single or dual port CLB distributed RAM) and registered output multiport RAMs (Xilinx single or dual port block RAM).

Low level optimisation at the netlist level explicitly for Xilinx includes use of the carry chain for wide logic gates, decoders and priority encoders, and CLB shift registers for some delay functions.

3PL supports any number of clocks and provides a means of resynchronising data between clock domains.

Intermediate code allows the use of pointers (both to immediate and to target variables) so that linked data structures can be constructed. Variables are created during immediate execution (interpretation) and continue to exist after their scopes have vanished, neatly solving the allocation problem (thus a function can return a pointer to a local variable which can continue to be used via the pointer after the function returns).

2.1 Variable modes

Variables in 3PL fall into 11 categories, referred to as **modes**. These are -

Immediate - an immediate variable only exists during execution of the 3PL interpreter. When a 3PL program is executed by the interpreter immediate mode variables are used in the same way as variables in languages such as C. Immediate mode variables do not generate FPGA code and are not represented in the FPGA configuration in any way. Immediate mode variables can be integer, boolean, floating point, string, array, struct, pointer, file or enumerated types. Immediate execution is discussed later.

A program containing only immediate mode variables will execute in a similar way to a C program and no FPGA code will be involved.

The remaining modes are all 'target' modes, that is they represent hardware devices in the FPGA.

Static - a static variable is a register in the FPGA. It can be integer, unsigned integer, fixed point, unsigned fixed point, floating point, boolean, array or struct types. All except the boolean type must have bit widths specified.

Queue - a variable which is a queue. If it is evaluated in a statement a datum will be popped. If it is assigned, a datum will be pushed. Otherwise it appears in statements and expressions in the same way as other variables. In any context a statement using a queue variable will wait to execute if the variable is not 'available', i.e. has a datum to pop or a free space in which to push a datum as appropriate.

Value - a value variable can represent any expression. The expression could be as simple as a single static or queue variable, in which case the value variable is a bit like a pointer.

Selectvalue - a variable which can select one of many inputs dynamically. In earlier FPGAs this would be implemented by 3-state buffers, but in later platforms the implementation uses combinatorial logic. These variables are used to construct buses internal to the FPGA.

Priority - a priority variable implements a priority encoder. It is combinatorial logic which appears as a boolean array where multiple simultaneous inputs are blocked except for that with the highest priority.

Input - these provide an interface for input pads.

Output - output variables drive 2-state or 3-state output pads.

Clock - clock variables provide clocks for the synchronous logic which 3PL produces. Connections between clock inputs, clock managers and clock buffers are also clock mode variables.

Rmemory - these create registered-output memory, implemented at the moment by block RAM.

Cmemory - these create combinatorial memory, implemented using CLB RAM.

2.2 Static mode variables

The most straightforward of the target modes is **static**. A static mode variable is equivalent to a normal variable in programming languages such as C or Java. It is implemented in the FPGA as a register using flip-flops. A static variable stores a value and therefore has state.

The following code shows the creation of a static variable and an assignment to it -

```
static(v, "int:8");
v <- 3;
```

The string "int:8" gives the type of the variable, which is an 8-bit integer.

The assignment operator for a static variable is <-. There are a number of different assignment operators for different variable modes, the intention being to make the modes immediately distinguishable on inspection of source code without the need in most cases to look for the statement creating the variable. Thus an assignment using <- immediately indicates that the variable on the left side is a static variable.

The assignment takes one clock cycle.

On assignment the widths of the left and right hand sides of the assignment need not be the same, however in that case padding or truncation will result. If the type is signed then padding will be by sign extension, otherwise zero padding is used.

3PL has the unusual characteristic that FPGA statements execute repeatedly, i.e. as if they were enclosed in infinite loops. The reason for this design decision is that FPGA statements are intended to operate on data streams and to do so with a high degree of parallelism. Thus every statement can potentially execute on every clock cycle and it is up to conditional code to control when statements execute. In a parallel data-driven system it seems more sensible to have execution on every clock cycle as the default with an overriding conditional mechanism to choose when to execute, rather than to enclose every statement in an explicit iteration loop.

In the example above the variable **v** will be assigned 3 on every clock cycle. This is not very useful, so we could use a conditional statement to exercise some control, e.g.

```
static(v, "int:8");
static(l, "log");
when (l)
  v <- 3;
```

Here we have introduced another static variable **l** which has a logical (boolean) type. When **l** is true, **v** is assigned 3. Now the **when** is executed repeatedly and if **l** is true the assignment is executed otherwise (since there is no **else** block) nothing is executed. Note that the **when** does not wait for **l** to be true, it simply evaluates **l** on every one of its continuous iterations and when true executes the code body, in this case a single assignment. To make the example more useful we could implement a controlled counter by the code -

```
static(v, "int:8");
static(l, "log");
when (l)
  v <- v + 1;
```

where **v** is incremented on clock cycles when **l** is true. 3PL allows the special case of addition or subtraction of one to/from a variable to be written using ++ and -- unary operators, e.g.

```
static(v, "int:8");
static(l, "log");
when (l)
  v++;
```

but note that these operators cannot be used on a static variable as part of an expression. We could make the counter bi-directional, e.g.

```

static(v, "int:8");
static({l, up}, "log");
when (l)
  when (up)
    v++;
  else
    v--;

```

where when **l** is true **v** will be incremented if **up** is true or decremented if **up** is false. As illustrated here many variable creation procedures allow lists of variables in braces where more than one variable of the same type is required.

The above example is quite correct, however it is preferable to avoid nested conditional statements where possible as this can lead to long path delays in the FPGA logic. We could recode the example as follows -

```

static(v, "int:8");
static({l, up}, "log");
when (l && up)
  v++;
when (l && !up)
  v--;

```

which achieves the same behaviour but reduces logic depth at the expense of logic width.

Static variables have default initial values which are 0 for numeric types and false for the logical type. A static variable can be given an initial value by means of the optional 3rd argument to procedure **static()**, e.g.

```
static(v, "int:8", 117);
```

will give static variable **v** an initial value of 117. If numeric the initial value can of course use hexadecimal (0x...), octal(0...) or binary (0b...) notation.

2.3 Input and output mode variables

We need to be able to get data into and out of the FPGA. For this purpose there are two variable modes, **input** and **output**.

An input mode variable represents signals coming into the FPGA via pads and input buffers. The following code latches input data into a static variable when an enable signal is high -

```

str(enable_pin, "D8");
var(din_pins, "[8]str", {"A1", "A2", "B1", "B2", "C1", "C2", "D1", "D2"});
input(d_in, "int:8", loc=din_pins);
input(enable, "log", loc=enable_pin);
static(d, type(d_in));
when (enable)
  d <- d_in;

```

Input mode variables are created by calling **input()** which has three or more arguments.

The first argument to **input()** is an identifier for the created variable.

The second argument to **input()** is the type, in this case one is an integer of 8 bits and the other is logical (boolean) and hence is implicitly one bit.

The third and following arguments are attributes for the input. This can include FPGA pins (required), a timing group identifier string, a flag to control the inserted input path delay, a flag to indicate that an IBUFG is to be used rather than an IBUF and finally flags for pullups or pulldowns. The attributes are passed in the form attribute=value.

In general arguments can be passed by position in the normal way or else by key=value where key is the name of the parameter for which the argument is intended (this is referred to in the 3PL manual as keymatch format). Keymatch format is useful where the parameters are many and the arguments are sparse. In 3PL arguments are optional and parameters can have default values specified for cases where no argument is passed to them.

In the code above the **loc** attribute must either be a string or an array or strings. These are used to generate FPGA pad location constraints. Note that array dimensions precede the array type.

A string immediate mode variable can be created by calling **str()**. The first argument is an identifier for the variable or else a list of identifiers in brackets. The second argument is optional and is the initial value of the string.

An array of strings is created by the **var()** procedure. This is the basic variable creation procedure for most modes of variable, procedures such as **static()**, being convenient equivalents. The code

```
static(l, "log");
str(s, "a_string");
```

is equivalent to

```
var(l, static, "log");
var(s, "str", "a_string");
```

If the second argument is not a mode identifier then it is interpreted as the variable type and the mode is assumed to be immediate. The **var()** procedure entry in the 3PL manual should be consulted for more details.

The input mode example above assigns the input mode variable **d_in** to the static variable **d** when the input mode variable **enable** is true (high).

Note the use of the function **type()** in the example - this simply returns the type of its argument, providing a simple way of creating a variable whose type is the same as another.

The above example would produce a fatal error in 3PL as the input variable **enable** is used to control conditional logic but is driven by an external source and has no timing information with respect to the current clock, so there could potentially be timing violations. The usual way to handle this is to register the inputs. We can also instruct the place-and-route software to place the register flip-flops in an I/O block by setting an attribute -

```
str(enable_pin, "D8");
var(din_pins, "[8]str", {"A1", "A2", "B1", "B2", "C1", "C2", "D1", "D2"});
input(d_in, "int:8", loc=din_pins);
input(enable_in, "log", loc=enable_pin);
static(d, type(d_in), IOB=true);
static(enable, "log", IOB=true);

static(dd, type(d));

d <- d_in;
enable <- enable_in;

when (enable)
    dd <- d;
```

With the inputs **d_in** and **enable_in** now latched into flip-flops within I/O blocks we minimise the delay between input pad and flip-flop D pins. The clock used for this logic must bear an appropriate phase relationship with the external data input such that setup and hold constraints are met.

We could expand the example by connecting the static variable **dd** to output pins -

```

str(enable_pin, "D8");
var(din_pins, "[8]str", {"A1", "A2", "B1", "B2", "C1", "C2", "D1", "D2"});
var(dout_pins, "[8]str", {"E1", "E2", "F1", "F2", "G1", "G2", "H1", "H2"});
input(d_in, "int:8", loc=din_pins);
input(enable_in, "log", loc=enable_pin);
static(d, type(d_in), IOB=true);
static(enable, "log", IOB=true);
output(, dd, loc=dout_pins);

static(dd, type(d));

d <- d_in;
enable <- enable_in;

when (enable)
    dd <- d;

```

The `output()` procedure has four or more arguments. The first is an identifier, but since in this case we have no need to refer to it we need no identifier so the argument is omitted. The second argument is the type, but the type can be inferred from the third argument so again the argument can be omitted. The third argument is optional and is the source to be used for the output (there are other ways of connecting an output mode variable to its source, in which case the third argument would be omitted but the first two would be required). The fourth and following arguments provide attributes as for the `input()` procedure. In this case they can include slew rate and drive current, the pin locations again being mandatory.

The `output()` statement connects the static variable `out` to the FPGA pads using output buffers. This example could be re-written to make the output assignment explicit -

```

str(enable_pin, "D8");
var(din_pins, "[8]str", {"A1", "A2", "B1", "B2", "C1", "C2", "D1", "D2"});
var(dout_pins, "[8]str", {"E1", "E2", "F1", "F2", "G1", "G2", "H1", "H2"});
input(d_in, "int:8", loc=din_pins);
input(enable_in, "log", loc=enable_pin);
static(d, type(d_in), IOB=true);
static(enable, "log", IOB=true);
static(dd, type(d));
output(out, type(dd),, loc=dout_pins);

d <- d_in;
enable <- enable_in;

when (enable)
    dd <- d;

out := dd;

```

note that the last assignment is immediate, not target. It is like assignment to a value variable and hence uses the same assignment operator. It is an unconditional assignment, i.e. it is connected continuously and not changed dynamically. It connects the value on the RHS to the output buffers described by the output mode variable on the LHS.

If we want to drive a three-state output pad one cycle after the assignment to `d` we need conditional output mode assignment. The code might be -

```

str(enable_pin, "D8");
var(din_pins, "[8]str", {"A1", "A2", "B1", "B2", "C1", "C2", "D1", "D2"});
var(dout_pins, "[8]str", {"E1", "E2", "F1", "F2", "G1", "G2", "H1", "H2"});
input(d_in, "int:8", loc=din_pins);
input(enable_in, "log", loc=enable_pin);
static(d, type(d_in), IOB=true);
static(enable, "log", IOB=true);
static(enable_del, "log");

```

```

static(dd, type(d));
output(out, type(dd),, loc=dout_pins);

d <- d_in;
enable <- enable_in;
enable_del <- enable;

when (enable)
    dd <- d;

when (enable_del)
    out |<- dd;

```

where the last statement is a conditional target assignment, i.e. a three-state assignment. There is a library function `delayline()` which we could use instead of the assignment to an extra variable -

```

str(enable_pin, "D8");
var(din_pins, "[8]str", {"A1", "A2", "B1", "B2", "C1", "C2", "D1", "D2"});
var(dout_pins, "[8]str", {"E1", "E2", "F1", "F2", "G1", "G2", "H1", "H2"});
input(d_in, "int:8", loc=din_pins);
input(enable_in, "log", loc=enable_pin);
static(d, type(d_in), IOB=true);
static(enable, "log", IOB=true);
static(dd, type(d));
output(out, type(dd),, loc=dout_pins);

d <- d_in;
enable <- enable_in;

when (enable)
    dd <- d;

when (delayline(enable, 1))
    out |<- dd;

```

2.4 Value mode variables

Consider the following code fragment -

```

static({a, b, c, d, e}, "int:8");
a <- c + d + e;
b <- c + d - e;

```

This creates five static variables, **a**, **b**, **c**, **d** and **e**, each of type "int:8", i.e. an 8-bit integer. The second statement assigns an expression containing **c**, **d** and **e** to **a** on every clock cycle. Similarly the third statement also assigns an expression to **b**. In each case expressions will be implemented by adders and subtractors but the 3PL compiler will not recognise the common sub-expression $c + d$. It is a waste of limited logic resources to duplicate the sub-expression so we need a way to extract the common sub-expression. A **value** mode variable provides the means to do this.

```

static({a, b, c, d, e}, "int:8");
value(v, "int:8");
v := c + d;
a <- v + e;
b <- v - e;

```

Here the value variable represents the expression $c + d$ which is thus only instantiated once, the output being shared by the two static assignments. The assignment operator for a value mode variable is `:=`. The assignment could be avoided by using the optional initialiser argument to the `value()` procedure -


```
static({a, b, c, d, e}, "int:8");
value(v, "int:8", c + d);
a <- v + e;
b <- v - e;
```

A value variable thus does not itself have state, i.e. there is no memory associated with a value variable. Note the following however -

```
static({a, b}, "int:8");
value(v, "int:8", b);
a <- v;
```

Here the value variable has as its value a single static variable. In this role it can be viewed as a pointer to that static variable (3PL does have pointer variables but these are not discussed here as they are seldom used in application code).

Value variables can be re-assigned at any time at which point they take on a new value. Value variables can also be evaluated prior to being assigned, in which case the later assignment is linked back by the compiler to the prior usage.

2.5 Parallel and sequential blocks

Parallel and sequential blocks provide the basic means of controlling execution order.

A sequential block is written as -

```
seq {
  statement
  ...
  ...
}
```

the enclosed statements being executed sequentially. The block terminates when the last statement in the block terminates. If the **seq** block has no enclosing conditional structure it will be executed repeatedly. If we wished to execute it once only, as in a conventional computer, we could terminate execution at the end of the block -

```
seq {
  statement
  ...
  ...
  stop();
}
```

A parallel block is coded in a similar way -

```
par {
  statement
  ...
  ...
}
```

the enclosed statements beginning execution simultaneously. The parallel block terminates when all enclosed statements have terminated, i.e. it terminates simultaneously with the longest executing statement(s). If the **par** block has no enclosing conditional structure it will be executed repeatedly.

These blocks may of course be nested.

Note that there is a difference between -

```

seq {
  a <- ...;
  b <- ...;
}
c <- ...;

```

and -

```

par {
  seq {
    a <- ...;
    b <- ...;
  }
  c <- ...;
}

```

In the first case the seq block executes repeatedly, taking 2 clock cycles each time. The single statement also executes repeatedly taking one clock cycle each time. In the second case the seq block and the single assignment start executing simultaneously but since the seq block takes two cycles the par block cannot terminate until the seq finishes. The par block then repeats, the seq block and the single assignment again starting simultaneously. Thus in the first case the two elements repeat independently whereas in the second case they repeat in step.

2.6 Target loops

Target loops, i.e. loops in the FPGA code, have the forms -

```

seq while (expression) {
  body
}

```

```

par while (expression) {
  body
}

```

```

seq do {
  body
} while (expression);

```

```

par do {
  body
} while (expression);

```

These function the same as conventional **while** and **do while** loops but with either sequential or parallel loop bodies. The braces are required even if the body is a single statement. Note that if the body is a single statement it does not matter whether the **seq** or **par** version is used.

These loops add no extra clock cycles on each iteration to the code body however there is one extra clock cycle required on loop termination.

2.7 Immediate execution

Thus far the language has been presented as a collection of statements to execute in the FPGA. It must appear a little odd that procedures are called to create variables, some of which (strings for example) do not appear to create anything in the FPGA. Where exactly do the procedures execute?

3PL is in fact a strictly sequential language - it is not a parallel language! 3PL source code is parsed into an internal form which is then executed by an interpreter. The language executes sequentially in the same way as languages such as C or Java. This execution is called **immediate** execution. Immediate assignment statements use the '=' operator. Immediate statements include **if()**, **for()**¹, **while()**, **do while()**, **switch**², **break**, **continue** and **return**. Assignments to value mode and priority mode variables and unconditional assignments to output mode variables are also immediate statements since they set up values within the interpreter and do not directly create in-line FPGA executable code. These all use the := assignment operator.

During immediate execution 3PL encounters target statements which generate target code, i.e. logic destined for the FPGA. The target logic thus generated executes in the FPGA.

The following simple code -

```
int(i, 3);
for (int(j) ; j<3 ; j++)
    println(i + j);
```

is wholly immediate. The first statement creates an immediate integer variable **i** with an initial value of 3. The next statement is a **for** loop which creates an immediate integer variable **j** within the scope of the loop and iterates three times. On each iteration a line is printed to standard output consisting of the sum of the two variables, giving the output -

```
3
4
5
```

When this code is executed (immediate execution) no target logic at all is created. Except for the fact that variables are created by calls to procedures rather than in declarations, the immediate language is largely identical to C.

The following code contains target statements (line numbers have been added as comments) -

```
int(i, 5); // 1
static({s1, s2}, "int:8"); // 2
par { // 3
    s1 <- i; // 4
    i += 7; // 5
    s2 <- i+2; // 6
} // 7
```

Immediate execution of line 1 creates an immediate integer variable **i** with an initial value of 5. Execution of line 2 creates two static variables, **s1** and **s2**, which are 8-bit integers. This will result in target logic for two registers. Execution of line 3 sets up a pending parallel code block for the target. Execution of line 4 creates target logic for a static assignment, which is added to the pending parallel block. The RHS of the assignment is an immediate variable, so it is evaluated and thus the constant integer 5 is to be assigned in the target. When immediate mode values are encountered in target statements their current values at that point are used. Execution of line 5 adds 7 to the value of variable **i** but generates no target code. Execution of line 6 generates an assignment to static variable **s2** of the constant integer 14. This assignment is also added to the pending parallel block. At line 7 the pending parallel block is closed and the target statements encountered within it are linked in as parallel target execution threads. Since the parallel block is not enclosed in any other control structures it will be embedded in an infinite loop as usual.

It can be seen that intermediate execution provides fine control over what target code is generated. Further than that, intermediate execution is essential where an algorithm is required to generate the target algorithm. An extreme example is the well known Batcher parallel sort algorithm where a complicated algorithm must be executed to lay out the logic for the parallel sorting network. A rather simple nonsense example illustrates the idea -

¹The comma operator is not implemented. The initial section may contain a procedure call to create the loop variable which will be in the scope of the loop body

²Cases do not flow on as they do in C. Cases allow string constants.

```

var(m, "[]int", {3, 7, 1, 5, 9, 8}); // 1
static({s1, s2}, "[6]int:8"); // 2
par { // 3
  for (int(i) ; i<6 ; i++) // 4
    s1[i] <- m[i] * s2[i] // 5
} // 6

```

Execution of line 1 creates an immediate integer array initialised with arbitrary values. Execution of line 2 creates two static variables which are arrays of 6 8-bit integers. Execution of line 3 sets up a pending parallel target code block. Lines 4 and 5 constitute an immediate loop which on each iteration creates a target static assignment which is added to the pending parallel target block. Thus immediate execution has been used to iterate through a number of target code generation statements which are collected together for target execution in a parallel block. This code is equivalent to having written -

```

var(m, "[]int", {3, 7, 1, 5, 9, 8});
static({s1, s2}, "[6]int:8");
par {
  s1[0] <- m[0] * s2[0]
  s1[1] <- m[1] * s2[1]
  s1[2] <- m[2] * s2[2]
  s1[3] <- m[3] * s2[3]
  s1[4] <- m[4] * s2[4]
  s1[5] <- m[5] * s2[5]
}

```

In the examples shown so far the variable types have been fixed. Since the variable types are usually entered as a string (there is also a type "type", e.g. that is what the type() function returns) a type can be a string-valued expression, so we could parameterise a type, e.g.

```

int(dim, 6);
int(width, 8);
static(s, "["+dim+"]int:"+width); // equivalent to "[6]int:8"

```

Here the type for static variable `s` is an expression constructed by concatenating string constants and variables (for strings '+' concatenates them). If the same type is to be used for several variables it could be assigned to a string variable, e.g.

```

int(dim, 6);
int(width, 8);
str(t1, "["+dim+"]int:"+width);
static(s1, t1);
static(sa, "[10]t1");

```

Note that a variable identifier can appear in a type string, as in the last line.

Immediate types do not have bit widths whereas target types must have a width. The exception is type "log" which does not have a width and can be used for immediate or target modes.

There is also a variable type "type". This can be created by several inbuilt procedures and functions. As mentioned the inbuilt function **type()** returns the type of a variable or expression (the function **typestring()** returns the type as a string). The inbuilt procedure **type()** (note that function and procedure call syntax is different so the same name can be used for these) creates a type from a string, so the above example could have been written -

```

int(dim, 6);
int(width, 8);
type(t1, "["+dim+"]int:"+width);
static(s1, t1);
static(sa, "[10]t1");

```

however `t1` is now not a string variable but a "type" variable.

Another procedure which creates a "type" variable is **struct()**. The call -

```

struct(s,
  f1, "log",
  f2, "[4]int"
);

```

creates a struct of immediate types, field **f1** being type "log" and field **f2** an array of four integers. This type can of course be nested, e.g.

```

struct(s,
  f1, "log",
  f2, "[4]int"
);
var(v, "[3]s");
v[0].f1 = false;
v[1].f2[3] = 5;

```

Structs can also be initialised, e.g.

```

struct(s,
  f1, "log",
  f2, "[4]int"
);
var(v, s, {true, {1, 3, 5, 7}});

```

The above example creates and initialises an immediate variable. The same can be done for a target mode variable, e.g.

```

struct(s,
  f1, "log",
  f2, "[4]uint:12"
);
static(v, s, {true, {1, 23, 570, 76}});

```

Where types are structs or arrays (or any nested combination) they may be assigned in one operation provided the LHS and RHS types match, including array dimensions. Thus the following is allowed -

```

struct(s,
  f1, "log",
  f2, "[4]int:3"
);
static(s1, "[6]s");
static(s2, "[3]s");
s2 <- s1[1..3];

```

where the notation [n..m] denotes a subrange, n and m being the limits. For a subrange the limits can be expressions. If the right hand limit is lower than the left hand limit the array members will be reversed.

Two other procedures which create types are **enum()** and **enumv()** which create enumerated types.

2.8 Target statements

Target statements create code which executes in the FPGA.

Target assignment statements use assignment operators which depend on the mode of the LHS variable. This is done so that the mode can be distinguished by sight rather than having to locate the variable creation procedure to determine its mode. These operators are -

- <- assign to a static mode variable
- <<- assign to a queue mode variable
- |- conditionally assign to an output mode variable

The target statement equivalent of an (immediate) **if ()** statement is **when ()**, which can also have an associated **else**. The true and false bodies can only be single target statements. These of course may be other target constructs such as **seq{} or par{}** , e.g.

```
when (enable)
    s <- 3;
else par {
    s <- 1;
    count++;
}
```

Different keywords for immediate constructs versus target constructs have been used to provide a visible distinction between immediate and target statements.

There is no target equivalent to the immediate **for ()** statement.

There are three target statement equivalents to an (immediate) **while ()** statement - **seq while () {}**, **par while () {}** and **sync while () {}**. The strange syntax is again intended to draw attention to the fact that these loops are target statements, not immediate. The **seq**, **par** and **sync** specify the loop body block type and the braces are required. As an example -

```
seq {
    count <- 10;
    par while (count != 0) {
        count--;
    }
}
```

will execute in 12 clock cycles. The initial assignment takes one cycle and a loop takes one cycle to exit in addition to the iterations (10 iterations in this case). Where the loop body has only one statement it does not matter whether **par** or **seq** is used.

There are also three **do while ()** loops - **seq do {} while ();**, **par do {} while ();** and **sync do {} while ();**, e.g.

```
seq {
    count <- 10;
    seq do {
        count--;
    } while (count != 0);
}
```

This loop will also take 12 clock cycles to execute.

2.9 Queue mode variables

A variable of mode **queue** is a queue whose default depth is two. A queue variable can be evaluated or assigned however any statement reading (evaluating) or writing (assigning) one or more queue variables will wait if a queue variable is not **available**. A queue variable is available for reading (evaluating) if it is not empty. A queue variable is available for writing (assigning) if it is not full. Otherwise queue variables are used in the same way as static variables. An expression can contain a mixture of static, value, queue and immediate mode variables. Any statement using that expression will wait if one or more queue reads within the expression are not available.

A queue write is done by a unique assignment operator -

```
queue({q1, q2}, "uint:8");
static(s, "uint:8");

q1 <<- q2 + s;
```

Queue variables may be any type.

Queue variables may be assigned at more than one place in the program however simultaneous assignments to the same element is erroneous and will give an unpredictable result³. Simultaneous assignments to different members of an array or different fields of a struct are usual. When a compound type (array or struct) of queue is assigned any fields or array members that are not explicitly assigned at the same time are assigned 0 for "int" or "uint" or false for "log".

Queue variables may be read at more than one place in the program and to describe the behaviour in that case we must first introduce the concept of a **module**.

A module is a block of code that has the following characteristics -

- it is a body of code that contains target execution threads each within its implied infinite loop
- it is a sink for queue reads

We looked earlier at the concept of statements being enclosed in infinite loops. It was stated that target statements that were not enclosed in any other structure were embedded in an infinite loop. A module is the environment in which all target execution threads originate. Each execution thread is initiated in an infinite loop, the loop iterating at the completion of its thread. Every 3PL program has a main module which is associated with the main source file, i.e. the source file passed to 3PL when it was called. This module⁴ contains the infinite loops enclosing statements in that source file.

It is crucial to establish sinks for queue reads, and a module is a queue sink. A queue which is read in more than one queue module will not pop the datum until all modules have executed a read. Thus if a queue is read in two modules if one module reads the queue but the other does not, the queue becomes unavailable in the first module and remains so until the second module has read the queue. When all modules have read a queue the datum is removed and if another datum is present in the queue the queue becomes available again to all modules. Thus a queue can have multiple sinks but each must read a datum before the queue pops that datum and makes the next (if any) available. This data synchronisation occurs automatically. Note that if all queue sinks (modules) read simultaneously as soon as read availability occurs then the queue can transfer data at the clock rate - there is no time penalty in this mechanism.

Within a queue sink (module) there can be multiple reads of the same queue but the effect these have on the queue depends on timing. If one or more queue reads of a queue occur simultaneously within a module that is regarded as a single read and each one receives the same datum. If multiple reads of queue within the same module occur on different clock cycles each of these is a separate read and they will read different data. Each read may then also wait for other queue sinks (modules) to complete their reads before advancing to the next datum.

There are two types of module -

- a named module which is called in the same way as a procedure, except that the call does not produce in-line code, and the definition has the keyword "module" instead of "procedure"
- an un-named module which is simply code enclosed in square brackets

A named module call or an un-named module instance do not produce code in-line at their location but instead establish a separate environment for execution threads. A module can contain any number of target statements and each statement will be enclosed in an infinite loop.

Named modules have their own local scope in the same way as procedures. Un-named modules also have their own scope, but they can also access the scope in which they are embedded. This is because they are not passed arguments so they need access to the local scope to access variables.

The following example shows the use of a named module with queue reads -

³There is a priority variable mode which can be used in arbitration code.

⁴it is called a **file** module which is described later

```

queue({q1, q2}, "uint:4");

q1 <<- .....

q2 <<- q1;
..... q2;

latchout(q1);

module
latchout (queue(p)) {
    var(opins, "[]str", {"A1", "A2", "B1", "B2"});
    static(s, type(p));
    output(, s, loc=opins);
    s <- p;
}

```

The main line program code is in the module associated with the source file (called a **file** module). This code writes some value to queue **q1**. Queue **q1** has two sinks - the current module, where it is read and assigned to **q2**, and the module **latchout** where it is read and assigned to a static variable which is connected to output pins. Thus the queue **q1** will expect reads from two modules, the file module and module **latchout** and will deliver the same datum to each regardless of when each sink decides to execute a read.

We could rewrite the above example using an un-named module -

```

queue({q1, q2}, "uint:4");

q1 <<- .....

q2 <<- q1;
..... q2;

[
    var(opins, "[]str", {"A1", "A2", "B1", "B2"});
    static(s, type(q1));
    output(, s, loc=opins);
    s <- q1;
]

```

Here we use variable **q1** directly where previously it was passed as an argument to the named module. We can do so because an un-named module can access variables in the scope in which it is enclosed (compare this with a seq block or a for loop where again statements within those constructs can access variables in the scope in which they are enclosed).

It can be seen that the timing of queue accesses is very important. The time at which queue reads or writes are attempted (depending on availability) may not be deterministic, i.e. it may depend on external input. To synchronise queue operations there is another target control structure **sync{}**. This is the same as **par{}** except that the parallel block will not begin execution until all queue reads and queue writes within the block can proceed. Thus a sync block containing multiple statements involving queue reads and queue writes guarantees that on entry these statements can execute in one clock cycle. A sync block can also be used to ensure that other target statements execute at the same time as queue operations, for example to count queue accesses -

```

queue(p, "uint:4");
static(count, "uint:32");

par {
    .... <- p;
    count++;
}

```

is not correct since the count may advance by one before the queue is available. If this is rewritten -


```

queue(p, "uint:8");
static(count, "uint:32");

sync {
  .... <- p;
  count++;
}

```

the count increment will occur in the same clock cycle as the queue read.

For target loop statements there is a **sync while** and **sync do while** where the loop bodies are sync blocks instead of par blocks or seq blocks.

Queue variables can be tested for availability. The unary operator < is true if its queue operand is available for reading. Similarly the unary operator > is true if its queue operand is available for writing. A queue can be examined instead of read by using the unary operator ?. This will return the current datum from the queue without popping the datum or interacting with any other queue sinks in any way. This can be used for example to select a datum from one of two queues without reading both, e.g.

```

queue({q1, q2, q3}, "uint:8");

sync {
  q1 <<- (?q2 > ?q3) ? q2 : q3;
}

```

which examines queues **q2** and **q3** and reads whichever has the largest datum, assigning it to queue **q1**. The sync ensures that **q2** and **q3** are available for reading and **q1** is available for writing before attempting to execute the statement (the ? unary queue examine operator does not wait for read availability and so could get a spurious value).

It can be seen that there are two, orthogonal, target execution control mechanisms in action. Statement execution is primarily initiated via threads within modules. For each statement, execution may be delayed until queues are available. Systolic code can be written where every statement has its own execution thread and execution depends solely on queue availability. This form of coding could perhaps be described as data flow. For example the following illustrates how this might appear -

```

type(t, "int:16");
var(ipins, "[3][8]str",
  { {"..", ....."},
    {"..", ....."},
    {"..", ....."}
});
var(iepins, "[3]str", {"..", .., ".."});
var(opins, "[8]str", {"..", .....});
str(oepin, "..");

queue({q1, q2, q3, p4, q5, q6, q7, q8}, "uint:8");

datain(q1 <- ipins[0], iepins[0]);
datain(q2 <- ipins[1], iepins[1]);
datain(q3 <- ipins[2], iepins[2]);

[p4 <<- q1 + q2;]
[q5 <<- q1 - q2;]
[q6 <<- p4 - q3;]
[q7 <<- q5 + q3;]
[q8 <<- q6 + q7;]

dataout(q8, opins, oepin);

module
datain (queue(p) <- var(pins, "[ ]str"), str(epin)) {
  input(data, type(p), loc=pins);
}

```

```

    input(enable, "log", loc=epin);
    static(d, type(p));
    static(e, "log");
    d <- data;
    e <- enable;
    when (e)
        p <<- d;
}

module
dataout (queue(p), var(pins, "[str]", str(epin)) {
    static(data, type(p));
    static(enable, "log");
    data <- p;
    enable <- <p;
    output(, data, loc=pins);
    output(, enable, loc=epin);
}

```

This example takes three 16 bit inputs and performs a series of pipeline operations upon them until they emerge combined into a single output. The inputs and outputs are latched using static variables - this is the safest way of clocking data on and off chip ⁵. It may not be strictly necessary but for safety each of the queue assignment statements has been placed in an un-named module to ensure that multiple queue reads are handled as separate queue sinks. In fact since queue reads would occur simultaneously on queue availability in each case this is not necessary, but is good practice and it saves having to think it through. Notice that no execution control structures were needed for this processing pipeline. Because the pipeline has data paths which diverge and then recombine there is the potential for parallel alternative data paths to deleteriously interact. Alternative pipeline data paths can interact if they do not have identical latency (in the example above they do). The following example has this problem -

```

queue({q1, q2, q3, p4}, "uint:8");

mod1(q2 <- q1); // assume this module has 3 pipeline stages
mod2(q3 <- q1); // assume this module has 6 pipeline stages
p4 <<- q2 + q3;

```

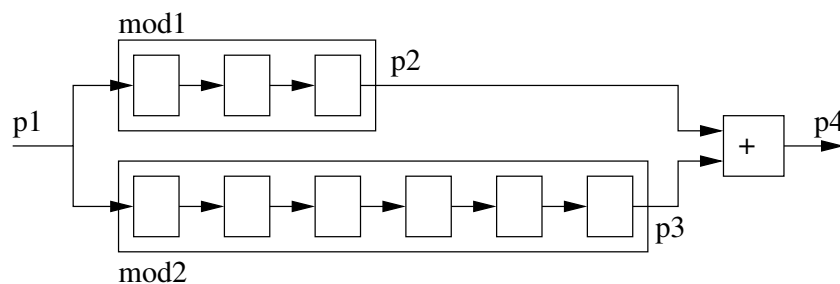


Figure 2.1: alternate pipeline paths

Here the first shorter path will fill queue **q2** after 5 clock cycles (assuming **q1** is always available). The last statement cannot execute because **q3** is not yet available, the second pipeline path being longer. The buffer in q2 is now full so it becomes unavailable for writing and the same will apply through the first pipeline. But this means that the first pipeline will stop reading **q1** which means that the second branch has not yet reached the output but cannot continue to read **q1**. The data already in the second pipeline will continue to move through the queue but no more input to that queue will occur. When the data in the second pipeline reaches the output the last assignment statement can now execute until it exhausts the second pipeline. The two pipeline branches are interacting and this will slow throughput to about half. This interaction can be eliminated by making the queue buffer depth of queue **q2** large enough to keep accepting data until the second pipeline fills. The example would be fixed by -

⁵The registers will be located in the I/O pads by default if the correct Xilinx map option is used. Alternatively the iob attribute for the static variables could be set true.

```

queue({q1, q2, q3, p4}, "uint:8");
attributes(q2, depth=16);

mod1(q2 <- q1); // assume this module has 3 pipeline stages
mod2(q3 <- q1); // assume this module has 6 pipeline stages
p4 <<- q2 + q3;

```

The buffer size is rounded up to various powers of two. Which powers of two, and the maximum allowed, depend on the FPGA - see the 3PL manual.

2.10 Target exceptions

3PL is designed for the generation of code which will process continuous data streams. If an exception condition arises it may prove necessary to prematurely exit from a conditional or loop block. In particular a statement within a block may be waiting for queue availability where the exception condition is such that this will not occur. A way is needed to force an exit from loops, code blocks or blocked statements.

The following target constructs may have an **unless** clause attached -

- **seq** {}
- **par** {}
- **sync** {}
- **when** ()
- **when** () else
- **seq while** () {}
- **par while** () {}
- **sync while** () {}
- **seq do** {} while ();
- **par do** {} while ();
- **sync do** {} while ();

For example a **seq** block may have some queue reads which we want to terminate when an exception condition occurs -

```

static({s1, s2}, ...);
queue({q1, q2}, ...);
value(e, "log", ...);

seq {
    s1 <- q1;
    s2 <- q2;
} unless (e)

```

The **seq** is at the top level and will execute in an infinite loop. If **e** is true the **seq** will immediately terminate on the next clock edge. If any assignments in the **seq** were ready to execute on that clock edge they will do so. Following termination the normal sequence of execution of the thread will continue, in this case the **seq** immediately executing again. This is probably not what we want - there is usually some corrective code we wish to execute. To do this there is an optional execution block associated with the **unless**, e.g. -

```

static({s1, s2}, ...);
queue({q1, q2}, ...);
value(e, "log", ...);

seq {
    s1 <- q1;
    s2 <- q2;
} unless (e) {
    reset(q1, q2);
}

```

The exception code body is in braces and may contain a number of statements, however it can only contain one target statement. That target statement may of course be a **seq {}**, **when ()**, **par while () {}** etc. The exception code block is executed after the terminated block, following which normal thread execution will continue.

Target statements with **unless** clauses cannot be nested⁶.

2.11 combinatorial output memory

Combinatorial output memory can be used as a combinatorial function on read, i.e. when an address is present at the input the memory output will reflect the value stored at that address after a short delay. A memory read does not require a clock. A memory write is clocked. In the case of Xilinx FPGAs combinatorial output memory is distributed CLB RAM.

Combinatorial output memory is accessed by means of variables whose mode is **cmemory**. These can only be created by the inbuilt procedure **cmemory()**, e.g.

```
int(ports, 2);
str(dtype, "uint:8");
str(atype, "uint:6");
var(init, "[uint", {..., .., .., .., .., ..}, ..});
cmemory(m1, ports, atype, dtype, init);
```

The first argument to **cmemory()** is the identifier for the created variable. The second argument is the number of ports which for Xilinx can be 1 or 2. The address type and data type follow. These can be any type at all but the address type is restricted to a maximum width of 9 bits. The last argument is initialisation data for the memory and is optional. The initialisation argument is shown in the example as an array variable but it could be a list of values in braces. Where an array variable is used it could of course contain values computed in an immediate execution loop, e.g. values implementing a mathematical function or a translation table.

If the memory has more than one port the ports can be accessed independently. In the case of Xilinx the second port, port 1, is read-only. The ports can be accessed at multiple places in the code but the code must be written to ensure that access on a port is mutually exclusive as access conflicts are not detected or resolved.

Because the memory behaves as a combinatorial element on reading, the memory output is accessed via an inbuilt function, e.g.

```
static(d, dtype);
static(a, atype);
static(l, "log");

when (l)
    d <- cmemoryread(m1, 0, a) + 3;
```

The memory data is available in the same clock cycle that the read occurs because it does not have to wait for a clock - the address propagates through the memory emerging as the stored data element at the output.

This example uses static variables. When **l** is true **d** is assigned the sum of the memory output from port 0 at address **a** and the constant 3. We could also use queue variables, e.g.

⁶Nesting of target statements with **unless** clauses may be allowed in the future if and when I can figure out the complicated logic to do it.

```

float(gamma, 0.45);
float(gain, pow(256.0, 1.0 - 1.0 / gamma));
str(dtype, "uint:8");
var(init, "[256]uint");
for (int(i) ; i<256 ; i++)
    init[i] = round(gain * pow((float)i, 1.0 / gamma));
cmemory(gamma_correct, 1, dtype, dtype, init);
queue({q1, q2}, dtype);

.
.
q2 <<- cmemoryread(gamma_correct, 0, q1);
.
.

```

where the memory read occurs whenever queues **q1** and **q2** are available (writing of **q1** and reading of **q2** are not shown in the code fragment). This example purports to show an 8-bit pixel gamma correction (I may have the function wrong, but you get the idea). The gamma correction lookup table is a **cmemory** mode variable with one port. The address input and data output are each 8 bits wide. The initialisation array has its contents computed during immediate execution using the value initially assigned to floating point variable **gamma**. The floating point values are rounded back to the nearest integer for assignment to the initialisation array⁷. A variation is to use a two-port **cmemory** and allow table entries to be overwritten from an external CPU when required. The ZYNQ platform and various boards with a serial interface provide this facility and the above code might then be written as -

```

class(cpu, cpu_api());
str(dtype, "uint:8");
cmemory(gamma_correct, 2, dtype, dtype); // 2 ports, separate for read and write
queue({wdata, waddr, q1, q2}, dtype);
cpu.write_queue(wdata <- 1);
cpu.write_queue(waddr <- 2);

cmemorywrite(gamma_correct, 0, waddr, wdata);

.
.
q2 <<- cmemoryread(gamma_correct, 1, q1);
.
.

```

The library module **cpu_write_queue()** allows a CPU coupled to the FPGA to write to a queue variable in the FPGA. The input arguments **1** and **2** are memory-mapped bus addresses for the bus interface to the FPGA. In this example the **cmemorywrite()** inbuilt procedure will write to the memory when queues **waddr** and **wdata** are both available.

Of course we could combine a more complicated execution thread with queue variables -

⁷3PL provides target floating point format but arithmetic operations are logic-intensive and must be pipelined, i.e. FP operations cannot be done in one step in one clock cycle. There is a library of pipelined floating point operations.

```

    struct(s,
        sum,      "uint:16",
        count,    "uint:16"
    );
    queue(dp, s);
    queue(ap, atype);
    static(count, "uint:16");
    static(accum, "uint:16");

    seq {
        sync while (ap != 0) {
            accum <- accum + cmemoryread(m1, 0, ap);
            count++;
        }
        sync {
            dp.sum <- accum;
            dp.count <- count;
            count <- 0;
            accum <- 0;
        }
    }
}

```

Here we loop reading memory and accumulating the sums of the values until the queued address is zero, at which point the sum and count are written to a queue (whose type is a struct containing fields for both values) and the static variables are cleared for the next iteration. The seq block then repeats.

In this last example if the input queue **ap** transmits all zeros the output data stream **dp** will be at the same rate, otherwise the output will be at a lower rate which depends on the length of non-zero sequences. This is a characteristic of a systolic network with buffered queues, compared with a normal signal processing lock-step approach, that each datum can proceed through the network at a rate determined by producers and consumers at each node of the network. Where all inputs to the network have data available and each node can execute in one clock cycle then data can flow at clock rate. Where data input is sporadic or at sub-clock-rate, or multiple sequential execution cycles are needed at some nodes, the data throughput will be slower.

2.12 registered output memory

Registered output memory uses a clock for both read and write. On executing a read the data appears at the output following the next clock edge. In the case of Xilinx FPGAs registered output memory is block RAM.

Because the memory output is delayed one clock cycle on a read, registered output memory reads must be pipelined, otherwise a read will take two clock cycles.

Registered output memory is accessed by means of variables whose mode is **rmemory**. These can only be created by the inbuilt procedure **rmemory()**.

In the previous section we had an example where data bytes were streamed through a look-up table for gamma correction -

```

    type(dtype, "uint:8");
    cmemory(gamma_correct, 2, dtype, dtype);
    queue({wdata, waddr, q1, q2}, dtype);

    .
    .
    q2 <- cmemoryread(gamma_correct, 1, q1);
    .
    .

```

We could convert that example to use registered output memory as follows -

```

type(dtype, "uint:8");
rmemory(gamma_correct, 2, dtype, dtype);
queue({wdata, waddr, q1, q2}, dtype);

.
.
seq {
  rmemoryread(gamma_correct, 1, q1);
  q2 <<- rmemoryout(gamma_correct, 1);
}
.
.

```

The seq loop will first attempt to execute a memory read from port 1 taking the address from queue **q1**. If **q1** is not available then the read will wait until it is. When the read has occurred the seq block will then execute the assignment of the current output value of the memory to queue **q2** (if **q2** is not available it will wait). The inbuilt function **rmemoryout()** gets the contents of the output register of the block RAM for the specified port. This value will remain in the output register until another memory read is executed. This code example does not pipeline the memory reads and hence takes two cycles per access.

Note that in this example **rmemoryread()** is a procedure, not a module, so the module in which **q1** is being read is the file module. Since there is only one sink for queue **q1** we do not need to worry about ensuring that bits of the code are in separate modules and hence all the code is in the file module. If queue **q1** was also read by other unrelated code we would have to ensure that the reads were in separate modules so that the independent reads were handled correctly.

To pipeline the above example the memory read must be separated from the following memory output value assignment. The assignment is dependent on a previous memory read so we need to signal that this has occurred. This could be done with another queue variable -

```

type(dtype, "uint:8");
rmemory(gamma_correct, 2, dtype, dtype);
queue({wdata, waddr, q1, q2}, dtype);
queue(read, "log");

.
.
sync {
  rmemoryread(gamma_correct, 1, q1);
  read <<- true;
}

when (read)
  q2 <<- rmemoryout(gamma_correct, 1);
.
.

```

but this code is not correct! If queue **q2** becomes unavailable for writing the assignment to it will wait but the next memory read may go ahead, overwriting the previous registered memory output value. We could acknowledge back via another queue but that would introduce an extra clock cycle.

We need to execute the memory read and assign the result in the one clock. Since the read must occur in a clock cycle previous to the assignment to **q2** the two operations can be staggered by doing an initial read prior to entering the loop.

```

type(dtype, "uint:8");
rmemory(gamma_correct, 2, dtype, dtype);
queue({wdata, waddr, q1, q2}, dtype);

.
.
seq {
  rmemoryread(gamma_correct, 1, q1);
  sync {
    q2 <<- rmemoryout(gamma_correct, 1);
    rmemoryread(gamma_correct, 1, q1);
  }
}
.
.

```

here -

- the first statement in the **seq** block executes the first read
- the second statement in the **seq** is an infinite loop
- the loop sync block will not execute until **q1** and **q2** are available (**q1** is not empty and **q2** is not full)
- **q2** is assigned the previous result and the next read are performed simultaneously
- the **seq** block never completes

Another approach is to create a module to handle pipelined rmemory reads -

```

type(dtype, "uint:8");
rmemory(gamma_correct, 2, dtype, dtype);
queue({wdata, waddr, q1, q2}, dtype);
queue(read, "log");

.
.
rread(q2 <- gamma_correct, 1, q1);
.
.

```

The code for the module might be -

```

global module
rread (queue(data) <- rmemory(m), int(port), queue(addr)) {
  static(hold, "log");
  when (!hold || >data)
    rmemoryread(m, port, addr);
  when (hold)
    data <<- cast(rmemoryout(m, port), type(data));
  hold <- <addr || hold && !>data;
}

```

We have declared the module as **global** so that it can be used anywhere - the scope of variables, modules, procedures and functions is discussed in a later section. In this module a static boolean variable **hold** is used to synchronise the memory reads and output queue assignments. When **hold** is true a value is held in the memory output register waiting to be assigned to the output queue. There are three statements in this module, each one executing independently in its own implicit infinite loop.

The first statement executes a memory read if the hold flag is false (there is currently no unassigned registered memory output value) or if the output queue is available for writing (the **>** unary prefix operator returns true if the queue variable to which it is applied is available for writing - there is a similar operator, **<** for read availability). If **<q1** is not true the statement will wait until it is true.

The second statement assigns the registered memory output value to the output queue when **hold** is true. If the output queue **data** is not available this statement will wait until it is available then execute the write.

The third statement continually updates the value of **hold** - it is assigned true if an input is available or it is already true and the output queue is not available, otherwise it is assigned false.

This module is already included in a 3PL library.

2.13 Clocks

A clock is another mode of variable - mode **clock**. A Clock variable can represent -

- a clock input from an input buffer driven from an FPGA pad
- a clock signal between a digital clock manager and a global clock buffer
- a clock signal distributed over a clock net driving FPGA synchronous logic

A clock mode variable is created by calling the inbuilt procedure **clock()**. For an input clock signal various attributes such as the frequency and the FPGA pin(s) must be supplied. For example -

```
str(c100_pin, "E17");
clock(c100, frequency=1.0e8, loc=c100_pin);
```

If the clock variable does not represent the output of an input clock buffer then there is no location attribute.

If a clock is passed as an argument to a module, procedure or function then the parameter is not given any attributes, e.g.

```
procedure
p (... <- clock(c)) {
    .
    .
}
```

To generate a clock a digital clock manager (DCM) is used. The following code illustrates this -

```
str(c100_pin, "E17");
clock(c100_in, frequency=100.0, loc=c100_pin, ibufg=true);
clock({global.c100, c100_raw});
clkdcn(clk0=c100_raw <- clkkin=c100_in, clkfb=c100);
clockbuffer(c100 <- c100_raw);
```

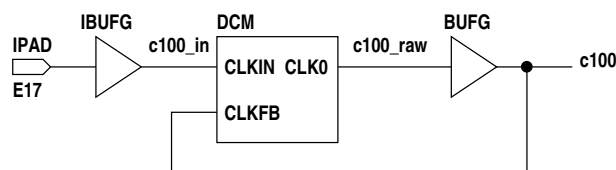


Figure 2.2: digital clock manager

Creating c100 with global scope ensures that it is globally accessible where **c100_in** and **c100_raw** may have local scope. Library procedures **clkdcn** and **clockbuffer** both propagate the frequency and period attributes from inputs to outputs so we don't have to assign these attributes to **c100_in** and **c100_raw**.

Before any logic involving synchronous elements is generated by 3PL the clock domain must be established. This is done by calling the inbuilt procedure **useclock()**, e.g.

```
useclock(c100);
```

When the clock domain is changed using **useclock()** the previous clock domain is pushed onto a stack. If **useclock()** is called with no argument the previous clock domain is popped off the stack and again becomes the current clock domain.

The clock domain used by variables is established when they are assigned or evaluated, not when they are created.

For a static variable the clock domain in use when the variable is assigned a value or is evaluated establishes the clock domain for the variable. Any other assignments or evaluations of that variable must be in the same clock domain.

For a queue variable the read and write clock domains may be different. The write clock domain is established by the clock domain in use when the first write is encountered and all subsequent writes must be in the same clock domain. Similarly the read clock domain is established by the clock domain in use on the first read of the queue and all subsequent reads must be in the same clock domain. If the read and write clock domains are the same, which is the usual case, the queue is called **synchronous** and the queue buffer depth will be 2 by default. If the read and write clock domains differ the queue is called **asynchronous** and the default buffer depth is 16. Resynchronisation of queue data between the two clock domains is handled automatically by the asynchronous queue.

In general in a statement the clock domain in use must agree with the clock domain of variables that are assigned or evaluated in that statement. Procedure and module calls are an exception in that the clock domains are checked in the body of the called module or procedure but not when arguments are evaluated and passed to parameters.

The correct way to pass data from one clock domain to another is via an asynchronous queue, however there are two functions, **accept()** and **sample()** which allow mixing of clock domains at the programmer's peril. **accept()** simply overrides the clock checks and can be used if data is seldom changed and the risk of data corruption is acceptable (safe for single bit data except for the very small probability of getting a metastable state. For single bits an inbuilt function **resync()** can be used to do this properly). **sample()** preserves data integrity by resampling it into the current clock domain, however since the clock frequencies differ a datum may be lost or duplicated (but will not be corrupted).

2.14 Modules

Modules and procedures were briefly mentioned in the context of queue variables. A module has the important property that it delineates a sink for queue variables. When a Queue has more than one sink each sink must execute a read on the queue before the datum is popped and the next becomes available. Multiple reads within a sink are treated as a single read if they occur simultaneously, otherwise they are separate reads. A queue has two sinks if the two code sections which read the queue are in different modules, For example -

```
queue({p, q1, q2}, t1);

.
p <<- .....;
m1(q1 <- p);
m2(q2 <- p);
.

module
m1 (queue(pout) <- queue(pin)) {
    pout <<- p - 3;
}

module
m1 (queue(pout) <- queue(pin)) {
    pout <<- p * 2;
}
```

Modules are declared with output parameters followed by input parameters, these lists being separated by <- (the left arrow can be omitted if there are no output parameters). Each parameter is a call to an inbuilt variable creation procedure. References to parameters in the module code body are replaced by the arguments passed in the call. If a parameter is not passed an argument then it becomes a local variable. For this reason parameters can have initial values which become the variable initial values if no matching argument is provided. There is an inbuilt procedure **matched()** which can be used to test if a parameter has been passed an argument. Arguments are normally matched to parameters by the order

in which both occur however an alternative is to use **parameter=argument** pairs in the call. This is useful where there are many parameters and only a few are to be passed arguments. A combination of both modes is allowed as long as no positional arguments follow any parameter=argument pairs.

Parameters can have target type widths missing, dimensions missing or even a type missing or a mode missing. The missing elements will be filled in from the argument. Where parameters have types and dimensions these will be checked against the argument type and dimensions. An exception is target numeric type widths which allow a mis-match, resulting in truncation or padding.

A module is a block of code which can be called. Module calls are only done during immediate execution. There is no code in the FPGA which executes any sort of shared code call! When a module is called however any target statements that it contains are established as independent execution streams. A module call would not usually be made within a target construct such as a **when** or one of the target loops. In contrast a procedure containing a target statement called within a target construct will result in that statement appearing in-line where the call is made.

Declaring a module and calling it can be verbose and inconvenient, especially if the code blocks are small and the modules are only called once. Instead an un-named module can be used. An un-named module is code enclosed in square brackets. It is not called as such but simply appears in-line, though it is executed as a distinct and separate module from that within which it is nested. The above code could be written

```
queue({p, q1, q2}, t1);

.
p <<- .....;
[q1 <<- p - 3;]
[q2 <<- p * 2;]
.
```

where now the assignments to queues **q1** and **q2** are in separate modules - the code in brackets. Un-named modules do not have calls and hence there are no arguments and parameters. The scope of an un-named module is like that of a code block in braces. Variables created inside the brackets are in the scope of the un-named module. But code within the un-named module can access the scope in which it is enclosed - in the case above **p**, **q1** and **q2** occur in the scope outside the two un-named modules but are accessible within the un-named modules.

Note that in this example the assignment to **q1** will occur when both **q1** and **p** are available. Similarly the assignment to **q2** will occur when both **q2** and **p** are available. If **q1** and **q2** become available at different times the assignments will not occur simultaneously, but each will read the same datum from **p** since **p** will not pop its datum until both reads have occurred (the first read to occur will not see **p** become available again until after the second read has occurred). If instead the code was

```
queue({p, q1, q2}, t1);

.
p <<- .....;
q1 <<- p - 3;
q2 <<- p * 2;
.
```

it would function correctly if all three queues were available, both **q1** and **q2** being assigned the datum from **p**. If however **p** and **q1** were available but **q2** was not, the assignment to **q1** would take place and the datum would be popped from **p**. When **q2** became available it would read the next datum from **p**. Another solution might be

```
queue({p, q1, q2}, t1);

.
p <<- .....;
sync {
    q1 <<- p - 3;
    q2 <<- p * 2;
}
```

```
}
.
```

which inhibits both assignments until all three queues are available.

2.15 Modules, procedures, functions and classes

3PL provides user-defined procedures and functions.

Briefly -

- module, procedure, function and class calls are only executed during immediate execution - the calls are not executed in the FPGA
- modules and procedures are called using the same syntax and semantics
- modules and procedures are declared the same way except for the keyword **module** or **procedure**
- a module may contain any number of target statements whereas a procedure can contain at most a single target statement (but which may be a seq, par sync etc.)
- each target statement in a module is established in its own execution thread whereas the single target statement in a procedure is executed in-line at the location of the call
- modules and procedures with no target statements are functionally identical
- modules delineate queue sinks but procedures do not
- functions can only appear within the context of an expression - they cannot be called using module or procedure call syntax
- functions can return immediate or target values
- functions cannot contain in-line target statements - they can contain module calls
- classes are called in a similar manner to modules or procedures but they return a value, a class instance
- a class cannot contain in-line target statements

A procedure call and definition are illustrated by the following example -

```
static(s, "uint:8");    // unsigned integer of 8 bits
static(v, "[2]uint:8"); // array of 2 unsigned integers of 8 bits

p(s <- v[0], v[1] + 2, 3);

procedure
p (static(os, "int:8") <- value({iv1, iv2}, "int:8"), int(i)) {
    os <- iv1 + (iv2 >> i);
}
```

This procedure, **p**, has one output parameter, the 8-bit integer static mode variable **os**, and three input parameters, two value variables and an immediate integer variable. Parameters are created by using the same variable creation procedures as before.

procedures have their own variable scope so any variables created in the body of the procedure (and any unmatched parameters) are in the local scope of the procedure and cannot be accessed from outside the procedure, including any deeper calls to procedures (or functions).

The convention for both procedure definitions and calls is that parameter and argument lists consist of outputs, <- and then inputs. If there are no output parameters in the definition the <- can be omitted. Similarly if there are no output arguments in the call the <- can be omitted. For value and immediate mode input parameters the argument is copied to the parameter by assignment, i.e. it is call-by-value. For other input parameter modes and for all output parameters the parameter 'points' indirectly to the argument variable, i.e. call-by reference.

A value mode input parameter can be passed an argument which is an expression, and that expression could be immediate mode (i.e. a constant in the target). Similarly an immediate mode input parameter can be passed an argument which is an expression, however that expression must be immediate mode, not a target mode. In the example above the first input parameter is value mode and has been passed one member of a static mode array. The second value mode parameter has been passed an expression which is the sum of a member of a static mode array and a constant. The body of the procedure adds the

first parameter to the second parameter right shifted by the third and assigns the result to the output static variable.

For the other cases these are call-by-reference and any reference to the parameter accesses the actual variable passed as an argument. Note that where the type is an array the argument may be a sub-array, e.g. -

```
static(v, "[6]uint:8");

p(v[1..3]);

procedure
p (static(p, "[3]int:8")) {
    ...
}
```

This applies to call-by-value and call-by-reference and to both input and output parameters.

If parameter dimensions are given the argument dimensions must be the same, however we could omit the parameter dimensions to make the procedure more flexible -

```
static(v, "[6]uint:8");

p(v[1..3]);

procedure
p (static(p, "[ ]int:8")) {
    println(dimensions(p)); // print the number of dimensions of p
    println(dimension(p, 0)); // print the first dimension of p
}
```

which would print

```
1
3
```

Arguments may be omitted in which case the unmatched parameters become local variables in the procedure. Input parameters may be given initial values so that these values will be used if no matching argument is passed.

Arguments are assigned to parameters in left to right order with inputs assigned before outputs. This means that when a parameter is created the type argument to the inbuilt variable creation procedure could use values passed to parameters to the left.

A parameter can be tested to determine if an argument has been passed to it by means of the inbuilt function **matched()** which returns true if an argument has been passed.

In addition to the usual matching of arguments to parameters by position the keymatch format may be used where each argument is identifier=value, the identifier being that of the parameter. The keymatch arguments can be in any order but arguments are assigned to parameters in parameter order. Positional arguments can precede keymatch arguments but cannot follow them.

Considerable flexibility is allowed in argument passing. Missing parameter dimensions can be inferred from the argument dimensions and target type widths need not be the same (truncation or padding will result if not).

Procedures may contain immediate statements however they may only contain at most a single in-line⁸ target statement! This statement may be a control structure containing other target statements nested to any depth. The following procedure definition -

⁸This qualification will be explained later.

```

procedure
p () {
  a <- 3; //
  b <- 4; // ILLEGAL!
}

```

is not allowed. It is not clear what is to be done with two target statements. If `p()` is called at the top level you might expect the assignments to be placed in their own infinite loops. If called in a `par` or `seq` block then perhaps they are added to the block. To avoid this uncertainty procedures can only contain one target statement but of course this can be a **par{}**, **seq{}**, **when**, **seq while**, **par while** etc.

Functions differ from procedures in that -

- there are no output parameters
- there are no in-line target statements
- a function call appears where a value is required, not on a line by itself as for a procedure call.
- there is a return statement to return a value - this can only be immediate mode or value mode

Functions have local variable scope in the same way as procedures.

Functions cannot have target statements but they can have target expressions and return values. The following illustrates a trivial function returning a target mode value -

```

function
add (value({a, b}, "int:8")) {
  value(v, "int:9")
  v := a + b;
  return(v);
}

```

Note that the declaration does not define the return type, so a function could have several different return types governed by conditional statements.

The trivial example above is verbose and could be recoded as -

```

function
add (value({a, b}, "int:8")) {
  value(v, "int:9", a + b)
  return(v);
}

```

or simply -

```

function
add (value({a, b}, "int:8")) {
  return(a + b);
}

```

It was previously stated that procedures could contain at most one in-line target statement and functions could not contain any. The term "in-line" excludes statements contained in modules within a procedure or functions. The reason is that a module is an independent block of code that originates target execution threads. If a procedure contains a target statement that statement is in-line, i.e. that statement will be inserted into the target execution thread in the context where the procedure is called. When a module is called it establishes a completely new context for establishing execution threads unrelated to the context in which it is called. As an example suppose we wish to write a function which returns a value which is the value of the argument delayed by a number of clock cycles. The function cannot contain target statements so it seems that we could not write this, but a function can contain a module which in turn can contain target statements. We could write the function as follows -

```

global function
del (value(in), int(n, 1)) {
  type(t, type(in));
  static(d, "["+n+"]t");

  [
    d[0] <- in;
    for (int(i, 1) ; i<n ; i++) // loop for i from 1 to n-1
      d[i] <- d[i-1];
  ]

  return(d[n-1]);
}

```

The first input parameter has no type specified so it will use the type of its argument. The second argument is immediate and is the delay count. It is set to default to 1 if it is not given an argument. Note that the array type is constructed as a string expression to insert the dimension gained from the second parameter. The value returned for the function is value mode and is the value of the last member of the static variable array. Note that the loop starts at index 1 as the loop variable *i* is initialised to 1.

The un-named module can access the scope of the function in which it is contained. Note that the assignments to variable *d* are not in a par block. Since we are at the top level of a module each assignment statement will execute in parallel in its own infinite loop at clock rate⁹.

Procedure and function calls can be recursive. This is a very powerful mechanism allowing very simple and concise code to be written. Recursion often offers solutions that cannot be written iteratively. Any target code is duplicated on each call.

We could instead write the above example recursively -

```

global function
del (value(in), int(n, 1)) {
  if (n == 0)
    return(in);
  static(d, type(in));
  [d <- in;]
  return(del(d, n-1));
}

```

Each call to the function delays zero or one clock cycles. If the count is non-zero it delays the input by one cycle then calls itself with the result and a count minus 1. The assignment again is within an un-named module but now it is a single statement. There will be *n* un-named modules created, each with one assignment.

When writing iterative code note that it can be convenient to use a value variable which is re-assigned on every iteration. The following function ORs the bits of its argument.

```

global function
orbits (value(v, "bits:")) {
  int(n, dimension(v));
  value(ret, "log", cast(getbit(v, 0), "log"));
  for (int(i, 1) ; i<n ; i++)
    ret := ret || cast(getbit(v, i), "log")
  return(ret);
}

```

The value variable *ret* is initially the LSB of the input but each time around the loop it is re-assigned the OR of itself and the next successive bit from the input. Finally it is returned as the value of the

⁹The code generator will eliminate the loop and simply control the enables to the register flip-flops. How and when these are asserted depends on how execution in the FPGA is started. It can be started by an external signal or it can start spontaneously after configuration - see target-execution-start in the manual index. A static variable can have its 'continuous' attribute set to true in which case the enables will be tied high.

function. Note that the function parameter has no data width - this is derived from the argument. We could make two changes to this example. First we could generalise it so that it would accept any type for the input. Secondly we could avoid the use of the library function `getbit()` by casting the input into an array of "log".

```
global function
orbits (value(v)) {
    int(n, size(v));
    value(vx,, forcecast(v, "["+n+"]log"));
    value(ret, "log", vx[0]);
    for (int(i, 1) ; i<n ; i++)
        ret := ret || vx[i]
    return(ret);
}
```

The library function `forcecast()` is a more extreme version of inbuilt function `cast()`. Where a value variable is given an initial value the type argument can be omitted since the type can be inferred from the initial value.

Module, procedure, function and class definitions may be nested, so a function or procedure may be defined within another function or procedure and hence can only be called by the parent function or procedure. By this means the scope and hence accessibility of functions and procedures can be limited and procedure and function identifiers can be re-used locally without ambiguity.

It may be apparent by now that modules, procedures, functions and classes are only called during immediate execution - they are not called within target code. There is no equivalent to a procedure or function call in the FPGA code. Indeed this would not make sense as it implies shared code and sequential access which defeats the purpose of using an FPGA for parallel execution. Procedures and functions are useful where the same code is needed multiple times, be it immediate code or target code, however the procedure and function calls only take place during immediate execution.

Module, procedures, functions and classes accept a "varargs" parameter which will match any number of arguments of any type or mode. This must follow normal parameters if used.

2.16 program structure and scope

There are a number of variable scopes -

- global scope
- file scope
- local scope
- block scope

Global scope is accessible anywhere. Variables are created in global scope if the variable identifier has "global." prepended when the variable creation inbuilt procedure is called. If the same variable identifier is used in more than one scope the global version can be explicitly selected by prepending "global.". Variables are never placed in global scope by default.

When the 3PL compiler begins to compile source code from the source file passed to it on the command line it creates a module to contain that code. That module is called a **file** module. A file module does not have an identifier since it is not explicitly called. Source code within that source file forms the code body of the file module and this is executed by the interpreter following compilation. Variables created immediately within the file module (i.e. not within nested blocks) by default are contained in the file module scope. If variables are created within code blocks nested within that file scope but their scope is required to be the file scope then their identifiers must have "file." prepended. Similarly if necessary a variable reference can be disambiguated by prepending "file.".

3PL allows other source files to be nested within the command line source file. There are two mechanisms for doing that.

A source file can be nested within another by the construction


```
source "file.3pl"
```

This simply inserts the text of the file at that point in the current file. This can be done to any depth. This is equivalent to `#include` in C. If instead the code

```
module "file.3pl"
```

is used the file is inserted as before however a new file module is created for which the new source file supplies the code body. again this can be done to any depth.

Any variables created immediately within the new file module are by default contained in the file scope for that module. Again this can be made explicit by prepending "file." to the variable identifier.

File scope is useful for hiding variables away from application code without the need to call a module for initialisation. For example a hardware interface is usually coded as a file module. The source file for this might be "gadget.3pl". By including this using the statement

```
module "gadget.3pl"
```

a file module will be created. Top level code within this file module will be interpreted at the point at which occurs in the enclosing source. This code will create variable, such as inputs, outputs, FPGA pin names etc. and by default these will be in the file module scope. The module will also define modules, procedures and functions which provide the interface to the gadget. These can access the variables in file scope but the callers to these cannot access the variables. Hiding variables in this way both protects the interface from incorrect usage and also avoids accidental conflicts with variable identifiers.

Within a module other than a file module, a procedure or a function variables created outside of nested code blocks are by default local to the module, procedure or function. They can be accessed anywhere within the module, procedure or function but not from any nested calls or from anywhere else.

Within a code block in braces any created variables are in the scope of the block. A problem arises where variables are created conditionally. The code

```
procedure
p (log(1)) {
  if (1)
    int(i);
}
```

does what one would expect - creates the integer `i` in the scope of the enclosing procedure. If this is changed to

```
procedure
p (log(1)) {
  if (1) {
    int(i);
    float(f);
  }
}
```

two variables are created, but they are in the scope of the `if` code block in braces, not in the scope of the procedure. To fix this the code should be changed to `procedure p (log(1)) if (1) int(local.i); float(local.f);` which creates the variables in the procedure scope.

Suppose one wishes to write a procedure which creates variables. The code

```
procedure
myvar (str(name), log(signed)) {
  if (signed)
    static(name->, "int:8");
  else
    static(name->, "uint:8");
}
```

will create a variable whose identifier is derived from the string argument by using the indirect operator. The problem is that the variable is created in the local scope of the procedure, not the scope in which it is called. To achieve the latter it can be changed to

```
procedure
myvar (str(name), log(signed)) {
    str(n, "calling." + name);
    if (signed)
        static(n->, "int:8");
    else
        static(n->, "uint:8");
}
```

or more simply

```
procedure
myvar (str(name), log(signed)) {
    if (signed)
        static(("calling." + name)->, "int:8");
    else
        static(("calling." + name)->, "uint:8");
}
```

or even more simply

```
procedure
myvar (str(name), log(signed)) {
    static(("calling." + name)->, signed ? "int:8" : "uint:8");
}
```

When the interpreter encounters a reference to a variable it first searches the current code block, then back out through nested code blocks until it reaches the local scope of the enclosing module, procedure or function. This behaviour is the same as exhibited by other common programming languages such as C or Java. If the variable has not been found the interpreter looks in the file (module) scope. Finally it looks in the global scope. If the variable identifier has "global.", "file.", "calling." or "local" prepended the interpreter explicitly expects to find the variable in the named scope.

When a non-inbuilt module, procedure or function is called, the interpreter searches from the current local scope back through the scopes of nested module, procedure and function **declarations** (NOT calls) until it finds a match. If that fails it looks in global scope. Thus, like variables, module, procedure and function identifiers can be used in different scopes as long as there is no conflict. A module, procedure or function definition occupies the scope of the module, procedure or function in which it is defined unless the keyword **global** precedes the **module**, **procedure** or **function** keyword, in which case it has global scope, i.e. it can be called from anywhere.

Nested declarations allow modules, procedures and functions to be limited in scope so that they can only be called locally. This also allows module, procedure and function identifiers to be re-used locally.

2.17 Selectvalue mode

Selectvalue mode creates busses using three-state buffers. A selectvalue variable can have a number of conditional assignments. The following is an example using this mode -

```

selectvalue(bus, "bits:8");
value(address, "uint:4");
value(write, "log");
static(data1, "int:5");
static(data2, "uint:7");

when ((address == 3) && write)
    bus |- cast(data1, type(bus));
when ((address == 7) && write)
    bus |- cast(data2, type(bus));

```

The assignment operator for a selectvalue is `|-` which is intended to look a little like a diagram of a connection to a bus. A conditional assignment to an output mode variable uses the same assignment operator because again a three-state drive is involved. When variable **address** is 3 and **write** is true, drive **data1** onto **bus**. When variable **address** is 7 and **write** is true, drive **data2** onto **bus**. The value of variable **bus** can be used in expressions the same way as other modes such as value or static. It is often the case that a bus must be shared by data of different types. This example illustrates this. The bus type has been made **bits** and variables **data1** and **data2** have been cast to that type. In both cases the variables have fewer bits than **bus** so **data2** will be zero padded to 8 bits and **data1** will be sign extended to 8 bits.

2.18 Priority mode

Priority mode creates a priority encoder. The type must be "`[]log`" where the array dimension is at least as large as the number of inputs (it may be larger and will be trimmed). The encoder is combinatorial and has the same number of outputs as inputs (or one extra, see below). If one or more inputs is true the output corresponding to the lowest subscript input which is true will be true, all other outputs being false. The following example arbitrates between two queue variables which are to be written to a common bus -

```

selectvalue(bus, "uint:8"); // bus
priority(arb, "[10]log"); // bus arbitration variable
queue({q1, q2}, "uint:8"); // bus sources

arb[0] := <q1; // q1 is available for reading
arb[1] := <q2; // q2 is available for reading

when (arb[0])
    bus |- q1;
when (arb[1])
    bus |- q2;

```

Since the lowest subscript applies to **q1** it will have priority over **q2** if both are available simultaneously. The priority variable is dimensioned 10 however the unused higher entries will be ignored. One additional output may be used, i.e. **arb[2]** in the case of the example. This extra output is true if none of the inputs is true.

2.19 More on types

Target variable types are -

bits: <i>n</i>	unspecified binary
uint: <i>n</i>	unsigned integer
int: <i>n</i>	signed integer
log	logical
fixed: <i>m .n</i>	fixed point
float: <i>m en</i>	floating point
	enumerated type
null	no data

Immediate types are -

bits	unspecified binary
uint	unsigned integer
int	signed integer
log	logical
str	string
float	floating point
->	pointer
type	type
list	a list
map	a map
file	a file
empty	take the type from an initialisation argument

In both target and immediate modes the types can be nested within arrays or structs. Immediate and target mode types cannot be mixed within a struct.

2.20 Inbuilt modules, procedures and functions

We have already seen the use of inbuilt procedures such as **static()** and **queue()**. There are very many such inbuilt procedures, even more inbuilt functions and a few inbuilt modules. The inbuilt functions include the usual numeric and string operations which are for immediate execution only.

2.21 Libraries

3PL libraries are source code files that are incorporated into the source stream by **source "..."** or **module "..."** statements. It is usual to incorporate a library called a *header file* as the first line of source code. This header file is specific to the FPGA platform and sets up FPGA package pins definitions and clock logic and provides interfaces to external devices on the platform. There are other libraries which are more general and contain useful modules, procedures and functions. Often these are already incorporated in the header file.

2.22 Petri nets

State machines are a convenient way of expressing a control algorithm. When implementing state machines in FPGAs it is usual to represent each state by a flip-flop state¹⁰. Exactly one state flip-flop should be set at any time. A state transition requires the current state flip-flop to be reset and another state flip-flop to be simultaneously set. A Petri net is a similar but more general scheme. The form

¹⁰The alternative is to represent the state by a counter, which is the usual way in software.

of Petri net implemented in 3PL would be described as "a finite capacity Petri net with a capacity of 1" which is a less general form. Although it is a contradiction in terms one can think of this form of Petri net as a state machine implementation where more than one state flip-flop can be set at a time. Transitions can then involve some flip-flops being cleared and others set. A simple state machine can be described by a Petri net.

In a Petri net we have *places* rather than *states* and each place can have a token (set) or no token (cleared). A transition involves taking tokens from one or more places and inserting them in one or more other places. Expressions describing when transitions are to take place are called *predicates*.

The following is an example of the syntax -

```

petrinet {
  initial q1, q2;
  q1, q2 -> q3, p4 when (a && (b != 0)) {s <- queue1;};
  q3 ->> q1 {l1 <- queue1;};
  q4 -> q2, q3 when (!l2);
}

```

Here there are places **q1**, **q2**, **q3** and **p4**. Each of these is a static variable of type "log". If they have not already been created the Petri net construct creates them. These variables cannot be assigned by any code, only by the Petri net mechanism itself, however their values can be used in the program code outside of the Petri net in the same way that ordinary variables are used.

In the example places **q1** and **q2** initially have tokens. When the expression $a \&\& (b \neq 0)$ (called the *predicate*) is true and **q1** and **q2** both contain tokens (are set), **q1** and **q2** will have their tokens removed (are cleared) and places **q3** and **p4** have tokens inserted (are set). The single right arrow indicates a weak transition, meaning that prior to the transition any tokens in the destination places are ignored. If a double right arrow is used this signifies a strong transition, meaning that the transition can only occur if the destination places do not have tokens.

In the example above the code block to the right of the first transition is executed when the transition occurs. It may contain only a single target statement (which can of course be a compound statement such as **seq{}**, **par{}** etc.). If a transition does not have a predicate then it occurs when the source places have tokens (and optionally destination places lack tokens). A transition code block may contain immediate statements but these would serve little purpose as they are only executed as the interpreter passes through the code during immediate execution.

To code a state machine as a Petri net the following apply -

- each transition has only one source and one destination place
- only one place is initialised
- transitions can be weak (it doesn't matter which are used, but weak transitions use fewer logic resources)

Chapter 3 - Variables, Constants, Types, Modes, Attributes, Operators and Expressions

3.1 Input format

The source code input may use either UNIX-style (newline character) or DOS/Windows-style (carriage return, line feed) line termination. Tokens are separated by white space (spaces, horizontal tabs or newlines). Good formatting dictates the use of indenting but the compiler treats indenting as simply white space. The last source code line need not be terminated by a newline character, but it is usually preferable to do so.

Comments may have the form `"/* ..... */`, in which case they may run over line breaks, or they may be indicated by `//.....` where they occupy the rest of the current line.

3.2 File inclusion

Source files may be inserted in-line by means of file inclusion statements. These are of two forms.

The first form

```
source "filename"
```

simply inserts the contents of the named file into the current file in place of the entire *source* statement line. A file may only be inserted once by a *source* statement.

The second form

```
module "filename"
```

also inserts the contents of the named file into the current file in place of the entire *module* statement line, but it differs from a *source* statement in that the inserted file establishes a new static module scope (for scope discussion see "Program source structure" [chapter 4](#) p91). A file may only be inserted once by a *module* statement.

White space (only) may precede the *source* or *module* on the line and separate it from the quoted file name. The remainder of the line following the quoted file name will be ignored. File inclusions using *source* or *module* statements may be nested to any depth.

The quoted file name may be replaced in either of the above by `directive("ident")`, i.e.

```
source directive("key")
```

or

```
module directive("key")
```

where the directive whose key is "key" contains the name of the file to be included. Since file inclusion is done during parsing the directive must be assigned using **presetdirective()** [section 9.50](#) p229 or by using the command line option -D.

The file inclusion may be made conditional by using one of the following forms -

```
source ifdirective("key") "ident"
```

or

```
module ifdirective("key") "ident"
```

where the directive "key" will be examined to determine whether to include the file "ident". If directive "key" does not exist or it has the boolean value **false** the file inclusion will be skipped, otherwise the file will be included. Note that if the directive is not boolean its existence will suffice, but if boolean its value will be used.

The above forms may also use a directive for the file identifier -

```
source ifdirective("key1") directive("key2")
```

or

```
module ifdirective("key1") directive("key2")
```

Here directive "key1" is used as before to determine whether to include the file and directive "key2" contains the name of the file. The same directive may be used for both purposes, e.g.

```
module ifdirective("lib1") directive("lib1")
```

will include the file whose name is contained in directive "lib1" if that directive is defined.

File names are used exactly as supplied so any file name extension must be included (use of the extension ".3pl" is recommended but not required).

Note that these file inclusions are performed during parsing, not during immediate execution, and so placing a *source* statement within an *if* statement does not make the inclusion conditional (and is syntactically incorrect).

Files included by either of these mechanisms must contain syntactically complete code, i.e. statements cannot be partially contained in an included file.

Since there are two mechanisms for inserting files, a file should be able to ensure that it is not inserted by the wrong mechanism. This can be done by including one of the lines

```
nosource
```

to preclude sourcing this file or

```
nomodule
```

to preclude calling this file as a module. These must have no leading characters before the keyword. The remainder of the line is ignored.

3.3 Single constants

Integer constants may be

- decimal
- hexadecimal (leading 0x or 0X)
- octal (leading 0)
- binary (leading 0b or 0B).

Their size is 64 bits.

Logical constants are **true** and **false**.

Floating point constants are decimal integers with a leading, trailing or embedded '.' and an optional exponent consisting of "E" or "e", an optional sign followed by an integer. If an exponent is supplied the '.' can be omitted where appropriate.

String constants have double quotes surrounding the string. The string may contain escape characters

- \n new line (line feed)
- \t horizontal tab
- \b backspace
- \r carriage return
- \f form feed
- \\ backslash
- \' single quote (single quotes do not need to be escaped)
- \" double quote
- \0nnn octal numeric escape where nnn is the 3 digit octal character code
- \xnn hexadecimal numeric escape where nn is the 2 digit hexadecimal character code

The global variables `math_pi` and `math_e` of type **float** are created by the compiler prior to immediate execution. These variables contain the values for π and e and are read-only.

3.4 Compound values

Compound values are (possibly nested) array or struct values.

- They can be used to initialise compound variables.
- They can be assigned to compound variables.
- They can be passed to compound immediate or value parameters in modules, procedures or functions.

A compound value is a comma-separated list enclosed in braces. Each list element is an immediate expression or a compound value. Each set of braces matches a top level or nested array or struct. Trailing elements for an array or struct may be omitted. Initial or intermediate elements may also be omitted, commas being placed to indicate this. Lists must not exceed the length of the array or struct they are intended to match.

A compound value is processed from left to right, thus the left-most item will appear at the lowest subscript or address.

Missing elements in a compound value are simply not assigned. Where a compound value with missing elements is used to initialise an immediate compound variable, or is assigned to it, the variable elements corresponding to the missing values will simply retain their previous or default values. Where a compound value is similarly used for a value variable, the variable elements corresponding to the missing constants will not be assigned, retaining their previous values (none if not previously initialised or assigned).

Examples -

```
var(i, "[3] [2]in", {{1, 2}, , {3}});
struct(s,
  f1,    "[3]int",
  f2,    "log",
  f3,    "str"
);
var(v, s);
v = {{3, 4, 5}, true, "fred"};
proc(a <- {7, 8, false});
```

Where a struct is initialised, assigned or passed an argument the field values may be specified by naming the fields rather than matching values to fields by position as in the example above, e.g.

```
struct(s,
      f1,    "[3]int",
      f2,    "log",
      f3,    "str"
);
var(v, s);
v = {f3="fred", f1={3, 4, 5}, f2=true};
proc(a <- {7, 8, flag=false});
```

When a struct is assigned values by name in this way no following position-matched values are allowed, for example the following is accepted -

```
struct(s,
      f1,    "[3]int",
      f2,    "log",
      f3,    "str"
);
var(v, s);
v = {{3, 4, 5}, f3="fred", f2=true};
```

but it is incorrect to do -

```
struct(s,
      f1,    "[3]int",
      f2,    "log",
      f3,    "str"
);
var(v, s);
v = {{3, 4, 5}, f2=true, "fred"};
```

because the last value is matched by position but is preceded by a value matched by name. This limitation makes sense in that values are matched by position in the list to fields in the struct until a match by name is encountered after which point the argument positions are no longer aligned with the fields.

An array initialised by a compound value may have some dimensions omitted, in which case the missing dimensions will be inferred from the compound value. Array dimensions cannot be omitted where a variable is not a module, procedure or function parameter and is not initialised.

Compound value entries for a string array may be types integer, boolean or floating point in which case they will be converted to strings.

Leaf elements in the compound value must be immediate primitive types (which includes pointers).

3.5 Identifiers

Identifiers begin with a letter or an underscore and contain any number of letters, digits or underscores. Alphabetic case is distinguished.

The following keywords may not be used as identifiers

- branch
- break
- case
- class
- conditionelse
- continue
- default
- do
- else

- for
- function
- global
- if
- ifcondition
- leadermodule
- module
- null
- par
- petrinet
- postmodule
- private
- procedure
- return
- seq
- source
- switch
- sync
- trailermodule
- unless
- varargs
- when
- while

Identifiers used for -

- variables
- struct fields
- modules and procedures
- functions
- directives

are stored in separate tables and hence the same identifier can be used for multiple purposes. However, this practice may render the code difficult to read.

3.6 Variables and modes

3.6.1 variable modes

There are eleven modes of variable -

- **immediate**
- **selectvalue**
- **value**
- **static**
- **queue**
- **priority**
- **input**
- **output**
- **cmemory**
- **rmemory**
- **clock**

Immediate variables only exist during program interpretation. Variables in the other ten modes are target variables which are active in the FPGA logic.

3.6.1.1 immediate variables

Immediate variables do not survive into target code except as constants (their current value at that point in the code during immediate execution). These variables must not have a width declared. Immediate integer variables are 64 bits and floating point variables are 64 bit double precision.

3.6.1.2 value variables

Variables of mode **value** represent the result of combinatorial expressions and hence do not have state (storage). An expression contains constants, variables and function calls joined by operators. An expression could consist of a single variable in which case the value variable to which it is assigned can be thought of as a pointer to that variable (or rather the value of the variable, i.e. its output, since a variable cannot be assigned (changed) via a value variable pointing to it).

The availability (see later) of a value variable depends on any queue reads in the expression it represents.

3.6.1.3 selectvalue variables

Variables of mode **selectvalue** represent buses or selectors.

The availability (see later) of a selectvalue variable depends on any queue reads in the equivalent expression.

3.6.1.4 input variables

Variables of mode **input** are external inputs.

An input variable is always (read) available.

3.6.1.5 output variables

Variables of mode **output** are external outputs.

An output variable is always (write) available.

3.6.1.6 static variables

Variables of mode **static** have state - they are registers. They have no associated signaling. They may be global, or local to modules, procedures, functions or code blocks. Static variables involve no signaling and their state changes are not implicitly synchronised with pipeline operations except in as much as module code may exercise local control.

Static variables containing arithmetic types may be any size but initialisation or assignment from immediate expressions or variables are restricted at the moment to a maximum of 64 bits for integer or 52.11 for floating point.

A static variable is always available.

3.6.1.7 queue variables

Variables declared **queue** are dataflow variables which are mainly used as pipeline connections between modules. They have associated signaling to indicate their availability for reading, examination or writing (assignment). When a datum is written to a queue, the datum becomes available at the output of the queue. A queue may contain more than one datum, in which case the data will be available in the order they were written. In a module the written end of a queue is referred to as an **output queue** since it is assigned to. The read or examined end of a queue is referred to as an **input queue**.

An input queue is available when it has a datum which can be read or examined (a datum which is read is removed from the queue whereas a datum which is examined remains in place), i.e. it is not empty.

An output queue is available when it can be written to, i.e. it is not full.

Queue variables containing arithmetic types may be any size but initialisation or assignment from immediate expressions or variables is limited to a maximum of 64 bits for integer or 52.11 for floating point.

A queue is 'synchronous' if it is written and read in the same clock domain.

A synchronous queue has a default buffer depth of 2. An asynchronous queue has a default buffer depth of 16. These depths can be explicitly set.

A queue may be type "null". This is useful for signalling as the queue can be written (null must be assigned) or read but no actual data storage is used.

3.6.1.8 cmemory and rmemory variables

Variables of mode **memory** represent RAM elements in the FPGA.

Memory variables can only be accessed via inbuilt procedures and functions and hence availability does not apply as they cannot be used in expressions or assignment statements.

3.6.1.9 priority variables

Variables of mode **priority** implement a priority encoder and are used for arbitration. The type must be a one dimensional array of "log" (if the type is omitted "[20]log" will be used). The array members are assigned combinatorial values in the same way as value mode variables. However, the resultant values of the logical array are not simply the input expressions but are arbitrated values, the lowest subscript having highest priority. Where one or more input expressions to the priority array are true, one and only one array value will be true and that one will correspond to the true input with the lowest subscript.

There is an inbuilt function, **prielse()**, which returns the value of the else case, i.e. a value which is true when no priority inputs are true.

The entire array need not be assigned. Unused entries will be ignored, but where an entry is evaluated (its output is used) it must have an assigned input value. The converse is not true - an entry that has been assigned need not be evaluated. Assignments need not be done in any particular order. This method of usage is designed to allow an array that is larger than required to be created after which entries can be assigned as immediate execution progresses. At the completion of immediate execution a priority encoder will be created in target code of a size which matches the entries in the array that have actually been used. This avoids prior knowledge of the required size of the encoder.

A priority variable array member may be assigned more than once but the default behaviour is that the final value assigned prior to termination of immediate execution prevails, all other previous assignments being lost. This can be changed by setting attribute "multiplein" to true in which case all assignments will be collected and ORed together on completion of immediate execution to form the input to that priority array member.

At the moment assignments to priority variables cannot contain queue reads. This restriction may be removed later.

A priority variable is always available.

3.6.1.10 clock variables

Variables of mode **clock** are used to represent signals. There may be a number of clocks to which program logic is synchronised, but there will also be associated intermediate clock signals used to connect together the hardware components used to generate these program clocks, such as input buffers, delay-locked loops and clock buffers.

3.6.2 variable creation

Variables are created during immediate execution by means of inbuilt procedures. The most general procedure used is **var(...)**. This has two to four arguments -

```
var(name, mode, type, init)
```

name is the identifier for the created variable.

Most variable creation inbuilt procedures allow creation of multiple variables of the same type, mode and initial value by replacing the *name* argument by an identifier list enclosed in braces {*v1*, *v2*,... }.

The **mode** argument is optional and must be one of the keywords **immediate**, **value**, **selectvalue**, **static** or **queue** - memory or input mode variables must be created by other procedures (see below). If the mode argument is omitted the mode will be **immediate**.

The type argument must be a type-valued expression or a string-valued expression describing the type. A type-valued expression can be a variable or variable member of type "type", the inbuilt functions **type()** or **struct()**, or a user-defined function returning type "type".

Type specifications in general and type variables in particular are described in a [section 3.7](#) p53.

The final argument to **var()** is optional and is an initialisation value. This may take one of three forms -

- a single immediate expression of appropriate type for a variable of primitive type
- a variable reference of matching type to the created variable
- a compound value

For the first two of the above cases where the mode is immediate the type argument may be the string "empty" in which case the variable type will be the type of the initialisation argument.

Where initialisation is provided and more than one variable is being created (using {...}) then all variables will be initialised identically.

Immediate, static and queue variables may be initialised. If they are a compound type the initialisation argument will usually be a compound value or a compound variable of suitable type, but a compound type may be initialised by a single immediate value in which case all members of the compound type will be initialised to that single value (the members of the compound type must all be the same primitive type appropriate to the initialisation value).

Uninitialised immediate and static variables are automatically initialised to 0, 0.0, false or "" as appropriate to the type.

Queue variables when initialised contain a single datum. Uninitialised queues have no value as they are not available. Initialised queues have a value and they are available.

A value mode variable can be initialised, but this does not preclude assignment of one or more other values later. A value mode variable can be initialised with an immediate constant in which case it behaves as a target constant unless assigned another expression. Uninitialised value variables have no default value. If the value variable is a compound type it may be initialised by a compound value.

Selectvalue variables cannot be initialised but their output when no input is selected can be specified.

Examples of variable creation -

```
var(byte, "int", 8);
var(sum, value, "uint:16");
var(even, static, "log");
var(pixel, queue, "uint:" + byte);
var(trigger, queue, "null");
```

The first has no mode declared, so is immediate. It is given an initial value of 8. The remainder are target variables. The second last example uses the '+' operator to concatenate an integer to a string to form the word size. The final example is a queue with no data which can be used as an event trigger. These examples all use string-valued expressions for the type argument. The following examples illustrate the use of type-valued arguments -

```

var(t1, type, "int:8");
var(v1, value, t1);
var(s2, static, type(v1));
var(t2, type, type(s2));
var(q1, queue, "[4]t2");
var(s3, static, "(f1, int:8, f2, [4]t1)");

```

For convenience the following procedures can be used instead of `var(...)` to create primitive immediate variables -

```

int(name, init);
log(name, init);
str(name, init);
file(name, init);
type(name, typespec);
struct(name, field1, type1, field2, type2, .....);

```

In each case a single variable of the type described by the procedure name is created. The initialisation arguments are optional.

These are further described in

- `int()` (section 9.40 p221)
- `log()` (section 9.42 p222)
- `str()` (section 9.77 p256)
- `file()` (section 9.32 p212)
- `type()` (section 9.81 p258)
- `struct()` (section 9.78 p257)

Similarly target variables can be created by-

```

static(name, type, init);
value(name, type, init);
queue(name, type, init);

```

which are equivalent to -

```

var(name, static, type, init);
var(name, value, type, init);
var(name, queue, type, init);

```

In all the above *init* is optional.

These are further described in

- `static()` (section 9.75 p253)
- `value()` (section 9.83 p259)
- `queue()` (section 9.57 p234)

Memory variables are created by the inbuilt procedures -

```

cmemory(name, ports, atype, dtype, init);
rmemory(name, ports, atype, dtype, init);

```

These create combinatorial output RAM (mode **cmemory**) and registered output RAM (mode **rmemory**) respectively. *name* is an identifier for the memory, *ports* is the number of ports, *dtype* is the data type, *atype* is the address type and *init* is an optional initialisation array (immediate array variable or compound value).

See `cmemory()` (section 9.13 p197) and `rmemory()` (section 9.62 p239),

Clock variables are created by the inbuilt procedure -

```

clock(name, ...attributes...);

```

See **clock()** (section 9.10 p194).

Input variables are created by the inbuilt procedure -

```
input(name, type, ...attributes...);
```

name is an identifier for the input variable. *type* is the variable type. *attributes* is optional and is either one or more key=value attribute arguments or a map of attributes. These arguments can be omitted and attributes added later by using inbuilt procedure **attributes()**.

See **input()** (section 9.39 p218).

Output variables are created by the inbuilt procedure -

```
output(name, type, source, attributes);
```

name is an optional identifier for the output variable. *type* is the variable type (also optional). **source** is an optional source value for unconditional assignment. *attributes* is an optional map of attributes. The argument can be a map containing the attributes or can be a compound value containing the attributes as key=value entries in one-dimensional list. This argument can be omitted and attributes added later by using inbuilt procedure **attributes()**.

See **output()** (section 9.47 p224).

Variables are normally created within the scope containing the call to the variable creation inbuilt procedure. However, if the variable identifier has a leading "global.", "file.", "calling." or "local." it is created in the specified scope.

Using "global.", global variables can be created from within any scope.

Using "file.", variables can be created in the current (deepest) file module scope.

Using "local." allows the conditional creation of variables in the scope of a module, procedure or function. For example the code -

```
module
m (...) {
  if (doit)
    static(s, "log");
}
```

will create static variable **s** in the module if variable **doit** is true. However, the following code -

```
module
m (...) {
  if (doit) {
    static(s, "log");
    int(j);
  }
}
```

creates variables **s** and **j** in the scope of the **if** block and not in the module scope. By changing the code to -

```
module
m (...) {
  if (doit) {
    static(local.s, "log");
    int(local.j);
  }
}
```


variables **s** and **j** are created in the scope of the called module.

Using "calling." allows the creation of variables in the scope of the previous module, procedure or function. This is useful for writing procedures to create variables, in which case the created variable needs to be in the calling scope, not the scope of the called procedure.

Variables may be created at any point in the code, but they must have been created prior to being referenced.

A variable reference where the identifier has a scope prepended will attempt to locate the variable only in that scope.

It is important to note that variables exist independently of the scope in which they are created. The scope only contains the map which ties the variable identifier to the variable, not the variable itself. That map disappears with the scope. If a pointer to a variable is kept then that variable can be referred to via the pointer even though the scope in which the variable was created has disappeared. For example the following code illustrates a variable created within a function which returns a pointer to that variable so that it can be used elsewhere -

```
var(p, "->");
p = make_static("uint:12");

when (1)
    p-> <- v;

function
make_static (str(t)) {
    static(s, t);
    return(&s);
}
```

Here a pointer variable **p** is assigned a pointer to a static variable created by function **make_static**. It can then be used by dereferencing the pointer; in the example the static variable is conditionally assigned a value **v** via the pointer **p**.

When creating any variable the identifier may be computed by using the indirect operator on a string variable rather than using an explicit identifier, e.g.

```
int(i, 23);
str(name, "global.var" + i);
log(name->);
```

will create a logical immediate variable in global scope whose identifier is **var23**.

3.7 Types

Immediate variables may be of primitive type

- bits
- uint
- int
- log
- str
- float
- ->
- type
- list
- map
- file
- empty

bits is a long unsigned integer (64 bit). Arithmetic operators +, -, *, / and % and relational operators <, <=, > and >= cannot be used on this type, but the bitwise logical operators and equality and inequality operators can be used.

int is a signed long integer (64 bit).

uint is an unsigned long integer (64 bit).

log is a boolean type whose value is 'true' or 'false'.

str is a character string type.

float is a floating point double precision type.

-> is a pointer to a variable, or components of a variable.

type is a type specification.

list is a list.

map is a sorted map with string keys.

file is a file.

empty for immediate mode specifies that the type will be taken from the variable initialisation argument.

Integer constants and expressions which are 0 or positive are automatically given type **uint**. Integer variables can be declared **uint** but they are implemented as **int** and no checks are made that the value of a uint remains non-negative.

Target variables may be of primitive type

- bits:*n*
- int:*n*
- uint:*n*
- log
- fixed:*m.n*
- ufixed:*m.n*
- float:*m.n*
- null

bits:*n* is a word of *n* bits. This is used for bit patterns that do not represent weighted binary arithmetic values, e.g. ASCII characters or gray codes. It can also be used where a variable is used at different times for different types, e.g. a communications bus in which case other types are cast in or out of it. This type cannot be used in arithmetic operations.

int:*n* and **uint:*n*** are integers or unsigned integers respectively whose size in bits is explicitly declared by *n*. The size *n* must be an immediate expression.

log is a logical type whose value is **true** or **false**.

fixed:*i.f* and **ufixed:*i.f*** are fixed point or unsigned fixed point types respectively with an *i* bit integer part and *f* bit fractional part. *i* or *f* may be zero (but not both). The maximum size of the fractional part is 63 bits.

float:*e.f* is a floating point type with an *f* bit mantissa and *e* bit biased exponent.

A **null** type can be used where a queue is being used only as an event trigger. It cannot be used for variables of other modes.

Variables may be of compound type

- struct
- array

A **struct** is a compound type consisting of fields named by an identifier. The fields may be any type, primitive or compound.

An array can be any type, including **structs**.

There is an enumerated type. This is effectively a primitive type but unlike the others is not specified the same way (see below).

3.7.1 type specifications

Where variable creation procedures require a type argument this can either be a string type description expression or a type variable ([section 9.81](#) p258).

If a string representation of a type is used, which is the most common way to specify types, this is just a string constant, a string expression or a string variable.

A type is specified by one of the following -

- A type keyword with optional trailing ":" and width where appropriate.
- The identifier of a type variable.
- The identifier of a string variable containing a type description string.
- A structure description.

Type keywords are

- **bits** - a word of arbitrary bit encoding
- **uint** - an unsigned integer
- **int** - a signed integer
- **ufixed** - an unsigned fixed point value
- **fixed** - a fixed point value
- **float** - a floating point value
- **log** - a boolean
- **str** - a string
- **->** - a pointer
- **type** - a type
- **class** - a class
- **list** - a list
- **map** - a map
- **file** - a file
- **null** - a dataless queue
- **empty** - the immediate type will be taken from the initialisation argument

An optional data width can only be appended to target mode numeric types (**uint**, **int**, **fixed** or **float**). The width specification is separated from the type by a colon. For **uint**, **int** a single width in bits is required. The **ufixed** and **fixed** types have integer part width followed by '.' and then fractional part width. For type **float** the width specification is given in the form *e.m* where *m* gives the number of bits in the mantissa and *e* the number of bits in the biased exponent (the total size is 1+n+m as there is also a sign bit). The actual mantissa fraction has an implied integer 1 preceding the fraction.

A type variable can be created by means of inbuilt procedures **type()** or **var()**. Inbuilt procedure **struct()** also creates a type variable.

Assuming we have created the variables -

```
int(i, 8);
str(s, "uint:");
```

the following are valid type strings -

```
"uint:8"
"uint:"+i
s+8
```

The first is a string constant. The second and third are string expressions, the "+" giving concatenation when one operand is a string (the other being converted to a string if necessary).

Any spaces in a type specification string are removed prior to its evaluation.

A type variable is a variable of type "type" and is created by the **type()** inbuilt procedure, e.g.

```
type(t, "uint:"+8);
```

The **type()** inbuilt procedure takes a type specification string and constructs a type variable containing that type value. There is an inbuilt function, **typestring()**, which will return a type description string of the type of the argument. These roughly complementary facilities allow manipulation of types. The type of a module, procedure or function argument can be turned into a string, modified, then used as the type of a new variable. Similarly types can be constructed initially using parameters.

The **struct()** inbuilt procedure also creates a variable of type "type".

The following example creates static variables a, b and c which are unsigned integers of width 8 bits -

```
int(i, 8);
str(s, "uint:");
var(a, static, "uint:8");
var(b, static, "uint:"+i);
var(c, static, s+8);
```

Note that types *uint* and *int* which can be used for both immediate and target mode variables **must** have a width specified for target mode variables but **must not** have a width specified for immediate variables.

Where a type key word is expected an identifier of a variable of type "str" or type **type** may be used. Such an embedded variable must not be followed by a width specification! If the embedded variable is type "str" then the string contained in the variable must itself be a valid type description string. The identifier may have a preceding scope such as **calling.**, **global.** etc. Thus type string expressions and type variables can be nested, e.g.

```
str(s1, "uint");
int(i, 8);
type(t, s1+i);
queue(p, "[4][3]t");
static(s, "[5]s1");
```

This nesting of string and type variables can be to any depth. The following is incorrect as an embedded variable cannot be followed by a width specification, i.e. it must be a complete type -

```
type(t, "uint");
queue(p, "t:8"); // ERROR!
```

Note that variables of type **type** can be assigned to variables of type "str" and vice versa and in each case a conversion will be applied. However a string assigned to a type variable must have valid type syntax or a fatal error occurs.

Note also that a string describing a type which contains an embedded identifier must be able to access that identifier when the string is used to create a type. If for example the string is passed to a module, procedure or function and is used within that code body as the type of a variable then the variable embedded within the type string must be accessible in the current scope. Where a string is passed as an argument to a type parameter the conversion occurs in the scope of the call, but if the string is passed to a string parameter and then used later then there may be a problem with the scope.

The following example works as expected -

```
struct(s,
  f1, "int",
  f2, "log"
);
p("[3]s");

procedure p(type(t)) {
  var(v, t);
  .
}
```

The argument is a string describing a type and contains the embedded type variable **s** (this could also have been a string variable). The procedure parameter **t** is type **type** and will be assigned the type derived from the string. When the argument is assigned to the procedure parameter the embedded variable **s** will be evaluated in the calling scope.

If however we change the example to -

```

struct(s,
    f1, "int",
    f2, "log"
);
p("[3]s");

procedure p(str(t)) {
    var(v, t); // cannot find embedded variable 's'!
    .
}

```

then the argument is passed unchanged and an attempt to use it as a type within the procedure body fails as embedded variable **s** is not accessible within the scope of the procedure body.

There are two ways this error could be overcome. We could explicitly specify the calling scope in the argument -

```

struct(s,
    f1, "int",
    f2, "log"
);
p("[3]calling.s");

procedure p(str(t)) {
    var(v, t);
    .
}

```

but this seems rather unsatisfactory. Alternatively we could force the conversion from string to type by using inbuilt function **strtotype()** [section 10.144](#) p333. The parameter is of type "str" so the assignment of the resulting type argument to the string parameter would convert the value back to string automatically.

```

struct(s,
    f1, "int",
    f2, "log"
);
p(strtotype("[3]s"));

procedure p(str(t)) {
    var(v, t);
    .
}

```

We could have also used an intermediate type variable -

```

struct(s,
    f1, "int",
    f2, "log"
);
type(it, "[3]s");
p(it);

procedure p(str(t)) {
    var(v, t);
    .
}

```

3.7.2 arrays

3.7.2.1 creating arrays

Arrays are created by prepending dimensions in brackets to the type, e.g.

```
var(pix, value, "[3][3]uint:8");
var(sum, static, "[10]int:16");
var(q1, queue, "[2]uint:8");
```

One or more dimensions may be omitted (empty brackets) for module, procedure or function parameters in which case they will be supplied from the matching argument. A parameter array which is initialised may also have some dimensions omitted, in which case the missing dimensions will be inferred from the argument if supplied or else by default from the initialisation. Either or both the argument or the initialisation may be a variable of matching compound type or a compound value.

If a parameter has one or more undimensioned arrays and it has no initialiser and is not passed an argument, the undimensioned arrays will be given dimensions of 0.

Array dimensions cannot be omitted where a variable is not a module, procedure or function parameter and is not initialised.

The array type may be an identifier of a type or string variable which specifies any type including an array.

Arrays may be used for modes **immediate**, **value**, **selectvalue**, **static** or **queue**. Variables of modes **memory** or **clock** do not have types and hence cannot be created as array types. Variables of mode **priority** must be arrays of *log* (default type if omitted).

Where array dimensions are given by immediate variables or expressions a type string is created by concatenating the dimensions and surrounding brackets, e.g.

```
int(dim1, 8);
int(dim2, 4);
var(a, queue, "["+dim1+"["+dim2+"]uint:8");
```

Care should be taken to parenthesize nested arithmetic expressions where necessary to ensure that they are evaluated first into numeric values prior to concatenation with strings, otherwise the evaluation order may be erroneous, for example

```
int(i, 2);
int(j, 3);
var(a, queue, "["+i+j+"]uint:8");
```

will give a dimension of 23 not 5.

3.7.2.2 referencing array members

An array subscript may be -

- a single subscript
- a subscript range
- missing

Trailing subscripts may be omitted entirely, implying that the array reference includes the full dimensions where the subscripts are omitted.

A subscript range selects a sub-array. If the first subscript of the pair is higher than the second the sub-array members will be in reverse order.

Where arrays have greater than one dimension subscripts are concatenated, each subscript being enclosed in brackets. Intermediate subscripts may be omitted by using empty brackets. Subscripts or subscript

ranges may be any expression whose type is *uint* or *int*. Subscripts may be immediate, value, selectvalue, static or queue mode.

The following are subscript examples -

```
a[2]          constant subscript
a[i]          variable subscript
a[2..j-1]     subscript range
a[]           missing subscript - equivalent to full range
a[2][i]
a[2..4][i..i+3]
```

Some caution should be exercised when using target subscripts. These employ considerable logic but more importantly the logic delays may prove difficult to fit within timing constraints. In particular delays can become large where the array dimensions are large or array references are nested. At low clock rates this is not a concern.

3.7.3 structs

Structures are usually created by using inbuilt procedure **struct()**, but they may also be specified by a type string.

3.7.3.1 using the struct() procedure

The first argument of the inbuilt procedure **struct(...)** is an identifier giving the struct name which will be created as a variable of type "type". This is followed by a number of argument pairs, the first component being an identifier giving the field name and the second an associated type string-valued expression.

If the first component of a pair is the keyword **extends** then the second component must be a struct name. The fields of that struct will be included in this struct at that point. The type of a field can of course be another struct in which case that struct is nested (rather than being extended).

The first component of a pair may be a comma-separated list of identifiers enclosed in braces in which case several fields of the same type will be created, one for each field identifier in the list.

The second component of a pair is the type of the field. This is a string specifying the type or is a type variable.

A field type cannot be, or contain, an undimensioned array.

Fields are stored in the order in which they are listed.

Examples -

```
struct(pixel,
  pix,    "uint:8",
  hr,     "log",
  vr,     "log"
);
```

This declares a pixel structure which has an 8-bit unsigned integer field called "pix", and horizontal and vertical reset indicators which are logical (1 bit). The arguments are all string constants but could be string variables or expressions.

```
struct(win,
  pixwin, "[3][3]pixel",
  edge,   "log"
);
```

This example contains a field which has as its type the previous structure. If we create a variable of this type -

```
var(a, queue, win);
```

then we can access the pixel value of one member of the window by -

```
...a[1][0].pix...
```

In the following case

```
struct(pixaddr,
      extends, pixel,
      address, "uint:19"
    );
```

we have created a structure which includes the fields of struct *pixel*. This is a single level struct and is not nested.

To create multiple fields of the same type a list of field identifiers may be used, enclosed in braces, e.g.

```
struct(pixaddr,
      f1,          "int",
      {f2, f3},    "log"
    );
```

Structs are created within the scope containing the call to the structure creation inbuilt procedure unless the struct identifier has a leading "global.", "file.", "calling." or "local." in which case it is created in the specified scope. Structs may be created at any point in the code, but they must have been created prior to being referenced.

A struct reference where the identifier has a leading "global.", "file.", "calling." or "local." will attempt to locate the struct only in the specified scope.

Note that structs must not mix types **uint**, **int**, **ufixed**, **fixed** or **float** such that some have a width specified and some do not. If the struct is to be used for immediate variables then there must be no widths specified. If the struct is to be used for target variables then widths must be given for these types. Types **log** and **enum** can be interpreted as either immediate or target mode so can be included in an immediate struct or a target struct. A target type can however be included in an immediate struct by using a pointer to the target type.

3.7.3.2 using a string specification

A struct can also be created using a string specification. This consists of outside parentheses enclosing field/type pairs, for example

```
"(a, log, b, [3]uint:8)"
```

Field identifiers cannot be enclosed in braces; if several successive fields have the same type they must still be specified separately. String or type variable identifiers can be included where a type would be expected. The following two statements are equivalent -

```
type(t, "(a, log, b, log, c, [3]uint:8)");
```

```
struct(t,
      {a, b}, "log",
      c      "[3]uint:8"
    );
```

Any spaces in the specification string are removed prior to its evaluation.

Structs may be used for modes **immediate**, **value**, **selectvalue**, **static** or **queue**. Structs cannot be used for modes **memory**, **clock** or **priority**.

3.7.3.3 referencing struct members

Struct members are accessed by using the struct name followed by "." and the field identifier, e.g.

```
v.f1
```

If the field has a compound type the field name may be followed by more field identifiers or subscripts, e.g.

```
v.f1[2].f2
```

Note that structs are interchangeable if their fields and associated types are identical. Thus the following code is legitimate -

```
struct(s1, f1, "int", f2, "log");
struct(s2, f1, "int", f2, "log");
var(v1, s1);
var(v2, s2);
v1 = v2;
```

though there seems no reason to do this and the practice is confusing and to be discouraged.

3.7.4 lists

Lists are created by inbuilt procedure **var()** with type string "list". An initialiser may be provided which must be another list variable or a compound value, in which case values will be copied into the new list. If multiple list variables are created using a variable list as the first argument then independent lists will be created (they are not pointers to the same list). If an initialiser is used when creating multiple list variables, each will be initialised with data copied from the initialising list. If the initialiser is a compound value the list must not contain nested lists, null entries or key=value entries.

List entries are always values, they are not pointers to other variables!

List entries must only be immediate, clock, cmemory or rmemory mode. Variables of other modes cannot be assigned to a list but pointers to them can be. ¹

List entries may be any type and entries do not have to be the same type.

List operations are summarised in the following table -

v = get(l, index);	get the list entry
... l[index]....	get the list entry
append(l, index, v);	append an item at the index position
appendall(l, index, l2);	append contents of another list
change(l, index, v);	change a list entry
l[index] = v;	change a list entry
remove(l, index);	remove a list entry
n = entries(l)	get the number of list entries
l = l2;	assign another list
l = { . . . };	copy a compound list
l = null;	clear the list

The inbuilt function **entries()** returns the number of elements in a list (or a map).

Lists may be nested within arrays, structs and other lists and maps.

Lists may contain arrays, structs and other lists and maps.

¹Those target modes are allowed as they do not have types and hence do not have subscripts or fields. The complexities of nested subscripts and fields preceding and following a list variable with its subscript (index) have been managed in the case of immediate mode variables but this has yet to be expanded to target modes. A future version of 3PL may allow all variable modes to be appended to lists.

Lists may contain null entries.

A list can be referenced via a pointer and a list entry can be selected using a subscript after the pointer dereference.

v = l[index] - get a list element using subscript notation

v = get(l, index) - get a list element using a function

The form using subscript notation was a later addition to 3PL. The inbuilt function has been retained as an alternative.

List items are indexed starting at 0. The index may range from 0 to **entries(l) - 1** inclusive. If **index** is outside this range a fatal error occurs.

A list item can also be accessed using a subscript. This can be done to any depth. If a list entry contains structs, arrays, lists or maps nested to any depth then trailing subscripts and fields can be used. In the case of lists the subscript again is the index into the list. In the case of maps the subscript may be a string (the key) or an integer (an index to a key).

In the case of the following -

```
struct (s,
  f1, "int",
  f2, "[4]list"
);
var(v, s);
var(va, "[3]int", {4, 6, 8});
append(v.f2[1], 0, va);
```

we could access the nested integer array by -

```
.....v.f2[1][2]....
```

If we wanted the whole array then -

```
.....v.f2[1]....
```

would be appropriate.

Note that where lists are assigned nested types the values are copies, not pointers to the original variable.

append(l, index, v) - append a list element

The item **v** is appended at the position indicated by **index** in list **l**. The index may range from 0 to **entries(l)** inclusive. If **index** is outside this range a fatal error occurs. An index of 0 inserts at the beginning of the list and an index equal to the list size appends to the end of the list. If the index argument is omitted the item will be appended to the end of the list. If the last argument is omitted null will be appended.

appendall(l, index, l2) - append list elements from another list

The contents of list **l2** are appended at the position indicated by **index** in list **l**. The index may range from 0 to **entries(l)** inclusive. If **index** is outside this range a fatal error occurs. An index of 0 inserts at the beginning of the list and an index equal to the list size appends to the end of the list. If the index argument is omitted the items will be appended to the end of the list.

l[index] = v - change a list element using subscript notation

change(l, index, v) - change a list element using an inbuilt procedure

The form using subscript notation was a later addition to 3PL. The inbuilt procedure has been retained as an alternative.

The item in list **l** at position **index** is replaced by item **v**. The index may range from 0 to **entries(l) - 1** inclusive. If **index** is outside this range a fatal error occurs. The new value does not have to be the same type as the previous value. If the last argument is omitted the list item will be changed to null.

A list item may be changed by assigning using a subscript, the subscript being the index into the list of the item to be changed. Where an item in a list has nested structs, arrays, lists or maps trailing subscripts and fields may be used to change items within those nested types.

In the case of the following -

```
struct (s,
  f1, "int",
  f2, "[4]list"
);
var(v, s);
var(va, "[8]int");
```

we could append an integer 3 to the nested list in *v* by -

```
append(v.f2[1], 0, 3);
```

We could then change that item to the value of variable *va* by -

```
v.f2[1] = va;
```

We could now assign to the nested integer array by -

```
v.f2[1][3] = 15;
```

remove(l, index) - remove a list element

The item in list *l* at position **index** is removed. The index may range from 0 to **entries(l) - 1** inclusive. If **index** is outside this range a fatal error occurs.

l = l2 - assign a list

If the LHS of an assignment is a list variable and the RHS is a list, the LHS list will be replaced by the RHS list. Unlike other immediate assignments the LHS is assigned the RHS value, not a copy of the RHS value. If a copy is required then the inbuilt function **copy()** [section 10.20](#) p277 can be used.

The RHS may be a one-dimensional compound value list, in which case it must not contain nested lists, null entries or key=value entries. The LHS list is cleared and values from the compound list are copied into the LHS list.

l = null - clear a list

If the LHS of an assignment is a list variable and the RHS is explicitly the keyword **null** (not a null value) the list is cleared.

3.7.5 maps

Maps are created by inbuilt procedure **var()** with type string "map". An initialiser may be provided which must be another map variable, or a compound value, in which case values will be copied into the new map. If multiple map variables are created using a variable list as the first argument then independent maps will be created (they are not pointers to the same map). If an initialiser is used when creating multiple map variables, each will be initialised with data copied from the initialising map. If the initialiser is a compound value the list must not contain nested lists or null entries and all entries must have key=value format.

Map keys are strings (type "str"). The elements in a map are sorted in the lexicographic order of their keys and remain so following additions or deletions.

Map entries must only be immediate, clock, cmemory or rmemory mode. Variables of other modes cannot be assigned to a map but pointers to them can be. ²

²Those target modes are allowed as they do not have types and hence do not have subscripts or fields. The complexities of nested subscripts and fields preceding and following a map variable with its subscript (key) have been managed in the case of immediate mode variables but this has yet to be expanded to target modes. A future version of 3PL may allow all variable modes to be included in maps.

Map entries may be any type and entries do not have to be the same type.

Map operations are summarised in the following table -

<code>v = get(m, key);</code>	get the map entry from a key
<code>v = m[key];</code>	get the map entry from a key
<code>v = get(m, index);</code>	get the map key at index
<code>v = m[index];</code>	get the map key at index
<code>put(m, key, v);</code>	put an item with a key
<code>m[key] = v;</code>	put an item with a key
<code>putall(m, m2);</code>	put items from another map
<code>remove(m, key);</code>	remove a map entry
<code>n = entries(m)</code>	get the number of map entries
<code>m = m2;</code>	assign a map
<code>m = { ... };</code>	copy a compound list
<code>m = null;</code>	clear the map

The inbuilt function **entries()** returns the number of elements in a map (or a list).

v = m[key] - get a map element using a key (subscript notation)

v = get(m, key) - get a map element using a key (inbuilt function)

The map item for the key is returned. If no item with that key exists then null is returned.

The use of a key as a subscript to access a map was a later addition to 3PL. The inbuilt function has been retained as an alternative.

A map item may also be referenced by using the string key as a 'subscript'. If the map is nested within an array, struct, list or map additional subscripts, fields, indices or keys as appropriate may precede the map key itself. If the map item is itself a nested type containing arrays, structs, lists or maps then trailing subscripts, fields, indices or keys as appropriate can be appended.

v = m[index] - get a map key using an index (subscript notation)

v = get(m, index) - get a map key using an index (inbuilt function)

The use of an integer index as a subscript to access a map was a later addition to 3PL. The inbuilt function has been retained as an alternative.

Map items are indexed starting at 0. Map entries are ordered according to the lexicographic order of the keys. The index may range from 0 to **entries(1) - 1** inclusive. If **index** is outside this range a fatal error occurs. The key at that position is returned. To get a value instead of a key using an index the calls must be nested, e.g. `get(get(m, index))` or alternatively `m[m[index]]`.

Access by means of an index allows iterative traversal of a whole map, e.g.

```
var(m, "map");
.
.
.
for(int(i) ; i<entries(m) ; i++)
    println(m[m[i]]);
```

The items will be accessed in lexicographic order.

m[key] = v) - put an element into a map using subscript notation

put(m, key, v) - put an element into a map using an inbuilt procedure

The item **v** is placed in the map **m** against key **key**. If that key is already in the map the value will be replaced. If the last argument is omitted null will be put against the key. The key should be type "str" but if it is type "int" or type "uint" it will be converted to a string when the entry is inserted into the map.

When using subscript form, subscripts, fields, indices and keys may precede or follow the key 'subscript' as appropriate where the map is itself is nested within a struct, array, list or map or the map value is itself of nested types.

putall(m, m2) - put all elements from another map into a map

All items from map **m2** are added to map **m**. Where a key already exists its value will be replaced.

v = remove(m, key) - remove a map element using a key

The map item for the key is removed from the map. If no item with that key exists then no change occurs.

m = m2 - assign a map

If the LHS of an assignment is a map variable and the RHS is another map, the RHS map is assigned to the LHS variable. Unlike other immediate assignments the LHS is assigned the RHS object, not a copy of the RHS object. Thus **m** and **m2** contain the same map and changes to map entries via either variable will be reflected in the other. If a copy is required then the inbuilt function **copy()** [section 10.20](#) p277 can be used.

If the LHS of an assignment is a map variable and the RHS is a compound value, the LHS map is cleared and the values in the RHS are entered. The compound value list must not contain nested lists or null entries and all entries must be key=value format.

m = null - clear a map

If the LHS of an assignment is a map variable and the RHS is explicitly the keyword **null** (not a null value) the map is cleared, i.e. all entries are removed.

Maps may be compared using the **==** operator. Two maps will appear equal if they are indeed the same map (one assigned to the other) or if the maps are independent but represent the same mappings.

3.7.6 enums

An enumerated type is usually created by using inbuilt procedure **enum()** or **enumv()**, but it may also be specified by a type string. Enumerated types have ordered values which are selected by an associated identifier. Each value has an associated unique ordinal which is not related to the ordering. The order of enumerated type values is that of their occurrence in the argument list of the procedures which create the type.

3.7.6.1 using the enum() and enumv() procedures

The first argument of the inbuilt procedure **enumv(...)** is an identifier giving the enum name which will be created as a variable of type "type". This is followed by a number of argument pairs, the first component being an identifier and the second an associated ordinal value which must be a positive integer expression.

Example -

```
enumv(e1,
      none, 3,
      typea, 7,
      typeb, 5
    );
```

The identifiers and ordinals must be unique within this enum, but the identifiers may be used in other enums or as variable identifiers without conflict.

If specific ordinals are not required the simpler procedure **enum()** can be used, e.g.

```
enum(e1, none, typea, typeb);
```

which is the same as the previous example except that the ordinals are now 0, 1, 2 etc.

3.7.6.2 using a string specification

An enum type can also be created using a string specification. This consists of outside braces enclosing identifier/ordinal pairs, for example

```
"{none, 3, typea, 7, typeb, 5}"
```

The ordinals must be explicitly specified.

Note that

```
type(e1, "{none, 3, typea, 7, typeb, 5}");
```

is equivalent to

```
enumv(e1, none, 3, typea, 7, typeb, 5);
```

Any spaces in the specification string are removed prior to its evaluation.

3.7.6.3 selecting enum values

An enum value is selected by appending a "." and the value identifier to the type identifier, e.g.

```
type(e1, "{none, 3, typea, 7, typeb, 5}");
var(v, e1);
if (v == e1.typea)
    v = e1.typeb;
```

Note that two enum types are interchangeable if their identifiers and ordinals are identical ordered lists. Thus for example if we create identical types -

```
type(e1, "{none, 3, typea, 7, typeb, 5}");
type(e2, "{none, 3, typea, 7, typeb, 5}");
```

then types e1 and e2 are interchangeable, so we could write

```
var(v, e1);
v = e2.typea
```

without error. However, there seems no legitimate reason to do this and to avoid confusion it is to be discouraged.

3.7.6.4 associated functions

There are a number of inbuilt functions associated with enumerated types -

ord(v) section 10.103 p314	the ordinal value of variable v
next(v) section 10.96 p310	the next value of variable v
prev(v) section 10.107 p315	the previous value of variable v
hasnext(v) section 10.65 p296	true if variable v has a next value (i.e. it is not the last value)
hasprev(v) section 10.67 p297	true if variable v has a previous value (i.e. it is not the first value)
first(x) section 10.56 p292	the first ordered value of variable or enum type x
last(x) section 10.79 p302	the last ordered value of variable or enum type x

Note that the order of enum values is not given by the magnitude of the ordinals but by their order in the argument list of the creation procedure.

The identifier of an enum value can be obtained by using the inbuilt function **tostring()** [section 10.157](#) p340.

3.7.7 files

Procedure **println()**([section 9.51](#) p230) can be used to write to standard output.

A file variable is created using the inbuilt procedure **file()**. To read or write a file it must be opened. This is done by the inbuilt procedure **open()**([section 9.46](#) p223), e.g.

```
file(f1, "file1");
open(f1, "r");
```

where the "r" specifies opened for reading.

A number of files are already open as global variables. **stdin** is the standard input, **stdout** and **stderr** are standard output and standard error, and **rpt** is the 3PL report file. The report file is **design_name.rpt** if 3PL was called with the -r option, otherwise it is **stdout**.

A file variable can be copied to another file variable, e.g.

```
file(f1, "file1");
file(f2, "file2");
f2 = f1;
```

after which f1 and f2 now share the same file descriptor.

A file opened for reading can be read by inbuilt function **readln()**([section 10.122](#) p323). This function has one argument, which is the file variable, and returns a line from the file as a string.

Formatted input lines can be read by **scanf()**([section 9.67](#) p245) or **fscanf()**([section 9.35](#) p214).

If a file read is attempted at the end-of-file position an empty string is returned. This can be tested by inbuilt function **hadeof()**([section 10.63](#) p295) which returns true if the previous read occurred at end-of-file.

A line can be written to a file opened for writing by inbuilt procedure **writeln()**([section 9.86](#) p264) where the first argument is the file variable and the second argument is the string to be written.

Formatted lines can be written using **printf()**([section 9.49](#) p228) or **fprintf()**([section 9.34](#) p213).

On completion the file can be closed by **close()**([section 9.12](#) p196) where its argument is the file variable.

There are a number of inbuilt functions which can be used to test files and directories. These do not require file variables but operate on path names.

filecanread() (section 10.40 p285)	file can be read
filecanwrite() (section 10.41 p286)	file can be written
filedelete() (section 10.42 p286)	delete file
fileexists() (section 10.43 p286)	file exists
filegetparent() (section 10.44 p287)	absolute path above file
fileisdir() (section 10.45 p287)	file is a directory
fileisfile() (section 10.46 p287)	file is a file
filelastmod() (section 10.47 p288)	time of last modification
filelength() (section 10.48 p288)	file length
filelist() (section 10.49 p289)	list of files and directories
filelistdirs() (section 10.50 p289)	list of directories
filelistfiles() (section 10.51 p289)	list of files
filemkdir() (section 10.52 p290)	create a directory
filerename() (section 10.53 p290)	rename a file or directory
filesetreadonly() (section 10.54 p291)	change permission to read-only

3.7.8 type conversion

Types may be converted using the **cast()** inbuilt function.

3.7.8.1 immediate type conversion

For immediate types, conversions between types *bits*, *uint*, *int* and *str* are allowed with the provisos -

- int to uint gives an error if the integer is -ve
- str to numeric gives an error if not correctly formatted

A string to be converted to numeric must be

- a string of 0's and 1's preceded by "0b" or "0B"
- a string of hexadecimal digits preceded by "0x" or "0X"
- a string of octal digits preceded by "0"
- a string of decimal digits whose 1st digit is not 0

An *int* may be converted to a *float* and vice versa.

3.7.8.2 target type conversion

Target mode type conversions allowed are shown in the following table -

		to							
		bits	uint	int	ufixed	fixed	log	float	enum
from	bits	az	az	as	az	as	az	az	-
	uint	az	az	as	cz	cs	az	-	az
	int	as	-	as	-	cs	-	-	-
	ufixed	az	bz	bs	dz	dz	-	-	-
	fixed	as	-	bs	-	ds	-	-	-
	log	az	az	-	-	-	a	-	-
	float	az	-	-	-	-	-	e	-
	enum	-	az	-	-	-	-	-	-

a copy right justified

b copy from left of binary point, ignoring fractional part

c copy to left of binary point and zero fractional part

d copy aligned with binary point truncating or zero padding least significant bits if size of fraction differs

e copy with adjustments to biased exponent and mantissa as required

z zero pad or truncate most significant end if width differs

s sign extend or truncate most significant end if width differs

If casting to or from compound types any types are allowed (including primitive types) as long as the numbers of bits are identical. Conversions into or out of compound types is carried out by mapping the bits in order without regard to component boundaries.

When casting between int or uint and fixed or ufixed the binary point is recognised and where appropriate the sign is propagated, i.e. the cast retains numeric sense.

When casting between bits and fixed or ufixed the binary point is not recognised and bits are simply right justified.

When casting between bits and int or fixed and the cast type is wider, sign extension is performed instead of zero padding.

Type float can only be cast to another float, adjusting the biased exponent and mantissa as required.

Type bits or uint can be cast to type log which will extract the rightmost bit as false (0) or true (1). Casting type log to bits or uint results in a single bit, zero padded where required.

Casting an enum to a uint provides the ordinal value (there is also an `ord()` built in function). Casting a uint to an enum selects the enum value given the ordinal.

The function call is

```
cast(expression, type)
```

where *type* is a string type specification or a type.

3.7.9 pointers

Pointers are immediate variables which can point to immediate or target variables, or components of compound variables. They can also point to classes, modules, procedures and functions. Target variables cannot themselves be pointers. Pointers can only be used during immediate execution, not target execution, i.e. all pointers are resolved before target code is generated. A pointer type is indicated by a `->` in the type string. The type may be an array of pointers, e.g.

```
var(ptr1, "->");
var(ptr2, "[10]->");
var(ptr3, immediate, "[10]->");
```

The first example is a single pointer, the second an array of pointers. The 3rd example shows the redundant use of the *immediate* keyword - unnecessary since omission of the variable mode argument means that the variable is immediate.

An unassigned pointer has a null value, which can be tested.

Pointers may point to compound types (arrays, structs, subarrays, struct fields etc.).

struct fields may be pointer types.

Note that the type of the variable pointed to is not specified. A pointer may point to any type of variable and type checking is done when the pointer is resolved.

A pointer variable is assigned using the immediate assignment statement syntax

```
ptr1 = &v1;
ptr2[i] = &v2;
ptr3 = &println;
```

The `&` operator gets a pointer to a variable, class, module, procedure or function. A pointer to a variable can be to the whole variable, a sub array, an array member or a struct field. A pointer can also point to a list or a map. However, a pointer cannot point to a list entry or a map entry or anything inside such an entry.

A pointer reference from the `&` operator can be -

- assigned to a pointer variable, array member or struct field
- passed as an argument to a pointer parameter
- used as an initialiser to a created pointer variable
- used as an initialiser within a compound value
- appear within a compound value on the RHS of an assignment statement
- appear within a compound value which is an argument in a module, procedure or function call

For modules, procedures and functions these can be either inbuilt or user-defined. Since variables and inbuilt and user-defined modules, procedures and functions have separate identifier tables, and across these tables identifiers may not be exclusive, the address operator `&` searches in the following order -

- variables
- user-defined classes (there are no inbuilt classes)
- inbuilt modules
- user-defined modules
- inbuilt procedures
- user-defined procedures
- inbuilt functions
- user-defined functions

and returns the first that it encounters.

A pointer may be explicitly assigned null by -

```
ptr = null;
```

Pointers may be tested for equality using the `==` and `!=` operators, e.g. -

```
ptr1 == ptr2
ptr3 != null
ptr4 == &var[1].f
```

the other relational operators and arithmetic operators cannot be used on pointers.

A pointer is resolved by using the operators `->`. If the pointer points to a compound type then subscripts or field identifiers follow the `->`. Subscripts and fields associated with the pointer variable are placed following the variable identifier and prior to the `->`.

A second version of the `->` pointer operator can be used to dereference pointers in arguments to modules, procedures or functions. This operator is `->>`. An attempt to dereference a pointer which is null using `->` results in a fatal error. A null pointer dereferenced using `->>` results in a null value. As an argument this is equivalent to omitting the argument. Thus if a module, procedure or function is called using arguments which are pointers dereferenced by using `->>` then any pointer which is null will simply omit the argument. Use of `->` in this case would give a fatal error. The two forms of pointer dereferencing are otherwise identical but the resulting null from the second form is not acceptable in other contexts so it would confer no advantage.

The following example shows a single pointer to a single variable

```
var(ptr, "->");
static(v1, "uint:8");
static(v2, "uint:8");

ptr = &v1;

ptr-> <- 3;      // equivalent to v1 <- 3
v2 <- ptr->;     // equivalent to v2 <- v1
```

Note that a pointer operator can be used on the left hand side of an assignment statement.

For a pointer to an array

```
var(ptr, "->");
static(v1, "[4]uint:8");
static(v2, "uint:8");
int(i, 1);

ptr = &v1[0];

ptr->[1] <- 3;    // equivalent to v1[1] <- 3
v2 <- ptr->[i];   // equivalent to v2 <- v1[i]
```

For a pointer to a struct

```
struct(s,
      f1,  "log",
      f2,  "uint:8"
);
static(v1, s);
static(v2, "log");
var(ptr, "->");

ptr = &v1;

ptr->.f2 <- 3;    // equivalent to v1.f2 <- 3
v2 <- ptr->.f2;   // equivalent to v2 <- v1.f2
```

For an array of pointers

```
struct(s,
      f1,  "log",
```

```

        f2,    "uint:8"
    );
    var(ptr, "[4]->");
    static(v1, "[4]s");
    static(v2, "uint:8");
    int(i, 3);

    ptr[0] = &v1[0].f1;
    ptr[1] = &v1[1];
    ptr[2] = &v1;

    ptr[0]-> <- false;
    v2 <- ptr[1]->.f2;
    when (ptr[2]->[i].f1)
        .....;

```

For pointer to pointers to pointers etc. the pointer dereferencing operator is simply repeated, e.g.

```

p1[1]->.f1->[i].f2->
p2->->->

```

Note that in the above examples where target variables are referenced using an immediate variable subscript, the value of the subscript at that time is substituted and thus the subscript in the target code is a constant. Target subscripts containing static and queue mode variables can be used for target variables and this is described later.

Pointers to modules and procedures are dereferenced again using the -> operator -

```

    var(p, "->");
    .
    .
    p = &buffer;
    .
    .
    p->(p1 <- p2);

```

A pointer variable **p** has been assigned a pointer to the inbuilt module **buffer()** and then the module has been called via the pointer. Note that we could also have initialised the pointer variable -

```

    var(p, "->", &buffer);
    .
    .
    p->(p1 <- p2);

```

Procedures are handled using the same syntax.

Pointers to functions are handled in a similar manner -

```

    var(p, "->", &exists);
    .
    .
    if (p->(v))
        println("variable ' + v + "' exists");

```

Where such a pointer is within an array or a struct subscripts and fields can be interposed -

```

    var(p, "[3]->", {, &exists});
    .
    .
    if (p[1]->(v))
        println("variable ' + v + "' exists");

```

Further indirection could be used -

```

var(p, "[3]->", {, &exists});
var(q, "[4]str", {, "p"});
.
.
if (q[2]->p[1]->(v))
    println("variable '" + v + "' exists");

```

Where a function returns a pointer the pointer can be dereferenced by applying the `->` operator to the function call, e.g.

```
c = f(a, b)->;
```

Subscripts and fields can be appear both before and after the `->` where appropriate.

3.8 Dimensionality of variable references

When assigning variables or passing variables as arguments to modules, procedures or functions compound types are allowed. Thus an array, sub-array or struct, or any legitimate combination of these, can be assigned or passed as an argument. Clearly when this is done the type structure must match across the assignment or argument/parameter match.

A constant subscript reduces the dimensionality of an array. For example if we declare an array by -

```
var(a, "[4]int");
```

then **a** is an array, **a[1..2]** is an array, but **a[1]** is not an array. Similarly for -

```
var(a, "[4] [3] [4]int");
```

a[2][1..2] is a 2-dimensional array since the constant subscript gets rid of one dimension. Trailing subscripts can be omitted without changing the result, so that **a[1..2][2]** is a 2-dimensional array as the omitted trailing subscript implies the full dimension.

3.8.1 compound assignment

Compound objects can be assigned as long as the dimensionality of the LHS and RHS match. For example -

```

var(a, "[4] [3] [4]int");
var(b, "[6] [4]int");
a[1..2][1] = b[3..4];

```

is valid as each side has dimensions **[2][4]**.

Structs can be nested within types but the same dimensionality consideration applies -

```

struct(s,
    a, "[2] [8]int",
    b, log
);
var(a, "[4] [3] [4]s");
var(b, "[6] [4]int");
a[1..2][1][2].a[1][2..5] = b[3..4];

```

here each side again has dimensions **[2][4]**.

If the RHS is a compound value with missing elements, LHS elements corresponding to those missing on the right will not be changed.

3.8.2 compound arguments

When a compound argument is passed to a module, procedure or function parameter which has no type, the parameter takes on a type which matches the dimensionality of the argument. Any subscripts and field identifiers attached to the parameter apply to the compound object passed to the parameter. Thus in the following example -

```
var(a, "[4][3][4]int");

.
proc(a[1..2][1]);
.

procedure
proc(var(p)) {
  ...p...           // 2-dimensional array
  ...p[0]...        // 1-dimensional array
  ...p[0][1..2]...  // 1-dimensional array
  ...p[] [0]...     // 1-dimensional array
  ...p[0][1]...     // single int
}
```

the procedure parameter is passed a 2-dimensional array. The matching parameter is now handled in the procedure as though declared as a 2-dimensional array with dimensions appropriate to the array passed. As with assignments the parameter can be referenced as a 2-dimensional array, a 1-dimensional array (one subscript given) or a single element (two subscripts given).

Again struct field identifiers may appear among the subscripts if structs are involved.

If a module, procedure or function parameter is created as a compound type then the type of the argument must match that of the parameter.

Between these extremes if a parameter contains undimensioned arrays then these will take on the dimensionality of the argument, but otherwise the types must match. If a parameter contains target types whose width specifiers are missing these will be supplied from the argument, but again the types must otherwise match. An undimensioned array parameter can derive missing array dimensions from the argument supplied or else from an initialiser if present. The argument and the initialiser may be variables or compound values. If a parameter containing undimensioned arrays is not initialised and is not passed an argument then the undimensioned arrays will be given dimensions 0.

If a parameter creates an undimensioned array because it is not passed an argument and has no initialiser then an execution error occurs. If a parameter has an undimensioned array type and the argument is optional then some initialisation must be supplied, e.g.

```
procedure
p1 (var(p, "[ ]int", {})) {
  ...
  ...
}
```

which will create an integer array of dimension 1. Since the initialisation entry is null no assignment is made and the single array member will have the default value, which in the case of an integer is 0.

Note that if a list of identifiers is supplied as the first argument to **var()**, rather than a single identifier, then missing dimensions in an array type cannot be inferred from arguments since all variables are constructed to be the same single type. However, missing dimensions may still be derived from an initialiser.

A value mode parameter may be passed an immediate argument. If the argument is a compound type then the parameter must have a target type equivalent to the immediate type of the argument. Arithmetic values will be checked to ensure that they fit within the size of the matching target sub-types. For example -

```

int(i, 4);
struct (st,
  ls, "[2]log",
  fi, "int:8"
);

p({{false, true}, i+30});

procedure p (value(v, st)) {
  .
  .
}

```

3.9 Arrays of queues or memories

Arrays of queues and arrays of memories are not supported in 3PL. However, when these are required there is a way to effectively create and access these. Queues and memories can be created within a function or procedure as local variables and pointers to them returned in a pointer array. For example to create a one-dimensional array of queues the following code would suffice (this is a simplified version as the library procedure of the same name in *stdlib.3pl* will handle any number of dimensions) -

```

procedure
queuearray (var(ptr) <- str(type)) {
  for (int(i) ; i<dimension(ptr) ; i++) {
    queue(p, type);
    ptr[i] = &p;
  }
}

```

The output parameter is a 1-dimensional array of pointers. The procedure creates as many queues as there are pointers in that array. The queues are created inside the *for* loop which has an immediate statement block. Each queue has the same identifier but they are unique as each iteration of the loop has a new scope. A pointer to each queue is assigned to a member of the pointer array after it is created (variables survive the termination of their scope but cannot be accessed directly via an identifier, only through a pointer).

Note also that the *for* loop has a statement block in brackets, which has its own scope. A single unbracketed variable creation procedure call in a *for* loop would create the variable in the scope containing the loop, so the following code -

```

for (int(i) ; i<dimension(ptr) ; i++)
  queue(p, type);

```

would produce a multiply-defined variable (the code would be useless anyway since we have not assigned pointers to the queues).

The queues in the array can now be used by the following constructions -

```

.
.
p[1]-> <<- ....;
.
.... <- p[i]->;
.
.

```

If the queues are compound (arrays or structs) then the subscripts and field identifiers follow the right arrow.

3.10 Directives

Directives are intended as a means by which immediate program execution can control the compiler. Thus far 40 directives are recognised by the compiler interpreter (see below) but more may be implemented in the future. Since directives can be created and evaluated by the application code as well as by the compiler interpreter they can be used in a limited way as an extra variable mechanism.

A directive is an entry in a global map. Each directive has a key and a value. The directive is stored in the map using the key to access the value. The key is a string. The value may be one of the 3PL immediate primitive types **int**, **log**, **float** or **str** (but not an enumerated type).

Directives are entered by using the inbuilt procedure **directive()** (section 9.16 p200) and are extracted by the inbuilt function **directive()** (section 10.28 p280). The inbuilt function **direxists()** (section 10.30 p281) can be used to check if a particular directive is present. The directive map can be obtained using the inbuilt function **directives()** (section 10.29 p281).

Directives may also be entered by using the -D command line option when calling 3PL (section 19.2 p624).

A directive can also be entered during parsing (prior to immediate execution) by the inbuilt procedure **presetdirective()** (section 9.50 p229).

Directives are global in scope.

The following directives are recognised by the compiler or set by the compiler itself -

- "languageVersion"
- "compileOnly"
- "compilerMakeDate"
- "author"
- "ALU"
- "continuous"
- "currentDirectory"
- "date"
- "designName"
- "family"
- "FIFO"
- "filePath"
- "forceExec"
- "gatesNotLUTs"
- "intermediateOnly"
- "intTruncWarning"
- "listNets"
- "listTDEs"
- "listTDEsSel"
- "locations"
- "netlistcomment"
- "netlistDisplay"
- "netlistFile"
- "optimiseConnect"
- "outputDirectory"
- "OS"
- "parentDirectory"
- "part"
- "postProcessTool"
- "QUEUEREG"
- "reportFile"
- "rptToFile"
- "shell"
- "sigList"
- "skipPostProc"
- "sourceFile"
- "sourceVersion"
- "srlAddrWidth"

- "unass_out"
- "verbose"

The compiler itself creates the following directives -

"languageVersion"	str	the compiler language version
"sourceVersion"	str	the current compiler source code version number
"compilerMakeDate"	str	the date the compiler was made. Note that the date and time that the 3PL compiler is executed can be obtained using inbuilt functions section 10.151 p336 time() for date and time elements and section 10.152 p337 timestring() for date and time as a string.
"OS"	str	the host computer operating system
"arch"	str	the host computer architecture
"srlAddrWidth"	int	the address width of CLB shift registers
"date"	str	the current date and time
"sourceFile"	str	the source file name specified on the command line (with ".3pl" appended by 3PL if absent)
"parentDirectory"	str	the full path name of the directory containing the main source file
"currentDirectory"	str	the full path name of the current directory
"outputDirectory"	str	the full path name of the output directory specified by the -o command line option, or the parent directory by default
"designName"	str	the design name derived from the source file name or specified by the -N command line option
"cellName"	str	the cell name to use if the netlist is a logic core produced by calling inbuilt procedure export() section 9.30 p211.
"family"	str	a string designating the FPGA family, e.g. XC5V

The compiler creates the following directives from command line options. Most of these directives can also be set by application code which will override any value derived from command line options.

"compileOnly"	log	true if the -c command line option has been used. 3PL will exit after compiling the source code and before execution. Cannot be set during execution.
"forceExec"	log	true if the -f command line option has been used. If true immediate execution and netlist code generation will be carried out regardless of the time of last modification of associated files. Cannot be set during execution.
"gatesNotLUTs"	log	true if the -G command line option has been used. If true this will generate netlist logic encoded as gates (AND, OR etc.) rather than as LUTs.
"intermediateOnly"	log	true if the -i command line option has been used. 3PL will exit after compiling the source code and completing execution but prior to netlist generation. It can be invoked if the aim is just to examine the TDE list rather than to produce a netlist.
"intTruncWarning"	log	true if the -w command line option has been used. 3PL will emit a warning to the report file for any target "int" or "uint" that has been assigned to a narrower destination. "fixed" and "ufixed" warnings are always given. If set during execution it will take effect from then on. Similarly can be cleared during execution.
"listNets"	log	true if the -n command line option has been used. This will produce a list of nets used in the netlist file with all aliases. The report is written to a .net file.
"listTDEs"	log	true if the -t or -T command line options have been used. Either of these options may optionally be followed by the digit 0, 1, 2 or 3 which will print the TDE list - 0 - after all optimisation 1 - before any optimisation 2 - after the 1st optimisation 3 - after family-specific optimisation. If no digit is encountered then the TDE list after all optimisation is printed.
"listTDEsSel"	int	this directive contains an integer value derived from the optional digit described in "listTDEs" above. It is 0 by default.
"locations"	log	true if the -l command line option has been used. If true listings of the TDE list will contain source code locations
"netlistDisplay"	log	true if the -d command line option has been used. If true the interactive netlist display window will pop up following netlist generation.
"optimiseConnect"	log	false if the -O command line option has been used; true by default. This disables code which attempts to eliminate CONNECT TDEs and connect them through the catalog instead. It can be used to diagnose some logic inconsistencies should they arise. It is only used during development or maintenance.
"rptToFile"	log	true if the -r command line option has been used - verbose mode is enabled and the report output is written to file design_name.rpt
"sigList"	log	true if the -T command line option has been used. It causes the TDE signal definition section of the TDE list to be printed along with the list itself. This is usually not useful. design_name.rpt
"skipPostProc"	log	true if the -s command line option has been used. It causes any post-processing 3PL code to be skipped.
"verbose"	log	true if the -v, -r, -t, -T or -P command line options have been used - a detailed report will be written to standard output by default or to file design_name.rpt if the -r option has been given. If set during execution some initial report output will be missing.

The remaining directives used by 3PL are set by application code -

String directive "**author**" is optional. If given a value (string) this will be inserted as an author field in the EDIF file (Xilinx claim this field should contain the company name, though there seems no reason

why it matters what it contains). The compiler interpreter uses the value of the directive (if any) at the completion of immediate code execution.

String directive "**ALU**" specifies that any static or selectvalue variables which are directly assigned the result of an arithmetic operation³ in two or more statements will have those operations combined into a single Arithmetic Logic Unit (ALU). Where a value to be added, subtracted or assigned is a constant, such constants are optimised to only appear once. This optimisation only applies to an operator whose result is directly assigned to the variable, not to operators in sub-expressions since these are evaluated in parallel and must have separate operator logic.

Where a number of increments, decrements, additions or subtractions are made to a static variable the use of an ALU avoids duplication of expressions for each instance using instead a single combined adder/subtractor. In simple cases this substantially reduces the use of logic gates and does not increase path delays. However if any values to be assigned, added to or subtracted from the variable require combinatorial logic of significant depth, i.e. they involve further arithmetic operators, then the use of a shared ALU may result in greater path delays which may exceed the clock domain timing constraint.

The effect of this directive can be changed for individual static variables by setting the **ALU** attribute. If the directive is not defined or is not true then the ALU attribute can be set true on individual static variables to implement an ALU if appropriate (as per the mixed assignment types described above). If the directive is defined and is true setting attribute ALU false will disable the effect on a particular static variable.

String directive "**continuous**" is defined and true (log) the following logic simplifications will be applied

- conditional static and queue assignments (within a **when** statement) will be enabled directly from the **when** test expression rather than indirectly via an execution signal through a **when** TDE. This optimisation occurs if the directive exists and is true on encountering a when during immediate execution.
- during optimisation following immediate execution all unconditional static and queue assignments will be removed from execution threads and enabled continuously. This achieves the same effect as setting the **continuous** attribute true on each static variable. This optimisation occurs if the directive exists and is true at completion of immediate execution.

To enable both optimisations universally "**continuous**" should be created and set true. This should be done using `presetdirective()` rather than `directive()` since it should be set prior to immediate execution - if set during immediate execution prior code may be generated without the directive which might introduce some subtle problems. The use of `directive()` for this purpose is not prohibited since it may be required to be set conditionally, in which case the conditional code should precede any target code. Use of this directive can significantly reduce logic path delays and to a small extent the logic resources used.

Directive "**continuous**" can also be set by supplying the 3PL command line argument `-D continuous`.

To restrict this optimisation to individual variables the "continuous" attribute can be set true on a static or queue mode variable instead of using the global directive. Alternatively the global effect can be removed for individual static or queue mode variables by setting the "continuous" attribute to false. The directive prevails if the attribute has not been set, otherwise the attribute is used.

Directive "**FIFO**" when set true prior to completion of immediate execution specifies that queue mode variables whose depth is such that block RAMs will be used (generally depth > 16) will be implemented using FIFO hardware associated with the block RAMs if the FPGA includes such hardware. This feature can be controlled on individual queue mode variables by attribute **FIFO** which if specified overrides the directive. The default behaviour if neither directive nor attribute are used is to NOT use the dedicated FIFO hardware.

Use of the FIFO hardware is a compromise between speed and logic usage. The FIFO hardware saves CLB gates and flip-flops, which would otherwise be used, but has greater path delays which may cause failure to meet timing constraints.

Directive "**filePath**" when true causes error messages to give file paths rather than file names. This may occasionally be useful to disambiguate variables of the same name in source files in different directories. This can also be set by the command line option `-F`.

³At the moment only + and -.

String directive **"gatesNotLUTs"** when set true prior to completion of immediate execution causes many LUTs normally produced by the code generated to be replaced with nested gates. The logic remains the same but the use of nested gates instead of explicit LUTs may change the way the vendor place-and-route software optimises and packs the logic which in turn may change the ease, or lack thereof, with which the timing constraints can be met. Thus when timing constraints fail to be met, trying the -G option may improve the situation. The directive can also be set using the -G option when calling 3PL

String directive **"netlistcomment"**, if defined, is copied as a comment to the EDIF netlist file and the NCF constraints file or the XDC constraints file. In the EDIF netlist file it appears within double quotes in a comment field. In the constraint files it appears following a "#". If the directive contains more than one line these will appear as separate comment lines in the constraint files, but in the netlist file the whole string will appear within a single pair of quotes, including the newlines.

String directive **"part"** must contain the full FPGA part specification (string). It is set within inbuilt procedure **part()** (section 9.48 p228). The part specification is not only inserted in the EDIF netlist file, it is also used to guide code generation to suit the FPGA type. The directive must be assigned a value prior to target code generation and its value cannot be changed during the remainder of immediate execution. An attempt to change the directive will result in a warning message. If this directive is not recognised a fatal error will be flagged and compilation will terminate. If this directive is not defined, at the moment (with single vendor implementation) the Xilinx UNISYMS generic part will be used.

Allowed values for directive "part" are -

- "XC95..." XC9500 CPLD, e.g. "XC9536PC44-5"
- "XC2C..." Coolrunner 2 CPLD, e.g. "XC2C32APC44-6"
- "XC2S..." Spartan 2 FPGA
- "XC3S..." Spartan 3 FPGA
- "XC6S..." Spartan 6 FPGA
- "XCV..." Virtex or Virtex E FPGA, e.g. "XCV300EBG432-7"
- "XC2VP..." Virtex 2 Pro FPGA, e.g. "XC2VP40FF1152-6"
- "XC2V..." Virtex 2 FPGA, e.g. "XC2V3000BF957-4"
- "XC4V..." Virtex 4 FPGA
- "XC5V..." Virtex 5 FPGA
- "XC6V..." Virtex 6 FPGA
- "XC7A..." Artix 7 FPGA
- "XC7K..." Kintex 7 FPGA
- "XC7V..." Virtex 7 FPGA
- "XC7Z..." Zynq FPGA

String directive **"QUEUEREG"** when set true prior to completion of immediate execution specifies that queue mode variables are to have a pipelined register added to their outputs. This reduces the clock-to-output time. The queue buffer logic incorporates the extra buffer with no change to functionality. The attribute **QUEUEREG** may be set (true or false) on individual queue mode variables, overriding the global directive.

String directive **"reportFile"** contains the absolute path name of the report file.

If the **shell()** section 9.70 p248 procedure is to be used and a shell other than the default shells **/usr/bin/bash** or **/bin/bash** is required then the string directive **shell** should be defined with the value being a string giving the shell path.

String directive **"unass_out"** controls the behaviour of output mode variables which have unassigned components. The directive type "log" -

- false - connect to ground
- true - connect to VCC

For individual output mode variables the default can be controlled by the output mode variable attribute **defaultout** (see section 3.11 p80). The action of this attribute overrides directive "unass_out". If neither specify a value an unassigned output mode variable component will result in a fatal error.

String directive **"postProcessTool"** is mandatory. It should have the value "ise" (for the Xilinx ISE tools) or "vivado" (for Xilinx Vivado). **It must be defined during parsing by presetdirective() or fatal termination of 3PL immediate execution will occur.** This should be done in the header file for each FPGA family member. If the directive is changed during immediate execution it must be remembered that post-processing 3PL code will use this directive and the default value for the FPGA

platform will prevail if post-processing is not preceded by main-code execution because the EDIF file is already up-to-date from a previous run.

The choice of place-and-route tools changes 3PL in two ways -

1. 3PL will create either an NCF constraint file for ISE or an XDC constraint file for Vivado, assuming that constraints have been generated
2. the form of element pins in the EDIF netlist for arrays is different, ISE requiring pin arrays of the form A0, A1, A2, A3 and Vivado A[0:3]

3.11 Attributes

Attributes are assigned to variables to

- control some aspects of code generation
- set associated parameters such as clock frequency or queue buffer depth
- pass constraints to the EDIF netlist generator

Case is ignored in attribute name strings. They are converted to lower case when used within 3PL.

Attributes are passed by means of a map type. This may be a variable of type "map" or it may be a one-dimensional compound value where every entry has the key=value format, the key being the attribute identifier and the value its associated value.

Among the attributes described below the **loc** attribute applies to several variable modes. This attribute is usually an array of strings ("[]str") but where a single location is involved it may also be a string ("str").

Attributes are usually simply omitted if not required, but they may be given default or inactive values if desired. Where an attribute is boolean (type "log") it must be present and true to take effect.

Attributes are described within the variable creation procedures to which they apply.

3.11.1 applying attributes

Attributes are applied to a variable in a map. The map keys are the attribute identifiers and the associated values are the attribute values. The entry values are mixed types, the type depending on the individual attribute.

The variable creation inbuilt procedures **input()**([section 9.39](#) p218), **output()**([section 9.47](#) p224) and **clock()**([section 9.10](#) p194) have an optional final argument which is an attribute map. The attribute map argument can be a variable of type "map" or it can be a compound value. The following code segments all achieve the same thing -

```
var(cmap, "map");
cmap["frequency"] = 100.0e6;
cmap["dutyicycle"] = 0.25;
clock(c);
attributes(c, cmap);
```

The **attributes()**([section 9.7](#) p192) inbuilt procedure applies an attributes map to a variable. This is very longwinded,

```
var(cmap, "map", {frequency=100.0e6, dutyicycle=0.25});
clock(c);
attributes(c, cmap);
```

is still rather verbose,

```
var(cmap, "map", {frequency=100.0e6, dutyicycle=0.25});
clock(c, cmap);
```

is shorter.

Attributes can instead be passed directly using key=value arguments, e.g.

```
clock(c, frequency=100.0e6, dutycycle=0.25);
```

is the most concise.

The **attributes()** inbuilt procedure will accept a bracketed list of variables as the first argument, applying the attributes to every variable in the list.

The **attributes()** procedure does not enter a new attributes map in the variable, it adds the entries in the argument map to that in the variable. Thus the **attributes()** procedure can be called multiple times to add one or more attributes to the variable. If the same attribute is entered again the new value replaces the old value. This can be used to enable and disable 3PL attributes that are used during immediate execution.

Netlist attributes can be deleted by assigning the value **null**. 3PL attributes cannot be deleted; 3PL attributes either relate to parameters that are mandatory such as clock domains, or are boolean in which case they can be disabled rather than deleted, so there is no reason to delete them.

Netlist attributes are only used at the termination of immediate execution so changing their values or deleting them during immediate execution is not an error.

Attributes **domain**, **readdomain** and **writedomain** can be assigned **null** and this sets the associated domain to the current clock domain.

Where a variable is passed as an argument to a module, procedure or function the associated parameter is implemented as an internal pointer to the argument variable (call by reference) and hence if attributes are changed in the parameter they are changed in the argument variable.

3.11.2 examining attributes

The attributes of a variable can be fetched by the inbuilt function **attributes()**([section 10.11](#) p271) which returns a map. If the second optional argument to the function is true an extended set of attributes is returned. These are -

- **identifier** - the abbreviated identifier
- **eidentifier** - the extended identifier
- **mode** - the mode
- **type** - the type
- **typestring** - the type as a string
- **decloc** - the source file location where created
- **inputparam** - true if this is an input parameter
- **outputparam** - true if this is an output parameter
- **matched** - true if this is a parameter matched by an argument
- **readonly** - true if read-only
- **assigned** - true if the variable has been assigned
- **used** - true if it is a target mode variable and its value has been used
- **clk_srcs** - for a clock variable the number of sources
- **clk_sinks** - for a clock variable the number of sinks
- **clk_CE** - for a clock variable the number of element clock inputs driven by this clock. This may be smaller than **clk_sinks** as that attribute includes clock inputs to buffers or clock managers where the clock is not triggering state changes.

The attributes in this list are read-only and cannot be assigned to the variable.

A single attribute can be fetched by the inbuilt function **attribute()**([section 10.10](#) p271).

Attributes can also be fetched for listing purposes as an array of structs by calling **listattributes()** [section 10.80](#) p303. The struct contains three string fields, key, the attribute key, mode, the mode of the value, and value which is a string version of the value or is the identifier of a clock domain. This array may be printed to standard output by calling library procedure **printattributes()** [subsubsection 11.13.3.5](#) p509.

Attribute **domain** for a value mode variable has a null value when read back regardless of any assignments to the variable.

For a selectvalue, static or priority mode variable attributes **domain**, **readdomain** and **writedomain** are the same. These attributes are only added to the variable when it is read or written.

For a queue mode variable, attributes **readdomain**, **writedomain** and **domain** are the same if the queue is synchronous. If the queue is asynchronous attributes **readdomain** and **writedomain** differ and attribute **domain** is not present in the attribute map. These attributes are only added to the variable when it is read (**readdomain**), written (**writedomain**) or both (**domain** if synchronous).

Attribute **identifier** is the variable identifier, i.e. the identifier used when the variable was created.

Attribute **eidentifier** is the expanded identifier. This is full unambiguous identifier used in the intermediate code. Both scope information and arbitrary appended integers are used to construct a unique identifier.

Attribute **mode** is the variable mode as a string.

Attribute **type** is the variable type returned as a "type" type.

Attribute **typestring** is the variable type returned as a string.

Attribute **decloc** is the source code location where the variable was created.

Attribute **inputparam** is a value of type "log" which is true if this variable is an input parameter to a module, procedure or function.

Attribute **outputparam** is a value of type "log" which is true if this variable is an output parameter to a module, procedure or function.

Attribute **matched** is a value of type "log" which is true if this variable is a parameter to a module, procedure or function and has been passed an argument.

Attribute **readonly** is a value of type "log" which is true if this variable has been made read-only.

Attribute **assigned** is a value of type "log" which is true if this variable has been assigned. The variable cannot be input mode, cmemory mode or rmemory mode.

For an immediate and value mode variables this attribute is true if the variable has been explicitly assigned a value during immediate execution - this does not include any initial value given to the variable on creation.

For target variables other than value mode this attribute is true if a target assignment statement has been processed during immediate execution where this variable appears on the left hand side - this excludes any initial values given to variables on creation.

For clock mode variables this attribute is true if -

- the clock variable was created using inbuilt procedure **clock()** where external input has been specified by supplying the location attribute
- the clock variable is a negative clock that has been created using inbuilt procedure **negclock()** and the positive clock from which it is derived is later assigned or has previously been assigned
- the clock variable has been assigned from another clock variable which is later assigned or has previously been assigned
- the clock variable is an input parameter to inbuilt procedure **elementoutput()** where it is connected to an output port or a 3-state port
- the clock variable is an output parameter to inbuilt procedure **clockfromexpr()**

Attribute **clk_sinks** for a clock variable is the number of places the clock is used to clock logic

Attribute **clk_srcs** for a clock variable is the number of places the clock is driven. If this is not 1 either the clock has no source or it has multiple sources; in either case an error.

3.12 Operators

The following operators have been implemented.

~	unary prefix	bitwise inversion
!	unary prefix	logical negation
-	unary prefix	arithmetic negation
<	unary prefix	queue read availability
>	unary prefix	queue write availability
&	unary prefix	address
++	unary prefix	increment
--	unary prefix	decrement
++	unary postfix	increment
--	unary postfix	decrement
?	unary prefix	queue examine
->	unary postfix	pointer dereference
->>	unary postfix	pointer dereference
*	binary	multiplication
/	binary	division
%	binary	remainder
+	binary	addition or string concatenation
-	binary	subtraction
<<	binary	left shift
>>	binary	right shift
^<<	binary	left rotate
>>^	binary	right rotate
<	binary	less than
>	binary	greater than
<=	binary	less than or equal to
>=	binary	greater than or equal to
==	binary	equal to
!=	binary	not equal to
&	binary	bitwise AND
^	binary	bitwise XOR
	binary	bitwise OR
&&	binary	logical AND
	binary	logical OR
^^	binary	logical XOR
+=	binary	add or concatenate right to left
-=	binary	subtract right from left
*=	binary	multiply left by right
/=	binary	divide left by right
%=	binary	remainder of left divide by right
&=	binary	bit AND right to right
=	binary	bit OR right to right
^=	binary	bit XOR right to right
&&=	binary	logical AND right to right
=	binary	logical OR right to right
^^=	binary	logical XOR right to right
<<=	binary	left shift left by right
>>=	binary	right shift left by right
?:	ternary	conditional

Note that the C ',' operator (comma), as sometimes used in **for** statements is not implemented in 3PL.

The binary and unary operators are listed in order of decreasing precedence in the following table -

* / %
+ -
== != < > <= >=
&
^
&&
^^
?:

Note that the relational operators have higher precedence than the logical operators so that an expression such as `a < 3 && b > 5` does not need parentheses.

The `++` and `--` operators can be used on **immediate** and **static** variables, but not **value**, **selectvalue**, **queue** or **priority** variables.

For **static** variables these operators can only be used in the form of increment and decrement statements and not in any other context, e.g. in an expression. Thus for example for static variable `t` -

```
t++;
```

is allowed but the forms

```
a <- t++;      // ERROR!
a[t++] <- b;   // ERROR!
func(t++);     // ERROR!
```

are not and would result in a fatal immediate execution error.

The pre-increment and post-increment forms behave identically since they form a statement and not part of an expression. The same applies to the decrements.

The `?` prefix unary operator allows evaluation of an expression containing queues without removing data from the queues. Unlike a queue read this operation proceeds regardless of queue availability. If an examined queue is not available the data is not valid. Use of this operator would normally be accompanied by an explicit queue availability test using the `<` prefix unary operator (next paragraph). The operand may be a single queue variable or any expression containing queues, but need not contain queues at all and may even be immediate. If the operand contains no queue reads then the value is simply returned.

The `<` prefix unary operator examines its expression argument for queue read availability and produces a logical result. This operator does not execute a read, i.e. a datum will not be removed from any queues. The expression may be a single queue variable or may contain a number of queues, but the expression need not contain queues and may even be immediate. Where the operand contains no queues `<` will return the constant *true*. It would be usual where the argument is an expression rather than a single queue variable for the expression to be passed via a value variable since the expression is presumably used elsewhere for actual evaluation and duplicating the code and then ignoring the result by applying the operator would be silly. Queue references within the operand of the `<` operator are ignored by any enclosing *sync*. The read availability operator can only be used in a module within which one or more reads of the queue or queues are also executed (read availability of a queue in a module which does not read the queue is not meaningful). The resulting logical value has the output (read) clock domain of the queue variable. If the output (read) clock domain has not yet been established then use of the availability logical value in a clocked context will establish the output (read) clock domain of the queue variable⁴.

The `>` prefix unary operator examines its variable argument for queue write availability and produces a logical result. This operator does not execute a write. Where the variable has one or more target

⁴If the queue read availability operator is used in an expression assigned to a value mode variable the output clock domain of the associated queue variable will not be established by that assignment. If the value variable is subsequently used on the RHS of an assignment to a clocked variable, or in another context in which a clock domain is required, then that will establish the clock domain of the associated queue variable if it is not already established.

subscripts the > operator requires them to exhibit read availability as well before it will return true. The resulting logical value has the input (write) clock domain of the queue variable. If the input (write) clock domain has not yet been established then use of the availability logical value in a clocked context will establish the input (write) clock domain of the queue variable.

Note that the > operator requires the queue variable assignment clock domain. It must be called in the same clock domain as assignments to the queue and if its use precedes any assignments in the source code it will set the assignment clock domain.

The postfix -> and ->> unary operators dereference a pointer. The second form is for use in a module, procedure or function argument where a null pointer results in omission of the argument rather than a fatal error. Note that this differs from its use in C in that if a struct is pointed to a field reference still requires a "." before the field identifier, e.g.

```
p->.field
```

The -> unary operator will also dereference a string variable where the string is a variable identifier. For example

```
str(s, "var1");
println(s->);
```

will print the value of variable **var1**. The identifier string may include scope prefixes such as "global".

The -> operator may be preceded by fields or subscripts to dereference a pointer or string contained in a compound type.

The -> operator may be followed by fields or subscripts when the type pointed to is compound.

The -> unary operator will also dereference an expression in parentheses, e.g.

```
int(i, 5);
println(("var" + i)->);
```

will print the value of variable **var5**. The expression may include function calls and can be of type "str" or type "->". In the latter case the expression can only be a function returning a pointer since arithmetic cannot be performed on pointers. The expression could be the address of a variable using the & operator but that would be singularly pointless.

The ? : ternary infix operator selects one of two values. The first argument must be type "log". The second and third arguments may be any type but must be the same type. If the test expression is immediate mode the availability of the result is the availability of the selected (2nd or 3rd) operand. If the test expression is target mode the availability of the result is the availability of the test expression ANDed with the logical result of the test expression and the availability of the selected 2nd or 3rd operand. Thus as long as the test expression is available and the selected 2nd or 3rd operand is available the result will be available even if the unselected operand is not available. This operator may be nested.

The >> and << binary infix shift operators can only be used on immediate types "uint" or "int" or target types "bits:", "uint:", "int:", "fixed:" or "ufixed:". The second argument may be an immediate variable of type "uint" or "int" (but cannot have a negative value), or a target variable of type "uint:". Right shifts on signed values extend the sign bit.

The >>^ and ^<< binary infix rotate operators can only be used on target mode values which can be of any type. The second argument must be an immediate variable of type "uint" or "int" and cannot have a negative value. These operators perform a simple bit rotation and take no account of the data type.

The following assignment symbols are not treated as operators, but rather as part of statement syntax.

=	immediate mode assignment
:=	value, priority or clock mode assignment
<-	static mode assignment
<<-	queue mode assignment
-	selectvalue or conditional output mode assignment

The ++ and – postfix operators can also be used as a form of assignment (see sections [section 5.1](#) p125 and [section 6.3](#) p134).

The following C operators have not been included (or in the case of -> the semantics are different).

,	binary	sequential evaluation
->	binary	field of variable pointed to
*	unary	value of variable pointed to

3.13 Expressions

An expression may contain

- constants
- variables of primitive type
- function calls returning primitive type
- operators
- parentheses

A primitive type may be a simple variable or a field or array member from a compound nested type which is a primitive type. (i.e. not arrays or structures as such). Variables may be **immediate**, **value**, **selectvalue**, **static**, **input** or **queue** mode.

3.13.1 expression evaluation

All operators are combinatorial. Binary logical operators && and || **are** short-circuit evaluated when the first operand is immediate. Binary logical operators && and || **are not** short-circuit evaluated when the first operand is a target mode since that would imply sequential evaluation.

If the first operand of the conditional ternary operator ?: is immediate mode, only one of the following two operands is evaluated, i.e. this operator is also short-circuit evaluated.

An expression is available when all queues referenced in the expression are available. Where the ternary operator ?: is used, all three operands must be available.

3.13.2 mixed types

The arithmetic operators '+', '-', '*', '/' and '%' accept the types shown on the axes of the following table, the table entries giving the result type. Types shown with an appended ":" are target types while those without are immediate.

	uint	int	float	str	uint:	int:	ufixed:	fixed:
uint	uint	int	float	str	uint:	int:	ufixed:	fixed:
int	int	int	float	str	int:	int:	fixed:	fixed:
float	float	float	float	str	-	-	-	-
str	str	str	str	str	-	-	-	-
uint:	uint:	int:	-	-	uint:	int:	ufixed:	fixed:
int:	int:	int:	-	-	int:	int:	fixed:	fixed:
ufixed:	ufixed:	fixed:	-	-	ufixed:	fixed:	ufixed:	fixed:
fixed:	fixed:	fixed:	-	-	fixed:	fixed:	fixed:	fixed:

The remainder operator '%' only accepts types **int**, **uint**, **int:** or **uint:**.

The addition operator '+' where at least one operand is a string concatenates the operands producing a string result. The other arithmetic operators do not accept string operands.

Arithmetic operators for target float types are not implemented⁵. Target floating point operations are carried out by pipelined library modules.

⁵Arithmetic expressions cannot use target floating point values because floating point operations are sufficiently complicated that they must be pipelined to achieve a worthwhile execution rate. By definition expression operators must be able to execute in a single clock cycle if the operands are available. Single cycle target floating point operators could be implemented, but the clock rate at which they were used would have to be very low.

Addition and subtraction of target fixed point operands with immediate floating point values is allowed, the floating point values being converted to fixed point with the same number of fractional bits, lesser significant bits being rounded off.

Logical operators only accept *log* operands. The result type is also *log*.

Relational operators accept *int*, *uint* and *str* operands. The `==` and `!=` operators will also accept *log*, *bits* and *type* operands and can also compare clock mode variables. The result is type *log*. Relational operators with arguments of type *str* perform a lexical comparison.

Shift operators accept *bits*, *int*, *uint fixed* and *ufixed* for the left operand but only *int* and *uint* for the right operand (shift count). The result type is the type of the left operand.

Bitwise operators accept *fixed*, *ufixed*, *int*, *uint*, *log* and *bits* for the operands. Where the operands are of different size the smaller will be expanded to match the larger, extending the sign bit in the case of signed types. The result width will be the same as that of the larger operand. The result type is determined by the following in order -

- If either operator is *fixed* then the other operator cannot be *fixed* or *ufixed* and the result type will be *fixed*.
- If either operator is *ufixed* then the other operator cannot be *fixed* or *ufixed* and the result type will be *ufixed*.
- If either operand is *int* the result is *int*.
- if either operand is *uint* the result is *uint*.
- if either operand is *bits* the result is *bits*.

3.13.3 expression width

Immediate integer variables and expressions have a 64 bit value. Immediate floating point variables and expressions have a 52.11 format (IEEE 754 double precision). Target integer and floating point values may be any size.

Where signed and unsigned operands are mixed the unsigned operand is zero padded by one bit to render it signed.

For mixed immediate and target modes the immediate operand will be converted to a target operand. The width will be the minimum which can contain the value, assuming a sign bit in the case of *int*.

For addition and subtraction the result width is the widest operand width plus one. For fixed point types the result point offset is equal to the largest or only operand point offset.

For multiplication the result width is the sum of the operand widths. For fixed point types the result point offset is the sum of the operand point offsets (integer types have zero point offset).

For division the result width is the sum of the operand widths. For fixed point types the result point offset is the first operand point offset plus the second operand width minus the second operand point offset (see example below).

For remainder the result is the same width and type as the second operand.

For `'>>'` the result width is the same as the first operand width. For `'<<'` the result width is the first operand width plus the shift count (immediate) or maximum possible shift count (target). For bitwise logical operators the result width is the maximum of the operand widths. The result type is the same as that of the left operand.

For `'?:'` the result width is the width of the widest of the other two operands.

For bitwise logical operators the operands may be any width, the the smaller being padded to the size of the larger (with sign extension if int). The result is the bitwise AND, OR or XOR of each matching pair of bits, one from each operand.

Fixed point divider example -

```
static(num, "fixed:8.5");
static(den, "fixed:10.4");
static(quot, "fixed:12.15");
quot <- num / den;
```

Note: this code really should use `divide(quot ← num, den)`⁶ The numerator width is 13 and the denominator width 14 so the result width will be 27. The result fixed point offset will be $5 + 14 - 4$ which is 15. Thus the quotient has been given the type "fixed:12.15". The type width and offset could be different to the division result type in which case the result would be truncated or padded/sign-extended as required.

For mixed types -

```
static(num, "ufixed:6.3");
static(den, "int:3");
static(quot, "fixed:7.6");
quot <- num / den;
```

The numerator width is 9 and the denominator width is 3. Because the denominator is signed the numerator will be converted to "fixed:7.3", making room for the extra sign bit. The result width will be the sum of these which is 13. The result fixed point offset will be $3 + 3 - 0$ which is 6. Thus the result type is "fixed:7.6".

3.13.4 operand clock domains

If an operator has two or three operands the operands must either have the same clock domain or have no clock domain. Immediate mode variables and expressions have no clock domain and value variables initialised to a constant have no clock domain.

3.13.5 assignment

The following table shows allowed right-hand-side types for each left-hand-side type. Target types have an appended ':

uint	uint	+int							
int	uint	int							
float	uint	int	float						
str				str	type				
type				str	type				
bits:						bits:			
uint:	uint	+int				uint:			
int:	uint	int				uint:	int:		
ufixed:	uint	+int	float			uint:		ufixed:	
fixed:	uint	int	float			uint:	int:	ufixed:	fixed:
float:	uint	int	float						float:

" +int" indicates that the immediate integer value must not be negative.

Fixed point types are truncated at the binary point when assigned to integer types.

The LHS and RHS widths, point offsets and floating point mantissa and biased exponent sizes may differ. Where the LHS type width is smaller than that of the RHS the most significant bits are truncated. If

⁶While division has been included in the set of arithmetic operations, current FPGAs do not provide division hardware elements. As a consequence 3PL implements division as a cascaded series of adder/subtractors resulting in a very great logic depth. For data to propagate through such a divider in one clock cycle would imply a very low clock rate. To avoid this problem `stdlib.3pl` includes a pipelined divide library `divlib.3pl` which contains a pipelined divide module. This module uses the same logic as produced for the divide operator but each stage is pipelined. The compiler itself cannot generate such a module implicitly since the language requires basic operations to be executed in one clock cycle.

the LHS width is greater than that of the RHS the value will be sign extended or zero padded at the most significant end as appropriate.

Note that types "str" and "type" can be assigned to each other. Assigning a type to a string variable produces the same string value as calling the nested inbuilt functions **typestring(typeval(t))**, i.e. the type is converted to the type specification string from which it can be generated. Assigning a string to a type variable is equivalent to calling inbuilt function **strtotype()** [section 10.144](#) p333.

target type "bits:" can only be assigned type "bits:", however the **cast()** inbuilt function can be used to cast any other type to "bits:".

The types allowed for assignment generally apply also to variable initialisation.

3.13.6 expression optimisation

The following special cases with mixed target mode and constant operands are handled -

- multiplication by 0 - value is constant 0
- multiplication by 1 - value is the other operand
- multiplication by a power of 2 - use a fixed left shift for a power greater than zero or a fixed right shift for a power less than zero
- division by 0 - a fatal error
- division by 1 - value is the other operand
- division of 0 - value is constant 0
- division by a power of 2 - use a fixed right shift for a power greater than zero or a fixed left shift for a power less than zero
- remainder by 0 - a fatal error
- remainder by 1 - value is constant 0
- remainder of 0 - value is constant 0
- remainder by a power of 2 - use an AND with a mask
- addition of 0 - value is the other operand
- subtraction of 0 - value is the other operand
- subtraction from 0 - converted to a unary - of the other operand
- logical AND of false - value is constant false
- logical AND of true - value is the other operand
- logical OR of false - value is the other operand
- logical OR of true - value is constant true
- logical XOR of false - value is the other operand
- logical XOR of true - converted to a logical NOT of the other operand
- int < 0 - value determined from the sign bit only
- int >= 0 - value determined from the sign bit only
- 0 <= int - value determined from the sign bit only
- 0 > int - value determined from the sign bit only

Bitwise operators with mixed target mode and constant operands are optimised during netlist generation.

3PL does no other expression optimisation. Target expressions should be written the way you want them implemented.

Chapter 4 - Program Source Structure

Module, procedure, function and class definitions are processed during parsing, not during immediate execution.

All statements are executed by the 3PL interpreter. Statements which when executed produce logic for the FPGA are called **target** statements. All other statements are called **immediate** statements.

Statements are referred to as **top level** if they are not enclosed in a control structure (if(), for(), while() etc.).

Top level immediate statements are executed in the order in which they are encountered. Top level statements may of course be control structures, inline modules or module or procedure calls in which case nested statements will be executed.

Module, procedure and function definitions may be distributed amongst top-level statements (but not at a deeper level of nesting) since they have been removed by the parser. Code is probably more readable though if definitions are not followed by statements.

4.1 scope

The **3PL** compiler is given a source file name via the command line. The file is treated as a **file module**, by which is meant that when code from that file is executed it is treated as though a module had been called, the module name being the file name. A file module has the property that it provides a static scope which is accessible to any code within the file, including within any inline modules or within procedures, functions or modules defined in the file module. That scope is called the **file scope**.

If a 3PL source file inserts another file by means of a **source** statement then that inserted file is expanded in place.

If a 3PL source file inserts another file by means of a **module** statement then that file is also expanded in place but it is treated as a file module, that is it behaves as though it has been called as a module at that point in the code, in this case a file module.

Nesting of **source** and **module** statements can be to any depth (but note that a given file can only be inserted once via a source or module statement (if source code is required in more than one place it should be in a named module or procedure)).

File scope has the property that it is accessible to any code within the file (including code inserted by **source** statements but not to nested file module statements). Every statement has an associated static file scope which is that of the file module within which it is contained.

If a class is created then its code body has a class scope and a common class scope. These are accessible within the class body and any modules, procedures and functions defined in that block. It is not accessible to any nested class blocks.

Finally there is a global scope and variables can be explicitly created in that global scope. Declared modules, procedures and functions can also be explicitly placed in global scope. Directives are only placed in global scope.

Scopes can be one of the following -

- global - accessible anywhere
- file - the scope of a module created by inserting a file, the main source file or a file imported via a module statement, and accessible anywhere within it (including within any defined modules, procedures or functions, but not within deeper nested file modules).
- class - the scope of the current class block if applicable
- local - the scope of a procedure, function or module.
- block - the scope of a code block in a control structure.
- calling - the scope of the calling module, procedure or function.

File scope is the scope of a file module, a module associated with the main source file or imported by a module statement as described previously. This is a static scope that can be accessed by any code within the file module, including code within modules, procedures or functions defined within that file module regardless of from where they are called.

Where file modules are nested, code within a nested scope can only access the nested file module scope, not that of enclosing file modules.

Global scope, file scope are static scopes, i.e. they are established during compilation and are not necessarily on the call stack.

Class scopes are created when class instances are created and are accessible during execution of a class code body or when a class variable or class module, procedure or function is called.

When a variable is referenced or evaluated the identifier is searched for through scopes in the following order (where they exist) -

- nested block scopes back up to the local scope
- local scope (module, procedure or function) - the local scope will be the file scope if we are in a file module
- the current class scope if applicable
- the current file scope
- global scope

The search completes when the first instance of the variable identifier is encountered.

In some cases the local scope will be the file scope. In-line statements in a file module will have a file module scope which is the same as the local scope for the module.

Note that inline modules are an exception to this rule. The scope search continues past the local scope of the module back to the scope of the containing named module, procedure or function. This gives in-line module statements access to the enclosing scope.

When a variable is created, by default it is created in the current scope. If within a code block in braces, the variables will be created in the scope of that block. Thus to conditionally create variables in the LOCAL scope they must be in separate conditional statements, e.g. -

```
if (1)
  int(i);
if (1)
  str(s);
```

or alternatively the calling scope can be specified -

```
if (1) {
  int(calling.i);
  str(calling.s);
}
```

Note that if "class" is specified as the scope this only accesses the class instance scope, not the class common scope.

The previous example shows that variables can be created in a specified scope by prepending a scope specifier as follows -

- local. - use the scope of the current module, procedure or function. (this may be file scope)
- calling. - use the scope of the calling module, procedure or function.
- class. - use the current class scope if applicable.
- file. - use the current static file module scope.
- global. - use the static global scope.

Similarly when evaluating variables the scope can be explicitly specified (for example to disambiguate variables with the same identifier but in different scopes, or in the case of calling scope to access variables not normally accessible).

As an example the source file s1.3pl contains -

```
int(s);
int(global.g1);
module "s2.3pl"

p();
[
    int(j);
]

procedure p () {
    int(l);
    if (true) {
        int(b);
    }
}

global procedure q() {
    .
    .
}
```

and source file s2.3pl -

```
int(s);
int(global.g2);
```

Execution begins in s1.3pl. It creates an integer **s** which is in the file module s1 scope, then an integer **g1** which is explicitly placed in global scope. Execution now enters source file s2.3pl which is a new file module. An integer **s** which is in file module s2 scope and an integer **g2** which is in global scope are created. Integers **g1** and **g2** can be accessed in either s1.3pl or s2.3pl as they are global, but each file module scope now contains a unique variable **s** which can only be accessed from within that file module.

We have reached the end of s2.3pl and now continue execution in s1.3pl. Integer **s** in file module s2 is now no longer accessible. Procedure **p** is called. This procedure is declared within the s1 file module so could only be called from within that file module, but note that procedure **q** could be called from anywhere as it has been declared global. Within procedure **p** integer **l** is created and that is in local scope, then integer **b** is created in the scope of the *if* block. Within the *if* block variables **g1**, **g2**, **l** and **b** are accessible and variable **s** in file module s1 is also accessible. On leaving the *if* block variable **b** would no longer be accessible (though we could still use it if we had a pointer to it). After returning from procedure **p** we enter an inline module in which integer **j** is created. **j** is local to the inline module and at this point **g1** and **g2** can still be accessed, as can integer **s** in file module s1.

The use of **source** and **module** statements and scope specifiers such as **global.** and **file.** provide a mechanism to achieve similar scope control to that provided by **static** and **extern** in **C**. The need for a different mechanism arises from the fact that variables in 3PL are created during immediate execution, not declared during parsing. For the same reason we cannot simply have a list of source files as in **C** but must use **source** and **module** statements to link source files together in such a way that variable creation occurs in the correct order.

File scope is intended to be used to hide variables such as I/O pins and signals or intermediate clock interconnections. All I/O pin locations strings, clocks, external memory interfaces and external bus

interfaces can be coded in a source file included using a module statement. File scope variables within that file will only be accessible from code within the file. Any variables that need to be accessible from elsewhere can be made global. The interface and other services provided by code within the file can be made available through module, procedure and function calls which are declared global.

Note that block scope only applies to control structures which have code blocks enclosed within braces (`{...}`). Conditional variable creation can be implemented by the use of an *if* containing a single statement, e.g.

```
if (condition)
    int(i);
```

which will create the integer variable *i* in the scope containing the *if* if logical variable *condition* is true. As shown in an earlier example it would be a mistake to write

```
if (condition) {
    int(i);
}
```

as the variable would be created in the scope within the *if* and would vanish when the *if* completed. This should be written -

```
if (condition) {
    int(local.i);
}
```

for the scope of the enclosing module, procedure or function or

```
if (condition) {
    int(calling.i);
}
```

for the scope immediately enclosing the *if*(*condition*). Clearly this is the way to conditionally create a group of variables where more than one variable creation call is needed.

If a procedure is to be written to create a variable in the current scope then the variable must be created in the calling scope, e.g.

```
procedure
my_var (str(name)) {
    name = "calling."+name;
    int(name->);
}
```

Here to make the example non-trivial we have passed the identifier of an integer we wish to create in the current (i.e. calling) scope. The procedure prepends the variable name string with keyword "calling." and then creates an integer using the indirect operator on the string (an indirect operator is normally used on a pointer variable but it can also be used on a string variable). Note that the procedure parameter is immediate mode so the argument will be passed by value, i.e. copied to variable *name* which is in effect a local variable. Thus we can change the value of the parameter within the procedure body.

Procedures, functions and named modules can be declared within any module, procedure or function. By default they can only be called within the same module, procedure or function or from within any module, procedure or function called from the current scope. A procedure, function or named module declared within a file module can be made globally accessible by preceding the **module**, **procedure** or **function** keyword by the keyword **global**. This cannot be done when they are declared at a deeper level.

Variables in various scopes can be printed by library procedure **printvars()** [subsection 11.13.3.7](#) p510 in **stdlib.3pl**. Among other parameters this lists the scopes in which variables occur. The scopes can be selected by an argument to the procedure.

Directives can be printed by library procedure **printdirs()** [subsection 11.13.3.6](#) p510 in **stdlib.3pl**.

4.2 immediate execution

3PL is parsed to produce a code tree for immediate execution. Any **source** statements encountered are expanded in place. Any **module** statements encountered are also expanded in place but as a form of in-line module called a **file module** where the file contents form the body of the module.

During parsing no variables are defined but module, procedure and function definitions are processed and entered into maps in the enclosing scopes in which the definitions are encountered.

During immediate execution statements are executed starting at the beginning of the main source file. **source** statements are no longer recognised as they have been expanded in place and are executed as though part of the enclosing source code. **module** statements and in-line modules are executed as though called at the point in the source code at which they occur. These implicit calls are without any parameters or arguments.

All variables are created during immediate execution.

Immediate execution begins with execution of the initial file module. Files included using **source** or **module** statements are executed at the point of inclusion.

During immediate execution modules, procedures and functions may be called. For immediate execution there is no distinction between modules and procedures, the calling syntax and semantics being the same. Both follow the same scope rules. The only distinction between modules and procedures is that modules initiate target code execution streams and control the way multiple readers of a queue are handled.

At the end of application code execution it is usual to have many housekeeping tasks to complete. For example a resource such as RAM or an external interface may have to be shared amongst multiple accesses. The logic to implement this can usually only be generated at the end of the application code. Similarly there are often variables created in different included files which ideally should be created very early in execution rather than when encountered during sequential immediate execution. The above can be achieved by declaring module definitions using keywords **leadermodule** or **trailermodule** (see [subsection 4.5.2](#) p104). The declaration includes a mandatory comment string and an optional ordering integer. Modules thus declared do not have an identifier or parameters as they are not called by name and cannot be called directly from source code. More than one leadermodule or trailermodule can be defined and they are placed in a list when encountered during parsing. When immediate execution begins the leader modules are executed in the order determined by the ordering integer or in which they were queued for the same ordering integer, the source code is then executed in sequential order, and finally the trailer modules are executed in the same way. Leader and trailer modules may call other modules, procedures or functions as usual. The leader or trailer module comment strings are printed to the report file when these modules are executed and this aids verification that they are executed in the desired order.

It is usual for an application program to first include a header file module which will define a trailer module. Housekeeping variables can thus be hidden in the scope of the header file module to be used later by the trailer module. The header file module usually defines FPGA part number and package pins and sets up clocks. Thus an application program is usually of the form

```
module "header.3pl"
.
.
.
```

where header.3pl sets up the environment for the application code and defines a trailer module to be called automatically at the completion of execution of the application code to tie up loose ends.

4.3 Target code execution

3PL is designed to process continuous data streams and the target execution control mechanism has been designed with this in mind. In particular it is convenient to be able to write assignment statements so that they are data driven. If we have a number of queue variables q1, q2 etc. it would be convenient to be able to write statements of the form

```
q1 <<- q2 + 5;
q4 <<- q1 - q3;
```

and have these execute on every datum that comes down each queue, rather than have to explicitly enclose each in an infinite loop

```
par {
  seq while (true) {
    q1 <<- q2 + 5;
  }
  seq while (true) {
    q4 <<- q1 - q3;
  }
}
```

4.3.1 the role of the module

Modules have two prime characteristics relevant to the target -

- they constitute single sinks for queues
- they initiate repeated execution of target code streams

The second of these is discussed in the next section.

Queues are allowed multiple readers. A queue datum cannot be popped from the queue until all readers have read. However, we may wish to have queue reads in several places, some of them conditional.

To simplify the rules for queue reading the module has been devised as a sink for queues.

A module may contain any number of places at which a queue is read. Any queue read within a module receives an acknowledgement of the read from that module. If a module executes two or more reads of a queue simultaneously the queue still receives only one acknowledgement and all reads extract the same datum. Queue reads within a module which are not simultaneous extract different data.

Where a queue is read in more than one module, each module must execute a read of a particular datum before it is popped and the next datum (if any) becomes available. Thus if the read within one module is delayed, other modules which have read the datum will now see that the queue is not available. When all modules have read the queue it may once again appear available to all modules. This ensures that all modules that contain a read operation for a queue get each datum regardless of when they execute the read. Note however that as described in the previous paragraph multiple reads of a queue within a module are not constrained in this way.

The behaviour described above for reads within multiple modules can be disabled by the **ignoremodules** queue variable attribute. If this is set to **true** then all reads of the queue will be treated as though within a single module. The purpose of this option is to allow code within multiple modules to read queues while looking after exclusive access where necessary by means of explicit user code. This can also be done also by ensuring that all the queue reads are contained within one module but this restriction can be inconvenient and may result in more obscure code.

A module is recognised as a reader for a queue if that module contains one or more value references for that queue. There are three cases which should be recognised as exceptions -

- queue examines via the `?` operator
- queue availability tests using the `<` or `>` operators
- passing a queue to a module as an argument

none of these make the associated module a queue sink.

A queue value reference is simply the appearance of a queue variable in any expression that is evaluated to read a queue datum. This might be the RHS of an assignment statement, a variable target array subscript or part of an expression tested for conditional execution.

A queue is written by a queue assignment statement, the queue identifier appearing on the LHS. Each assignment to a queue inserts one datum into its queue. If the datum is compound (an array or struct)

then the whole type will be written when any assignment is made. Multiple array members or struct fields can be written by separate assignment statements as long as these are executed simultaneously. If two assignments to the same array member or field are made simultaneously that is an execution error¹.

Queue reads and writes from multiple parallel statements can be synchronised by the use of a **sync {...}** block described in section [section 6.14](#) p151.

Unsynchronised queue writes may be arbitrated by using the **waitpri ()** control structure, section [section 6.17](#) p155

4.3.2 target execution streams

Procedures do not exist as far as the target is concerned. Procedure calls are done during immediate execution and leave no trace in the target code. We can conceptually eliminate the influence of procedure calls on target code by regarding procedure definitions as being expanded in place of the call with parameters replaced by call arguments.

The same applies to function calls. These are again done during immediate execution and any resulting target code generated is substituted in an expression in place of the call (a function can also create an independent module which drives a value returned by the function).

It was mentioned above that modules -

- constitute single sources and sinks for queues
- initiate repeated execution of target code streams

The second of these characteristics is of interest here. We can ignore procedure calls as discussed above and consider any target code within a procedure as substituted for the call.

Any top level target statement in a module (remembering that we have conceptually eliminated procedures) is effectively inside its own infinite loop, that is it executes repeatedly forever. This applies regardless of the statement type be it a static or queue assignment (value, selectvalue, pointer and priority assignments are immediate) a seq, sync or par block, a when etc.

Note that execution of each statement within its own infinite loop is not the same as repeated execution of a parallel block containing those statements. In a parallel block each statement in the block is executed once, but the next repetition cannot take place until every statement in the block has executed. For example one of the statements in the par block might be quite a long seq block, or an assignment statement might wait quite a long time for queue availability . With statements each in their own infinite loop they each repeat at their own maximum rates.

Thus each top level statement within a module executes repeatedly forever and does so independently of other top level statements.

If a 3PL target code stream is to execute only once in the way conventional programming languages dictate then a stream can be simply terminated with a call to inbuilt procedure **stop()**.

An execution stream can proceed through a number of control structures. A **when** conditionally selects one of two code blocks according to the value of a logical expression. There are also target **while** and **do while** loops. Statements can be enclosed in a **seq** block to execute them sequentially or in a **par** or **sync** block to execute them in parallel. All of the above may of course be nested in any way. Sequential execution causes the execution stream to pass from statement to statement in order. Parallel execution causes the execution stream to diverge into multiple streams, the block waiting until all streams have completed before it completes.

Execution streams can be prematurely terminated by means of the **unless** construct. This can be used with **when**, **branch**, **seq**, **par**, **sync**, **seq while**, **par while**, **sync while**, **seq do while**, **par do while** and **sync do while** constructs.

As well as an execution stream or streams there is an orthogonal control mechanism - queue availability. An assignment statement will only execute when

- execution control reaches it in its stream

¹at the moment target execution errors are not detected, but this facility may be added as a compiler option later

- all queues that it reads or writes are available

If any referenced queues are not available execution of that stream is suspended until they are. Note that this is different to a target conditional statement such as a **when** which will simply choose a true or false branch of the stream but will not pause of its own volition (it may wait if its test expression contains queue reads).

4.3.3 starting target execution streams

Execution threads start in an implied infinite loop initiated within a module. If no special provision is made every execution thread will start as soon as GWE is asserted after the configuration has been loaded. Every clock domain has its own start pulse initiating every thread in that clock domain. These start pulses are initiated at the same time.

The start of execution threads may be delayed in the following ways -

- after a specified number of clock pulses
- after a specified time
- when a target expression is true

In the last case the target expression must only contain signals that are external inputs or are derived from FPGA elements such as DCM lock signals since all 3PL target variables of any mode with state are still frozen in their initial conditions.

Specification of the start condition is done using inbuilt procedure `start()` [section 9.73](#) p251).

The start conditions are specified relative to one clock domain and the start pulse in that clock domain is resynchronised into all other clock domains to start them.

4.4 Clocks

3PL generates synchronous logic and supports any number of separate clocks. There are two aspects of clocks with which a program must deal -

- selection of the clock currently in force
- generation of clock signals

4.4.1 selecting a clock

A clock variable is created by

```
clock(cv, ...attributes...);
```

The attributes arguments are optional, but if clock frequency or period attributes have not been set by the end of immediate execution no timing constraints will be generated for logic using this clock. Only one of frequency or period attributes should be given, the other being derived automatically. If the duty cycle attribute is omitted 50% will be assumed. In the case of an external clock signal there are additional attributes.

The clock attributes are usually omitted when -

- the clock variable is a module, procedure or function parameter, in which case it will reference the argument
- the clock variable is to be assigned from another clock variable using `:=` assignment (see below)
- the clock variable is to be an argument to a library module or procedure which will set the attributes

A clock variable may be assigned another clock variable following which it becomes an indirect reference to that variable, for example

```
clock(cv);
cv := c;
```

where **cv** is now a reference to **c** (note: this does not require the use of the pointer operator). Since this is an indirect reference **cv** can now be used in place of **c**, including operands to equality comparisons.

The clock signal currently in force is changed by calling the **useclock()** procedure. For example

```
useclock(c100);
```

would select **c100** as the current clock, pushing the previous current clock onto a stack. The call

```
useclock();
```

pops the previous clock off the stack, i.e restores the previous current clock. If there are no previous clocks the current clock becomes null. Clocks can be pushed on the stack to any depth.

The clock signal currently in force can only be changed at the top level in a module. This precludes a clock change within the infinite loops enclosing execution streams. An execution stream must have a single clock controlling all statements within it. Changing the clock in a module between one top level statement and the next poses no problem.

4.4.2 clock domains of variables

Variables do not have established clock domains when created. The clock domain of a variable can be established in one of three ways -

- when a variable is assigned its domain is set to the current clock domain, but some modes are exceptions - see below
- when a variable is finally evaluated for use in an execution thread its clock domain is set to the current clock domain
- the clock domain of a variable may be explicitly established by calling inbuilt procedure **attributes()** or by specifying the attribute in the variable creation procedure call [section 9.7](#) p192

Where the clock domain of a variable is already established the above three cases check the clock domain instead, causing a fatal error if the check fails.

A value variable itself has no clock domain, but the values it contains may have clock domains. When used as part of an evaluated expression the clock domains of the values assigned to the variable are used. If at that time they have no clock domains their clock domains are set to the current clock domain. Where a value variable is of compound type different elements may be assigned expressions with different clock domains. The following code illustrates this behaviour -

```
value(v, "[2]log");
static({s1, s2, s3, s4}, "log");
v[0] := s1;
v[1] := s2;

useclock(c100);
s3 <- v[0];

useclock(c50);
s4 <- v[1];
```

The assignments to value variable **v** do not establish the clock domains of **s1** or **s2**. Variable **v**, being value mode, does not itself have a clock domain. When the assignment **s3 <- v[0]** is encountered during immediate execution the current clock domain (**c100**) will be established for **s3** (by assignment) and **s1** (by evaluation of the expression containing **v[0]**). Similarly the assignment **s4 <- v[1]** will set **s4** and **s2** to clock domain **c50**. The purpose of this example is to explain that assignment of expressions to a value variable does not establish the clock domain of variables in that expression.

If an expression assigned to a value variable contains variables whose clock domains are already established then when the value variable is evaluated in a clocked context the clocks will be checked for identity with the current clock.

A value variable which is evaluated prior to its assignment is given a value that has the current clock domain. Later assignment to the variable must be in the same clock domain.

Selectvalue, static and priority mode variables have a single clock domain in which both assignment and value reference are done. These variables have no clock domain when created and the clock domain is established from the current clock domain when the variable is first assigned or its value is used in a clocked context, e.g. an assignment value. All subsequent assignments or value references must be done in the same (current) clock domain.

Input mode variables have only an output (read) clock domain since they can be evaluated but can not be assigned. Input mode variables have no clock domain when created and the clock domain is established from the current clock domain when the value of the variable is first used. All subsequent value references must be done in the same clock domain.

Output mode variables have only an input (write) clock domain since they cannot be evaluated, only assigned. Output mode variables have no clock domain when created and the clock domain is established from the current clock domain when the value of the variable is first assigned. All subsequent assignments must be done in the same clock domain. Note that if an output mode variable is assigned an expression containing value mode variables which have not been assigned values yet, the interim values will be set to the current clock domain. When ultimately these value variables are assigned values the values must have the same clock domain.

Queue mode variables have separate read and write clocks. Again queue variables have no clock domains when created. The write clock domain is set from the current clock domain on the first write (assignment) to the queue after which all future writes must be in the same clock domain. Similarly the read clock domain is set from the current clock domain on the first queue read or examine and all further reads or examines must be done in that clock domain. Evaluation of expressions containing queue availability operators in a clocked context (i.e. but not when assigned to a value mode variable) will also establish or check the associated queue clock domain (read or write).

The clock domain(s) of a variable can be forced by setting attributes **domain**, **readdomain** or **write-domain** (see [section 3.11](#) p80) as appropriate. This pre-empts clock domain assignment resulting from variable assignments or evaluations. Once a variable clock domain has been established it cannot be changed and all further accesses to that variable must adhere to the established clock domain(s).

For selectvalue, static, queue and priority mode variables the value to be assigned must either be the same clock domain as the current clock domain or must have no clock domain (external inputs with no clock domain or constants). The clock domain of the assigned variable (write clock in the case of queues) must not differ from the current clock domain - if it has no clock domain the assignment will set it to the current domain.

Note that where variables, or expressions containing variables, of mode selectvalue, static, queue or priority are assigned to value variables, the output clock domain of the RHS variables is not resolved until the value variable is actually directly used. The same consideration applies to arguments passed to module, procedure or function parameters. Clock domain resolution of variables is delayed as late as possible until they are actually assigned to or their values are finally required. Their clock domains remain unresolved through assignment to value variables or argument passing.

Where an assigned value violates the clock domain it may be converted by the inbuilt function **sample()** [section 10.128](#) p326 or acceptance may be forced by the inbuilt function **accept()** [section 10.2](#) p265. **accept()** simply makes its argument appear to be in the current clock domain (or another clock domain or no clock domain) but otherwise has no effect. **sample()** resamples its argument into the current clock domain. The use of **accept()** may result in data corruption unless it is known that the clocks are suitably related in phase and frequency. The arguments to **accept()** and **sample()** must not contain queue reads.

A queue which has identical read and write clock domains is termed a **synchronous queue**. If it has different read and write clock domains it is called an **asynchronous queue**.

For a queue variable the clock domain for writing is established by -

- the first write (assignment) to the queue variable (or component of the variable if the type is compound)
- the use of the write availability operator > on the queue

- a call to **queuewritestatus()** on the queue
- a call to **reset()** on the queue

If the write clock domain is not yet established in any of these situations then it will be set to the current clock domain, otherwise it must be the same as the current clock domain.

The clock domain for queue reading is established by -

- the first read or examine of the variable (or component of the variable if the type is compound)
- the use of the read availability operator < on the queue
- a call to **queureadstatus()** on the queue

If the read clock domain is not yet established in any of these situations then it will be set to the current clock domain, otherwise it must be the same as the current clock domain.

The inbuilt procedure **domain()** may be used to explicitly set the clock domain of a selectvalue, static, queue or priority variable or a value variable which is an external input. On calling **domain()** the clock domain of the variable must be null (in which case it is set) or else must already have that clock domain. In the case of a queue variable the input and output clock domains will both be set, i.e. it will be a synchronous queue.

The inbuilt procedure **readdomain()** may be used to explicitly set the output (read) clock domain of a queue variable (for other modes it functions the same way as **domain()**).

The inbuilt procedure **writedomain()** may be used to explicitly set the input (write) clock domain of a queue variable.

Memory variables have a clock associated with each access port. The clock associated with a port is determined by the clock that is current when the first read or write procedure or function call for that port is made during immediate execution. All subsequent calls to read or write procedures or functions for that port must be in the same clock domain.

A clock variable may be used effectively as a logical variable, but this can only be done by assigning it to a value mode variable of type log and then that value variable can be used in the normal way. The assignment may be explicit, e.g.

```
clock(clk, ....);
value(v, "log");
v := clk;
```

or it may be done via initialisation

```
clock(clk, ....);
value(v, "log", clk);
```

A clock may be driven by internal logic by use of the inbuilt procedure **clockfromexpr()**, e.g.

```
clock(clk, frequency=....);
value(v, "log");
clockfromexpr(clk <- v);
```

If an internal clock signal is to be connected to an output pad this can be done by passing the clock variable as a source argument (third argument) to inbuilt procedure **output()** [section 9.47](#) p224.

4.4.3 checking clock domains

The current clock domain and the previous clock domain can be checked by inbuilt functions **currentclock()** [section 10.23](#) p278 and **prevclock()** [section 10.108](#) p316 respectively. The returned clock domain can be checked against a known clock, e.g.

```
if (currentclock() != c100)
    .....
```

can be checked for null, e.g.

```
if (currentclock() == null)
    .....
```

or can have its identifier printed -

```
println(identifier(currentclock()));
```

The returned clock can also be assigned to a clock variable -

```
clock(c);
c := prevclock();
```

but note that a null assignment is not allowed.

The clock domain of variables can be determined by inbuilt functions `domain()` [section 10.31](#) p282, `readdomain()` [section 10.120](#) p322 and `writedomain()` [section 10.172](#) p349. These will return null if the domain has not yet been established. The domain can be tested, printed etc as illustrated in the previous paragraph.

4.4.4 setting up clocks

Clock devices are highly part-specific. In general implementing a clock requires the following -

- The clock signal and any intermediate signals must be declared to 3PL and connected via an FPGA input buffer using the inbuilt procedure **clock()**.
- A clock driver such as a DLL or DCM must be created. This is done via library procedures (see below).
- An FPGA global clock buffer must be provided, for which there is a library procedure **clockbufg()**.
- Additional timing constraints between clock groups may be needed, entered by using the library procedure **maxdelay()**.

The inbuilt procedure **clock()** is used to create clock variables, including any input and intermediate clock signals associated with the clock generation hardware in the FPGA. The frequency or period attributes must be supplied and optionally the duty cycle attribute can be specified. For an existing clock these parameters can be determined by the inbuilt functions **frequency()**, **period()** and **dutycycle()**.

Library procedures `clkdll()`, `clkdlle()`, `clkdllef()`, `clkdcn()`, `clkmmcm()` and `pll()` create the hardware elements which take an input or internal clock and produce a related output clock. That output clock may be simply the input clock conditioned by a delay-locked loop to achieve zero delay or it may be a multiple, submultiple or phase-shifted version of the input clock. The arguments passed to this procedure and the capabilities it accesses differ with the FPGA family. It can usually be called via library procedures which offer a simpler interface for common situations.

By default if the clock variable is used for any target statements a timing group identifier will be generated for the clock variable. This identifier is used to automatically generate a clock period timing constraint for all logic driven by the clock. If the clock variable has global scope the timing group identifier will be the variable identifier converted to upper case. For any other scope the timing group identifier will be the full hierarchical variable name converted to upper case. The latter case is necessary to avoid any possible ambiguity but results in a long identifier, and this is impossible to use in any other timing constraint since it has arbitrary integers embedded to disambiguate specific instances. Where a clock variable is not global and its timing group identifier is to be used in generating other timing constraints, e.g. between clock groups, a timing group identifier must be supplied to replace the default id. This is done by setting the attribute **tnm** in the clock variable.

Note that the intermediate clock signals interconnecting the input buffers, clock managers and global clock buffers are all variables of mode **clock** and potentially can all be used by 3PL to control logic. However, Clock variables that can actually be used to clock logic devices are restricted to those driven by appropriate FPGA devices such as global clock buffers. The limitations on which clock signals can be used to drive logic are FPGA specific and 3PL itself makes no distinction.

Where clock variables are assigned using the `:=` operator and are also used to set the clock domain, note that logic using that clock will be connected to the clock source determined by the value the clock variable has when immediate execution terminates. Changing the value of a clock variable by assignment, even if accompanied by passing it to `useclock()`, will not result in associated code having different clock domains - the code will all be driven by the clock source given by the final value of the clock variable.

Further details may be obtained in the sections on the header files for the individual boards.

4.5 Modules, procedures, functions and classes

4.5.1 Module, procedure, function and class calls

Procedures, named modules and classes are called by

```
name (output arguments <- input arguments);
```

and if there are no output parameters -

```
name (input arguments);
```

Functions have no output parameters and are called by

```
name (arguments)
```

A function call appears within an expression (it may be the entire expression itself).

When a class, named module, procedure or function is called the current scope, file scope and global scope are searched in that order.

Class calls are similar to function calls, the returned result being a class instance which is an immediate value of type "class". Unlike functions class calls may contain output arguments.

For classes, procedures, named modules, functions or classes if there are no arguments the parentheses are present but empty.

Note that the separation of arguments into input and output groupings is mostly a convention, but for some parameter modes the mechanism by which an argument is passed to a parameter differs depending on whether it is in the input group or the output group (see [subsection 4.5.3](#) p106 below).

Arguments are matched against parameters by position in the argument list, but for declared modules, procedures or functions (i.e. not inbuilt ones) they may also be paired explicitly by using

```
parameter = argument
```

for each argument in the list where *parameter* is the parameter identifier. This is referred to as "key match" argument passing. The effect of using the '=' to match an argument to a parameter is simply to re-order the argument list so that the arguments appear in the order of the parameters, arguments being omitted where parameters have not been matched. This means that arguments will be evaluated and assigned to parameters in parameter order, left to right, not necessarily in argument order.

Key-match argument/parameter matching is useful where arguments are sparse since it avoids runs of commas and resulting mistakes in positioning arguments. It can also make code more readable by giving clues to the purpose of each argument, but argument lists then consume more space.

Null arguments cannot appear among key-match arguments. If they do the fatal error "too many input arguments" will occur, or else in the case of a **varargs** parameter (see [subsubsection 4.5.3.8](#) p109) "null argument to input varargs" or "null argument to output varargs" errors.

The constant **null** may be passed to a parameter. If the parameter is immediate mode and its type is ">", i.e. it is a pointer, then the pointer is assigned the null pointer. For any other parameter mode

and type the assignment is skipped. Thus except for the case of pointers, the argument **null** can be used to conditionally skip passing an argument to a parameter. This applies both to argument passing by position or by key=value.

Non-key-match arguments cannot appear among key-match arguments unless the last parameter is **varargs** (see [subsection 4.5.3.8](#) p109). If they do the fatal error "too many input arguments" will occur.

Argument matching to parameters by position and by key match can be mixed in one call but position matched arguments must precede any key matched arguments, i.e. no position matched arguments may appear after a key matched argument (unless **varargs** is used). The occurrence of the first key-match argument in a list signals the end of matching arguments to parameters by position.

Key-match argument passing cannot be used with inbuilt modules, procedures or functions at the moment since these do not have parameters as such. This may be changed in the future.

Key-match argument passing can also be used to pass arguments to a **varargs** list - see [subsection 4.5.3.8](#) p109.

Procedure and function calls may be recursive. Because calls are made during immediate execution, not target execution, recursion does not involve code in the FPGA. In principle class and module calls can also be recursive but it is hard to envisage a case where this would be useful.

4.5.2 Module, procedure, function and class declarations

A named module is declared by -

```
module
name (output parameters <- input parameters)
  body
}
```

or

```
module
name (input parameters) {
  body
}
```

if there are no output parameters. Similarly a procedure declaration appears as -

```
procedure
name (output parameters <- input parameters) {
  body
}
```

or

```
procedure
name (input parameters) {
  body
}
```

A function is declared by -

```
function
name (parameters) {
  body
}
```

and has no output parameters.

A class is declared by -

```

class
name (output parameters <- input parameters)
  body
}

```

or

```

class
name (input parameters) {
  body
}

```

if there are no output parameters.

A leader module is declared by -

```

leadermodule (s, n) {
  body
}

```

Argument **s** is a comment string which is printed in the report file when the module is executed. Optional argument **n** is an ordering integer. If omitted its default value is 0. The new module will be placed in the leader module list after any modules with the same or lower order number.

Note that for leader, trailer or post-processing modules both arguments must be literal constants, not variables or expressions, as they are extracted from tokens during parsing, not evaluated during immediate execution.

Since it is not called directly it has no identifier. As each leader module declaration is encountered during parsing it is added to the leader module list. Modules in this list are executed in order prior to beginning normal immediate execution.

A trailer module is declared by -

```

trailermodule (s, n) {
  body
}

```

Argument **s** is a comment string which is printed in the report file when the module is executed. Optional argument **n** is an ordering integer. If omitted its default value is 0. The new module will be placed in the trailer module list after any modules with the same or lower order number.

Since it is not called directly it has no identifier. As each trailer module declaration is encountered during parsing it is added to the trailer module list. Modules in this list are executed in order after normal immediate execution terminates.

If a trailer module is defined within a class then the class variables, common block variables, modules, procedures and functions will be accessible to the trailer module.

A post-processing module is declared by -

```

postmodule (s, n) {
  body
}

```

Argument **s** is a comment string which is printed in the report file when the module is executed. Optional argument **n** is an ordering integer. If omitted its default value is 0. The new module will be placed in the post module list after any modules with the same or lower order number.

Since it is not called directly it has no identifier. As each post-processing module declaration is encountered during parsing it is added to the post-processing module list. Modules in this list are executed in order after netlist and constraint file generation has been completed. This code can invoke place-and-route tools and any other post-processing operations which can again be conditional on the time of last modification of related files.

See also [subsection 11.7.2.1](#) p438 and [section 19.3](#) p627.

Module, procedure, function and class declarations may be nested. A module, procedure, function or class can be called from within the module, procedure, function or class in which it is declared or from any nested module, procedure, function or class.

In the case of declarations in a file module the keywords **module**, **procedure**, **function** or **class** may be preceded by the keyword **global** which makes the module, procedure or function globally accessible. The keyword **global** cannot be used on declarations nested deeper than the file scope.

Identifiers fall into four categories -.

- variables
- directives
- modules, procedures and classes
- functions and classes

The same identifier may be used in each of these categories as the syntax disambiguates the multiple occurrences. The overlap with classes is because a class is called in a similar manner to a function but can also be called as for a procedure, the returned value being ignored. It is thus preferable identifiers for modules, procedures, functions and classes are mutually unique.

The same identifier may also be used in different scopes in which case the search order determines which is used. In the case of modules, procedures and functions the most deeply nested will be found first.

For procedures, named modules, functions or classes if there are no parameters the parentheses are present but empty.

A **leadermodule** or **trailermodule** can create variables in the global scope (*global.name*) or in the file scope in which it is declared (*file.name*).

4.5.3 parameters and arguments

Parameters are declared using the same inbuilt variable creation procedure calls as normal variables (they cannot be user-defined procedures). These procedure calls are enclosed in the parentheses following the module, procedure or function identifier. The type of a parameter must be the same as that of the corresponding call argument but word widths need not be the same. Where word widths differ the assignment rules described previously (truncation or zero/sign padding) apply.

Parameter variables which are not matched by corresponding arguments become local variables. Parameters can be initialised so that they have useful default values should they have no matching arguments in a call. While obviously useful for input parameters this also applies to output parameters.

Parameters may be any mode.

The argument to an immediate input parameter can be an immediate expression, which includes the cases of a single immediate variable or constant. An immediate mode parameter is always assigned an argument by value, even if the argument is itself an immediate variable, so if an immediate input parameter is assigned within a module, procedure or function its local value is simply changed and the argument is not affected.

The argument to a value input parameter can be an expression, including the cases of single target or immediate variables or constants. An immediate argument will be converted to a target constant. Value input parameters are call-by-value if the argument is not itself a single value variable. If the argument is a single value variable the argument is passed by reference.

A value input parameter may be passed an immediate compound type in which case the parameter must be declared with a matching target compound type. Any immediate value which exceeds the size of the corresponding target type will cause a fatal error.

A value argument may be passed to an input, selectvalue, static, queue or priority output parameter. This causes the argument value variable to refer to the parameter variable which itself is now a local variable.

Precluding the cases listed above -

- immediate mode input parameter
- value mode input parameter passed anything except a value mode variable
- an output parameter which is input, selectvalue, static, queue or priority mode and is passed a value mode variable

which are call-by-value, all other parameters are call-by-reference and can only be passed variables of the same mode as arguments. Parameters which are Call-by-reference actually use the argument variables instead of the parameter variables.

Where a parameter would normally be assigned an argument value by value rather than by reference, as described above, passing by reference can be forced by using the inbuilt procedure **ref()** [section 9.58 p237](#) instead of **var()**, **int()**, **str()** etc. In that case an argument which is an expression rather than a variable will cause a fatal execution error.

Note that the separation of parameters into input and output groups is mostly a convention intended to make the code more readable. Apart from the differences in the way arguments are passed to parameters described above, there are no other semantic differences between input and output parameters. In particular input parameters may be assigned values (but note that the effect depends on whether they are call-by-value or call-by reference) and output parameters may be used in expressions. However, it is preferable to adhere to this scheme where possible. An exception is where a variable is to be passed to a module or procedure within which code appears both to assign to it and to use its value. In that case it is best to pass it to an output parameter.

A value parameter is a very general way of passing target arguments to a module, procedure or function. The argument to a value parameter can be a constant (or compound value), an expression or a variable of any mode except *memory* (an immediate variable or expression will pass a constant). In the case of a floating point target mode argument to a value parameter the parameter type must either match the argument type exactly or else must have the type omitted. Target floating point types cannot have the mantissa or biased exponent sizes omitted and hence no adjustment to the argument type is possible. If the parameter type is given it must match the argument type exactly. If the parameter type is missing it will take the type from the argument. In the case where the argument is an immediate floating point type it will be converted to a target type "float:23.8" when passed in the call.

An argument to a call-by-reference parameter may be a partial variable reference, for example an array member, a sub-array or a struct field. This is allowed as long as there is no type mismatch with the associated parameter. If the parameter is not typed or has an array dimensions omitted then the type or array dimensions can be taken from the argument. The use of a partial variable as an argument applies to the queue mode as well as all other modes.

Wherever a call-by-reference parameter is used in the code body, the argument variable that it points to will be used instead. If a parameter has not been passed an argument then it becomes simply a local variable within the scope of the module, procedure or function.

Since call-by-reference parameters use the argument variable in place of the parameter variable the clock domain of the argument variable is relevant. Similarly a value parameter inherits the clock domain or domains of its argument (a compound value may have elements of differing clock domains).

Note that it is important in which module a queue is read. This is regardless of whether it is passed as an argument to a module, procedure or function where it appears as a parameter. Ultimately a queue variable, even if referenced via a parameter (or chain of arguments/parameters), is read within a module. Each module in which a queue is read constitutes a queue destination. Each queue destination **must** read a datum before it can be popped from the queue. When a destination module reads a datum from a queue the queue will appear to it to be unavailable unless and until all other destination modules execute a read (they can of course all read simultaneously). Reads in multiple locations in a destination module read the same datum if they execute in the same clock cycle, otherwise they will read different data.

Where a parameter is static or queue mode the parameter will have the clock of the matching argument variable.

4.5.3.1 missing parameter target primitive widths

A target mode parameter which is type *bits*, *uint* or *int*, or is an array of one of these types, may have the width missing following the colon, or may have width 0. In that case the type match with the supplied

argument will be checked and the width will be taken from the argument. If the parameter variable width is supplied it must agree with the width of the argument variable. Note that the colon is still necessary in the type string to indicate a target mode type. If the argument is a constant the width will be the minimum necessary to contain the constant taking into account the sign bit if type *int*.²

4.5.3.2 missing parameter dimensions

A parameter type specification may have missing array dimensions (empty '[]') in which case the type match with the supplied argument will be checked and missing dimensions will be copied from the argument type. Note that this cannot be done if the parameter is created as one of a bracketed list, e.g.

```
procedure
p (var({a, b}, "[]int")) {
}
```

is not allowed as both **a** and **b** must be the same type, but the array dimension is not supplied and the arguments may be of different dimensions. What is allowed is a common initialiser, e.g.

```
procedure
p (var({a, b}, "[] []int", {{3, 5, 1}, {4}})) {
}
```

as here both **a** and **b** will be two-dimensional arrays of type "[2][3]int" and both will be initialised.

Where a parameter type has one or more missing dimensions and both an initialiser and an argument are supplied the dimension of the argument will be used. If an argument is not supplied the dimension of the initialiser will be used.

4.5.3.3 missing parameter types

If a parameter is declared without a type at all, the type of the call argument will be used. A parameter variable with missing type must be created singly, not as one of a bracketed list.

4.5.3.4 missing parameter modes

If a parameter is declared without a mode, the mode of the call argument will be used. A parameter variable with missing mode must be created singly, not as one of a bracketed list.

4.5.3.5 type conversions

In the particular case of a parameter of type "str" and argument of type "type" the argument will be converted to string representation of the type (the same as invoking unbuilt function **typestring()**). For a parameter of type "type" and argument of type "type" the string will be converted to a type.

The same conversion occurs where a variable is created as one of these types and an initialisation argument is of the other type.

4.5.3.6 parameter/argument processing order

For procedures and named modules the input parameter list is processed first followed by the output parameter list. Parameter lists are processed left to right. Within this order later parameters can use characteristics derived from those already processed.

²At the moment a target parameter array variable cannot derive the type width from an argument which is a compound value. In that case the target variable width will be zero, giving a fatal error. This may be fixed in the future.

4.5.3.7 parameter testing

Parameters can be checked to determine if they were supplied an argument by the use of the inbuilt function **matched()**. Undimensioned array parameters can have their dimensions determined after argument matching by using the inbuilt functions **dimensions()** and **dimension()**. Untyped parameters can have their types determined after argument matching by using the inbuilt function **type()**;

4.5.3.8 variable number of arguments

Where the input or output arguments are unknown in number or mode the input or output parameter list (or both) may contain the keyword **varargs** to match a number of arguments. **varargs** must be the last item in an input or output parameter list i.e. it cannot be followed by any variable creation procedure calls.

When **varargs** appears in a parameter list local array variables **inargs** and **outargs** as appropriate will be created automatically. For each argument an invisible parameter of appropriate mode will be created. For arguments which are variables the mode will be the same as the variable. For a target expression the parameter variable will be mode *value*. These **inargs** and **outargs** arrays are pointers to the created invisible parameters. Immediate code can interrogate these invisible parameters via the pointer arrays to determine mode, type, dimensions etc. using inbuilt functions **mode()**, **type()**, **typestring()** (and possibly **dimensions()** and **dimension()**, though **typestring()** returns this information in string form). Inbuilt function **dimension()** can be used to get the number of arguments from **inargs** or **outargs**. The invisible parameters are accessed via the pointers, e.g.

```
a <- inargs[1]->;
outargs[3]-> <<- a + b;
```

If an argument has fields or array members then field identifiers or array subscripts can follow the '->', e.g.

```
a <- inargs[1]->[2];
outargs[3]->data <<- a + b;
```

A function, module or procedure whose parameter list contains **varargs** can be passed key-match arguments. Where the key-match arguments identify explicit parameters the arguments will be assigned to the associated parameter variables as usual. If the key does not match a parameter identifier the key-argument pair will be assigned to the **varargs** list. For this reason the **varargs** pointer arrays **inargs** and **outargs** have associated arrays **inkeys** and **outkeys** which are arrays of strings, and **inmap** and **outmap** which are maps containing pointers to arguments addressed by the key string.

String arrays **inkeys** and **outkeys** contain the keys from key-matched arguments. Where an argument to **varargs** is not key-matched the associated key array entry is "". These arrays of keys are always present when **varargs** is used but if there are no key-matched arguments then the key arrays will contain empty strings.

In addition to the above, key-match arguments are also entered into maps, **inmap** and **outmap**, where the argument key is the map key and the associated value is a pointer to the argument value.

Maps **inmap** and **outmap** only contain key-match entries, not positional arguments. Map access using a key that is not present in the map will return a null.

As an example, the procedure -

```
procedure
p (int({a, b}), varargs) {
}
```

might be called by -

```
p(, 2, k=3, a=5, n=4);
```

The first parameter **a** is skipped and the second parameter **b** is positionally assigned 2. The next three arguments are key-match, but the second of these will actually assign 5 to parameter **a**. Thus the varargs array **inargs** will contain pointers to immediate integer variables containing 3 and 4 and the corresponding array **inkeys** will contain "k" and "n".

Note that we could also have non-key-match arguments (but not null arguments) among the key-match arguments passed to varargs, e.g.

```
p(2, k=3);

procedure
p (varargs) {
}
```

would result in array **inargs** containing pointers to immediate integer variables containing 2 and 3 and the string array **inkeys** will contain "" and "k".

If we did not have **varargs** as the last parameter then non-key-match arguments and null arguments cannot appear among key-match arguments. If they do the fatal error "too many input arguments" will occur.

A compound value cannot be passed to a **varargs** parameter as there is no type available with which to parse the compound value.

These argument passing mechanisms are intended to provide

- argument matching to parameter by position or parameter identifier
- variable length, type and mode argument list
- extension of key=argument to variable argument lists

It is envisaged that module, procedure and function calls would use one of the following scenarios for fixed parameter lists, i.e. **varargs** not used -

- positional arguments only
- key-match arguments only
- first few arguments positional, the remainder key-match

For variable argument lists, i.e. where a **varargs** parameter is used -

- parameters with final varargs - positional matching arguments to parameters, non-key-match and key-match arguments to varargs
- varargs parameter only - non-key-match and key-match arguments

The situation where some key-match arguments match parameters and others are assigned to varargs and some positional arguments match parameters and yet others are assigned to varargs is quite legal but is probably much too confusing.

Variable arguments can be passed on to nested calls by using inbuilt function **arglist()** [section 10.4 p268](#). Note however that this cannot be used for calls to inbuilt modules, procedures or functions.

As an example -

```
procedure q1 (varargs) {
    q2(arglist(null, inargs));
}
```

In this case the input arguments to procedure q1 will be passed to procedure q2. If inargs is simply passed directly it will appear as a one-dimensional array of pointers. By passing via arglist() the values pointed to will appear as separate arguments.

4.5.4 target code

During immediate execution modules, procedures, functions and classes exist as single code bodies even though perhaps called repeatedly. In that sense they function in the same way as normal programming languages. However, any target code that they generate is not shared in any way. Target code is instantiated afresh on every call.

For example

```

static(a, "[4]int:8");
static(b, "[4]int:8");
var(double, "[4]int");

for (int(i) ; i<4 ; i++)
    reverse(a, double[i] <- b, i);

procedure
reverse (static(o, "[]int:8"), int(d) <- static(i, "[]int:8"), int(j) {
    d = j * 2;
    o[j] <- i[3-j];
}

```

Here an immediate loop calls a procedure four times. Each time the immediate integer argument *i* is doubled and stored in the argument array *double*. However, the procedure also generates a target assignment on each call. It is important to note here that we have generated four target assignment statements as a result of immediate execution. It is as though we wrote -

```

double[0] = 0 * 2;
a[0] <- b[3];
double[1] = 1 * 2;
a[1] <- b[2];
double[2] = 2 * 2;
a[2] <- b[1];
double[3] = 3 * 2;
a[3] <- b[0];

```

Execution control of these target statements must be considered. As written the above code must be immediately within a module to be correct. Each of the generated target assignment statements will be wrapped in its own infinite loop to execute on every clock cycle (see next section). However, suppose that the immediate loop above is contained within a *when* statement. In that case we must enclose the four target assignments in a parallel or sequential block, e.g.

```

static(a, "[4]int:8");
static(b, "[4]int:8");
var(double, "[4]int");

when (1) par {
    for (int(i) ; i<4 ; i++)
        reverse(a, double[i] <- b, i);
}

procedure
reverse (static(o, "[]int:8"), int(d) <- static(i, "[]int:8"), int(j) {
    d = i * 2;
    o[j] <- i[3-j];
}

```

which is equivalent to -

```

when (1) par {
    double[0] = 0 * 2;
    a[0] <- b[3];
    double[1] = 1 * 2;
    a[1] <- b[2];
    double[2] = 2 * 2;
    a[2] <- b[1];
    double[3] = 3 * 2;
    a[3] <- b[0];
}

```

Here immediate execution of statements within the *par* is carried out sequentially. Each immediate assignment is carried out as it is encountered. Each target assignment is appended to the parallel block code body to be eventually executed in parallel in the FPGA. The **when** itself will be within an infinite loop.

4.6 Procedures

A procedure is simply a way of writing code which can be used in one or more places with the calling arguments effectively substituted for the parameters. It also provides a way of hiding variables which would otherwise be duplicated (variables local to the procedure).

During immediate execution a procedure may execute any number of immediate or target statements, but it must only create at most a single in-line target statement. If a procedure created more than one in-line target statement the behaviour of these statements would depend on the calling context, i.e. the behaviour would be different if called from the top level inside a module (each statement would then be in its own infinite loop) or elsewhere (it must be inside a **seq**, **par** or **sync** in which case these environments would control the way the procedure body executed. To achieve consistency in the way a procedure body behaves the single in-line target statement rule has been imposed. The procedure body can now explicitly use a **seq**, **par** or **sync** block for the code body or it can use an inline module so that the enclosed statements are in individual infinite loops.

Any queues which are used within a procedure, whether they be global or file scope or passed via arguments, have the calling module as their destination module for signaling purposes.

A procedure may return from within its code block by the statement **return**;. If there is no return statement then the procedure will return after completing execution.

4.7 Functions

A function code block may only contain

- immediate statements
- value, selectvalue, pointer or priority mode assignments
- module calls
- inline modules

It may contain procedure or function calls if they also observe the same constraints. The reason for this restriction is that a function can only contain code which references variables or expressions. It cannot contain code which is one or more elements in a target execution stream. It is assumed that the function will execute when required (subject to any contained queues) and it cannot contribute any extra execution cycles. However, the function may contain one or more modules since they each contain one or more independent execution streams. The function in that case may simply return a value which is a variable, or an expression containing one or more variables, modified by the module.

A function can return a value of any mode.

A function returns a value by means of the statement **return(...)**;

For example a function to double a value mode integer could be written -

```
function
double (value(p, "int:8")) {
    return(p + p);
}
```

This simply substitutes the expression for the call with the parameter *p* replaced by the value of the argument.

A function code body cannot contain in-line executable target code statements. The above example could be legitimately rewritten as -

```

function
double (value(p, "int:8")) {
    value(v, "int:8");
    v := p + p;
    return(v);
}

```

since the use of value mode variables does not generate in-line target code. Value mode variables simply represent target mode expressions. If however we attempted to change the example to -

```

function
f (value(p, "int:8")) {
    static(a, "int:8");
    a <- p + p; // ERROR! in-line target code
    return(a);
}

```

This would give a fatal immediate execution error. This contains an executable assignment statement which generates in-line target code. It is not even clear exactly what this function should do. The implication is perhaps that the evaluation of the function, when it is called from an expression, will require a delay of one clock cycle while the static assignment takes place. Target code execution signaling does not allow this to be done this way! Note that removal of the static assignment into a procedure would not make any difference, e.g.

```

function
f (value(p, "int:8")) {
    static(a, "int:8");
    x(a <- p);
    return(a);
}

procedure
x (static(o, "int:8") <- value(i, "int:8")) {
    o <- i + i;
}

```

is still erroneous for the same reason - the procedure call effectively just substitutes its body in place of the call so we still get the previous code.

A target mode function can contain in-line target statements if it isolates them within a module. That way those statements are quite separate from the function body. For example, suppose that we wish to use a static variable which has been delayed by one clock cycle. We could write a function for that as follows -

```

function
delay (value(p, "int:8")) {
    static(a, "int:8");
    [
        a <- p;
    ]
    return(a);
}

```

The function body is now legal as it does not itself contain any in-line target statements but generates a separate module which does contain such statements. The function simply returns the value of a static variable. That static variable is assigned in the independent module (a named module call could have been used instead of an un-named module).

Note that in the above example the assignment of the static variable **a** is unconditional and thus occurs every clock cycle regardless of when the return value of the function is used. In some cases we may need to ensure that the assignment only occurs when we use the value of the function. This can be done by using the inbuilt function **used()** [section 10.165](#) p345. This function can only appear in the body of a

function definition so to use it in an associated module it must be passed as a value mode variable to the module -

```
function
f (value(p)) {
  static(a, type(p));
  value(used, "log", used());

  [
    when (used)
      a <- p;
  ]

  return(a);
}
```

As an aside there are two simplifications in this version. Firstly the function parameter type has been omitted. It will take the type of the argument, which is fine in this case. Secondly the type of static mode variable **a** has been copied from the function parameter rather than being explicitly given. These make the code more succinct and also provide some generality in the function. The type of the argument to the function could be any target type however complex.

Note that any source statement containing a call to this function, and there may be any number, will create a separate instance of the entire function code and its nested module. This is true even if there are multiple calls within the same statement; each generates a separate instance.

As a note of caution suppose we were to call function **f()** with an argument which is a queue or an expression containing queue reads?

```
static(s, "uint:8");
queue(q, "uint:8");

.
.
.
s <- f(q);
.
.
```

The behaviour of this code would probably not be what we would want. The module would repeat execution of the **when** until the function value is used at which time it will attempt to do the assignment to **a**. If all queues in **p** are available (not empty) then the assignment will occur and the function will return the value of **a** as before. If however all queues in **p** are not available that assignment will wait until they are available. Meanwhile the function has no dependence on **p** and will simply return the value of **a**. The value of **a** will belatedly change when **p** becomes available. An attempt to fix the problem using a **sync** will also not work as the queue read is actually in a different module to that in which the **sync** appears -

```
.
.
sync {           // WONT WORK!
  s <- f(q);     // q is read in different module
}
.
.
```

The solution is to abandon the attempt to use a function and a separate module and instead simply code it in the following straightforward way -

```
static(s, "uint:8");
queue(q, "uint:8");
static(a, type(q));
```

```

.
.
sync {
  a <- q;
  s <- a;
}
.
.

```

The declaration of a could be moved into the sync block since it is not used outside that block -

```

static(s, "uint:8");
queue(q, "uint:8");

.
.
sync {
  static(a, type(q));
  a <- q;
  s <- a;
}
.
.

```

and the whole sync block could be placed in a procedure -

```

static(s, "uint:8");
queue(q, "uint:8");

.
.
p(s <- q);
.
.
procedure
p (static(ss) <- queue(qq)) {
  sync {
    static(a, type(qq));
    a <- qq;
    ss <- a;
  }
}

```

A function call may be followed by subscripts and field identifiers where the function returns an arrays, struct or class (arbitrarily nested).

If a function returns a pointer then it may be dereferenced using the `->` operator, e.g.

```
a = f()->;
```

The `->` operator may be followed by subscripts or field identifiers, e.g.

```
a = f()->[i].f2;
```

These function call constructions may also appear on the left hand side of an assignment statement, e.g.

```
f()->[i].f2 = 5;
```

A function call can also appear as an incrementing or decrementing statement. In that case it will be followed by the `->` operator and a `++` or `--`, e.g.

```
f()->[i].f2++;
```

If the variable pointed to by the function return is static then this will generate a target statement.

The example could be further complicated by a multiply indirect function call and the return of an array of pointers, e.g.

```
a[2]->b.f1[j]->().f3[4]->[i].f2++;
```

4.8 Modules

A module is a subroutine whose main purpose is to create execution threads in the FPGA.

Modules are similar to procedures but differ in two respects -

- modules are single destinations for queues whereas procedures have no particular significance when reading and writing queues
- module top level in-line target statements are each enclosed in individual infinite loops, but a procedure can only generate one in-line target statement which is inserted in-line back at the calling location

Thus modules form the base of target code execution by initiating execution threads whereas procedures are simply a means of re-using immediate code or improving readability by subdivision. An execution thread thus initiated can only terminate if it encounters a **stop()** procedure call ([section 9.76](#) p256) around which there is no parallel execution path.

Module definitions can be nested but scope rules determine where they can be called.

Module and procedure calls can be nested but executable target statements directly within each module have independent infinite loops, i.e. execution threads. Where modules are nested each individual module is a separate queue destination regardless of the nesting. For example -

```
m1(q);

module m1 (queue(q1)) {
    ... <- q1;
    .
    .
    m2(q1);
    .
}

module m2 (queue(q2)) {
    .
    ... <- q2;
    .
}
```

Module m1 is called and in turn calls module m2. Each of the modules reads the queue and is thus a destination for the queue. Thus to pop an entry from the queue both reads must execute, whether simultaneously or at different times.

Modules usually communicate with each other mainly by means of queues, but variables of other target modes can also be used - a module need not use queues at all. A module need not contain any target executable statements at all, in which case it is effectively a procedure.

Queues to be read or written by a module may be passed in the call via parameters (named module) or else may be in the global scope or in the same file scope (file or inline modules). There may be several places in a module where a queue is written. These queue writes will either be mutually exclusive events or will simultaneously write to different parts of a compound type. Writes to words within a queue constitute separate queue writes if they occur at different times (clock cycles) but are a single queue write if they occur simultaneously (the same clock cycle)³.

³Simultaneous writing to the same word in a queue is an undetected execution error. Where this is not precluded by the logic it can be avoided by arbitration using the waitpri control structure [section 6.17](#) p155

Every un-nested target statement within a module is placed within an infinite loop. Thus if a file module (source file) contained only -

```
static(s, "uint:8");

s++;
```

then the only target executable statement, `s++`, would execute on every clock cycle. If the statement were nested, say in a **when** statement,

```
value(l, "log");
static(s, "uint:8");

when (l)
s++;
```

then the top level statement, the **when**, would execute on every clock cycle but the nested increment statement would only execute on the clock cycles where `l` is true. Similarly a sequential thread would be created by

```
seq {
    .
    ... target statements
    .
}
```

where the sequential block would repeat indefinitely but the enclosed target statements would execute sequentially. The sequential execution could be initiated by a boolean -

```
value(l, "log");

when (l) seq {
    .
    ... target statements
    .
}
```

While `l` is false the **when** will execute on every clock cycle. When `l` is true the **when** will execute the sequential block, which will take a number of clock cycles. On completion of the sequential block the when will again execute as before.

Any nested modules within a module operate as separate modules, independent except for any shared variables.

Another simple type of module is an in-line module. This is simply a code block enclosed in brackets, e.g.

```
[
    .
    ... target statements
    .
]
```

Here again each un-nested statement will execute in its own infinite loop.

An in-line module is "called" when immediate execution reaches it. Thus it can be called multiple times (it can even be enclosed in a loop) and each occurrence produces a separate instance of the target code within the module.

A named module is a declared module. It has an identifier and optional input and output parameters, e.g.

```

module m1 (value(out, "uint:16" <- int(i), value(in, "uint:8"))) {
  .
  ... statements
  .
}

```

Again un-nested statements are enclosed in infinite loops.

There are six types of module -

- file modules are created by a **module** statement which imports the source code for the module from a source file. The main source file also forms a file module.
- named modules are declared in the source code in a similar way to procedures and functions. They are invoked by name.
- inline modules occur as source code enclosed in brackets and are called implicitly at the point at which they occur.
- leader modules are created by a **leadermodule()** statement and are called automatically once prior to execution of the main source code.
- trailer modules are created by a **trailermodule()** statement and are called automatically once following completion of execution of the main source code.
- post processing modules are created by a **postmodule()** statement and are called automatically once following completion of execution of any trailer modules.

Named modules can be invoked as often as required since they are called by name. Inline modules can be invoked more than once by being enclosed in a loop. File modules can only be invoked once.

4.8.1 file modules

When the main source file is read by 3PL that source file becomes a module, called a file module. Additional file modules are imported by a module statement, e.g.

```

module "spi.3pl"

```

A file module can have only one instance, i.e. it can only be imported once.

A file module has no parameters and thus cannot be passed arguments.

A feature of a file module is that it has its own scope and can thus be used to hide variables, limiting their access to code within the file module, including nested modules, procedures, functions and classes.

4.8.2 named modules

Named modules are declared and invoked as described in the previous section. Each call creates a new instantiation of a module.

A module call will normally be made at the top level with respect to target code but this need not be the case. In particular there is occasional need to include a module call within a function body. For example a function to delay a value by one clock cycle cannot have target executable statements in its body; it can only return expressions. However, a module to create the delay can be called within the function body since the module code executes independently. The function would return an output value from the module.

It is allowed, but not good practice, to call modules from within target structures such as **when**, **branch**, **seq while**, **par while**, **sync while**, **seq do while**, **par do while** or **sync do while**. Since the module target code does not execute in-place but in a completely separate execution environment a module call within these structures will appear misleading. In particular a module call within a target loop structure will not be subject in any way to the loop iterations. Note that a module call within an immediate loop will indeed be called subject to loop iterations.

A module may return from within its code block by the statement **return**; (this includes file modules and inline modules). If there is no return statement a module code body simply terminates as usual at the end. For the initial file module, whether it executes a return or simply executes to the last statement, any trailer modules are executed after it terminates.

4.8.3 inline modules

In-line modules are intended for very simple modules where the source code would be more verbose were they to be declared and called in the usual way.

An inline module is written in-line within square brackets, i.e.

```
[
    body
]
```

This construct both defines the module and also represents the call, i.e. when immediate code execution reaches the in-line module the module is 'called'. An alternative way to look at it is that immediate code flows into and through the in-line module and if a return statement is encountered this is similar to a continue statement in a loop, i.e. it skips to the end of the block. The square brackets delineate the code block as a separate source or destination for any queue reads or writes that it contains, as for all modules.

An in-line module has no parameters and thus cannot be passed arguments.

An inline module is a statement in a code body, but while it is allowed to appear within target control structures such as *when*, *sync* etc. it does not make sense to do so as the code for the module is not generated in-line but as an independent entity. It is normal to include inline modules within immediate control structure such as *if*, *for* etc. to make the calls conditional or to replicate them, during immediate execution, for example -

```
for (int(i) ; i<6 ; i++)
[
    body
]
```

which would instantiate six copies of the module. Clearly to make sense the content of the module would differ in some way on each iteration.

An inline module may read and write queues. These queues may be created outside the module but in an accessible scope or may be created inside the module. there are no module parameters or arguments so queues cannot be passed to the module by that means.

An inline module has a block scope similar to braced code blocks in control structures. The module can access variables back to the enclosing procedure, function or named module. It can also access its file scope and global scope.

An inline module code block may return conditionally prior to the end of the block by the statement **return;**, otherwise it will return after the last statement.

4.8.4 leader modules

Leader modules are declared by a **leadermodule()** statement. They are created during parsing, not during immediate execution. They are invoked prior to the start of immediate execution. An optional argument to the statement can be used to determine the order of execution of leader modules.

A leader module has two parameters, a comment string and an optional order integer. The comment string is used to report execution of the leader module in the report file if present. The order integer is used to order the execution of multiple leader modules.

4.8.5 trailer modules

Trailer modules are declared by a **trailermodule()** statement. They are created during parsing, not during immediate execution. They are invoked following completion of immediate execution of the main

program. An optional arguments to the statement can be used to determine the order of execution of trailer modules.

A trailer module has two parameters, a comment string and an optional order integer. The comment string is used to report execution of the trailer module in the report file if present. The order integer is used to order the execution of multiple trailer modules.

Trailer modules are used to finalise target code that by its nature remains incomplete during execution. This is usually external interfaces to other devices such as CPUs, memory or other FPGAs. For example a multiplexed interface may have a number of ports created during immediate execution which are unknown in number until main program execution has completed. Trailer code can then generate the necessary connecting logic.

4.8.6 post-processing modules

Post-processing modules are declared by a **postmodule()** statement. They are created during parsing, not during immediate execution. They are invoked following completion of immediate execution of the main program and any trailer modules. An optional arguments to the statement can be used to determine the order of execution of trailer modules.

A post-processing module has two parameters, a comment string and an optional order integer. The comment string is used to report execution of the post-processing module in the report file if present. The order integer is used to order the execution of multiple post-processing modules.

Post-processing modules are intended to invoke place-and-route tools following successful generation of netlist and constraint file(s). This can be done by using calls to inbuilt procedures **exec()** or **shell()**.

4.9 Classes

The class structure is provided for complicated interfaces where multiple ports may be added to an interface during immediate execution and the logic to tie it all together is added at the end. For most straightforward applications classes are not needed and indeed 3PL existed for two decades before the need for classes was finally recognised.

A class provides a wrapper around a collection of variables and associated modules, procedures, functions and nested classes. In its simplest form it provides functionality similar to a struct -

```
class c () {
    int(i, 34);
    static(s, "uint:8");
}
```

An instance of this class can be created by -

```
var(c1, "class");
c1 = c();
```

or more simply -

```
var(c1, "class", c());
```

or even more simply -

```
class(c1, c());
```

Thus a class is called in a similar way to a function but differs in that -

- A call to a class only ever returns an immediate value of type "class".
- Unlike a function call, a class call can have output arguments.

Variables within the above class instance **cl** can be accessed as in the following examples -

```
j = cl.i;
cl.s++;
```

Each class instance also has a variable **this** of type "class" defined which is the current class instance.

Like modules, procedures and functions, variables are re-created on each call and hence their values can be different in different instances of the class.

It should be noted that parameters created in the call to the class are local variables (as is the case with modules, procedures and functions) and so they can also be accessed in the same way as the other class variables.

There are cases where it would be convenient to share data between class instances and this can be done by using a common block -

```
class c () {
  common {
    value(v, "log");
  }
  int(i, 34);
  static(s, "uint:8");
}
```

here variable **v** is created once only when the class is called to create the first instance of the class. All further instances will share the same variable and its value. Common block variables can be both evaluated and assigned in any instance of the class and any changes will be reflected in all instances of the class. Note that variables are only created in the class common block on the first call, they cannot be created by later calls.

The common block must be at the top level within the class, i.e. it cannot be enclosed within any other control structure.

If common class variables are to be given initial values derived from arguments to the first class call then it is probably good practice to call the class once initially to create the common class variables and employ conditional statements to return immediately having done that. If the initialisation call is not intended to create an instance then the return value can be discarded by calling the class using a procedure call syntax. Later calls to the class will automatically ignore the common block. Using key/value argument/parameter assignments makes it easy to hide this once-only code, e.g.

```
class(cl);           // create variable of type "class"

c(v1=x);             // call to initialise common
cl = c(i1=5, i2=7)); // call to create a class instance

// class definition
class c (int({i1, i2}), value(v1, "log")) {
  common {
    value(v, "log", v1); // copy v1 to common v
  }
  if (matched(v1)
    return; // is initial call - ignore the rest.

  int(i, 34);
  static(s, "uint:8");
  .
  i = i1 + i2;
  .
  .
}
```

The above could also have instead -

```

if (!matched(i1) && !matched(i2))
    return;

```

A class can contain definitions of modules, procedures, functions and classes and these can be called as in the following example -

```

class(cl, c());
int(j);
static(ss, "uint:8");

cl.p(j <- 4);    // assigns 4 * cl.i to j
ss <- cl.f();    // FPGA code assigning cl.s + 5 to ss

class c () {
    int(i, 34);
    static(s, "uint:8");

    procedure p (int(out) <- int(in)) {
        out = in * i;
    }

    function f () {
        return(s + 5);
    }
}

```

Here the first assignment statement is immediate, calling class procedure **p()**. Parameter **in**, whose argument is 4, is multiplied by class variable **i** and the result is assigned via parameter **out** to **j**.

The second assignment statement is target (FPGA) code. The class function **f()** is called and returns the sum of the static 8-bit unsigned integer class variable **s** and constant 5 and the returned value is assigned to static variable **ss**.

A class cannot contain in-line target statements in its code body, either directly or via a procedure call.

Modules, procedures and functions can have their own local variables as usual but also have read and write access to variables in the class in which they are defined (as in the example above). They also can call other modules, procedures, functions and classes defined within the class.

A module, procedure, function or class defined within a class may be preceded by the keyword **private**, e.g.

```

class c () {
    .
    .
    .

    private
    procedure p (int(out) <- int(in)) {
        .
        .
    }

    function f () {
        .
        .
    }
}

```

which means that the module etc can only be called from within the class, not via class instance. For example in -

```

class(cl, c());
.
cl.p();
i = cl.f();

class c () {
  .
  .

  private
  procedure p (int(out) <- int(in)) {
    out = in * i;
  }

  function f () {
    return(s + 5);
  }
}

```

the call **cl.p()** will fail whereas **i = cl.f()** will succeed.

Chapter 5 - Immediate Statements

Immediate statements generate no target code execution threads.

An immediate variable is one which has been declared with no mode, or explicitly with immediate mode.

5.1 immediate mode assignment

Immediate assignment is assignment to an immediate variable. An immediate assignment statement has the forms -

```
variable = expression;           // simple assignment
variable += expression;          // additive assignment
variable -= expression;          // subtractive assignment
variable *= expression;          // multiplicative assignment
variable /= expression;          // divisional assignment
variable %= expression;          // remainder assignment
variable &&= expression;          // logical AND assignment
variable ||= expression;          // logical OR assignment
variable ^= expression;          // logical XOR assignment
variable &= expression;          // bitwise AND assignment
variable |= expression;          // bitwise OR assignment
variable ^= expression;          // bitwise XOR assignment
variable <<= expression;          // left shift assignment
variable >>= expression;          // right shift assignment
variable++;                       // increment statement
variable--;                       // decrement statement
```

Where a variable is a structure or an array, members can be assigned by -

```
variable.field = expression;
variable[subscript] = expression;
variable.field++;
variable.field--;
variable[subscript]++;
variable[subscript]--;
```

For simple assignment whole structs or arrays can be assigned by

```
variable = variable;
variable = {...}; // compound value
```

For simple assignment if the LHS is type "str" and the RHS is type "type" the RHS will be converted to string representation of the type (the same as invoking unbuilt function **typestring()**). If the LHS is type "type" and the RHS is type "str" the string will be converted to a type.

For the other assignment types the RHS type must be primitive. Types allowed are -

	LHS	RHS
+=	int, uint float float str	int, uint float int, uint str, int, uint, float
--	int, uint float float	int, uint float int, uint
*=	int, uint float float	int, uint float int, uint
/=	int, uint float float	int, uint float int, uint
%=	int, uint float	int, uint int, uint
&&=	log	log
 =	log	log
^^=	log	log
&=	int, uint, bits	int, uint, bits
 =	int, uint, bits	int, uint, bits
^=	int, uint, bits	int, uint, bits
<<=	int, uint, bits	int, uint
>>=	int, uint, bits	int, uint

Note that += can be used for string concatenation where the RHS can be one of several types.

The logical assignments &&= and ||= do not perform short-circuit evaluation unlike the equivalent binary operators.

RHS expressions may contain only immediate variables and constants and related operators and function calls returning immediate values.

LHS variable references may have side effects, for example incremented or decremented subscripts or subscripts returned by calls to functions within which enduring state is changed.

As a special case a value mode expression whose value is a constant may be assigned to an immediate mode variable. A value mode expression may be tested by calling the inbuilt function **istargconst()** [section 10.77](#) p301. These features may be used where either a constant of a variable is passed as an argument to a parameter and immediate conditional code needs to distinguish which is the case.

5.2 value mode assignment

Assignment to a value mode variable. is an immediate assignment, it is not executed in the target. It may however create target logic. A value assignment statement has the form -

```
variable := expression;
```

Whole structs or arrays can be assigned by

```
variable := variable;
```

RHS expressions may contain constants, both immediate and target variables and function calls and operators.

A value variable is a target variable which represents the value of a target expression. Constants will be converted to equivalent signals in the target logic.

A value variable can also be assigned an initial value when created (see **value()** [section 9.83](#) p259)

Whether the assignment is of a primitive or a compound type the widths of individual words may differ as long as the primitive types match. Bits and unsigned integers will be padded or truncated as needed. Signed integers will be truncated or sign extended. Signed integers may not be assigned to unsigned integers. Unsigned integers may be assigned to signed integers, but since size adjustments are made by truncation or zero padding the sign of the result may be incorrect unless the LHS is at least one bit larger than the RHS.

A value mode variable may be created as type "empty", in which case it can be assigned any type. Following assignment the variable will retain the type of the RHS value assigned to it. Any further assignments to that variable will be subject to type checking as normal. A value variable can be cleared and its type changed to "empty" by calling inbuilt procedure **emptyvalue** [section 9.25](#) p207.

A value variable is usually combinatorial in that it does not itself have intrinsic state in the target device but may explicitly create state devices. For example in the following

```
value(v, "log");

v := a && b || c;
```

variable *v* represents the value of the logical expression and as such is combinatorial.

But in the following example

```
value(v, "log");

v := a;
```

v is effectively a pointer to variable *a* (though not one that can be used with the indirect operator). The state, or storage, is in *a*, not in *v*. If *a* changes, the value of *v* will also change since it simply reflects the value of *a*.

Value variables are used mainly as 'pointers' to combinatorial expressions that are to be shared. 3PL will not attempt to find common subexpressions and instantiate these once, indeed that can be counter-productive for speed. Value variables as parameters are also a way of passing a variable or expression to or from a module, procedure or function.

When a value variable is a compound type each element of the type is separate. Thus a value array can be linked to quite distinct sources, e.g.

```
value(a, "[3]uint:8");
static(b, "uint:8");
queue(c, "uint:8");
value(d, "[2]uint:8");

d[1] := a + 2;
a[0] := b;
a[1] := c;
a[2] := 2 * d[1];
```

The value array *a* could now be assigned in total to another value array, whole or in part -

```
value(e, "[5]uint:8");

e[1..3] := a;
```

Each component of the left hand side value variable array is assigned the value of the corresponding right hand side value variable.

Value variables can be re-assigned at any time. When a value variable is referenced its current value is used. If a value variable is referenced prior to its first assignment, a temporary link will be put in place and this will be replaced by the first assigned value after which the value variable behaves as described

above. This pre-assignment usage is provided to avoid problems associated with cycles of dependence which occur frequently.

If the value of a value variable is used prior to assignment the value will be given the current clock domain and the value assigned later must have the same clock domain.

Where a value variable is assigned a value following evaluation of that variable, i.e. it has been used prior to assignment, the value assigned must not contain queue reads.

Where a value variable is assigned a value following evaluation of that variable, i.e. it has been used prior to assignment, the value assigned must not contain the function **cmemoryread()** [section 10.17 p275](#).¹

A value must not be assigned an expression containing **cmemoryread()** [section 10.17 p275](#) if the value is never used as this generates a read access to the memory variable which is never connected to appropriate control logic. This restriction may be removed in future.

There is a special case of value assignment which allows a clock variable to be used as a logic value. A clock variable can be assigned to a value mode variable of type **log**.

A value variable as a whole has a no clock domain, but the components of a value will each have a clock domain. That clock domain will be determined when the variable is evaluated by examining the clock domains of all ultimate source variables in expressions, i.e. variables that have state such as static mode and queue mode. The components, i.e. array members or structure fields, need not have the same clock domain, though that would be unusual. When a value mode variable is evaluated the value will have the clock domain of the component. If a number of components are accessed in one evaluation, for example a sub-array, they must all have the same clock domain.

A value variable may have the clock domains of its components set by passing the **domain** attribute in the variable creation procedure call or by setting it by calling inbuilt procedure **attributes()**. This will establish the clock domains of any components that thus far have no clock domain, all the way back to variables with state such as static mode or queue mode variables. Where such variables already have a clock domain the domain will be compared and if different a fatal error will result.

For a discussion of clock domains see [subsection 4.4.2 p99](#).

5.3 priority mode assignment

Priority mode variables are always arrays of type **log**. An assignment to a member, or members, of a priority variable array forms an input to the priority encoder where subscript 0 has the highest priority.

Assignment of priority mode variable array members has the same syntax as for value variables -

```
variable[i] := expression;
```

The assigned expression, which is an input, must not contain queue reads, but it may contain queue examines and queue availability tests.

A priority variable array member may be assigned more than once but the default behaviour is that the final value assigned prior to termination of immediate execution prevails, all other previous assignments being lost. This can be changed by setting attribute "multiplein" to true in which case all assignments will be collected and ORed together on completion of immediate execution to form the input to that priority array member.

A priority variable inherits the current clock domain. Values assigned to it must either have the same clock domain. If the value to be assigned has a different clock domain it must be converted by one of the inbuilt functions **sample()** [section 10.128 p326](#) or **accept()** [section 10.2 p265](#).

¹This restriction is imposed because of the need to link the execution signal for a statement containing a *cramread()* (or *cmemoryread()*) to the memory variable itself. The execution signal is used to multiplex the read addresses where multiple reads are involved. Post assignment to a value variable occurs after that execution signal was available. This limitation may be removed in the future

A priority variable has a single clock domain for reading and writing. When a priority variable is created using the inbuilt procedure **priority()** [section 9.52](#) p230 it has no associated clock domain. The clock domain is established by

- the first assignment to the variable
- the first use of the value of the variable

Note that evaluation of the variable in any context, for example in inbuilt or library modules, procedures or functions, will establish a clock domain if the variable does not yet have a clock domain. This will always be the current clock domain (unless the code has been explicitly written to establish the clock domain of its argument to something other than the current clock domain, in which case this will presumably be well documented).

When a priority variable is assigned, the clock domain of the assigned value must be the same as the current clock domain or null.

When a priority variable is assigned, If it has no clock domain the current clock domain will be established as the clock domain of the variable. If it has a clock domain it must be the same as the current clock domain.

When the value of a priority variable is used that value will have the clock domain of the variable.

Note that control structures **waitpri** and **waitprilock** use priority variables for mutual exclusion and provide a more convenient interface than use of priority variables directly [section 6.17](#) p155.

5.4 output mode unconditional assignment

An output variable represents a connection off-chip via output buffers and pads. Unconditional assignment of output mode variables is used where 3-state output is not required.

Unconditional assignment has the same syntax as for value mode variables -

```
variable := expression;
```

Any component of an output mode variable may only be assigned once, either by default via the 3rd argument to inbuilt procedure **output()** or explicitly by a conditional or unconditional assignment statement. Components such as structure fields or array members may be assigned in separate statements and may be a mixture of conditional and unconditional.

In the Xilinx implementation if the source of the output is a static variable then the associated flip-flops can be located in the IOBs. This can be done by a suitable flag to the Xilinx tools but is best done by explicitly setting the IOB attribute to true on the static variable since the 3PL netlist optimiser is then aware of this. If the static variable flip-flops go to other destinations as well as the output buffers then they will be replicated by the 3PL netlist optimiser, the replicants being sourced in parallel and located in IOBs while the originals stay in CLBs. The advantage of this is that the propagation time from flip-flop to pad will be the minimum achievable (controlled also by slew rate and drive current attributes). If this is not the case then a timing constraint from the clock domain to the output pads may need to be provided.

If a source for an output mode variable is specified when it is created (3rd argument to **output()**, see [section 9.47](#) p224) that is equivalent to an initial unconditional assignment of that source to the output mode variable.

The assigned expression must not contain queue reads, but it may contain queue examines and queue availability tests. The LHS may have array subscripts, subscript ranges or struct fields. If the value to be assigned has a different clock domain it must be converted by one of the inbuilt functions **sample()** [section 10.128](#) p326 or **accept()** [section 10.2](#) p265.

An output variable has a write clock domain but no read clock domain. When an output variable is created using the inbuilt procedure **output()** [section 9.47](#) p224 it has no associated clock domain. Its clock domain is established by the first assignment to the variable (or component of the variable if the type is compound) and is the current clock domain. The RHS expression must also have the current clock domain as its output (read) clock domain. The RHS expressions may be immediate or target mode.

An output mode variable must have at least a **loc** attribute, providing a location for each pad.

5.5 clock mode assignment

Assignment of clock mode variables has the same syntax as for value variables -

```
variable := variable;
```

but the RHS can only be a clock variable or a function returning a clock variable. An assignment of null is not allowed.

The LHS clock variable now becomes an implicit indirect reference to the RHS clock variable. The two variables are now equivalent in all contexts (but the netlist identifier is the RHS variable).

These assignments may be chained.

Where clock variables are assigned and are also used to set the clock domain, note that logic using that clock will be connected to the clock source determined by the value the clock variable has when immediate execution terminates. Changing the value of a clock variable by assignment, even if accompanied by passing it to **useclock()**, will not result in associated code having different clock domains - the code will all be driven by the clock source given by the final value of the clock variable.

5.6 the if control structure

The **if** has the format -

```
if (expression)
  body
```

which executes the body if the test expression is true.

An alternative form is

```
if (expression)
  true_body
else
  false_body
```

which provides an additional body to be executed when the test expression is false.

The expression must be immediate and must evaluate to a logical result.

In the first form the body will be executed if the expression is true. In the second form one of the code bodies will be executed as selected.

A body may be a single statement or a statement block in braces (`{...}`).

Any target statements generated by execution of one of the bodies will replace this *if* statement in the target code. If there are multiple target statements generated then this *if* must be enclosed in a *seq*, *par* or *sync* block to control their execution.

5.7 the while control structure

The **while** control structure iterates while a test expression is true. The format is -

```
while (expression)
  body
```

which executes the body while the test expression remains true.

The expression must be immediate and must evaluate to a logical result.

The body may be a single statement or a statement block in braces (`{...}`).

Any target statements generated by execution of the body will accumulate as a list of statements to replace this *while* statement in the target code. If there are multiple target statements generated then this *while* must be enclosed in a *seq*, *par* or *sync* block to control their execution.

5.8 the do while control structure

The **do while** control structure iterates while a test expression is true. The format is -

```
do {
    body
} while (expression);
```

which executes the body while the test expression remains true. This differs from the **while** in that the body is executed at least once.

The expression must be immediate and must evaluate to a logical result.

Any target statements generated by execution of the body will accumulate as a list of statements to replace this *do while* statement in the target code. If there are multiple target statements generated then this *do while* must be enclosed in a *seq*, *par* or *sync* block to control their execution.

5.9 the for control structure

The **for** control structure iterates while a test expression is true and provides initialisation and iteration control. The format is -

```
for (initialisation ; expression ; iteration)
    body
```

The initialisation statement must be a single immediate assignment statement or an inbuilt variable creation procedure call. If the latter the resulting variable will be in a local scope containing the loop.

The expression must be immediate and must evaluate to a logical result.

The iteration statement must be a single immediate assignment statement or increment or decrement statement as the C ',' (comma) operator is not implemented in 3PL.

The body may be a single statement or a statement block in braces ({...}).

Any target statements generated by execution of the body will accumulate as a list of statements to replace this *for* statement in the target code. If there are multiple target statements generated then this *for* must be enclosed in a *seq*, *par* or *sync* block to control their execution.

5.10 the switch control structure

The **switch** statement selects a code block by testing the value of an expression against a number of cases. The format is -

```
switch (expression) {
case expression_or_list1:
    body1
case expression_or_list2:
    body2
default:
    bodyd
}
```

each *expression_or_list* may be a single expression or a comma-separated list of expressions enclosed in braces. The value derived from *expression* is compared with the values in each of the case statements and equality results in the associated code body being executed. **expression** may be any immediate primitive type, including float². The case values must be of the same type as the test expression, but

²floating point equality comparisons require caution as rounding errors in computed values often result in values differing from the exact

they need not evaluate to constants. The case values, whether constant or not, need not be unique. If they are repeated the first one encountered (from the top) will be used.

Note that unlike languages such as C or Java the code associated with a case value does not flow on to the next case, i.e. break statements are ignored by the switch and may be used within a switch case code body to exit from an enclosing loop.

Any target statements generated by execution of a body will replace this *switch* statement in the target code. If there are multiple target statements generated then this *switch* must be enclosed in a *seq*, *par* or *sync* block to control their execution.

5.11 the break statement

A **break** statement can only occur within immediate **while**, **do while** or **for** loops. It switches immediate execution to the end of the enclosing iterative statement, exiting that statement.

5.12 the continue statement

A **continue** statement can only occur within immediate **while**, **do while** or **for** loops. It switches immediate execution to the end of the enclosing iterative statement to conditionally execute the next iteration.

5.13 the return statement

A **return** statement returns from a call to a module, procedure or function. It may also be used to terminate execution of a file module (which is executing via an implicit call) or to terminate execution of an inline module. On returning from the initial file module any trailer modules will then be executed.

For a module or procedure the syntax is -

```
return;
```

For a function the syntax is -

```
return(v);
```

where the argument is an expression giving the returned value of the function.

A module or procedure will of course return to the point at which it was called if code execution simply proceeds past the last statement in its code block. Functions must use **return** to return a value.

A **return** statement cannot appear in the following target control structures -

- branch
- par
- seq
- sync
- sync when
- seq do while
- par do while
- sync do while
- seq while
- par while
- sync while
- unless
- waitpri
- when

Chapter 6 - Target Statements

Target statements are copied to a target code structure during the immediate execution. The immediate code selects which statements are copied and may duplicate target code with variations introduced by immediate expressions.

Note that

value mode assignment,

priority mode assignment ,

clock mode assignment and

unconditional output mode assignment

do not generate in-line target code and so have been included under immediate assignment above [chapter 5](#) p125.

6.1 expression availability and operators

An expression is available when the contained queue variables are available. This means that the conditional operator **?:** cannot be used to select an available subexpression where the alternative is not available. For example in the following case -

```
q1 <<- 1 ? q2 : q3;
```

where **q1** to **q3** are queues, all three queues must be available for the statement to execute even though only one of queues **q2** and **q3** will be read. Similarly if this statement is enclosed in a **sync** control structure the sync body will not be executed until all three queues are available regardless of conditional selection by variable **l**.

6.2 selectvalue mode assignment

A selectvalue assignment statement has the form -

```
variable |- expression;
```

Whole structs or arrays can be assigned by

```
variable |- variable;
```

6.2.0.1 immediate execution

On immediate execution this generates one in-line target statement.

6.2.0.2 target execution

Selectvalue assignment is assignment to a selectvalue variable.

RHS expressions may be immediate or target mode.

A selectvalue variable represents a bus, each assignment being a drive onto the bus¹. This is a target assignment in that the assignment is only valid for one clock cycle - the bus is driven for one cycle when this statement executes. If the RHS contains one or more queue reads execution may be delayed waiting for queue availability. The assignment can be held if required by repeated execution.

For compound types assignments can be made to individual fields or array members. On any clock cycle when a component of the variable is not assigned the output for that component will be the default value specified when the selectvalue variable was created.

A selectvalue variable has a single clock domain for reading and writing. When a selectvalue variable is created using the inbuilt procedure **selectvalue()** [section 9.66](#) p244 it has no associated clock domain. The clock domain is established by

- the first assignment to the variable (or component of the variable if the type is compound)
- the first use of the value of the variable or one of its components

The clock domain of the variable will be set to the current clock domain.

If a value to be assigned to a selectvalue variable has a different clock domain it must be converted by one of the inbuilt functions **sample()** [section 10.128](#) p326 or **accept()** [section 10.2](#) p265. Note that evaluation of the variable in any context, for example in inbuilt or library modules, procedures or functions, will establish a clock domain if the variable does not yet have a clock domain. This will always be the current clock domain.

If the RHS expression contains queue variables their output (read) clock domain must be the current clock domain (if this is the first read or examine for a queue its output domain will be established by the assignment).

Whether the assignment is of a primitive or a compound type the widths of individual words may differ as long as the primitive types match. Bits and unsigned integers will be padded or truncated as needed. Signed integers will be truncated or sign extended. Signed integers may not be assigned to unsigned integers. However, unsigned integers may be assigned to signed integers, since size adjustments are made by truncation or zero padding the sign of the result may be incorrect unless the LHS is at least one bit larger than the RHS.

6.3 static mode assignment

Assignment to a static variable has the forms

```
variable <- expression;
variable +<- expression;
variable -<- expression;
variable *<- expression;
variable++;
variable--;
```

Where a variable is a structure or an array, fields or members can be assigned by

```
variable.field <- expression;
variable[subscript] <- expression;
variable.field +<- expression;
variable[subscript] +<- expression;
variable.field -<- expression;
variable[subscript] -<- expression;
```

¹CLB logic is used by default but where the target device has internal 3-state buffers these can be used to implement selectvalue variables. The attribute "threestate" is used to force the use of 3-state buffers.

```
variable.field *← expression;
variable[subscript] *← expression;
variable.field++;
variable.field--;
variable[subscript]++;
variable[subscript]--;
```

or any depth of nesting of fields and members can be used. Whole structures or arrays can be assigned by

```
variable ← variable;
```

The increment and decrement statement forms can only be applied to types "int:" and "uint:". The +←, -← and *← forms can only be applied to types "int:", "uint:", "fixed:" and "ufixed:". Type conversions in assignment are allowed in many cases and these are described in [subsection 3.13.5](#) p88.

6.3.0.1 immediate execution

On immediate execution this generates one in-line target statement.

6.3.0.2 target execution

RHS expressions and variables may contain queues or queue fields which are read or examined.

Regardless of any context or enclosing constructs which control execution, a static assignment can only execute if the RHS expression is available.

If the LHS variable is a struct or an array several assignment statements may be needed. These assignment statements may be grouped to execute simultaneously.

Where assignment to a static variable which is a struct or array is incomplete, the unassigned fields or array members will be unchanged.

Whether the assignment is of a primitive or a compound type the widths of individual words may differ as long as the primitive types match. Bits and unsigned integers will be padded or truncated as needed. Signed integers will be truncated or sign extended. Signed integers may not be assigned to unsigned integers. However, unsigned integers may be assigned to signed integers, since size adjustments are made by truncation or zero padding the sign of the result may be incorrect unless the LHS is at least one bit larger than the RHS.

In any one execution cycle the same array member or struct field may not be assigned to by more than one assignment statement. Such conflicts cannot be identified in general by the compiler. If an assignment conflict occurs the resulting value is undefined. ²

Optimisation note: if an immediate value, or a variable which is value mode and is constant, is assigned to a static variable or one of its components, the constant is checked against the initial value for that variable or component. If they are the same the variable or component is reset to its initial value rather than being assigned. This has benefits in both gate usage and speed -

- with one fewer assignment the data input logic is simplified using fewer gates
- simpler data input logic means that it will usually be faster
- reset logic is inherently simpler, using fewer gates than data input logic
- being simpler reset logic is inherently faster

A static variable, or one or more components of a variable, can be explicitly reset by using the inbuilt procedure **reset()** [section 9.61](#) p238

A static variable has a single clock domain for reading and writing. When a static variable is created using the inbuilt procedure **static()** [section 9.75](#) p253 it has no associated clock domain. The clock domain is established by -

²At the moment the compiler does not detect any assignment conflicts. It may in a future version. Assignment conflicts could also be detected at execution time and cause exceptions to be thrown, but that has not been implemented yet.

- the first assignment to the variable (or component of the variable if the type is compound)
- the first use of the value of the variable or one of its components

The clock domain of the variable will be set to the current clock domain.

Note that evaluation of the variable in any context, for example in inbuilt or library modules, procedures or functions, will establish a clock domain if the variable does not yet have a clock domain. This will always be the current clock domain (unless the code has been explicitly written to establish the clock domain of its argument to something other than the current clock domain, in which case this will presumably be well documented).

If the `reset()` inbuilt procedure is used on a static variable -

- if the static variable has a clock domain established, it must be the same as the current clock domain
- if the static variable does not have a clock domain established, it will be given the current clock domain

An assignment executes on one clock cycle unless the RHS expression contains queue reads on queues which are not available.

6.4 queue mode assignment

Assignment to a queue variable is of the form

```
variable <<- expression;
```

Where a variable is a structure or an array, fields or members can be assigned by

```
variable.field <<- expression;
variable[subscript] <<- expression;
```

or any depth of nesting of fields and members can be used. Whole structures or arrays can be assigned by

```
variable <<- variable;
```

6.4.0.1 immediate execution

On immediate execution this generates one in-line target statement.

6.4.0.2 target execution

The `<<-` assignment writes to a queue. The datum becomes available on the queue output.

An assignment executes on one clock cycle unless the RHS expression contains queues which are not available (are empty) or the LHS queue is not available (is full).

A queue variable has a clock domain for reading and another clock domain for writing. When a queue variable is created using the inbuilt procedure `queue()` [section 9.57](#) p234 it has no associated clock domains.

The clock domain for writing is established by -

- the first write (assignment) to the queue variable (or component of the variable if the type is compound)
- the use of the write availability operator `>` on the queue
- a call to `queuewritestatus()` on the queue
- a call to `reset()` on the queue

If the write clock domain is not yet established in any of these situations then it will be set to the current clock domain, otherwise it must be the same as the current clock domain.

The clock domain for reading is established by -

- the first read or examine of the variable (or component of the variable if the type is compound)
- the use of the read availability operator < on the queue
- a call to **queuereadstatus()** on the queue

If the read clock domain is not yet established in any of these situations then it will be set to the current clock domain, otherwise it must be the same as the current clock domain.

The read and write clock domains will usually be the same, in which case the queue variable is called **synchronous**.

When a queue variable is written (assigned), If it has no write clock domain the current clock domain will be established as the write clock domain of the variable. If it has a write clock domain it must be the same as the current clock domain.

When a queue variable is written (assigned), the clock domain of the assigned value must be the same as the current clock domain or null.

If the **reset()** inbuilt procedure is used on a queue variable -

- if the queue variable has a write clock domain established, it must be the same as the current clock domain
- if the queue variable does not have a write clock domain established, it will be given the current clock domain

When a queue variable read or examined the resulting value will have the read clock domain of the queue variable.

RHS expressions and variables may contain queues or queue fields, which may be read or examined.

Regardless of any context or enclosing constructs which control execution, a queue assignment can only execute if the LHS queue is available and the RHS expression is available.

When a queue of compound data type is accessed the entire compound datum is read or written. On a read any unused struct fields or array members are ignored. On a write any unassigned struct fields or array members will be clear (0, false etc.)³.

Compound types or subtypes can be assigned in a single assignment statement where the RHS can be expressed as a single variable or part of a variable. For example

```
queue(p, "[4][2]uint:8");
static(s, "[4][4]uint:10");

p <<- s[1..2];
```

will assign a subarray of static variable *s* to queue *p* and will truncate the 10 bit unsigned integers to 8 bit.

Members of a compound data type can be accessed in separate assignment statements as long as those statements are executed simultaneously so that a single read or write occurs. Simultaneity can be enforced by enclosing these statements in a *sync* block. For example in the following

```
struct(s,
  f1,    "log",
  f2,    "uint:8"
);
queue(q1, s);
queue(q2, s);

q1.f1 <<- q2.f1;
q1.f2 <<- q2.f2 + 3;
```

³For an enumerated type the value with an ordinal of 0 will be assigned. If the enumerated type has been created using `enumv()`, rather than `enum()`, such that there is no value for ordinal 0 then the assigned value will be invalid!

the two statements would execute simultaneously since the queue dependencies are the same (q1 and q2). Where different queues are used we need to synchronise the statements, e.g.

```

struct(s,
  f1,    "log",
  f2,    "uint:8"
);
queue(q1, s);
queue(q2, s);
queue(q3, s);

sync {
  q1.f1 <<- q2.f1;
  q1.f2 <<- q3.f2;
}

```

In any one execution cycle the same array member or struct field may not be assigned to by more than one assignment statement. Such conflicts cannot be identified in general by the compiler. If an assignment conflict occurs the resulting value is undefined (actually the data are ORed - also see the footnote in the discussion of assignment conflicts for static assignments in the previous section.) It is usual to rely on the logic of the code and its timing to ensure mutual exclusion but in the case of data resulting from multiple asynchronous external sources this may not be possible. In this case the waitpri() control structure, section [section 6.17](#) p155, can be used to arbitrate.

A queue may have data type "null", e.g.

```
queue(trigger, "null");
```

in which case it will have no associated data. Such a queue can be used to signal an event. A queue whose data type is *null* can be written by

```
variable <<- null;
```

and can be read by

```
null <<- variable;
```

A queue of type null can be assigned to another queue of type null.

A queue of type null cannot return a value on reading, however it can be used as if it were type "log" by casting it to "log", in which case the value returned will always be **true**.

In the case of the target conditional statements **when** and **sync when** the conditional expression can be a single queue mode variable of type "null" which will be treated as though of type "log" whose value will always evaluate as TRUE.

Thus a wait on a null queue can be explicit as in -

```

when (trigger) sync {
  .
  .
  .
}

```

Alternatively an assignment to null can be used. In the following example it is enclosed in a sync block so will wait until the queue is available and will then explicitly acknowledge (read) the queue by assignment to null -

```

sync {
  null <<- trigger;
  .
  .
  .
}

```

In each case null queue **trigger** will be waited upon and then read.

In other contexts a null queue may be cast to type "log" where again its value will always be true.

To acknowledge queues whose values have been extracted by the examine operator ? assignment to null can be used (read and ignore the result) -

```
null <<- expression;
```

where if the expression contains any queues they will be acknowledged without their values being read. It would be usual for the RHS to be either a queue variable or a value variable representing an expression. In the latter case any queue reads contained in the expression would occur and be acknowledged. An expression, rather than a variable, occurring on the RHS would generate unused code (though this would be discarded by the code generator optimiser) and hence is pointless, though not erroneous.

Whether the assignment is of a primitive or a compound type the widths of individual words may differ as long as the primitive types match. Bits and unsigned integers will be padded or truncated as needed. Signed integers will be truncated or sign extended. Signed integers may not be assigned to unsigned integers. Unsigned integers may be assigned to signed integers, but since size adjustments are made by truncation or zero padding the sign of the result may be incorrect unless the LHS is at least one bit larger than the RHS.

6.5 output mode conditional assignment

Conditional assignment of output mode variables is used where 3-state output is required.

Unconditional assignment to an output mode variable was described previously ([section 5.4](#) p129).

Conditional output mode assignment can have the form -

```
when (condition)
    variable |- expression;
```

where the **when** is at the top level, i.e. not nested within any other control structure except a module. It is preferable that the condition be simply a static variable of type "log" with the **direct** and **iob** attributes set to true since in that case the static variable flip-flops can be placed in the IO blocks for better timing.

A conditional assignment may also be executed within any normal execution thread, e.g.

```
seq {
    .
    .
    variable |- expression;
    .
    .
}
```

In each of the above the pad output will be in high impedance state except for the clock cycle during which the conditional assignment statement executes.

The LHS variable may have a compound type and the assignment may reference all or part of that variable. Assignments to components of a compound type may be conditional, unconditional or a mixture of both.

Conditional or unconditional assignment to a component of a variable can only be done once.

An output mode variable must have at least a **loc** attribute, providing a location for each pad.

6.5.0.1 immediate execution

On immediate execution this generates one in-line target statement.

6.5.0.2 target execution

RHS expressions may be immediate or target mode. If the value to be assigned has a different clock domain it must be converted by one of the inbuilt functions **sample()** [section 10.128](#) p326 or **accept()** [section 10.2](#) p265.

A RHS expression cannot contain a queue read (but can contain a queue examine or a queue availability operator).

An output variable represents a connection off-chip via output buffers and pads. The buffers may be 2-state or 3-state.

In the Xilinx implementation if the source of the output is a static variable or an examined synchronous queue variable then the associated flip-flops can be located in the IOBs. This can be done by a suitable flag to the Xilinx tools but is best done by explicitly setting the IOB attribute on the static variable since the 3PL netlist optimiser is then aware of this. If the flip-flops go to other destinations as well as the output buffers then they will be replicated, the replicants being connected in parallel and located in IOBs while the originals stay in CLBs. Similarly if the expression controlling conditional output is a static variable or an examined synchronous queue variable then the associated flip-flop will be replicated for each output bit, the replicants being connected in parallel and located in IOBs. Placement of output flip-flops in IOBs removes the necessity to apply timing constraints to these outputs. In light of the above, in the following code -

```
when (1)
  o |- v;
```

l and **v** should be static variables if possible and **o** should have attribute **direct** set. As described above in that case the flip-flops for the variable will be located in the IOBs and the propagation time from flip-flop to pad will be the minimum achievable (controlled also by slew rate and drive current attributes). If those conditions are not met then a timing constraint from the clock domain to the output pads will usually need to be applied.

Whether the assignment is of a primitive or a compound type the widths of individual words may differ as long as the primitive types match. Bits and unsigned integers will be padded or truncated as needed. Signed integers will be truncated or sign extended. Signed integers may not be assigned to unsigned integers. Unsigned integers may be assigned to signed integers, but since size adjustments are made by truncation or zero padding the sign of the result may be incorrect unless the LHS is at least one bit larger than the RHS.

An output variable has a write clock domain but no read clock domain. When an output variable is created using the inbuilt procedure **output()** [section 9.47](#) p224 it has no associated clock domain. Its clock domain is established by the first assignment to the variable (or component of the variable if the type is compound) and is the current clock domain. The RHS expression must also have the current clock domain as its output (read) clock domain.

6.6 the seq, par and sync control structures

These are of the form -

```
seq {
  body
}
```

or

```
par {
  body
}
```

or


```

sync {
    body
}

```

The body is executed sequentially during immediate execution but any target statements encountered in the body generate code for the target which will execute them sequentially (*seq*) or in parallel (*par* and *sync*).

Target code in a **seq** {...} block will be executed sequentially in the target. It may contain any blocks or statements. Each component of the body will execute when the previous one has completed. Any components may wait on queues. In the target the block will complete when the last component has finished execution.

Target code in a **par** {...} block will be executed in parallel. It may contain any blocks or statements. All components will execute once. Any component may wait on queues. The block will complete when all components have executed.

Target code in a synchronised block **sync** {...} will be executed in parallel when all queues read or written (but not examined or availability tested) anywhere in the block are available. This includes any queues read or written in procedures or functions called. Note that if a queue read is conditional this is not taken into account and the sync will still wait on read availability.

Note that the *sync* only controls initial entry into the parallel block and not any subsequent behaviour. For that reason it would be unusual, though not erroneous, to include other blocks or loops within a *sync* block.

Using a *sync* block can improve code by taking the queue availability test away from individual statements in the block and using a single test instead. In many cases the compiler recognises that code is within a *sync* block and omits the queue availability test. This is done with care since later queue reads in say a *seq* block within a *sync* block need to check queue availability since another read may have been done since the *sync* availability test.

6.7 the when control structure

The **when** has the formats -

```

when (expression)
    body

```

and

```

when (expression)
    true_body
else
    false_body

```

The expression must evaluate to a logical result (see exception below) and is usually target mode, but it may be immediate mode. If the test expression is a constant value mode variable, or is immediate mode, then the test will be omitted in the target code and the true or false block will be created as appropriate. If the test expression is not a constant value mode variable then it must have a single clock domain which is the same as the current clock domain.

The test expression may be a queue of type "null" in which case it will be treated as though of type "log" whose value is true. This is treated as a special case and hence a null queue cannot occur within a logical expression without casting it to type "log".

The code bodies are a single statement, which may be a *seq*, *par* or *sync* block. A code body may be a plain block {...} as long as it does not contain more than one target statement.

Either of the above forms of **when** may be followed by an exception termination

```
unless (exception_expression) {}
```

or

```
unless (exception_expression) {
    exception_body
}
```

Exception termination constructs are allowed on several control structures - see [section 6.16](#) p153.

6.7.1 immediate execution

On immediate execution this generates an in-line target statement. Immediate sequential execution is carried out on the code body or bodies of the *when*.

6.7.2 target execution

In the first form the target body will be executed if the expression is true. In the second form one of the target code bodies will be executed as selected. Following either of the above the **when** will complete execution.

If the test expression contains queue variables the test expression may wait on queue availability. If the test expression reads a queue it will remove a datum. If a statement within an enclosed code body also reads the same queue, care must be taken to ensure that these reads occur simultaneously, otherwise two data will be read rather than one (see examples below). In some cases it may be best to use an *examine* rather than a read on a queue variable within a test expression for a control structure.

The *when* test adds no execution cycles so the *when* executes in the number of cycles taken by the code body executed, with the exception that a missing code body will execute in one clock cycle (for example if there is no *else* body and the test is false then the *when* will take one clock cycle).

The following is an example of the correct use of queues -

```
when (p == 3)
    s <- p;
```

The **when** test will wait on the availability of queue **p**. Once **p** is available its value will be tested. If the value of the queue variable is 3, the assignment to the static variable **s** will be performed. If the test fails queue **p** will nevertheless have been read during evaluation of the test expression. If the test succeeds, the queue read in the test expression and the queue read in the enclosed assignment statement will take place in the same clock cycle and hence only a single read of one datum will take place.

The following example is incorrect -

```
when (q1 == 3)
    s <- q1 + q2;
```

The **when** test will wait on the availability of queue **q1**. Once **q1** is available its value will be tested and if successful the assignment will be executed. The assignment statement may now wait on the availability of queue **q2**, but the read of queue **q1** in the test expression will have gone ahead regardless. When queue **q2** becomes available and queue **q1** again becomes available **q1** will be read again. This example is easily fixed by enclosing the **when** in a sync block -

```
sync {
    when (q1 == 3)
        s <- q1 + q2;
}
```

which will wait to enter the **when** until all enclosed queues are available, at which time the **when** will execute both its test evaluation and conditionally the enclosed statement in one clock cycle. Alternatively it could have been written as -

```
when (<q1 && ?q1 == 3)
  s <- q1 + q2;
```

which would do an examine of **q1**, rather than a read, in the test expression (having first verified availability of **q1**). This is less elegant than the previous solution and is less efficient.

The compiler recognises cases where a **when()** contains queue reads or writes in one of the code bodies and also reads some but not all of these queues in the test expression and issues a warning.

Note that **when** statements pass execution control through the test expression within the same cycle that the body executes. There are two consequences of this that should be noted.

Firstly the signal delay through the test expression may reduce the timing margins for the first statement (seq block) or statements (par block) in the code bodies. In some cases it may for example be possible to change the order of statements in a seq block so that a complex statement is not the first.

Secondly nested **when** statements concatenate the control delay through each of their test expressions. For this reason the depth to which it is prudent to nest **when** statements depends on the clock frequency. At high clock frequencies it is preferable to use multiple parallel **when** statements, for example -

```
when (11)
  when (12)
    statement1
  else
    statement2
```

may be better written as -

```
when (11 && 12)
  statement1
when (11 && !12)
  statement2
```

assuming that these are in a parallel execution context (and assuming that the clock frequency is too high for the nested example). Note that if a logical expression is complex, e.g. -

```
when (i < 4567)
  when (12)
    statement1
  else
    statement2
```

its duplication can be avoided by -

```
value(1, "log", i < 4567);
when (1 && 12)
  statement1
when (1 && !12)
  statement2
```

If an exception termination construct is appended, execution within the **when** can be prematurely terminated - see [section 6.16](#) p153.

6.8 the sync when control structure

The **sync when** has the format -

```

sync when (expression) {
  body
}

```

The expression must evaluate to a logical result (see exception below) and is usually target mode, but it may be immediate mode. If the test expression is a constant value mode variable, or is immediate mode, then the test will be omitted in the target code and the code block will be created or not as appropriate. If the test expression is not a constant value mode variable then it must have a single clock domain which is the same as the current clock domain. The test expression cannot contain unbuffered queue reads.

The test expression may be a queue of type "null" in which case it will be treated as though of type "log" whose value is true. This is treated as a special case and hence in general a null queue cannot occur within a logical expression without casting it to type "log".

The code body in braces is a parallel block as in the case of **sync** and **sync while**. The code block cannot contain unbuffered queue reads or writes.

The **sync when** may be followed by an exception termination

```

unless (exception_expression) {}

```

or

```

unless (exception_expression) {
  exception_body
}

```

Exception termination constructs are allowed on several control structures - see [section 6.16](#) p153.

6.8.1 immediate execution

On immediate execution this generates an in-line target statement. Immediate sequential execution is carried out on the code body of the *sync when*.

6.8.2 target execution

The **sync when** will execute the enclosed code block if the test expression is true and all queue variables in the test expression and code block are available. If not, execution of the statement terminates. This differs from a **when** inside a **sync** where the **sync** will wait on queue availability. If the **sync when** fails execution of the thread continues with no delay, i.e. the statement takes no clock cycles to execute.

It is important to note that the **sync** takes no account of conditional access to queues within its block, i.e. -

- queue reads or writes within a **when** statement
- queue reads within operands to the ternary conditional operator ?:

If queue reads or writes are located within the block the **sync** will test for availability on all of these before execution of the target code in the block, even though some of these queue accesses may not eventuate due to conditional execution.

For example in the following case -

```

queue({q1, q2, "log"};
sync when (q1) {
  seq {
    q1 <<- q2;
    q2 <<- q3;
  }
}

```

the statement will only succeed if queue q2 contains a single datum since it has the default depth of 2. If q2 is empty <q2 will be false and if q2 is full >q2 will be false. No account is taken of the fact that q2 is read before it is written.

The target code body will be executed if the expression is true and all queues read in the test expression or read or written anywhere in the code body are available. Statements in the code body are executed in parallel. If any queues to be read in the test expression, or any queues in the code body, are not available the **sync when** will not evaluate the test expression and hence will not execute the code body and execution of the **sync when** will finish. Similarly if evaluation of the test expression yields a false result the code body will not be executed.

If the test expression contains queue variables the test expression will not wait on their availability. No queues in the test expression will be read unless all queues to be read in the test expression are available and all queues to be read and written in the code block are available.

The *sync when* test adds no execution cycles so the *sync when* executes in the number of cycles taken by the code body.

Note that **sync when** statements pass execution control through the test expression within the same cycle that the body executes. The signal delay through the test expression may reduce the timing margins for the first statement (seq block) or statements (par block) in the code bodies. In some cases it may for example be possible to change the order of statements in a seq block so that a complex statement is not the first.

If an exception termination construct is appended, execution within the **sync when** can be prematurely terminated - see [section 6.16](#) p153.

The **sync when** is intended to be used where queue waits are to be avoided. For example to control queue assignment one might write -

```
when (go)
  q2 <<- q1;
```

The disadvantage of this is that if **go** is true but **q1** is not available for reading, or **q2** is not available for writing, then the assignment statement will hang waiting for queue availability. If we attempt to avoid entering the **when** by enclosing it in a **sync** -

```
sync {
  when (go)
    q2 <<- q1;
}
```

the **sync** itself will hang waiting for availability. We could change the first attempt to -

```
when (>q2 && <q1 && go)
  q2 <<- q1;
```

which achieves what we want. The **when** will not execute the assignment unless the two queues are available and **go** is true. If these conditions are not met the **when** finishes and passes execution on.

Using a **sync when** we could instead write -

```
sync when (go) {
  q2 <<- q1;
}
```

The **sync when** achieves the same result as the final example above while avoiding the necessity of finding all queue reads and writes in the code block and explicitly testing for their availability.

If in the above example the variable **go** was a queue then the last two examples are no longer equivalent. A queue read encountered in the test expression of a **when** will cause the **when** to hang waiting for the queue to become available before it can evaluate its test expression. The **sync when** will abandon any attempt to evaluate the test expression in that case since it can see that the queue is not available. It could notionally be coded as -

```

when (<go)
  when (go && >q2 && <q1)
    q2 <<- q1;

```

While this is functionally equivalent to the use of **sync when** as listed above, it generates logic which is much less efficient, both in path delays and resource usage.

6.9 the branch control structure

The **branch** has the format -

```

branch (expression) {
  case cexpr1:
    body1
  case cexpr2:
    body2
  default:
    body3
}

```

This branches to one of the case code bodies as a result of testing the value of the expression, which must be a target mode. The case expressions must be immediate mode. A case body target code does not flow on to the next case body target code.

A case may have a comma separated list of immediate expressions within braces, e.g.

```

branch (expression) {
  case {cexpr1, cexpr2, cexpr3}:
    body1
  case cexpr4:
    body2
}

```

in which case if the branch expression matches any of the case expression values the code body will be executed.

Case values are immediate expressions which evaluate to constants for the target code. The resulting case values must be unique. Primitive types allowed are "bits", "uint", "int" and "enum".

The branch statement terminates when the selected code body terminates.

The default case is optional. If no cases are matched then no target statements are executed and the branch statement terminates.

The branch expression and the case expressions must be compatible types. The test expression must have a single clock domain which is the same as the current clock domain.

The code bodies are single statements, which may of course be *seq*, *par* or *sync* blocks.

The **branch** may be followed by an exception termination

```

unless (exception_expression) {}

```

or

```

unless (exception_expression) {
  exception_body
}

```

Exception termination constructs are allowed on several control structures - see [section 6.16](#) p153.

6.9.1 immediate execution

On immediate execution this generates an in-line target statement. Immediate sequential execution is carried out through all the case expressions and the associated code bodies in order.

6.9.2 target execution

The branch test expression is evaluated and compared simultaneously with all case values (the case values are constant in the target code but may have been evaluated from expressions during immediate execution). If the branch test expression value matches a case value then the associated target statement is executed. The target statement may be a control structure containing nested target statements.

If the test expression contains queue variables the test expression may wait on queue availability. If the test expression reads a queue it will remove a datum. If a statement within an enclosed code body also reads the same queue, care must be taken to ensure that these reads occur simultaneously, otherwise two data will be read rather than one. In some cases it may be best to use an examine rather than a read on a queue variable within a test expression for a control structure if it is known that the queue variable is available for reading.

The case selection takes no clock cycles but the branch test expression evaluation and parallel case comparison delay is added to the execution delay of every case block. If a case is not matched the *branch* statement will take one clock cycle).

6.10 the seq while, par while and sync while control structures

This statement creates a **while** loop in target code. The body of the loop is enclosed in a **seq** block, a **par** block or a **sync** block.

The format is -

```
seq while (expression) {
    body
}
```

or

```
par while (expression) {
    body
}
```

or

```
sync while (expression) {
    body
}
```

The expression must evaluate to a logical result. It is usually target mode but may be immediate mode. If the expression is immediate and false the body will not be subject to immediate execution and no target code will be generated for the statement. If the expression is immediate and true an infinite target loop will be created. Similarly the test expression may be a constant value mode variable in which case the behaviour is the same as for an immediate expression. If the expression is target mode and is not a constant value mode variable then it must have a single clock domain which is the same as the current clock domain.

The code body is one or more statements.

Any of the above forms of **while** may be followed by an exception termination

```
unless (exception_expression) {}
```

or

```
unless (exception_expression) {
    exception_body
}
```

Exception termination constructs are allowed on several control structures - see [section 6.16](#) p153.

6.10.1 immediate execution

On immediate execution this generates an in-line target statement. Immediate sequential execution is carried out on the code body of the *while* as long as the test expression is not immediate mode and false in value.

6.10.2 target execution

The **seq while**, **par while** and **sync while** control structures repeatedly execute the code body while the test expression is true.

When the expression and body are both available and the expression evaluates to true, control passes to the body.

When the expression is available and evaluates to false the *while* will terminate.

If the test expression contains queue variables the test expression may wait on queue availability . If the test expression reads a queue it will remove a datum. If a statement within the enclosed code body also reads the same queue, care must be taken to ensure that these reads occur simultaneously, otherwise two data will be read rather than one. See the examples in the section on **when()** ([section 6.7](#) p141 above) which also pertain to the target **while()**.

The compiler recognises cases where a target **while()** contains queue reads or writes in the code body and also reads some but not all of these queues in the test expression, and in such cases issues a warning.

The target *while* adds no extra cycles for each loop iteration test, but termination of the loop, even when there are no iterations, takes one extra clock cycle.

Note that **seq while**, **par while** and **sync while** control structures pass execution control through the test expression within the same cycle that the body executes. As a consequence the signal delay through the test expression may reduce the timing margins for the first statement (seq block) or statements (par block) in the code body. In some cases it may for example be possible to change the order of statements in a seq block so that a complex statement is not the first.

If an exception termination construct is appended, execution within the **while** can be prematurely terminated - see [section 6.16](#) p153.

6.11 the seq do while, par do while and sync do while control structures

This statement creates a **do while** loop in target code. The body of the loop is enclosed in a **seq** block, a **par** block or a **sync** block.

The format is -

```
seq do {
    body
} while (expression);
```


or

```
par do {
  body
} while (expression);
```

or

```
sync do {
  body
} while (expression);
```

The expression must evaluate to a logical result and must be target mode.

The code body is one or more statements.

Any of the above forms of **do while** may be followed by an exception termination

```
unless (exception_expression) {}
```

or

```
unless (exception_expression) {
  exception_body
}
```

Exception termination constructs are allowed on several control structures - see [section 6.16](#) p153.

6.11.1 immediate execution

On immediate execution this generates an in-line target statement. Immediate sequential execution is carried out on the code body of the *do while*.

6.11.2 target execution

The **seq do while**, **par do while** and **sync do while** control structures repeatedly execute the body while the test expression remains true. This differs from the **while** in that the body is executed at least once.

If the code body contains queue reads and the test expression also reads the same queues care must be taken to ensure that where the code body and test expression are intended to read the same datum simultaneously the intent is not thwarted by unexpected waits in the body code. In particular if the test expression contains queue reads then the body code should either not read those queues or else should not contain any other queue reads or any queue writes. The test expression must have a single clock domain which is the same as the current clock domain.

Note that in the following -

```
par do {
  s <- p;
} while (p);
```

queue **p** will be read on the initial execution of the body. In the next clock cycle the test expression will again read **p** and if the result is true the body will execute on the same clock cycle reading queue **p** simultaneously with the test expression.

The compiler recognises cases where a target **do while()** contains queue reads or writes in the code body and also reads some but not all of these queues in the test expression, and in such cases issues a warning.

The target *while* adds no extra cycles for each loop iteration test, but one extra clock cycle is required for loop termination.

If an exception termination construct is appended, execution within the **do while** can be prematurely terminated - see [section 6.16](#) p153.

6.12 the `seq{...}` sequential block

The format is -

```
seq {
    body
}
```

The code body is one or more statements.

The **seq** may be followed by an exception termination

```
unless (exception_expression) {}
```

or

```
unless (exception_expression) {
    exception_body
}
```

Exception termination constructs are allowed on several control structures - see [section 6.16](#) p153.

6.12.1 immediate execution

On immediate execution this generates an in-line target statement. Immediate sequential execution is carried out on the code body of the *seq*.

6.12.2 target execution

The **seq** control structure executes the in-line target statements within it sequentially. The control structure finishes when the last contained statement finishes execution.

The target *seq* adds no extra cycles to the time taken by the code body target statements.

If an exception termination construct is appended, execution within the **seq** can be prematurely terminated - see [section 6.16](#) p153.

6.13 the `par{...}` parallel block

The format is -

```
par {
    body
}
```

The code body is one or more statements.

The **par** may be followed by an exception termination

```
unless (exception_expression) {}
```

or

```
unless (exception_expression) {
    exception_body
}
```

Exception termination constructs are allowed on several control structures - see [section 6.16](#) p153.

6.13.1 immediate execution

On immediate execution this generates an in-line target statement. Immediate sequential execution is carried out on the code body of the *par*.

6.13.2 target execution

The **par** control structure executes the in-line target statements within it in parallel. The control structure finishes when the last contained statement finishes execution.

The target *par* adds no extra cycles to the time taken by the longest code body target statement.

If an exception termination construct is appended, execution within the **par** can be prematurely terminated - see [section 6.16](#) p153.

6.14 the `sync{...}` synchronised parallel block

The format is -

```
sync {
    body
}
```

The code body is one or more statements.

The **sync** may be followed by an exception termination

```
unless (exception_expression) {}
```

or

```
unless (exception_expression) {
    exception_body
}
```

Exception termination constructs are allowed on several control structures - see [section 6.16](#) p153.

6.14.1 immediate execution

On immediate execution this generates an in-line target statement. Immediate sequential execution is carried out on the code body of the *sync*.

6.14.2 target execution

The **sync** control structure executes the in-line target statements within it in parallel but it will not begin parallel execution of the body until all queues read or written within the code body are available. The control structure finishes when the last contained statement finishes execution.

It is important to note that the **sync** takes no account of conditional access to queues within its block, i.e. -

- queue reads or writes within a **when** statement
- queue reads within operands to the ternary conditional operator **?:**

If queue reads or writes are located within the block the **sync** will wait for availability on all of these before execution of the target code in the block, even though some of these queue accesses may not eventuate due to conditional execution.

The *sync* adds no extra cycles to the time taken by the code block unless it must wait for queue availability.

Because the **sync** block will not execute the code block within it until any queues referenced in that block are available, many statements which would otherwise test for queue availability before executing do not do so if they are within a **sync** block. For example assignment statements immediately within a **sync** block omit the queue availability test. Although the validity of execution is not affected by these considerations, using a **sync**, rather than a **par**, to create a parallel block can result in savings in both logic components and path delays. The omission of queue availability tests within a **sync** block does not extend -

- beyond the initial entry test of a **seq while(){} or par while(){} block**
- into a **seq do while(){} or par do while(){} block**
- beyond the first target statement of a **seq block**

The omission of queue availability tests within a **sync** block does extend into -

- when blocks
- par and sync blocks
- for, while and do while blocks

Generally it is good practice to use a **sync** instead of a **par** when queues are used and it is not necessary to wait on individual availability of those queues. It is also preferable to enclose a **when** in a **sync** rather than use **sync** blocks within the **when**, e.g.

```
sync {
  when (...) par {
    a <<- c;
    b <<- d;
  } else par {
    a <<- e;
    b <<- f;
  }
}
```

or

```
sync {
  when (...) sync {
    a <<- c;
    b <<- d;
  } else sync {
    a <<- e;
    b <<- f;
  }
}
```

are preferable to

```

when (...) sync {
  a <<- c;
  b <<- d;
} else sync {
  a <<- e;
  b <<- f;
}

```

In the last case each assignment statement will contain logic to test for the availability of the queues concerned, whereas the previous two cases will test only once for all queues before executing the **when**. However, note that this might not be desirable where the alternative blocks in the **when** handle different queues. In the last case the **when** might select the true block and execute it when queues *a*, *b*, *c* and *d* are available regardless of the availability of queues *e* and *f*. The other two cases would not enter the outer sync unless all six queues were available even though two of them would not be used.

If an exception termination construct is appended, execution within the **sync** can be prematurely terminated - see [section 6.16](#) p153.

6.15 the [...] inline module

The format is -

```

[
  body
]

```

The code body is one or more statements.

6.15.1 immediate execution

On immediate execution this generates an inline module. Immediate sequential execution is carried out on the code body.

6.15.2 target execution

The inline module created is separate from the module in which it is created though it shares the scope of the module, procedure or function in which it is created. It does not generate any in-line target code in the module, procedure or function in which it is created.

6.16 the exception termination construct

The control structures -

- when
- branch
- seq
- par
- sync
- sync when
- seq while
- par while
- sync while
- seq do while
- par do while

- sync do while

may be followed by an exception termination

```
unless (exception_expression) {}
```

or

```
unless (exception_expression) {
    exception_body
}
```

The exception body is a single target statement but this might be a **seq**, **par** or other type of target code block statement.

A control structure with an exception termination cannot be nested within another control structure with an exception termination, either in a control structure code block or an exception code block⁴.

6.16.1 immediate execution

Following immediate execution of the code within the control structure the code in the exception body, if present, is executed. An exception body cannot execute a **break**, **continue** or **return** (but a **break** or **continue** could be nested within a **for** loop in the exception body for example).

6.16.2 target execution

When the exception expression is true, any target code executing within the associated control structure will cease, including execution within any nested target control structures. The clock cycle in which the exception expression is true will override any statements executing in the control structure. If there is an exception code block, that code will then be executed. The control structure then terminates as though it had executed to completion normally.

An exception termination takes a minimum of one cycle.

Note that if code within a control structure has executed a **stop()** it would never ordinarily complete. An exception termination will force completion of that control structure regardless of the **stop()**, i.e. the code block will skip to the end despite any intervening **stop()** even if the code has already reached the **stop()**.

An exception block may contain a **stop()** in which case execution of the exception block will cease at that point and the associated control structure will not complete.

If the exception expression does not contain queue reads, it may remain true for an indefinite time. Execution of the associated control structure will cease immediately and any exception code will be executed. Completion of the associated control structure will then occur if the exception expression is now false, otherwise completion will be delayed until the exception expression becomes false. By prolonging the exception expression true value completion of the associated control structure can be delayed. Any excursions of the exception expression value during execution of the exception block will be ignored.

If the exception expression contains queue reads then the exception condition will occur when a queue read occurs and the resulting value is true. False values are read and ignored. Execution of the associated control structure will cease immediately, any exception code will be executed and completion of the associated control structure will then occur. If the exception expression is available on completion the expression will again be evaluated and if true another exception will occur.

An exception code block may contain queue reads and writes. These will wait on queue availability as usual.

⁴This restriction may be removed in the future.

6.16.3 notes

Cessation of execution on an exception termination includes any statements waiting on queues or **waitpri** statements waiting on priority. The queue variables themselves are not affected, a pending read or write simply failing to execute. A single assignment involving queues can be terminated on an exception by enclosing it in a **seq**, e.g.

```
seq {
  q1 <<- q2 + 2;
} unless (reset) {}
```

but this example is pointless as the statement would simply continue with another execution after the reset. A more likely scenario would be to clear a processing pipeline when the exception occurs, e.g.

```
seq {
  q1 <<- q2 + 2;
} unless (reset) {
  par {
    reset(q1, q2);
    .
    . other resetting code
    .
  }
}
```

The pending assignment will be killed, the queues cleared and other reset code executed, then, if at the top level, the **seq** will resume execution. This might alternatively have been written

```
seq {
  q1 <<- q2 + 2;
} unless (reset) {}

when (reset) par {
  reset(q1, q2);
  .
  . other resetting code
  .
}
```

Note that in this case if **reset** is true for one cycle the **seq** will execute again on the following cycle if it is at the top level, since there is no exception code block to delay completion, but the queues will have been cleared on the same clock cycle as the reset prior to continued execution of the **seq**.

6.17 the waitpri () construct

The **waitpri** control structure allows arbitration between conflicting statements, usually but not necessarily involving queue writes or memory reads or writes. It takes the forms -

```
waitpri (pv, p)
statement
```

or

```
waitprilock (pv, p)
statement
```

where **pv** is a priority mode variable and **p** is an immediate mode expression with an integer value. **p** can have a non-zero positive value where lower values have a higher priority (zero is reserved). **p** has a

maximum value which is one less than the dimension of the priority variable array (default dimension is 20). Priority values for a particular priority variable do not have to be contiguous but must be unique.

The priority variable must not be assigned anywhere (the **waitpri** () and **waitprilock** perform assignments to the priority variable). The values of the priority variable boolean array may be used elsewhere, each member being true when the associated statement executes. The first entry in the priority array, subscript 0, is used to lock out priority requests when execution has been granted by **waitprilock**. This output will go high on the clock cycle following execution granting at any priority level and will remain high until inbuilt target procedure **priunlock**() is called.

In the case of **waitprilock** if there are no enclosed target statements (or perhaps no statements at all, i.e. just a ";") then a single cycle **nop**() is provided. The **waitpri** must have a single target statement (which might be a **sync** statement or a **sync when** statement containing multiple target statements).

A **waitpri** requests granting of execution of its enclosed statement via the priority variable and the specified priority level. Where queue availability is required this is waited upon if necessary prior to requesting priority. Priority is only held for one clock cycle.

The **waitprilock** is the same as **waitpri** but it continues to hold priority. Priority is released by executing the inbuilt inline target code procedure **priunlock**() [section 9.53](#) p232. **priunlock**() takes one clock cycle to execute but can usually be executed in parallel with the last statement to be executed under mutual exclusion.

The same priority variable may be used by both **waitpri** and **waitprilock** statements.

By default each priority level for a priority variable may be used only once. A situation may occur where it is desired to have two or more waitpri or waitprilock statements waiting on the same priority where it is known that the execution logic ensures that they are mutually exclusive. In that case the default can be overridden by setting the attribute "multiplein" true on the priority variable.

The **waitpri** or **waitprilock** will delay execution until a clock cycle on which -

- all queue variables in the statement are available
- the priority variable grants execution, i.e. no higher priority request has pre-empted it

These conditions are not latched; queue availability may come and go cycle by cycle as may priority pre-emption. If these conditions are met immediately then execution is not delayed, i.e. no clock cycle overheads are imposed unless queue non-availability or priority pre-emption intervenes.

The statement enclosed within a **waitpri** or **waitprilock** may only be -

- a target assignment statement
- a **sync** block
- a **par** block which contains no queue reads or writes
- **cmemorywrite**()
- **rmemoryread**()
- **rmemorywrite**()
- **nop**() (restricted to single cycle)
- **assert**()
- **reset**()

Inbuilt inline target procedure **stop**() is not allowed.

Nested sync or par blocks in turn must also only contain statements from the same restricted list.

The **waitpri** only guarantees priority for a single cycle so execution must occur in one cycle. Thus the enclosed statement must be controlled at a single point. At this point in execution a request is made for priority to be granted. Thus a single assignment is OK and a **sync** block also controls execution of an enclosed block at a single point (where it waits on queue availability). A **par** block will also execute in a single clock cycle as long as it does not read or write a queue mode variable.

These restrictions are enforced by the compiler.

Note that for a waitprilock these restrictions only apply to the statement associated with the waitprilock, not to following statements up to the priunlock() call!

A **waitpri** or **waitprilock** should not be nested within -

- a **sync** block

- a **sync when** block

however it is not an error. A **sync** block is intended to ensure that its enclosed statements (or at least the first of a nested **seq**) can execute simultaneously. An enclosed **waitpri** or **waitprilock** might not execute because of priority pre-emption. While this would not be an error it rather defeats the purpose of the **sync** since the delay means that queue availability would again have to be tested.

A **waitpri** or **waitprilock** cannot be nested within another **waitpri** or **waitprilock** as the statement enclosed in the outer one must be able to execute when enabled but the inner one may preclude that. Again the compiler prohibits this.

Where a **sync** is nested within a **waitpri** or **waitprilock** the **waitpri** or **waitprilock** actually passes control of priority seeking to the enclosed **sync** so that the **sync** waits on both queue availability (if there are queue reads or writes in its body) and priority granting.

The following example arbitrates writes to a queue -

```
priority(pv);
queue({p, q1, q2}, t);

waitpri (pv, 1)
  p <<- q1;
waitpri (pv, 2)
  p <<- q2;
```

Data appearing in queues **q1** or **q2** will be written to queue **p**. Where they arrive simultaneously **q1** will prevail. If multiple assignments are required then a **sync** block may be used -

```
priority(pv);
queue(p, "[2]t");
queue({q1, q2, q3}, t);
static(s, t);

waitpri (pv, 1)
  sync {
    p[0] <<- q1;
    p[1] <<- q2;
  }
waitpri (pv, 2)
  sync {
    p[0] <<- q3;
    p[1] <<- s;
  }
```

If sequential statements are to be executed while holding priority then a **waitprilock** must be used -

```
priority(pv);
queue(p, t);
queue({q1, q2, q3}, t);
static(s, t);

seq {
  waitprilock (pv, 1)
    p <<- q1;
    p <<- q2;
    sync {
      p <<- q3;
      priunlock(pv);
    }
}
```

A **waitprilock** may be used as an execution gate to inhibit processing of queue data until downstream processing of the previous datum has finished. In that case the **priunlock()** may be executed in a different execution thread -

```

priority(pv);
queue(p, "[4]uint:8");
queue(q1, "log");

seq {
  static(x, "[4]uint:8");
  static({count, addr, i}, "uint:3");

  waitprilock (pv, 1)
  sync {
    x <- p;
    count <- 4;
    addr <- 0;
    i <- 0;
  }
  par while (count != 0) par {
    rmemorywrite(m, 0, addr, x[i]);
    addr++;
    count--;
    i++;
  }
  q1 <- true;
}

seq {
  when (q1)
  .
  .
  .
  rmemoryread(m, 1, 0);
  par while (...) {
    ... <- r memoryout(m, 1);
    rmemoryread(m, 1, a);
  }
  .
  .
  priunlock(pv);
}

```

The first **seq** will attempt to repeatedly execute as long as queues **p** and **q1** are available. The second **seq** repeats, waiting on queue **q1** each time. The assumption here is that the first **seq** alters state that is shared with the second **seq** and this requires that data be received singly. In this example each repeat of the first **seq** will wait at the **waitprilock()** until the second **seq** executes the **priunlock()** at the end of processing the previous datum.

If the data sources are not queues, conditional statements may be used -

```

priority(pv);
queue(p, t);
static({s1, s2}, t);
input(l1, "log", .....);
input(l2, "log", .....);

when (l1)
  waitpri (pv, 1)
  p <- s1;
when (l2)
  waitpri (pv, 2)
  p <- s2;

```

Note that a **when** cannot be nested within a **waitpri**.

Waitpri is also useful for arbitration among memory accesses where the arguments to the inbuilt procedure call may or may not involve queues -

```

priority(pv);
rmemory(m, 1, atype, dtype);
static(addr1, atype);
queue(addr2, atype);
static(data, dtype);
value(1, "log");

when (1)
    waitpri (pv, 1)
        rmemorywrite(m, 0, addr1, data);
waitpri (pv, 2)
    rmemoryread(m, 0, addr2);

```

Waitpri may also be used for mutually exclusive control of a bus -

```

priority(pv);
selectvalue(bus, t);
static(s, t);
value(1, "log");
queue(p, t);

when (1)
    waitpri (pv, 1)
        bus <<- s;
waitpri (pv, 2)
    bus <<- p;

```

Here when **1** is true static variable **s** will be written to the bus for one clock cycle. In this case no queue waits are involved. When queue **p** is available it will be written to the bus when not pre-empted by the other write.

6.18 the petrinet construct

A Petri net contains a number of nodes known as *places*. Each place can hold one token (a *finite capacity* Petri net where each place has a capacity of 1). A single token can be taken from one or more source places and a single token put in one or more destination places as a result of a transition (an *ordinary* petri net). A transition can also additionally depend on a logical *predicate*. The Petri net must be *marked* to initially place tokens in some places.

A Petri net where only one place is marked and each transition has one source place and one destination place is a simple state machine.

The Petri net is effectively a module in that the generated logic is not in-line and the Petri net acts as a single queue destination for queue acknowledgement.

The syntax is illustrated by the following example -

```

petrinet (r) {
    initial q1, q2;
    q1, q2 -> q3, q4 when (a && (b != 0)) {i = 1; s <- queue1;};
    q3 -> q1 {l1 <- queue1;};
    q4 -> q2, q3 when (!l2);
}

```

Here the Petri net has four places, **q1** to **q4**. If these identifiers have already been created they must be made static and type log and must not have been previously assigned. If they have not been created **petrinet** will create them. Normal scope rules apply to these variables. Place variables cannot be assigned values except via the Petri net mechanism. The values of place variables can be used as for normal variables.

The first line contains (**r**) which is an optional (parenthesised) reset expression. If present a logical true value of this expression will reset the Petri net to its initial marking. This will override any concurrent transitions. The reset will not affect any transition code blocks that are currently executing.

The second line must be present and lists the place marking. In this case places **q1** and **q2** will initially contain tokens (will be true).

The following lines describe allowed transitions. A transition is a list of initial (source) places, a right single or double arrow, a list of final (destination) places, an optional predicate and finally an optional code block to be executed on transition. Each transition statement is terminated by a semicolon.

A single right arrow specifies a *weak* transition. The transition occurs when the source places each contain a token (are true) and the predicate, if present, is true. The transition results in the source places losing their tokens (all become false) and the destination places each receiving a token (all become true). With a weak transition a destination place may already have a token but since it can only hold one token it still contains a single token after the transition.

A double right arrow specifies a *strict* transition. This has the additional constraint over a weak transition that the transition can only occur if none of the destination places have a token.

When a transition occurs an associated code block will be simultaneously executed. If the code block takes longer than one clock cycle to execute then the transition associated with the code block cannot occur again until the code block has finished execution. This does not inhibit any other transitions, including transitions which affect the same places.

The optional predicate is an additional condition which must be met for a transition to fire. It may contain any logical expression and that expression may include place variables. Note that all place variables are created prior to analysing the transitions hence they can be used in transition predicates ahead of any other occurrences.

A transition may be duplicated if the predicates differ. This effectively ORs the predicates but allows each transition to execute a separate code block.

Predicates may contain queue reads but these must be used with caution. The **when** clause specifying the predicate executes every clock cycle and thus will execute any queue reads whenever those queues are available regardless of whether a transition is possible. It only makes sense to use a queue read in a predicate if the transition is always ready to fire when the queue becomes available.

Transition execution blocks may also contain queue reads (or writes). The Petri net as a whole is a module and so queue reads wherever they occur within the Petri net will be acknowledged individually. The Petri net will act as a single queue destination and a queue source will wait until a read has occurred within the Petri net before advancing to the next datum. Queue reads occurring anywhere within the Petri net, either in predicates or code blocks, will constitute a single read if they occur simultaneously or as separate reads otherwise.

A transition code block may contain any number of immediate statements but must contain exactly one in-line target statement. The latter of course may be a **seq**, **par** etc. which nest other target statements. Immediate statements within code blocks are executed sequentially from top to bottom of the Petri net construct. Care must be taken if a transition code block executes for more than two cycles. A transition cannot occur if an associated transition code block is executing (because of a previous occurrence of the same transition).

Chapter 7 - Memory, Buses, Priority Encoders, I/O, Floating Point, unbuffered queues

7.1 Memory access

Since memory cannot be accessed in parallel it is not regarded as an array and is treated as a serial device which is read or written through one or more ports. Memory is generally available in one of two forms

- combinatorial output
- registered output

Combinatorial output memory behaves when read as a combinatorial device - the read address is presented to the memory and the data at the referenced address appears at the output.

Registered output memory only reads on the clock edge following both presentation of the desired address and the execution of a read command. Registered output memory behaves like a static variable in that its value (the registered output) can be used anywhere (though accessed by means of a function) but that value reflects the most recent read operation.

Both types of memory employ clocked writes, where the address and data are presented to the memory and execution of a write command causes the write to occur.

Memory variables are created by the inbuilt procedures -

```
cmemory(name, ports, atype, dtype, init);
rmemory(name, ports, atype, dtype, init);
```

These create combinatorial output RAM and registered output RAM respectively. *name* is an identifier for the memory, *ports* (immediate) is the number of ports, *dtype* (immediate) is the data type, *atype* (immediate) is the address type and *init* is an optional initialisation array (immediate array variable or compound value). The initialisation data is a one or more dimensional immediate variable or structured constant whose first dimension is no larger than the depth of the memory. It may be smaller in which case the memory is zero padded. The types and structure of the initialisation variable or structured constant after the first dimension must match the memory data type. Immediate constants are padded to fit target elements but must not exceed the target width.

If the initialisation argument is omitted the memory will be initialised to all zero. Address width is constrained by the 3PL implementation and by the FPGA. The implementation philosophy has been to provide an unrestricted data width but limit the address width since combining memory elements for greater depth requires more logic and introduces more path delays than for greater width.

Where a memory variable is a parameter to a module, procedure or function only the first argument (the identifier) to *cmemory* or *rmemory* is needed.

Particular FPGAs may limit the generality of some operations, e.g. one port may be read-only. Within this limitation multiple reads and writes on a port are allowed and the address and data inputs will

be multiplexed, however there is no access arbitration and it is up to the application code to ensure exclusive access. Coincident accesses will result in data corruption.

Each memory port may have a separate clock. The port clock is determined by the current clock domain when the first read or write inbuilt procedure/function call is executed for that port.

7.1.1 Combinatorial output memory

A write is performed by use of the procedure *cmemorywrite*. This is called by -

```
cmemorywrite(m, port, addr, data);
```

m is the memory mode variable identifier, *port* (immediate) is the memory port, *addr* is the address and *data* is the data to be written. *addr* and *data* may be immediate, value, selectvalue, static or queue modes.

The data and address may be any type, including compound types. The types must match those given when the memory variable was created, but primitive widths may differ in which case padding or truncation will occur.

If the address or data arguments contain queue variables the write will wait on queue availability.

A read is performed by use of the function *cmemoryread*. This is called by -

```
cmemoryread(m, port, addr)
```

m is the memory mode variable identifier, *port* (immediate) is the memory port and *addr* (immediate, value, selectvalue, static or queue mode) is the address.

The address may be any type, including a compound type. This type must match that given when the memory variable was created, but primitive widths may differ in which case padding or truncation will occur.

The value of the function is the data output from the read. The type of the function return is the data type given when the memory variable was created

There may be calls to read and/or write a memory port at multiple points in the program, but the program logic must ensure that these calls are mutually exclusive (the compiler cannot determine that). Failure to observe this will result in data errors. Mutual exclusion can be enforced by the use of the **waitpri** control structure [section 6.17](#) p155.

If both the following conditions hold -

- a port has only a single call to *cmemoryread* and no calls to procedure *cmemorywrite*
- the address expression contains no queues

then the function output will continuously (statically) reflect the memory contents at the requested address. This is because for a single read access only the address input is permanently connected and hence the output remains continuously valid. If the port has more than one access point then the address must be multiplexed and is only connected for the execution cycle of the statement in which the function call appears. For a single read access expressions containing a call to this function can be passed as arguments to modules, procedures or functions or can be assigned to value variables.

If the above is not the case then the function output will only be valid in the execution cycle of the assignment statement in which the function call is part of the RHS expression, and expressions containing a call to this function must **not** be passed as arguments to modules, procedures or functions or be assigned to value variables (the compiler will detect these and flag a fatal error).

Note that **cmemoryread()** may appear in an expression assigned to a value mode variable and that variable may then be used in the RHS of a subsequent assignment statement. There are two limitations to the use of **cmemoryread()** in this context but -

- the value variable must be used at least once in the RHS of an assignment otherwise the control logic for the memory read will have no inputs

- the value variable must be assigned the expression containing **cmemoryread()** prior to its occurrence in the RHS of an assignment - post-assignment cannot be used in this instance

These cases are recognised during immediate execution are fatal errors.

A call to **cmemorywrite()** (section 9.14 p198) or **cmemoryread()** (section 10.17 p275) sets the clock domain for the memory port to the current clock domain. All further calls to either of these inbuilt procedures for the same port of the same combinatorial output memory must be in the same clock domain (or a fatal error results).

7.1.1.1 Xilinx implementation

In the case of Xilinx FPGAs combinatorial output memory is distributed CLB RAM. This can have one or two access ports. One logic slice can hold 1 bit x 32 words single port or 1 bit x 16 words dual port.

Combinatorial output memory can be declared with an address width ranging from 2 (4 words) to 9 (512 words) and with any data width.

Port 0 can be read or written (or both). Port 1 is read-only.

7.1.2 Registered output memory

A write is performed by use of the procedure *rmemorywrite*. This is called by -

```
rmemorywrite(m, port, addr, data);
```

m is the memory mode variable identifier, *port* (immediate) is the memory port, *addr* is the address and *data* is the data to be written. *addr* and *data* may be immediate, value, selectvalue, static or queue modes.

The data and address may be any type, including compound types. The types must match those given when the memory variable was created, but primitive widths may differ in which case padding or truncation will occur.

If the address or data arguments contain queue variables the write will wait on queue availability.

By default a write to a registered output memory port also writes the value into the output register, overwriting the previous value. ¹

A read is performed by use of the procedure *rmemoryread*. This is called by -

```
rmemoryread(m, port, addr);
```

m is the memory mode variable identifier, *port* (immediate) is the memory port and *addr* (immediate, value, selectvalue, static or queue mode) is the address.

The address may be any type, including a compound type. This type must match that given when the memory variable was created, but primitive widths may differ in which case padding or truncation will occur.

If the address argument contains queue variables the read will wait on queue availability.

A registered output memory port read latches the result in the output register of the memory port, so the result can be used after the read has been executed (not in the same execution cycle). The value of the memory variable port register can be accessed by the function

```
rmemoryout(m, port)
```

¹Some FPGAs support different behaviour in this regard which will be incorporated into 3PL at a later date.

where *m* is the memory mode variable identifier and *port* (immediate) is the memory port. This value is continuously valid.

A call to **rmemorywrite()** (section 9.64 p242) or **rmemoryread()** (section 9.63 p242) sets the clock domain for the memory port to the current clock domain. All further calls to either of these inbuilt procedures for the same port of the same registered output memory must be in the same clock domain (or a fatal error results).

There may be calls to read and/or write a memory port at multiple points in the program, but the program logic must ensure that these calls are mutually exclusive (the compiler cannot determine that). Failure to observe this will result in data errors. Mutual exclusion can be enforced by the use of the **waitpri** control structure section 6.17 p155.

7.1.2.1 Xilinx implementation

In the case of Xilinx FPGAs registered output memory is block RAM. This can have one or two access ports.

For SPARTAN 2, VIRTEX and VIRTEX E the memory formats supported are -

address width	data width	words
8	16	256
9	8	512
10	4	1024
11	2	2048
12	1	4096

For SPARTAN3, VIRTEX 2, VIRTEX 2 Pro or Virtex 4 the memory formats supported are -

address width	data width	words
9	36	512
10	18	1024
11	9	2048
12	4	4096
13	2	8192
14	1	16384

For Virtex 5, Virtex 6, series 7 the memory formats supported are -

address width	data width	words
10	36	1024
11	18	2048
12	9	4096
13	4	8192
14	2	16384
15	1	32768

Registered output memory will employ the smallest address width which will encompass the address type. The data may be any width, multiple blocks being used if the data width exceeds the block data width associated with the address width. Both ports have the same address and data widths (the physical devices allow these to differ).

Either port can be read or written (or both).

7.2 Using selectvalue variables

Selectvalue mode variables implement what is usually called a bus in hardware parlance. A *selectvalue* variable is combinatorial, that is it does not store data. A *selectvalue* variable can be assigned values in

a number of places in the source code but each assignment is only valid for one execution cycle. Multiple assignments must be mutually exclusive.

A *selectvalue* variable is created using the *selectvalue()* inbuilt procedure (or the *var()* inbuilt procedure). The current clock is not relevant when the variable is created.

A *selectvalue* variable is used where one of a number of values is to be selected.

Selection among alternative values can be done by multiple mutually exclusive assignments to static or queue variables, but there are two reasons why this might not be satisfactory. Firstly assignment to variables of these modes involves a delay - the data does not become available in the current execution cycle. Secondly the assignment logic for these variable modes requires significant logic resources and introduces data propagation delays in carrying out the assignment which may exceed the time available in the clock period. These delays increase with the number of separate assignment statements to the same primitive component.

Selection among alternative values can also be done by a *value* mode variable whose RHS expression carries out the selection, either by primitive boolean logic or by nested conditional operators (?:). This avoids the one or more execution cycle delay inherent with static or queue variables, since value variables are combinatorial, but the selection logic again requires significant logic resources and introduces data propagation delays which grow with the number of alternatives.

A bus is one or more signal lines driven from one selected source (usually among others). If two such assignments occur in the same execution cycle then that is an execution error and the result is undefined. If in any particular execution cycle no assignment is made to the variable then its value is undefined.

CLB logic is used to implement *selectvalue* variables, but if the device contains 3-state buffers then these can be used instead by setting the *selectvalue* variable attribute "threestate" true². Signal propagation delays through these buffers are not large and are independent of the number of parallel buffers, but in practice it has been found preferable to use CLB logic for minimum delay.

A *selectvalue* assignment only has effect for one execution cycle, but it can be extended indefinitely by simply controlling it by a conditional (*when*) or a loop (*while* or *do while*). Although a *selectvalue* variable is not itself clocked it must have an associated clock domain. The variable output is defined when assigned a value. That assignment occurs within an execution thread for which there is an associated clock. Similarly the value of the *selectvalue* variable will be used in expressions which must be evaluated in that same clock domain. A *selectvalue* variable has a clock domain established on the first assignment or evaluation encountered during immediate execution. All further assignments or evaluations must be in the same clock domain.

The following example selects from among 4 inputs which are a mixture of variables and constants -

```
str(dtype, "uint:8");
var(busd, selectvalue, dtype);
static(a, dtype);
queue(b, dtype);
static(count, "uint:3");
value({l1, l2}, "log");

when (l1)
    busd |- a;
when (l2)
    busd |- 3;
busd |-b;
while (count != 0) par {
    busd |- 0;
    count--;
}
```

For any execution cycles when value variable *l1* is true variable *busd* will have the value of static variable *a*. For any execution cycles when value variable *l2* is true variable *busd* will have the value constant 3. For

²The use of dedicated 3-state buffers and associated routing resources can improve utilisation of resources in the FPGA, but these specialised resources are not nearly as numerous as common logic blocks and hence they may be exhausted by over enthusiastic use of buses.

any execution cycles when queue variable *p* is available variable *busd* will have the value read from that queue. For any execution cycles when static variable *count* is non-zero *busd* will have the value constant 0 and *count* will be decremented. Any of these assignments may execute for a number of consecutive execution cycles, though in the case of the queue each execution cycle would involve a different datum. We assume here that these assignments will be mutually exclusive.

A bus may be used to share data of a number of different types. in that case the bus could be created as type *bits* and all assignments cast to that type, e.g.

```
str(btype, "bits:8");
var(busd, selectvalue, btype);
static(a, "int:6");
value(b, "uint:4");

when (...)
    a |← cast(a, dtype);
when (...)
    a |← cast(b, dtype);
```

Note that since the RHS of each assignment has fewer bits than the LHS the values will be padded, in the signed case with the sign bit.

7.3 Using priority variables

Priority encoders are implemented in 3PL by using variables of mode *priority*. A priority variable must be created using the *priority()* or *var()* inbuilt procedures. The type must be a one-dimensional array of type *log*. The array may be larger than required, unused array members being ignored.

Priority mode variables are used by control structures **waitpri** and **waitprilock** - section 6.17. These control structures probably cover most applications of priority mode variables and are easier to use than the methods described below. The following explanation and examples are provided for an understanding of how these variables work.

Although a *priority* variable is not itself clocked it must have an associated clock domain. Inputs assigned to the priority variable will have clock domains (they may have a null clock domain, but that would be unusual) and these must all have the same clock domain. Similarly the value of the priority variable will be used in expressions which must be evaluated in that same clock domain. A priority variable has a clock domain established on the first assignment or evaluation encountered during immediate execution. All further assignments or evaluations must be in the same clock domain.

A priority variable is combinatorial in that its inputs are mapped to its outputs with no stored state, and the outputs are a function of the inputs within the current execution cycle (after a propagation delay).

Inputs to the decoder are simply assigned in the same way as value variables using the `:=` assignment statement. This connects the RHS value to the encoder input selected by the subscript. The priority outputs are continuously available as the priority variable values. Inputs may be assigned in any order. The priority encoder is created at the completion of 3PL program execution hence if more than one assignment is made to any input, by default the last one will be used. This can be changed by setting attribute "multiplein" to true in which case all assignments will be collected and ORed together on completion of immediate execution to form the input to that priority array member.

Members of the priority variable array whose values are used must be assigned. Partial use of a priority variable array has been designed to cater for the case where the number of members is not known prior to immediate execution of the 3PL program. Priority variables are usually used for arbitration among multiple accesses to a resource where mutual exclusion is required but not observed.

For a first example suppose we wish to take the outputs of four queues and write them to another queue. If more than one queue is available we want to take them in some priority order. We could create a module waiting on each source queue, each one writing to the destination queue. Arbitration could be carried out by a global priority variable.

```

queue({sq1, sq2, sq3, sq4}, type);
queue(dq, type);
priority(pri); // default type is "[20]log"
int(pri_index);
.
.
write_queue(sq1);
write_queue(sq2);
write_queue(sq3);
write_queue(sq4);
.
.
module
write_queue (queue(p)) {
    pri[pri_index] := <p;
    sync when (pri[pri_index]) {
        dq <-- p;
    }
    pri_index++;
}

```

Each immediate call to *write_queue* creates a module. The first (immediate) statement assigns the source queue availability to the next array member of the priority variable. The second (target) statement reads the source queue when the priority variable array member is true. The final (immediate) statement increments the priority variable index ready for the next call. The source queue priorities will be highest for *sq1* to lowest for *sq4*. If we wished to specify the priority on each call we could pass it as an argument

```

queue({sq1, sq2, sq3, sq4}, type);
queue(dq, type);
priority(pri);
.
.
write_queue(sq1, 5);
write_queue(sq2, 0);
write_queue(sq3, 2);
write_queue(sq4, 9);
.
.
module
write_queue (queue(p), int(priority)) {
    pri[priority] := <p;
    sync when (pri[priority]) {
        dq <-- p;
    }
}

```

7.4 Input/output

7.4.1 input and output attributes

Variables of mode **input**, **output** and **clock** have associated attributes. The attributes are such things as package pin locations (loc), timing names (tnm) and output current drive (drive). These are set as trailing arguments to inbuilt procedures **input()** (section 9.39 p218), **output()** (section 9.47 p224) or **clock()** (section 9.10 p194).

When attributes are passed to **input()**, **output()** or **clock()** as trailing arguments they may be in the form attribute=value, e.g.

```
input(vin, type(data), loc=mezz_pins[3..7], tnm="IREG");
```

Alternatively the trailing arguments may be maps of attributes, e.g.

```
var(inattr, "map", {loc=mezz_pins[3..7], tnm="IREG"});
input(vin, type(data), inattr);
```

The map variable could of course have entries assigned rather than initialised, or could be a combination of both, e.g.

```
var(inattr, "map", {loc=mezz_pins[3..7]});
inattr["tnm"] = "IREG";
input(vin, type(data), inattr);
```

The use of a map means that common attributes may be collected in a map which can be used for multiple input or output mode variables but attributes such as **loc** which are individual to particular calls to `input()` or `output()` can be passed separately, e.g.

```
var(inattr, "map", {tnm="IREG", ifddelay="none"});
input(vin1, type(data), loc=mezza_pins[3..7], inattr);
input(vin2, type(data), loc=mezzb_pins[10..14], inattr);
```

Note that the inbuilt procedure **attributes()** can be used to assign or change the attributes of a variable. With the exception of pad locations most attributes are only used when generating logic for input, output and clock mode variables at completion of immediate execution so they can be changed at any time during immediate execution. An exception is the **direct** attribute for output mode variables which is used whenever the variable is assigned.

7.4.2 input connections

Input connections are made by creating an input mode variable using inbuilt procedure **input()** (section 9.39 p218), e.g. -

```
iopins(pins, "[4]str", {"A1", "B1", "B2", "B3", "C1", "C2", "D1"});
input(vin, "uint:4", loc=pins[2..5]);
```

pins is created as a string array of pin names from which four are selected, matching the data width.

The first argument to **input()** is the identifier of the created input mode variable.

The second argument is the type of the input mode variable.

The third and following arguments are optional and are attributes to be set. The attributes may be passed as attribute=value pairs or within one or more maps. The maps have the attribute names as keys and the associated values are the attribute values. Attribute values may be integers, strings or booleans but these are all converted to upper case strings. If an attribute is given a single value then this attribute value will be applied to every pin. Alternatively the attribute value may be a one-dimensional array whose dimension matches the number of pins. The array may be within a variable or it may be a compound constant in braces, e.g.

```
var(a1, "[4]str", "LVC MOS33");
input(vin, "uint:4", loc=pins[2..5], iostandard=a1);
```

or

```
input(vin, "uint:2", loc=pins[2..3], iostandard={"LVC MOS33", "LVC MOS25"});
```

The **loc** attribute must be supplied and the number of pins must match the number of bits in the data type. The attribute can be any nested type as long as the primitives contained therein are only "str". The pins are in order LSB to MSB of the target variable. If the input is differential then the pins are assigned in successive pairs.

By default all input buffers with an associated flip-flop have a small delay inserted to avoid a hold constraint on the clocked input data. This delay can be removed by setting the attribute **ifddelay** to 0. If there is no input flip-flop extra delay can be inserted in the signal path by setting attribute **ibufdelay**.

A timing group identifier attribute **tnm** can be set to a string value to put the input pads in a timing group. This can be used to set up a timing constraint between the input and some other group. For example to remove the input delay and set up a 10ns timing constraint between the input and a clock we could write -

```
iopins(pins, "[]str", {"A1", "B1", "B2", "B3", "C1", "C2", "D1"});
input(vin, "uint:8", loc=pins[2..5], tnm="VPAD");
maxdelay("VPAD", "CLK", "10.0");
```

where **CLK** is the timing group identifier for components driven by the clock in question. A timing constraint identifier will be generated by concatenating the two timing group names separated by an underscore, e.g. **TS_VPAD_CLK** in this case.

If an input register is required the input variable can be assigned to a static variable on every clock cycle, e.g.

```
iopins(pins, "[]str", {"A1", "B1", "B2", "B3", "C1", "C2", "D1"});
input(vin, "uint:4", loc=pins[2..5]);
static(v_del, "bits:4"); attributes(v_del, continuous=true, iob=true);
v_del <- vin;
```

The **continuous** attribute removes the assignment from any execution thread and connects the static variable (register) clock enable to logic high. The **iob** attribute tells the place-and-route software that the flip-flops comprising the static variable (register) should be placed in I/O blocks along with their associated pads and input buffers

Assigning an external input variable to a static variable in this way is the preferred method for handling external input. The static variable will be clocked by a clock synchronised with the external data, i.e. the external clock must be the current clock when the static assignment is encountered during immediate execution. It is not necessary to place a timing constraint between the input pad and the static variable since the routing is the fastest available, being simply the direct connection inside the I/O block. A timing constraint could be applied just to verify that the propagation delay is sufficiently short to meet specifications. Note that Xilinx provide an optional (but default) delay in external data connections to ensure that there is a zero hold requirement on input flip-flops.

It is difficult to decide on a specification for an acceptable input timing constraint. The external data source has changed the bus data some time after the clock. The data and clock arrive at the FPGA with perhaps some skew between them (the data change would usually be expected to follow the clock edge after some delay). The clock arrives at the FPGA and, if a DLL is used, is distributed with very little delay within the FPGA. The clock then clocks the I/O block flip-flops.

If external data input is to be used directly without being assigned to a static variable then unless the clock frequency is very low (say less than 5MHz) timing constraints should be used. The following is an example where the input data is passed through an adder before being assigned to a static variable.

```
iopins(pins, "[]str", {"A1", "B1", "B2", "C3", "C4", "D1", "D2", "E1"});
input(data_in, "bits:8", loc=pins, tnm="DPADS");
useclock(c);
static(v, "bits:8"); attributes(v, continuous=true);
v <- data_in + 3;
maxdelay("DPADS", "C", 20.0);
```

Here "C" is the timing group identifier for clock *c*, the internal clock derived from the external clock to which the external data is synchronised. The timing group identifier can be specified by supplying a

"tnm" attribute, otherwise it is generated automatically and is simply an upper case version of the clock identifier.

The timing constraint says that the data paths from the input connections through the adder up to the static variable (actually to any logic clocked by clock *c*, which will just be the flip-flops making up the static variable) must be 20ns or less (including flip-flop setup time).

7.4.3 output connections

Output connections are made by creating an output mode variable. As an example -

```
iopins(pins, "[str]", {"A16", "B15", "A17", "C17"});
value(v, "bits:4");
value(enable, "log");
output(vout, type(v),, loc=pins);
when (enable)
    vout | $\leftarrow$  v;
```

connects the 4 bit value *v* to the four pins specified. The logical value variable **enable** drives the output when it is true and puts it in the high-impedance state when false.³ If we do not want 3-state output we could simplify the code to -

```
iopins(pins, "[str]", {"A16", "B15", "A17", "C17"});
value(v, "bits:4");
output(vout, type(v),, loc=pins);
vout := v;
```

or more simply

```
iopins(pins, "[str]", {"A16", "B15", "A17", "C17"});
value(v, "bits:4");
output(, v, loc=pins);
```

The inbuilt procedure **output()**([section 9.47](#) p224) has at least four arguments. The first is a variable identifier, if any, for the created output mode variable - it need not have an identifier if the third argument is provided.

The second argument is the type, which also can be omitted if the third argument is provided.

The third argument is an optional unconditional source for the output. This can be used to avoid an explicit assignment to the output mode variable.

The fourth and following arguments are optional and are attributes to be set and are passed in asimilar way to those described above for input mode variables.

Note that if the source variable to output() is static its value should preferably not be referenced elsewhere as that will preclude its implementation as pad flip-flops, internal flip-flops being used instead. This will adversely affect the delay between clock edge and external data output and will probably also lead to some data skew. To avoid this 3PL will automatically duplicate any flip-flops whose output are connected to output buffers and also to other logic. One of the duplicates drives the output buffer and the other the internal logic. Clocks, clock enables and sets/resets are connected in parallel. This duplication can be suppressed by adding a **noshadow** attribute with a true value.

The preferred scenario is to drive outputs from a static variable. The flip-flops implementing the static variable will be located in the I/O blocks as long as those flip-flops are not referenced anywhere else in the program. This arrangement means that the static variable is changed on the clock edge and the data propagation time to the output pads via output buffers is minimal. It is not necessary to impose

³In the example it would be preferable if variables **enable** and **v** were static, each with the 'iob' attribute set to true, and variable **vout** with 'direct' attribute true. That would result in the static variables, or shadow versions thereof, being placed in IO blocks rather than in CLBs. This provides the smallest clock to output delays.

timing constraints from the clock to the output PADS since the propagation time cannot be changed, but timing constraints could be used to simply verify that the data propagation time is low enough.

An example is-

```
iopins(pins, "[str", {"A1", "B1", "B2", "C3", "C4", "D1", "D2", "E1"});
static(data_out, "bits:8"); attributes(data_out, iob=true);
output(, data_out, loc=pins, tnm="PADS");
```

which creates a static variable *data_out* and connects its output to the 8 FPGA package pins specified. The output source is supplied via the third argument so the output will be directly driven i.e. 2-state not 3-state. The drive current and slew rate will be the default values and the output pads will be added to a timing group called "PADS". The **iob** attribute tells the place-and-route software that the flip-flops comprising the static variable (register) should be placed in I/O blocks along with their associated output buffers and pads.

The following example illustrates a 3-state output -

```
iopins(pins, "[str", {"A1", "B1", "B2", "C3", "C4", "D1", "D2", "E1"});
static(data_out, "bits:8"); attributes(data_out, iob=true);
value({l1, l2}, "log");
output(o, type(data_out),, loc=pins, tnm="PADS");
when (l1 && l2)
  o |<- data_out;
```

Here the output drive is controlled by logical static variables *l1* and *l2* - if both are true the output will be driven, otherwise the output will be high-impedance.

If the external output is a selectvalue variable or is a value variable at the output of combinatorial logic then the timing relationship between clock driven data sources and the output pads may need to be constrained. The following code illustrates this -

```
useclock(c);

iopins(pins, "[str", {"A1", "B1", "B2", "C3", "C4", "D1", "D2", "E1"});
value(data_out, "bits:8");
static(v, "bits:8");
output(, data_out, loc=pins, tnm="PADS");

v <- .....;
data_out := v + 3;
maxdelay("C", "PADS", 20.0);
```

The value variable *data_out* is created and made an external output with pads entered into a timing group called "PADS". The variable is assigned a combinatorial expression whose operands are a static variable whose clock domain is *c* and an immediate (constant). A timing constraint is generated which limits the time from any logic clocked by clock *c* to the output pads to 20ns. The only flip-flops (or other clocked state devices) that have combinatorial paths to these output pads are the ones that constitute the static variable *v*.

7.4.4 bi-directional connections

Bi-directional I/O is simply a combination of input and output. The input and output variables are separate and are connected to the pads by separate calls -

```
iopins(pins, "[str", {"A16", "B15", "A17", "C17"});
type(t, "uint:4");

input(vin, t, loc=pins);
.
.
```

```

value(enable, "log");
value(data, t);
output(vout, t,, loc=pins);
when (enable)
    vout | $\leftarrow$  data;

```

assignments or initialisation for **enable** and **v** are not shown.

Again it is preferable to use registers on input and output -

```

iopins(pins, "[]str", {"A16", "B15", "A17", "C17"});
type(t, "uint:4");

static({in, out}, t); attributes({in, out}, iob=true);

input(vin, t, loc=pins);
in <- vin;

value(enable, "log");
output(vout, t,, loc=pins);
when (enable)
    vout | $\leftarrow$  out;

```

If we wanted to change the above example to include timing constraints it might appear as follows -

```

useclock(c);

iopins(pins, "[]str", {"A16", "B15", "A17", "C17"});
type(t, "uint:4");

static({in, out}, t); attributes({in, out}, iob=true);
attributes(in, continuous=true);

input(vin, t, loc=pins, tnm="DPADS");
in <- vin;

value(enable, "log");
output(vout, type(v),, loc=pins);
when (enable)
    vout | $\leftarrow$  out;

maxdelay("C", "DPADS", 10.0);
maxdelay("DPADS", "C", 15.0);

```

The tnm attribute could be applied to either the input mode variable or the output mode variable (if both, only one constraint file entry will be generated).

This code applies an output constraint of 10ns and an input constraint of 15ns. Note that clock variables that have been used as clock domains have clock timing group names assigned to them automatically. These timing group names are derived from the clock identifier forced to upper case, thus a clock **clk** would have a timing group identifier **CLK** associated with it.

7.4.5 a note on timing constraints

The examples above have applied timing constraints between I/O pads and state devices such as flip-flops and memories using a specified clock (via the clock timing group identifier).

A clock period (or frequency) is specified as an attribute to the clock variable. A timing group identifier may also be specified for a clock as an attribute, otherwise a timing group identifier will be generated automatically using an upper case version of the clock variable identifier. All state devices to which this clock is connected become members of the associated timing group. The clock period is applied as a

constraint for all data paths between devices in that timing group. A timing constraint for data paths between devices in different timing groups can be applied by using the library procedure *maxdelay*. This library procedure is in the header file "xilinx/common.3pl". It is possible to place a static variable in another timing group so that different timing constraints can be applied. This is done by setting the attribute *tnm*. We could change the bi-directional interface code above by -

```
iopins(pins, "[str", {"A16", "B15", "A17", "C17"});
type(t, "uint:4");

static({in, out}, t);
attributes(in, tnm="groupo", iob=true);
attributes(out, tnm="groupi", iob=true);

input(vin, t, loc=pins, tnm="DPADS");
in <- vin;

value(enable, "log");
output(vout, type(v),, loc=pins);
when (enable)
    vout |~ out;

maxdelay("DPADS", "group0", 15.0);
maxdelay("group1", "DPADS", 10.0);
```

Now the timing constraints are explicitly between the "PADS" group and each of the static variables.

7.4.6 the specification of I/O voltage standard

The examples so far have not specified the I/O voltage standard. The place-and-route software seems to accept I/O pads and buffers with no voltage standard specified, but it is preferable to include it. The voltage standard is another attribute which can be set for a clock, input or output mode variable.

The attribute is "iostandard" and common values are "LVCMOS33" or "LVCMOS25" for 3.3v and 2.5v CMOS respectively.

Another attribute commonly used when setting up timing constraints between I/O pads and internal state devices is "tnm".

The attributes arguments can often be rather long. For example they might appear as -

```
loc=mezz_pins[3..7], iostandard="LVCMOS33", tnm="OREG"
```

This is somewhat verbose. A more serious difficulty is that the I/O standard is supplied explicitly in each case where it is usually really a property of the associated I/O pin. To make this process more convenient a number of related library procedures and functions have been provided in library "xilinx/common.3pl".

I/O pins are usually specified in string or string array variables, e.g.

```
iopins(bpins, "[str", {"A1", "B1", "A2", "C1"});
```

but any type can be used as long as all the primitives contained therein are type "str".

The same can now be done by calling library procedure **iopins()** [subsection 11.2.2.19](#) p388 with an additional argument supplying the I/O standard, e.g.

```
str(v33, "LVCMOS33");
iopins(global.apins, "[str", {"A1", "B1", "A2", "C1"}, iostandard=v33);
iopins(bpins, "[2][2]str", {"A3", "B4"}, {"A4", "C5"}, iostandard=v33);
iopins(cpin, "str", "D1", iostandard=v33);
```

These calls will create string or string array variables exactly as before but in addition a map is created in which each pin designation is a key whose value is the voltage standard string.⁴

Library procedures `clock()`, `input()` and `output()` call inbuilt procedures `clock()`, `input()` or `output()` respectively to create the variable and then take the supplied attributes, generate constraints, then add the attributes to the variable. Some of the attributes are used by 3PL in code generation and do not result in constraints being generated. Those that cause constraints to be generated usually are not further used by 3PL but are still stored in the variable for any possible later reference (they can be examined by inbuilt functions `attributes()` (section 10.10 p271 and `attribute()` (section 10.11 p271)).

Where a pad is bi-directional and a voltage standard is specified for both the input and the output only one constraint for this will be generated.

Note that the inference of I/O standard is done from the actual pin designation strings, not from the variable containing these string, thus copies of sub-arrays, or even string constants, can be used as long as the pins have been originally entered into the I/O standards map by using `iopins()`.

7.4.7 differential input and output

The following code illustrates differential input for 2-bit data -

```
iopins(pins, "[str", {"A16", "B15", "A17", "C17"});
type(t, "uint:2");
static(in, t);
input(in, t, loc=pins, differential=true);
```

For each bit two successive pins are used from the `loc` attribute. The first of the pair is for the +ve input and the second for the -ve input.

7.4.8 clock input

For an input clock a clock mode variable is created using library procedure `clock()` (for internal clocks the inbuilt procedure `input()` can be used instead).

Clock mode variables represent all clock signals. A clock variable driven by a global clock buffer can be used as the clock domain for logic. Other clock variables represent the input clock signal from a pad and intermediate clock signals connecting clock managers and clock buffers. These are not used for clock domains since they are not connected to clocked devices such as flip-flops or RAMs.

The following code creates a 100MHz input clock signal -

```
iopins(c100_pin, "A16");
clock(c100_in, frequency=100.0e6, loc=c100_pin, ibufg=true);
```

The frequency attribute is set to 100MHz. The `ibufg` selects whether an IBUFG input clock buffer is to be used, the alternative being an IBUF⁵.

In an FPGA the input clock can only be used as an input to either a DLL/DCM or a global clock buffer. To go directly to a clock buffer (and hence omit any phase correction) the library procedure `clockbufg()` is used. This library procedure is defined in one of the shared platform libraries, `xilinx/xc2sv.3pl` or `xilinx/xc3s2v4v.3pl`.

```
iopins(c100_pin, "A16");
clock(c100_in, frequency=100.0e6, loc=c100_pin, ibufg=true);
clock(global.c100, frequency=frequency(c100_in), tnm="C100");
clockbufg(c100 <- c100_in);
```

⁴The use of a string variable for the voltage standard argument rather than a string constant not only shortens the line but also means that a single value is stored (pointed to) for each entry in the map. If the same constant is used repeatedly each will be a separate value which will use more memory.

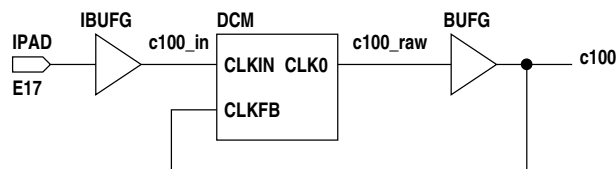
⁵It is not clear if this attribute is necessary as the Xilinx software may automatically use an IBUFG on a clock input pad.

The clock **c100** can now be used as a clock domain. Note that it is made global to ensure that it can be accessed anywhere (of course clock domains can be in a restricted scope if desired). A timing group name "C100" (same as clock variable identifier but with alphabetic characters changed to upper case) is automatically defined in the constraints file and all logic clocked with this clock will be included in that timing group. The clock frequency attribute will be set for the timing group in the constraints file from which the place-and-route software will apply a timing constraint.

In the above example it is not necessary to set the frequency attribute of c100 as the procedure `clockbufg()` will set the frequency attribute of its output argument to that of its input argument.

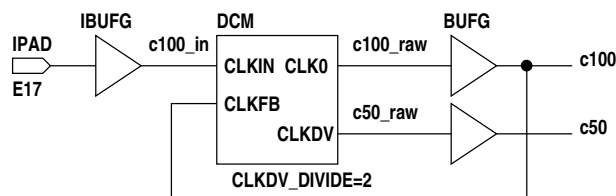
To align the internal clock phase with the input clock a digital clock manager is used. The following code implements one of these.

```
iopins(c100_pin, "E17");
float(c100_freq, 100.0e6);
clock(c100_in, frequency=c100_freq, loc=c100_pin, ibufg=true);
clock(global.c100, frequency=c100_freq, tnm="C100");
clock(c100_raw, frequency=c100_freq);
clkdcn(clk0=c100_raw <- clkkin=c100_in, clkfb=c100);
clockbufg(c100 <- c100_raw);
```



To generate a divided clock as well -

```
iopins(c100_pin, "E17");
float(c100_freq, 100.0e6);
float(clock_ratio, 2.0);
clock(c100_in, frequency=c100_freq, loc=c100_pin, ibufg=true);
clock({global.c100, c100_raw}, frequency=c100_freq);
clock({global.c50, c50_raw}, frequency=c100_freq/clock_ratio);
attributes(c100, tnm="C100");
attributes(c50, tnm="C50");
clkdcn(
  clk0=c100_raw,
  clkdv=c50_raw
  <-
  clkkin=c100_in,
  clkfb=c100,
  clkdv_divide=tostring(clock_ratio)
);
clockbufg(c100 <- c100_raw);
clockbufg(c50 <- c50_raw);
```

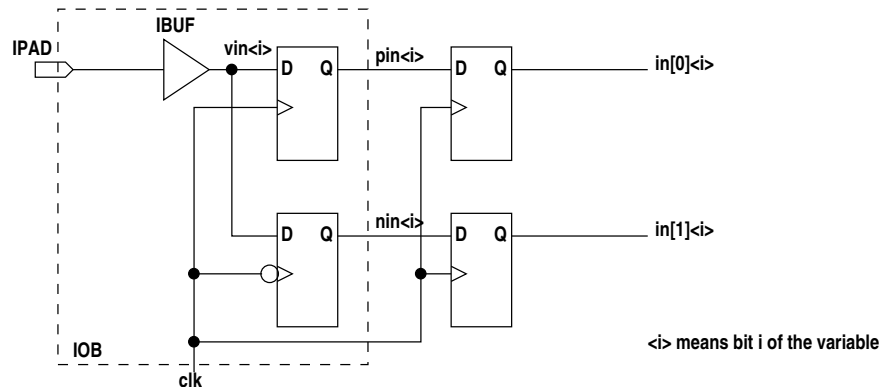


Note that in the above example the tnm attributes have been assigned using inbuilt procedure `attributes()` rather than as an argument to `clock()`.

In a CPLD the input clock can be used directly as a clock domain.

7.4.9 DDR input

For double data rate input two flip-flops are used in each IOB, one clocked directly and the other clocked via an inverter. The output of the -ve edge flip-flop must then be clocked on the next +ve edge into another flip-flop so that both bits are available in one clock domain.



At present the Xilinx place-and-route tools accept the following code, but a more direct approach will be implemented shortly.

```

iopins(pins, "[]str", {"A1", "A2", "A3", "A4"});
negclock(nclk, clk);
input(vin, "bits:4", loc=pins);
static(pin, "bits:4"); attributes(pin, continuous=true);
static(nin, "bits:4"); attributes(nin, continuous=true);
static(in, "[2]bits:4"); attributes(in, continuous=true);

useclock(clk);
pin <- vin;

useclock(nclk);
nin <- accept(vin);

useclock(); // back to previous clock domain, clk
in[0] <- pin;
in[1] <- accept(nin);

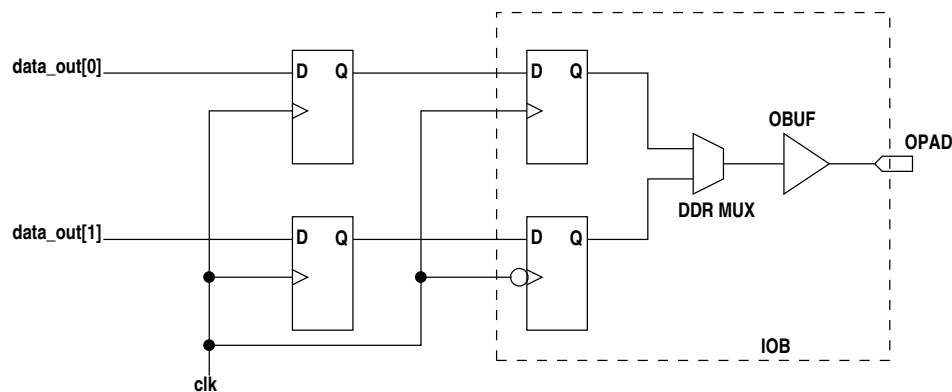
```

When a negative clock is created using inbuilt procedure **negclock()** the positive clock is still used but it is flagged such that an inverter is inserted at every logic element which uses it. Clock inverters are built in to all Xilinx clocked elements.

Note the use of `accept()` to allow the assignment despite the different clock domains. It is not clear if the Xilinx place-and-route tools will generate and apply a half-period constraint between flip-flops clocked on opposite clock edges. These paths contain no logic, only routing, and hence tight timing constraints can be expected to be easily met.

7.4.10 DDR output

For double data rate input two flip-flops are used in each IOB, one clocked directly and the other clocked via an inverter. The outputs of the two flip-flops are multiplexed to the output buffer and pad (using the positive clock I assume). The data must be clocked from two preceding flip-flops clocked with the positive clock.



At present DDR output is not implemented, but will be shortly.

7.5 Floating point

Immediate floating point values are stored in 64 bit IEEE 754 single precision form with a biased exponent of 8 bits and a mantissa of 23 bits. The usual arithmetic can be used with these to form expressions.

Target floating point values are stored as a struct with fields for the sign, biased exponent and mantissa. Thus target floating point types are not actually primitive.

Apart from the creation of floating point variables target floating point is not explicitly supported by 3PL. Since target floating types are represented by associated structs they can be assigned to variables and passed as arguments to parameters. Since they are handled as structs the following limitations apply

- Target floating point variables cannot be used in expressions. This restriction is imposed because the complexity of floating point operations dictates that pipelining should be used to achieve a reasonable throughput. Since operators in expressions must be capable of executing in one cycle if the operands are available, internal pipelining cannot be used by expression operators. If target floating point operators were implemented (which is easy enough to do) a very low clock frequency would have to be used to allow time for complete operations to be executed. The alternative adopted is to use library modules to carry out target floating point operations, these being pipelined to sustain a high throughput at the cost of latency.
- Assignment of target floating point values cannot perform size adjustment and hence the left and right sides must match exactly in type. Where floating point sizes are to be changed library functions exist to do this.
- Floating point parameters in modules, procedures or functions must either match their arguments exactly in type or else must have their type omitted, in which case they will be given the type of the argument.
- Where immediate floating point arguments are passed to target floating point parameters the immediate values are converted to target floating point values with constant fields. These are structs of equivalent type "float:23.8". The parameters must either be exactly this type or else must have the type omitted.

For target floating point arithmetic operations there is a library, **float.3pl** (see [section 11.16](#) p530), which contains modules implementing pipelined add, subtract, multiply and divide.

7.6 Unbuffered queues

An unbuffered queue is one whose depth has been set to zero (by 'depth=0' as an attribute passed to inbuilt procedure queue() or using the **attributes()** [section 9.7](#) p192 inbuilt procedure to set it afterwards). For queue creation see **queue()** [section 9.57](#) p234. These are used to synchronise two execution threads, optionally passing a datum in the process. A thread writing to an unbuffered queue will wait until a read is executed on that queue and vice versa. The situation is symmetrical with respect

to the reader and writer, each waiting on the other. The read and write will execute on the same clock cycle.

Unbuffered queues can only be synchronous, not asynchronous.

Unbuffered queues cannot be initialised.

An unbuffered queue may be of type "null" if no data is to be transmitted.

An unbuffered queue cannot be read or written within -

- a **waitpri** statement.
- a **waitprilock** statement.
- a **sync** statement.
- a **sync when** statement.

An unbuffered queue can be read or written within other control structures.

An unbuffered queue can not be an argument to the ?: ternary conditional operator.

A waitprilock can be used to avoid simultaneous writes to an unbuffered queue as long as an unbuffered queue access is not contained within the subject statement of the waitprilock, for example -

```
queue(ubq, "uint:8", depth=0);
static(v, "uint:8");
priority(p);

seq {
  .
  .
  waitprilock(p, 1)
    ; // null subject statement
  ubq <<- 3;
  priunlock(1);
  .
}
```

The above restrictions are required because an unbuffered queue uses signalling between reader and writer which is not compatible with the signalling used for buffered queues. When a read or write to an unbuffered queue is initiated that raises the available flag for the complementary write or read. When the complementary write or read occurs the opposite availability flag is briefly raised and then the read and write transaction is completed. If both the read and the write to an unbuffered queue were waiting on queue availability before proceeding then deadlock would occur since availability requires the initiation of the other side of the transaction. The second read or write operation of a transaction pair could employ normal synchronisation with other queues, but to allow this would provide too many opportunities for deadlock and hence the above restrictions have been imposed and enforced.

An unbuffered queue can have its read or write availability tested using the > or < unary operators. Read availability will be true when a write is pending. In the case of a chain of assignments containing unbuffered queues read availability will be true when all upstream assignments are ready to execute and have availability themselves. When an unbuffered queue has read availability its value can be examined using the (? unary operator). Write availability will be true when a read is pending. In the case of a chain of assignments containing unbuffered queues write availability will be true when all downstream assignments are ready to execute and have write availability themselves.

An assignment statement may have an unbuffered queue on the LHS and multiple buffered and unbuffered queues on the right hand side. The assignment will wait for the RHS buffered queues to be available and the RHS unbuffered queues to have a write pending. Reads from multiple unbuffered queues in an expression will wait until all unbuffered queues have writes able to proceed.

An unbuffered queue may be read in more than one place and may be written in more than one place. In the case of multiple writers they must be mutually exclusive. waitprilock() statements may be used to ensure this. Any pending reads of an unbuffered queue will wait until a write to that queue after which they will continue along their execution threads. Unlike buffered queues the modules in which unbuffered queue reads are situated are not significant, i.e. unbuffered queues behave as though the 'ignore_modules' attribute was set (the 'ignore_modules' attribute itself is ignored).

In general unbuffered queues may be passed to modules, procedures and functions. Unbuffered queues cannot however be passed to inbuilt procedures which produce inline target code such as `cmemorywrite()`, `rmemoryread()` and `rmemorywrite()`.

Unbuffered queues are intended for synchronisation of execution threads usually, but not necessarily, within loops. Such loops may be implicit as with top level **seq** or **par** blocks or may be explicit (target **while** or target **do while**). The following illustrates a simple case -

```
queue(ubq, "uint:8", depth=0);
static(v, "uint:8");

seq {
  .
  .
  ubq <- 3;
  .
  .
}

seq {
  .
  .
  v <- ubq;
  .
  .
}
```

The first sequential thread to reach the queue access will wait for the other. After the write and read take place simultaneously the two threads will continue sequential execution and then repeat.

The master-slave concept described earlier can be used to implement an initiation and acknowledgment scheme. A master write can be issued to initiate some process. The slave process waits using the read availability operator. When this returns true the slave can examine the data using the examine operator and then execute the controlled operation. Meanwhile the write will hang waiting for availability. When the slave side has finished it can then execute a read to release the master write. If data is to be transferred in the other direction on acknowledgement then the master would be a read and the slave a write. An example of the former is given below -

```
queue(ubq, "uint:8", depth=0);

// master thread
static(status, type(ubq));
seq {
  ubq <- ...;           // initiate the slave process
  .                     // continue the thread
  .
  .
}

// slave thread
when (<ubq) seq {
  branch (?ubq) { // perform operations based on the examined value
    case .....
    .
    .
  }
  null <- ubq;         // do a read to release the write in the master thread
```

By checking read availability the slave thread detects the write by the master thread. It then examines the written value rather than reading it, thus leaving the master write suspended. Finally the slave thread releases the master thread by doing a read. Since the data value is not now needed it can be dumped into null.

Note that this mechanism is valid for chains of unbuffered queue assignments and applies to any point along that chain.

Where an acknowledge status is to be returned the master reads and the slave writes -

```
queue(ubq, "uint:8", depth=0);

// master thread
static(status, type(ubq));
seq {
  status <- ubq; // initiate the slave process waiting for return
  .             // do something with the returned status
  .
  .
}

// slave thread
when (>ubq) {
  .             // perform operations
  .
  .
  ubq <<- ...; // return a status
}
```

The master thread will initiate a read, but the queue is unavailable as there is no matching write from the slave. The slave however sees that there is write availability as a result of the master read. It goes ahead with thread execution and on completion writes a status value. The write-read transaction now completes and the master thread continues.

An assignment may have multiple buffered and unbuffered queues on its RHS, for example -

```
type(t, "uint:8");
queue({ubq1, ubq2}, t, depth=0);
queue({q1, q2}, t);
static(v, t);

seq {
  .
  q1 <<- ...;
  .
}

seq {
  .
  q2 <<- ...;
  .
}

seq {
  .
  ubq1 <<- ...;
  .
}

seq {
  .
  ubq2 <<- ...;
  .
}

seq {
  .
  v <- q1 + ubq1 + q2 + ubq2;
  .
}
```


}

The assignments to buffered queues q1 and q2 will occur when encountered (if these queues are not full). The assignment at the bottom will wait until q1 and q2 are not empty and writes are pending on ubq1 and ubq2. The writes to ubq1 and ubq2 will also wait until the last assignment statement is able to execute. When that happens that last seq block will move on as will the two seq blocks containing the writes to ubq1 and ubq2.

Chapter 8 - Inbuilt Modules

The following called modules are built into the compiler. User called modules or procedures must not use identifiers that conflict with these.

The following abbreviations are used to denote modes -

I	immediate
IN	input
V	value
SV	selectvalue
S	static
Q	queue
M	memory
PR	priority
C	clock

The following abbreviations are used to denote types -

bits	bits
uint	unsigned integer
int	integer or unsigned integer
fixed	fixed point
float	floating point
log	logical (boolean)
str	string
type	type
any	any type
prim	a primitive type
comp	a compound type (arrays, structs or any combination)
[]	array of type

Modules with 'yes' to 'Parameter=argument allowed' can have arguments matched to parameters by position in the argument list or by parameter=argument (keymatch) or a mixture as long as positional arguments do not follow keymatch arguments. Otherwise these keymatch arguments are not allowed. This may be because -

- the module has a single input or output argument in which case no positional or keymatch parameter association is needed
- the module has an indefinite number of arguments and hence there are no 'named' parameters with which to associate arguments

The keymatching may be restricted to just input or just output.

8.1 del - variable length delay line

Call:

```
del(out <- in, len, init, step, varlen, reset);
```

Output parameters:

param	mode	type	
out	V	any	output

Input parameters:

param	mode	type	
in	V	any	input
len	I	int	optional line fixed delay
init	I	any	optional initialisation
step	V	log	optional line step
varlen	V	uint	optional line variable delay
reset	V	log	optional line reset

Parameter=argument allowed:

Input arguments only.

Description:

This module is a delay line implemented using CLB shift registers or flip-flops. On each clock cycle the input is written to the line and the contents move one item right. The rightmost item appears as the out value. This module can function as a sequence generator if the output is connected back to the input.

This module is used extensively internally by 3PL to generate single cycle delays in execution threads in which context it is implemented as a single flip-flop.

For large delays this uses a lot of CLBs and the alternative library function **delay()** [subsubsection 11.13.4.7](#) p514 in library stdlib.3pl might be preferable. That library function will call **del()** in any case if the delay value is small enough.

This has 1 to 6 input arguments and 1 output argument. Input arguments **len** and **init** are immediate mode. **step** may be immediate or target mode as described below. The other input arguments must not be modes **immediate**, **queue**, **memory** or **clock** and value mode arguments must not contain queue reads.

None of the arguments to this module can contain queue reads. This limitation could be easily removed but it is felt that the resulting behaviour of this module could become unduly complex. In limited circumstances this limitation can be worked around and there are examples below to illustrate this.

The 1st input argument, **in**, is the delay line data input. It can be any target type.

The optional input argument, **len**, specifies a fixed delay. This must be an immediate int. If this argument is 0 or omitted and the 5th argument **varlen** is also omitted, the output will simply be connected to the input to give zero delay.

The optional input argument, **init**, is an initialisation array. This must be immediate mode and an array of the same type as **in**. It may be smaller than the delay line length in which case the input end of the delay line will be zero initialised. The low order entries in the initialisation array are loaded in the output end of the delay line. Thus for example a delay of 5 initialised to {1, 2, 3} will output 1 2 3 0 0. If initialisation values are too large their high-order bits will be truncated.

The optional input argument, **step**, is a line step enable. This must be type log. If this argument is used the line will only advance when **step** is true, otherwise the line will advance on every clock cycle.

The optional input argument, **varlen**, is a variable line length. This must be type uint. This argument may be limited or unavailable depending on the type of FPGA. For Xilinx the variable length is limited to values between 0 and 15 giving delays of 1 to 16 respectively. A wider argument will be truncated and a narrower one zero padded. Note that when the line length is changed data may be skipped or repeated. If the line length is changed while stepping is inhibited, the line length argument is an address with which the contents of the line can be non-destructively examined.

The optional input argument, **reset**, returns the line to its initial state when true.

The output argument, **out**, is the delay line data output. This argument must be value mode and must be the same type as the data input **in**.

If the delay line is initialised, these initial values appear at the output during the initial n line steps, where n is the fixed delay (or the initial delay selected by **varlen** if used).

For Xilinx the following apply -

- For FPGAs prior to Virtex5 the variable length argument is limited to values between 0 and 15 inclusive, giving line lengths of 1 to 16 respectively. From Virtex5 onwards the lengths are 1 to 32.
- If the reset is used, the variable length argument cannot be used.
- If the reset is used or the variable length parameter is not used and the delay is 3 or less, the delay line will be implemented using flip-flops, otherwise it will be implemented using CLB LUT shift registers.

Implementation using CLB LUT shift registers minimises resources, requiring only one CLB LUT for every 16-bit delay line.

Note: the fixed delay parameter *len* gives the number of cycles to be delayed but the variable delay parameter *varlen* is the number of cycles delay minus one!

In Xilinx devices there is a substantial delay between the clock edge and the arrival of the shift register signal at the output of the logic block (2.8nS to 3.5nS for a -6 speed XC2VP and about 1.15 to 1.43nS for a -2 speed XC5V). The delay for a flip-flop is very much smaller. For this reason a delay line with fixed length is implemented using a flip-flop for the last delay stage. If a variable length delay line is required, this flip-flop is not inserted and hence the significant output delay must be considered. It is preferable in that case to assign the delay line output directly to a variable, i.e. with no intervening combinatorial logic, to avoid introducing any extra path delay.

The limitation stated above that inputs cannot contain queue reads can be overcome in some cases. For example to use a queue as data input we can use the examine operator as long as a queue wait controls the delay step. For example to form a running average of length 8 -

```
queue(q, "uint:8");
static(rsum, "uint:11");
queue(rav, "uint:8");
value(out, "uint:8");

del(out <- ?q, 8,, <q);    // there is no queue read here!

seq {
  rsum <- rsum + q - out; // there is a queue read here
  rav <- rsum / 8;
}
```

Here a data stream is written to queue **q** and a running average appears as a stream through queue **rav**. The data input does not contain a queue read as the examine operator simply looks at the queue output, valid or not, and does not pop a datum. It is important to note that the read availability of **q** is tested in two places, the reading of **q** and the expression **<q**. Although **del()** is a distinct module **<q** is evaluated as an argument to **del()** in the current scope, not the scope of **del()**, so there is only a single destination to **q**, or rather both availability tests are in the same destination module.

Another approach is used in library function **del()** in xilinx/common.3pl [subsubsection 11.2.3.2](#) p396 which calls inbuilt module **del()**.

8.2 resync - resynchronise a level

Call:

```
resync(out <- in);
```

Output parameters:

param	mode	type	
out	V	log	output pulse

Input parameters:

param	mode	type	
in	V SV S PR	log	target value pulse or level

Parameter=argument allowed:

No

Description:

in is an expression or a value, selectvalue, static, input or priority variable of type log.

out is a value of type log.

When the input rises and is sampled by the clock associated with that variable, the output will produce a single cycle pulse synchronised with the current clock domain.

If the input is an input, selectvalue, static or priority mode variable and it has not yet been referenced or assigned then it will be given the current clock domain. An unassigned value variable will also be given the current clock domain. A value variable that has been assigned but has a null clock domain will result in error termination.

Chapter 9 - Inbuilt Procedures

The following procedures are built into the compiler. User called modules or procedures must not use identifiers that conflict with these.

The following abbreviations are used to denote modes -

I	immediate
V	value
SV	selectvalue
S	static
Q	queue
CM	combinatorial output memory
RM	registered output memory
PR	priority
IN	input
O	output
C	clock

The following abbreviations are used to denote types -

bits	bits
uint	unsigned integer
int	integer or unsigned integer
fixed	fixed point
float	floating point
log	logical (boolean)
str	string
enum	enumerated type
type	type
file	file
map	map
list	list
any	any type
prim	a primitive type
comp	a compound type (arrays, structs or any combination)
[]	array of type

Procedures with 'yes' to 'Parameter=argument allowed' can have arguments matched to parameters by position in the argument list or by parameter=argument (keymatch) or a mixture as long as positional arguments do not follow keymatch arguments. Otherwise these keymatch arguments are not allowed. This may be because -

- the procedure has a single input or output argument in which case no positional or keymatch parameter association is needed

- the procedure has an indefinite number of arguments and hence there are no 'named' parameters with which to associate arguments

The keymatching may be restricted to just input or just output.

Procedures with 'yes' to in-line target execution will produce in-line code in the FPGA. This code is controlled by the execution stream in which it is embedded and by the availability of any queues in the arguments. Other procedures do not produce in-line code in the FPGA. However, they may produce target components which are accessed by in-line target code elsewhere in the source code.

9.1 addmodeattribute - add attribute to the mode attribute map

Call:

```
addmodeattribute(mode, key, isarray, type);
```

Output parameters:

none

Input parameters:

param	mode	type	
mode	I	str	mode
key	I	str	attribute
isarray	I	log	is an array
type	I	str	primitive type

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

Add an attribute to the attribute map for a variable mode. 3PL contains an internal map initialised with entries for allowed attributes for each mode. These attributes are internal to 3PL and affect the netlist generation. Attributes from which constraints are generated are not handled internally by 3PL but by library code appropriate to the FPGA platform. To allow centralised vetting of attributes in calls to inbuilt procedures input(), output() and clock() this procedure is used to add these constraint attributes to the map.

mode is a string identifying the mode and is "input", "output" or "clock"

key is the attribute name.

isarray is true if the attribute value is an array (one member for each bit). If not an array the value will be duplicated for each bit.

type is the attribute value primitive type and can be "uint", "float", "log" or "str" as appropriate.

9.2 append - append a list entry

Call:

```
append(l, index, v);
```

Output parameters:

none

Input parameters:

param	mode	type	
l	I	list	a list
index	I	int	an index
v	I V C	any	the value

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

The item **v** is appended at the position indicated by **index** in list **l**. The index may range from 0 to **entries(l)** inclusive. If **index** is outside this range a fatal error occurs. An index of 0 inserts at the beginning of the list and an index equal to the list size appends to the end of the list. If the index argument is omitted the item will be appended to the end of the list. If the last argument is omitted null will be appended.

The appended item may be immediate, clock, cmemory or rmemory mode.

9.3 appendall - append entries from another list

Call:

appendall(l, l2);

Output parameters:

none

Input parameters:

param	mode	type	
l	I	list	a list
l2	I	list	another list

Parameter=argument allowed:

No

In-line target execution:

No

Description:

The contents of list **l2** are appended at the position indicated by **index** in list **l**. The index may range from 0 to **entries(l)** inclusive. If **index** is outside this range a fatal error occurs. An index of 0 inserts at the beginning of the list and an index equal to the list size appends to the end of the list. If the index argument is omitted the items will be appended to the end of the list.

9.4 argadjust - adjust arguments for arithmetic operations

Call:

argadjust(out1, out2, rt <- in1, in2, op, matchleft);

Output parameters:

param	mode	type	
out1	V	-	adjusted first operand
out2	V	-	adjusted second operand
rt	I	type	result type

Input parameters:

param	mode	type	
in1	V	-	first operand
in2	V	-	second operand
op	I	str	operation
matchleft	I	log	use left offset

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure takes two value mode arguments and an operator and supplies two adjusted values and a result type. The input operands (the 1st and 2nd arguments) must be type "uint:", "int:", "ufixed:" or "fixed:". The 3rd input argument, the operator type, must be one of the strings "+", "*" or "/".

This procedure accesses code within the 3PL interpreter used for adjusting target variables in arithmetic expressions.

The "+" operator is used for addition and "-" for subtraction but the action is identical. The operand with the smaller offset will be adjusted to match the other, i.e. the binary points will be aligned by left shifting the operand with fewer fractional bits. The type assigned to the 3rd output argument will have the same binary point offset as the now-adjusted operands and the integer part will be one bit wider than the operand with the largest integer part.

If the 4th input argument is supplied and is true, the result offset will be the same as the first input operand and the second input operand will be adjusted by left or right shifting as required. This feature is included for the case where the first operand is the result of an intermediate hardware operation and is not accessible for adjustment (for example in an XC5V DSP block where a multiplication can be followed by an addition or subtraction but the intermediate product cannot be adjusted).

The "*" operator is used for multiplication. The operands are not adjusted and are returned as the output values. The output result type fraction width is the sum of the operand fraction widths and the integer width is the sum of the operand integer widths.

The "/" operator is used for division. The operands are not adjusted and are returned as the output values. The output result type fraction width is the sum of the numerator fraction width and the denominator integer width. The result integer width is the sum of the numerator integer width and the denominator fraction width.

9.5 assert - assert a value mode variable for a clock cycle

Call:

```
assert(v <- o);
```

Output parameters:

param	mode	type	
v	V	log	value carrying the pulse

Input parameters:

param	mode	type	
o	I	log	if true, OR with previous value

Parameter=argument allowed:

No

In-line target execution:

yes

Description:

This procedure asserts a logical value variable true for one clock cycle when this statement executes. The value is true for the clock cycle leading up to the clock edge on which execution occurs.

If the input argument is provided and is true the output value variable will be assigned the OR of its previous value and the execution pulse. If it has no previous assigned value it is assigned the execution pulse. Using this optional input argument a signal pulsed from more than one place can be generated.

Note: when ORed in this way, use of the value of the output variable must follow the last call to assert() since each call re-assigns the value mode variable!

If the ORing option is not selected and multiple calls to **assert()** are made with the same output argument, each call will re-assign the output value variable, which may be what is wanted or may not!

The following code illustrates some points to consider -

```
value({1, v1, v2}, "log");
static(v, "log");

when (1) seq {
    ....
    .
    .
    par {
        assert(v1 <-);
        v2 <- true;
    }
    v2 <- false;
    .
}
```

'v1' and 'v2' will both be true for one clock cycle, but 'v1' will be asserted one clock cycle earlier than 'v2'. Use of **assert()** has the potential disadvantage that since it is actually the execution signal it suffers any delays inherent in the logic path of any conditional and hence might introduce timing constraint difficulties.

Note that the following usage is allowed -

```
value(v, "log");

when (v)
    ....
.
.
assert(v <-);
.
```

Here the call to **assert()** occurs only once. 3PL notices when assigning a value to **v** within the call to **assert()** that **v** has been referenced previously at which time it had not been assigned a value, so it takes care of the post-assignment (see discussion of post-assignment in section 5.2). The following would not work in the same way -

```
value(v, "log");

when (v)
    ....
.
.
assert(v <-);
.
assert(v <- true);
.
```

The second call to **assert()** would indeed OR the two execution signals together and assign the result to variable **v**, but the **when ()** would use the value of **v** after the first call to **assert()**. Since with parallel code the order of statements is not significant, it would clearly be better to use a policy of referencing the variable following all calls to **assert()**.

see also inbuilt function **used()** [section 10.165](#) p345.

9.6 attributeprocedure - register a procedure to handle attributes

Call:

```
attributeprocedure(p);
```

Output parameters:

none

Input parameters:

param	mode	type	
p	-	-	procedure identifier

In-line target execution:

No

Description:

This procedure registers the argument procedure as a user-defined procedure to handle input, output and clock mode attributes. The registered procedure will be called after every input, output and clock mode variable is created. The called procedure will be passed a single argument which is a pointer to the newly-created variable.

This facility is intended to allow attributes to be attached to I/O pin specifications and have them transferred to relevant input, output or clock mode variables when those variables are created. It can also provide any necessary re-formatting of attributes, either those brought in from pin associations or those passed as arguments to inbuilt procedures `input()`, `output()` or `clock()`.

There is a Xilinx library procedure for this purpose; `processPinAttributes()` [subsection 11.2.2.24 p393](#), in `xilinx/common.3pl`.

`attributeprocedure()` should be called prior to calls to inbuilt procedures `input()`, `output()` or `clock()`.

9.7 attributes - assign attributes to a variable

Call:

```
attributes(v, attr);
```

Output parameters:

none

Input parameters:

param	mode	type	
v	any	-	variable identifier
attr	I	map	attributes map

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure applies attributes to one or more variables. The first argument may be a compound value containing a number of identifiers for created variables. Identifiers may have fields or subscripts appended but these will be ignored, the attribute(s) being applied to the entire variable.

The optional second argument can be

- a map of attributes
- a one-dimensional compound constant containing key=value entries
- a key=value argument

In the last case this may be followed by more key=value arguments. If this is not supplied the attributes can be added later using inbuilt procedure `attributes()` [section 9.7 p192](#). For a description of applicable attributes see [section 3.11 p80](#).

Attributes are applied to a variable in a map. The map keys are the attribute identifiers and the associated values are the attribute values. The entry values are mixed types, the type depending on the individual attribute.

A compound value can be used for the attributes argument. It must be a one-dimensional list and every entry must be key=value format.

The attributes() procedure does not enter a new attributes map in the variable, it adds the entries in the argument map to that in the variable. Thus the attributes() procedure can be called multiple times to add one or more attributes to the variable. If the same attribute is entered again the new value replaces the old value. This can be used to enable and disable 3PL attributes that are used during immediate execution.

Netlist attributes can be deleted by assigning the value **null**. 3PL attributes cannot be deleted; 3PL attributes either relate to parameters that are mandatory such as clock domains, or are boolean in which case they can be disabled rather than deleted, so there is no reason to delete them.

3PL attributes **domain**, **readdomain** and **writedomain** may be assigned null which is interpreted to mean that the current clock domain is to be used.

Note that for input, output and clock mode variables many of the attributes are ignored by 3PL itself and must be processed by 3PL source code. These modes are handled by inbuilt procedures input(), output() and clock() which call a library procedure named by attributeprocedure() section 9.6 p191 to process the attributes. These attributes will also be added to the variable but 3PL itself will ignore them.

Attributes are described in the sections pertaining to the relevant variable creation procedures.

9.8 change - change a list entry

Call:

change(l, index, v);

Output parameters:

none

Input parameters:

param	mode	type	
l	I	list	a list
index	I	int	an index
v	I V C	any	the value

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

The item in list **l** at position **index** is replaced by item **v**. The index may range from 0 to **entries(l) - 1** inclusive. If **index** is outside this range a fatal error occurs. The new value does not have to be the same type as the previous value. If the last argument is omitted the list item will be changed to null.

The new item may be immediate, clock, cmemory or rmemory mode.

9.9 class - create a variable of type "class"

Call:

class(c, init);

Output parameters:

None.

Input parameters:

init is an immediate value of type "class".

Parameter=argument allowed:

No

In-line target execution:

no

description:

This procedure creates one or more variables of type "class".

The first argument is the variable identifier, or it can be a comma-separated list of identifiers enclosed in braces.

The second argument is optional and is an initial value for the variable. It is only allowed where a single variable is being created. The "class" value may be from another variable of the same type but more usually it would be the returned value from a call to a class, for example the following creates a variable **cl** of type "class" and assigns to it an instance of the class **c** returned from a call to **c()**.

```
class(cl, c());
.
.

class c () {
    .
    .
    .
}
```

9.10 clock - create one or more clock variables

Call:

clock(ident,);

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2... }	-	-	variable identifier(s)
....	-	-	optional attributes

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create a clock variable.

Create one or more clock mode variables.

ident is the variable identifier. If more than one variable is to be created the list of variables must be in braces {*id1*, *id2*, ... }. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection [3.6.2](#) p53.

When a **loc** attribute is provided only a single clock variable can be created by this procedure.

When **clock()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

Any following arguments specify attributes. Each argument may specify an attribute by keymatch format, i.e. attribute=value where 'attribute' is an attribute identifier and 'value' is an appropriate

mode and type. Alternatively an argument can be a map containing attributes where the key (string) is the attribute identifier and the associated value is the attribute value.

Attributes may also be associated with the input pin or pin pair. This is described in chapter [chapter 24](#) p663 **Constraints, Properties and Attributes** where other aspects of processing attributes and constraints for input, output and clock mode variables are also described.

Attributes can be added by using inbuilt procedure **attributes()** [section 9.7](#) p192.

Attributes are required for two main purposes; control of netlist generation and constraint file entries. Attributes giving input pin locations and input buffer types directly control netlist generation. I/O standards and timing group names exercise control of place-and-route via a constraint file.

The applicable attributes are shown in the following table.

These attributes are not case-sensitive but will be converted to lower case.

The attribute values will be applied to all pins.

clock mode			
frequency	3PL	float	frequency, Hz
dutycycle	3PL	float	duty cycle
differential	netlist	log	differential
ibufg	netlist	log	buffer type
loc	netlist	str or [2]str	locations

Attributes **frequency** and **dutycycle** specify these parameters for clock nets and any associated clock domains. **Note that at present dutycycle is not used. It has been included for future use.**

The following attribute is read-only and is created when inbuilt procedure **clock()** is called and can be retrieved from the clock mode variable if required by the function call attribute("ids").

clock mode			
ids	3PL	[]str	element IDs

The following attributes only apply to a clock variable if it is a clock input from a pad. If the **loc** attribute is supplied to a clock variable an input pad and buffer will be created, otherwise the clock variable source is internal to the FPGA.

loc specifies one or two pad locations (FPGA pins) for the clock input - two locations if it is differential (+ve, -ve order). For a differential clock input this attribute will be an array of two strings.

differential indicates that the clock input is differential - two locations must be supplied and the buffer will be an IBUFDS or IBUFGDS rather than the default IBUF or IBUFG. This attribute is used internally by 3PL for code generation.

ibufg if true specifies that the input buffer must be an IBUFG or IBUFGDS instead of the default IBUF or an IBUFDS. This attribute is used internally by 3PL for code generation. It is not clear from Xilinx documentation that an IBUFG needs to be explicitly specified, or if instead this can be inferred from the package pin(s). other software.

The following attributes are required for constraint generation and are handled at completion of immediate execution by library procedure **generateconstraints()** [subsection 11.2.2.18](#) p388. The attribute values are single values which will be applied to the pin or pins.

clock mode constraint attributes			
tnm	constraint	str	timing group id
diff_term	constraint	log	differential 100Ω termination
iostandard	constraint	str	voltage standard

tnm is the name of a timing group in which all state devices clocked by this clock are to be included for timing constraint purposes. All clocks which drive logic clock inputs (as distinct from inputs to buffers of clock managers) will have a period constraint generated by generate_constraints() in xilinx/common.3pl. By default a timing group name will be constructed from the clock identifier translated to upper case, but if a "tnm" attribute is provided then this string will be used instead.

diff.term indicates that a 100Ω termination resistance should be applied to a differential input.

iostandard is a string specifying one of the allowed I/O standards (see Xilinx documentation).

The following attribute is read-only and is added when target statement variables are assigned or evaluated and the clock domain is established or verified. It can be retrieved from the clock mode variable if required by the function call attribute("valid").

clock mode			
valid	3PL	log	is a valid logic clock

Any attributes that are not listed above will result in a fatal error.

Where the call to **clock()** is to create a parameter to a module, procedure or function, only the first argument may be given and it must be a single identifier. The clock frequency, period and duty cycle are copied from the matching argument.

Where a clock variable is an external input the **loc** attribute must be supplied. This would be an array of two location strings for a differential clock input or a single location string otherwise.

As far as the compiler is concerned any clock variable may be used to control logic (by calling inbuilt procedure **useclock()**). However, many clock variables are used as routing to connect clock managers, input pads, buffers etc together and these interconnect clock variables must not be passed to **useclock()**.

Note: for each pin a netlist port "PORT_pin" is created where 'pin' is the pin identifying string. This port name is also the netlist identifier of the netlist from the output of the pad to the input of the IBUFG. This convention can be used when generating associated constraints.

9.11 clockfromexpr - drive a clock from an expression

Call:

```
clockfromexpr(clk <- expr);
```

Output parameters:

clk is a clock variable.

Input parameters:

expr is an expression of type log which is to drive the clock.

Parameter=argument allowed:

No

In-line target execution:

no

description:

This procedure connects a clock signal to an expression. That clock signal would normally then be buffered using a global clock buffer created by library procedure **clockbufg**. The argument may be input, selectvalue, value, static or priority mode but must be type log and must contain no queue references.

9.12 close - close a file

Call:

```
close(file);
```

Output parameters:

none

Input parameters:

param	mode	type	
file	I	file	file variable

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure closes an open file.

9.13 cmemory - create a combinatorial output memory

Call:

cmemory(name, ports, atype, dtype, init, ...);

Output parameters:

none

Input parameters:

param	mode	type	
name	-	-	memory identifier
ports	I	int	number of ports
atype	I	int	address type
dtype	I	int	data type
init	I	[]int	optional initialisation (variable or compound constant)
...	I	-	optional attributes

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

Create a combinatorial output RAM.

name is an identifier for the memory. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection 3.6.2 p53.

ports is the number of ports.

atype is the address type. This can be any target type with the constraint that the total number of bits must not exceed 9.

dtype is the data type. This can be any target type.

init is an optional initialisation array. The initialisation data is a one-dimensional variable or array which is no larger than the depth of the memory. It may be smaller in which case the memory is zero padded. The types and structure of the initialisation variable or structured constant after the first dimension must match the memory data type. Immediate constants are padded to fit target elements but must not exceed the target width. If the initialisation argument is omitted the memory will be initialised to all zero.

An initial value may also be set later by calling inbuilt procedure **attributes()**

Additional trailing arguments may be passed to set attributes. These must be of the form attr=value.

This procedure may create a module, procedure or function parameter in which case only the first argument is needed (the memory identifier).

In the case of Xilinx FPGAs combinatorial output memory is distributed CLB RAM. This can have one or two access ports. One logic slice can hold 1 bit x 32 words single port or 1 bit x 16 words dual port.

Port 0 can be read or written (or both). Port 1 is read-only.

Architectures supported -

Spartan-II, Spartan IIE	YES
Spartan-3	YES
Virtex, Virtex-E	YES
Virtex-II, Virtex-II Pro	YES
XC9500, XC9500XV, XC9500XL	NO
Coolrunner XPLA3, Coolrunner-II	NO

The range of memory depths that have been implemented is shown in the following table. The depth required is determined from the number of bits in the address type. Memories with depths smaller than the minimum listed are still implemented, but the physical memory will be the minimum size shown and the smaller depth will be achieved by tying higher address bits low. The column marked ROM is for memories that are initialised and are read but not written -

	ROM	1 port	2 port
Spartan-II, Spartan IIE	16 - 256	16 - 256	16 - 128
Spartan-3	16 - 2048	16 - 512	16 - 128
Virtex	16 - 256	16 - 256	16 - 256
Virtex-E	16 - 256	16 - 256	16 - 256
Virtex-II	16 - 2048	16 - 1024	16 - 512
Virtex-II Pro	16 - 2048	16 - 1024	16 - 512
Virtex 4	16 - 2048	16 - 512	16 - 128
Virtex 5	16 - 4096	16 - 4096	16 - 2048

The data may be any width.

Applicable attributes are shown in the following table. The attributes can be passed as arguments to this procedure or may be applied to the variable using the inbuilt procedure **attributes()** [section 9.7 p192](#).

When **cmemory()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

cmemory mode attributes			
init	3PL		initial value
domain	3PL	clock	force clock domain for all ports

These attributes are case-insensitive.

init sets an initial value for the memory. Any previous initial value will be overwritten. The initial value must be a compound type and the nested type must be an immediate type equivalent to the memory data type, i.e. same type but without a width field. The initial value is not used until completion of immediate execution when variables are processed into target code and hence it can be changed at any time during immediate execution.

domain forces the clock domain (both input and output) for all ports to the specified clock domain. If assigned **null** the current clock domain is selected. If the variable already has a clock domain as a result of having been assigned or evaluated then that clock domain is compared with the intended domain. If the assigned clock domain agrees with that already in place then there is no further action, otherwise a fatal error occurs.

9.14 cmemorywrite - write to a combinatorial output memory

Call:

```
cmemorywrite(m, port, addr, data);
```

Output parameters:

none

Input parameters:

param	mode	type	
m	M	-	memory being written
port	I	int	port
addr	I V SV S Q PR	any	memory address
data	I V SV S Q PR	any	data to be written

Parameter=argument allowed:

Yes

In-line target execution:

yes

Description:

Write to a port of a combinatorial output memory.

m is the memory mode variable identifier.

port is the memory port.

addr is the address.

data is the data to be written.

The data and address may be any type, including compound types, but the types must match the types given when the combinatorial output memory was created (see **cmemory()** [section 9.13](#) p197) but primitive component widths may differ in which case padding or truncation will occur.

Xilinx FPGAs do not allow writes to port 1. Within this limitation multiple reads and writes on a port are allowed and the address and data inputs will be multiplexed, however there is no access arbitration and it is up to the application code to ensure exclusive access. Coincident accesses will result in data corruption.

A call to **cmemorywrite()** ([section 9.14](#) p198) or **cmemoryread()** ([section 10.17](#) p275) sets the clock domain for the memory port to the current clock domain. All further calls to either of these inbuilt procedures for the same port of the same combinatorial output memory must be in the same clock domain (or a fatal error results).

9.15 designname - change the design name

Call:

```
designname(name);
```

Output parameters:

none

Input parameters:

param	mode	type	
name	I	str	design name string

Parameter=argument allowed:

No

In-line target execution:

no

Description:

Change the design name, overriding the default derived from the file name. This can be overridden by the command line option **-N**. It is also overridden if inbuilt procedure **export()** is called.

9.16 directive - enter a directive

Call:

directive(key, value);

Output parameters:

none

Parameter=argument allowed:

No

Input parameters:

param	mode	type	
key	I	str	directive key
value	I	any	directive value (optional)

In-line target execution:

No

Description:

There is a single global directive map. The key is a string. The value may one of the 3PL immediate primitive types **int**, **log**, **float** or **str**. If the second argument is not provided the entry will be type **log** and value **true**.

If the directive is already present its value will be changed but the type must be the same or a fatal error will occur.

see also inbuilt procedures

presetdirective() [section 9.50](#) p229

removedirective [section 9.60](#) p238.

and inbuilt functions

directive() [section 10.28](#) p280.

direxists() [section 10.30](#) p281.

directives() [section 10.29](#) p281.

9.17 divrem - combined quotient and remainder

Prototype:

```
procedure
divrem (value(quot), value(rem) <- value(num), value(denom)) {}
```

Output parameters:

quot is the quotient.

rem is the remainder.

Input parameters:

num is the numerator.

denom is the denominator.

Parameter=argument allowed:

Yes

In-line target execution:

no

description:

This procedure takes numerator and denominator values and computes the quotient and remainder. The

results appear on output value variables since this procedure is combinatorial. The input and output types must be all *int* or all *uint* (widths may be different). The inputs may be immediate.

Note: this procedure uses cascaded adders and subtractors and will have a long data path delay. However, the code is optimised for some cases where one operand is constant - for division of 0 (return 0:0), division by 1 (return n:0) and division by power of 2 (quotient by shifting, remainder by masking).

9.18 element - add an element to the EDIF netlist

Call:

element(name, id);

Output parameters:

none

Input parameters:

param	mode	type	
name	I	str	element name
id	I	str	optional unique block identifier

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure adds an element to the EDIF netlist. This is intended for special elements, perhaps particular to a specific FPGA family. For example global clock buffers are not considered generic enough to be included as fundamental logic elements so they are included using this procedure. A family-specific example would be the high-speed serial interfaces. Note that this is a low level process that bypasses 3PL execution streams. It is rather similar to embedding assembly code within a high level language.

The first argument is the Xilinx library element name, e.g. **element("BUFG")**.

The second argument is an optional unique block identifier. If this argument is omitted then an identifier of the form "e123" will be allotted. **Explicit identifiers must avoid the forms "e123" and "b123" to avoid conflicts with 3PL internally generated element identifiers.**

Regardless of explicit use of the block identifier or not, the element constructed by this procedure remains as the current element until another call to this procedure. Connections to the element are made by the inbuilt procedures **elementinput()**, **elementclockinput()**, **elementoutput()** or **elementtsoutput()** (see links below). The construction of an element must be completed by calling inbuilt procedure **elementend()** [section 9.20](#) p203. Properties for the element are added by calling inbuilt procedure **elementproperty()** [section 9.23](#) p205. Associated entries to the NCF file for the element can be made using inbuilt procedure **elementaddncf()** [subsection 11.2.2.5](#) p378.

Note that this procedure can also be used to add standard elements to the netlist, e.g. **element("FDRE")**.

Elements defined by using **element()** are added to cell definitions in the EDIF file. If another element of the same cell name is created they will share the same cell definition. The ports in the cell definition will be the set of all input, output or three-state ports specified in all instances of the element. Thus an instance of an element may not specify all the available ports for the element but the cell definition will accumulate all ports specified in the instances of the element. Any available ports not specified will be omitted from the cell definition. In the case of array ports where the port width changes from instance to instance the largest dimension will be used in the cell definition.

For further description see [chapter 23](#) p659.

See also -

elementend() [section 9.20](#) p203,

elementproperty() [section 9.23](#) p205,
elementinput() [section 9.21](#) p203,
elementclockinput() [section 9.19](#) p202
elementoutput() [section 9.22](#) p204
elementtsoutput() [section 9.24](#) p206.

9.19 elementclockinput - connect a clock input signal to an EDIF netlist element

Call:

elementclockinput(id, port_name, signal);

Output parameters:

none

Input parameters:

param	mode	type	
id	I	str	optional block identifier
signal	V SV S Q I	any	optional signal
port_name	I	str	port name

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure connects a clock input signal to an element (see **element()** [section 9.18](#) p201).

The block identifier **id** specifies the element to which a port is to be added. If this argument is omitted then the most recent element begun by a call to **element** is assumed.

Creation of multiple elements may be interleaved, but it is preferable to avoid interleaving and to complete an element before starting the next.

port_name is a string identifying the element port to be connected.

signal is the signal to be connected to the element port. It must be clock mode or else an immediate type "log" whose value is false, in which case the clock input will be connected to GND. This argument may also be an immediate mode string in which case it refers to a single FPGA port (PAD), the string value being the pin designation. For a string "AB12" the port name will be "PORT_AB12". The element port must be a single bit.

If the **signal** argument is missing or null the port will be connected to ground.

This procedure should only be used for clock inputs to elements where the input functions as an edge trigger. Thus for inputs to clock buffers or the source inputs to clock managers **elementinput()** [section 9.21](#) p203 should be used instead. This distinction was once required so that 3PL could keep track of clock destinations for timing constraint generation, however this is no longer the case. A remnant remains in the **clk_sinks** and **clk_CE** read-only attributes of clock mode variables.

See also -

element() [section 9.18](#) p201,
elementend() [section 9.20](#) p203,
elementproperty() [section 9.23](#) p205,
elementinput() [section 9.21](#) p203,
elementoutput() [section 9.22](#) p204,
elementtsoutput() [section 9.24](#) p206.

9.20 elementend - complete construction of an EDIF netlist element

Call:

elementend(id);

Output parameters:

none

Input parameters:

param	mode	type	
id	I	str	optional block identifier

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure is called to complete entry of a netlist element. The optional argument gives the block identifier of the element to be completed. If omitted, the most recent uncompleted element is assumed.

9.21 elementinput - connect an input signal to an EDIF netlist element

Call:

elementinput(id, port_name, signal, arrayformat);

Output parameters:

none

Input parameters:

param	mode	type	
id	I	str	optional block identifier
port_name	I	str	port name
signal	V SV S Q I	any	signal
arrayformat	I	int	array format indicator

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure connects an input signal to an element (see **element()** [section 9.18](#) p201).

The block identifier **id** specifies the element to which a port is to be added. If this argument is omitted then the most recent element begun by a call to **element** is assumed.

Creation of multiple elements may be interleaved, but it is preferable to avoid interleaving and to complete an element before starting the next.

port_name is a string identifying the element port to be connected. The port width is determined by the signal width.

signal is the signal to be connected to the element port. It may be value, selectvalue, or static and may be any type. If value mode the value must not contain any queue reads. This argument may also be an immediate mode string in which case it refers to a single FPGA port (PAD), the string value being the

pin designation. For a string "AB12" the port name will be "PORT_AB12". If the signal is more than one bit wide an array format argument must be provided.

If the **signal** argument is missing or null no port connection is made.

arrayformat indicates the form of ports which are arrays. If this argument is 1 it indicates that array ports should be expanded to individual pins of the form A0, A1, A2 etc. If the value is 2 then the port will be a formal EDIF array. ISE will accept format 1 for all elements. Some versions of ISE (e.g. 10) give errors when format 2 is used for block RAM elements where different port sizes are used in the same configuration. Vivado requires format 2 for port arrays. Since Vivado cannot be used for earlier FPGAs, block RAMs for these (up to XCS6, XCV6) are defined using format 1. Series 7 block RAMs are defined using format 2 since early versions of ISE do not support these devices. A single bit port can have a format type of 1, but not type 2.

In the case of clock mode inputs this procedure should only be used for inputs to elements where the input does not function as an edge trigger, e.g. for inputs to clock buffers or the source inputs to clock managers. For edge trigger inputs **elementclockinput()** [section 9.19](#) p202 should be used instead. This distinction was once required so that 3PL could keep track of clock destinations for timing constraint generation, however this is no longer the case. A remnant remains in the **clk_sinks** and **clk_CE** read-only attributes of clock mode variables.

As an example of port input, the code -

```
input(is, "log", loc="AB12", tnm="IC");
```

could be effectively replaced by -

```
value(is, "log");

ibuf(is <- "AB12", "IC");

module
ibuf (value(out) <- str(pin), str(tnm)) {
  element("IBUF");
  elementinput("I", pin);
  elementoutput(out <- , "O");
  elementStrProperty("loc", pin);
  elementend();
  IOconstraint(pin, "tnm", "IC");
}
```

though note that the variable **is** would now be value mode, not input mode.

See also -

element() [section 9.18](#) p201,
elementend() [section 9.20](#) p203,
elementproperty() [section 9.23](#) p205,
elementclockinput() [section 9.19](#) p202,
elementoutput() [section 9.22](#) p204,
elementtsoutput() [section 9.24](#) p206.

9.22 elementoutput - connect an output signal to an EDIF netlist element

Call:

```
elementoutput(signal <- id, port_name, arrayformat);
```

Output parameters:

param	mode	type	
signal	V SV S Q I	any	signal

Input parameters:

param	mode	type	
id	I	str	optional block identifier
port_name	I	str	port name
arrayformat	I	int	array format indicator

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure connects an output signal to an element (see **element()** [section 9.18](#) p201).

The block identifier **id** specifies the element to which a port is to be added. If this argument is omitted then the most recent element begun by a call to **element** is assumed.

Creation of multiple elements may be interleaved, but it is preferable to avoid interleaving and to complete an element before starting the next.

signal is the signal to be connected to the element port. It must be value mode, of any type, or clock mode. This argument may also be an immediate mode string in which case it refers to a single FPGA port (PAD), the string value being the pin designation. For a string "AB12" the port name will be "PORT_AB12" (see input example in **elementinput()** [section 9.21](#) p203). If the signal is more than one bit wide an array format argument must be provided.

port_name is a string identifying the element port to be connected. The port width is determined by the signal width.

arrayformat indicates the form of ports which are arrays. If this argument is 1 it indicates that array ports should be expanded to individual pins of the form A0, A1, A2 etc. If the value is 2 then the port will be a formal EDIF array. ISE will accept format 1 for all elements. Some versions of ISE (e.g. 10) give errors when format 2 is used for block RAM elements where different port sizes are used in the same configuration. Vivado requires format 2 for port arrays. Since Vivado cannot be used for earlier FPGAs, block RAMs for these (up to XCS6, XCV6) are defined using format 1. Series 7 block RAMs are defined using format 2 since early versions of ISE do not support these devices. A single bit port can have a format type of 1, but not type 2.

See also -

element() [section 9.18](#) p201,
elementend() [section 9.20](#) p203,
elementproperty() [section 9.23](#) p205,
elementinput() [section 9.21](#) p203,
elementclockinput() [section 9.19](#) p202
elementtsoutput() [section 9.24](#) p206.

9.23 elementproperty - set an element property

Call:

```
elementproperty(id, property, value);
```

Output parameters:

none

Input parameters:

param	mode	type	
id	I	str	optional block identifier
property	I	str	property name
value	I	str	property value

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure sets a property value for an element. The optional argument **id** gives the block identifier of the element to be completed. If omitted, the most recent uncompleted element is assumed.

elementproperty() only takes strings as value arguments so a number of library procedures are available in xilinx/common.3pl to format other types. These are -

elementLogProperty() [subsubsection 11.2.2.11](#) p383,
elementIntProperty() [subsubsection 11.2.2.10](#) p382,
elementBinProperty() [subsubsection 11.2.2.6](#) p379,
elementHexProperty() [subsubsection 11.2.2.9](#) p381,
elementFloatProperty() [subsubsection 11.2.2.8](#) p381,
elementStrProperty() [subsubsection 11.2.2.12](#) p384.

These procedures also provide limit checks or valid argument lists.

See also -

element() [section 9.18](#) p201,
elementend() [section 9.20](#) p203,
elementinput() [section 9.21](#) p203,
elementclockinput() [section 9.19](#) p202
elementoutput() [section 9.22](#) p204.

9.24 elementtsoutput - connect a 3-state output signal to an EDIF netlist element

Call:

```
elementtsoutput(signal <- id, port_name, arrayformat);
```

Output parameters:

param	mode	type	
signal	V SV S Q I	any	signal

Input parameters:

param	mode	type	
id	I	str	optional block identifier
port_name	I	str	port name
arrayformat	I	int	array format indicator

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure connects a 3-state output signal to an element (see **element()** [section 9.18](#) p201).

The block identifier **id** specifies the element to which a port is to be added. If this argument is omitted then the most recent element begun by a call to **element** is assumed.

Creation of multiple elements may be interleaved, but it is preferable to avoid interleaving and to complete an element before starting the next.

signal is the signal to be connected to the element port. It must be selectvalue mode of any type. If the signal is more than one bit wide an array format argument must be provided. This argument may

also be an immediate mode string in which case it refers to a single FPGA port (PAD), the string value being the pin designation. For a string "AB12" the port name will be "PORT_AB12".

port_name is a string identifying the element port to be connected. If the signal is more than one bit wide an array format argument must be provided. The port width is determined by the signal width.

arrayformat indicates the form of ports which are arrays. If this argument is 1 it indicates that array ports should be expanded to individual pins of the form A0, A1, A2 etc. If the value is 2 then the port will be a formal EDIF array. ISE will accept format 1 for all elements. Some versions of ISE (e.g. 10) give errors when format 2 is used for block RAM elements where different port sizes are used in the same configuration. Vivado requires format 2 for port arrays. Since Vivado cannot be used for earlier FPGAs, block RAMs for these (up to XCS6, XCV6) are defined using format 1. Series 7 block RAMs are defined using format 2 since early versions of ISE do not support these devices. A single bit port can have a format type of 1, but not type 2.

See also -

element() [section 9.18](#) p201,
elementend() [section 9.20](#) p203,
elementproperty() [section 9.23](#) p205,
elementinput() [section 9.21](#) p203,
elementclockinput() [section 9.19](#) p202
elementoutput() [section 9.22](#) p204.

9.25 emptyvalue - clear a value-mode variable

Call:

```
emptyvalue(name);
```

Output parameters:

none

Input parameters:

param	mode	type	
name	V	-	value variable identifier

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure clears the value of a value-mode variable and changes its type to "empty". This allows the variable to be subsequently assigned a value of any type. This is particularly useful where a value variable is to be supplied as an argument to an output parameter where the type of that parameter is not known prior to the call.

9.26 enum - create an enumerated type

Call:

```
enum(name, id1, id2,...);
```

Output parameters:

none

Input parameters:

param	mode	type	
name			type identifier
id1	I		1st value identifier
id2	I		2nd value identifier
.	I		...

Parameter=argument allowed:

No

In-line target execution:

No

Description:

The **enum()** procedure creates an enumerated type where the ordinal values are 0, 1, 2 etc (see also [section 9.27](#) p208).

The first argument, **name**, is an identifier giving the enum name which will be created as a variable of type "type". This is followed by a number of value identifiers which will have associated ordinal values 0, 1, 2 etc.

The identifiers must be unique within this enum, but the identifiers may be used in other enums or as variable identifiers without conflict.

Enumerated type variables are created using the enumerated type, e.g.

```
enum(e1, none, typea, typeb);
var(v1, e1);
static(s1, e1);
```

The enumerated type can be used for immediate, value, selectvalue, static or queue mode variables.

Enumerated type values are accessed by using the type name and the value identifier, e.g.

```
when (s1 == e1.typea)
  s1 <- e1.typeb;
```

There are a number of inbuilt functions for handling enumerated types -

ord(v) section 10.103 p314	the ordinal value of variable v
next(v) section 10.96 p310	the next value of variable v
prev(v) section 10.107 p315	the previous value of variable v
hasnext(v) section 10.65 p296	true if variable v has a next value (i.e. it is not the last value)
hasprev(v) section 10.67 p297	true if variable v has a previous value (i.e. it is not the first value)
first(x) section 10.56 p292	the first ordered value of variable or enum type x
last(x) section 10.79 p302	the last ordered value of variable or enum type x

The identifier of an enum value can be obtained by using the inbuilt function **tostring()** [section 10.157](#) p340.

9.27 enumv - create an enumerated type

Call:

```
enumv(name, id1, v1, id2, v2,...);
```

Output parameters:

none

Input parameters:

param	mode	type	
name			type identifier
id1			1st value identifier
v1	I	int	1st value ordinal
id2			2nd value identifier
v2	I	int	2nd value ordinal
.			...

Parameter=argument allowed:

No

In-line target execution:

No

Description:

The **enumv()** procedure creates an enumerated type where the ordinal values are explicitly given (see also [section 9.26](#) p207).

The first argument, **name**, is an identifier giving the enum name which will be created as a variable of type "type". This is followed by a number of pairs of identifiers and associated ordinal values. The ordinal value arguments must be immediate expressions of type int which are not negative.

The identifiers and ordinals must be unique within this enum, but the identifiers may be used in other enums or as variable identifiers without conflict.

Enumerated type variables are created using the enumerated type, e.g.

```
enum(e1,
     none, 3,
     typea, 7,
     typeb, 5
);
var(v1, e1);
static(s1, e1);
```

The enumerated type can be used for immediate, value, selectvalue, static or queue mode variables.

Enumerated type values are accessed by using the type name and the value identifier, e.g.

```
when (s1 == e1.typea)
  s1 <- e1.typeb;
```

There are a number of inbuilt functions for handling enumerated types -

ord(v) section 10.103 p314	the ordinal value of variable v
next(v) section 10.96 p310	the next value of variable v
prev(v) section 10.107 p315	the previous value of variable v
hasnext(v) section 10.65 p296	true if variable v has a next value (i.e. it is not the last value)
hasprev(v) section 10.67 p297	true if variable v has a previous value (i.e. it is not the first value)
first(x) section 10.56 p292	the first ordered value of variable or enum type x
last(x) section 10.79 p302	the last ordered value of variable or enum type x

The identifier of an enum value can be obtained by using the inbuilt function **tostring()** [section 10.157](#) p340.

9.28 exec - execute a command

Call:

```
exec(status, out, err <- command, environ, dir, in);
```

Output parameters:

param	mode	type	
status	I	int, uint	status returned from command
out	I	str	standard output from command
err	I	str	standard error from command

Input parameters:

param	mode	type	
command	I	[]str	command with arguments to be executed
environ	I	map	optional environment variables
dir	I	str	optional working directory
in	I	str	optional string to standard input to command

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure executes a command with arguments, given by the first input parameter **command**. This is an array of strings, the first giving the command path and the remainder supplying arguments.

Note that if the command path is not absolute, the **PATH** variable in the environment on the call to 3PL will be use to find the executable file, *not* that in the optional **environ** input parameter.

No shell is executed. See **shell()** [section 9.70](#) p248 if this is desired.

The optional second input parameter, **environ**, is a map of environment variables. For each entry the key is the environment variable and the associated map value is the variable value. Value types are restricted to "int", "uint", "str" and "float", other types resulting in a fatal error. If no argument is supplied for this parameter the environment on the call to 3PL is passed to the command.

The optional third input parameter, **dir**, is a string specifying the working directory. If no argument is supplied for this parameter the current directory on the call to 3PL is used.

The optional fourth input parameter, **in**, is a string to be copied to standard input of the command process.

If the first output parameter, **status**, is given an argument, the status returned from the command is copied to this variable. The argument must be immediate mode and type "int" or "uint".

If the second output parameter, **out**, is given an argument, the standard output from the command is copied to this variable. The argument must be immediate mode and type "str". The resulting string is usually one or more lines, each terminated by a newline character.

If the third output parameter, **err**, is given an argument, the standard error from the command is copied to this variable. The argument must be immediate mode and type "str". The resulting string is usually one or more lines, each terminated by a newline character.

If the command cannot be found, or cannot be executed, a fatal error occurs.

This procedure waits until the command exits. If the command is signalled then this procedure continues as for a normal exit.

See also **shell()** [section 9.70](#) p248.

9.29 exit - terminate execution

Call:

```
exit(message);
```

Output parameters:

none

Input parameters:

param	mode	type	
message	I	str	optional message

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Terminate 3PL immediate execution. If the argument is provided a message is printed on standard out.

See also **textit()** [section 9.79](#) p257, **trace()** [section 9.80](#) p258.

9.30 export - export a logic 'core'

Call:

`export(OP1=o1, OP2=o2, ... <- IP1=o1, IP2=i2, ...);`

Output parameters:

See description

Input parameters:

See description

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure signals the compiler that the logic is a 'core' (or macro) to be exported so that its netlist can be linked into the netlists from other programs (not necessarily from 3PL programs).

The `-e` option must be used in the call to 3PL to supply the export core name, otherwise a fatal error will occur. This string will also be used to name the netlist and constraint output files.

The input arguments are signal inputs to the 'core'. These arguments may be passed using the key-matching syntax. The identifier on the LHS of the '=' is the port name used when linking the netlists. The RHS of the '=' is a 3PL variable to be an input from the port. If the argument is a single identifier then it is taken as both the port identifier and the identifier of a variable driving the port.

Input port variables may be **value** or **clock** mode. **Value** mode variables may be any target type. Input port variables which are mode **value** must have their clock domains established by assigning attribute **domain** (see [section 3.11](#) p80) before the call to **export()** which passes them as input arguments.

Output arguments are signal outputs from the 'core'. These arguments may also be passed using the key-matching syntax. The identifier on the LHS of the '=' is the port name used when linking the netlists. The RHS of the '=' is a 3PL variable or expression to drive the port.

Output port variables can be mode **value**, **selectvalue**, **static**, **priority** or **clock**. A **selectvalue** mode output port variable is used for a 3-state port. Non-clock output port variables may be any target type. These variables must not contain queue reads. If the argument is a single identifier then it is taken as both the port identifier and the identifier of a variable driving the port.

9.31 external - declare a signal as an EDIF port

Call:

`external(portname, var, direction);`

Output parameters:

None.

Input parameters:

portname is the identifier to be used for the EDIF port name ("str")

var is a target mode variable providing the signal to the external port.

direction is "in" or "out" to select the port direction.

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure connects a signal to an external EDIF port in the netlist file.

For an input port the variable mode must be VALUE or CLOCK.

For an output port the variable mode must be VALUE, STATIC or CLOCK. For a value mode variable the call to **external()** must occur after the variable is assigned.

9.32 file - create file variables

Call:

file(id, path);

Output parameters:

none

Input parameters:

param	mode	type	
ident	-	-	variable identifier
path	I	str	file path string

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure creates a variable of type **file**. **ident** is the variable identifier. The path name may be relative to the current directory or may be absolute (a leading \).

A file must be opened by calling inbuilt procedure **open()**([section 9.46](#) p223) before it can be read or written.

A number of files are already open as global variables. **stdin** is the standard input, **stdout** and **stderr** are standard input and standard error, and **rpt** is the 3PL report file. The report file is **design_name.rpt** if the 3PL was called with the -r option, otherwise it is **stdout**.

Files are read as characters by
readln()([section 10.122](#) p323)
scanf()([section 9.67](#) p245)
fscanf()([section 9.35](#) p214)
 and are read as bytes by
readbyte()([section 10.121](#) p323).

Files are written as characters by
writeln()([section 9.86](#) p264)
printf()([section 9.49](#) p228)
fprintf()([section 9.34](#) p213)
 and are written as bytes by
writebyte()([section 9.85](#) p263).

The two modes cannot be mixed; a file read in one mode cannot be subsequently read in the opposite mode, and similarly for writing.

The open file can be closed by calling inbuilt procedure **close()**([section 9.12](#) p196).

See also inbuilt functions -

filecanread() ([section 10.40](#) p285)
filecanwrite() ([section 10.41](#) p286)
filedelete() ([section 10.42](#) p286)
fileexists() ([section 10.43](#) p286)
filegetparent() ([section 10.44](#) p287)
fileisdir() ([section 10.45](#) p287)
fileisfile() ([section 10.46](#) p287)

filelastmod() (section 10.47 p288)
filelength() (section 10.48 p288)
filelist() (section 10.49 p289)
filelistdirs() (section 10.50 p289)
filelistfiles() (section 10.51 p289)
filemkdir() (section 10.52 p290)
filerename() (section 10.53 p290)
filesetreadonly() (section 10.54 p291)
hadeof() (section 10.63 p295)
readbyte() (section 10.121 p323)
readln() (section 10.122 p323)

See also inbuilt procedures -

fprintf() (section 9.34 p213)
fscanf() (section 9.35 p214)

9.33 float - create one or more immediate floating point variables

Call:

float(ident, init);

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2... }	-	-	variable identifier(s)
init	I	float	optional initial value

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create one or more immediate floating point variables. If more than one variable is to be created the list of variables must be in braces ({...}). An integer or floating point expression may be provided as the second argument to initialise the variables, otherwise they will be initially zero.

9.34 fprintf - print a formatted line to an output file

Call:

fprintf(file <- fmt, v1, v2,...);

Output parameters:

param	mode	type	
file	I	file	output file

Input parameters:

param	mode	type	
fmt	I	str	format string
v1	I	any	input variable
v2	I	any	input variable
.	I	any	input variable
.	I	any	input variable

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure is equivalent to `fprintf()` in C. The format string is that used by the Java Formatter class.

The format argument may be followed by a single argument which is an array of pointers, for example **inargs** in the calling module, procedure or function. In that case the pointers are resolved and provide the input variable list to be printed.

The output file must have already been opened.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

See also -

printf() [section 9.49](#) p228

sprintf() [section 9.72](#) p250

writeln() [section 9.86](#) p264

writebyte() [section 9.85](#) p263.

9.35 fscanf - scan a line from a file into variables

Call:

```
fscanf(v1, v2,... <- file, fmt);
```

Output parameters:

param	mode	type	
v1	I	any	output variable
v2	I	any	output variable
.	I	any	output variable
.	I	any	output variable

Input parameters:

param	mode	type	
file	I	str	an input file
fmt	I	str	format string

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure scans a line from an input file extracting values from fields and assigning these to the output variables.

The input file must have already been opened.

The scanned line must consist of fields (tokens) separated by delimiters of commas or spaces (any number of either, i.e. the delimiter pattern is `[, ]+`).

The format string is not the same as that used in `scanf`, `sscanf` or `fscanf` in C. It is a string of single characters, one per conversion. Allowed conversions are -

'd'	integer
'u'	unsigned integer
'x'	hexadecimal integer
'o'	octal integer
'f'	floating point value
's'	string
'.'	convert according to the variable type
'*'	skip the token
' '	ignored
tab	ignored

The format string must not contain any other characters. Spaces or tabs can be included anywhere if desired. Because the scanner cannot match explicit fields from the format string and there are no conversion parameters such as field widths or precisions there is no need for a '%' escape character. The number of actual conversions (i.e. not including the skip) must be equal to the number of output variables.

The format argument may be null or "" in which case tokens will be converted and assigned to the variables based on the variable types.

Tokens that are to be converted as hexadecimal values must not have a leading 0x. Similarly a leading 0 will not be interpreted as indicating an octal value. The conversion character alone decides which radix will be used.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

See also -

scanf() [section 9.67](#) p245

sscanf() [section 9.68](#) p246

readln() [section 10.122](#) p323

readbyte() [section 10.121](#) p323

9.36 import - import a 'core'

Call:

```
import(OP1=o1, OP2=o2, ... <- name, bname, IP1=o1, IP2=i2, ...);
```

Output parameters:

See description

Input parameters:

See description

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure signals the compiler that an external 'core' (or macro) is to be linked into this program. The core is not necessarily one created by a 3PL program.

The first input argument must be a string expression giving the name of the 'core'.

The second argument is immediate mode and type "str" and is taken as an optional netlist block name for the imported core element. This identifier must begin with an alphabetic character and contain only alphabetic characters, digits and underscores. In addition it should not consist of 'b' followed by a string of digits since that is the block naming scheme used by the compiler to generate unique block names and hence the use of such a string in this context may result in a clash. If this argument is omitted the block will be labeled with a string of the form "bnnnn".

The remaining input arguments are signal inputs to the 'core'. These arguments may be passed using the key-matching syntax. The identifier on the LHS of the '=' is the port name used when linking the netlists. The RHS of the '=' is a 3PL variable or expression to drive the port. If the argument is a single identifier then it is taken as both the port identifier and the identifier of a variable driving the port.

Input port variables can be mode **input**, **value**, **selectvalue**, **static**, **priority** or **clock**. Input port variables may be any target type.

Output arguments are signal outputs from the 'core'. These arguments may be passed using key-matching syntax. The identifier on the LHS of the '=' is the port name used when linking the netlists. The RHS of the '=' is a 3PL variable to be an output from the port. Value variables must not contain queue reads. If the argument is a single identifier then it is taken as both the port identifier and the identifier of a variable driven by the port.

Output port variables may be mode **value**, **selectvalue** (3-state output) or **clock**. Value and select-value mode variables may be any target type. Output port variables which are mode **value** or mode **selectvalue** will have null clock domains. they may have their clock domains established by assigning attribute **domain** (see [section 3.11](#) p80) after the call to **import()** where they are passed as output arguments.

The number of bits in each argument must exactly match the associated core port width.

All input ports must be connected but connections to unwanted output ports may be omitted.

9.37 ident - extract an identifier from an argument

Call:

`ident(v);`

Output parameters:

none

Input parameters:

param	mode	type	
v	-	-	variable identifier

Parameter=argument allowed:

No

In-line target execution:

No

Description:

ident() can only be used to create an input parameter to a module, procedure or function. If it is called in any other context it generates a fatal error. **ident()** creates a variable **v** containing the identifier passed as a matching argument. The argument matching **ident()** must be an identifier with no appended subscripts or fields. The identifier may have a scope prepended. It is not relevant if the identifier matches a previously created variable.

The created variable **v** is a struct with two fields; **id**, which is the argument identifier as a string, and **context** which is a string "global", "file", "calling", "local" or "default" describing the scope context of the argument identifier.

This procedure is intended to enable the writing of procedures which create variables, e.g. the following procedure creates an initialised one-dimensional string array -

```
procedure
strarray (ident(v), var(init, "[]str")) {
    int(dim, dimension(init));
    str(name);
    switch (v.context) {
    case "global":
        name = "global."+v.id;
```

```

    case "file":
        name = "file."+v.id;
    case "calling":
        trace();
        fatal("strarray() - cannot have 'calling' context");
    case "local":
        trace();
        fatal("strarray() - cannot have 'local' context");
    case "default":
        name = "calling."+v.id; // create in calling scope
    }
    var(name->, "["+dim+"]str", init);
}

```

This procedure first gets the dimension of the required array from the dimension of the second argument, the array initialisation compound value. It then creates a string variable and this is assigned the required identifier string, **v.id**, with a scope keyword prepended. The **switch** makes sure that the scope requested, **v.context**, is sensible. "global" and "file" are OK and the default will create the variable in the scope of the caller. Scope "calling" would imply the scope prior to the calling scope which is not accessible. Scope "local" would be useful in case this procedure is called within a nested code block, but unfortunately that scope cannot be accessed. Finally a call to **var()** creates the required variable by using the indirect operator on local string variable **name**. The second argument to **var()** is the type (string array) and the third argument supplies the initial values of the array.

If called by

```
strarray (sar, {"s1", "s2"});
```

it would create a variable **sar** which is an array of "[2]str" initialised with entries "s1" and "s2". It is created in the scope of the call. Alternatively

```
strarray (global.sar, {"s1", "s2"});
```

would create **sar** in global scope and

```
strarray (file.sar, {"s1", "s2"});
```

would create **sar** in the current file module scope.

Unfortunately as discussed above we cannot use scope "local", e.g.

```

if (...) {
    ...
    strarray (local.sar, {"s1", "s2"});
}

```

because we already need scope "calling" in the procedure to create the variable in the calling scope. The call

```

if (...) {
    ...
    strarray (sar, {"s1", "s2"});
}

```

will create the variable in the scope of the code block of the **if**.¹

¹3PL could be changed to allow an extra step back up the scope chain beyond "calling" ("callingcalling"?) but that might be regarded as excessive!

9.38 includedir - add directory for source/module path

Call:

```
includedir(dir);
```

Output parameters:

none

Parameter=argument allowed:

No

Input parameters:

param	mode	type	
dir	I	str	directory string

Parameter=argument allowed:

No

In-line target execution:

No

Description:

The string argument is added to the list of directories to be searched for source/module files. If the string has a leading "/" the directory path will be absolute, otherwise it will be relative to the directory containing the initial source file. A trailing "/" on the string is optional.

9.39 input - create an input mode variable

Call:

```
input(ident, type, ....);
```

```
input(ident, ident2, type, ....);
```

Output parameters:

none

Input parameters:

param	mode	type	
ident		any	variable identifier
ident2		any	variable identifier
type	I	str or type	type or type string
....			attributes

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure creates an input mode variable.

An input variable represents a connection from off-chip via input pads and buffers.

The created variable has no clock domain. When an input variable is referenced in an assignment its clock domain will be established. The clock domain may also be established by assigning attribute **readdomain** (see [section 3.11](#) p80) in which case references to the variable must be in that clock domain (unless it is an argument to inbuilt function **accept()**).

For bi-directional pads both **input()** and **output()** are used with the same pins.

The resulting input mode variable cannot be assigned.

ident is the input mode variable identifier. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection 3.6.2 p53.

ident2 is an optional identifier for a differential output. This is only applicable for devices which support the IBUFDS_DIFF_OUT element. In that case **ident** is the positive output and **ident2** provides an inverted output. **ident2** is created as a value mode variable.

type is the variable type which can be specified by a type-mode variable or by a string-valued expression. Any type is allowed.

Any following arguments specify attributes. Each argument may specify an attribute by keymatch format, i.e. attribute=value where 'attribute' is an attribute identifier and 'value' is an appropriate mode and type (modes are all immediate except for clock domains which are mode clock). Alternatively an argument can be a map containing attributes where the key (string) is the attribute identifier and the associated value is the attribute value.

Attributes may also be associated with the input pins. This is described in chapter chapter 24 p663 **Constraints, Properties and Attributes** where other aspects of processing attributes and constraints for input, output and clock mode variables are also described.

Attributes can be added by using inbuilt procedure **attributes()** section 9.7 p192.

Attributes are required for two main purposes; control of netlist generation and constraint file entries. Attributes giving input pin locations and input buffer types directly control netlist generation. I/O standards and timing group names exercise control of place-and-route via a constraint file. A clock domain specification is an exception and is used to specify the clock domain of the input should that be necessary.

The applicable attributes are shown in the following table.

These attributes are not case-sensitive but will be converted to lower case.

The following attributes are used by **input()** itself.

input mode attributes			
differential	netlist	log	differential
loc	netlist	str or []str	locations

Where an attribute value is an array the dimension must be equal to the number of pins and each array member will apply to the corresponding output pad or buffer. A single attribute value will be applied to all pins.

differential indicates that the input pads are differential.

The **loc** attribute is mandatory. The array dimension must exactly match the number of bits in the variable type. If the input is differential the array dimension must be twice the number of bits, the locations being used in pairs for the +ve and the -ve inputs (in that order).

The following attribute is read-only and is created when **input()** is called and can be retrieved from the input mode variable if required by using inbuilt function **attribute()**.

input mode			
ids	3PL	[]str	element IDs

ids is a string array that contains the netlist identifiers of the input buffer elements. This attribute is read-only and is initialised when the input variable is created. This identifier can be used should there be a necessity to apply a constraint (other than location which is done via the port name) to the input buffer.

The following attributes are required for constraint generation and are handled at completion of immediate execution by library procedure **generateconstraints()** subsection 11.2.2.18 p388. The attribute values are single values which will be applied to all input bits.

input mode			
ibuf_delay_value	constraint	int or []int	extra delay to IBUF
ifd_delay_value	constraint	int or []int	extra delay to IFD
tnm	constraint	str or []str	timing group id
pullup	constraint	log or []log	pullup
pulldown	constraint	log or []log	pulldown
diff_term	constraint	log or []log	differential 100Ω termination
iostandard	constraint	str or []str	voltage standard

ibufdelay specifies a delay to be inserted in the non-registered output path from an input buffer. The values (integers) are 0 to 12 inclusive (default 0).

ifddelay specifies a delay to be inserted in the registered path within an input buffer. The values (integers) are 0 to 6 inclusive (default is a value generated automatically, see device databook).

tnm is the name of a timing group in which the input pads are to be included for timing constraint purposes.

If **pullup** or **pulldown** are true the input will be given a weak pullup or pulldown. It is not allowed for both these attributes to be true.

diff_term indicates that a 100Ω termination resistance be applied to a differential input.

iostandard is a string specifying one of the allowed I/O standards (see Xilinx documentation).

The following attributes are tolerated but ignored. They may be included as attributes for the output side of a bi-directional pin.

input mode			
slew	constraint	str or []str	timing group id
drive	constraint	uint or []uint	drive current

Any attributes that are not listed above will result in a fatal error.

Note: for each pin a netlist port "PORT_pin" is created where 'pin' is the pin identifying string. This port name is also the netlist identifier of the netlist from the output of the pad to the input of the IBUF or IBUFG. This convention can be used when generating associated constraints.

In the Xilinx implementation if a static variable is assigned from an input variable, and has no other assignments (except for assignment of a constant equal to its initial value, which is implemented as a reset), then the associated flip-flops will be located in IOBs. This means that the propagation delay from pad to flip-flop is the minimum achievable (also controlled by attribute **ifddelay**). If an input variable is used in an expression rather than being assigned directly to a static variable then a timing constraint from the input pads to the clock domain will usually need to be applied.

input() may be used as a parameter in a module, procedure or function definition. It must be passed an argument which is mode INPUT. The call to **input()** to create the parameter must have only a single argument, the parameter identifier and all references to the parameter will apply to the argument variable. That variable will point indirectly to the argument INPUT mode variable. Although it is conventional to utilise this for an input parameter, using **input()** for an output parameter/argument will nevertheless function correctly.

9.39.1 input() as a parameter

input() may be used as a parameter in a module, procedure or function definition. It must be passed an argument which is mode INPUT.

The call to **input()** to create the parameter need not have an identifier (1st argument), but in that case it cannot be referred to in the module, procedure or function body.

The type argument (2nd argument) is optional. If provided it must match the type of the argument exactly. If not provided the parameter type will be determined from the argument variable type.

When **input()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

All operations on the created parameter variable will be applied instead to its matching argument variable. If there is no argument to the input mode parameter then it will be created as a local variable.

Although it is conventional to utilise this for an input parameter, using **input()** for an output parameter/argument will nevertheless function correctly.

Note that the changes to attributes in the argument variable will take place even if the parameter input mode variable is not used!

9.40 int - create one or more immediate integer variables

Call:

```
int(ident, init);
```

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2... }	-	-	variable identifier(s)
init	I	int	optional initial value

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create one or more immediate mode variables of type INT.

ident is the variable identifier. If more than one variable is to be created the list of variables must be in braces *id1, id2, { ... }*. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection [3.6.2](#) p53.

An integer expression may be provided as the second argument to initialise the variables, otherwise they will be initially zero. Integers are 64 bits so the initialisation value is restricted to a range appropriate to that size.

9.41 liststacks - list the current state of the body and scope stacks

Call:

```
liststacks();
```

Output parameters:

none

Input parameters:

none

Parameter=argument allowed:

No

In-line target execution:

No

Description:

List the state of the body and scope stacks on standard output. **This procedure is intended only as an aid to 3PL interpreter development and has no other use.**

9.42 log - create one or more immediate logical variables

Call:

log(ident, init);

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2... }	-	-	variable identifier(s)
init	I	log	optional initial value

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create one or more immediate mode variables of type LOG.

ident is the variable identifier. If more than one variable is to be created the list of variables must be in braces *id1*, *id2*, {...}. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection [3.6.2](#) p53.

A logical expression may be provided as the second argument to initialise the variables, otherwise they will be initially false.

9.43 msg - print to standard out and the report file

Call:

msg(s, t);

Output parameters:

none

Input parameters:

param	mode	type	
s	I	str	message string
t	I	log	print a traceback

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Print a message to standard output and to the optional report file if it is active.

s is the message string.

t (optional) if true requests a stack trace.

See also -

println() [section 9.51](#) p230

texit() [section 9.79](#) p257

rpt() [section 9.65](#) p243.

9.44 nop - a no-operation (delay)

Call:

`nop(n);`

Output parameters:

none

Input parameters:

param	mode	type	
n	I	int	number of cycles

Parameter=argument allowed:

No

In-line target execution:

Yes

Description:

Do nothing for one or more cycles.

n is the number of cycles to delay. If omitted, the default is 1.

9.45 negclock - create a negative clock

Call:

`negclock(ident, c);`

Output parameters:

none

Input parameters:

param	mode	type	
ident	-	-	identifier of new -ve clock
c	C	-	reference clock variable

Parameter=argument allowed:

No

In-line target execution:

Yes

Description:

Creates a negative-edge clock where **ident** is the identifier of the new clock. The clock is a negative-edge version of the the clock **c** (which must not itself be a negative clock). The new clock will have the frequency and period parameters of the reference clock and will have 1.0 minus its duty cycle. If the frequency, period or duty cycle of either clock is changed the parameters of the associated clock will also change as appropriate.

A negative clock is not actually generated. Instead the created negative clock variable is linked to the related positive clock variable but is flagged as inverted and the signal identifier in the intermediate code list has an "_" appended. When a negative clock signal is encountered during generation of the EDIF netlist the positive clock signal is used followed by an inverter. Xilinx clocked elements have an inbuilt inverter so that is used rather than an explicit external inverter element. This avoids the use of a separate negative clock network.

9.46 open - open a file

Call:

`open(ident, rwa);`

Output parameters:

none

Input parameters:

param	mode	type	
var		any	variable identifier
rwa	I	str	read, write or append specifier

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure opens a file for reading or writing. The first argument is the file immediate variable identifier. The second argument is "r" for reading, "w" for writing or "a" for appending.

9.47 output - create an output mode variable

Call:

```
output(ident, type, src, ...);
```

Output parameters:

none

Input parameters:

param	mode	type	
var		any	optional variable identifier
type	I	str or type	optional type or type string
src	T	str	optional source
...			attributes

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure creates an output mode variable.

An output mode variable has no initial clock domain. The clock domain is determined by the first assignment. Any subsequent assignments, whether to the whole variable or a component of a compound type, must have the same clock domain. The clock domain may be established by assigning attribute **writedomain** (see attributes below). If the value to be assigned has a different clock domain it must be converted by one of the inbuilt functions **sample()** [section 10.128](#) p326 or **accept()** [section 10.2](#) p265.

The resulting output mode variable cannot be evaluated, i.e. it cannot appear on the RHS of an assignment statement, or anywhere else a value is expected.

For bi-directional pads both **input()** and **output()** are used with the same pins (locations).

ident is the variable identifier. If parameter **src** is matched then this argument is optional since there may be no reason to refer to the variable anywhere. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection [3.6.2](#) p53.

type is the variable type which can be specified by a variable of type "type" or by a string-valued expression. Any type is allowed except "map" or "list". If parameter **src** is matched then this argument is optional since the type can be extracted from the third argument, but if the third argument is immediate mode the type argument must be provided unless the third argument is type "log" (a target type "log"

can be unambiguously inferred from immediate type "log" whereas other types require a width to be specified for the target type).

src is an optional data source for the output. Its use avoids the need for assignments to the output mode variable to establish the data source (see 5.4). This can only be used for unconditional output, i.e. where 3-state output is not required. The argument may be immediate mode. If later assignment to the output variable is made this source is overridden. If this argument is not used an explicit assignment must be made to this output variable. If this argument is provided the type argument (second argument) can be omitted if the source is a target mode or if it is immediate mode type "log". If this argument is immediate mode and any type except "log" then the type argument must also be provided. The argument to parameter **src** cannot be queue, cmemory, rmemory or output mode.

src may also be a 1-dimensional list of variables and constants of type "log" in braces.

Note: if **src** contains value mode variables which have not been assigned values yet, the interim values will be set to the current clock domain. When ultimately these value variables are assigned values the values must have the same clock domain. Other than the above specific case no attempt is made by **output()** to establish the output clock domain of argument variables to parameter **src**.

Any following arguments specify attributes. Each argument may specify an attribute by keymatch format, i.e. attribute=value where 'attribute' is an attribute identifier and 'value' is an appropriate mode and type (modes are all immediate except for clock domains which are mode clock). Alternatively an argument can be a map containing attributes where the key (string) is the attribute identifier and the associated value is the attribute value.

Attributes may also be associated with the output pins. This is described in chapter [chapter 24](#) p663 **Constraints, Properties and Attributes** where other aspects of processing attributes and constraints for input, output and clock mode variables are also described.

Attributes can be added by using inbuilt procedure **attributes()** [section 9.7](#) p192.

Attributes are required for two main purposes; control of netlist generation and constraint file entries. Attributes giving input pin locations and output buffer types directly control netlist generation. I/O standards and timing group names exercise control of place-and-route via a constraint file. A clock domain specification is an exception and is used to specify the clock domain of the output should that be necessary.

When **output()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

The applicable attributes are shown in the following table.

These attributes are not case-sensitive but will be converted to lower case.

The following six attributes are used by **output()** itself.

output mode			
direct	3PL	log	enable direct from test
differential	netlist	log or []log	differential
defaultout	3PL	log or []log	default for unconnected outputs
noshadow	3PL	log or []log	don't duplicate output flip-flop
loc	constraint	str or []str	locations

Where an attribute value is an array the dimension must be equal to the number of pins and each array member will apply to the corresponding output pad or buffer. A single attribute value will be applied to all pins.

direct must have been set prior to any assignment statements using the output mode variable. The other attributes are not used until completion of immediate execution (excluding postprocessing).

direct specifies that if possible the output 3-state enable will be driven directly from a conditional signal, not through an execution thread and WHEN block. This can significantly simplify the logic and reduce the probability of failing to meet timing constraints locally. Note that when this attribute is in effect the 3-state enable is immediately active following configuration loading regardless of any 3PL startup mechanisms - see **start()**([section 9.73](#) p251).

direct cannot be passed via `iopins()` as it cannot be expanded to an array, being a single attribute applying to the whole variable and not to individual outputs.

differential indicates that the output pads are differential. Two locations must be supplied for each data bit (see **loc** above).

defaultout specifies that unconnected outputs variables, or parts of output variables, should be tied high or low. If this attribute is not supplied then the action on encountering unconnected outputs depends on the global directive "unass_out" (see [section 3.10](#) p75). The values must be **true** to tie the outputs high, or **false** to tie low.

noshadow when true inhibits the duplication of flip-flops whose output is connected to an output buffer. By default any flip-flop whose output is connected to an output buffer is placed in the IO pad containing the buffer. If the output of the flip-flop is used elsewhere then the flip-flop is duplicated in a CLB with all inputs in parallel with the pad flip-flop and the output of the duplicate used for internal logic. If **noshadow** is true then duplication will not be done, the flip-flop will of necessity reside in a CLB and its output will be routed to the output buffer. This will result in much worse clock-to-output time. Setting this attribute ensures that the pad output genuinely reflects the flip-flop state and not that of a duplicate but at the cost of delayed output.

The **loc** attribute is mandatory. The array dimension must exactly match the number of bits in the variable type. If the output is differential the array dimension must be twice the number of bits, the locations being used in pairs for the +ve and the -ve inputs (in that order).

The following attribute is read-only and is created when **output()** is called and can be retrieved from the output mode variable if required by using inbuilt function **attribute()**.

output mode			
ids	3PL	[]str	element IDs

ids is a string array that contains the netlist identifiers of the output buffer elements. This attribute is read-only and is initialised when the output variable is created. This identifier can be used should there be a necessity to apply a constraint (other than location which is done via the port name) to the output buffer.

The following attributes are required for constraint generation and are handled at completion of immediate execution by library procedure **generateconstraints()** [subsection 11.2.2.18](#) p388. The attribute values are single values which will be applied to all output bits.

output mode attributes		
slew	str or []str	slew rate
drive	int or []int	drive current
tnm	str or []str	timing group id
pullup	log or []log	pullup
pulldown	log or []log	pulldown
iostandard	str or []str	voltage standard

The **slew** attribute is the string "fast" or "slow" and selects one of two slew rates for Xilinx output buffers (see Xilinx documentation).

The **drive** attribute is an integer which specifies the output drive current for Xilinx output buffers (see Xilinx documentation).

tnm is the name of a timing group in which the output pads are to be included for timing constraint purposes.

If **pullup** or **pulldown** are true the output will be given a weak pullup or pulldown. It is not allowed for both these attributes to be true.

iostandard is a string specifying one of the allowed I/O standards (see Xilinx documentation).

The following attributes are tolerated but ignored. They may be included as attributes for the input side of a bi-directional pin.

input mode			
ibuf_delay_value	constraint	int	extra delay to IBUF
ifd_delay_value	constraint	int	extra delay to IFD

Any attributes that are not listed above will result in a fatal error.

There are two forms of assignment to output variables -

Unconditional assignment is used where 3-state output is not required. This has the form

```
output(ov, type,, attr);
ov := value;
```

If the type is compound then separate assignments can be made to different components, i.e. array members or struct fields. Output variables can have multiple assignments but the last assignment prior to termination of immediate execution is the one which will take effect. Any prior assignments have no effect.

For unconditional assignment it is usually more succinct to include the source as the third argument to output() rather than performing a separate explicit assignment, e.g. the case above could instead be written -

```
output(, value, attr);
```

Here the first argument is omitted since the identifier is not needed as the variable is not explicitly assigned. The second argument is omitted because the source is a target mode from which the type can be inferred. The third argument can also be a list of values of type "log", e.g.

```
static(1, "log");
output(, {false, 1, true}, loc=pins[3..5]);
```

where the list values can be immediate or allowed target modes, constants or variables, as long as they are type "log".

Conditional assignment is used to exercise 3-state control over the output. This has the form

```
output(ov, type,, attr);
when (enable)
    ov |- value;
```

Conditional assignment can only be done once² and it cannot follow an unconditional assignment. Similarly an unconditional assignment cannot follow a conditional assignment. A conditional assignment must assign the whole variable, not just some components of a compound type. Thus an output mode variable appearing in an unconditional assignment cannot have subscripts or struct fields. If fields or array members are to be assigned from different sources a value variable of the same type as the output mode variable can be used to collect up the disparate sources into a single variable for assignment.

Note: for each pin a netlist port "PORT_pin" is created where 'pin' is the pin identifying string. This port name is also the netlist identifier of the netlist from the output of the OBUF to the input of the pad. This convention can be used when generating associated constraints.

Note also that where an output mode variable does not have an identifier (the first argument is omitted) an identifier of the form OUPUTnnn is created for the variable where nnn is just an integer in sequence from 0.

9.47.1 output() as a parameter

output() may be used as a parameter in a module, procedure or function definition. It must be passed an argument which is mode OUTPUT.

²This restriction could be relaxed if warranted

The call to **output()** to create the parameter need not have an identifier (1st argument), but in that case it cannot be referred to in the module, procedure or function body.

The type argument (2nd argument) is optional. If provided it must match the type of the argument exactly. If not provided the parameter type will be determined from the argument variable type. If there is no argument the type can be inferred from the source argument type, described below.

The source argument (3rd argument) if supplied will overwrite any source in the argument variable. The type must match that of the variable as described above, or it may determine the variable type if it cannot be inferred otherwise.

Attributes may be supplied as the 4th and following arguments. These attributes will be added to or supplant those of the argument output mode variable.

All operations on the created parameter variable will be applied instead to its matching argument variable. If there is no argument to the output mode parameter then it will be created as a local variable.

Although it is conventional to utilise this for an output parameter, using **output()** for an input parameter/argument will nevertheless function correctly.

Note that the changes to a source value or to attributes in the argument variable will take place even if the parameter output mode variable is not used!

9.48 part - specify the FPGA

Call:

part(p);

Output parameters:

none

Input parameters:

p is the part specification (string).

Parameter=argument allowed:

No

In-line target execution:

no

description:

This procedure notifies the compiler of the FPGA part specification. The procedure creates directive "part", which contains the part specification string, and also creates a directive "family" which has the values "XC2S", "XCV", "XC3S", "XC2V", "XC2V" etc as appropriate.

The part specification must be as expected by the vendor software, e.g. "xcv300ebg432-7".

9.49 printf - print a formatted line to stdout

Call:

printf(fmt, v1, v2,...);

Output parameters:

none

Input parameters:

param	mode	type	
fmt	I	str	format string
v1	I	any	input variable
v2	I	any	input variable
.	I	any	input variable
.	I	any	input variable

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure is equivalent to `printf()` in C. The format string is that used by the Java Formatter class.

The format argument may be followed by a single argument which is an array of pointers, for example **inargs** in the calling module, procedure or function. In that case the pointers are resolved and provide the input variable list to be printed.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

See also -

fprintf() [section 9.34](#) p213

sprintf() [section 9.72](#) p250

writeln() [section 9.86](#) p264

writebyte() [section 9.85](#) p263.

9.50 presetdirective - enter a directive during parsing

Call:

`presetdirective(key, value);`

Output parameters:

none

Input parameters:

param	mode	type	
key	I	str	directive key
value	I	any	directive value (optional)

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Note: this procedure is executed when encountered during parsing. It is not an inbuilt procedure that can be called during immediate execution.

This procedure creates a directive.

It is intended to facilitate conditional file inclusion, see [section 3.2](#) p43.

The procedure call can only be preceded on the line by white space. The call syntax may include white space for formatting. The remainder of the line following the semicolon is ignored. These limitations arise because the statement is not recognised by the parser but is intercepted by the source file reader, the line not being passed to the token generator and parser.

The key is a string. The value may one of the 3PL immediate primitive types **int**, **log**, **float** or **str**. If the second argument is not provided the entry will be type **log** and value **true**.

The key and value are expressions which cannot contain variables or calls to user-defined functions. They can contain constants, calls to inbuilt functions, evaluations of global directives via inbuilt function **directive()** (See [section 10.28](#) p280) and expression operators. Binary integer constants of the form 0b.... or 0B.... are not allowed in this context, but hexadecimal and octal forms are accepted.

Because this inbuilt procedure is recognised prior to the token generator and parser an erroneous call may shadow a prior error. For example

```
module "nofile"
  presetdirective("name", 3.x);
```

would result in a fatal error for the second statement even when the first statement fails to open file "nofile". The first error is passed to the parser as a token but the second error causes an exception in the file reader before the parser has processed the error token.

see also inbuilt procedures

directive() [section 9.16](#) p200

removedirective [section 9.60](#) p238.

and inbuilt functions

directive() [section 10.28](#) p280.

direxists() [section 10.30](#) p281.

directives() [section 10.29](#) p281.

9.51 println - print a line

Call:

println(v);

Output parameters:

none

Input parameters:

param	mode	type	
v	I	int, log, str	value to be printed

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Print a line to standard out. The argument may be types int, uint, log, str or an enumerated type (the value identifier is printed).

See also **msg()** [section 9.43](#) p222.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

9.52 priority - create a priority variable

Call:

priority(ident, type, init, ...);

or

priority(ident, size, init, ...);

Output parameters:

none

Input parameters:

param	mode	type	
ident	-	-	variable identifier
size	I	type uint	optional array size
type	I	type str	optional type or type string
init	I	log	optional initialisation
...	I	-	optional attributes

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create a priority variable. Its type will be an array of "log".

size is an optional 2nd argument which is the dimension of the "[log]" array. Note that if a priority mode variable is created by using the inbuilt procedure **var()** [section 9.84](#) p261 and the array size is to be specified then a proper type argument must be used (see below).

type is a type variable or a string expression giving the variable type explicitly. The type must be an array of log or a fatal error will occur.

The array may be larger than required in which case unused entries, whether trailing or interspersed, will be pruned.

The size/type argument may be omitted and for a variable which is not a parameter it will be given the default type "[20]log". In the case of a parameter the variable will take the type of the argument.

The optional third argument, **init**, allows the inputs to the priority array to be assigned initially rather than by the use of explicit assignment statements later.

Additional trailing arguments may be passed to set attributes. These must be of the form attr=value.

A priority variable can be assigned or re-assigned at any time, the assigned variables only taking effect at the conclusion of immediate execution.

A priority variable has a single clock domain for reading and writing. When a priority variable is created using the inbuilt procedure **priority()** it has no associated clock domain. The clock domain is established by

- the first assignment to the variable
- the first use of the value of the variable

Note that evaluation of the variable in any context, for example in inbuilt or library modules, procedures or functions, will establish a clock domain if the variable does not yet have a clock domain. This will always be the current clock domain (unless the code has been explicitly written to establish the clock domain of its argument to something other than the current clock domain, in which case this will presumably be well documented).

When a priority variable is assigned, the clock domain of the assigned value must be the same as the current clock domain or null.

When a priority variable is assigned, If it has no clock domain the current clock domain will be established as the clock domain of the variable. If it has a clock domain it must be the same as the current clock domain.

When the value of a priority variable is used that value will have the clock domain of the variable.

Applicable attributes are shown in the following table. The attributes can be passed as arguments to this procedure or may be applied to the variable using the inbuilt procedure **attributes()** [section 9.7](#) p192.

When **priority()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

priority mode attributes			
domain	3PL	clock	force the clock domain
readonly	3PL	boolean	set to read only
multiplein	3PL	boolean	OR multiple inputs

domain forces the clock domain (both input and output) to the specified clock domain. If assigned **null** the current clock domain is selected. If the variable already has a clock domain as a result of having been assigned or evaluated then that clock domain is compared with the intended domain. If the assigned

clock domain agrees with that already in place then there is no further action, otherwise a fatal error occurs.

readonly true specifies that from now on this variable cannot be assigned to. This attribute cannot be assigned false.

Normally an assignment to a member of a priority variable array overwrites any previous assignment, the final assignment prior to completion of immediate execution prevailing. If attribute **multiplein** is set true then all assignments to a member of a priority variable array are collected and on completion of immediate execution these are ORed to form the input to that array member.

9.53 priunlock - unlock priority execution

Call:

```
priunlock(ident);
```

Output parameters:

none

Input parameters:

param	mode	type	
ident	Q	-	priority variable identifier

Parameter=argument allowed:

No

In-line target execution:

yes

Description:

This procedure creates in-line code which unlocks a priority variable previously locked by a **waitprilock** statement. It executes in one clock cycle.

For a description of usage, along with control structures **waitpri** and **waitprilock**, see section [section 6.17](#) p155.

9.54 pulse - generate a pulse - DEPRECATED!

Call:

```
pulse(v <- o);
```

Output parameters:

param	mode	type	
v	V	log	value carrying the pulse

Input parameters:

param	mode	type	
o	I	log	if true, OR with previous value

Parameter=argument allowed:

No

In-line target execution:

yes

Description:

Pulse a logical value variable for one clock cycle when this statement executes. The value is true for the clock cycle leading up to the clock edge on which execution occurs.

This procedure has been replaced by the inbuilt procedure **assert()**([section 9.5](#) p190). The latter procedure is identical in function and the name was changed as it was felt that 'pulse' was too hardware-oriented. In addition this procedure may be executed on several successive clock cycles in which case it

really could be viewed as a level rather than a pulse. The important thing is that it is asserted true in time for the next clock edge.

If the input argument is provided and is true the output value variable will be assigned the OR of its previous value and the execution pulse. If it has no previous assigned value it is assigned the execution pulse. Using this optional input argument a signal pulsed from more than one place can be generated.

Note: when execution pulses are ORed in this way, use of the value of the variable must follow the last call to pulse() since each call re-assigns the value variable!

Note that if the ORing option is not selected and multiple calls to **pulse()** are made with the same output argument, each call will re-assign the output value variable, which may be what is wanted or may not!

9.55 put - put a map entry

Call:

```
put(m, key, v);
```

Output parameters:

none

Input parameters:

param	mode	type	
m	I	map	a map
key	I	str, int, uint	a string key
v	I V C	any	the value

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

The item **v** is placed in the map **m** against key **key**. If that key is already in the map the value will be replaced. If the last argument is omitted null will be put against the key. The key should be type "str" but if it is type "int" or type "uint" it will be converted to type "str".

9.56 putall - put entries from another map

Call:

```
putall(m, m2);
```

Output parameters:

none

Input parameters:

param	mode	type	
m	I	map	a map
m2	I	map	another map

Parameter=argument allowed:

No

In-line target execution:

No

Description:

All items from map **m2** are added to map **m**. Where a key already exists its value will be replaced.

9.57 queue - create one or more queue variables

Call:

```
queue(ident, type, init, ...);
```

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2... }	-	-	variable identifier(s)
type	I	type str	optional type or type string
init	I	any	optional initialisation
...	I	-	optional attributes

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create one or more queue variables. If more than one variable is to be created the list of variables must be in braces ({...}). **type** is a type variable or a string expression giving the variable type. This may be omitted in the case of parameter creation in which case the variable will take the type of the argument.

The optional third argument allows the queue to be created with an initial datum, i.e. the queue will be initially available. This argument must be immediate mode. If the type is compound the initialisation argument will usually be a compound value, but it may be initialised by a single immediate value in which case all members of the compound type will be initialised to that value.

Additional trailing arguments may be passed to set attributes. These must be of the form attr=value.

A queue variable has a clock domain for reading and another clock domain for writing. When a static variable is created using the inbuilt procedure **queue()** it has no associated clock domains.

The clock domain for writing is established by -

- the first write (assignment) to the queue variable (or component of the variable if the type is compound)
- the use of the write availability operator > on the queue
- a call to **queuewritestatus()** on the queue
- a call to **reset()** on the queue

If the write clock domain is not yet established in any of these situations then it will be set to the current clock domain, otherwise it must be the same as the current clock domain.

The clock domain for reading is established by -

- the first read or examine of the variable (or component of the variable if the type is compound)
- the use of the read availability operator < on the queue
- a call to **queureadstatus()** on the queue

If the read clock domain is not yet established in any of these situations then it will be set to the current clock domain, otherwise it must be the same as the current clock domain.

The read and write clock domains will usually be the same, in which case the queue variable is called **synchronous**.

When a queue variable is written (assigned), If it has no write clock domain the current clock domain will be established as the write clock domain of the variable. If it has a write clock domain it must be the same as the current clock domain.

When a queue variable is written (assigned) the clock domain of the assigned value must be the same as the current clock domain or null.

If the **reset()** inbuilt procedure is used on a queue variable -

- if the queue variable has a write clock domain established, it must be the same as the current clock domain
- if the queue variable does not have a write clock domain established, it will be given the current clock domain

When a queue variable read or examined the resulting value will have the read clock domain of the queue variable.

Architectures supported -

Spartan-II, Spartan IIE	YES
Spartan-3	YES
Virtex, Virtex-E	YES
Virtex-2, Virtex-2 Pro	YES
Virtex-4, Virtex-5, Virtex-6	YES
XC9500, XC9500XV, XC9500XL	NO
Coolrunner XPLA3, Coolrunner-II	NO

Applicable attributes are shown in the following table. The attributes can be passed as arguments to this procedure or may be applied to the variable using the inbuilt procedure **attributes()** [section 9.7](#) p192.

When **queue()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

queue mode attributes			
depth	3PL	int	queue buffer depth
mindepth	3PL	int	minimum queue buffer depth
buffersize	3PL	int	queue buffer depth
minbuffersize	3PL	int	minimum queue buffer depth
readdomain	3PL	clock	force the read (output) clock domain
writedomain	3PL	clock	force the write (input) clock domain
domain	3PL	clock	force both the read and write clock domains
alu	3PL	boolean	use an ALU if appropriate
dsp	3PL	boolean	use a DSP block
fifo	3PL	boolean	use a hardware FIFO
queuereg	3PL	boolean	extra output register to improve timing
continuous	3PL	boolean	no startup dependence
ignoremodules	3PL	boolean	all reads treated as if in same module
readonly	3PL	boolean	set to read only
queuefieldzero	netlist	boolean	clear unassigned fields

These attributes are case-insensitive.

depth sets the depth of the queue buffer. The default depth is 2 for a synchronous queue and 16 for an asynchronous queue. The values allowed depend on the FPGA family.

	using cmemory	using rmemory
Spartan-II, Spartan IIE	16	256 to 4096
Spartan-3	16	512 to 16384
Virtex	16 to 32	256 to 4096
Virtex-E	16 to 32	256 to 4096
Virtex-II	16 to 64	512 to 16384
Virtex-II	16 to 64	512 to 16384
Virtex 4	16	512 to 16384
Virtex 5	16 to 128	1024 to 32768

A depth of 0 may also be set. This is a special case, creating an unbuffered queue. The use of unbuffered queues is very restricted - see [section 7.6](#) p177.

buffersize alternative older attribute name for **depth**.

mindepth sets the depth of the queue buffer but rounds the supplied depth up to the next allowed value.

minbuffersize alternative older attribute name for **mindepth**.

readdomain forces the read (output) clock domain to the specified clock domain. **writedomain** forces the write (input) clock domain to the specified clock domain. **domain** forces both the read (output) clock domain and the write (input) clock domain to the specified clock domain. If **null** is assigned, the current clock domain is selected. If the variable already has a clock domain as a result of having been assigned or evaluated then that clock domain is compared with the intended domain. If the assigned clock domain agrees with that already in place then there is no further action, otherwise a fatal error occurs.

alu specifies that assignments of expressions containing arithmetic operators '+' and '-' as the final or only operation can be combined into a single common adder/subtractor using CLBs. If the **alu** attribute is false it can override the global directive **ALU** to suppress the use of an ALU for an individual static variable.

The **alu** attribute can save logic and incurs no time penalty for the operations drawn in to the adder/-subtractor, but all other assignments suffer an increase in propagation time since assigned values must now traverse the shared adder/subtractor. This tradeoff must be considered in each case.

The **alu** attribute cannot be added or changed once the variable has been assigned.

The behaviour of the **dsp** attribute depends on the FPGA -

- For Virtex-5 and Virtex-6 it has a similar effect to the **alu** option in that it draws arithmetic and bitwise logic operations in assignments to the variable into a single ALU, but using a DSP block rather than CLBs.
- For other FPGAs the attribute has no effect.

The **dsp** attribute will not use a Virtex-5 or Virtex-6 DSP block as an ALU if the word width exceed 48 bits (multiplier operands are restricted to 25 by 18 bits).

Note that for Virtex-5 or Virtex-6 if the **alu** attribute is true but not the **dsp** attribute then arithmetic operators '+' and '-' as the final or only operation will be combined into a single common adder/subtractor using CLBs.

The **fifo** attribute when true specifies that the queue buffer will use inbuilt block RAM FIFO logic if possible. It is ignored if the FPGA does not have this logic or the queue variable is asynchronous and uses the read or write status outputs. This attribute cannot be used if the queue is type "null", has the default depth of 2, or has a default depth which such that CLB RAM is used (depth <= 16). In these cases an informative message is produced and the attribute is ignored. FIFO hardware can be enabled globally by setting directive **global.FIFO** in which case this attribute can be assigned false to counteract the directive, i.e. the attribute when set overrides the directive. The FIFO directive is false by default.

Use of the FIFO hardware is a compromise between speed and logic usage. The FIFO hardware saves CLB gates and flip-flops, which would otherwise be used, but has greater path delays which may cause failure to meet timing constraints.

The **queuereg** attribute when true causes an extra pipelined output register to be added. This reduces the clock-to-output time. The queue buffer logic incorporates the extra buffer with no change to functionality.

Continuous true specifies that if possible assignment will occur as soon as the FPGA is operational after configuration. This can increase operating speed by eliminating clock-enable logic. There are two situations where this can occur -

- the queue variable has a non-conditional assignment - the assignment operates continuously following configuration loading regardless of any 3PL startup mechanisms.
- a **when** statement has only unconditional assignments to variables in its nested code block(s) - the assignments are enabled directly from the **when** test expression (or its inverse) and are not controlled from execution threads initiated by the 3PL startup mechanisms.

see **start()** (section 9.73 p251). Note that if directive "continuous" is defined as type log with value true at the end of immediate execution, all queue variables will be treated as though they had the continuous attribute true.

ignoremodules true specifies that all queue reads will be treated as though from within the same module. Thus regardless of which module a queue read is contained within it will pop the queue when executed and the availability of the next datum will be seen within all modules in which the queue is read. This is distinct from the default behaviour of queue reads from multiple modules where a queue read from each relevant module must be executed before the next datum (if any) is seen as available to all the modules. This attribute is ignored for unbuffered queues.

The purpose of this option is to allow code within multiple modules to read queues while looking after exclusive access where necessary by means of explicit user code. This can also be done also by ensuring that all the queue reads are contained within one module but this restriction can be inconvenient and may result in more obscure code.

readonly true specifies that from now on this variable cannot be assigned to. This attribute cannot be assigned false.

queuefieldzero false inhibits the clearing of unassigned fields or array members when a queue variable is assigned. The default is **true**. If this attribute is made false for a queue variable which has a non-primitive type and an incomplete assignment is made, i.e. not all struct fields or array members are assigned, the unassigned elements for the datum will be undefined instead of explicitly zero. The value in making this attribute false is that it saves logic resources and reduces the delays on some data paths. Where a queue variable has a primitive type, or the whole variable is assigned in one statement or in multiple statements a par or sync block, this attribute has no effect.

9.58 ref - create one or more variables

Call:

ref(ident, mode, type, init);

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2...}	-	-	variable identifier(s)
mode	-	-	optional mode keyword
type	I	type str	optional type or type string
init	I	-	optional initial value

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create a variable. This procedure behaves the same as **var()** section 9.84 p261 except where used to create an input parameter to a module, procedure or function. In that case its argument is passed by reference and not by value. This situation arises where an immediate or value mode variable is passed as an argument and a pointer to the variable via the parameter is required within the module, procedure or function. The argument passed to a **ref()** parameter must be a variable and not an expression, but it may have subscripts or fields.

See **var()** section 9.84 p261 for details.

9.59 remove - remove a list entry

Call:

```
remove(l, index);
```

Output parameters:

none

Input parameters:

param	mode	type	
l	I	list	a list
index	I	int	an index

Parameter=argument allowed:

No

In-line target execution:

No

Description:

The item in list **l** at position **index** is removed. The index may range from 0 to **entries(l) - 1** inclusive. If **index** is outside this range a fatal error occurs.

9.60 removedirective - remove a directive

Call:

```
removedirective(key);
```

Output parameters:

none

Input parameters:

param	mode	type	
key	I	str	directive key

Parameter=argument allowed:

No

In-line target execution:

No

Description:

A directive is removed from the directive map. If the directive is not found a fatal error occurs.

see also inbuilt procedures

directive() [section 9.16](#) p200

presetdirective() [section 9.50](#) p229

and inbuilt functions

directive() [section 10.28](#) p280.

direxists() [section 10.30](#) p281.

directives() [section 10.29](#) p281.

9.61 reset - reset a static or queue variable

Call:

```
reset(v1, v2, ...);
```

Output parameters:

none

Input parameters:

param	mode	type	
v	S Q	any	variable reference
-	- -	-	-
-	- -	-	etc

Parameter=argument allowed:

No

In-line target execution:

Yes

Description:

This procedure resets one or more static or queue variables.

For a static variable reset it to its initial value. The argument may have immediate or target subscripts or field identifiers, i.e. can be a variable or any components of a variable (any array members or struct fields). A reset overrides any simultaneous assignment. A reset uses different inputs to the static variable than assignments and can use less logic and be faster than an assignment.

For a queue, clear the queue buffer making the queue unavailable. The argument must not have subscripts or field identifiers. It would be unusual to reset a single queue variable, the procedure normally being used to reset all queue variables in a pipeline to return to initial conditions. Because a queue variable can buffer more than one datum it is not possible to ascertain how many data are lost when a queue variable is reset, but resetting an entire pipeline results in a deterministic state.

A static or queue variable can be reset at more than one place in the program, the resets being ORed. A reset overrides any simultaneous assignment.

If **reset()** is used on a static variable -

- if the static variable has a clock domain established, it must be the same as the current clock domain
- if the static variable does not have a clock domain established, it will be given the current clock domain

If **reset()** is used on a queue variable -

- if the queue variable has a write clock domain established, it must be the same as the current clock domain
- if the static variable does not have a write clock domain established, the write clock domain will be set to the current clock domain

A queue reset will remove the write availability during the reset blocking any write. If the queue is synchronous the reset takes one clock cycle. If the queue is asynchronous the reset takes one write clock cycle plus a following read clock cycle plus another following write clock cycle.

Note that if an immediate value, or a variable which is value mode and is constant, is assigned to a static variable or one of its components, the constant is checked against the initial value for that variable or component. If they are the same the variable or component is reset to its initial value rather than being assigned. Thus **reset()** need not be used explicitly since assigning a value equal to the initial value will implicitly invoke a reset rather than an assignment.

9.62 rmemory - create a registered output memory

Call:

```
rmemory(name, ports, atype, dtype, init, ...);
```

Output parameters:

none

Input parameters:

param	mode	type	
name	-	-	memory identifier
ports	I	int	number of ports
atype	I	int	address type
dtype	I	int	data type
init	I	[]int	optional initialisation (variable or compound value)
...	I	-	optional attributes

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

Create a registered output RAM.

name is an identifier for the memory. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection 3.6.2 p53.

ports is the number of ports.

atype is the address type. This can be any target type with the constraint that the total number of bits must not exceed the maximum block RAM address width for the FPGA type. For Virtex, Virtex E and Spartan 2 this is 12 bits. For Virtex2, Virtex2 PRO and Spartan 3 this is 14 bits.

dtype is the data type. This can be any target type.

init is an optional initialisation array. The initialisation data is a one-dimensional variable or array which is no larger than the depth of the memory. It may be smaller in which case the memory is zero padded. The types and structure of the initialisation variable or structured constant after the first dimension must match the memory data type. Immediate constants are padded to fit target elements but must not exceed the target width. If the initialisation argument is omitted the memory will be initialised to all zero.

An initial value may also be set later by calling inbuilt procedure **attributes()**

Additional trailing arguments may be passed to set attributes. These must be of the form attr=value.

section 9.7 p192 with key "init". A number of constraint strings or string arrays may be appended. Each constraint will be applied to each component block RAM, in each case with "INST *name*" prepended where *name* is the netlist identifier of each block RAM.

This procedure may create a module, procedure or function parameter in which case only the first argument is needed (the memory identifier).

In the case of Xilinx FPGAs registered output memory is block RAM. This can have one or two access ports.

Either port can be read or written (or both).

Architectures supported -

Spartan-II, Spartan-IIE	YES
Spartan 3, Spartan 6	YES
Virtex, Virtex-E	YES
Virtex-II, Virtex-II Pro	YES
Virtex 4, Virtex 5, Virtex 6	YES
series 7, Zynq	YES
XC9500, XC9500XV, XC9500XL	NO
Coolrunner XPLA3, Coolrunner-II	NO

The range of memory depths that have been implemented is shown in the following table. The depth required is determined from the number of bits in the address type. Memories with depths smaller than

the minimum listed are still implemented, but the physical memory will be the minimum size shown and the smaller depth will be achieved by tying higher address bits low.

	1 port	2 port
Spartan-II, Spartan IIE, Virtex, Virtex-E	256 - 4096	256 - 4096
Spartan-3, Virtex-II, Virtex-II Pro, Virtex 4,	512 - 16384	512 - 16384
Spartan 6, Virtex 5, Virtex 6, series 7, Zynq	1024 - 32768	1024 - 32768

Registered output memory will employ the smallest address width which will encompass the address type. The data may be any width, multiple blocks being used if the data width exceeds the block data width associated with the address width. Both ports have the same address and data widths (the physical devices allow these to differ).

Applicable attributes are shown in the following table. The attributes can be passed as arguments to this procedure or may be applied to the variable using the inbuilt procedure **attributes()** [section 9.7](#) p192.

When **rmemory()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

rmemory mode attributes			
init	3PL		initial value
continuous	3PL	boolean	continuous enable
writemode	netlist	string	write behaviour
writemodea	netlist	string	write behaviour for port A
writemodeb	netlist	string	write behaviour for port B
domain	3PL	clock	force clock domain for all ports
domain0	3PL	clock	force clock domain port0
domain1	3PL	clock	force clock domain port1

These attributes are case-insensitive.

init sets an initial value for the memory. Any previous initial value will be overwritten. The initial value must be a compound type and the nested type must be an immediate type equivalent to the memory data type, i.e. same type but without a width field. The initial value is not used until completion of immediate execution when variables are processed into target code and hence it can be changed at any time during immediate execution.

continuous true specifies that the block RAM enables are tied high instead of being gated by the read enable ORed with the write enable. This can gain speed and reduce routing resource usage but at the cost of increased noise and power consumption (and hence also FPGA temperature).

write_mode allows the default write mode to be changed for block RAMs on Xilinx XC3S (Spartan 3), XC2V (Virtex 2) and XC2VP (Virtex 2 PRO) and XC4V (Virtex 4). The default is "write_first". Other write modes are "read_first" and "no_change". **write_mode_a** and **write_mode_b** select the write mode on separate ports of dual-port block RAM.

domain forces the clock domain (both input and output) for all ports to the specified clock domain. If assigned **null** the current clock domain is selected. If the variable already has a clock domain as a result of having been assigned or evaluated then that clock domain is compared with the intended domain. If the assigned clock domain agrees with that already in place then there is no further action, otherwise a fatal error occurs.

domain0 and **domain1** force the clock domain (both input and output) for port 0 or 1 of registered-output memory to the specified clock domain. If assigned **null** the current clock domain is selected. If the variable already has a clock domain as a result of having been assigned or evaluated then that clock domain is compared with the intended domain. If the assigned clock domain agrees with that already in place then there is no further action, otherwise a fatal error occurs. These attributes cannot be set for combinatorial-output memory.

9.63 rmemoryread - read a registered output memory

Call:

```
rmemoryread(m, port, addr);
```

Output parameters:

none

Input parameters:

param	mode	type	
m	M	-	memory being read
port	I	int	port
addr	I V SV S Q PR	any	memory address

Parameter=argument allowed:

Yes

In-line target execution:

Yes

Description:

Read from a port of a registered output memory.

m is the memory mode variable identifier.

port is the memory port.

addr is the address.

The address may be any type, including a compound type, but the type must match the type given when the registered output memory was created (see **rmemory()** [section 9.62](#) p239) but primitive component widths may differ in which case padding or truncation will occur.

If the address input argument contains queue reads execution of the write will wait on queue read availability.

The value resulting from the read will be available from the following clock cycle at the output of the memory. This value can be accessed via the inbuilt function **rmemoryout** [section 10.127](#) p326.

Multiple reads and writes on a port are allowed and the address and data inputs will be multiplexed, however there is no access arbitration and it is up to the application code to ensure exclusive access. Coincident accesses will result in data corruption.

A call to **rmemorywrite()** ([section 9.64](#) p242) or **rmemoryread()** ([section 9.63](#) p242) sets the clock domain for the memory port to the current clock domain. All further calls to either of these inbuilt procedures for the same port of the same registered output memory must be in the same clock domain (or a fatal error results).

9.64 rmemorywrite - write to a registered output memory

Call:

```
rmemorywrite(m, port, addr, data);
```

Output parameters:

none

Input parameters:

param	mode	type	
m	M	-	memory being written
port	I	int	port
addr	I V SV S Q PR	any	memory address
data	I V SV S Q PR	any	data to be written

Parameter=argument allowed:

Yes

In-line target execution:

Yes

Description:

Write to a port of a registered output memory. The registered output of the port is also changed to the written value.

m is the memory mode variable identifier.

port is the memory port.

addr is the address.

data is the data to be written.

The data and address may be any type, including compound types, but the types must match the types given when the combinatorial output memory was created (see **rmemory()** [section 9.62](#) p239) but primitive component widths may differ in which case padding or truncation will occur.

If the address input or data input arguments contain queue reads execution of the write will wait on queue read availability.

Multiple reads and writes on a port are allowed and the address and data inputs will be multiplexed, however there is no access arbitration and it is up to the application code to ensure exclusive access. Coincident accesses will result in data corruption.

A call to **rmemorywrite()** ([section 9.64](#) p242) or **rmemoryread()** ([section 9.63](#) p242) sets the clock domain for the memory port to the current clock domain. All further calls to either of these inbuilt procedures for the same port of the same registered output memory must be in the same clock domain (or a fatal error results).

9.65 rpt - print to the report file

Call:

```
rpt(s);
```

Output parameters:

none

Input parameters:

param	mode	type	
s	I	str	message string

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Print a message to the optional report file if it is active.

s is the message string.

See also -

println() [section 9.51](#) p230

texit() [section 9.79](#) p257

msg() [section 9.43](#) p222.

9.66 selectvalue - create one or more selectvalue variables

Call:

selectvalue(ident, type, default, ...);

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2... }	-	-	variable identifier(s)
type	I	str type	type or type string
default	I		default value
...	I	-	optional attributes

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create one or more selectvalue variables. If more than one variable is to be created the list of variables must be in braces ({...}).

type is a type variable or a string expression giving the variable type. This may be omitted in the case of parameter creation in which case the variable will take the type of the argument.

default is optional and specifies the value the selectvalue mode variable should have when no input is selected. If omitted, 0 is used ³.

Additional trailing arguments may be passed to set attributes. These must be of the form attr=value.

A selectvalue variable has no clock domain when created. The first assignment to the variable establishes its clock domain as the current clock domain. A selectvalue variable which is evaluated prior to any assignment is given the clock domain current when the evaluation occurs. Later assignments to the variable must be in the same clock domain.

Applicable attributes are shown in the following table. The attributes can be passed as arguments to this procedure or may be applied to the variable using the inbuilt procedure **attributes()** [section 9.7 p192](#).

When **selectvalue()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

selectvalue mode attributes			
domain	3PL	clock	force the clock domain
direct	3PL	boolean	enable direct from test
readonly	3PL	boolean	set to read only
alu	3P	boolean	use an ALU if appropriate
dsp	3PL	boolean	use a DSP block
threestate	3PL	boolean	use 3-state buffers rather than CLB logic

domain forces the clock domain (both input and output) to the specified clock domain. If assigned **null** the current clock domain is selected. If the variable already has a clock domain as a result of having been assigned or evaluated then that clock domain is compared with the intended domain. If the assigned clock domain agrees with that already in place then there is no further action, otherwise a fatal error occurs.

direct true specifies that if possible the output enable will be driven directly from a conditional signal, not through an execution thread and WHEN block. This can significantly simplify the logic and reduce

³If the "threestate" attribute is true the default value is implemented by the use of pullups or pulldowns. The specific FPGA platform may require a pullup explicitly on each signal, i.e. the default value may be constrained to be all 1s.

the probability of failing to meet timing constraints locally. Note that when this attribute is in effect the selectvalue is operative following configuration loading regardless of any 3PL startup mechanisms - see **start()**([section 9.73](#) p251).

readonly true specifies that from now on this variable cannot be assigned to. This attribute cannot be assigned false.

alu specifies that assignments of expressions containing arithmetic operators '+' and '-' as the final or only operation can be combined into a single common adder/subtractor using CLBs. If the **alu** attribute is false it can override the global directive **ALU** to suppress the use of an ALU for an individual static variable.

The **alu** attribute can save logic and incurs no time penalty for the operations drawn in to the adder/- subtracter, but all other assignments suffer an increase in propagation time since assigned values must now traverse the shared adder/subtractor. This tradeoff must be considered in each case.

The **alu** attribute is only used at completion of immediate execution.

The behaviour of the **dsp** attribute depends on the FPGA -

- For Virtex-5 and Virtex-6 it has a similar effect to the **alu** attribute in that it draws arithmetic and bitwise logic operations in assignments to the variable into a single ALU, but using a DSP block rather than CLBs.
- For other FPGAs the attribute has no effect.

The **dsp** attribute will not use a Virtex-5 or Virtex-6 DSP block as an ALU if the word width exceed 48 bits (multiplier operands are restricted to 25 by 18 bits).

Note that for Virtex-5 or Virtex-6 if the **alu** attribute is true but not the **dsp** attribute then arithmetic operators '+' and '-' as the final or only operation will be combined into a single common adder/subtractor using CLBs.

threestate true specifies that three state buffers are to be used rather than CLB logic. If the device family does not support three state drivers then CLB logic will be used regardless of this attribute. Devices which have internal three-state buffers are Spartan-II, Spartan IIE, Virtex, Virtex-E, virtex-II, Virtex-II Pro, XC9500, XC9500XV and XC9500XL.

9.67 scanf - scan a line from stdin into variables

Call:

```
scanf(v1, v2,... <- fmt);
```

Output parameters:

param	mode	type	
v1	I	any	output variable
v2	I	any	output variable
.	I	any	output variable
.	I	any	output variable

Input parameters:

param	mode	type	
fmt	I	str	format string

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure scans a line from standard input extracting values from fields and assigning these to the output variables.

The scanned line must consist of fields (tokens) separated by delimiters of commas or spaces (any number of either, i.e. the delimiter pattern is `[, ]+`).

The format string is not the same as that used in `scanf`, `sscanf` or `fscanf` in C. It is a string of single characters, one per conversion. Allowed conversions are -

'd'	integer
'u'	unsigned integer
'x'	hexadecimal integer
'o'	octal integer
'f'	floating point value
's'	string
'.'	convert according to the variable type
'*'	skip the token
' '	ignored
tab	ignored

The format string must not contain any other characters. Spaces or tabs can be included anywhere if desired. Because the scanner cannot match explicit fields from the format string and there are no conversion parameters such as field widths or precisions there is no need for a '%' escape character. The number of actual conversions (i.e. not including the skip) must be equal to the number of output variables.

The format argument may be null or "" in which case tokens will be converted and assigned to the variables based on the variable types.

Tokens that are to be converted as hexadecimal values must not have a leading 0x. Similarly a leading 0 will not be interpreted as indicating an octal value. The conversion character alone decides which radix will be used.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

See also -

fscanf() [section 9.35](#) p214

sscanf() [section 9.68](#) p246

readln() [section 10.122](#) p323

readbyte() [section 10.121](#) p323

9.68 sscanf - scan a string into variables

Call:

`sscanf(v1, v2,... <- line, fmt);`

Output parameters:

param	mode	type	
v1	I	any	output variable
v2	I	any	output variable
.	I	any	output variable
.	I	any	output variable

Input parameters:

param	mode	type	
line	I	str	line to be scanned
fmt	I	str	format string

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure scans an input string extracting values from fields and assigning these to the output variables.

The scanned line must consist of fields (tokens) separated by delimiters of commas or spaces (any number of either, i.e. the delimiter pattern is [,]+).

The format string is not the same as that used in scanf, sscanf or fscanf in C. It is a string of single characters, one per conversion. Allowed conversions are -

'd'	integer
'u'	unsigned integer
'x'	hexadecimal integer
'o'	octal integer
'f'	floating point value
's'	string
'.'	convert according to the variable type
'*'	skip the token
' '	ignored
tab	ignored

The format string must not contain any other characters. Spaces or tabs can be included anywhere if desired. Because the scanner cannot match explicit fields from the format string and there are no conversion parameters such as field widths or precisions there is no need for a '%' escape character. The number of actual conversions (i.e. not including the skip) must be equal to the number of output variables.

The format argument may be null or "" in which case tokens will be converted and assigned to the variables based on the variable types.

Tokens that are to be converted as hexadecimal values must not have a leading 0x. Similarly a leading 0 will not be interpreted as indicating an octal value. The conversion character alone decides which radix will be used.

See also -

scanf() [section 9.67](#) p245

fscanf() [section 9.35](#) p214

9.69 setdirective - enter a directive - DEPRECATED!

Call:

```
setdirective(key, value);
```

Output parameters:

none

Input parameters:

param	mode	type	
key	I	str	directive key
value	I	any	directive value (optional)

Parameter=argument allowed:

No

In-line target execution:

No

Description:

DEPRECATED - this procedure has been renamed **directive()** but is otherwise unchanged. Procedure setdirective() has been retained temporarily for backward compatibility.

There is a single global directive map. The key is a string. The value may one of the 3PL immediate primitive types **int**, **log**, **float** or **str**. If the second argument is not provided the entry will be type **log** and value **true**.

see also inbuilt procedures

directive() [section 9.16](#) p200.

presetdirective() [section 9.50](#) p229

removedirective [section 9.60](#) p238.

and inbuilt functions

directive() [section 10.28](#) p280.

direxists() [section 10.30](#) p281.

directives() [section 10.29](#) p281.

9.70 shell - execute a command in a shell

Call:

shell(status, out, err <- command, environ, dir, in);

Output parameters:

param	mode	type	
status	I	int, uint	status returned from shell
out	I	str	standard output from shell
err	I	str	standard error from shell

Input parameters:

param	mode	type	
command	I	str	command to be executed
environ	I	map	optional environment variables
dir	I	str	optional working directory
in	I	str	optional string to standard input to command

Parameter=argument allowed:

Yes

In-line target execution:

No

Description:

This procedure executes the command string passed as an argument using a shell. The default shell is **/usr/bin/bash**, or if not found **/bin/bash**, but this may be overridden by setting directive **shell** to a string giving the path of the required shell.

The first input parameter **command** is a string containing the command path with any command line options and arguments.

The optional second input parameter, **environ**, is a map of environment variables. For each entry the key is the environment variable and the associated map value is the variable value. Value types are restricted to "int", "uint", "str" and "float", other types resulting in a fatal error. If no argument is supplied for this parameter the environment on the call to 3PL is passed to the shell.

Note the shell may read user configuration files (such as **.bash_profile**) which may alter or replace **environ** before **command** is searched for and executed. To avoid this possibility consider using **exec()** [section 9.28](#) p209.

The optional third input parameter, **dir**, is a string specifying the working directory. If no argument is supplied for this parameter the current directory on the call to 3PL is used.

The optional fourth input parameter, **in**, is a string to be copied to standard input of the shell process. Note that if the shell or command closes its standard input stream, a fatal error occurs.

If the first output parameter, **status**, is given an argument, the status returned from the shell is copied to this variable. The argument must be immediate mode and type "int" or "uint".

If the second output parameter, **out**, is given an argument, the standard output from the shell is copied to this variable. The argument must be immediate mode and type "str". The resulting string is usually one or more lines, each terminated by a newline character.

If the third output parameter, **err**, is given an argument, the standard error from the shell is copied to this variable. The argument must be immediate mode and type "str". The resulting string is usually one or more lines, each terminated by a newline character.

If the shell cannot be found, or cannot be executed, a fatal error occurs.

This procedure waits until the shell exits. If the shell, or the command executed by the shell, is signalled then this procedure continues as for a normal exit.

This procedure assumes that the operating system is a version of Unix, Linux or MacOS.

See also **exec()** [section 9.28](#) p209.

9.71 special - vendor-specific special functions

Call:

`special(...);`

Output parameters:

none

Input parameters:

see description

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure is used to implement special functions particular to the FPGA vendor. The first argument is an integer selecting the required special function. At the moment there are only two special functions as described below. Others could be added if required. Output arguments are not at the moment accepted as the existing special functions do not use any.

9.71.1 special 0 - add a line to the Xilinx NCF file

special(0, format_string, port, target_arguments)

The optional target arguments may be value, static, clock or selectvalue mode. Each target argument must be a single bit.

The format string is written to the ISE NCF file. The format string may contain %n or %N conversions at which points netlist identifiers are to be inserted. The netlist identifiers are obtained from the input signals which are matched to the conversions in order. There will usually be only one of these though the format allows any number. %n is replaced by the identifier of the next input signal argument. %N is similar to the above but changes the identifier to upper case and removes any leading "GLOB_" string. This is used for generating clock timing group names from the clock signal identifier. A % followed by any other character is unchanged. The target mode signal arguments are matched to the %n and %N escape sequences left to right. The number of target mode arguments must be sufficient for the number of %n and %N escape sequences, any excess nets will be ignored.

The port argument is optional and may be null (or missing if there are no target arguments to follow). If supplied it is a port name to which the constraint applies and no target arguments are required. If the port is ultimately unconnected to logic then the constraint will be omitted.

This can also be called via library procedure **addncfline(str(s))** [subsection 11.2.2.2](#) p377 in **common.3pl**, which only allows a single optional target argument, however this meets most requirements.

9.71.2 special 1 - add a line to the Xilinx XDC file

special(1, format_string, port, target_arguments)

The optional target arguments may be value, static, clock or selectvalue mode. Each target argument must be a single bit.

The format string is copied to the XDC file. It must have a trailing ';' but does not need a newline character. The format string and input signal arguments are processed as above for the ISE NCF file constraints.

The port argument is optional and may be null (or missing if there are no target arguments to follow). If supplied it is a port name to which the constraint applies and no target arguments are required. If the port is ultimately unconnected to logic then the constraint will be omitted.

This can also be called via library procedure **addxdcline(str(fmt), value(sig))** [subsection 11.2.2.3 p377](#) in **common.3pl**, which only allows a single optional target argument, however this meets most requirements.

9.72 sprintf - copy a formatted line to a string variable

Call:

```
sprintf(s <- fmt, v1, v2,...);
```

Output parameters:

param	mode	type	
s	I	str	output string

Input parameters:

param	mode	type	
fmt	I	str	format string
v1	I	any	input variable
v2	I	any	input variable
.	I	any	input variable
.	I	any	input variable

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure is equivalent to sprintf() in C. The format string is that used by the Java Formatter class.

The format argument may be followed by a single argument which is an array of pointers, for example **inargs** in the calling module, procedure or function. In that case the pointers are resolved and provide the input variable list to be formatted.

See also -

printf() [section 9.49](#) p228

fprintf() [section 9.34](#) p213

writeln() [section 9.86](#) p264

writebyte() [section 9.85](#) p263

and library function **sprintf()** [subsection 11.13.5.1](#) p522

9.73 start - set the start condition for execution threads

Call:

start(v, clk);

Output parameters:

none

Input parameters:

param	mode	type	
v	V,I	log, uint	start condition
clk	C	optional clock	

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure sets up the start of all execution threads for FPGAs. It cannot be used for XC2C or XC95 CPLDs.

Every execution thread is started by a start pulse in its clock domain.

These start pulses can be generated in four ways -

1. If no call is made to start(), each clock domain will have an associated start pulse generated as soon as GWE is asserted after the configuration has been loaded.
2. if the argument to start() is value mode or input mode and is type "log" then a start pulse will be generated for every clock domain when that input is asserted.
3. if the argument to start() is immediate mode and type "uint" or "int" then the start will be delayed by the number of clock cycles specified by that integer 1st argument. The clock domain to which the count is to apply is specified by the optional 2nd argument or else is the current clock. Following that delay start pulses will be generated in all clock domains.
4. if the argument to start() is immediate mode and type "float" then the start will be delayed by the time in second specified by the floating point 1st argument. The time will be in periods of the relevant clock domain, either the optional 2nd argument or else the current clock domain. Note that choosing the slowest clock will use the smallest counter.

In all cases nothing occurs until the assertion of GWE.

For case 2 the global execution start signal must be derived from external signals. The start level will usually either be a single external input signal or else an expression decoding an external bus or other communications interface. Any expression must not depend on any state devices which are clocked (e.g. static variables, queue variables, memory variables etc.) since these will not operate until the primary start pulse has occurred.

When start() is called for all except case 1 a clock domain is required. This will be the current clock domain by default or it can be specified in the optional 2nd argument. If the clock domain is null a fatal error will occur.

In the case of Spartan 6, Virtex 4 and all later FPGAs there is a START hardware element available which provides an output signal that is asserted high on completion of configuration startup. This can be used where the configuration startup waits for DCMs and PLLs to lock. The start of execution of 3PL threads can be delayed until this signal is asserted, for example for Spartan 6 -

```
value(s1, "log");
element("STARTUP_SPARTAN6");
elementoutput(s1 <- , "EOS");
elementend();
start(s1);
```

9.74 statemachine - create a state machine - DEPRECATED!

Call:

```
statemachine(sf1, st1, e1, sf2, st2, e2, ..., reset);
```

Output parameters:

none

Input parameters:

param	mode	type	
sf1	S	log	from state variable
st1	S	log	to state variable
e1	I, V, SV, S, PR	log	state transition expression
sf2	S	log	from state variable
st2	S	log	to state variable
e2	I, V, SV, S, PR	log	state transition expression
.	.		.
reset	V, SV, S, PR		optional reset

Parameter=argument allowed:

No

In-line target execution:

No

Description:

No in-line code is created (the state machine is effectively a module) but the state variables are in the scope in which this procedure is called. Although not an error, it should not be called from within a target control structure such as *when*, *while* or *do while* as it does not generate in-line code and hence would not appear within the body of the control structure or be mediated by it in any way.

States are represented by static logical variables, one and only one of which is true at any time (this is called 'one-hot encoding').

The procedure has a multiple of 3 input arguments, with an optional extra trailing input argument, and no output arguments. The input arguments are triples, each of which describes a state transition. The 1st and 2nd arguments of a triple are the previous and next state of the transition and the 3rd argument of the triple is the logical expression causing the transition. The optional trailing input argument is a reset logical expression which, when it evaluates as true, will reset the state machine to its initial state. This reset overrides all other state transitions and resets all state variables to initial states (the first state true and all other states false).

The states are unsubscripted variable identifiers. For each state appearing in a transition triple a target static variable of type log will be created in the current scope. If such a variable already exists it will be used instead (this allows variables to be created in advance for use in previous expressions prior to the call which creates the state machine). State variable identifiers must not have been previously used in any accessible scope except as described above. In addition if a state variable is pre-declared it must not be assigned a value or be initialised. State variables created by this procedure can never be assigned by user code but otherwise can be used as normal static logical variables. State variables for one state machine cannot be used as state variables in another state machine (the same identifiers can be used where the scope is not shared).

The transition expressions may be immediate or target expressions. They may contain queue reads. Transition expressions must evaluate to type log.

The optional trailing reset expression must evaluate to type log and must not contain queues.

The first state encountered in the transition list is the initial state and hence will be initialised to true. This applies even when the initial state variable is pre-declared in which case this procedure will initialise that variable.

At the moment there is little checking of the validity of the state transitions. The only check is that all states can be entered. Invalid state transitions may result in multiple or no state variables being true. Validation checking may be improved later.

This procedure has been deprecated. Use the language construct *petrinet*.

9.75 static - create one or more static variables

Call:

static(ident, type, ...);

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2...}	-	-	variable identifier(s)
type	I	type str	optional type or type string
init	I		optional initial value
...	I	-	optional attributes

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create one or more value variables. If more than one variable is to be created the list of variables must be in braces ({...}). **type** is a type variable or a string expression giving the variable type. This may be omitted in the case of parameter creation in which case the variable will take the type of the argument.

The third argument is an optional initial value which must be immediate mode. If the type is compound the initialisation argument will usually be a compound value, but it may be initialised by a single immediate value in which case all members of the compound type will be initialised to that value. If more than one variable is created all will be initialised to the same value (the compound type members would have to be the same primitive type). An initial value may also be set later by calling inbuilt procedure **attributes()** [section 9.7](#) p192 with key "init". If no initial value has been given the initial value will be 0, false etc. as appropriate. If no initial value has been provided then attribute "init" will not appear in the attributes map returned by inbuilt function **attributes()** [section 10.11](#) p271.

Additional trailing arguments may be passed to set attributes. These must be of the form attr=value.

A static variable has a single clock domain for reading and writing. When a static variable is created using the inbuilt procedure **static()** it has no associated clock domain. The clock domain is established by

- the first assignment to the variable (or component of the variable if the type is compound)
- the first use of the value of the variable or one of its components

Note that evaluation of the variable in any context, for example in inbuilt or library modules, procedures or functions, will establish a clock domain if the variable does not yet have a clock domain. This will always be the current clock domain (unless the code has been explicitly written to establish the clock domain of its argument to something other than the current clock domain, in which case this will presumably be well documented).

When a static variable is assigned, the clock domain of the assigned value must be the same as the current clock domain or null.

When a static variable is assigned, If it has no clock domain the current clock domain will be established as the clock domain of the variable. If it has a clock domain it must be the same as the current clock domain.

If the **reset()** inbuilt procedure is used on a static variable -

- if the static variable has a clock domain established, it must be the same as the current clock domain

- if the static variable does not have a clock domain established, it will be given the current clock domain

When the value of a static variable is used that value will have the clock domain of the variable.

Applicable attributes are shown in the following table. The attributes can be passed as arguments to this procedure or may be applied to the variable using the inbuilt procedure **attributes()** [section 9.7 p192](#) and may be read back using inbuilt function **attributes()** [section 10.11 p271](#).

When **static()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

static mode attributes			
init	3PL		initial value
domain	3PL	clock	force the clock domain
continuous	3PL	boolean	no startup dependence
readonly	3PL	boolean	set to read only
alu	3PL	boolean	use an ALU if appropriate
dsp	3PL	boolean	use a DSP block
loc	UC:C	string(s)	locations
iob	UC:C	boolean	locate in I/O blocks
ids	UC:C	string(s)	element IDs

3PL - 3PL uses the attribute itself for control logic or optimisation.

UC:C - 3PL does not use the attribute - it must be used by 3PL source code to generate constraint file entries.

These attributes are case-insensitive.

init sets an initial value for the variable. Any previous initial value will be overwritten. If the static variable type is compound the initialisation value will usually be a compound value, but it may be initialised by a single immediate value in which case all members of the compound type will be initialised to that value (the compound type members would have to be the same primitive type). The initial value is not used until completion of immediate execution when variables are processed into target code and hence it can be changed at any time during immediate execution.

domain forces the clock domain (both input and output) to the specified clock domain. If assigned **null** the current clock domain is selected. If the variable already has a clock domain as a result of having been assigned or evaluated then that clock domain is compared with the intended domain. If the assigned clock domain agrees with that already in place then there is no further action, otherwise a fatal error occurs.

Continuous true specifies that if possible assignment will occur as soon as the FPGA is operational after configuration. This can increase operating speed by eliminating clock-enable logic. There are two situations where this can occur -

- the static variable has a non-conditional assignment - the assignment operates continuously following configuration loading regardless of any 3PL startup mechanisms.
- a **when** statement has only unconditional assignments to variables in its nested code block(s) - the assignments are enabled directly from the **when** test expression (or its inverse) and are not controlled from execution threads initiated by the 3PL startup mechanisms.

see **start()** ([section 9.73 p251](#)). Note that if directive "continuous" is defined as type log with value true, all static variables which do not have this attribute set will be treated as though they had the continuous attribute true. The directive prevails if the attribute has not been set, otherwise the attribute is used.

The "continuous" attribute should be set prior to assignments to the variable to have effect. Similarly the "continuous" directive should be set as early as possible in immediate execution or else prior to immediate execution using **presetdirective()**.

readonly true specifies that from now on this variable cannot be assigned to. This attribute cannot be assigned false.

alu specifies that assignments of expressions containing arithmetic operators '+' and '-' as the final or only operation can be combined into a single common adder/subtractor using CLBs. If the **alu** attribute

is false it can override the global directive **ALU** to suppress the use of an ALU for an individual static variable.

The **alu** attribute can save logic and incurs no time penalty for the operations drawn in to the adder/-subtractor, but all other assignments suffer an increase in propagation time since assigned values must now traverse the shared adder/subtractor. This tradeoff must be considered in each case.

The **alu** attribute can only be applied to a static variable of primitive type; it will be ignored following a warning message otherwise.

The **alu** attribute is only used at completion of immediate execution.

The behaviour of the **dsp** attribute depends on the FPGA -

- For Virtex-5 and Virtex-6 it has two effects -
 - it causes the static variable to be implemented using the output register of a DSP block instead of flip-flops.
 - For Virtex-5 and Virtex-6 it has a similar effect to the **alu** attribute in that it draws arithmetic and bitwise logic operations in assignments to the variable into a single ALU, but using a DSP block rather than CLBs.
- For Virtex-2, Virtex-2Pro and Virtex-4⁴ it causes the static variable to be implemented using the output register of a multiplier block instead of flip-flops.
- For other FPGAs the attribute has no effect.

The **dsp** attribute will not implement the static variable using a multiplier or DSP register in the following cases -

- the variable is initialised to a non-zero value
- the variable has the **iob** attribute

The attribute will be ignored and a warning message issued in these cases.

Note that the use of the **dsp** attribute reduces the number of flip-flops required by using multiplier or DSP block output registers instead, but this may make timing constraints harder to meet because -

- the clock edge to output time of these may be worse than that for CLB flip-flops
- the bits can no longer be distributed among CLBs and routing must instead converge on one or more multiplier or DSP blocks

In addition the **loc** attribute if supplied will be ignored (with a warning message). For Virtex-2 and Virtex-2 PRO each multiplier element has a 36 bit register. For Virtex-5 and Virtex-6 the output register is 48 bits. For variables whose type is not primitive each component will be implemented in a separate multiplier or DSP register. For components larger than the hardware register size sufficient DSP registers will be used to accommodate the component (but as noted above registers larger than 48 bits cannot use the Virtex-5 or Virtex-6 DSP block as an ALU). In the case of Virtex-5 or Virtex-6, setting this attribute true also sets the **alu** attribute true and the ALU logic will be implemented in the same DSP block. Note that if the limitations listed above are not met the **dsp** attribute will be cleared (an associated warning message will be printed), the variable will be implemented using CLB flip-flops and if an ALU is generated it will use CLB logic.

Note that for Virtex-5 or Virtex-6 if the **alu** attribute is true but not the **dsp** attribute then the static variable will be implemented using flip-flops and arithmetic operators '+' and '-' as the final or only operation can be combined into a single common adder/subtractor using CLBs.

loc provides absolute locations for the CLB flip-flops comprising the static variable (register). Locations are matched to flip-flops starting at the least significant bit. For a single bit static this attribute may be a string, otherwise it must be a string array. Each string must have the location format expected by the vendor for the FPGA type. 3PL does not at present provide support for relative placement of flip-flops in static variables though this could easily be added if required.

iob specifies that the flip-flops implementing the static variable are to be placed in IOB blocks rather than CLB blocks (this can also be requested by using the -pr option in the Xilinx map utility). If the value of the variable is used in addition to being connected to output buffers then the flip-flops will be duplicated in parallel, one set in the I/O blocks and the other in CLBs.

⁴Virtex-4 has a DSP block but for reasons of local (non)usage the full capabilities of Virtex4 are not supported by 3PL. The DSP block is treated as a simple multiplier block.

ids is a string array that contains the identifiers of the netlist elements (in this case D-type flip-flops) that implement the static mode variable (register). This attribute is read-only and is initialised when the variable is created. It can be used to apply constraints, such as absolute or relative location, to these elements. The array is ordered from least to most significant bit. For example to place a static variable in specific FPGA slice locations (for ISE) -

```
static(s, "uint:8");
var(slocs, "[8]str", {"", "", "", .....}); // array of slice locations
var(sids, "[8]str", attribute(s, ids));      // array of netlist element identifiers
for (int(i) ; i<8 ; i++)
    addncfline("INST \"\"+ sids[i] + \"\" LOC=" + slocs[i]);
```

For Vivado -

```
static(s, "uint:8");
var(slocs, "[8]str", {"", "", "", .....}); // array of slice locations
var(sids, "[8]str", attribute(s, ids));      // array of netlist element identifiers
for (int(i) ; i<8 ; i++)
    addxdcline("set_property LOC " + slocs[i] + "[ get_cells " + sids[i] + "]);
```

9.76 stop - stop a target execution stream

Call:

stop();

Output parameters:

none

Input parameters:

none

Parameter=argument allowed:

No

In-line target execution:

yes

Description:

Stop execution of the containing execution stream. Ultimately this will stop the module infinite loop in which it is contained. Parallel execution streams will continue to execute until their mutual completion point is reached. A stop() can be skipped or cancelled by an **unless** construct on the containing block which will force the code to skip to the end of the block.

9.77 str - create one or more immediate string variables

Call:

str(ident, init);

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2... }	-	-	variable identifier(s)
init	I	log	optional initial value

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create one or more immediate variables of type STR.

ident is the variable identifier. If more than one variable is to be created the list of variables must be in braces *id1, id2, {...}*. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection 3.6.2 p53.

A string expression may be provided as the second argument to initialise the variables, otherwise they will be initially "". If the type of the initialisation argument is INT, LOG or FLOAT it will be converted to a string.

9.78 struct - create a struct

Call:

```
struct(ident, f1, t1, f2, t2, ...);
```

Output parameters:

none

Input parameters:

param	mode	type	
ident	-	-	struct identifier
f1 or {fa, fb...}	-	-	field identifier(s)
t1	I	str type	field type or type string
f2 or {fc, fd...}	-	-	field identifier(s)
t2	I	str type	field type or type string
.	.	.	etc.

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create a struct.

ident is the identifier of the struct.

f1 is the identifier of the first field. More than one field of the same type can be specified by including a list of field identifiers in braces (*{fa, fb, ...}*).

t1 is the type or string type description of the first field.

Optionally other argument pairs, follow.

If a field identifier is the keyword **extends** then the second component of the argument pair must be a struct identifier. The fields of that struct will be included in this struct at that point.

The type of a field can of course be another struct in which case that struct is nested (rather than being extended).

See subsection 3.7.3 p59 for a description of structs and also an alternative string specification for creating them.

9.79 textit - print stack trace and terminate execution

Call:

```
textit(message);
```

Output parameters:

none

Input parameters:

param	mode	type	
message	I	str	optional message

Parameter=argument allowed:

No

In-line target execution:

No

Description:

If the argument is provided, print the message on standard out, prepended by the source code location. Print a stack trace to standard out, then terminate immediate execution. This lists all module, procedure and function calls and the source code file names and line numbers at which they occurred.

See also -

exit() [section 9.29](#) p210**trace()** [section 9.80](#) p258

9.80 trace - print a stack trace

Call:

trace();

Output parameters:

none

Input parameters:

none

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Print a stack trace to standard out. This lists all module, procedure and function calls and the source code file names and line numbers at which they occurred.

See also -

exit() [section 9.29](#) p210**texit()** [section 9.79](#) p257

9.81 type - create a type variable

Call:

type(v, descr);

Output parameters:

none

Input parameters:

param	mode	type	
v	-	-	created immediate variable
descr	I	str	type description string

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure creates a variable of type "type". The first argument is the identifier of the created variable. The second argument is the type specification string or another type variable or type returned from a function.

A type variable can be used as the type argument to the various variable creation procedures instead of a string argument. It can also be passed as an argument to modules, procedures or functions where the matching parameter is also type "type".

Note that a type variable can also be simply assigned a type description string which will result in the type being generated from the string.

9.82 useclock - set the current clock

Call:

```
useclock(c);
```

Output parameters:

none

Input parameters:

param	mode	type	
c	C	-	clock mode variable

Parameter=argument allowed:

No

In-line target execution:

No

Description:

The procedure call sets the current clock domain, given by the clock mode variable argument, and pushes the previous current clock domain onto a stack. If there is no argument the previous clock domain is restored from the stack, or if that stack is now empty there is now no current clock domain.

To explicitly set a null current clock domain the argument can be the keyword **null**. The previous clock domain can later be restored by calling **useclock()** with no argument. Another way to get a null current clock domain is to call **useclock()** with no argument enough times to empty the clock stack.

The clock cannot be changed within an execution stream. This procedure can only be called at the top level in a module (i.e. not nested within target control structures). Calls elsewhere will result in a fatal 3PL immediate execution error.

9.83 value - create one or more value variables

Call:

```
value(ident, type, init, ...);
```

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2... }	-	-	variable identifier(s)
type	I	str type	optional type or type string
init	I	str	optional initial value
...	I	-	optional attributes

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create one or more value variables. If more than one variable is to be created the list of variables must be in braces ({...}).

type is a type variable or a string expression giving the variable type. This may be omitted in the case of parameter creation in which case the variable will take the type of the argument. It may also be omitted if an initialisation argument is given in which case it uses the type of that argument. In either case for an immediate constant the type is constructed from the size of the constant and its sign.

A value mode variable may be created as type "empty", in which case it can be assigned any type. Following assignment the variable will retain the type of the RHS value assigned to it. Any further assignments to that variable will be subject to type checking as normal. A value variable can be cleared and its type changed to "empty" by calling inbuilt procedure **emptyvalue** [section 9.25](#) p207.

The third argument is an optional initial value which may be a target expression or an immediate expression, i.e. a value may be assigned an initial constant -

```
static(a, "uint:10");
value(v1, "uint:8", 5);           // constant initial value
value(v2, "uint:12", a + 37);    // target expression initial value
```

If the value is an array it may be initialised by a single immediate value in which case all members of the array will be initialised to that value, otherwise it can be initialised by a compound constant or an immediate mode variable with a matching type.

```
var(n, "[3]int", {9, 6, 3});
value(v1, "[3]uint:8", {5, 7, 9}); // constant initial array
value(v2, "[3]uint:8", n);         // variable initial array
value(v3, "[10]uint:12", 37);      // single value for array
```

For a compound type the initialisation does not have to initialise every member of the type, e.g.

```
var(n, "[3]int", {9, 6, 3});
value(v1, "[6]uint:8", {5, 7, 9}); // constant smaller initial array
value(v2, "[6]uint:8", {5,, 9});  // constant initial array with gap
value(v2, "[6]uint:8", n);        // smaller variable initial array
```

Where an initial value is provided the type argument can occasionally be omitted, in which case the value variable will take the type of the initial value, e.g.

```
static({s1, s2}, "uint:8");
value(v, , s1 + s2);
```

Here the initial expression is the sum of two unsigned 8-bit integers which will give a 9-bit result, hence **v** will have type "uint:9". For this to be accepted the initial expression must have an unambiguous target type. If the initialisation argument is a compound value then the type argument must be given.

In the call to **value()** additional trailing arguments may be passed to set attributes. These must be of the form attr=value.

A value variable has no initial default clock domain. A value variable which is evaluated prior to its assignment is given an initial value which has the clock domain that was current when the evaluation occurred. Later assignment to the variable must be in the same clock domain. A clock domain may be established by setting the domain attribute (see below).

Applicable attributes are shown in the following table. The attributes can be passed as arguments to this procedure or may be applied to the variable using the inbuilt procedure **attributes()** [section 9.7](#) p192.

When **value()** is used to create a module, procedure or function parameter it cannot be passed attributes. The created parameter variable will adopt the attributes of the associated argument.

value mode attributes			
domain	3PL	clock	resolve the clock domains of assigned values
readonly	3PL	boolean	set to read only

These attributes are case-insensitive.

domain resolves the clock domains of any values assigned to the value variable to the specified clock domain. If assigned **null** the current clock domain is selected. If the values are already resolved their clock domains are compared with the intended domain. If the assigned clock domain agrees with that within the values then there is no further action, otherwise a fatal error occurs. If a value has not been assigned, the clock domain will be set and a future assignment must have the same clock domain. The attribute will affect all variables contained in the value expression. Setting the clock domain has a particular use for value variables which are outputs of an **import()** [section 9.36](#) p215 procedure. Note that if this attribute is read back a null is returned.

readonly true specifies that from now on this variable cannot be assigned to. This attribute cannot be assigned false.

9.84 var - create one or more variables

Call:

```
var(ident, mode, type, init);
```

Output parameters:

none

Input parameters:

param	mode	type	
ident or {id1, id2...}	-	-	variable identifier(s)
mode	-	-	optional mode keyword
type	I	type str	optional type or type string
init	I	-	optional initial value

Parameter=argument allowed:

No

In-line target execution:

No

Description:

Create a variable.

ident is the variable identifier. If more than one variable is to be created the list of variables must be in braces *id1, id2, {...}*. The variable identifier can also be constructed from a string by using the indirect operator on a string variable or parenthesised expression, see subsection [3.6.2](#) p53.

mode is an optional mode keyword. This keyword must be one of

- immediate
- value
- selectvalue
- static
- queue
- priority

The mode may be omitted (completely - no ",," is needed) in which case the mode will be **immediate** by default unless **var** is being used to create a module, procedure or function parameter in which case it will use the mode of the matching argument (this can only be done for a single variable, not a variable list).

var() cannot create a variable whose mode is rmemory or cmemory but it can create a module, procedure or function parameter which is passed an rmemory or cmemory mode variable as argument. In that case the resulting parameter variable will reference the argument variable.

See further description at [subsection 4.5.3](#) p106.

Variables of other modes cannot be created by this procedure. The inbuilt procedures for creating these are -

mode	procedure
input	input() section 9.39 p218
output	output() section 9.47 p224
cmemory	cmemory() section 9.13 p197
rmemory	rmemory() section 9.62 p239
clock	clock() section 9.10 p194

type is a type variable or a string expression giving the variable type. This may be omitted in the case of parameter creation in which case the variable will take the type of the argument (this can only be done for a single variable, not a variable list). For priority mode the type must be an array of log.

Where the first argument is a list of identifiers all of the created variables will have the same type.

init is an optional initialisation expression, or compound value, matching the type. For a compound type the initialisation may be a single immediate value in which case all members of the compound type will be initialised to that value. If omitted the variables, or their components, except for a value mode variable, will be initialised by default to 0, false "" etc. The initialisation argument cannot be used if the type argument is omitted, but the type argument may be "empty" in which case the type will be derived from the initialisation. Note that the initialisation argument may be IMMEDIATE, VALUE or CLOCK mode and the created variable will have the same mode if the mode argument is omitted (otherwise the modes must agree).

Where the type is an array with missing dimensions the dimensions can be derived from an initialisation array, be it a variable or a compound value. For multiple variables all will be initialised to the same array values but note that the arrays are distinct - see [subsection 3.8.2](#) p73. If an argument is supplied the parameter will take its dimensions from the argument instead of the initialisation array.

Where the variable created is a parameter to a module, procedure or function, missing type dimensions can be derived from the dimensions of the argument. If no argument is supplied the dimensions can still be derived from a default initialiser. If the call to **var()** creates more than one parameter, i.e. the first argument is a bracketed identifier list, missing dimensions cannot be supplied from associated arguments, but they can still be derived from an initialiser. Remember that where an identifier list is supplied all the variables created will have the same type, though the dimensions for that type may have been derived from an initialiser. This means that the arguments matching those parameter variables must have the same dimensions. If parameters are to match arguments which are arrays of different dimensions then separate calls to **var()** must be used for each parameter.

Value mode variables can be initialised to an immediate or target mode expression.

For convenience there are specific variable creation inbuilt procedures for some modes or types -

mode	type	procedure	description	section
immediate	int	int()	immediate integer variable	section 9.40 p221
immediate	log	log()	immediate logical variable	section 9.42 p222
immediate	str	str()	immediate string variable	section 9.77 p256
immediate	float	float()	immediate floating point variable	section 9.33 p213
immediate	type	type()	immediate type variable	section 9.81 p258
immediate	enum	enum() enumv()	immediate enumerated type variable	section 9.26 p207 section 9.27 p208
immediate	struct	struct()	immediate struct variable	section 9.78 p257
immediate	class	class()	immediate class variable	section 9.9 p193
value	any	value()	value variable	section 9.83 p259
selectvalue	any	selectvalue()	selectvalue variable	section 9.66 p244
static	any	static()	static variable	section 9.75 p253
queue	any	queue()	queue variable	section 9.57 p234
priority	any	priority()	priority variable	section 9.52 p230
cmemory	any	cmemory()	combinatorial output memory variable	section 9.13 p197
rmemory	any	rmemory()	registered output memory variable	section 9.62 p239
clock	-	clock()	clock variable	section 9.10 p194

Target mode variables have no associated clock domain when they are created. For a description of the establishment of clock domains see the specific target variable creation inbuilt procedures **value()** [section 9.83](#) p259, **selectvalue()** [section 9.66](#) p244, **static()** [section 9.75](#) p253, **queue()** [section 9.57](#) p234 and **priority()** [section 9.52](#) p230 or the descriptions of selectvalue variable assignment [section 6.2](#) p133 value variable assignment [section 5.2](#) p126 static variable assignment [section 6.3](#) p134 queue variable assignment [section 6.4](#) p136 or priority variable assignment [section 5.3](#) p128.

See also **var()** [section 9.58](#) p237 where a module, procedure or function input parameter needs to ensure that its argument is passed by reference and not by value. This situation arises where an immediate or value mode variable is passed as an argument and a pointer to the variable via the parameter is required within the module, procedure or function.

9.85 writebyte - write a byte to a file

Call:

writebyte(file, e);

Output parameters:

none

Input parameters:

param	mode	type	
file	I	file	file
e	I	any	value to print

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure writes a byte to an output file. The file must have already been opened as an output file. The immediate value may be types **bits**, **uint**, **int** or **log**. A log argument will write 0 or 1. The other types must have a value between 0 and 255 or a fatal error will occur.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

See also -

writeln([section 9.86](#) p264

printf()[section 9.49](#) p228
 fprintf()[section 9.34](#) p213
 sprintf()[section 9.72](#) p250.

9.86 writeln - write a line to a file

Call:

```
writeln(file, e);
```

Output parameters:

none

Input parameters:

param	mode	type	
file	I	file	file
e	I	any	value to print

Parameter=argument allowed:

No

In-line target execution:

No

Description:

This procedure writes a line to an output file. The file must have already been opened as an output file. The immediate value may be types **bits**, **uint**, **int**, **log**, **str** or **float**.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

See also -

printf()[section 9.49](#) p228
 fprintf()[section 9.34](#) p213
 sprintf()[section 9.72](#) p250
 writebyte()[section 9.85](#) p263.

Chapter 10 - Inbuilt Functions

The following functions are built into the compiler. User functions must not use identifiers that conflict with these.

10.1 abs - absolute value function

Call:

abs(v)

parameters:

param	mode	type	
v	any	int, fixed or float	

Parameter=argument allowed:

No

return mode:

immediate or target

return type:

int, fixed or float as per argument

Description:

Returns the absolute value of the argument. The mode of the input may be IMMEDIATE, VALUE, SELECTVALUE, STATIC or QUEUE.

An IMMEDIATE mode input may only be type "int" or "float" and the result will be the same type.

Target mode inputs may only be types "int:", "fixed:." or "float:." and the result will be the same type.

10.2 accept - ignore clock domain differences

Call:

accept(v, c)

parameters:

param	mode	type	
v	V	any	variable or expression
c	C	-	optional clock variable or null

Parameter=argument allowed:

Yes

return mode:

value

return type:

type of argument

Description:

This function returns a value mode result which is the argument variable or expression with its clock domain changed. If no second argument is provided the returned value is the same as the first argument but with the current clock domain. If the second argument is provided it should be a clock mode variable identifier and the returned value is the same as the first argument but with the designated clock domain. If the second argument is **null** or the comma is present but the argument is omitted, the returned value is the same as the first argument but with no clock domain. The clock domain of the argument is neither checked nor resolved (assigned).

If the first argument clock domain is the same as the current clock domain the argument is simply returned as the value.

The first argument cannot contain queue reads and cannot be a memory or clock mode variable.

This function is used where it is desired to use the value of a variable or expression in a clock domain that differs from that of the argument without synchronising it in any way.

This function should only be used in the following circumstances -

- the frequencies of the clocks of the two clock domains have a simple integer relationship, the clocks are locked in phase and the lower frequency clock edges are coincident with an edge of the higher frequency clock
- the argument variable or expression only changes occasionally and the occurrence of corruption on a small proportion of data is acceptable

In the former case a time constraint from the first clock domain to the second clock domain should be imposed, the value being the shortest time interval between a clock edge in the source domain and the next edge in the destination domain. In the latter case a time constraint would be meaningless.

If Clocks are not suitably related and corrupted data is not acceptable then the inbuilt function **sample()** [section 10.128](#) p326 should be used instead.

WARNING:

This function must be used with great care! It introduces the possibility that data may be changing at the time of the local clock transition, resulting in data corruption. The consequences can be serious if this occurs in a test value in a *when*, target *while* or target *do while* structure in which case execution control can be unpredictably and permanently disrupted¹. The use of *accept()* in such contexts has not been prohibited since there are situations where it is safe. This is usually in the context of external asynchronous signals and requires the programmer to have (and correctly apply) detailed knowledge of signal timing.

10.3 acos - trigonometric arc cosine function

Call:

acos(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

¹This can occur because the control logic usually clocks the single bit test value into two flip-flops simultaneously. The routing delays to the flip-flops will differ and if the signal is glitching during setup the flip-flops may be set inconsistently. This could result in failure to execute either part of a when statement (which will kill the whole execution thread) or execution of both parts. Even worse disruption could occur in the case of loops.

return mode:

immediate

return type:

float

Description:

Returns the arc cosine of the immediate argument. The result is the angle in radian.

10.3.1 addsub - conditionally add or subtract

Call:

```
addsub(in1, in2, add, gate, xci)
```

parameters:

param	mode	type	
in1	V,SV,S,Q	uint, int	
in2	V,SV,S,Q	uint, int	
add	V,SV,S,Q	log	
gate	V,SV,S,Q	log	
xc	V,SV,S,Q	log	

Parameter=argument allowed:

Yes

return mode:

value

return type:

uint, int, ufixed, fixed

Description:

This function returns the sum or difference of its operands. If the input parameter **add** is true, the value returned is **in1** + **in2**, otherwise it is **in1** - **in2**.

If the optional input parameter **gate** has an argument then when the value of that argument is false the second input argument is gated to zero and thus the result is the first input argument regardless of whether the operation is add or subtract. When **gate** is false both **add** and **xc** have no effect.

If the optional input parameter **xc** has an argument then it is XORed with the carry in. This has the effect of incrementing the sum in the case of an addition or decrementing the difference in the case of a subtraction. Using the carry in this way is much more efficient than using a second adder to increment the sum or a second subtracter to decrement the difference.

The following table shows the result type for allowed input types. Types with a trailing ":" are target types.

	uint	int	uint:	int:	ufixed:	fixed:
uint	uint	int	uint:	int:	ufixed:	fixed:
int	int	int	int:	int:	fixed:	fixed:
uint:	uint:	int:	uint:	int:	ufixed:	fixed:
int:	int:	int:	int:	int:	fixed:	fixed:
ufixed:	ufixed:	fixed:	ufixed:	fixed:	ufixed:	fixed:
fixed:	fixed:	fixed:	fixed:	fixed:	fixed:	fixed:

If the operands are mixed signed/unsigned then the unsigned operand is increased in width by one by appending a zero bit at the most significant end. Integer operands are matched in size by padding the smaller operand. If the operands are mixed integer/fixed point the operands are aligned at the fixed point and padded at either end to match their sizes.

The returned value has the same number of significant bits as the maximum of the widths of **in1** and **in2** plus one (extra MSB).

If one or both operands are fixed point then the operand with the smallest fixed point offset is left shifted to align the binary points prior to adding or subtracting. The result width is the maximum of the aligned operand widths plus one (extra MSB) and the result binary point offset is the same as the aligned operand binary point offset.

Any or all operands may contain queue reads, in which case the availability of the function value will depend on the availability of the operands which contain queue reads.

10.4 arglist - pass lists, arrays or maps of arguments

Call:

arglist(k, v)

or

arglist(m)

parameters:

param	mode	type	
k	I	array or list	
v	I	array or list	
m	I	map	

Parameter=argument allowed:

No

return mode:

immediate

return type:

-

Description:

This function allows lists or maps of arguments to be inserted at any point within an input or output argument list. When called with two arguments the first is an array or a list of key strings. This argument may be explicitly 'null' (i.e. not omitted) if there are no key=value style arguments. The second argument is an array or list of arguments to be inserted. If both arguments are present the two arrays/lists must be the same size. The key array/list may have leading null entries as long as the complete resulting argument list to the enclosing module, procedure or function does not have key=value arguments preceding normal arguments (this is not illegal but is very confusing and should be avoided).

If this function is called with a single argument then it must be type "map" The map entries are inserted as key=value arguments.

Amongst other uses this function can be used to pass varargs arguments to deeper subroutines, e.g.

```
procedure p(varargs <- varargs) {
  p2(arglist(null, outargs) <- arglist(null, inargs));
}
```

or

```
procedure p(varargs <- varargs) {
  p2(arglist(outkeys, outargs) <- arglist(inkeys, inargs));
}
```

or more generally arguments from a procedure call may be inserted amongst other arguments, e.g.

```
procedure p(varargs) {
  .
  .
  p2(arglist(a, b, arglist(null, inargs), c);
  .
}
```


This function should only be called as an argument to a library or user-defined class, module, procedure or function. It should not be called as an argument to an inbuilt class, module, procedure or function. In that case it will result in error termination.

10.5 arraytostr - convert an array of integers to a string

Call:

arraytostr(ia)

parameters:

param	mode	type	
ia	I	[]uint	integer array

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Converts an array of integers, whose values are ASCII codes, into a string.

See also -

[section 10.138](#) p331 `strtoarray()` converting a string to an integer array.

10.6 arraytype - array type of a variable or expression

Call:

arraytype(v_or_e)

parameters:

param	mode	type	
v_or_e	-	-	variable or expression

Parameter=argument allowed:

No

return mode:

immediate

return type:

type

Description:

Return the array type of a variable or expression. If the type is not an array null is returned.

See also -

`type()` [section 10.160](#) p341

`typestring()` [section 10.162](#) p343

`primetypestring()` [section 10.110](#) p317

`imtotargettype()` [section 10.70](#) p298

`targetointype()` [section 10.150](#) p336

`typeval()` [section 10.163](#) p343

`typevalstring()` [section 10.164](#) p344

`typeisequal()` [section 10.161](#) p342

10.7 asin - trigonometric arc sine function

Call:

asin(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the arc sine of the immediate argument. The result is the angle in radian.

10.8 atan - single argument trigonometric arc tangent function

Call:

atan(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the arc tangent of the immediate argument. The result is the angle in radian.

10.9 atan2 - two argument trigonometric arc tangent function

Call:

atan2(f1, f2)

parameters:

param	mode	type	
f1	I	float	
f2	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the arc tangent of the 1st argument divided by the 2nd argument. The arguments must both be immediate mode. The result is the angle in radian.

10.10 attribute - get an attribute from a variable

Call:

attribute(v, attr)

parameters:

param	mode	type	
v	I	any	
attr	I	str	

Parameter=argument allowed:

No

return mode:

immediate

return type:

type of the attribute

Description:

Returns an attribute from the attributes map for the variable. For a list of attributes see [section 3.11 p80](#). The function will return attributes which are in the extended category.

The first argument is the variable identifier.

The 2nd argument is a string giving the attribute identifier.

If the specified attribute is not present in the attribute map then null is returned.

10.11 attributes - get the attributes of a variable

Call:

attributes(v, extended)

parameters:

param	mode	type	
v	I	any	
extended	I	log	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the attributes map for the variable. The returned map is not a copy (except when 'extended' is true) so any changes o the map will be reflected in the map withion the variable itself. For a list of attributes see [section 3.11 p80](#).

If the 2nd argument is present and true, some internal attributes of the variable will also be returned. These extra attributes are -

- **identifier** - the abbreviated identifier
- **eidentifier** - the extended identifier

- **mode** - the mode
- **type** - the type
- **typestring** - the type as a string
- **decloc** - the source file location where created
- **inputparam** - true if this is an input parameter
- **outputparam** - true if this is an output parameter
- **matched** - true if this is a parameter matched by an argument
- **readonly** - true if read-only
- **assigned** - true if the variable has been assigned
- **used** - true if it is a target mode variable and its value has been used
- **indirect** - true if the variable is indirect. This is the case for some parameter/argument situations and is also the case for clocks assigned to other clocks or clocks which are inverted.
- **clk_sinks** - if it is a clock mode variable the number of destinations to which the clock is connected
- **clk_srcs** - if it is a clock mode variable the number of sources from which the clock is driven (should be 1)

Note that these attributes cannot be set so this returned map should not be applied to a variable. If the second argument is omitted or false the extended attributes will be omitted and the returned attributes map is suitable for application to a variable.

Note also that an extended attributes map is a new map containing attributes copied from the attribute map of the variable. Any changes to the map will not affect the variable attributes.

10.12 biasedexponent - target float biased exponent

Call:

biasedexponent(arg)

parameters:

param	mode	type	
arg	V SV S Q IN	float	target float value

Parameter=argument allowed:

No

return mode:

value

return type:

uint

Description:

This function returns the biased exponent of a target floating point value. The biased exponent is return as type uint: whose width is equal to the width of the biased exponent.

10.13 biasedexponentwidth - width of target float biased exponent

Call:

biasedexponentwidth(arg)

parameters:

param	mode	type	
arg	I V SV S Q IN	type float	target float type or value

Parameter=argument allowed:

No

return mode:

immediate

return type:

uint

Description:

This function returns the number of bits in the biased exponent of a target floating point value. The argument may be a target floating point value or it may be a target floating point type.

10.14 bitsfor - number of bits to represent an integer - DEPRECATED!

Call:

bitsfor(arg)

parameters:

param	mode	type	
arg	I	uint or int	integer value

Parameter=argument allowed:

No

return mode:

immediate

return type:

uint

Description:

This function returns the number of bits needed to represent the +ve or -ve integer argument. For a +ve integer the sign bit is not included.

The function *size()* [section 10.132](#) p328 should be used instead. It will return the same value, and is more general in that it will accept target primitives as well as immediate arguments.

10.15 cast - cast a value to a another type or type width

Call:

cast(value, type)

parameters:

param	mode	type	
value	I, V, SV, S, Q	any	value to be cast
type	I	str, type	type string or type

Parameter=argument allowed:

No

return mode:

mode of first argument

return type:

type of second argument

Description:

For immediate types conversions between types *bits*, *uint*, *int* and *str* are allowed with the provisos -

- int to uint gives an error if the integer is -ve
- str to numeric gives an error if not correctly formatted

A string to be converted to numeric must be

- a string of 0's and 1's preceded by "0b" or 0B"
- a string of hexadecimal digits preceded by "0x" or "0X"

- a string of octal digits preceded by "0"
- a string of decimal digits whose 1st digit is not 0

Target mode type conversions allowed are shown in the following table -

		to							
		bits	uint	int	ufixed	fixed	log	float	enum
from	bits	az	az	as	az	as	az	az	-
	uint	az	az	az	cz	cz	az	-	az
	int	as	as	as	cs	cs	-	-	-
	ufixed	az	bz	bz	dz	dz	-	-	-
	fixed	as	-	bs	ds	ds	-	-	-
	log	az	az	-	-	-	a	-	-
	float	az	-	-	-	-	-	e	-
	enum	-	az	-	-	-	-	-	-

- a** copy right justified
- b** copy from left of binary point, ignoring fractional part
- c** copy to left of binary point and zero fractional part
- d** copy aligned with binary point truncating or zero padding least significant bits if size of fraction differs
- e** copy with adjustments to biased exponent and mantissa as required
- z** zero pad or truncate most significant end if width differs
- s** sign extend or truncate most significant end if width differs

If casting to or from compound types any types are allowed (including primitive types) as long as the numbers of bits are identical. Conversions into or out of compound types is carried out by mapping the bits in order without regard to component boundaries.

When casting between int or uint and fixed or ufixed the binary point is recognised and where appropriate the sign is propagated, i.e. the cast retains numeric sense.

When casting between bits and fixed or ufixed the binary point is not recognised and bits are simply right justified.

When casting between bits and int or fixed and the cast type is wider, sign extension is performed instead of zero padding.

Type float can only be cast to another float, adjusting the biased exponent and mantissa as required.

Type bits or uint can be cast to type log which will extract the rightmost bit as false (0) or true (1). Casting type log to bits or uint results in a single bit, zero padded where required.

Casting an enum to a uint provides the ordinal value (there is also an `ord()` built in function). Casting a uint to an enum selects the enum value given the ordinal.

As a special case a queue of type "null" can be cast to a value of type "log".

See also -

round() [section 10.126](#) p325)

trunc() [section 10.159](#) p341)

ceil() [section 10.16](#) p274)

floor() [section 10.58](#) p292)

ord() [section 10.103](#) p314)

10.16 ceil - round float to nearest integer towards $+\infty$

Call:

`ceil(f)`

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Rounds the floating point argument to the nearest integer towards $+\infty$. An integer type is returned

10.17 cmemoryread - read a combinatorial output memory

Call:

cmemoryread(mem, port, address)

parameters:

param	mode	type	
mem	M	-	memory identifier
port	I	int	port
address	V SV S Q	any	address value

Parameter=argument allowed:

Yes

return mode:

value

return type:

data type of memory

Description:

mem is the memory mode variable identifier.

port is the memory port.

addr is the address.

The address may be any type, including a compound type, but the type must match the type given when the registered output memory was created (see **cmemory()** [section 9.13](#) p197) but primitive component widths may differ in which case padding or truncation will occur.

The value of the function is the data output from the read. The type of the function return is the same type as the data type given when the combinatorial output memory was created.

There may be calls to read and/or write a memory port at multiple points in the program, but the program logic must ensure that these calls are mutually exclusive (the compiler cannot determine that). Failure to observe this will result in data errors.

If both the following conditions hold -

- a port has only a single call to cmemoryread and no calls to procedure cmemorywrite()
- the address expression contains no queues

then the function output will continuously (statically) reflect the memory contents at the requested address. This is because for a single read access only the address input is permanently connected and hence the output remains continuously valid. If the port has more than one access point then the address must be multiplexed and is only connected for the execution cycle of the statement in which the function call appears. For a single read access expressions containing a call to this function can be passed as arguments to modules, procedures or functions or can be assigned to value variables.

If the above is not the case then the function output will only be valid in the execution cycle of the assignment statement in which the function call is part of the RHS expression. In this case an expression

containing **cmemoryread()** cannot be assigned to a value variable after that value has been evaluated (this is detected and results in a fatal error).

A call to **cmemorywrite()** (section 9.14 p198) or **cmemoryread()** (section 10.17 p275) sets the clock domain for the memory port to the current clock domain. All further calls to either of these inbuilt procedures/functions for the same port of the same combinatorial output memory must be in the same clock domain (or a fatal error results).

Note that **cmemoryread()** may appear in an expression assigned to a value mode variable and that variable may then be used in the RHS of a subsequent assignment statement. However, there are two limitations to the use of **cmemoryread()** in this context -

- the value variable must be used at least once in the RHS of an assignment otherwise the control logic for the memory read will have no inputs and a code generation error will occur
- the value variable must be assigned the expression containing **cmemoryread()** prior to its occurrence in the RHS of an assignment - post-assignment cannot be used in this instance

These errors are not recognised during immediate execution but produce indecipherable errors during code generation (proper execution checks will be implemented later).

10.18 componentpointer - get a pointer to a variable component

Call:

componentpointer(v, i)

parameters:

param	mode	type	
v	S	any	static variable
i	I	int or uint	index

Parameter=argument allowed:

No

return mode:

immediate

return type:

->

Description:

Returns a pointer to a component of a value, static or queue mode variable. This is provided to allow access to a single component of a variable of compound type which has nested structs and/or arrays by means of a single integer index instead of field names and arrays subscripts. It has two arguments. The first argument is a variable reference, i.e. it may have fields or subscripts. The second argument is an integer index. If the variable type is primitive the index must be zero. The index must be non-negative and less than the number of components in the type.

The increment and decrement operators cannot be used on a dereferenced pointer returned from componentpointer().

This function is intended for assigning or evaluating value, static or queue mode variable fields or array members from outside the FPGA, e.g. from an attached CPU. A request to assign to a variable field or array member can be transmitted to the FPGA using a simple integer index to select the component to be assigned.

When determining the index value to be used the following rules are pertinent -

- struct fields are packed in the order they are listed in the type
- arrays are packed with last subscript varying most rapidly

See also **componenttype()** section 10.19 p277.

10.19 componenttype - get the type of a variable component

Call:

componenttype(v, i)

parameters:

param	mode	type	
v	S	any	static variable
i	I	int or uint	index

Parameter=argument allowed:

No

return mode:

immediate

return type:

type

Description:

Returns the type of a component of a value, static or queue mode variable. This is provided to allow access to a single component of a variable of compound type which has nested structs and/or arrays by means of a single integer index instead of field names and arrays subscripts. It has two arguments. The first argument is a variable reference, i.e. it may have fields or subscripts. The second argument is an integer index. If the variable type is primitive the index must be zero. The index must be non-negative and less than the number of components in the type.

This function is a companion to inbuilt function **componentpointer()** [section 10.18](#) p276.

10.20 copy - copy a list or map

Call:

copy(v)

parameters:

param	mode	type	
v	I	list or map	subject list or map

Parameter=argument allowed:

No

return mode:

immediate

return type:

list or map

Description:

Returns a **shallow** copy of the argument list or map. Most immediate assignments are by value, i.e. the RHS values are copied to the LHS, but lists and maps are not copied on assignment. Following assignment the LHS and RHS reference the same object. This function is provided for occasions where a copy is required.

10.21 cos - trigonometric cosine function

Call:

cos(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the cosine of the immediate argument. The argument is the angle in radian.

10.22 cspan - span a head string selected by characters not in a set

Call:

cspan(s1, s2)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	search string

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:Returns the length of the head string from the first argument whose characters are **not** contained in the set of characters given by the second argument string.

10.23 currentclock - get the current clock variable

Call:

currentclock()

parameters:

none

return mode:

clock

return type:

none

Description:

Gets the current clock variable. If there is no current clock a fatal error occurs. Null will be returned if there is no current clock domain or the clock domain is explicitly null.

10.24 currentelement - get the current element identifier

Call:

currentelement()

parameters:

none

return mode:

immediate

return type:

str

Description:

Gets the identifier of the current element. The current element is that element currently being constructed, or that was most recently constructed, using the inbuilt procedure **element()** [section 9.18 p201](#). Null will be returned if that procedure has not yet been called.

10.25 deflate - compress a string

Call:

deflate(s)

parameters:

param	mode	type	
s	-	-	string to be compressed

return mode:

immediate

return type:

[]uint

Description:

Returns the compressed string as an array of unsigned 32-bit integers. This function was specifically created for storing compressed .info text in rmemory (block RAM) so that it can later be read by the CPU and uncompressed (inflated). It avoids the necessity of running gzip in a shell from within 3PL and then reading the resulting file, thus greatly simplifying the code.

The compressed bytes are packed as 32-bit unsigned integers such that reading them into CPU memory will result in the correct byte order.

The first word is a checksum of the remainder of the block. The second word is a count of the number of bytes of uncompressed data and the third word a count of the number of compressed bytes. The compressed data that follows these counts contains a 2-byte header relevant to zlib which is 0x78 (compression method) and 0x9C (flags). The last word is zero-padded as required.

It would be preferable for the compression output to be in an array of byte, but 3PL does not have this type. It would not be appropriate for the output to be a string type.

10.26 dimension - get a dimension of an array or size of a map or list

Call:

dimension(ident, index)

parameters:

param	mode	type	
ident	-	-	variable identifier
index	I	int	optional dimension index

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns one dimension of an array variable. The optional second argument selects which dimension to report. If the 2nd argument is omitted the first or only dimension will be reported. If the variable type is not an array, 0 is returned. If the index equals or exceeds the number of dimensions a fatal error occurs.

It will also return the number of entries in a map or list. In that case the second argument must be omitted.

10.27 dimensions - get the number of dimensions of an array

Call:

dimensions(ident)

parameters:

param	mode	type	
ident	-	-	variable identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the number of dimensions of a variable. If the variable type is not an array, 0 is returned.

10.28 directive - get the value of a directive

Call:

directive(key)

parameters:

param	mode	type	
key	I	str	directive key

Parameter=argument allowed:

No

return mode:

immediate

return type:

primitive

Description:

Returns the value of a directive. If the directive is not found a fatal error occurs (use *exists()* to check that a directive exists if unsure).

The key cannot have a scope prepended - directives are all global.

see also inbuilt functions -

directives() [section 10.29](#) p281.

direxists() [section 10.30](#) p281.

and inbuilt procedures

directive [section 9.16](#) p200.

removedirective [section 9.60](#) p238.

10.29 directives - get the map of directives

Call:

directives()

parameters:

none

return mode:

immediate

return type:

map

Description:

Returns a map of directives.

see also inbuilt functions -

directive() [section 10.28](#) p280.

direxists() [section 10.30](#) p281.

and inbuilt procedures

directive [section 9.16](#) p200.

removedirective [section 9.60](#) p238.

10.30 direxists - determine if a directive exists

Call:

direxists(key)

parameters:

param	mode	type	
key	I	str	key string

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Search currently accessible scopes and determine if a directive with the key specified by the argument string exists.

The key cannot have a scope prepended - directives are all global.

see also inbuilt functions -

varexists() [section 10.166](#) p346.

directives() [section 10.29](#) p281.

and inbuilt procedures

directive [section 9.16](#) p200.

removedirective [section 9.60](#) p238.

10.31 domain - get the clock domain for a variable

Call:

domain(ident)

parameters:

param	mode	type	
ident	-	-	variable identifier

Parameter=argument allowed:

No

return mode:

clock

return type:

none

Description:

Returns the clock domain for a variable. If the variable has not had its value referenced or been assigned so far, null will be returned as the variable will not yet have had its clock determined. The variable may be selectvalue, value, static or priority mode. It cannot be queue, memory or clock mode.

See also **writedomain()** [section 10.172](#) p349 which gets the write (input or assignment) clock and **readdomain()** [section 10.120](#) p322 which gets the read (output or evaluation) clock.

For a queue the read and write clocks may be the same or may differ. For selectvalue, value, static and priority mode variables there is only one clock and both inbuilt functions will return that clock.

10.32 dutycycle - get the duty cycle of a clock variable

Call:

dutycycle(v)

parameters:

param	mode	type	
v	C	-	clock variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the duty cycle of a clock variable. This is the proportion of time the clock is high.

See also inbuilt functions -

frequency() [section 10.60](#) p293

period() [section 10.104](#) p314

10.33 elapsedtime - get the elapsed time

Call:

elapsedtime(start)

parameters:

param	mode	type	
start	I	log	3PL start time

Parameter=argument allowed:

No

return mode:

immediate

return type:

"float"

Description:

Returns the elapsed time in second since the start of immediate execution(argument is missing or false) or the elapsed time since the start of 3PL compilation of the source code (argument is true).

The time granularity and accuracy depends on the operating system.

See also -

[section 10.151](#) p336 **time()** for date and time elements and

[section 10.152](#) p337 **timestring()** for date and time as a string.

10.34 endswith - a string is the tail of another

Call:

endswith(s1, s2)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	test string

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Returns true if the 2nd argument string is the tail of the 1st argument string.

10.35 entries - get the number of entries in a list or map - DEPRECATED

Call:

entries(ident)

parameters:

param	mode	type	
ident	-	-	list or map identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the number of entries in a list or map.

This function has been deprecated. **dimension()** now includes the same function.

10.36 env - get the environment variables

Call:

env()

parameters:

None.

return mode:

immediate

return type:

map

Description:

Returns the environment variables as a map. The map keys are the environment variable identifiers and the associated map values are the values of the variables. The values are strings.

10.37 exists - determine if an identifier is recognised - DEPRECATED!

Call:

exists(ident)

parameters:

param	mode	type	
ident	-	-	identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Search currently accessible scopes and determine if an identifier is in any maps, returning an integer code. Integer values are -

0	unknown
1	variable
2	directive

In the case of directive maps the identifier is tried as a string key. If the identifier has **global.**, **file.**, **calling.** or **local.** prepended then the search will be restricted to the specified scope.

This function has been deprecated as the use of an integer returned value is clumsy. The builtin functions **varexists()** [section 10.166](#) p346 or **direxists()** [section 10.30](#) p281 should be used instead to test explicitly for variables or directives.

10.38 exp - e to a power

Call:

exp(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns Euler's number e raised to the power of the immediate argument.

10.39 family - string identifying platform

Call:

family()

parameters:

None.

return mode:

immediate

return type:

str

Description:

Returns a string identifying the platform, e.g. "xc5v" for Virtex5.

10.40 filecanread - check if a file is readable

Call:

filecanread(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Return true if file exists and is readable. The argument must be immediate mode and may be type "str" or "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

10.41 filecanwrite - check if a file is writable

Call:

filecanwrite(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Return true if file exists and is writable. The argument must be immediate mode and may be type "str" or "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

10.42 filedelete - delete a file

Call:

filedelete(f)

parameters:

param	mode	type	
f	I	str	file name

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Delete a file. Return true if successful.

10.43 fileexists - check if a file exists

Call:

fileexists(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Return true if the file exists. The argument must be immediate mode and may be type "str" of "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

10.44 filegetparent - get the path to the file**Call:**

filegetparent(f)

parameters:

param	mode	type	
f	I	str	file name

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Get the path to the file. Return the path string.

10.45 fileisdir - check if a file is a directory**Call:**

fileisdir(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Return true if the file is a directory. The argument must be immediate mode and may be type "str" of "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

10.46 fileisfile - check if a file is a file**Call:**

fileisfile(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Return true if the file is a file and not a directory. The argument must be immediate mode and may be type "str" or "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

10.47 filelastmod - get the time of last modification

Call:

filelastmod(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Get the time of last modification of a named file. The time is returned as an int representing the time the file was last modified, measured in milliseconds since the epoch (00:00:00 GMT, January 1, 1970), or 0 if the file does not exist or if an I/O error occurs.

The argument must be immediate mode and may be type "str" or "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

10.48 filelength - get the file length in bytes

Call:

filelength(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Get the length of a file. The length is the return value. Returns 0 if the file does not exist.

The argument must be immediate mode and may be type "str" or "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

10.49 filelist - get a list of files and directories

Call:

filelist(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

[]str

Description:

Get a list of files and directories in a directory.

f is the directory and must be immediate mode and may be type "str" of "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

The names of the files and directories are returned in an array of strings. The file and directory names are relative to the specified directory, they are not absolute paths.

10.50 filelistdirs - get a list of directories

Call:

filelistdirs(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

[]str

Description:

Get a list of directories in a directory.

f is the directory and must be immediate mode and may be type "str" of "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

The names of the directories are returned in an array of strings. The directory names are relative to the specified directory, they are not absolute paths.

10.51 filelistfiles - get a list of files

Call:

filelistfiles(f)

parameters:

param	mode	type	
f	I	str or file	file

Parameter=argument allowed:

No

return mode:

immediate

return type:

[]str

Description:

Get a list of files in a directory.

f is the directory and must be immediate mode and may be type "str" of "file". The former is a path name string, relative or absolute. The latter is a file variable specifying the file.

The names of the files are returned in an array of strings. The file names are relative to the specified directory, they are not absolute paths.

10.52 filemkdir - create a directory

Call:

filemkdir(f)

parameters:

param	mode	type	
f	I	str	file name

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:Create a directory. Return true if successful. **f** is the directory name.

10.53 filerename - rename a file or directory

Call:

filerename(fold, fnew)

parameters:

param	mode	type	
fold	I	str	file name
fnew	I	str	file name

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Rename a file or directory. Return true if successful. **fold** is an existing file or directory. **fnew** is the new name.

10.54 filesetreadonly - set a file or directory read-only

Call:

filesetreadonly(f)

parameters:

param	mode	type	
f	I	str	file name

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

set a named file or directory read-only. Return true if successful. **f** is the file or directory name.

10.55 find - find a substring using a regular expression

Call:

find(s, re, start)

parameters:

param	mode	type	
s	I	str	subject string
re	I	str	regular expression string
start	I	int	optional starting index

Parameter=argument allowed:

No

return mode:

immediate

return type:

```
struct (
    start, "int",
    end,   "int"
);
```

Description:

find() attempts to match a regular expression string against a subject string and returns a struct containing the starting and ending indices of the match in the subject string. If there is no match the returned indices are zero.

The first argument **s** is the subject string.

The second argument **re** is a Java regular expression - see [Appendix B](#) p673.

The third argument **start** is an optional starting index for the search and is 0 by default.

See also **match()** [section 10.86](#) p306 for a more complex string search.

10.56 first - get the first value of an enum

Call:

first(v)

parameters:

param	mode	type	
v	I V SV S Q	enum or type	enumerated type value

Parameter=argument allowed:

No

return mode:

immediate

return type:

enum

Description:

Returns the first value of an enumerated type. The first value is the first in the order the values had as arguments to the procedure which created the enum type.

The argument may be either an enum type variable or a variable which is an enum type, e.g. for the following variables -

```
enum(e, val1, val2, val3);
static(s, e);
```

the argument to **first()** could be either **e** or **s** and the result would be the value labelled by identifier **val1** of enumerated type **e** in immediate mode.

The argument may be immediate, value, selectvalue, static or queue mode. The returned result is immediate mode and the type is the same as the argument.

10.57 floatisnegative - sign of target float

Call:

floatisnegative(arg)

parameters:

param	mode	type	
arg	V SV S Q IN	float	target float value

Parameter=argument allowed:

No

return mode:

value

return type:

log

Description:

This function returns the sign bit of a target floating point value as type "log".

10.58 floor - round float to nearest integer towards $-\infty$

Call:

floor(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Rounds the floating point argument to the nearest integer towards $-\infty$. An integer type is returned

10.59 format - format values into a string

Call:

format(fmt, v1, v2,...)

parameters:

param	mode	type	
fmt	I	str	format string
v1	I	-	1st value to be converted
v2	I	-	2nd value to be converted
.	I	-	...

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Processes a format string consuming the following input parameters as it encounters conversion specifications. See Java documentation for `java.lang.String.format`.

10.60 frequency - get the frequency of a clock variable

Call:

frequency(v)

parameters:

param	mode	type	
v	C	-	clock variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the frequency of a clock variable in Hz. If the frequency is variable this is the maximum frequency.

See also inbuilt functions -

period() [section 10.104](#) p314

dutycycle() [section 10.32](#) p282

10.61 get - get a list or map entry

Call:

get(l, index)

get(m, key)

get(m, index)

parameters:

param	mode	type	
l	I	list	a list
m	I	map	a map
index	I	int	a list or map index
key	I	str	a map key

Parameter=argument allowed:

No

return mode:

value

return type:

type of list or map entry

Description:

This function can be used to get a list or a map entry.

For a list **get(l, index)** returns the item at the position in the list given by the index. The index must be in the range 0 to entries(l) - 1 inclusive.

For a map **get(m, key)** returns the item matching the string key. If there is no such entry, null is returned. The items are always stored in the lexicographic order of the keys. The call **get(m, index)** returns the key at the position in the ordered keys given by the index. The index must be in the range 0 to entries(l) - 1 inclusive. If the index is out of range a fatal error occurs. To get the value instead of the key at an index use **get(get(m, index))**.

10.62 getfield - get the nth field from a struct

Call:

getfield(v, n)

parameters:

param	mode	type	
v	I V SV S Q	struct	variable
n	I	int	field

Parameter=argument allowed:

No

return mode:

mode of the struct

return type:

type of the field

Description:

This function returns the value of the field specified by the second argument from the struct given as the first argument.

10.63 hadeof - had EOF on the last file read

Call:

hadeof(file)

parameters:

param	mode	type	
file	I	file	open input file

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

This function returns true if the most recent read was at end-of-file.

10.64 hasfield - check if a struct has a specific field

Call:

hasfield(v, f)

parameters:

param	mode	type	
v	I V SV S Q	-	variable
f	-	-	field

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

This function returns true if the field identifier given by the second argument is included in the struct given as the first argument. The first argument may be a struct variable (which is a variable of type **type**) or it may be an immediate or target variable that is a struct type. For example in the code -

```

struct(s,
    f1, "log",
    f2, "int"
);
var(v, s);
...hasfield(s, f1).....
...hasfield(v, f1).....

```

the first argument to **hasfield()** can be either the struct variable itself or the variable that is the struct type.

If the first argument is not an appropriate type **hasfield()** will simply return false.

10.65 hasnext - is there a next value of an enum

Call:

hasnext(v)

parameters:

param	mode	type	
v	I V SV S Q	enum	enumerated type value

Parameter=argument allowed:

No

return mode:

immediate or value

return type:

log

Description:

Returns true if there is a next value of an enumerated type. The next value is the successor in the order the values had as arguments to the procedure which created the enum type. An alternative view is that the function returns false if the argument value is the last value in the ordering of the enumerated type, otherwise it returns true.

If the argument is immediate mode the returned result is immediate mode and the type is log.

If the argument is a target mode (value, selectvalue, static or queue) the returned result is value mode and the type log. For an argument which is a queue the result will only be available when the queue is available.

10.66 hasqueureads - does an argument contain queue reads

Call:

hasqueureads(arg)

parameters:

param	mode	type	
arg	-	any	expression

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

This function returns true if the argument is a target variable or expression containing queue reads (not queue examines). If the argument only contains constants or immediate variables it returns false. If the argument is a clock, cmemory or rmemory mode variable the function will return false.

This function is used to determine if a value argument to a module, procedure or function contains any queue reads to avoid further calls to inbuilt or library modules, procedures or functions which prohibit queues in their arguments.

If called with no arguments this function checks all input parameters to the current module, procedure or function and returns true if any are queues or contain queue reads.

10.67 hasprev - is there a previous value of an enum

Call:

hasprev(v)

parameters:

param	mode	type	
v	I V SV S Q	enum	enumerated type value

Parameter=argument allowed:

No

return mode:

immediate or value

return type:

log

Description:

Returns true if there is a previous value of an enumerated type. The previous value is the predecessor in the order the values had as arguments to the procedure which created the enum type. An alternative view is that the function returns false if the argument value is the first value in the ordering of the enumerated type, otherwise it returns true.

If the argument is immediate mode the returned result is immediate mode and the type is log.

If the argument is a target mode (value, selectvalue, static or queue) the returned result is value mode and the type log. For an argument which is a queue the result will only be available when the queue is available.

10.68 head - get the head of a string

Call:

head(s1, i2)

parameters:

param	mode	type	
s1	I	str	subject string
i2	I	int	length of head

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns the head of the string first argument, the length of the head string being given by the integer second argument. If the length requested is greater than the string length a fatal error occurs.

10.69 identifier - get the identifier of a variable

Call:

identifier(ident, o)

parameters:

param	mode	type	
ident	-	-	identifier or identifier function
o	I	str	optional type

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Return a string which is the identifier of the variable. This has 1 or 2 arguments. The 1st or only argument is the variable. The optional 2nd argument selects the form of the returned identifier string.

The argument may also be one of the functions which returns a variable identifier such as **getclock()**, **currentclock()**, **prevclock()**, **domain**, **readdomain** and **writedomain**. A null return from one of these functions will result in the string "NULL".

o selects the form in which the identifier string is returned as follows

o omitted - the literal identifier string is returned

o = "sliteral" - the identifier is prepended by the identifier of the module, procedure or function in which it has been created, except where the variable was created within a nested code block. In the case of file variables the display format is *file:var* and for global variables *glob:var*.

o = "ext" - the complete unambiguous variable name is returned.

o = "real" - for a parameter the argument variable identifier will be returned. For an assigned clock variable the identifier of the RHS variable of the assignment will be returned. Otherwise the variable identifier string is returned.

o = "chain" - as for "real" but both the parameter identifier and the argument identifier will be returned separated by ->. The identifiers are preceded by the name of the module, procedure or function in which they were created.

10.70 imtotargettype - convert an immediate type to a target type

Call:

imtotargettype(expr)

parameters:

param	mode	type	
expr	any	any	value

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Return the type of an immediate variable as a target type. The type is a string in the same format as is used for type declarations. The type will be converted to target type using widths derived from the immediate variable value(s).

THIS FUNCTION HAS NOT YET BEEN COMPLETED AND IS A TEMPLATE WHICH AT PRESENT WILL GIVE A FATAL ERROR!

See also -

`type()` [section 10.160](#) p341
`arraytype()` [section 10.6](#) p269
`typestring()` [section 10.162](#) p343
`primetypestring()` [section 10.110](#) p317
`targetointype()` [section 10.150](#) p336
`typeval()` [section 10.163](#) p343
`typevalstring()` [section 10.164](#) p344
`typeisequal()` [section 10.161](#) p342

10.71 includedirs - get the directories searched for module and source files

Call:

`includedirs()`

parameters:

None

Parameter=argument allowed:

No

return mode:

immediate

return type:

[]str

Description:

Return a string array containing the absolute paths of directories searched for module and source included files.

10.72 inttohexstring - get a hex string representing an integer

Call:

`inttohexstring(v, l)`

parameters:

param	mode	type	
v	I	bits, int	value
l	I	int	minimum length

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Return a hexadecimal string which represents the value of the first argument. The first argument must be type bits or int. The optional second argument gives the minimum length of the string, the result being zero-padded on the left if necessary.

10.73 isnull - determine if null

Call:

isnull(v)

parameters:

param	mode	type	
v	V	any	variable or function

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Returns true if the argument is null. This function provides a null test for values which might not be primitive. The use of operators == and != causes a fatal error if one of the operands is not a primitive type.

10.74 isread - determine when a queue variable is read

Call:

isread(v)

parameters:

param	mode	type	
v	Q	any	queue mode variable

Parameter=argument allowed:

No

return mode:

value

return type:

log

Description:

Returns a target value of type "log" which is true on the clock cycle in which a read from a queue variable occurs. The argument must be a queue variable.

The queue variable argument must not have subscripts or fields as a read from a queue variable always reads all components of that variable so it would be meaningless to specify some components of the queue variable.

The argument variable read domain must be the same as the clock domain of the call to **isread()**. If a queue mode variable does not yet have a (read) clock domain established then the call to **isread()** will establish that domain.

See also -

iswritten() [section 10.78](#) p302

waswritten() [section 10.169](#) p347

10.75 issigned - determine if a type is signed

Call:

issigned(arg)

parameters:

param	mode	type	
arg	V SV S Q IN I	any	

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

This function returns true if the type of the argument is "INT:" or "FIXED:". If the argument is immediate mode its type must be "type" in which case its type value will be tested for "INT:" or "FIXED:".

10.76 isstruct - determine if a variable type is a struct

Call:

isstruct(v)

parameters:

param	mode	type	
v	I V SV S Q	struct	variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

This function returns true if the type of the argument is a struct.

10.77 istargconst - determine if a value mode variable is a constant

Call:

istargconst(v)

parameters:

param	mode	type	
v	V	prim	value mode variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Returns true if the value mode variable **v** is a target constant.

10.78 iswritten - determine when a static or queue variable is written

Call:

iswritten(v)

parameters:

param	mode	type	
v	S Q	any	static or queue mode variable

Parameter=argument allowed:

No

return mode:

value

return type:

log

Description:

Returns a target value of type "log" which is true on the clock cycle in which a write to a static or queue variable or a part of a static variable occurs. The argument must be a static or queue variable.

A static variable argument may have struct fields or array subscripts in which case the returned value will be true following a write to any field or array member within those specified by the argument.

A queue variable argument must not have subscripts or fields as a write to a queue variable always writes all components of that variable so it would be meaningless to specify some components of the queue variable.

The argument variable write domain must be the same as the clock domain of the call to **iswritten()**. If a static or queue mode variable does not yet have a (write) clock domain established then the call to **iswritten()** will establish that domain.

This function should be used with caution. It must not be used in a conditional assignment to the variable which is the argument to the function as this is a logical contradiction and would lead to a closed loop in the logic!

See also inbuilt function

waswritten() [section 10.169](#) p347 to signal on the cycle following the one on which the write occurs.
isread() [section 10.74](#) p300 can be used to signal on the cycle a queue is read.

10.79 last - get the last value of an enum

Call:

last(v)

parameters:

param	mode	type	
v	I V SV S Q	enum or type	enumerated type value

Parameter=argument allowed:

No

return mode:

immediate

return type:

enum

Description:

Returns the last value of an enumerated type. The last value is the last in the order the values had as arguments to the procedure which created the enum type.

The argument may be either an enum type variable or a variable which is that enum type, e.g. for the following variables -

```
enum(e, val1, val2, val3);
static(s, e);
```

the argument to **first()** could be either **e** or **s** and the result would be the value labelled by identifier **val3** of enumerated type **e** in immediate mode.

The argument may be immediate, value, selectvalue, static or queue mode. The returned result is immediate mode and the type is the same as the argument.

10.80 listattributes - get a list of variable attributes

Call:

listattributes(v)

parameters:

param	mode	type	
v	any	any	variable identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

[n](key, str, mode, str, value, str)

Description:

Returns a list of attributes for a value. The returned value is an array, each member being a struct. The struct has three fields, *key* is a string giving the attribute name, *mode* is a string giving the mode of the attribute value and *value* is a string representation of the value or the identifier in the case of a clock domain.

The argument is the variable identifier.

The struct returned by the function can be used without requiring the structure definition by the following -

```
var(ld, "empty", listattributes());
```

after which **ld[i].key**, **ld[i].mode** etc can be used.

See also library procedure **printattributes()** [subsubsection 11.13.3.5](#) p509.

10.81 listvars - get a list of variables

Call:

listvars(scope, allsegs)

parameters:

param	mode	type	
scope	I	str	optional scope
allsegs	I	str	optional flag

Parameter=argument allowed:

No

return mode:

immediate

return type:`[n](scope, str, ident, str, var, ->)`**Description:**

Returns a list of variables, their scope and type. The returned value is an array, each member being a struct. The struct has three fields, *scope* is a string giving the scope, *ident* is a string giving the identifier and *var* is a pointer to the variable.

The optional first argument, **scope**, is a string limiting a scope. By default all block and local variables, the file scope variables and global scope variables will be listed.

Argument "global" causes only global variables to be listed.

Argument "file" causes only file scope variables to be listed.

Argument "files" causes variables in all file scopes to be listed.

Argument "local" causes only local variables to be listed, i.e. only those declared within the current module, procedure or function, and not variables in nested control structures.

Argument "calling" causes only variables in the calling scope to be listed.

Argument "" selects the default, i.e. all block and local variables, the file scope variables and global scope variables will be listed.

The optional second argument, **allsegs**, controls which file scopes are examined. If true, variables in all file scopes are listed otherwise only variables in the current file scope are listed.

The variables within each scope are listed in lexicographic order.

The struct returned by the function can be used without requiring the structure definition by the following -

```
var(lv, "empty", listvars());
```

after which `lv[i].scope`, `lv[i].ident` etc can be used.

See also library procedure `printvars()` [subsection 11.13.3.7](#) p510.

10.82 location - get the current source location

Call:`location()`**parameters:**

none

return mode:

immediate

return type:

str

Description:

Returns the current location in the source code as a string. The location gives the source file name, line number and column number.

See also inbuilt procedure `trace()` [section 9.80](#) p258 which prints the location and stack trace to standard out.

10.83 log - natural logarithm

Call:

log(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the natural logarithm (base e) of the immediate argument.

10.84 mantissa - target float mantissa

Call:

mantissa(arg)

parameters:

param	mode	type	
arg	V SV S Q IN	float	target float value

Parameter=argument allowed:

No

return mode:

value

return type:

uint

Description:

This function returns the mantissa of a target floating point value. The mantissa is return as type uint: whose width is equal to the width of the mantissa.

10.85 mantissawidth - width of target float mantissa

Call:

mantissawidth(arg)

parameters:

param	mode	type	
arg	I V SV S Q IN	type float	target float type or value

Parameter=argument allowed:

No

return mode:

immediate

return type:

uint

Description:

This function returns the number of bits in the mantissa of a target floating point value. The argument may be a target floating point value or it may be a target floating point type.

10.86 match - find substrings using a regular expression

Call:

match(s, re, start)

parameters:

param	mode	type	
s	I	str	optional subject string
re	I	str	optional regular expression string
start	I	int	optional starting index

Parameter=argument allowed:

No

return mode:

immediate

return type:

"list"

Description:

match() attempts to match a regular expression string against a subject string and returns a list of the groups of matched substrings. If there is no match the returned list is empty.

The first argument **s** is the subject string.

The second argument **re** is a Java regular expression - see [Appendix B](#) p673.

The third argument **start** is an optional starting index for the search and is 0 by default.

To perform further matches on the same subject with the same pattern from the end of the previous match the function is called again with but with no arguments.

Group 0 in the returned list is the entire substring matched. Following groups are substrings matched by parenthesised expressions in the regular expression.

If a match fails an empty list is returned.

See also **find()** [section 10.55](#) p291 for a simpler string search.

10.87 matched - determine if a parameter has an argument

Call:

matched(ident)

parameters:

param	mode	type	
ident	-	-	identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Returns true if the identifier is that of a module, procedure or function parameter which has been passed an argument. Also returns true if the identifier is that of a variable which is not a module, procedure or function parameter.

Returns false if the identifier is that of a parameter which has not been passed an argument.

Note that a parameter which is not passed an argument becomes a local variable. If that parameter is passed as an argument to a further call there is no indication in the deeper scope that the argument two levels back was omitted. To resolve this problem inbuilt function **nullarg()** [section 10.97](#) p311 can be used.

A fatal error occurs if the identifier is not recognised.

10.88 max - maximum of integers

Call:

max(i1, i2, ...)

parameters:

param	mode	type	
i1	I	int	integer value
i2	I	int	integer value
.	I	int	etc

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the maximum of the integer arguments. The arguments may be negative.

10.89 maxvalue - maximum value of a target type

Call:

maxvalue(v)

parameters:

param	mode	type	
v	target	uint, int, enum	type

Parameter=argument allowed:

No

return mode:

value

return type:

type specified by argument

Description:

Returns the maximum value of the variable argument. The argument must specify a primitive target type. It may be a structured type with fields and/or subscripts appended such that it represent a primitive type. The argument mode cannot be CLOCK, CMEMORY or RMEMORY mode. The maximum type size is 63 bits.

The argument may be a target-mode variable in which case the maximum value the variable can represent will be returned.

The variable may also be an immediate mode of type "type" in which case the maximum value of the type will be returned. A type variable may also have fields and/or subscripts where the type is structured.

The argument may also be an immediate variable, expression or literal constant of type "str" in which case it will be interpreted as a type.

The types currently allowed are "int:", "uint:", "fixed", "ufixed", "float" and "enum". In the latter case the largest ordinal is returned.

See also **minvalue()** [section 10.93](#) p309 for the minimum value.

10.90 memoryaddresstype - get the address type of a memory variable

Call:

memoryaddresstype(v)

parameters:

param	mode	type	
v	M	-	cmemory or rmemory mode variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

type

Description:

Returns the address type of a cmemory or rmemory mode variable.

See also inbuilt function **memorydatatype()** [section 10.91](#) p308.

10.91 memorydatatype - get the data type of a memory variable

Call:

memorydatatype(v)

parameters:

param	mode	type	
v	M	-	cmemory or rmemory mode variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

type

Description:

Returns the data type of a cmemory or rmemory mode variable.

See also inbuilt function **memoryaddresstype()** [section 10.90](#) p308.

10.92 min - minimum of integers

Call:

min(i1, i2, ...)

parameters:

param	mode	type	
i1	I	int	integer value
i2	I	int	integer value
.	I	int	etc

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the minimum of the integer arguments. The arguments may be negative.

10.93 minvalue - minimum value of a target type

Call:

minvalue(v)

parameters:

param	mode	type	
v	target	uint, int, enum	type

Parameter=argument allowed:

No

return mode:

value

return type:

type specified by argument

Description:

Returns the minimum value of the variable argument. The argument must specify a primitive target type. It may be a structured type with fields and/or subscripts appended such that it represent a primitive type. The argument mode cannot be CLOCK, CMEMORY or RMEMORY mode. The maximum type size is 63 bits.

The argument may be a target-mode variable in which case the minimum value the variable can represent will be returned.

The variable may also be an immediate mode of type "type" in which case the minimum value of the type will be returned. A type variable may also have fields and/or subscripts appended where the type is structured to reduce to a primitive type.

The argument may also be an immediate variable, expression or literal constant of type "str" in which case it will be interpreted as a type.

The types currently allowed are "int:", "uint:", "fixed", "ufixed", "float" and "enum". In the latter case the largest ordinal is returned.

See also **maxvalue()** [section 10.89](#) p307 for the maximum value.

10.94 mod - modulus of integer or float

Call:

mod(v)

parameters:

param	mode	type	
v	I	int, float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

int or float

Description:

Returns the modulus of the integer or floating point argument. The result is integer or floating to match the argument type.

10.95 mode - the mode of a variable

Call:

mode(ident)

parameters:

param	mode	type	
ident	-	-	variable identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Return the mode of a variable as an string. Values are -

"immediate"
"value"
"selectvalue"
"static"
"queue"
"priority"
"input"
"output"
"cmemory"
"rmemory"
"clock"

10.96 next - get the next value of an enum

Call:

next(v)

parameters:

param	mode	type	
v	I V SV S Q	enum	enumerated type value

Parameter=argument allowed:

No

return mode:

immediate or value

return type:

enum

Description:

Returns the next value of an enumerated type. The next value is the successor in the order the values had as arguments to the procedure which created the enum type.

If the argument is immediate mode the returned result is immediate mode and the type is the same as the argument. If there is no next value a fatal error occurs.

If the argument is a target mode (value, selectvalue, static or queue) the returned result is value mode and the type is the same as the argument. For an argument which is a queue the result will only be available when the queue is available. If there is no next value a value with an ordinal of 0 is returned.

10.97 nullarg - omit an unmatched parameter

Call:

nullarg(ident)

parameters:

param	mode	type	
ident	-	-	identifier

Parameter=argument allowed:

No

return mode:

any

return type:

any

Description:

This function is used to wrap an argument to a module, procedure or function where the argument is itself a module, procedure or function parameter. The argument to nullarg() must be a parameter. It may have subscripts, fields or pre or post increment or decrement operators.

If the argument to nullarg() is a parameter which has been matched by an argument then the argument is passed in place of the call to nullarg().

If the argument to nullarg() is a parameter which has not been matched by an argument then no argument is passed in place of the call to nullarg(), i.e. the argument is omitted.

This is used for nested calls where a parameter is an argument to another call. If the parameter has not been passed an argument it is often preferable to similarly omit it in the deeper call rather than pass it with its default value. This is particularly the case when a parameter gets its mode and type from an argument. If it is not passed an argument it has no mode or type and cannot in turn be passed as the argument to another call.

10.98 numfields - get the number of fields in a struct

Call:

numfields(v)

parameters:

param	mode	type	
v	I V SV S Q	struct	variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

This function returns the number of fields in the struct given as the argument.

10.99 numinargs - number of input arguments

Call:

numinargs()

parameters:

none

return mode:

immediate

return type:

int

Description:

Returns the number of input arguments to the current module, procedure or function.

See also **numwords()** [section 10.101](#) p313 which returns the number of words in a variable, expression or type.

10.100 numoutargs - number of output arguments

Call:

numoutargs()

parameters:

none

return mode:

immediate

return type:

int

Description:

Returns the number of output arguments to the current module procedure or function (returns 0 for a function).

10.101 numwords - get the number of words in a variable, expression or type

Call:

numwords(v)

parameters:

param	mode	type	
v	any	any	variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

This function returns the number of words in the variable or expression given as the argument. The argument may be any mode except clock, cmemory or rmemory. If the argument has fields or subscripts the number of words in the reference is returned. If the variable is an immediate type "type" the number of words in the type is returned.

The number of words is the number of fields or array members in the type, including any nested types. In the following example -

```
struct(s,
    f1, "log",
    f2, "[2][3]int:8",
    f3, "bits:4"
);
static(ss, s);
println(numwords(s));
println(numwords(ss));
println(numwords(ss.f2));
```

would print

```
8
8
6
```

See also `numfields()` [section 10.98](#) p312 which returns the number of fields in a struct.

10.102 offset - point offset of a fixed point value

Call:

offset(arg)

parameters:

param	mode	type	
arg	V SV S Q IN	fixed ufixed	fixed point value

Parameter=argument allowed:

No

return mode:

immediate

return type:

uint

Description:

This function returns the binary point offset of a fixed point value, i.e. the number of bits to the right of the fixed binary point.

10.103 ord - get the ordinal of an enum or a character

Call:

ord(v)

parameters:

param	mode	type	
v	I V SV S Q	enum	enumerated type value or string

Parameter=argument allowed:

No

return mode:

immediate or value

return type:

int or uint

Description:

Returns the ordinal of an enumerated type value or a character. The ordinal is a positive integer.

If the argument is immediate mode enum type the returned result is immediate mode and the type is uint.

If the argument is a target mode enum type (value, selectvalue, static or queue) the returned result is value mode type uint. For an argument which is a queue the result will only be available when the queue is available.

If the argument is immediate mode string type it must be of length 1 (a single character). The ordinal value of that character is returned (the ASCII code).

10.104 period - get the period of a clock variable

Call:

period(v)

parameters:

param	mode	type	
v	C	-	clock variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the period of a clock variable in ns. If the clock period is variable this is the minimum clock period.

See also inbuilt functions -
frequency() [section 10.60](#) p293
dutycycle() [section 10.32](#) p282

10.105 portdomain - get the port clock domain for a memory variable

Call:

readdomain(ident, port)

parameters:

param	mode	type	
ident	-	-	variable identifier
port	-	-	port number

Parameter=argument allowed:

No

return mode:

clock

return type:

none

Description:

Returns the clock domain for a port of a cmemory or rmemory mode variable. If the memory port has not had a read or write procedure called thus far then null will be returned.

10.106 pow - raise to a power

Call:

pow(f1, f2)

parameters:

param	mode	type	
f1	I	float, uint, int	
f2	I	float, uint, int	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float or int

Description:

Returns the first argument raised to the power of the second argument. Both arguments must be immediate. If an argument is "int" or "uint" type it is converted to "float" type. If neither arguments were type "float" and the second argument is not negative the result is returned as type "int".

10.107 prev - get the previous value of an enum

Call:

prev(v)

parameters:

param	mode	type	
v	I V SV S Q	enum	enumerated type value

Parameter=argument allowed:

No

return mode:

immediate or value

return type:enum **Description:**

Returns the previous value of an enumerated type. The previous value is the predecessor in the order the values had as arguments to the procedure which created the enum type.

If the argument is immediate mode the returned result is immediate mode and the type is the same as the argument. If there is no previous value a fatal error occurs.

If the argument is a target mode (value, selectvalue, static or queue) the returned result is value mode and the type is the same as the argument. For an argument which is a queue the result will only be available when the queue is available. If there is no previous value a value with an ordinal of 0 is returned.

10.108 prevclock - get the previous clock variable

Call:

prevclock()

parameters:

none

return mode:

clock

return type:

none

Description:

Gets the previous clock variable, i.e. the one in use prior to the most recent **useclock()** call. Null will be returned if there was no previous clock domain or the previous clock domain was explicitly null.

10.109 prielse - the else case value for a priority variable

Call:

prielse(p)

parameters:

param	mode	type	
p	PR	[]log	priority variable identifier

Parameter=argument allowed:

No

return mode:

value

return type:

log

Description:

Return the else case of a priority variable. The returned VALUE variable is true if all the priority variable inputs are low.

10.110 **primetypestring** - type of a primitive variable as a string

Call:

primetypestring(ident)

parameters:

param	mode	type	
ident	-	-	variable identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Return the type of a primitive variable or expression as a string description. If the variable is a target type the colon and width are omitted from the string. If the variable is not a primitive type, e.g. it is an array or struct, then "none" is returned. If the variable is an enum type then the string for that type will be returned.

For modes **cmemory** or **rmemory** "none" is returned. For mode **clock** "log" is returned.

Note that if the argument is the identifier for a type variable or a variable created by calling struct(), or is the return value from inbuilt function type(), then **primetypestring()** will simply return "type" .

See also -

type() [section 10.160](#) p341

arraytype() [section 10.6](#) p269

imtotargettype() [section 10.70](#) p298

targetointype() [section 10.150](#) p336

typestring() [section 10.162](#) p343

typeval() [section 10.163](#) p343

typevalstring() [section 10.164](#) p344

typeisequal() [section 10.161](#) p342

10.111 **queuedepth** - get the depth of a queue

Call:

queuedepth(v)

parameters:

param	mode	type	
v	Q	-	queue variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the depth of a queue variable buffer. The depth is set by setting attributes **depth** or **mindepth** for the queue variable.

The number of entries in a synchronous queue during target execution can be determined by the value returned from inbuilt function **queuewords()** (section 10.117 p320). For asynchronous queues the library function **asyncqueuewords()** (subsubsection 11.13.4.3 p513) will give a guaranteed minimum instead of the actual count. Either of these this can only be called from within a module which reads from the queue.

The number of free spaces in a synchronous queue can be determined by the value returned from inbuilt function **queuespaces()** (section 10.116 p320). For asynchronous queues the library function **asyncqueuespaces()** (subsubsection 11.13.4.2 p512) will give a guaranteed minimum instead of the actual count. Either of these this can only be called from within a module which writes to the queue.

10.112 queueisempty - determine if a queue is empty

Call:

queueisempty(v)

parameters:

param	mode	type	
v	Q	-	synchronous queue variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Returns true if the queue is empty. This function cannot be used with an asynchronous queue. The advantage of this function over simply the inverse of the read availability (operator <) is that it can be used in a module which does not read from the queue, for example a module which writes to a queue can tell if all entries have been read.

The number of entries in a synchronous queue during target execution can be determined by the value returned from inbuilt function **queuewords()** (section 10.117 p320). For asynchronous queues the library function **asyncqueuewords()** (subsubsection 11.13.4.3 p513) will give a guaranteed minimum instead of the actual count. Either of these this can only be called from within a module which reads from the queue.

The number of free spaces in a synchronous queue can be determined by the value returned from inbuilt function **queuespaces()** (section 10.116 p320). For asynchronous queues the library function **asyncqueuespaces()** (subsubsection 11.13.4.2 p512) will give a guaranteed minimum instead of the actual count. Either of these this can only be called from within a module which writes to the queue.

10.113 queuemaxdepth - get the maximum allowed depth of a queue

Call:

queuemaxdepth()

parameters:

none

return mode:

immediate

return type:

int

Description:

Returns the maximum allowed depth of a queue variable buffer.

10.114 **queuemindepth** - get the minimum allowed depth of an asynchronous queue

Call:

queuemindepth()

parameters:

none

return mode:

immediate

return type:

int

Description:

Returns the minimum allowed depth of an asynchronous queue variable buffer. This is also the minimum allowed depth of a synchronous queue variable buffer above depth 2 (the default depth).

10.115 **queuereadstatus** - get the read status of an asynchronous queue

Call:

queuereadstatus(v)

parameters:

param	mode	type	
v	Q	-	queue variable

Parameter=argument allowed:

No

return mode:

value

return type:

struct

Description:

Returns the read status of an asynchronous queue, i.e. a queue which is assigned in a different clock domain to that in which it is read. The status is a struct with the following fields of type "log" -

etoq	empty to 1/4 full
ttoh	2 words to 1/2 full
qtot	1/4 full + 2 to 3/4 full
htof	1/2 full + 2 to full
ttof	3/4 full + 2 to full

The five fields are mutually exclusive and one of them is always true.

Note that queuereadstatus() is similar to a queue read in that it is synchronised to the queue output clock. It must be called in the same clock domain as reads of the queue and if the call precedes any reads in the source code it will set the read clock domain.

The read status has no latency with respect to the read clock. The empty condition can be monitored with the read availability operator.

There is a complementary inbuilt function **queuewritestatus()** (section 10.118 p321) to get the status in the write clock domain.

Where a queue is synchronous, i.e. assignments to it occur in the same clock domain in which it is read, the **queuereadstatus()** function cannot be used (this is checked). Instead the **queueworlds()** (section 10.117 p320) function can be used to get the number of buffer entries and **queuespaces()** (section 10.116 p320) returns the number of available spaces.

There is a library function **asyncqueueworlds()** (subsubsection 11.13.4.3 p513) which can be used to compute a guaranteed minimum word count from the read status. Similarly there is a library function **asyncqueuespaces()** (subsubsection 11.13.4.2 p512) which can be used to compute a guaranteed minimum space count from the write status.

For any queue variable **queuedepth()** (section 10.111 p317) gives the depth of the queue buffer.

10.116 queuespaces - get the number of free spaces in a queue

Call:

queuespaces(v)

parameters:

param	mode	type	
v	Q	-	queue variable

Parameter=argument allowed:

No

return mode:

value

return type:

uint

Description:

Returns the number of free entries in the buffer of a synchronous queue, i.e. a queue which is assigned in the same clock domain to that in which it was created.

A call to **queuespaces()** must be in the same clock domain as all reads and writes of the queue. If **queuespaces()** is called prior to any queue reads or writes it will set the queue clock domain to the current domain.

For an asynchronous queue, i.e. a queue which is assigned in a different clock domain to that in which it is read, the occupancy status can only be approximately determined by using inbuilt functions **queuereadstatus()** (section 10.115 p319) and **queuewritestatus()** (section 10.118 p321). There is a library function **asyncqueuespaces()** (subsubsection 11.13.4.2 p512) which can be used to compute a guaranteed minimum space count from the write status. Similarly there is a library function **asyncqueueworlds()** (subsubsection 11.13.4.3 p513) which can be used to compute a guaranteed minimum word count from the read status.

To get the number of occupied entries use **queueworlds()** (section 10.117 p320).

See section 3.11 p80 (queue mode attributes) for details on the depth of queue buffers. To get the queue buffer depth use **queuedepth()** (section 10.111 p317).

For a synchronous queue the inbuilt function **queueisempty()** (section 10.112 p318) can be called. Unlike the read availability operator this can be used in a module which does not read from the queue.

10.117 queueworlds - get the number of entries in a queue

Call:

queueworlds(v)

parameters:

param	mode	type	
v	Q	-	queue variable

Parameter=argument allowed:

No

return mode:

value

return type:

uint

Description:

Returns the number of occupied entries in the buffer of a synchronous queue, i.e. a queue which is assigned in the same clock domain to that in which it was created.

A call to **queuwords()** must be in the same clock domain as all reads and writes of the queue. If **queuwords()** is called prior to any queue reads or writes it will set the queue clock domain to the current domain.

For an asynchronous queue, i.e. a queue which is assigned in a different clock domain to that in which it is read, the occupancy status can only be approximately determined by using inbuilt functions **queureadstatus()** (section 10.115 p319) and **queuwritestatus()** (section 10.118 p321). There is a library function **asyncqueuwords()** (subsubsection 11.13.4.3 p513) which can be used to compute a guaranteed minimum word count from the read status. Similarly there is a library function **asyncqueuespaces()** (subsubsection 11.13.4.2 p512) which can be used to compute a guaranteed minimum space count from the write status.

To get the number of free entries use **queuespaces()** (section 10.116 p320).

See section 3.11 p80 (queue mode attributes) for details on the depth of queue buffers. To get the queue buffer depth use **queuedepth()** (section 10.111 p317).

10.118 queuwritestatus - get the write status of an asynchronous queue

Call:

queuwritestatus(v)

parameters:

param	mode	type	
v	Q	-	queue variable

Parameter=argument allowed:

No

return mode:

value

return type:

struct

Description:

Returns the read status of an asynchronous queue, i.e. a queue which is assigned in a different clock domain to that in which it is read. The status is a struct with the following fields of type "log" -

etoq	empty to 1/4 full
ttoh	2 words to 1/2 full
qtot	1/4 full + 2 to 3/4 full
htof	1/2 full + 2 to full
ttof	3/4 full + 2 to full

Parameter=argument allowed:

No

The five fields are mutually exclusive and one of them is always true.

Note that `queuewritestatus()` is similar to an assignment in that it is synchronised to the queue input clock. It must be called in the same clock domain as assignments to the queue and if the call precedes any assignments in the source code it will set the assignment clock domain.

The write status has no latency with respect to the write clock. The full condition can be monitored using the write availability operator.

There is a complementary inbuilt function `queuereadstatus()` (section 10.115 p319) to get the status in the read clock domain.

Where a queue is synchronous, i.e. assignments to it occur in the same clock domain in which it was created the `queuewritestatus()` function cannot be used (this is checked). Instead the `queuewords()` (section 10.117 p320) function can be used to get the number of buffer entries and `queuespaces()` (section 10.116 p320) returns the number of available spaces.

There is a library function `asyncqueuespaces()` (subsubsection 11.13.4.2 p512) which can be used to compute a guaranteed minimum space count from the write status. Similarly there is a library function `asyncqueuwords()` (subsubsection 11.13.4.3 p513) which can be used to compute a guaranteed minimum word count from the read status.

For any queue variable `queuedepth()` (section 10.111 p317) gives the depth of the queue buffer.

10.119 random - a random number

Call:

`random()`

parameters:

none.

return mode:

immediate

return type:

float

Description:

Returns an immediate random number. The random number n is in the range $0.0 \leq n < 1.0$.

10.120 readdomain - get the read clock domain for a variable

Call:

`readdomain(ident)`

parameters:

param	mode	type	
ident	-	-	variable identifier

Parameter=argument allowed:

No

return mode:

clock

return type:

none

Description:

Returns the read (output) clock domain for a variable. If the variable has not had its value referenced so far null will be returned as the variable will not yet have had its output clock determined. The variable may be selectvalue, value, static, queue or priority mode. It cannot be memory or clock mode.

See also **writedomain()** [section 10.172](#) p349 which gets the write (input or assignment) clock and **domain()** [section 10.31](#) p282 which gets the single clock for selectvalue, value, static and priority mode variables.

For a queue the read and write clocks may be the same or may differ. For selectvalue, value, static and priority mode variables there is only one clock and all three inbuilt functions will return that clock.

10.121 readbyte - read a byte from an input file

Call:

readbyte(file)

parameters:

param	mode	type	
file	I	file	open input file

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

This function reads a byte from an input file. The argument is an open input file variable. If a read beyond the end-of-file is attempted -1 is returned and the inbuilt function **hadeof()** will return true.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

See also -

scanf() [section 9.67](#) p245

fscanf() [section 9.35](#) p214

sscanf() [section 9.68](#) p246

readln() [section 10.122](#) p323.

10.122 readln - read a line from an input file

Call:

readln(file)

parameters:

param	mode	type	
file	I	file	open input file

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

This function reads a line of text from an input file. The argument is an open input file variable. If a read beyond the end-of-file is attempted an empty string is returned and the inbuilt function **hadeof()** will return true.

See the note on character and byte reads and writes in the description of inbuilt procedure **file()** [section 9.32](#) p212.

See also -

scanf() [section 9.67](#) p245

fscanf() [section 9.35](#) p214

sscanf() [section 9.68](#) p246

readbyte() [section 10.121](#) p323.

10.123 replace - replace left substring

Call:

replace(s1, s2, s3)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	selection string
s3	I	str	replacement string

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns the string supplied by the first argument with the leftmost substring matched by the second argument replaced by the third argument string. If the substring is not found the first argument is returned unchanged.

The second argument string is a Java regular expression - see [Appendix B](#) p673.

10.124 replaceall - replace all substrings

Call:

replaceall(s1, s2, s3)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	selection string
s3	I	str	replacement string

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns the string supplied by the first argument with all substrings matched by the second argument replaced by the third argument string. If the substring is not found the first argument is returned unchanged.

The second argument string is a Java regular expression - see [Appendix B](#) p673.

10.125 resetoutput - get the reset output of a queue

Call:

resetoutput(v)

parameters:

param	mode	type	
v	Q	-	queue variable

Parameter=argument allowed:

No

return mode:

value

return type:

log

Description:

Returns a value of type log which is true for one clock cycle when the queue is reset.

Note that resetoutput() is similar to a queue read in that it is synchronised to the queue output clock. It must be called in the same clock domain as reads of the queue and if the call precedes any reads in the source code it will set the read clock domain.

For an asynchronous queue the reset output is synchronised to the output clock (read clock domain) and follows the reset input to the queue which is synchronised to the input clock (write clock domain).

The purpose of this function is to allow queue resets of pipelines with multiple clock domains to be chained from the source end of a pipeline. The source end of a pipeline is explicitly reset and each clock domain section of the pipeline downstream derives its reset signal from the asynchronous queue reset output at the head of the section.

10.126 round - round float or fixed to nearest integer

Call:

round(f)

parameters:

param	mode	type	
f	I	float	
	V	fixed, ufixed	

Parameter=argument allowed:

No

return mode:

immediate or value

return type:

int or uint

Description:

For an immediate argument of type float this rounds the floating point argument to the nearest integer. An int type is returned.

For a value argument of type fixed or ufixed this rounds the fixed point argument using round-to-even. This method rounds a fraction which is exactly 1/2 to the nearest even integer to avoid a statistical bias. Type int is returned for fixed and uint for ufixed.

Note that rounding of an integer target type at a selected bit position can be accomplished by first casting it to a fixed or ufixed type with a fractional part of appropriate size then applying the round() function.

10.127 rmemoryout - get the value of a registered output memory

Call:

rmemoryout(mem, port)

parameters:

param	mode	type	
mem	M	-	memory identifier
port	I	int	port

Parameter=argument allowed:

Yes

return mode:

value

return type:

memory data type

Description:

Get a value which is the output of a registered memory port. This value is continuously valid. The type is the same as the data type argument provided when the memory was created.

A call to **rmemoryout()** sets the clock domain for the memory port to the current clock domain. All calls to **rmemoryout()** ([section 10.127](#) p326), **rmemoryread()** ([section 9.63](#) p242) or **rmemorywrite()** ([section 9.64](#) p242) for the same port of the same registered output memory must be in the same clock domain (or a fatal error results).

10.128 sample - sample an expression

Call:

sample(e)

parameters:

param	mode	type	
e	-	any	target expression

Parameter=argument allowed:

No

return mode:

value

return type:

type of argument

Description:

The argument must be a static variable or a value (an expression or a value variable) containing static variables or a value variable which has been passed as an output argument to the **import()** inbuilt procedure. The argument must not contain queue reads and cannot be a memory or clock mode variable.

The argument value will be sampled in the current clock domain and the resulting value returned as the value of the function. The argument must have a clock domain or a fatal error will occur. It will

have one if has already been evaluated previously (in a clock domain). If evaluation occurs later in the source code then the clock domain of the argument must have been explicitly forced prior to calling **sample()**. This can be done by using inbuilt procedure **domain()** or **readdomain()** on any variables in the argument. The argument cannot be a constant.

The simplest case where this is used is where a static variable in one clock domain is assigned to a static variable (or a queue) in another clock domain. The sampling avoids problems with violations of setup and hold time which might otherwise be encountered. If the two clocks have an exact frequency ratio with a fixed phase then sampling may not be required as long as an appropriate clock-to-clock timing constraint is imposed. If the source variable changes very seldom and an occasional transient data error can be tolerated then again sampling can be omitted (see inbuilt function **accept()** [section 10.2](#) p265).

This is a simple device to avoid the use of a full dual-port resynchronising queue buffer where the data is known to be sparse or where the data is sampled rather than pipelined. If the input clock frequency is the higher of the two then some data will not be sampled. If the input clock frequency is the lower of the two then data will be multiply sampled. If the two clock frequencies are very close both missing samples and duplicated samples may occur. Note that regardless of missing or duplicated samples the sampling process maintains data order, i.e. data are never out of sequence.

A typical use of this function is when the source static variable seldom changes, for example a controlling parameter assigned via the CPU bus, but the value is used in another clock domain to the interface static variable.

sample() should not be used for detecting single cycle pulses in another clock domain - in that case see inbuilt module **resync()** [section 8.2](#) p185 or library function **resync()** [subsection 11.13.4.20](#) p521 which guarantee a single cycle pulse regardless of relative clock frequencies.

The sampled data may be delayed up to two clock cycles in the current clock domain.

If the argument clock domain is the same as the current clock domain the argument is simply returned as the value.

10.129 selecta - a combinatorial selector from arrays

Call:

selecta(sel, data)

parameters:

param	mode	type	
sel	V	☐ log	selector array
data	V	☐ any	data array

Parameter=argument allowed:

No

return mode:

value

return type:

type of data inputs

Description:

An inbuilt selector function using gate logic. This has 2 arguments. The 1st argument is an array of type **log**. The 2nd argument is an array of any type whose 1st or only dimension is the same as that of the 1st argument array.

Each member of the selector array is a logical value which selects the corresponding data element of the data array.

The function has a defined value when at most one select input is available and true and the associated data input is available, the value reflecting the selected data input. If more than one input is selected the value is undefined. If no inputs are selected, the value is zero or false depending on data type(s).

If none of the function inputs contain queues, the function value is always available. If any function inputs contain queues the function value is available when an input selector and its associated data input are available and the selector is true. Note that if some inputs contain queues but one input selector and its associated data do not, the function will be available when that selector is true. Note that the use of this function as a gate with a clear value when no selectors are true is only possible if the inputs contain no queues since any queues in any inputs will render the function value unavailable when no input is selected.

See also library function `select()` [subsection 11.13.5](#) p521 where the inputs are select-data pairs.

10.130 signum - sign function

Call:

`signum(v)`

parameters:

param	mode	type	
v	I	int or float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

int or float as per argument

Description:

Returns 0 or 0.0 if the argument is zero, 1 or 1.0 if the argument is +ve, or -1 or -1.0 if the argument is -ve..

10.131 sin - trigonometric sine function

Call:

`sin(f)`

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the sine of the immediate argument. The argument is the angle in radian.

10.132 size - get the size in bits

Call:

`size(ve)`

parameters:

param	mode	type	
ve	V SV S Q PR	-	variable or expression

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the size in bits.

The argument cannot be cmemory, rmemory or clock mode.

The argument type cannot be type **str**, **->**, **file** or immediate **enum**.

If the argument is a target mode (other than memory or clock mode) the size in bits will be returned. If the type is primitive it will return the size of the primitive type in bits. If the type is structured it will return the total size of the type in bits.

If the argument is immediate mode -

- int or uint - the minimum number of bits required to represent the primitive type is returned (for a +ve integer no allowance is made for a sign bit!) but for arguments 0 or 1, 1 is returned and for argument -1, 2 is returned
- log - 1 is returned
- float - 64 is returned
- type - the size of the type value is returned (the type must be target type(s))

In the case of a structured type the argument may have fields or subscripts in which case the size of the field or array member will be returned.

The argument may be type "type" in which case the size of the type will be returned. If the type is a structured type then fields or subscripts may be appended and the size of these fields or array members will be returned. If the structured type is immediate rather than target then the size returned will be 1 for each element since sizes of immediates can only be derived from values, not types.

10.133 span - span a head string selected by characters in a set

Call:

span(s1, s2)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	search string

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the length of the head string from the first argument whose characters are contained in the set of characters given by the second argument string.

10.134 split - split a string into substrings

Call:

split(s1, s2)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	split point regular expression string

Parameter=argument allowed:

No

return mode:

immediate

return type:

[]str

Description:

This function splits a string into substrings, returning them as an array of strings. The split point is indicated by a regular expression, supplied as the second argument. The substrings within the first argument are separated by sections which match the regular expression. If the regular expression does not match any section of the subject string then the returned array has a dimension of one, the single entry containing the whole subject string.

The second argument string is a Java regular expression - see [Appendix B](#) p673.

10.135 sqrt - an immediate square root

Call:

sqrt()

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the square root of the immediate argument.

10.136 stacktrace - get the current stack trace

Call:

stacktrace()

parameters:

none

return mode:

immediate

return type:

str

Description:

Returns a multi-line string listing the source locations of all module, procedure and function calls in the stack.

See also inbuilt procedure **trace()** [section 9.80](#) p258 which prints the location and stack trace to standard out.

10.137 startswith - a string is the head of another

Call:

startswith(s1, s2)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	test string

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Returns true if the 2nd argument string is the head of the 1st argument string.

10.138 strtocarray - convert a string to an array of integers

Call:

strtocarray(s)

parameters:

param	mode	type	
s	I	str	subject string

Parameter=argument allowed:

No

return mode:

immediate

return type:

[]uint

Description:

Returns an array of integers whose values are the ASCII codes representing the argument string.

Note that this function is usually used as an initialiser for an array variable and this must be type "[]uint" rather than "[]int", e.g.

```
var(a, "[]uint", strtocarray(s));
```

This strict typing in this context may be relaxed in the future.

See also -

[section 10.5](#) p269 **arraytostr()** converting an integer array to a string.

10.139 strlen - find the length of a string

Call:

strlen(s)

parameters:

param	mode	type	
s	I	str	subject string

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the length the argument string.

10.140 strpos - find leftmost substring position

Call:

strpos(s1, s2)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	search string

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the leftmost position of an occurrence of the second argument string in the first argument string. If not found, -1 is returned.

10.141 strrpos - find rightmost substring position

Call:

strrpos(s1, s2)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	search string

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the rightmost position of an occurrence of the second argument string in the first argument string. If not found, -1 is returned.

10.142 strtod - convert a string to a float

Call:

strtod(s)

parameters:

param	mode	type	
s	I	str	subject string

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the floating point value represented by the string.

10.143 strtoint - convert a string to an integer

Call:

strtoint(s)

parameters:

param	mode	type	
s	I	str	subject string

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the integer value represented by the string. The string may begin with 0x or 0X for hexadecimal, 0b or 0B for binary or 0 for octal.

10.144 strtotype - convert a string to an type

Call:

strtotype(s)

parameters:

param	mode	type	
s	I	str	subject string

Parameter=argument allowed:

No

return mode:

immediate

return type:

type

Description:

Returns the type value represented by the string.

A string describing a type can be assigned to a type variable or passed as an argument to a parameter which is a type variable. The type string may contain the identifier of another type variable, but if the string is used in a scope in which the embedded identifier is not accessible then the conversion of string-to-type will fail. This function converts a string to a type, evaluating an embedded string variable in the process in the current scope after which the type can be used remote from that embedded variable.

10.145 substitute - substitute characters

Call:

substitute(s1, s2, s3)

parameters:

param	mode	type	
s1	I	str	subject string
s2	I	str	selection character set
s3	I	str	substitute character set

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns the string supplied by the first argument with selected characters replaced. The second argument is a string whose characters constitute the set of selection characters. The third argument is a string whose characters constitute the matching set of replacement characters. If the 3rd argument string is shorter than the 2nd then those trailing characters in the 1st set will be eliminated. If the 3rd argument string is longer than the 2nd then those trailing characters in the 2nd set will be ignored. If a character is repeated in the 2nd argument string the left-most substitution will occur.

10.146 substring - get a substring

Call:

substring(s, i1, i2)

parameters:

param	mode	type	
s	I	str	subject string
i1	I	int	starting position
i2	I	int	optional substring length

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns a substring of the first argument. The second argument gives the starting position of the substring and the optional third argument gives its length. If the third argument is missing the substring extends to the end of the subject string. If the starting position is not within the subject string or the length extends beyond the subjects string a fatal error occurs.

10.147 tail - get the tail of a string

Call:

tail(s1, i2)

parameters:

param	mode	type	
s1	I	str	subject string
i2	I	int	length of tail

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns the tail of the string first argument, the length of the tail string being given by the integer second argument. If the length requested is greater than the string length a fatal error occurs.

10.148 tan - trigonometric tangent function

Call:

tan(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Returns the tangent of the immediate argument. The argument is the angle in radian.

10.149 targconsttoimmed - convert a value mode constant to an immediate

Call:

targconsttoimmed(v)

parameters:

param	mode	type	
v	V	prim	value mode variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

bits, uint, int or log

Description:

Return the immediate value of a value variable which is a target constant.

10.150 targtoimtype - convert a target type to an immediate type

Call:

targtoimtype(expr)

parameters:

param	mode	type	
expr	any	any	value

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Return the type of a target variable as an immediate type. The type is a string in the same format as is used for type declarations. The type will be converted to immediate type by removing width fields and converting "bits" to "uint".

See also -

type() [section 10.160](#) p341

arraytype() [section 10.6](#) p269

typestring() [section 10.162](#) p343

primetypestring() [section 10.110](#) p317

imtotargettype() [section 10.70](#) p298

typeval() [section 10.163](#) p343

typevalstring() [section 10.164](#) p344

typeisequal() [section 10.161](#) p342

10.151 time - get the current millisecond time

Call:

time(start)

parameters:

No arguments

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the current millisecond time, i.e. the time in millisecond since midnight, January 1, 1970 UTC.

See also -

[section 10.152](#) p337 `timestring()` for date and time as a string, [section 10.33](#) p283 `elapsedtime()` for elapsed time.

10.152 timestring - get the date and time as a string

Call:`timestring(fmt)`**parameters:**

param	mode	type	
fmt	I	str	optional format string

Parameter=argument allowed:

No

return mode:

immediate

return type:

"str"

Description:

Returns a string giving the current date and time. If the optional argument is supplied it must be a format string for the date/time. The format required is described below. The default is "**yyyy-MM-dd HH:mm:ss Z**".

The following description of the optional format argument is taken from the Java API.

Date and time formats are specified by date and time pattern strings. Within date and time pattern strings, unquoted letters from 'A' to 'Z' and from 'a' to 'z' are interpreted as pattern letters representing the components of a date or time string. Text can be quoted using single quotes (') to avoid interpretation. "'" represents a single quote. All other characters are not interpreted; they're simply copied into the output string during formatting or matched against the input string during parsing.

The following pattern letters are defined (all other characters from 'A' to 'Z' and from 'a' to 'z' are reserved):

param	mode	type	
fmt	I	str	optional format string

Letter	Date or Time Component	Presentation	Examples
G	Era designator	Text	AD
y	Year	Year	1996; 96
M	Month in year	Month	July; Jul; 07
w	Week in year	Number	27
W	Week in month	Number	2
D	Day in year	Number	189
d	Day in month	Number	10
F	Day of week in month	Number	2
E	Day in week	Text	Tuesday; Tue
a	Am/pm marker	Text	PM
H	Hour in day (0-23)	Number	0
k	Hour in day (1-24)	Number	24
K	Hour in am/pm (0-11)	Number	0
h	Hour in am/pm (1-12)	Number	12
m	Minute in hour	Number	30
s	Second in minute	Number	55
S	Millisecond	Number	978

Pattern letters are usually repeated, as their number determines the exact presentation:

Text: For formatting, if the number of pattern letters is 4 or more, the full form is used; otherwise a short or abbreviated form is used if available. For parsing, both forms are accepted, independent of the number of pattern letters.

Number: For formatting, the number of pattern letters is the minimum number of digits, and shorter numbers are zero-padded to this amount. For parsing, the number of pattern letters is ignored unless it's needed to separate two adjacent fields.

Year: If the formatter's Calendar is the Gregorian calendar, the following rules are applied. For formatting, if the number of pattern letters is 2, the year is truncated to 2 digits; otherwise it is interpreted as a number. For parsing, if the number of pattern letters is more than 2, the year is interpreted literally, regardless of the number of digits. So using the pattern "MM/dd/yyyy", "01/11/12" parses to Jan 11, 12 A.D.

For parsing with the abbreviated year pattern ("y" or "yy"), SimpleDateFormat must interpret the abbreviated year relative to some century. It does this by adjusting dates to be within 80 years before and 20 years after the time the SimpleDateFormat instance is created. For example, using a pattern of "MM/dd/yy" and a SimpleDateFormat instance created on Jan 1, 1997, the string "01/11/12" would be interpreted as Jan 11, 2012 while the string "05/04/64" would be interpreted as May 4, 1964. During parsing, only strings consisting of exactly two digits, as defined by Character.isDigit(char), will be parsed into the default century. Any other numeric string, such as a one digit string, a three or more digit string, or a two digit string that isn't all digits (for example, "-1"), is interpreted literally. So "01/02/3" or "01/02/003" are parsed, using the same pattern, as Jan 2, 3 AD. Likewise, "01/02/-3" is parsed as Jan 2, 4 BC. Otherwise, calendar system specific forms are applied. For both formatting and parsing, if the number of pattern letters is 4 or more, a calendar specific long form is used. Otherwise, a calendar specific short or abbreviated form is used. **Month:** If the number of pattern letters is 3 or more, the month is interpreted as text; otherwise, it is interpreted as a number.

[section 10.151](#) p336 **time()** for date and time elements,
[section 10.33](#) p283 **elapsedtime()** for elapsed time.

10.153 toarray - convert to a one-dimensional array

Call:

toarray(v, n)

parameters:

param	mode	type	
v	I	-	
n	I	int	

Parameter=argument allowed:

No

return mode:

immediate

return type:

array

Description:

Converts the 1st argument to a one-dimensional array. The argument must be a primitive type or a compound type where all members are of the same primitive type. Thus a primitive would be converted to an array of dimension 1. A compound type would be converted to a 1-dimensional array whose dimension equals the number of elements in the compound type.

If the optional 2nd argument is provided it specifies the array size. Where the 1st argument is primitive or an array of dimension 1 it will be expanded to a one dimensional array of dimension n. For other types n must match the size of the compound type to be converted.

10.154 todegree - convert radian to degree

Call:

todegree(f)

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Converts the immediate argument from radian to degree.

10.155 tolower - convert string to lower case

Call:

tolower(s)

parameters:

param	mode	type	
s	I	str	subject string

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns the argument string with any upper case characters converted to lower case.

10.156 toradian - convert degree to radian

Call:

toradian()

parameters:

param	mode	type	
f	I	float	

Parameter=argument allowed:

No

return mode:

immediate

return type:

float

Description:

Converts the immediate argument from degree to radian.

10.157 tostring - string value

Call:

tostring(a)

parameters:

param	mode	type	
a	I	any	

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns a string representation of the immediate mode argument. Argument types accepted are str, int, uint, float, log, bits or an enumerated type. In the case of an enumerated type the string value returned is the value identifier.

10.158 toupper - convert string to upper case

Call:

toupper(s)

parameters:

param	mode	type	
s	I	str	subject string

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Returns the argument string with any lower case characters converted to upper case.

10.159 trunc - truncate float or fixed to nearest integer towards zero

Call:

trunc(f)

parameters:

param	mode	type	
f	I	float	
	V	fixed, ufixed	

Parameter=argument allowed:

No

return mode:

immediate or value

return type:

int or uint

Description:

Truncates an immediate floating point or target fixed point argument to the nearest integer towards zero. If the argument is **immediate** mode the returned value is also **immediate** mode. If the argument is mode **selectvalue**, **value**, **static** or **queue** the returned value is **value** mode. Other modes are not allowed.

For immediate mode input type "int" is returned. For target modes the returned value is type "int:" or "uint:" for input types "fixed:" and "ufixed:" respectively. If the argument is type "int:" or "uint:" then it is simply returned as the function value.

10.160 type - type of a variable or expression

Call:

type(v_or_e)

parameters:

param	mode	type	
v_or_e	-	-	variable or expression

Parameter=argument allowed:

No

return mode:

immediate

return type:

type

Description:

Return the type of a variable or expression.

Note that if the argument is a variable of type **type** then type **type** will be returned, not the type contained in the type variable. For that use inbuilt function **typeval()**.

type() [section 10.160](#) p341
arraytype() [section 10.6](#) p269
typestring() [section 10.162](#) p343
primetypestring() [section 10.110](#) p317
imtotargettype() [section 10.70](#) p298
targetointype() [section 10.150](#) p336
typeval() [section 10.163](#) p343
typevalstring() [section 10.164](#) p344
typeisequal() [section 10.161](#) p342

10.161 typeisequal - compare two types for equality

Call:

typeisequal(t1, t2, strict)

parameters:

param	mode	type	
t1	-	-	variable or expression
t2	-	-	variable or expression
strict	I	log	variable or expression (optional)

Parameter=argument allowed:

Yes

return mode:

immediate

return type:

log

Description:

Compare the types of the first two arguments. These may be immediate or target mode and may be variables or expressions. If an argument is of type **type** the type value will be used rather than the type of the variable or expression. The argument types may be immediate or target but must be the same mode except in the case of type **log**.

Types **int** and **uint** are treated as equivalent.

If either of the first two arguments are type **type** then they will be regarded as equal regardless of their values. To compare type variables they must be passed via inbuilt function **typeval()** which will unwrap the contained type.

If the third argument is provided and is true the comparison will be strict, i.e. the types must be identical, except that **int** and **uint** are treated as equivalent (but the number of bits must be the same).

If the third argument is omitted or is false and the first and second arguments are primitive types then they may vary in bit widths and still be regarded as equal. Thus the non-strict comparison returns true if an expression of the second type could be assigned to a variable of the first type.

Where the first two arguments are compound types the third argument is ignored and the comparison will be strict (including bit widths).

type() [section 10.160](#) p341
arraytype() [section 10.6](#) p269
typestring() [section 10.162](#) p343
primetypestring() [section 10.110](#) p317
imtotargettype() [section 10.70](#) p298
targetointype() [section 10.150](#) p336
typeval() [section 10.163](#) p343
typevalstring() [section 10.164](#) p344
typeisequal() [section 10.161](#) p342

10.162 **typestring** - type of a variable as a string

Call:

typestring(ident)

parameters:

param	mode	type	
ident	-	-	variable identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

str

Description:

Return the type of a variable or expression as a string description. The type string is the same as that used for variable creation.

If the argument is a variable which has a struct type or an enum type a string description of the struct or enum will be returned - see [subsection 3.7.3](#) p59 for a description of the string specification for structs and [subsection 3.7.6](#) p65 for a description of the string specification for enums.

If the argument is a variable of type "type" then this function will simply return "type", not a string representing the nested type. In that case use the inbuilt function **typevalstring()** [section 10.164](#) p344.

Note that a type string can also be formed from a type by simply assigning the type variable to a string variable or initialising the string variable with a type variable or the inbuilt function **type()**.

type() [section 10.160](#) p341

arraytype() [section 10.6](#) p269

typestring() [section 10.162](#) p343

primetypestring() [section 10.110](#) p317

imtotargettype() [section 10.70](#) p298

targetointype() [section 10.150](#) p336

typeval() [section 10.163](#) p343

typevalstring() [section 10.164](#) p344

typeisequal() [section 10.161](#) p342

10.163 **typeval** - value of a variable of type "type"

Call:

typeval(v)

parameters:

param	mode	type	
v	-	-	variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

type

Description:

Return the value of a variable or expression which is of type "type". This is used where the type value of a variable or expression of type "type" is required.

`type()` [section 10.160](#) p341
`arraytype()` [section 10.6](#) p269
`typestring()` [section 10.162](#) p343
`primetypestring()` [section 10.110](#) p317
`imtotargettype()` [section 10.70](#) p298
`targettoimtype()` [section 10.150](#) p336
`typeval()` [section 10.163](#) p343
`typevalstring()` [section 10.164](#) p344
`typeisequal()` [section 10.161](#) p342

10.164 typevalstring - value of a variable of type "type" as a string

Call:

`typevalstring(v)`

parameters:

param	mode	type	
v	-	-	variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

string

Description:

Return the value of a variable or expression which is of type "type" as a string. This is used where the type value of a variable or expression of type "type" is required as a string. For example the code -

```
struct(s,
    f1, "log",
    f2, "int"
);
println(typestring(s));
```

will print "type" since variable `s` has type "type". To get the type contained in the variable use `typeval(s)`, thus the code

```
struct(s,
    f1, "log",
    f2, "int"
);
println(typevalstring(s));
```

will print "(f1,log,f2,int)".

`type()` [section 10.160](#) p341
`arraytype()` [section 10.6](#) p269
`typestring()` [section 10.162](#) p343
`primetypestring()` [section 10.110](#) p317
`imtotargettype()` [section 10.70](#) p298
`targettoimtype()` [section 10.150](#) p336
`typeval()` [section 10.163](#) p343

`typevalstring()` [section 10.164](#) p344
`typeisequal()` [section 10.161](#) p342

10.165 used - target value which is "true" when a function executes

Call:

`used()`

parameters:

None.

return mode:

value

return type:

log

Description:

This function returns a target signal of type "log" which is true when the function code actually executes (is evaluated).

The following example uses a function which returns successive values in a sequence. It should advance the sequence only when the target assignment in which it is called executes.

```
seq while (..) {
    .
    a <- s();
    .
}
.
.
.
function
s () {
    var(v, "[5]int", {5, 7, 9, 3, 1});
    value(ret, "int:4");
    value(step, "log", used());
    del(ret <- ret, dimension(v), v, step);
    return(ret);
}
```

Here the call to `del()` creates a module which provides the sequence (since it is a module it does not constitute executable statements within the function definition - a function cannot contain inline target statements). The module steps to the next member of the sequence when its last argument **step** is true. Here that argument reflects the inbuilt function `used()` which will only be true on clock cycles on which the function value is being used (an intermediate value variable has been used here only to make the code more readable since passing `used()` as an argument to `del()` would not indicate the purpose of that argument).

Note that the same instantiation of a function call can be used in more than one statement, for example

```
value(sv, "int:4", s());
seq while (..) {
    .
    a <- sv;
    .
    b <- sv;
}
```

where the two assignments will get successive members of the sequence, which is quite different to -

```
seq while (..) {
.
  a <- s();
.
  b <- s();
}
```

where each function call is a separate (duplicate) instance and hence they operate independently and produce the same values.

see also inbuilt `proceduremassert()` [section 9.5](#) p190.

10.166 varexists - determine if a variable exists

Call:

`varexists(ident)`

parameters:

param	mode	type	
ident	-	-	identifier

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Search currently accessible scopes and determine if a variable matching the identifier argument exists. If the identifier has **global.**, **file.**, **calling.** or **local.** prepended then the search will be restricted to the specified scope.

To test for the existence of a directive see `direxists()` [section 10.30](#) p281.

10.167 varlist - return a list of target mode variables

Call:

`varlist(o)`

parameters:

param	mode	type	
o	I	log	selection option

Parameter=argument allowed:

No

return mode:

immediate

return type:

list

Description:

This function returns a list of the currently declared target mode variables. If option **o** is true then only INPUT, OUTPUT and CLOCK mode variables are included in the list.

This function is of use when generating constraint files at the end of immediate execution. Attributes can be read from INPUT, OUTPUT and CLOCK mode variables and used to generate the constraints. Library procedure **generateConstraints()** [subsubsection 11.2.2.18](#) p388 in xilinx/common.3pl does this in the case of Xilinx platforms, the call being placed last in trailer code.

10.168 wasassigned - determine if variable has been assigned

Call:

wasassigned(v)

parameters:

param	mode	type	
v	-	-	

Parameter=argument allowed:

No

return mode:

immediate

return type:

log

Description:

Returns true if the argument variable has been assigned. Assignment does not include initialisation. The argument must be a variable, can be of any type and can be any mode except INPUT, CMEMORY or RMEMORY.

10.169 waswritten - determine when a static or queue variable has been written

Call:

waswritten(v)

parameters:

param	mode	type	
v	S Q	any	static or queue mode variable

Parameter=argument allowed:

No

return mode:

value

return type:

log

Description:

Returns a target value of type "log" which is true on the clock cycle following a write to a static or queue variable or a part of a static variable. The argument must be a static or queue variable.

A static variable argument may have struct fields or array subscripts in which case the returned value will be true following a write to any field or array member within those specified by the argument.

A queue variable argument must not have subscripts or fields as a write to a queue variable always writes all components of that variable so it would be meaningless to specify some components of the queue variable.

The argument variable write domain must be the same as the clock domain of the call to **iswritten()**. If a static or queue mode variable does not yet have a (write) clock domain established then the call to **iswritten()** will establish that domain.

See also inbuilt function

iswritten() [section 10.78](#) p302 to detect the cycle on which the write occurs.

isread() [section 10.74](#) p300 can be used to signal on the cycle a queue is read.

10.170 width - get the width of a primitive type - DEPRECATED!

Call:

width(value)

parameters:

param	mode	type	
value	I V SV S Q PR	-	variable or expression

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the width in bits of an immediate or target primitive type. The argument can be an expression or a variable. If the type is not a primitive or the mode is memory or clock a fatal error occurs. For an immediate argument the type must be int or log. For a +ve integer the sign bit is not included.

This function has been deprecated. The inbuilt function **size()** [section 10.132](#) p328 should be used instead.

10.171 words - get the size in words

Call:

words(v)

parameters:

param	mode	type	
v	V SV S Q PR	-	variable

Parameter=argument allowed:

No

return mode:

immediate

return type:

int

Description:

Returns the number of words.

The argument cannot be cmemory, rmemory or clock mode.

The function returns the number of words in the type of the variable, a word being a component of primitive type. The argument may have subscripts and fields in which case the number of words in the referenced part of the variable is returned. If the variable is type "null", 0 is returned. If the variable is type "->", "list" or "map", 1 is returned (use inbuilt function **entries()** [section 10.35](#) p283 for the number of entries in a list or map). If the variable is type "type", the number of words in the type is returned.

10.172 writedomain - get the write clock domain for a variable

Call:

writedomain(ident)

parameters:

param	mode	type	
ident	-	-	variable identifier

Parameter=argument allowed:

No

return mode:

clock

return type:

none

Description:

Returns the write (input or assignment) clock domain for a variable. If the variable has not been assigned so far null will be returned as the variable will not yet have had its input clock determined. The variable may be selectvalue, static, queue or priority mode. It cannot be memory or clock mode.

See also -

readdomain() [section 10.120](#) p322 which gets the read (output or evaluation) clock and **domain()** [section 10.31](#) p282 which gets the single clock for selectvalue, value, static and priority mode variables.

For a queue the read and write clocks may be the same or may differ. For selectvalue, value and priority mode variables there is only one clock and all three inbuilt functions will return that clock.

Chapter 11 - Libraries

3PL libraries are source files containing modules, procedures, functions and global structs and variables which can be accessed by application code and other libraries. They are inserted in the application source code via a **module** statement or a **source** statement (see "file inclusion" [section 3.2](#) p43). These library files are searched for by the 3PL compiler in the order -

- current directory
- directories listed in -I options to the 3PL command line
- directories listed in environment variable **THREEPL_INCLUDE_DIRS**

3PL libraries may be -

- generic
- interfaces to hardware specific to FPGA families
- platform-specific

Libraries are arranged in the following directory structure -

```
boards/
  avnet/
    aes_sp3a_eval400.3pl
    microzed_7010.3pl
    microzed_7020.3pl
  dangerous_prototypes
    breakout_xc2c64a.3pl
    breakout_xc9572x1.3pl
  diligent
    zybo.3pl
  gadgetfactory
    papilio_one_100.3pl
    papilio_one_250.3pl
    papilio_one_500.3pl
    papilio_pro.3pl
  memec/
    v4fx12lc.3pl
    v4fx12mm.3pl
  xilinx/
    ml505.3pl
capture.3pl
ctypes.3pl
divlib.3pl
DS2431.3pl
float.3pl
json_type.3pl
multilib.3pl
qdr2_ram.3pl
spi.3pl
```

```

stdlib.3pl
uart.3pl
uarts.3pl
xilinx/
  clocks.3pl
  common.3pl
  cpu.3pl
  dsp48.3pl
  gtp_dual.3pl
  io.3pl
  mgt.3pl
  postprocess.3pl
  ppc405lib.3pl
  ppplib.3pl
  ps7.3pl
  uart_cpu.3pl
  xc2s.3pl
  xc2v.3pl
  xc2vp.3pl
  xc3s.3pl
  xc4v.3pl
  xc5v.3pl
  xc6s.3pl
  xc6v.3pl
  xc7.3pl
  xcv.3pl
  zynq_axi.3pl
zbt_ram.3pl

```

The following are generic libraries -

- capture.3pl - capture a data stream
- ctypes.3pl - generate C types from 3PL structs and enums
- divlib.3pl - pipelined integer divider
- float.3pl - pipelined floating point modules
- json.3pl - conversion of a 3PL type to a JSON string
- multilib.3pl - pipelined integer multiplier
- qdr2_ram.3pl - external quad data rate synchronous static RAM interface
- stdlib.3pl - useful elements such as pulse generators, pulse stretchers, adder/subtractors, incrementer/decrementers.
- uart.3pl - generalised UART modules
- uarts.3pl - simple UART modules
- spi.3pl - SPI interface
- zbt_ram.3pl - external zero bus turnaround synchronous static RAM interface

DS2431.3pl provides a module to read and write the DS2431 single wire EPROM chip.

The libraries in directory **xilinx** contain elements specific to FPGA families. The same elements may be contained in several libraries with perhaps detail differences in implementation. These libraries are -

- clocks.3pl - clock managers and clock buffers
- common.3pl - low-level elements and functions common to all FPGA families
- cpu.3pl - interface between the FPGA and a CPU
- dsp48.3pl - Virtex4 and later DSP blocks
- gtp_dual.3pl - a GTP dual multi-gigabit transceiver tile
- io.3pl - input/output functions for Virtex5 and later
- mgt.3pl - an MGT transceiver
- postprocess.3pl - a module to generate the configuration file from the EDIF file using ISE or VIVADO
- ppc405lib.3pl - an embedded PowerPC instance with instruction and data RAMs
- ppplib.3pl - interface for an embedded PowerPC
- uart_cpu.3pl - low level UART CPU interface

- xc2s.3pl - Spartan 2 elements
- xc3s.3pl - Spartan 3 elements
- xc6s.3pl - Spartan 6 elements
- xcv.3pl - Virtex and Virtex-E elements
- xc2v.3pl - Virtex2 elements
- xc2vp.3pl - Virtex2Pro elements
- xc4v.3pl - Virtex 4 elements
- xc5v.3pl - Virtex 5 elements
- xc6v.3pl - Virtex 6 elements
- xc7.3pl - Artix 7, Kintex 7, Virtex 7 and Zynq 7000 elements
- zynq-axi.3pl - Zynq PL instance and AXI interface to CPU

Directory **boards** contains header files for a limited number of commercial FPGA boards. There are not very many included as most of our work in CSIRO has been on substantial applications implemented on large in-house FPGA platforms. The libraries for these have not been included in this distribution as they would not be of general interest.

There are four subdirectories, avnet, xilinx, memec and gadgetfactory.

The **avnet** directory contains header files for Xilinx boards supplied by AVNET. At the moment there are four -

- aes_sp3a_eval400.3pl - Avnet Xilinx Spartan3A evaluation kit
- microzed.7010.3pl - Avnet Xilinx xc7010 microzed Zynq evaluation kit
- microzed.7020.3pl - Avnet Xilinx xc7020 microzed Zynq evaluation kit

The **digilent** directory contains one header file for the Digilent zybo Zynq evaluation board.

- zybo.3pl - Digilent Zybo Zynq evaluation kit

The **xilinx** directory contains header files for Xilinx boards supplied from Xilinx. At the moment there is one -

- ml505.3pl - Xilinx Virtex-5 evaluation board

This file does not support all features of the board.

The **memec** directory contains header files for Xilinx boards supplied from Memec. At the moment there are two -

- v4fx12lc.3pl - Xilinx Virtex4 demonstration board
- v4fx12mm.3pl - Xilinx Virtex4 mini module board

These files so far are only the equivalent of "hello world" and provide only the main clock, an optional divided clock and a few pin definitions.

The **gadgetfactory** directory contains header files for papilio boards supplied by the Gadget Factory. At the moment there are four -

- papilio_one_100.3pl - Papilio board with XC3S100E Spartan 3E
- papilio_one_250.3pl - Papilio board with XC3S250E Spartan 3E
- papilio_one_500.3pl - Papilio board with Xc3S500E Spartan 3E
- papilio_pro.3pl - Papilio board with XC6SLX9 Spartan

Note that these library files are often also referred to as "header" files since they define FPGA pinouts and establish clocks as well as defining modules, procedures and functions.

As an example the header file for a GadgetFactory Papilio 250 is -

```

nosource

// 3pl board support for GadgetFactory Papilio One 250k

// Initial setup
part("XC3S250EVQ100-4");    // FPGA part number
module "xilinx/xc3s.3pl"    // include file for Spartan 3

// default IO is 3.3V on this board
str(lvcmos33, "LVCMOS33");

// 32MHz clock input pin
iopins(c32_pin, "str", "P89", frequency=32.0e6, ibufg=true, iostandard=lvcmos33);

```

```

// 48 I/O pins across A (wing 1A), B (wing 1B), C (wing 2A)
iopins(global.A_pins, "[16]str", {
    "P18", "P23", "P26", "P33", "P35", "P40", "P53", "P57",
    "P60", "P62", "P65", "P67", "P70", "P79", "P84", "P86"
}, iostandard=lvcmos33
);
iopins(global.B_pins, "[16]str", {
    "P85", "P83", "P78", "P71", "P68", "P66", "P63", "P61",
    "P58", "P54", "P41", "P36", "P34", "P32", "P25", "P22"
}, iostandard=lvcmos33
);
iopins(global.C_pins, "[16]str", {
    "P91", "P92", "P94", "P95", "P98", "P2", "P3", "P4",
    "P5", "P9", "P10", "P11", "P12", "P15", "P16", "P17"
}, iostandard=lvcmos33
);

// FTDI USB channel B serial receive and transmit.
// Labelling is reversed to schematic!
// 'tx' should be an FPGA output, 'rx' an input
iopins(global.serial_rxd_pin, "str", "P88", iostandard=lvcmos33);
iopins(global.serial_txd_pin, "str", "P90", iostandard=lvcmos33);

// create 32MHz clock
clock(global.c32_in, loc=c32_pin);
simpleclock(global.c32, c32_in);
useclock(c32);

directive("comms_baud_rate", 115200); // define default serial device baud rate
module "xilinx/uart_cpu.3pl" // include serial interface

```

An application program using this board simply has to begin with -

```
module "boards/gadget_factory/papilio_one_250.3pl"
```

(for example see [chapter 21](#) p645

Platform-specific module files

There is a module file for each Xilinx platform device that should be included in source code.

These files must be incorporated by using a **module** statement, e.g. -

```
module "xilinx/xc2s.3pl"
```

Each of these module files includes other module files appropriate to the platform device. This avoids the necessity to include these files explicitly in the source code.

Note that libraries exist for MGT and GTP serial channels, but not yet for GTX, GTH or GTZ. Also the GT11 element for Virtex4 has not been implemented.

module file	device	includes module files
xilinx/xc2s.3pl	Spartan2	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl
xilinx/xc3s.3pl	Spartan3	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl
xilinx/xc6s.3pl	Spartan6	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl xilinx/io.3pl
xilinx/xcv.3pl	Virtex and Virtex-E	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl
xilinx/xcv2.3pl	Virtex2	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl
xilinx/xcv2p.3pl	Virtex2 Pro	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl xilinx/ppclib.3pl xilinx/mgt.3pl
xilinx/xcv4.3pl	Virtex4	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl xilinx/io.3pl xilinx/dsp48.3pl xilinx/ppclib.3pl
xilinx/xcv5.3pl	Virtex5	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl xilinx/io.3pl xilinx/dsp48.3pl xilinx/ppclib.3pl xilinx/gtp_dual.3pl
xilinx/xcv6.3pl	Virtex6	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl xilinx/io.3pl xilinx/dsp48.3pl xilinx/ppclib.3pl xilinx/gtp_dual.3pl
xilinx/xc7.3pl	Artix7 Kintex7 Virtex7	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl xilinx/io.3pl xilinx/dsp48.3pl xilinx/gtp_dual.3pl
xilinx/zynq_axi.3pl	Zynq7000	stdlib.3pl xilinx/common.3pl xilinx/clocks.3pl xilinx/io.3pl xilinx/dsp48.3pl xilinx/gtp_dual.3pl

11.1 Clock Manager and Clock Buffer Library File - xilinx/clocks.3pl

This library creates any clock managers or clock buffers for all Xilinx FPGAs.

This library must be incorporated by using a **module** statement -

```
module "xilinx/clocks.3pl"
```

This library is already incorporated by all FPGA family header files.

The procedures in this library attempt to propagate clock frequencies and periods from inputs to outputs using the parameters controlling clock division or synthesis. For that reason most take an error exit if an input clock does not have a frequency-period attribute.

At the moment clkmmcm(), plladv() are exceptions to the last paragraph. They will be changed to conform ASAP.

Where clock input pads, clock buffers and clock manager are interconnected clock mode variables are used (required by parameter modes). Where a clock is derived from CLB logic the inbuilt function **clockfromexpr()** [section 9.11](#) p196 can be used to assign a value mode variable to a clock mode variable. Where input delays are required for an input clock path (see **idelay()** [subsection 11.5.2.2](#) p426) the input should be input mode rather than clock mode. The output from the delay can then be convert to clock mode by using **clockfromexpr()**.

11.1.1 library modules

None.

11.1.2 library procedures

11.1.2.1 clkdcn - create a Spartan3, Virtex2, Virtex2Pro, Virtex4 or Virtex5 DCM digital clock manager

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clkdcn (
    clock({clk0, clk90, clk180, clk270, clk2x, clk2x180, clkdv, clkfx, clkfx180}),
    value({locked, psdone, drdy}, "log"), value(dout, "uint:16"), value(status, "uint:8")
    <-
    str(loc), log({clkdiv_by_2, duty_cycle_correction, startup_wait}),
    str(clkdv_divide),
    float(clkin_period), int({clkfx_divide, clkfx_multiply, phase_shift}),
    str({clk_feedback,
    clkout_phase_shift, dfs_frequency_mode, dll_frequency_mode, deskew_adjust}),
    clock({clkkin, clkfb, psclk}), value({reset, psincdec, psen, dwe, den}, "log"),
    value(din, "bits:16"), value(daddr, "uint:7")
) {}
```

Output parameters:

	mode	type	
clk0	clock		output clock
clk90	clock		output clock shifted 90deg
clk180	clock		output clock shifted 180deg
clk270	clock		output clock shifted 270deg
clk2x	clock		output clock at double frequency
clk2x180	clock		output clock at double frequency shifted 180deg
clkdv	clock		divided output clock
clkfx	clock		frequency synthesised output clock
clkfx180	clock		frequency synthesised output clock shifted 180deg
locked	value	log	DLL is locked
status	value	uint:8	status
psdone	value	log	phase shift has finished
drdy	value	log	DCM ready for reconfiguration
dout	value	bits:16	status and reconfiguration data output

Input parameters:

	mode	type	
loc	immediate	str	optional location
clkdiv_by_2	immed	log	DLL clock output at $f_{in}/2$
duty_cycle_correction	immediate	log	correct duty cycle to 50%
startup_wait	immediate	log	delay startup on lock
clkdv_divide	immed	str	n for clock output at f_{in}/n
clkdiv_period	immediate	float	input clock period, ns
clkfx_divide	immed	int	m for DFS clock output at $f_{in} \times n/m$
clkfx_multiply	immed	int	n for DFS clock output at $f_{in} \times n/m$
phase_shift	immediate	int	fixed phase shift value
clk_feedback	immediate	str	"1x", "2x", "none"
clkout_phase_shift	immediate	str	"none", "fixed", "variable"
dfs_frequency_mode	immediate	str	"high" or "low"
dll_frequency_mode	immediate	str	"high" or "low"
des skew_adjust	immediate	str	"system_synchronous", "source_synchronous", or 0 - 15
clkdiv	clock		input clock
clkfb	clock		clock feedback
psclk	clock		phase shift clock
reset	value	log	reset
psincdec	value	log	phase shift increment or decrement
psen	value	log	phase shift enable
dwe	value	log	reconfiguration data write enable
den	value	log	select reconfiguration data or status output
din	value	log	reconfiguration data input
daddr	value	log	reconfiguration address input

In-line target execution:

no

description:

Creates a Virtex4 or Virtex5 DCM digital clock manager. See Xilinx documentation for signal and attribute details. The first group of input parameters are immediate mode and are attributes. The remaining input parameters are input signals, clock mode for clocks and value mode for control signals.

Attributes

loc is an optional location string from the following list - "DLL0P", "DLL1P", "DLL2P", "DLL3P", "DLL0S", "DLL1S", "DLL2S", "DLL3S". These select from the four primary and four secondary delay-locked loops. If "" it is ignored.

clkdiv_by_2 - see Xilinx documentation.

If **duty_cycle_correction** is true or missing then duty cycle correction is enabled, giving a DLL output duty cycle of 50%. If this attribute is false then the duty cycle will be the same as that of the input.

startup_wait if true causes startup following configuration to be delayed until the DCM achieves lock.

clkfx_divide is the frequency divider for the DFS (1 to 32).

clkfx_multiply is the frequency multiplier for the DFS (2 to 32).

phase_shift is the phase shift value which must be in the range -255 to +255 inclusive. A unit value represents 1/256 of a cycle.

clkdv_divide indicates that the divided frequency output is being used. The division ratio is a string argument which must be one of "1.5", "2.0", "2.5", "3.0", "3.5", "4.0", "4.5", "5.0", "5.5", "6.0", "6.5", "7.0", "7.5", "8.0", "9.0", "10.0", "11.0", "12.0", "13.0", "14.0", "15.0" or "16.0".

clkin_period is the input clock period. This must be specified if the DFS is used.

clk_feedback must be "2X" if the clock feedback is at twice input frequency. By default the feedback is at the input frequency, or this attribute can be explicitly "1X" to achieve the same.

clkout_phase_shift is a string selecting the mode of the phase shifter. If omitted or "none" the outputs are not phase shifted. If this argument is "fixed" a clock skew is added to the input clock shifting the phase of all clock outputs. The phase shift is specified by the next input argument. If "variable" is specified the phase shift is initially set as specified by the next argument, but it can be dynamically incremented or decremented - see the 17th to 19th input arguments. On reset the phase shift returns to the initial value.

dfs_frequency_mode indicates that the DFS is on the high frequency range. The low and high frequency ranges are given in the data sheets.

dll_frequency_mode indicates that the DLL is in the high frequency range (see data sheets). In this mode the 90 degree and 270 degree outputs and the 2X frequency outputs are not available.

deskew_adjust controls the amount of delay in the feedback path. Its values can be "system_synchronous" (default), "source_synchronous" or "0 to 15".

dcm_performance_mode optimises the DCM for high frequency low jitter ("max_speed", the default) or low frequency and maximum phase shift range ("max_range").

input signals

din, **daddr**, **dwe**, **den** and **dclk** are only available on Virtex4 and Virtex5.

clkin is the input clock. This argument must be supplied and is the first target mode input argument.

clkfb is the clock feedback. This argument must be supplied. This is usually a buffered version of the output clock at phase 0.

reset is an optional reset. This resets the DLL, DFS and phase shifter and re-establishes lock.

psincdec is true if a phase increment is to be executed or false if a phase decrement is to be executed. Execution is controlled by the following two inputs.

psen is the phase increment/decrement enable. This must be held true for one period of the phase shift adjustment clock (the next input argument).

psclk is the phase increment/decrement clock.

din is data input for a dynamic reconfiguration write.

daddr is the address input for a dynamic reconfiguration write.

dwe is driven high for one **dclk** cycle along with **den** to execute a dynamic reconfiguration write.

den is driven high for one **dclk** cycle to execute a dynamic reconfiguration read or write.

dclk is a clock controlling dynamic reconfiguration operations.

output signals

drdy and **dout** are only available on Virtex4 and Virtex5.

status is only available on Spartan3, Virtex2 and Virtex2Pro.

clk0, **clk90**, **clk180** and **clk270** are clock outputs at the input frequency. These are at phases 0, 90, 180 and 270 degree respectively relative to the input clock.

clk2x and **clk2x180** are at twice the input frequency at phases 0 and 180 degree respectively.

clkdv is the divided clock frequency where the division ratio is given by the 3rd input argument.

clkfx and **clkfx180** are the frequency synthesised clock at phases 0 and 180 degree respectively. The frequency is the input multiplied by the value of **clkfx_multiply** and divided by the value of **clkfx_divide**.

locked is high when the DLL and DFS are locked to the input clock. This signal has a null clock domain.

status is an 8 bit status (see Xilinx unified library documentation). This signal has a null clock domain.

psdone goes high when a phase shift adjustment has been done. It remains high for one period of the phase adjustment clock. Note that this signal has a null clock domain, but it is synchronous with **psclk**.

drdy goes high when the dynamic configuration port finishes an operation. On a read the data on **do** must be captured on the next rising **dclk** edge. When writing the next operation can commence on the same clock cycle in which **drdy** goes high.

dout is the data output when reading dynamic configuration data. The data on **do** must be captured on the rising **dclk** edge when **drdy** is high.

11.1.2.2 clkdll - create a Spartan2, Virtex or VirtexE DLL delay-locked loop clock manager

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clkdll (
  clock({clk0, clk90, clk180, clk270, clk2x, clkdv}),
  value(locked, "log")
  <-
  str(loc), log({duty_cycle_correction, startup_wait}),
  int(clkdv_divide),
  clock({clkln, clkfb}), value(reset, "log")
) {}
```

Output parameters:

	mode	type	
clk0	clock		output clock
clk90	clock		output clock shifted 90deg
clk180	clock		output clock shifted 180deg
clk270	clock		output clock shifted 270deg
clk2x	clock		output clock at double frequency
clkdv	clock		divided output clock

Input parameters:

	mode	type	
loc	immediate	str	optional location
clkin_divide_by_2	immediate	log	divide clock by 2
duty_cycle_correction	immediate	log	correct duty cycle
startup_wait	immediate	log	delay startup on lock
clkdv_divide	immediate	int	clock division ratio
clkin	clock		input clock
clkfb	clock		clock feedback
reset	value	log	reset

In-line target execution:

no

description:

Creates a Virtex DLL delay-locked loop clock manager. See Xilinx documentation for signal and attribute details. The first group of input parameters are immediate mode and are attributes. The remaining input parameters are input signals, clock mode for clocks and value mode for control signals.

attributes

loc is an optional location string from the following list - "DLL0P", "DLL1P", "DLL2P", "DLL3P", "DLL0S", "DLL1S", "DLL2S", "DLL3S". These select from the four primary and four secondary delay-locked loops. If "" it is ignored.

clkin_divide_by_2 - see Xilinx documentation.

If **duty_cycle_correction** is true or missing then duty cycle correction is enabled, giving a DLL output duty cycle of 50%. If this attribute is false then the duty cycle will be the same as that of the input.

startup_wait if true causes startup following configuration to be delayed until the DLL achieves lock.

clkdv_divide indicates that the divided frequency output is being used. The division ratio is a string argument which must be an integer from 1 to 32 inclusive.

input signals

clkin is the input clock. This argument must be supplied and is the first target mode input argument.

clkfb is the clock feedback. This argument must be supplied. This is usually a buffered version of the output clock at phase 0.

reset is an optional reset. This resets the DLL, DFS and phase shifter and re-establishes lock. operations.

output signals

clk0, **clk90**, **clk180** and **clk270** are clock outputs at the input frequency. These are at phases 0, 90, 180 and 270 degree respectively relative to the input clock.

clk2x is at twice the input frequency.

clkdv is the divided clock frequency where the division ratio is given by the 3rd input argument.

locked is high when the DLL and DFS are locked to the input clock. This signal has a null clock domain.

11.1.2.3 clkdll - create an extended Spartan2, Virtex or VirtexE DLL delay-locked loop clock manager

Library: xilinx/clocks.3pl

Prototype:

```

global procedure
clkdlle (
    clock({clk0, clk90, clk180, clk270, clk2x, clk2x180, clkdv}),
    value(locked, "log")
    <-
    str(loc, log({duty_cycle_correction, startup_wait}),
    int(clkdv_divide),
    clock({clkin, clkfb}), value(reset, "log")
) {}

```

Output parameters:

	mode	type	
clk0	clock		output clock
clk90	clock		output clock shifted 90deg
clk180	clock		output clock shifted 180deg
clk270	clock		output clock shifted 270deg
clk2x	clock		output clock at double frequency
clk2x180	clock		output clock at double frequency shifted 180deg
clkdv	clock		divided output clock

Input parameters:

	mode	type	
loc	immediate	str	optional location
clkin_divide_by_2	immediate	log	divide clock by 2
duty_cycle_correction	immediate	log	correct duty cycle
startup_wait	immediate	log	delay startup on lock
clkdv_divide	immediate	int	clock division ratio
clkin	clock		input clock
clkfb	clock		clock feedback
reset	value	log	reset

In-line target execution:

no

description:

Creates a Virtex DLLE extended delay-locked loop clock manager. See Xilinx documentation for signal and attribute details. The first group of input parameters are immediate mode and are attributes. The remaining input parameters are input signals, clock mode for clocks and value mode for control signals.

attributes

loc is an optional location string from the following list - "DLL0P", "DLL1P", "DLL2P", "DLL3P", "DLL0S", "DLL1S", "DLL2S", "DLL3S". These select from the four primary and four secondary delay-locked loops. If "" it is ignored.

clkin_divide_by_2 - see Xilinx documentation.

If **duty_cycle_correction** is true or missing then duty cycle correction is enabled, giving a DLL output duty cycle of 50%. If this attribute is false then the duty cycle will be the same as that of the input.

startup_wait if true causes startup following configuration to be delayed until the DLL achieves lock.

clkdv_divide indicates that the divided frequency output is being used. The division ratio is a string argument which must be an integer from 1 to 32 inclusive.

input signals

clkin is the input clock. This argument must be supplied and is the first target mode input argument.

clkfb is the clock feedback. This argument must be supplied. This is usually a buffered version of the output clock at phase 0.

reset is an optional reset. This resets the DLL, DFS and phase shifter and re-establishes lock. operations.

output signals

clk0, **clk90**, **clk180** and **clk270** are clock outputs at the input frequency. These are at phases 0, 90, 180 and 270 degree respectively relative to the input clock.

clk2x and **clk2x180** are at twice the input frequency at phases 0 and 180 degree respectively.

clkdv is the divided clock frequency where the division ratio is given by the 3rd input argument.

locked is high when the DLL and DFS are locked to the input clock. This signal has a null clock domain.

11.1.2.4 clkdlhf - create a high frequency Spartan2, Virtex or VirtexE DLL delay-locked loop clock manager

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clkdlle (
    clock({clk0, clk180, clkdv}),
    value(locked, "log")
<-
    str(loc), log({duty_cycle_correction, startup_wait}),
    int(clkdv_divide),
    clock({clkin, clkfb}), value(reset, "log")
) {}
```

Output parameters:

	mode	type	
clk0	clock		output clock
clk180	clock		output clock shifted 180deg
clkdv	clock		divided output clock

Input parameters:

	mode	type	
loc	immediate	str	optional location
clkin_divide_by_2	immediate	log	divide clock by 2
duty_cycle_correction	immediate	log	correct duty cycle
startup_wait	immediate	log	delay startup on lock
clkdv_divide	immediate	int	clock division ratio
clkin	clock		input clock
clkfb	clock		clock feedback
reset	value	log	reset

In-line target execution:

no

description:

Creates a Virtex DLL high frequency delay-locked loop clock manager. See Xilinx documentation for signal and attribute details. The first group of input parameters are immediate mode and are attributes. The remaining input parameters are input signals, clock mode for clocks and value mode for control signals.

attributes

loc is an optional location string from the following list - "DLL0P", "DLL1P", "DLL2P", "DLL3P", "DLL0S", "DLL1S", "DLL2S", "DLL3S". These select from the four primary and four secondary delay-locked loops. If "" it is ignored.

clkdiv_divide_by_2 - see Xilinx documentation.

If **duty_cycle_correction** is true or missing then duty cycle correction is enabled, giving a DLL output duty cycle of 50%. If this attribute is false then the duty cycle will be the same as that of the input.

startup_wait if true causes startup following configuration to be delayed until the DLL achieves lock.

clkdv_divide indicates that the divided frequency output is being used. The division ratio is a string argument which must be an integer from 1 to 32 inclusive.

input signals

clk is the input clock. This argument must be supplied and is the first target mode input argument.

clkfb is the clock feedback. This argument must be supplied. This is usually a buffered version of the output clock at phase 0.

reset is an optional reset. This resets the DLL, DFS and phase shifter and re-establishes lock. operations.

output signals

clk0 and **clk180** are clock outputs at the input frequency. These are at phases 0 and 180 degree respectively relative to the input clock.

clkdv is the divided clock frequency where the division ratio is given by the 3rd input argument.

locked is high when the DLL and DFS are locked to the input clock. This signal has a null clock domain.

11.1.2.5 simpleclock - create a clock of specified frequency

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
simpleclock (ident(v), clock(in), float(freq), float(tol)) {}
```

Output parameters:

none.

Input parameters:

	mode	type	
v	-	-	new clock variable identifier
in	clock	-	input clock
freq	immediate	float	requested frequency
tol	immediate	float	frequency tolerance

In-line target execution:

no

description:

Creates a clock variable using a clkdcn and clock buffer as its source. **At present it can only be used on FPGA platforms that contain DCMs.**

v is an identifier for the new clock mode variable.

in is a clock that can be used to drive the DCM.

freq is the requested frequency in Hz. If this has no argument then the clock frequency will be the same as that of the input clock. This can be used to simply add a DCM and clock buffer to an input clock from an input buffer.

tol is the allowable tolerance on the requested frequency in Hz.

If a frequency within the specified tolerance cannot be generated then a fatal error occurs.

This procedure does not attempt to optimise the use of DCMs to generate multiple clocks. It uses a new DCM for each call.

The procedure determines the multiplier and divisor which will produce the closest synthesiser frequency to that requested. If the requested frequency is lower than the input frequency the procedure also attempts to find a clock division ratio close to that required to achieve the requested frequency. The closest match from the above is then chosen. If the resulting frequency is not within tolerance a fatal error occurs.

The clock variable will have the generated frequency attribute, not that of the requested frequency.

Example of simple clock input.

```
iopins(c32_pin, "str", "P94", frequency=32.0e6, ibufg=true, iostandard=lvcmos33);

clock(c32_in, loc=c32_pin);      // clock input from pin and input buffer
simpleclock(global.c32, c32_in); // global clock
```

Example of multiple global clocks generated from c32 clock.

```
simpleclock(global.c16, c32, 16.0e6, 0.1); // 16 MHz clock
simpleclock(global.c25, c32, 25.0e6, 0.1); // 25 MHz clock
```

11.1.2.6 clkmmcm_adv - create an advanced MMCM multi-mode clock manager

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clkmmcm_adv (
    clock({ clkout0, clkout1, clkout2, clkout3, clkout4, clkout5, clkout6,
            clkout0b, clkout1b, clkout2b, clkout3b, clkfbout, clkfboutb}),
    value({clkfbstopped, clkinstopped, drdy, locked, psdone}, "log"),
    value(dout, "bits:16")
    <-
    str(loc),

    log({ clkfbout_use_fine_ps, clkout0_use_fine_ps, clkout1_use_fine_ps,
          clkout2_use_fine_ps, clkout3_use_fine_ps, clkout4_use_fine_ps,
          clkout5_use_fine_ps, clkout6_use_fine_ps, clock_hold, startup_wait}),
    int(divclk_divide),
    float({ clkfbout_phase, clkfbout_mult_f,
            clkout0_divide_f,
            clkout0_duty_cycle, clkout0_phase, clkout1_divide_f,
            clkout1_duty_cycle, clkout1_phase, clkout2_divide_f,
            clkout2_duty_cycle, clkout2_phase, clkout3_divide_f,
            clkout3_duty_cycle, clkout3_phase, clkout4_divide_f,
            clkout4_duty_cycle, clkout4_phase, clkout5_divide_f,
            clkout5_duty_cycle, clkout5_phase, clkout6_divide_f,
            clkout6_duty_cycle, clkout6_phase, ref_jitter1, ref_jitter2}),
    str(compensation)
    clock({clkfb, clkin1, clkin2, dclk, psclk})),
```

```

    value({clkinsel, den, dwe, psen, psincdec, pwrdown, reset}, "log"),
    value(din, "bits:16"), value(daddr, "uint:7")
  ) {}

```

Output parameters:

	mode	type	
clkout0	clock		output clock 0
clkout1	clock		output clock 1
clkout2	clock		output clock 2
clkout3	clock		output clock 3
clkout4	clock		output clock 4
clkout5	clock		output clock 5
clkout6	clock		output clock 6
clkout0b	clock		output inverted clock 0
clkout1b	clock		output inverted clock 1
clkout2b	clock		output inverted clock 2
clkout3b	clock		output inverted clock 3
clkfbout	clock		output clock feedback
clkfboutb	clock		output inverted clock feedback
clkfbstopped	value	log	clock feedback stopped indicator
clkinstopped	value	log	input clock stopped indicator
drdy	value	log	DRP ready
locked	value	log	MMC locked
psdone	value	log	phase shift done
dout	value	log	DRP data output

Input parameters:

	mode	type	
loc	immediate	str	optional location
clkfbout_use_fine_ps	immediate	log	enable feedback phase shift fine counter
clkout0_use_fine_ps	immediate	log	enable output 0 phase shift fine counter
clkout1_use_fine_ps	immediate	log	enable output 1 phase shift fine counter
clkout2_use_fine_ps	immediate	log	enable output 2 phase shift fine counter
clkout3_use_fine_ps	immediate	log	enable output 3 phase shift fine counter
clkout4_use_fine_ps	immediate	log	enable output 4 phase shift fine counter
clkout5_use_fine_ps	immediate	log	enable output 5 phase shift fine counter
clkout6_use_fine_ps	immediate	log	enable output 6 phase shift fine counter
clock_hold	immediate	log	not documented
startup_wait	immediate	log	delay startup on lock
divclk_divide	immediate	int	divider ratio, 1 to 128
clkfbout_phase	immediate	float	clock feedback phase, -360.0 to 360.0
clkfbout_mult_f	immediate	float	clock feedback multiplier, 1.0 to 64.0
clkout0_divide_f	immediate	float	clock 0 divider, 1.0 to 128.0
clkout0_duty_cycle	immediate	float	clock 0 duty cycle, 0.001 to 0.999
clkout0_phase	immediate	float	clock 0 phase, -360.0 to 360.0
clkout1_divide_f	immediate	float	clock 1 divider, 1.0 to 128.0
clkout1_duty_cycle	immediate	float	clock 1 duty cycle, 0.001 to 0.999
clkout1_phase	immediate	float	clock 1 phase, -360.0 to 360.0
clkout2_divide_f	immediate	float	clock 2 divider, 1.0 to 128.0
clkout2_duty_cycle	immediate	float	clock 2 duty cycle, 0.001 to 0.999
clkout2_phase	immediate	float	clock 2 phase, -360.0 to 360.0
clkout3_divide_f	immediate	float	clock 3 divider, 1.0 to 128.0
clkout3_duty_cycle	immediate	float	clock 3 duty cycle, 0.001 to 0.999
clkout3_phase	immediate	float	clock 3 phase, -360.0 to 360.0
clkout4_divide_f	immediate	float	clock 4 divider, 1.0 to 128.0
clkout4_duty_cycle	immediate	float	clock 4 duty cycle, 0.001 to 0.999
clkout4_phase	immediate	float	clock 4 phase, -360.0 to 360.0
clkout5_divide_f	immediate	float	clock 5 divider, 1.0 to 128.0
clkout5_duty_cycle	immediate	float	clock 5 duty cycle, 0.001 to 0.999
clkout5_phase	immediate	float	clock 5 phase, -360.0 to 360.0
clkout6_divide_f	immediate	float	clock 6 divider, 1.0 to 128.0
clkout6_duty_cycle	immediate	float	clock 6 duty cycle, 0.001 to 0.999
clkout6_phase	immediate	float	clock 6 phase, -360.0 to 360.0
ref_jitter1	immediate	float	input clock 1 jitter, 0.000 to 0.999
ref_jitter2	immediate	float	input clock 2 jitter, 0.000 to 0.999
compensation	immediate	str	see Xilinx documentation
clkfb	clock		clock feedback input
clkin1	clock		input clock 1
clkin2	clock		input clock 2
dclk	clock		DRP clock input
pclk	clock		phase shift clock
clkinsel	value	log	input clock selector
den	value	log	DRP enable
dwe	value	log	DRP write enable
psen	value	log	phase shift enable
psincdec	value	log	phase shift increment or decrement
pwrdown	value	log	power down
reset	value	log	reset
din	value	bits:	DRP data input
daddr	value	uint:	DRP address input

In-line target execution:

no

description:

Creates an advanced Virtex6 or series 7 mixed mode clock manager. See Xilinx documentation for signal and attribute details. The first group of input parameters are immediate mode and are attributes. The remaining input parameters are input signals, clock mode for clocks and value mode for control signals.

11.1.2.7 clkmmcm_base - create a basic MMCM multi-mode clock manager

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clkmmcm_base (
    clock({ clkout0, clkout1, clkout2, clkout3, clkout4, clkout5, clkout6,
            clkout0b, clkout1b, clkout2b, clkout3b, clkfbout, clkfboutb}),
    value(locked, "log")
    <-
    str(loc),
    str(bandwidth),
    float(clkfbout_mult_f),
    float(clkfbout_phase),
    float(clkin1_period),
    int({clkout1_divide, clkout2_divide, clkout3_divide,
         clkout4_divide, clkout5_divide, clkout6_divide}),
    float(clkout0_divide_f),
    float(clkout0_duty_cycle, clkout1_duty_cycle,
           clkout2_duty_cycle, clkout3_duty_cycle,
           clkout4_duty_cycle, clkout5_duty_cycle,
           clkout6_duty_cycle),
    float(clkout0_phase, clkout1_phase, clkout2_phase,
           clkout3_phase, clkout4_phase, clkout5_phase,
           clkout6_phase),
    int(divclk_divide),
    float(ref_jitter1),
    log(startup_wait),
    clock({clkin, clkfb}),
    value({pwrdown, reset}, "log")
) {}
```

Output parameters:

	mode	type	
clkout0	clock		output clock 0
clkout1	clock		output clock 1
clkout2	clock		output clock 2
clkout3	clock		output clock 3
clkout4	clock		output clock 4
clkout5	clock		output clock 5
clkout6	clock		output clock 6
clkout0b	clock		output inverted clock 0
clkout1b	clock		output inverted clock 1
clkout2b	clock		output inverted clock 2
clkout3b	clock		output inverted clock 3
clkfbout	clock		output clock feedback
clkfboutb	clock		output inverted clock feedback

Input parameters:

	mode	type	
loc	immediate	str	optional allocation
bandwidth	immediate	str	bandwidth
clkfbout_mult_f	immediate	float	output multiplier
clkfbout_phase	immediate	float	output phase
clkin1_period	immediate	float	input clock period
clkout0_divide_f	immediate	float	clock 0 divider, 1.0 to 128.0
clkout1_divide	immediate	int	clock 1 divider, 1 to 128
clkout2_divide	immediate	int	clock 2 divider, 1 to 128
clkout3_divide	immediate	int	clock 3 divider, 1 to 128
clkout4_divide	immediate	int	clock 4 divider, 1 to 128
clkout5_divide	immediate	int	clock 5 divider, 1 to 128
clkout6_divide	immediate	int	clock 6 divider, 1 to 128
clkout0_duty_cycle	immediate	float	clock 0 duty cycle, 0.001 to 0.999
clkout1_duty_cycle	immediate	float	clock 1 duty cycle, 0.001 to 0.999
clkout2_duty_cycle	immediate	float	clock 2 duty cycle, 0.001 to 0.999
clkout3_duty_cycle	immediate	float	clock 3 duty cycle, 0.001 to 0.999
clkout4_duty_cycle	immediate	float	clock 4 duty cycle, 0.001 to 0.999
clkout5_duty_cycle	immediate	float	clock 5 duty cycle, 0.001 to 0.999
clkout6_duty_cycle	immediate	float	clock 6 duty cycle, 0.001 to 0.999
clkout0_phase	immediate	float	clock 0 phase, -360.0 to 360.0
clkout1_phase	immediate	float	clock 1 phase, -360.0 to 360.0
clkout2_phase	immediate	float	clock 2 phase, -360.0 to 360.0
clkout3_phase	immediate	float	clock 3 phase, -360.0 to 360.0
clkout4_phase	immediate	float	clock 4 phase, -360.0 to 360.0
clkout5_phase	immediate	float	clock 5 phase, -360.0 to 360.0
clkout6_phase	immediate	float	clock 6 phase, -360.0 to 360.0
divclk_divide	immediate	int	divider ratio, 1 to 128
startup_wait	immediate	log	delay startup on lock
clkin	clock		input clock
clkfb	clock		clock feedback input
pwrdown	value	log	power down
reset	value	log	reset

In-line target execution:

no

description:

Creates a basic Virtex6 or series 7 mixed mode clock manager. See Xilinx documentation for signal and attribute details. The first group of input parameters are immediate mode and are attributes. The remaining input parameters are input signals, clock mode for clocks and value mode for control signals.

11.1.2.8 pll_base - create a series-7 basic PLL

Library: xilinx/clocks.3pl

Prototype:

```

global procedure
pll_base (
  clock({clkout0, clkout1, clkout2, clkout3, clkout4, clkout5, clkfbout}),
  value(locked, "log")
  <-
  str(loc),
  int({ clkout0_divide, clkout1_divide, clkout2_divide, clkout3_divide,
        clkout4_divide, clkout5_divide, clkfbout_mult, clkfbout_phase,
        divclk_divide}),
  str({compensation, bandwidth}),

```

```

float({clkout0_phase, clkout1_phase, clkout2_phase, clkout3_phase, clkout4_phase,
      clkout5_phase, clkout0_duty_cycle, clkout1_duty_cycle, clkout2_duty_cycle,
      clkout3_duty_cycle, clkout4_duty_cycle, clkout5_duty_cycle, ref_jitter,
      clkin_period}),
clock({clkin, clkfbin}), value(reset, "log")
) {}

```

Output parameters:

	mode	type	
clkout0	clock		output clock 0
clkout1	clock		output clock 0
clkout2	clock		output clock 0
clkout3	clock		output clock 0
clkout4	clock		output clock 0
clkout5	clock		output clock 0
clkfbout	clock		feedback output
locked	value	log	PLL is locked

Input parameters:

	mode	type	
loc	immediate	str	optional location
clkout0_divide	immediate	int	division ratio for output 0
clkout1_divide	immediate	int	division ratio for output 1
clkout2_divide	immediate	int	division ratio for output 2
clkout3_divide	immediate	int	division ratio for output 3
clkout4_divide	immediate	int	division ratio for output 4
clkout5_divide	immediate	int	division ratio for output 5
clkfbout_mult	immediate	int	multiplication factor for all outputs
clkfbout_phase	immediate	int	phase shift for feedback
divclk_divide	immediate	int	division ration for all outputs
compensation	immediate	str	
bandwidth	immediate	str	
clkout0_phase	immediate	float	phase shift for output 0
clkout1_phase	immediate	float	phase shift for output 1
clkout2_phase	immediate	float	phase shift for output 2
clkout3_phase	immediate	float	phase shift for output 3
clkout4_phase	immediate	float	phase shift for output 4
clkout5_phase	immediate	float	phase shift for output 5
clkout0_duty_cycle	immediate	float	duty cycle for output 0
clkout1_duty_cycle	immediate	float	duty cycle for output 1
clkout2_duty_cycle	immediate	float	duty cycle for output 2
clkout3_duty_cycle	immediate	float	duty cycle for output 3
clkout4_duty_cycle	immediate	float	duty cycle for output 4
clkout5_duty_cycle	immediate	float	duty cycle for output 5
ref_jitter	immediate	float	reference clock jitter
clkin_period	immediate	float	input clock period, ns
clkin	clock		input clock
clkfbin	clock		clock feedback
reset	value	log	reset

In-line target execution:

no

description:

Creates a Virtex5 PLL phase locked loop. See Xilinx documentation for signal and attribute details. The first group of input parameters are immediate mode and are attributes. The remaining input parameters are input signals, clock mode for clocks and value mode for control signals.

The PLL is PLL_BASE, i.e. only a single input is provided. The dual switchable input PLL, PLL_ADV, is not supported.

attributes

loc is an optional location string from the following list - "DLL0P", "DLL1P", "DLL2P", "DLL3P", "DLL0S", "DLL1S", "DLL2S", "DLL3S". These select from the four primary and four secondary delay-locked loops. Ignored if "".

clkin_period is the input clock period. This must be specified if the DFS is used.

compensation specifies the PLL phase compensation. The values are "system_synchronous" (default) or "source_synchronous".

bandwidth specifies the PLL programming algorithm affecting jitter, phase margin and other characteristics. Values are "high", "low" and "optimised" (default).

clkout0_divide to **clkout5_divide** provides a PLL division ratio for each clock output. The values range from 1 (default) to 128.

clkout0_phase to **clkout5_phase** specifies the PLL output phase relationship (relative to what?). Values are in the range 0.01 to 360.0 but the default is listed as 0.0.

clkout0_duty_cycle to **clkout5_duty_cycle** specifies the PLL duty cycle of each clock output. Values range from 0.01 to 0.99 with a default of 0.5.

clkfbout_mult specifies a common frequency multiplier for all PLL clock outputs. This works in combination with the individual frequency divisors. The values range from 1 to 64 with a default of 1.

divclk_divide specifies a frequency division ration for all PLL clock outputs. It is documented how this interacts with the individual divisors. Range is 1 to 52 with a default of 1.

clkfbout_phase specifies a phase offset for the PLL clock feedback output. Values range from 0.0 to 360.0 with a default of 0.0.

ref_jitter specifies the PLL reference clock jitter. This is documented as "specified in terms of the UI which is a percentage of the reference clock". Allowed values are 0.000 to 0.999 with a default of 0.100.

input signals

clkin is the input clock.

clkfbin is the feedback clock input, which is normally connected to **clkfbout** or a buffered version of an output clock.

reset is the reset input.

output signals

clkout0, **clkout1**, **clkout2**, **clkout3**, **clkout4** and **clkout5** are separate clock outputs for a PLL. Each of these has its own frequency divisor, phase and duty cycle.

clkfbout is the clock feedback output.

locked is high when the DLL and DFS are locked to the input clock. This signal has a null clock domain.

11.1.2.9 clockbufg - create a BUFG clock buffer

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clockbufg (clock(out) <- clock(in), str(loc), log(invert)) {
```

Output parameters:

out is the clock variable for the distributed clock.

Input parameters:

in is the internal clock variable for the buffer input clock.

loc is optional and is specified by a string of the form "BUFGMUX2S". Generally the location argument may be omitted or "", allowing the Xilinx software to select a clock buffer.

invert specifies an optional inverter. Ignored if "".

In-line target execution:

no

description:

This procedure creates a BUFG clock buffer which is common to all Xilinx FPGAs. The number of clock buffers available depends on the particular FPGA.

If **invert** is true an inverter is inserted at the input.

The frequency, minimum frequency, maximum frequency, period and duty cycle of the output clock are derived from the input clock overriding any previous values.

11.1.2.10 clockbufio - a local clock buffer

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clockbufio (clock(out) <- clock(in), str(loc)) {}
```

Output parameters:

out is the clock variable for the distributed clock.

Input parameters:

in is the internal clock variable for the buffer input clock.

loc is optional and is specified by a string of the form "BUFGMUX2S". Generally the location argument may be omitted or "", allowing the Xilinx software to select a clock buffer.

In-line target execution:

no

description:

This procedure creates a local clock buffer. The number of clock buffers available depends on the particular FPGA.

The frequency, period and duty cycle of the output clock are derived from the input clock overriding any previous values.

11.1.2.11 clockmuxbuffer - a clock mux/buffer

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clockmuxbuffer (clock(out) <- clock({in0, in1}), value(select, "log"), str(loc)) {}
```

Output parameters:

out is the clock variable for the distributed clock.

Input parameters:

in0 is the internal clock variable for the first buffer input clock.

in1 is the internal clock variable for the second buffer input clock.

select is a logical value which selects the first clock input if false and the second clock input if true.

loc is optional and is specified by a string of the form "BUFGMUX2S". Generally the location argument may be omitted, allowing the Xilinx software to select a clock buffer.

In-line target execution:

no

description:

This procedure creates a clock buffer with multiplexed input. The number of multiplexed clock buffers available depends on the particular FPGA.

The frequency, period and duty cycle of the output clock are derived from the input clock overriding any previous values.

11.1.2.12 clockmuxbufferdisable - a clock mux/buffer with disable

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clockmuxbufferdisable (clock(out) <- clock(in), value(disable, "log"), str(loc)) {
```

Output parameters:

out is the clock variable for the distributed clock.

Input parameters:

in is the internal clock variable for the first buffer input clock.

disable is a logical value which disables the clock output if true.

loc is optional and is specified by a string of the form "BUFGMUX2S". Generally the location argument may be omitted or "", allowing the Xilinx software to select a clock buffer.

In-line target execution:

no

description:

This procedure creates a clock buffer with enabled output. If the argument to parameter **disable** goes true the clock output is disabled low. The output clock is enabled and disabled without glitches or shortened high half-periods.

The frequency, period and duty cycle of the output clock are derived from the input clock overriding any previous values.

11.1.2.13 clock_params - derive clock synthesis parameters

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
clock_params (
    int({mul, div}),
    float(af)
    <-
    float({fin, freq}),
    int({mmin, mmax, dmin, dmax}),
    float(tol)
) {}
```

Output parameters:**mul** is the synthesis multiplier**div** is the synthesis divisor**af** is the resulting frequency**Input parameters:****fin** is the input frequency**freq** is the required frequency**mmin** is the minimum value of the multiplier**mmax** is the maximum value of the multiplier**dmin** is the minimum value of the divisor**dmax** is the maximum value of the divisor**tol** is an optional frequency tolerance to minimise the search**In-line target execution:**

no

description:

This procedure derives an integer multiplier and an integer divisor for clock synthesis.

The procedure searches through combinations of multiplier and divisor to find the nearest generated frequency that is not greater than the requested frequency. The optional tolerance argument when supplied greatly reduces the execution time. Frequencies may be in any units, e.g. MHz, KHz, Hz, since only ratios are being examined, but hence the units must be consistent. It is nevertheless preferable to use frequencies in Hz to be consistent in all contexts.

If the requested frequency cannot be generated within the constraints of the arguments supplied the procedure returns an error code.

mul and **div** are the frequency synthesis parameters, **af** is the actual frequency achieved.

A fatal error occurs if the frequency cannot be synthesised because the tolerance is too small or the requested frequency is too low (the error message will specify which).

To avoid the fatal error the related library function of the same name can be used instead. The function returns a structure containing the result parameters along with an error code.

11.1.2.14 lowskew - request low skew secondary clock routing

Library: xilinx/clocks.3pl

Prototype:

```
global procedure
lowskew (clock(clk)) {}
```

Output parameters:

none

Input parameters:**clk** is a clock variable.**In-line target execution:**

no

description:

This procedure places a low skew constraint on the routing of a clock signal. This is required if the clock cannot be distributed by a global clock net (from a global clock buffer). There are only 4 global clock buffers.

11.1.3 library functions

11.1.3.1 clock_params - derive clock synthesis parameters

Library: xilinx/clocks.3pl

Prototype:

```
global function
clock_params (
    float({fin, freq}),
    int({mmin, mmax, dmin, dmax}),
    float(tol)
) {}
```

Input parameters:

fin is the input frequency

freq is the required frequency

mmin is the minimum value of the multiplier

mmax is the maximum value of the multiplier

dmin is the minimum value of the divisor

dmax is the maximum value of the divisor

tol is an optional frequency tolerance to minimise the search

Return mode and type:

immediate, struct

In-line target execution:

no

description:

This function derives an integer multiplier and an integer divisor for clock synthesis.

The function searches through combinations of multiplier and divisor to find the nearest generated frequency that is not greater than the requested frequency. The optional tolerance argument when supplied greatly reduces the execution time. Frequencies may be in any units, e.g. MHz, KHz, Hz, since only ratios are being examined, but hence the units must be consistent. It is nevertheless preferable to use frequencies in Hz to be consistent in all contexts.

If the requested frequency cannot be generated within the constraints of the arguments supplied the procedure returns an error code.

The return value has the following structure -

```
struct(rv,
    {mul, div}, "int",
    af,         "float",
    exit,       "int"
);
```

where **mul** and **div** are the frequency synthesis parameters, **af** is the actual frequency achieved and **exit** is an exit code which is 0 for normal return, 1 for a frequency tolerance which is too small and 2 for a requested frequency which is too low.

11.1.4 library classes

None.

11.2 Common Xilinx Procedures Library File - xilinx/common.3pl

This library provides common modules, procedures and functions for all Xilinx FPGAs.

This library must be incorporated by using a **source** statement -

```
source "xilinx/common.3pl"
```

All Xilinx family header files include this library.

11.2.1 library modules

11.2.1.1 sequence - generate a sequence

Library: xilinx/common.3pl

Prototype:

```
global module
sequence (value(out) <- var(s), value(step, "log", true)) {}
```

Output parameters:

out is the output value variable.

Input parameters:

s is the sequence array.

step is an optional input to control data movement.

description:

This module generates a cyclic sequence of values, these appearing on the output value variable. The sequence is determined by the input argument **s** which must be an array of immediate values. The type of these values must be an immediate equivalent to the type of the output argument and may be of compound type. The 2nd input argument, **step**, is optional and is used to control when the sequence advances, otherwise this occurs on every clock edge.

11.2.2 library procedures

11.2.2.1 IOconstraint - add an I/O constraint to the constraints file

Library: xilinx/common.3pl

Prototype:

```
global procedure
IOconstraint (str(loc), str(con), var(val)) {}
```

Output parameters:

none

Input parameters:

loc is the I/O pad designation

con is the constraint identifier

val is the constraint value

In-line target execution:

no

description:

This procedure adds a line to the NCF (ISE) or XDC (Vivado) constraint file as determined by directive "postProcessTool" ([section 3.10](#) p75).

con is a string specifying the constraint. Valid constraints are -

"diff_term":	log	input differential termination
"drive":	int	output drive current
"ibuf_delay_value":	int	input direct delay
"ifd_delay_value":	int	input flip-flop delay
"iostandard":	log	I/O standard
"pullup":	log	pullup
"pulldown":	log	pulldown
"slew":	str	output slew rate
"tnm":	str	timing name

The constraint value **val** may be type "str", "log" or "int" as appropriate.

Note that if the pad is bi-directional and an I/O standard constraint is provided separately for both input and output the second will be ignored.

11.2.2.2 addncfline - add line to NCF file

Library: xilinx/common.3pl

Prototype:

```
global procedure
addncfline (str(s)) {}
```

Output parameters:

none

Input parameters:

s is the line to be added to the NCF file

In-line target execution:

no

description:

This procedure adds a line to the NCF file. (Note that a trailing semicolon is not appended.)

See also -

append_ncf_file() [subsection 11.2.2.4](#) p378

elementaddncf() [subsection 11.2.2.5](#) p378

addxdcline() [subsection 11.2.2.3](#) p377

maxdelay() [subsection 11.2.2.20](#) p391

period_mul() [subsection 11.2.2.23](#) p393

period_div() [subsection 11.2.2.23](#) p393

netaddncf() [subsection 11.2.2.21](#) p392.

11.2.2.3 addxdcline - add line to XDC file

Library: xilinx/common.3pl

Prototype:

```
global procedure
addxdcline (str(fmt), value(sig)) {}
```

Output parameters:

none

Input parameters:

fmt is a format string of the line to be added to the XDC file. **sig** is an optional signal to insert into the line.

In-line target execution:

no

description:

This procedure adds a line to the XDC file. (Note that a trailing semicolon is not appended.) **fmt** is a format string which may contain a **%n** conversion to insert the optional signal identifier **sig**.

See also -

addncfline() [subsection 11.2.2.2](#) p377
append_ncf_file() [subsection 11.2.2.4](#) p378
elementaddncf() [subsection 11.2.2.5](#) p378
maxdelay() [subsection 11.2.2.20](#) p391
period_mul() [subsection 11.2.2.23](#) p393
period_div() [subsection 11.2.2.23](#) p393
netaddncf() [subsection 11.2.2.21](#) p392.

11.2.2.4 append_ncf_file - append a file to the NCF file

Library: xilinx/common.3pl

Prototype:

```
global procedure
append_ncf_file (str(filename)) {}
```

Output parameters:

none

Input parameters:

filename is the input file name.

In-line target execution:

no

description:

This procedure reads a file and appends it to the NCF file (Netlist Constraints File).

See also -

addncfline() [subsection 11.2.2.2](#) p377
elementaddncf() [subsection 11.2.2.5](#) p378
addxdcline() [subsection 11.2.2.3](#) p377
maxdelay() [subsection 11.2.2.20](#) p391
period_mul() [subsection 11.2.2.23](#) p393
period_div() [subsection 11.2.2.23](#) p393
netaddncf() [subsection 11.2.2.21](#) p392.

11.2.2.5 elementaddncf - add line to NCF file

Library: xilinx/common.3pl

Prototype:

```
global procedure
elementaddncf (str({id, attr, val})) {}
```

Output parameters:

none

Input parameters:**id** is the netlist block identifier (optional).**attr** is the attribute or constraint name or else a line to be added to the NCF file for the element.**val** is the value of the attribute or constraint if the previous argument is an attribute or constraint name.**In-line target execution:**

no

description:

This procedure adds a line to the Xilinx NCF file for an element. If the block identifier is given then the NCF entry applies to that element, otherwise it applies to the element most recently constructed by inbuilt procedure **element()**. Note that in either case for the element concerned its construction must have been completed by calling inbuilt procedure **elementend()** before calling the current library procedure.

The constraint string will have something like "INST block234" prepended, where "block234" is the identifier for the element, and will also have a trailing ";" appended.

If the argument to **val** is omitted then the string **attr** will be used verbatim. If the argument to **val** is present then **attr** will be used as a key to a table of constraints for the element and **val** will be the associated value. In that case when the attribute or constraint is written to the constraint file a "=" will be inserted between the attribute/constraint name and the key. Thus an attribute may be entered by -

```
elementaddncf("b123", "REG=3");
```

in which case the entire second argument becomes the key and the value is null. The key only is copied to the constraint file. The limitation of this is that the attribute value cannot be changed. If on the other hand the form

```
elementaddncf("b123", "REG", "3");
```

is used then the attribute can be changed later if desired.

See also -

addncfline() [subsection 11.2.2.2](#) p377

append_ncf_file() [subsection 11.2.2.4](#) p378

addxdcline() [subsection 11.2.2.3](#) p377

maxdelay() [subsection 11.2.2.20](#) p391

period_mul() [subsection 11.2.2.23](#) p393

period_div() [subsection 11.2.2.23](#) p393

netaddncf() [subsection 11.2.2.21](#) p392.

11.2.2.6 elementBinProperty - add a property of type "int" to an element in binary format

Library: xilinx/common.3pl

Prototype:

```
global procedure
elementBinProperty (str(id), str(prop), int({n, min, max}), int(digits)) {}
```

Output parameters:

none

Input parameters:**id** is the element identifier or null.

prop is the property identifier.

n is the property value.

min is the property minimum value.

max is the property maximum value.

digits is the number of binary digits required.

In-line target execution:

no

description:

This procedure adds a property of type "int" to an element in binary format. **id** is a unique identifier for the element. If the argument is omitted then the current element is used, i.e. the one most recently created using inbuilt procedure **element()** [section 9.18](#) p201.

prop is the property identifier.

n is the value of the property. If no argument is passed then this procedure does nothing and returns. Note that if this procedure is called within another module or procedure and the property value is a parameter of that enclosing procedure then the inbuilt function **nullarg()** [section 10.97](#) p311 may be used to enclose the argument to this procedure so that an omitted argument is ignored rather than the default parameter value of the enclosing procedure being used.

min and **max** are the minimum and maximum allowed values of the property. If one of these is exceeded then a fatal error will be forced.

digits specifies the number of binary digits with which to express the property value. The binary digit string is not preceded by any base indicator (e.g. there is **no** leading "0b").

See also -

elementLogProperty() [subsection 11.2.2.11](#) p383 for logical properties, **elementIntProperty()** [subsection 11.2.2.10](#) p382 for integer properties, **elementHexProperty()** [subsection 11.2.2.9](#) p381 for hexadecimal properties, **elementFloatProperty()** [subsection 11.2.2.8](#) p381 for floating point properties, **elementStrProperty()** [subsection 11.2.2.12](#) p384 for string properties.

11.2.2.7 elementcheck - check that a Xilinx element is allowed in the current device family

Library: xilinx/common.3pl

Prototype:

```
global procedure
elementCheck(str({element, name})) {}
```

Output parameters:

none

Input parameters:

element is the name of a Xilinx library element.

name is the name of a library module, procedure or function which is attempting to create the library element.

In-line target execution:

no

description:

This procedure checks that the Xilinx library element named by the first argument is available for the current Xilinx device family. Allowed elements for each device family are kept in an abbreviated map in xilinx/common.3pl, i.e. only a subset of elements are included, namely those used explicitly in 3PL libraries. The map can be expanded when required.

If the check fails, i.e. the element is not available in the current device family, then 3PL exits with an error message and traceback. **name** is provided solely for use in the error message.

See also library function **elementcheck()** [subsection 11.2.3.3](#) p398 which can check non-fatally.

11.2.2.8 elementFloatProperty - add a property of type "float" to an element

Library: xilinx/common.3pl

Prototype:

```
global procedure
elementFloatProperty (str(id), str(prop), float({n, min, max}), int(digits)) {}
```

Output parameters:

none

Input parameters:

id is the element identifier or null.

prop is the property identifier.

n is the property value.

min is the property minimum value.

max is the property maximum value.

digits is the number of digits required after the decimal point.

In-line target execution:

no

description:

This procedure adds a property of type "float" to an element. **id** is a unique identifier for the element. If the argument is omitted then the current element is used, i.e. the one most recently created using inbuilt procedure **element()** [section 9.18](#) p201.

prop is the property identifier.

n is the value of the property. If no argument is passed then this procedure does nothing and returns. Note that if this procedure is called within another module or procedure and the property value is a parameter of that enclosing procedure then the inbuilt function **nullarg()** [section 10.97](#) p311 may be used to enclose the argument to this procedure so that an omitted argument is ignored rather than the default parameter value of the enclosing procedure being used.

min and **max** are the minimum and maximum allowed values of the property. If one of these is exceeded then a fatal error will be forced.

digits specifies the number of digits to appear to the right of the decimal point.

See also -

elementLogProperty() [subsubsection 11.2.2.11](#) p383 for logical properties, **elementIntProperty()** [subsubsection 11.2.2.10](#) p382 for integer properties, **elementBinProperty()** [subsubsection 11.2.2.6](#) p379 for binary properties, **elementHexProperty()** [subsubsection 11.2.2.9](#) p381 for hexadecimal properties, **elementStrProperty()** [subsubsection 11.2.2.12](#) p384 for string properties.

11.2.2.9 elementHexProperty - add a property of type "int" to an element in hexadecimal format

Library: xilinx/common.3pl

Prototype:

```
global procedure
elementHexProperty (str(id), str(prop), int({n, min, max}), int(digits)) {}
```

Output parameters:

none

Input parameters:

id is the element identifier or null.

prop is the property identifier.

n is the property value.

min is the property minimum value.

max is the property maximum value.

digits is the number of hexadecimal digits required.

In-line target execution:

no

description:

This procedure adds a property of type "int" to an element in binary format. **id** is a unique identifier for the element. If the argument is omitted then the current element is used, i.e. the one most recently created using inbuilt procedure **element()** [section 9.18](#) p201.

prop is the property identifier.

n is the value of the property. If no argument is passed then this procedure does nothing and returns. Note that if this procedure is called within another module or procedure and the property value is a parameter of that enclosing procedure then the inbuilt function **nullarg()** [section 10.97](#) p311 may be used to enclose the argument to this procedure so that an omitted argument is ignored rather than the default parameter value of the enclosing procedure being used.

min and **max** are the minimum and maximum allowed values of the property. If one of these is exceeded then a fatal error will be forced.

digits specifies the number of hexadecimal digits with which to express the property value. The hexadecimal digit string is not preceded by any base indicator (e.g. there is **no** leading "0x").

See also -

elementLogProperty() [subsubsection 11.2.2.11](#) p383 for logical properties, **elementIntProperty()** [subsubsection 11.2.2.10](#) p382 for integer properties, **elementBinProperty()** [subsubsection 11.2.2.6](#) p379 for binary properties, **elementFloatProperty()** [subsubsection 11.2.2.8](#) p381 for floating point properties, **elementStrProperty()** [subsubsection 11.2.2.12](#) p384 for string properties.

11.2.2.10 elementIntProperty - add a property of type "int" to an element

Library: xilinx/common.3pl

Prototype:

```
global procedure
elementIntProperty (str(id), str(prop), int({n, min, max})) {}
```

Output parameters:

none

Input parameters:

id is the element identifier or null.

prop is the property identifier.

n is the property value.

min is the property minimum value.

max is the property maximum value.

In-line target execution:

no

description:

This procedure adds a property of type "int" to an element. **id** is a unique identifier for the element.

If the argument is omitted then the current element is used, i.e. the one most recently created using inbuilt procedure **element()** [section 9.18](#) p201.

prop is the property identifier.

n is the value of the property. If no argument is passed then this procedure does nothing and returns. Note that if this procedure is called within another module or procedure and the property value is a parameter of that enclosing procedure then the inbuilt function **nullarg()** [section 10.97](#) p311 may be used to enclose the argument to this procedure so that an omitted argument is ignored rather than the default parameter value of the enclosing procedure being used.

min and **max** are the minimum and maximum allowed values of the property. If one of these is exceeded then a fatal error will be forced.

See also -

elementLogProperty() [subsection 11.2.2.11](#) p383 for logical properties, **elementBinProperty()** [subsection 11.2.2.6](#) p379 for binary properties, **elementHexProperty()** [subsection 11.2.2.9](#) p381 for hexadecimal properties, **elementFloatProperty()** [subsection 11.2.2.8](#) p381 for floating point properties, **elementStrProperty()** [subsection 11.2.2.12](#) p384 for string properties.

11.2.2.11 elementLogProperty - add a property of type "log" to an element

Library: xilinx/common.3pl

Prototype:

```
global procedure
elementLogProperty (str(id), str(prop), log(l)) {}
```

Output parameters:

none

Input parameters:

id is the element identifier or null.

prop is the property identifier.

l is the property value.

In-line target execution:

no

description:

This procedure adds a property of type "log" to an element. **id** is a unique identifier for the element. If the argument is omitted then the current element is used, i.e. the one most recently created using inbuilt procedure **element()** [section 9.18](#) p201.

prop is the property identifier.

l is the value of the property. If no argument is passed then this procedure does nothing and returns. Note that if this procedure is called within another module or procedure and the property value is a parameter of that enclosing procedure then the inbuilt function **nullarg()** [section 10.97](#) p311 may be used to enclose the argument to this procedure so that an omitted argument is ignored rather than the default parameter value of the enclosing procedure being used.

See also -

elementIntProperty() [subsection 11.2.2.10](#) p382 for integer properties, **elementBinProperty()** [subsection 11.2.2.6](#) p379 for binary properties, **elementHexProperty()** [subsection 11.2.2.9](#) p381 for hexadecimal properties, **elementFloatProperty()** [subsection 11.2.2.8](#) p381 for floating point properties, **elementStrProperty()** [subsection 11.2.2.12](#) p384 for string properties.

11.2.2.12 elementStrProperty - add a property of type "str" to an element

Library: xilinx/common.3pl

Prototype:

```
global procedure
elementStrProperty (str(id), str({prop, v}), varargs) {}
```

Output parameters:

none

Input parameters:

id is the element identifier or null.

prop is the property identifier.

v is the property value.

varargs matches a number of string arguments listing the allowed property values.

In-line target execution:

no

description:

This procedure adds a property of type "str" to an element. **id** is a unique identifier for the element. If the argument is omitted then the current element is used, i.e. the one most recently created using inbuilt procedure **element()** [section 9.18](#) p201.

prop is the property identifier.

v is the value of the property. If no argument is passed then this procedure does nothing and returns. Note that if this procedure is called within another module or procedure and the property value is a parameter of that enclosing procedure then the inbuilt function **nullarg()** [section 10.97](#) p311 may be used to enclose the argument to this procedure so that an omitted argument is ignored rather than the default parameter value of the enclosing procedure being used.

Optional arguments following **prop** are strings specifying the allowed values for the property. If the property value is not identical to one of these strings then a fatal error will be forced. If the optional arguments are not supplied then the property will be defined unconditionally.

See also -

elementLogProperty() [subsection 11.2.2.11](#) p383 for logical properties, **elementIntProperty()** [subsection 11.2.2.10](#) p382 for integer properties, **elementBinProperty()** [subsection 11.2.2.6](#) p379 for binary properties, **elementHexProperty()** [subsection 11.2.2.9](#) p381 for hexadecimal properties, **elementFloatProperty()** [subsection 11.2.2.8](#) p381 for floating point properties,

11.2.2.13 fdre - Create an FDRE flip-flop

Library: xilinx/common.3pl

Prototype:

```
global procedure
fdre (
    value(q, "log")
    <-
    value(d, "log", false),
    clock(c),
    value(ce, "log", true),
    value(r, "log", false),
    str(init, "R")
) {
```

Output parameters:

q is the flip-flop output.

Input parameters:

d is the data input, default low.

c is the clock input.

ce is the optional clock enable input, default high.

r is the optional synchronous reset input, default low.

init is the initial value, default reset.

In-line target execution:

no

description:

This procedure creates a D-type flip-flop with clock enable and synchronous reset. The **d**, **ce** and **r** inputs must have the same clock domain as the input clock. The **q** output is set to the same clock domain.

See also -

fdse() [subsection 11.2.2.15](#) p386

fdrse() [subsection 11.2.2.14](#) p385

fdce() [subsection 11.2.2.16](#) p386

fdpe() [subsection 11.2.2.17](#) p387.

11.2.2.14 fdrse - Create an FDRSE flip-flop

Library: xilinx/common.3pl

Prototype:

```
global procedure
fdrse (
    value(q, "log")
    <-
    value(d, "log", false),
    clock(c),
    value(ce, "log", true),
    value(r, "log", false),
    value(s, "log", false),
    str(init, "R")
) {
```

Output parameters:

q is the flip-flop output.

Input parameters:

d is the data input, default low.

c is the clock input.

ce is the optional clock enable input, default high.

r is the optional synchronous reset input, default low.

s is the optional synchronous set input, default low.

init is the initial value, default reset.

In-line target execution:

no

description:

This procedure creates a D-type flip-flop with clock enable and synchronous reset and set. The **d**, **ce**, **r** and **s** inputs must have the same clock domain as the input clock. The **q** output is set to the same clock domain.

See also -

fdre() [subsection 11.2.2.13](#) p384

[fdse\(\)](#) [subsection 11.2.2.15](#) p386
[fdce\(\)](#) [subsection 11.2.2.16](#) p386
[fdpe\(\)](#) [subsection 11.2.2.17](#) p387.

11.2.2.15 fdse - Create an FDSE flip-flop

Library: xilinx/common.3pl

Prototype:

```

global procedure
fdse (
    value(q, "log")
    <-
    value(d, "log", false),
    clock(c),
    value(ce, "log", true),
    value(s, "log", false),
    str(init, "S")
) {

```

Output parameters:

q is the flip-flop output.

Input parameters:

d is the data input, default low.

c is the clock input.

ce is the optional clock enable input, default high.

s is the optional synchronous set input, default low.

init is the initial value, default set.

In-line target execution:

no

description:

This procedure creates a D-type flip-flop with clock enable and synchronous set. The **d**, **ce** and **s** inputs must have the same clock domain as the input clock. The **q** output is set to the same clock domain.

See also -

[fdre\(\)](#) [subsection 11.2.2.13](#) p384
[fdrse\(\)](#) [subsection 11.2.2.14](#) p385
[fdce\(\)](#) [subsection 11.2.2.16](#) p386
[fdpe\(\)](#) [subsection 11.2.2.17](#) p387.

11.2.2.16 fdce - Create an FDCE flip-flop

Library: xilinx/common.3pl

Prototype:

```

global procedure
fdce (
    value(q, "log")
    <-
    value(d, "log", false),
    clock(c),
    value(ce, "log", true),
    value(clr, "log", false),
    str(init, "R")
) {

```

Output parameters:

q is the flip-flop output.

Input parameters:

d is the data input, default low.

c is the clock input.

ce is the optional clock enable input, default high.

clr is the optional asynchronous clear input, default low.

init is the initial value, default reset.

In-line target execution:

no

description:

This procedure creates a D-type flip-flop with clock enable and asynchronous clear. The **d**, **ce** and **clr** inputs must have the same clock domain as the input clock. The **q** output is set to the same clock domain.

See also -

fdre() [subsection 11.2.2.13](#) p384

fdse() [subsection 11.2.2.15](#) p386

fdrse() [subsection 11.2.2.14](#) p385

fdpe() [subsection 11.2.2.17](#) p387.

11.2.2.17 fdpe - Create an FDPE flip-flop

Library: xilinx/common.3pl

Prototype:

```
global procedure
fdpe (
    value(q, "log")
    <-
    value(d, "log", false),
    clock(c),
    value(ce, "log", true),
    value(pre, "log", false),
    str(init, "S")
) {
```

Output parameters:

q is the flip-flop output.

Input parameters:

d is the data input, default low.

c is the clock input.

ce is the optional clock enable input, default high.

pre is the optional asynchronous set input, default low.

init is the initial value, default set.

In-line target execution:

no

description:

This procedure creates a D-type flip-flop with clock enable and asynchronous set. The **d**, **ce** and **pre** inputs must have the same clock domain as the input clock. The **q** output is set to the same clock domain.

See also -

fdre() [subsection 11.2.2.13](#) p384

fdse() [subsection 11.2.2.15](#) p386

fdrse() [subsection 11.2.2.14](#) p385

fdce() [subsection 11.2.2.16](#) p386.

11.2.2.18 generateConstraints - generate constraint file entries from I/O mode variable attributes

Library: xilinx/common.3pl

Prototype:

```
global procedure
generateConstraints () {}
```

Output parameters:

none

Input parameters:

none

In-line target execution:

no

description:

This procedure traverses all input, output and clock mode variables and from their attribute maps generates constraint file entries. The procedure obtains a list of pointers to these variables by calling inbuilt function `varlist()` [section 10.167](#) p346. This procedure must be called once when there are no further calls to `input()`, `output()` or `clock()`. It is best to include the call to it in a late-order trailer module.

11.2.2.19 iopins - create FPGA pin specifications

Library: xilinx/common.3pl

Prototype:

```
global procedure
iopins(ident(id), type(t), var(pins, t), varargs) {}
```

Output parameters:

none

Input parameters:

id is the identifier for a string or array of strings to be created.

t is the type of the following 'pins' argument.

pins is a collection of strings specifying FPGA pins.

In-line target execution:

no

description:

This procedure attaches attributes to a collection of FPGA pin strings. Optionally the procedure can create a variable containing the pin strings.

For each pin string a map of attributes is created. This map is itself contained in a single map whose key is the pin string and value the attribute map for that pin.

Parameter **id** is an optional identifier which if given an argument will be used to create an immediate variable whose value is the collection of strings designating the FPGA pins.

Parameter **t** is the type of the pin argument. The argument may be type **type** or it may be a type string. The type may be simply "str" for one pin, or more usually nested arrays and structs whose primitive types are only "str".

Parameter **pins** is the collection of pin strings, the argument to which may be either a compound constant or a variable of the designated type.

Following these parameters a number of additional arguments may be provided to specify attributes associated with the pins. These additional arguments may be keymatch, i.e. attribute=value, or may be simple arguments. Where an argument is not keymatch it must be type "map" whose keys are attributes (strings) and values are attribute values.

The arguments are processed left to right and where an attribute appears more than once the right-most value will prevail.

Each attribute value must be either a single value which will be added to each pin map.

This procedure may be called multiple times, perhaps with different but overlapping collections of pins. Each time attributes will be attached to the attribute map of each pin. Where attribute entries already exist they will be overwritten. The variable identifier must be different in each case or may be omitted.

It is important to note that the prime function of iopins() is to establish attributes associated with pins. As an optional byproduct a variable can be created which contains the collection of pins and this would be used to pass locations to library procedures input(), output() or clock() for external connections. The pin locations can however be passed to these procedures quite independently of iopins since only the pin strings are needed as the attributes are extracted from the attribute maps associated with the pins. It is of course usual to use the variable optionally created by iopins() to pass the location pins to the I/O procedures.

The pin attributes established by iopins() are used by library procedures input(), output() and clock() but attributes can also be passed directly to these procedures as arguments to them in a similar way. It is a matter of choice whether to associate attributes directly with pins or instead to pass them directly to the input and output procedures. This may be partly decided by whether pins are connected to fixed external components or else to hardware which varies with the application.

Attributes passed directly to input(), output() or clock() will override any attributes passed via iopins(). An attribute associated with a pin via iopins() that is passed to input(), output() or clock() must not conflict with that attribute on another pin in that call, though that attribute need not be on all pins.

The attributes attached to pins by iopins() may be unused. Attributes are only used where pins are processed by library procedures input(), output() or clock() and then only those relevant to that procedure. Unrecognised attributes will result in a fatal error.

The following table lists the allowed attributes that are recognised by iopins(). These attributes are not case-sensitive but will be converted to lower case.

iopins()			
differential	netlist	log	differential
diff_term	constraint	log	differential input termination
drive	constraint	int	drive current
dutycycle	3PL	float	clock duty cycle
frequency	3PL	float	clock frequency
ibuf_delay_value	constraint	log	input buffer delay
ibufg	netlist	log	use an IBUFG instead of an IBUF
ifd_delay_value	constraint	int	input flip-flop delay
iostandard	constraint	str	voltage standard
noshadow	3PL	log	don't duplicate output flip-flop
period	3PL	float	clock period
pulldown	constraint	log	pulldown
pullup	constraint	log	pullup
slew	constraint	str	slew rate
tnm	constraint	str	timing name

diff_term if true indicates that a 100Ω termination resistance be inserted between differential inputs

differential indicates that the pads are differential. Two locations must be supplied for each data bit.

The **drive** attribute is an integer which specifies the output drive current in mA for Xilinx output buffers (see Xilinx documentation).

dutycycle specifies the duty cycle of a clock input.

frequency specifies the frequency of a clock input.

ibufdelay gives the delay, 0 to 16, to be inserted in an input buffer path. The value has no particular units.

ibufg specifies that an IBUFG be used instead of an IBUF.

ifddelay gives the delay, 0 to 8, to be inserted in an input flip-flop input path. The value has no particular units.

iostandard is a string specifying one of the allowed I/O standards (see Xilinx documentation).

noshadow when true inhibits the duplication of flip-flops whose output is connected to an output buffer. By default any flip-flop whose output is connected to an output buffer is placed in the IO pad containing the buffer. If the output of the flip-flop is used elsewhere then the flip-flop is duplicated in a CLB with all inputs in parallel with the pad flip-flop and the output of the duplicate used for internal logic. If **noshadow** is true then duplication will not be done, the flip-flop will of necessity reside in a CLB and its output will be routed to the output buffer. This will result in much worse clock-to-output time. Setting this attribute ensures that the pad output genuinely reflects the flip-flop state and not that of a duplicate but at the cost of delayed output.

period specifies the period of a clock input.

If **pullup** or **pulldown** are true the output will be given a weak pullup or pulldown. It is not allowed for both these attributes to be true.

pullup

The **slew** attribute is the string "fast" or "slow" and selects one of two slew rates for Xilinx output buffers (see Xilinx documentation).

tnm is the name of a timing group in which the output pads are to be included for timing constraint purposes.

Typical usage might be -

```
static(led, "bits:4");
var(tnms, "[2]str", {"l1", "l2"});
iopins(led_pins, "[4]str", {"A1", "B1", "A2", "C2"}, iostandard="lvcmos33", pullup=true);
iopins( "[2]str", {"B1", "A2"}, tnm=tnms);
output(, led, loc=led_pins, drive=24);
```

which will create a variable **led_pins** in the current scope which is an array of 4 strings. The I/O standard "lvcmos33" will be established for each of the four pins (inputs or outputs) and a pullup resistor is specified for each. For two of the pins separate timing name attributes have been added.

The following example would produce an error -

```
static(led, "bits:4");
iopins(led_pins, "[4]str", {"A1", "B1", "A2", "C2"}, pullup=true);
iopins( "[2]str", {"B1", "A2"}, pullup=false);
output(, led, loc=led_pins);
```

In this example the four pins initially have attribute pullup true then on the next call to iopins() two of them have the pullup attribute changed to false. The error occurs when these four pins are passed to output() as the pullup attribute is now not the same for all pins. If the pullup attribute was passed directly to the call to output() then the pullup attribute values associated with the pins via iopins() would be ignored, e.g.

```
static(led, "bits:4");
iopins(led_pins, "[4]str", {"A1", "B1", "A2", "C2"}, pullup=true);
iopins( "[2]str", {"B1", "A2"}, pullup=false);
output(, led, loc=led_pins, pullup=true);
```

would not give an error.

This procedure is used in conjunction with inbuilt procedures **input()** [section 9.39](#) p218, **output()** [section 9.47](#) p224 and **clock()** [section 9.10](#) p194.

11.2.2.20 maxdelay - specify max delay between timing groups

Library: xilinx/common.3pl

Prototype:

```
global procedure
maxdelay (str({group1, group2}), float(time)) {}
```

Output parameters:

none

Input parameters:

group1 is a string giving the timing name of a netlist group.

group2 is a string giving the timing name of a netlist group.

time is the maximum delay in second allowed for signals propagating from signal sources in group1 to signal destinations in group2.

In-line target execution:

no

description:

This procedure sets up the timing constraints for paths between components in different timing groups.

It will generate a **TIMESPEC** constraint in the ISE NCF file or a corresponding **set_max_delay** constraint in the Vivado XDC file.

For input delay constraints -

group1 is timing group name for the input ports and **group2** is the timing name for the clock.

For ISE **time** specifies the maximum delay in ns for all signal paths from all the relevant input ports to elements clocked by the clock. This is a maximum delay time constraint from port to clock internal to the FPGA. This maximum delay must be met by ISE to satisfy the constraint.

For Vivado **time** specifies the maximum delay in the external signals from the clock edge to arrival at the input ports. This is the actual maximum delay time from external clock edge to input ports external to the FPGA where the notional external clock is phase aligned with the internal clock. If the actual external clock is skewed relative to the internal clock then this skew must be adjusted for in the specified time. This delay time is used by Vivado along with the clock period to determine the maximum time delay between pads and clocked elements that must be achieved during place-and-route.

For output delay constraints **group1** is timing name for the clock and **group2** is the timing group name for the output ports.

For ISE **time** specifies the maximum delay in ns for all signal paths from elements clocked by the clock to all the relevant output ports. This is a maximum delay time constraint from clock to port internal to the FPGA. This maximum delay must be met by ISE to satisfy the constraint.

For Vivado **time** specifies the maximum delay in the external signals from the output ports to the external clock edge . This is the actual maximum delay time from output ports to external clock edge external to the FPGA where the notional external clock is phase aligned with the internal clock. If the actual external clock is skewed relative to the internal clock then this skew must be adjusted for in the specified time. This delay time is used by Vivado along with the clock period to determine the maximum time delay between clocked elements and pads that must be achieved during place-and-route.

maxdelay() only adds a struct to a list of delay structs in common.3pl. A later call to generateConstraints() builds the constraint elements for ISE or Vivado as appropriate.

See also -

addncfline() [subsection 11.2.2.2](#) p377
append_ncf_file() [subsection 11.2.2.4](#) p378
elementaddncf() [subsection 11.2.2.5](#) p378
addxdcline() [subsection 11.2.2.3](#) p377
period_mul() [subsection 11.2.2.23](#) p393
period_div() [subsection 11.2.2.23](#) p393
netaddncf() [subsection 11.2.2.21](#) p392.

11.2.2.21 netaddncf - add constraint to NCF file for a net

Library: xilinx/common.3pl

Prototype:

```
global procedure
addncf (str(s), value(sig)) {}
netaddncf (value(sig), str({attr, val})) {}
```

Output parameters:

none

Input parameters:

sig is .

attr is .

val is .

In-line target execution:

no

description:

An attribute or constraint is applied to a net. Argument /bf sig is a target variable that identifies the subject net. The attribute/constraint /bf attr and the value /bf val are both strings.

See also -

addncfline() [subsection 11.2.2.2](#) p377
append_ncf_file() [subsection 11.2.2.4](#) p378
elementaddncf() [subsection 11.2.2.5](#) p378
addxdcline() [subsection 11.2.2.3](#) p377
maxdelay() [subsection 11.2.2.20](#) p391
period_mul() [subsection 11.2.2.23](#) p393
period_div() [subsection 11.2.2.23](#) p393.

11.2.2.22 period_div - timing group for divided clock

Library: xilinx/common.3pl

Prototype:

```
global procedure
period_div (str(newgroup, group), float(ratio)) {}
```

Output parameters:

none

Input parameters:

newgroup is a new timing group identifier.

group is a timing group identifier.
ratio is a period ratio.

In-line target execution:

no

description:

This procedure creates a new timing group whose name is given by **newgroup**. The period of the new group is the period of the old timing group **group** divided by **ratio**.

See also -

addncfline() [subsection 11.2.2.2](#) p377
append_ncf_file() [subsection 11.2.2.4](#) p378
elementaddncf() [subsection 11.2.2.5](#) p378
addxdcline() [subsection 11.2.2.3](#) p377
maxdelay() [subsection 11.2.2.20](#) p391
period_mul() [subsection 11.2.2.23](#) p393
netaddncf() [subsection 11.2.2.21](#) p392.

11.2.2.23 period_mul - timing group for divided clock

Library: xilinx/common.3pl

Prototype:

```
global procedure
period\_mul (str(newgroup, group), float(ratio)) {}
```

Output parameters:

none

Input parameters:

newgroup is a new timing group identifier.
group is a timing group identifier.
ratio is a period ratio.

In-line target execution:

no

description:

This procedure creates a new timing group whose name is given by **newgroup**. The period of the new group is the period of the old timing group **group** multiplied by **ratio**.

See also -

addncfline() [subsection 11.2.2.2](#) p377
append_ncf_file() [subsection 11.2.2.4](#) p378
elementaddncf() [subsection 11.2.2.5](#) p378
addxdcline() [subsection 11.2.2.3](#) p377
maxdelay() [subsection 11.2.2.20](#) p391
period_div() [subsection 11.2.2.23](#) p393
netaddncf() [subsection 11.2.2.21](#) p392.

11.2.2.24 processPinAttributes - process pin attributes

Library: xilinx/common.3pl

Prototype:

```
global procedure
processPinAttributes (var(ptr, "->")) {}
```

Output parameters:

none

Input parameters:**ptr** is a pointer to an input, output or clock mode variable.**In-line target execution:**

no

description:

This procedure is called following the creation of every input, output or clock mode variable. This procedure must have been previously registered with the 3PL system by calling

```
attributeprocedure(processPinAttributes);
```

prior to any calls to inbuilt procedures `input()`, `output()` or `clock()`. This is best done by placing the call in a leader module, ordered within leader modules such that it occurs early enough to precede the above variable creation calls.

The argument is a pointer to an input, output or clock mode variable. This procedure accesses the attribute map of the variable concerned and extracts the 'loc' attribute if present. From the 'loc' attribute it consults global map **pinAttributes** which has been created by library procedure **iopins()** [subsection 11.2.2.19](#) p388 using each pin (string) as a key. The value from that map is another map containing attributes for that pin. These attributes are added to the attributes map of the variable if they are not already present since attributes passed as arguments to the variable creation procedure should override attributes derived from pins.

When an attribute is extracted from pin attribute maps it must be added to the input, output or clock variable attribute map as an array with one entry per pin since an attribute may have different values amongst the pins.

The procedure also expands most attributes from the variable attribute map to arrays if they are single values (some attributes, such as frequency or readdomain, are required to remain single valued as they apply to the variable itself, not the input/output buffers or pads).

In the case of an input clock with differential input attributes attached to the pair of pins are redundant since the clock attributes except **loc** are required to be single values. It is reasonable to expect that attributes associated with pins will be a pair of identical values and `processPinAttributes()` may use either (it will actually extract them in turn and the 2nd one will prevail).

Note that `processPinAttributes()` ignores attributes that it does not recognise. At present `iopins()` has a list of attributes that it will accept, which it checks for type. It will give a fatal error on an unrecognised attribute (though `input()`, `output()` and `clock()` will accept unrecognised attributes for addition to their attribute maps without comment). This may be relaxed in the future so that additional arbitrary attributes can be added for software use.

For information on attributes see **input()** [section 9.39](#) p218, **output()** [section 9.47](#) p224 or **clock()** [section 9.10](#) p194.

See also `attributeprocedure()` [section 9.6](#) p191.

11.2.2.25 pullup - connect pullups to a bus

Library: xilinx/common.3pl

Prototype:

```
global procedure
pullup (value(in)) {}
```

Output parameters:

none

Input parameters:**in** is a selectvalue mode variable.**In-line target execution:**

no

description:

This procedure connects pullup resistors to a selectvalue variable (an internal bus).

11.2.2.26 setiostandard - set the I/O voltage standard for a pin or pins

Library: xilinx/common.3pl

Prototype:

```
global procedure
setiostandard (str(iostandard), varargs) {}
```

Output parameters:

none

Input parameters:**iostandard** is a string giving the I/O voltage standard.

For second argument see description.

In-line target execution:

no

description:

This procedure established I/O voltage standards for one or more FPGA inputs or outputs. The first argument is a string giving the voltage standard. The second argument is a string or a string array of any number of dimensions, the strings being the FPGA pin designations. The voltage standard will be established for the specified pins (it is stored in a map where each pin is a key and the associated value is the standard string). This procedure may be called again to change the standard established for pins on a previous call since the I/O standards are only used at the completion of immediate execution.

This procedure is used in conjunction with library function **getiostandard()** [subsection 11.2.3.4](#) p399 and library procedure **iopins()** [subsection 11.2.2.19](#) p388.

11.2.3 library functions**11.2.3.1 clockused - determine if a clock variable has been used**

Library: xilinx/common.3pl

Prototype:

```
global function
clockused (c) {}
```

Return mode and type:

immediate, log

Input parameters:**c** is a clock variable**description:**

This function returns true if the clock variable passed as argument has been used to drive logic.

The function can be used to avoid generating timing constraints, and even the clock generator itself, if a clock variable is not used to drive logic. This determination can only be made at the completion of application code and so would normally be done in a trailer module.

11.2.3.2 del - variable length delay line

Library: xilinx/common.3pl

Prototype:

```
global function
del (
    value(in),
    int(len, 1),
    var(init, "[]int"),
    value(step, "log", true),
    value(varlen, "uint:4"),
    value(reset, "log"),
) {}
```

Return mode and type:

value, type(in)

Input parameters:**in** is the input value. It may be any target type.**len** is the length of the delay line.**init** is an optional initialisation.**step** is an optional step control.**varlen** is an optional variable length**reset** is an optional reset.**description:**

This function is a delay line implemented using CLB shift registers. On each clock cycle the input is written to the line and the contents move one item right. The rightmost item appears as the out value. This calls inbuilt module **del()** [section 8.1](#) p184. This function can operate as a sequence generator if the output is connected back to the input.

This has up to 6 input arguments. Input arguments **len** and **init** are immediate mode. **step** may be immediate or target mode as described below. The other input arguments must not be modes **immediate**, **queue**, **memory** or **clock** and value mode arguments must not contain queue reads.

The argument to **step** must be type "log" and may be as follows -

- no argument - the delay will advance on each clock cycle
- an immediate mode value (or constant value mode) which is **true** - the delay will advance on each clock cycle
- an immediate mode value (or constant value mode) which is **false** - the delay will advance on clock cycles in which the function is evaluated in the target code, e.g. on used in assignment
- a static, selectvalue or value mode value - the delay will advance on clock cycles when **step** is true.

For example -

```
when (1)
    a <- del(b, 10, false);
```


will shift the current value of **b** into the delay line every time the assignment to **a** is made (**l** is true), the assigned value being popped off the end.

Whereas -

```
when (l)
  a <- del(b, 10);
```

will shift **b** into the delay line every clock cycle, popping off the value at the other end. When **l** is true the value popped off will also be assigned to **a**.

```
when (l)
  a <- del(b, 10, true);
```

will behave the same way.

Where **step** is variable during target execution the shifting is controlled solely by **step** while the assignment, controlled by **l**, assigns the end value of the delay line to **a** regardless of whether a shift has occurred or not.

```
when (l)
  a <- del(b, 10, step);
```

The limitation stated above that inputs cannot contain queue reads can be overcome in some cases. For example to use a queue as data input we can use the examine operator as long as a queue wait controls the delay step. For example to form a running average of length 8 -

```
queue(q, "uint:8");
static(rsum, "uint:11");
queue(rav, "uint:8");

seq {
  rsum <- rsum + q - del(?q, 8, false);
  rav <- rsum / 8;
}
```

The data input to the **del()** function is a queue examine, not a queue read. The examine operator provides the value of the top of a queue regardless of the validity (e.g. it might be empty). This is fine if the statement only executes when we have separately determined that the queue is available for reading, which is the case here. The **del()** function as noted above will only step when the function is evaluated if the step argument is immediate constant **false**. In this example the function will not be evaluated until **q** is available to read as it is included in the RHS expression along with the function call. At that point the function will be evaluated and the delay line will step.

For large delays this uses a lot of CLBs and the alternative library function **delay()** [subsubsection 11.13.4.7 p514](#) in library `stdlib.3pl` might be preferable. That library function will call **del()** in any case if the delay value is small.

This has 1 to 6 input arguments and 1 output argument. Input arguments **len** and **init** are immediate mode. The other input arguments must not be modes **immediate**, **queue**, **memory** or **clock** and expression arguments must not contain queue reads.

The 1st input argument, **in**, is the delay line data input. It can be any target type.

The optional input argument, **len**, specifies a fixed delay. This must be an immediate int. If this argument is 0 and the 5th argument **varlen** is also omitted, the output will simply be connected to the input to give zero delay.

The optional input argument, **init**, is an initialisation array. This must be immediate mode and an array of the same type as **in**. It may be smaller than the delay line length in which case the output end of the delay line will be zero initialised. The low order entries in the initialisation array are loaded in the input end of the delay line. If initialisation values are too large their high-order bits will be truncated.

The optional input argument, **step**, is a line step enable. This must be type log. If this argument is used the line will only advance when **step** is true, otherwise it will advance on every clock cycle.

The optional input argument, **varlen**, is a variable line length. This must be type uint. This argument may be limited or unavailable depending on the type of FPGA. For Xilinx the variable length is limited to values between 0 and 15 giving delays of 1 to 16 respectively. A wider argument will be truncated and a narrower one zero padded. Note that when the line length is changed data may be skipped or repeated. If the line length is changed while stepping is inhibited, the line length argument is an address with which the contents of the line can be non-destructively examined.

The optional input argument, **reset**, returns the line to its initial state when true.

The delay line data output is the value of the function which is the same type as the data input **in**.

If the delay line is initialised, these initial values appear at the output during the initial *len* line steps, where *len* is the fixed delay (or the initial delay selected by **varlen** if used).

For Xilinx the following apply -

- For FPGAs prior to Virtex5 the variable length argument is limited to values between 0 and 15 inclusive, giving line lengths of 1 to 16 respectively. From Virtex5 onwards the lengths are 1 to 32.
- If the reset is used, the variable length argument cannot be used.
- If the reset is used or the line length is 1, the delay line will be implemented using flip-flops, otherwise it will be implemented using CLB LUT shift registers.

Implementation using CLB LUT shift registers minimises resources, requiring only one CLB LUT for every 16-bit delay line.

Note: the fixed delay parameter *len* gives the number of cycles to be delayed but the variable delay parameter *varlen* is the number of cycles delay minus one!

In Xilinx devices there is a substantial delay between the clock edge and the arrival of the shift register signal at the output of the logic block (2.8nS to 3.5nS for a -6 speed XC2VP and about 1.15 to 1.43nS for a -2 speed XC5V). The delay for a flip-flop is very much smaller. For this reason a delay line with fixed length is implemented using a flip-flop for the last delay stage. If a variable length delay line is required, this flip-flop is not inserted and hence the significant output delay must be considered. It is preferable in that case to assign the delay line output directly to a variable, i.e. with no intervening combinatorial logic, to avoid introducing any extra path delay.

See also library function `delay()` [subsection 11.13.4.7](#) p514 which constructs a fixed-length delay line but for long delays provides an option of using block RAM rather than shift registers.

11.2.3.3 elementcheck - check that a Xilinx element is allowed in the current device family

Library: xilinx/common.3pl

Prototype:

```
global function
elementCheck(str(element)) {}
```

Return mode and type:

immediate, log

Input parameters:

element is the name of a Xilinx library element.

description:

This function checks that the Xilinx library element named by the argument is available for the current Xilinx device family. Allowed elements for each device family are kept in an abbreviated map in xilinx/-common.3pl, i.e. only a subset of elements are included, namely those used explicitly in 3PL libraries. The map can be expanded when required.

The result of the check is returned, true if the element is available in the current device family.

See also library procedure `elementcheck()` [subsection 11.2.2.7](#) p380 which will issue a fatal error message on failure, rather than returning the check result.

11.2.3.4 getiostandard - get the I/O standard for a pin or pins

Library: xilinx/common.3pl

Prototype:

```
global function
getiostandard (varargs) {}
```

Return mode and type:

immediate, str

Input parameters:

See description.

description:

This function returns a string giving the I/O standard for one or more FPGA inputs or outputs. The argument must be a string or a string array of any number of dimensions, the strings being the FPGA pin designations. Multiple pins must have the same I/O standard or a fatal error occurs.

The I/O standard for one or more pins is established or changed by calling library procedure **setiostandard()** [subsection 11.2.2.26](#) p395.

This function is used in conjunction with library procedures **setiostandard()** [subsection 11.2.2.26](#) p395 and **iopins()** [subsection 11.2.2.19](#) p388.

11.2.3.5 reverse - reverse a value

Library: xilinx/common.3pl

Prototype:

```
global function
reverse (value(in)) {}
```

Return mode and type:

value, type(in)

Input parameters:

in is the operand.

description:

This function returns input operand with bits reversed. Since this does a simple bit-wise reversal it only makes sense if the input type can be sensibly reversed.

11.2.3.6 sequence - generate a sequence

Library: xilinx/common.3pl

Prototype:

```
global function
sequence (var(s), type(stype), value(step, "log", true)) {}
```

Return mode and type:

value, "bits:"

Input parameters:

s is the sequence array.

stype is the data type.

step is an optional input to control data movement.

description:

This function generates a cyclic sequence of values, these appearing on the output value variable. The sequence is determined by the input argument **s** which must be an array of immediate values. The type of these values is given by the 2nd argument, **stype**, and may be of compound type. The 3rd input argument, **step**, is optional and is used to control when the sequence advances, otherwise this occurs on every clock edge. The sequence is produced starting with subscript 0 of the sequence data and moving up the array.

11.2.4 library classes

None.

11.3 Spartan6, Virtex4, Virtex5, Virtex6 and series 7 DSP48 block Library File - xilinx/dsp48.3pl

This library creates a DSP48 block for Spartan6, Virtex4, Virtex5, Virtex6 and series 7 FPGAs.

This library must be incorporated by using a **module** statement -

```
module "xilinx/dsp48.3pl"
```

11.3.1 library modules

11.3.1.1 dspaddsub - DSP add/subtract

Library: xilinx/dsp48.3pl

Prototype:

```
global module
dspaddsub (
    value(out),
    value(pat, "log")
    <-
    value(a),
    value(b),
    value(add, "log", true),
    value(gate, "log", true),
    int(areg, breg, preg),
    int(pattern),
    int(mask, 0xfffffffffff)
) {}
```

Output parameters:

out result.

pat pattern matched.

Input parameters:

a first operand.

b second operand.

add add or subtract.

gate gate second operand.

areg number of registers in A input.

breg number of registers in B input.

preg number of registers on DSP48E output.

pattern pattern to match against result.

mask mask to select pattern bits.

description:

This module uses the DSP48 to add or subtract two operands. The operands may be immediate, input, value, static or queue mode and may be of type "uint:", "int:", "ufixed:", or "fixed:". The types may be mixed. The operands may be up to 48 bits signed or 47 bits unsigned and need not be the same size. The output result type is given by the following table -

	uint:	int:	ufixed:	fixed:
uint:	uint:	int:	ufixed:	fixed:
int:	int:	int:	fixed:	fixed:
ufixed:	ufixed:	fixed:	ufixed:	fixed:
fixed:	fixed:	fixed:	fixed:	fixed:

The output result binary point offset, i.e. number of fraction bits, is equal to the largest binary point offset of the two operands. The output result integer part width is equal to the largest integer part width of the two operands plus one. The full width of the result cannot exceed 48 bits. The result will be cast to fit the type of the argument passed to the first output parameter **out**.

The argument supplied to the first output parameter, **out**, is a value mode variable which should have the type "empty" to allow it to take the result type.

The third input argument **add** is type "log" and is true for addition and false for subtraction. If omitted, the operation is addition. An immediate mode argument may be supplied for a fixed operator or a target value may be used to select the operator dynamically.

The fourth input argument **gate** is type "log" and is true for addition or subtraction of the second operand and false to pass the first operand, i.e. add or subtract 0. If omitted, the operation is addition/subtraction. An immediate mode argument may also be supplied.

The fifth and sixth input parameters **areg** and **breg** are type "int" and specify the number of registers to insert in the **a** and **b** inputs respectively. Allowed values for **areg** are 0 (default), 1 and 2 and for **breg** 0 (default), and 1.

The seventh input parameter **preg** specifies the number of registers on the result and pattern match outputs. Allowed values are 0 (default) or 1. If **preg** is 0 and the outputs are assigned to static variables those variables may be subsumed into the DSP48 output register.¹

The eighth parameter **pattern** is an optional bit pattern to match against the arithmetic result. If they are identical the second output argument **pat** will be true. The last parameter **mask** selects which bits from ANDing the result and the pattern are to be used. All bits passed by the mask must be true for the match result to be true. If **mask** is not passed an argument then all bits are matched to the pattern.

The second output parameter **pat** provides the result of the pattern match, if used.

If no internal registers are used the input arguments may contain queue reads, in which case the arguments to output parameters **out** and **pat** will be available for reading when all input arguments are available. If any inserted registers are specified (**areg**, **breg**, **creg** or **preg** are non-zero) the inputs cannot contain queue reads and the code will exit with a fatal error.

11.3.1.2 dspcounter - DSP counter

Library: xilinx/dsp48.3pl

Prototype:

```
global module
dspcounter (
    value(out),
    value(pat, "log")
    <-
    value(step, "log", true),
    value(reset, "log", false),
    value(pattern, "bits:48", 0),
    int(mask, 0xfffffffffff),
    value(up, "log", true)
) {}
```

Output parameters:

out counter value.

¹this will only be done if the static variables are not initialised (0 is allowed), are not assigned any other value but a DSP48 output (can be assigned 0/false or reset) and if both outputs are used are assigned (or reset) in the same way (have a common clock-enable and reset). This is because the DSP48 output registers for the arithmetic output and the pattern match output share a clock enable and reset and they cannot be initialised). An assignment of 0/false to these static variables becomes a reset since by default it is the assignment of the initial values. Any other assignments would require an input multiplexer between the DSP48 output and the registers.

pat pattern matched.

Input parameters:

step advance the count.

reset clear the count.

pattern pattern to match against count.

mask mask to select pattern bits.

up up or down.

description:

This module uses the DSP48 as a 48 bit counter. The count will be cast to fit the type of the argument passed to the first output parameter **out**.

The first input parameter, **step**, is a boolean which will advance the count when true. If it is not passed an argument then by default the counter will advance on every clock cycle.

The second input parameter, **reset**, is a boolean which will synchronously reset the counter to zero when true.

The third parameter **pattern** is an optional bit pattern to match against the count. If they are identical the second output argument **pat** will be true. The pattern may be either immediate mode (constant) or target mode to compare against a target variable or expression. The fourth parameter **mask** selects which bits from ANDing the count and the pattern are to be used. All bits passed by the mask must be true for the match result to be true. If **mask** is not passed an argument then all bits are matched to the pattern.

The last input parameter determines the count direction, true for up and false for down. It may of course have an immediate or target argument depending on whether the count direction is to be changed dynamically or fixed. If the argument is omitted the direction is up.

The second output parameter **pat** provides the result of the pattern match, if used.

The input arguments must not contain queue reads.

11.3.1.3 dsplog - DSP logical operation

Library: xilinx/dsp48.3pl

Prototype:

```
global module
dsplog (
    value(out),
    value(pat, "log")
    <-
    value(a),
    value(b),
    value(op, "uint:3"),
    int(pattern),
    int(mask, 0xfffffffffff)
) {}
```

Output parameters:

out counter value.

pat pattern matched.

Input parameters:

a first input operand.

b second input operand.

op operator.

pattern pattern to match against result.

mask mask to select pattern bits.

description:

This module uses the DSP48 as a 48 bit counter. The result is a logical operation on the two input operands.

Input parameters **a** and **b** are the operands and **op** is an operation code. The operation codes are -

4	0b000100	$a \wedge b$
5	0b000101	$\sim(a \wedge b)$
6	0b000110	$\sim(a \wedge b)$
7	0b000111	$a \wedge b$
12	0b001100	$a \& b$
13	0b001010	$a \& \sim b$
14	0b001110	$\sim(a \& b)$
15	0b001111	$\sim a \mid b$
36	0b100100	$\sim(a \wedge b)$
37	0b100101	$a \wedge b$
38	0b100110	$a \wedge b$
39	0b100111	$\sim(a \wedge b)$
44	0b101100	$a \mid b$
45	0b101101	$a \mid \sim b$
46	0b101110	$\sim(a \wedge b)$
47	0b101111	$\sim a \& b$

The argument to **op** may be immediate mode for a fixed operation or may be target mode to dynamically select the operation.

The operands can be any width up to 48 bits and can be any type.

The fourth parameter **pattern** is an optional bit pattern to match against the result. If they are identical the second output argument **pat** will be true. The last parameter **mask** selects which bits from ANDing the result and the pattern are to be used. All bits passed by the mask must be true for the match result to be true. If **mask** is not passed an argument then all bits are matched to the pattern.

The second output parameter **pat** provides the result of the pattern match, if used.

The input arguments may contain queue reads in which case the arguments to output parameters **out** and **pat** will be available for reading when all input arguments are available.

11.3.1.4 dspmul - DSP multiply

Library: xilinx/dsp48.3pl

Prototypes:

```
global module
dspmul (
    value(out), value(pat, "log")
    <-
    value(a),
    value(b),
    int(areg, breg, mreg, preg),
    int(pattern),
    int(mask, 0xfffffffffff)
) {}
```

Output parameters:

out result.

pat pattern matched.

Input parameters:

a first operand.

b second operand.

areg number of registers in A input.

breg number of registers in B input.

mreg number of registers on multiplier output.

preg number of registers on DSP48 output.

pattern pattern to match against result.

mask mask to select pattern bits.

description:

These modules use the DSP48 to multiply two signed operands. The operands may be immediate, input, value, static or queue mode and may be of type "uint:", "int:", "ufixed:", or "fixed:". The types may be mixed. The result type is determined by the input operand type nearest the top of the following table -

fixed
ufixed
int
uint

The first operand, **a**, is 25 bits maximum (24 bits if unsigned) and the second operand, **b**, is 18 bits maximum (17 bits if unsigned).

The output result binary point offset, i.e. number of fraction bits, is equal to the sum of the offsets of the operands. The product integer part width is equal to the sum of the integer widths of its two operands. The result will be cast to fit the type of the argument passed to the first output parameter **out**.

The argument supplied to the first output parameter, **out**, is a value mode variable which may have the type "empty" to allow it to take the result type without casting.

The third and fourth input parameters **areg** and **breg** are type "int" and specify the number of registers to insert in the **a** and **b** inputs respectively. Allowed values for **areg** and **breg** are 0 (default), 1 and 2.

The fifth input parameter **mreg** is type "int" and specifies the number of registers to insert between the multiplier and the adder/subtractor which in this case simply passes the product. Allowed values are 0 (default) or 1.

The sixth input parameter **preg** specifies the number of registers on the result and pattern match outputs. Allowed values are 0 (default) or 1. If **preg** is 0 and the outputs are assigned to static variables those variables may be subsumed into the DSP48 output register.²

The seventh parameter **pattern** is an optional bit pattern to match against the arithmetic result. If they are identical the second output argument **pat** will be true. The last input parameter **mask** selects which bits from ANDing the result and the pattern are to be used. All bits passed by the mask must be true for the match result to be true. If **mask** is not passed an argument then all bits are matched to the pattern.

The second output parameter **pat** provides the result of the pattern match, if used.

If no internal registers are used the input arguments may contain queue reads, in which case the arguments to output parameters **out** and **pat** will be available for reading when all input arguments are available. If any inserted registers are specified (**areg**, **breg**, **mreg** or **preg** are non-zero) the inputs cannot contain queue reads and the code will exit with a fatal error.

²this will only be done if the static variables are not initialised (0 is allowed), are not assigned any other value but a DSP48 output (can be assigned 0/false or reset) and if both outputs are used are assigned (or reset) in the same way (have a common clock-enable and reset). This is because the DSP48 output registers for the arithmetic output and the pattern match output share a clock enable and reset and they cannot be initialised). An assignment of 0/false to these static variables becomes a reset since by default it is the assignment of the initial values. Any other assignments would require an input multiplexer between the DSP48 output and the registers.

11.3.1.5 dspmuladd - DSP multiply and add**11.3.1.6 dspmulsub - DSP multiply and subtract****11.3.1.7 dspsubmul - DSP subtract and multiply**

Library: xilinx/dsp48.3pl

Prototypes:

```
global module
dspmuladd (
    value(out), value(pat, "log")
    <-
    value(a),
    value(b),
    value(c),
    int(areg, breg, creg, mreg, preg),
    int(pattern),
    int(mask, 0xfffffffffff)
) {}
```

```
global module
dspmulsub (
    value(out), value(pat, "log")
    <-
    value(a),
    value(b),
    value(c),
    int(areg, breg, creg, mreg, preg),
    int(pattern),
    int(mask, 0xfffffffffff)
) {}
```

```
global module
dspsubmul (
    value(out), value(pat, "log")
    <-
    value(a),
    value(b),
    value(c),
    int(areg, breg, creg, mreg, preg),
    int(pattern),
    int(mask, 0xfffffffffff)
) {}
```

Output parameters:**out** result.**pat** pattern matched.**Input parameters:****a** first operand.**b** second operand.**c** third operand.**areg** number of registers in A input.**breg** number of registers in B input.**creg** number of registers in A input.**mreg** number of registers on multiplier output.**preg** number of registers on DSP48 output.**pattern** pattern to match against result.

mask mask to select pattern bits.

description:

These modules use the DSP48 to combine three operands. The operands may be immediate, input, value, static or queue mode and may be of type "uint:", "int:", "ufixed:", or "fixed:". The types may be mixed. The result type is determined by the input operand type nearest the top of the following table -

fixed
ufixed
int
uint

The operations performed are -

dspmuladd	$a * b + c$
a	25 bits signed
b	18 bits signed
c	48 bits signed

dspmulsub	$a * b - c$
a	25 bits signed
b	18 bits signed
c	48 bits signed

dspsubmul	$a - b * c$
a	48 bits signed
b	25 bits signed
c	18 bits signed

Unsigned arguments must be 1 bit smaller.

The output result binary point offset, i.e. number of fraction bits, is equal to the sum of the offsets of the operands to the multiplier; if the remaining operand has a larger offset it is trimmed at the least significant end. This differs from the way 3PL normally handles expressions where significant bits are retained. The product integer part width is equal to the sum of the integer widths of its two operands. The output result integer part width is equal to the largest integer part width of the product and the remaining operand plus one. The full width of the result cannot exceed 48 bits. The result will be cast to fit the type of the argument passed to the first output parameter **out**.

The argument supplied to the first output parameter, **out**, is a value mode variable which may have the type "empty" to allow it to take the result type without casting.

The fourth to sixth input parameters **areg**, **breg** and **creg** are type "int" and specify the number of registers to insert in the **a**, **b** and **c** inputs respectively. Allowed values for **areg** and **breg** are 0 (default), 1 and 2. **creg** may be 0 (default) or 1.

The seventh input parameter **mreg** is type "int" and specifies the number of registers to insert between the multiplier and the adder/subtractor. Allowed values are 0 (default) or 1.

The eighth input parameter **preg** specifies the number of registers on the result and pattern match outputs. Allowed values are 0 (default) or 1. If **preg** is 0 and the outputs are assigned to static variables those variables may be subsumed into the DSP48 output register.³

The ninth parameter **pattern** is an optional bit pattern to match against the arithmetic result. If they are identical the second output argument **pat** will be true. The last input parameter **mask** selects which bits from ANDing the result and the pattern are to be used. All bits passed by the mask must be true for the match result to be true. If **mask** is not passed an argument then all bits are matched to the pattern.

³this will only be done if the static variables are not initialised (0 is allowed), are not assigned any other value but a DSP48 output (can be assigned 0/false or reset) and if both outputs are used are assigned (or reset) in the same way (have a common clock-enable and reset). This is because the DSP48 output registers for the arithmetic output and the pattern match output share a clock enable and reset and they cannot be initialised). An assignment of 0/false to these static variables becomes a reset since by default it is the assignment of the initial values. Any other assignments would require an input multiplexer between the DSP48 output and the registers.

The second output parameter **pat** provides the result of the pattern match, if used.

If no internal registers are used the input arguments may contain queue reads, in which case the arguments to output parameters **out** and **pat** will be available for reading when all input arguments are available. If any inserted registers are specified (**areg**, **breg**, **creg**, **mreg** or **preg** are non-zero) the inputs cannot contain queue reads and the code will exit with a fatal error.

11.3.2 library procedures

11.3.2.1 dsp48 - create a DSP48 block for Virtex4 FPGAs

Library: xilinx/dsp48.3pl

Prototype:

```
global procedure
dsp48 (
    value(bcout, "bits:18"),
    value(pcout, "bits:48"),
    value(p, "bits:48")
    <-
    str({loc, b_input}),
    int({areg, breg, creg, mreg, preg, opmodereg,
        carryinreg, subtractreg, carryinselreg}),

    clock(clk),
    value(a, "bits:18"),
    value(b, "bits:18"),
    value(c, "bits:48"),
    value(opmode, "uint:7"),
    value(subtract, "log"),
    value(carryinsel, "uint:2"),
    value(bcinc, "bits:18"),
    value(pcin, "bits:48"),
    value({carryin, cea, ceb, cec, cem, cep,
        cectl, ceinsub, cecarryin, rsta, rstb, rstc, rstm,
        rstp, rstctl, rstcarryin}, "log")
) {}
```

Output parameters:

bcout - 18 bit B carry out

pcout - 48 bit P carry out

p - 48 bit P out

Input parameters:

AttributesS -

loc - optional location.

b_input - select B input cascade.

areg - 0, 1 or 2 for number of pipeline registers.

breg - 0, 1 or 2 for number of pipeline registers.

creg - 0 or 1 for number of pipeline registers.

mreg - 0 or 1 for number of pipeline registers.

preg - 0 or 1 for number of pipeline registers.

opmodereg - 0 or 1 for number of pipeline registers.

carryinreg - 0 or 1 for number of pipeline registers.

subtractreg - 0 or 1 for number of pipeline registers.

carryinselreg - 0 or 1 for number of pipeline registers.

Input signals -

clk - clock input.
a - 18 bit a input.
b - 18 bit b input.
c - 48 bit c input.
opmode - 7 bit operation mode.
subtract - select add or subtract.
carryinsel - select carry source.
bcin - 18 bit B carry in.
pcin - 48 bit P carry in.
carryin - carry input.
cea - A register clock enable.
ceb - B register clock enable.
cec - C register clock enable.
cem - M register clock enable.
cep - P register clock enable.
cectrl - opmodereg and carryinreg clock enable.
ceinsub - subtract input register clock enable.
cecarryin - carry input register clock enable.
rsta - reset A register.
rstb - reset B register.
rstc - reset C register.
rstm - reset M register.
rstp - reset P register.
rstctrl - reset control register.
rstcarryin - reset carryin register.

In-line target execution:

no

description:

This procedure creates a Virtex4 DSP48 block. The first 11 input parameters are attributes, the remaining inputs being input signals.

11.3.2.2 dsp48a - create a DSP48A block for Spartan6 FPGAs

Library: xilinx/dsp48.3pl

Prototype:

```

global procedure
dsp48a (
    value(bcout, "bits:18"),
    value(pcout, "bits:48"),
    value(m, "bits:36"),
    value(p, "bits:48"),
    value(carryout, "bits:1"),
    value(carryoutf, "bits:1")
    <-
    str(loc),
    str(carryinsel, "CARRYIN"),
    str(rsttype, "SYNC"),
    int({a0reg, b0reg, }, 0),
    int({a1reg, b1reg, creg, dreg, mreg, preg, opmodereg,
        carryinreg, carryoutreg}, 1)

    clock(clk),
    value(a, "bits:18"),
    value(b, "bits:18"),
    value(c, "bits:48"),
    value(d, "bits:18"),

```

```

    value(opmode, "uint:7"),
    value(pcin, "bits:48"),
    value(carryin, "bits:1"),
    value({cea, ceb, cec, ced, cem, cep, ceopmode,
           cecarryin, rsta, rstb, rstc, rstd, rstm,
           rstp, rstopmode, rstcarryin}, "log")
  ) {}

```

Output parameters:

Input parameters:

In-line target execution:

no

description:

This procedure creates a Spartan6 DSP48A block. The first 14 input parameters are attributes, the remaining inputs being input signals.

11.3.2.3 dsp48e - create a DSP48E block for Virtex5 FPGAs

Library: xilinx/dsp48.3pl

Prototype:

```

global procedure
dsp48e (
  value(acout, "bits:30"),
  value(bcout, "bits:18"),
  value(pcout, "bits:48"),
  value(p, "bits:48"),
  value(carryout, "[4]log"),
  value({carrycascout, multsignout, patterndetect, patternbdetect,
         overflow, underflow}, "log")

  <-
  str(loc, a_input, b_input, use_mult, use_simd, sel_pattern, sel_mask,
       sel_rounding_mask, autoreset_pattern_detect_optinv, use_pattern_detect),
  int(areg, acascreg, breg, bcascreg, creg, mreg, preg, opmodereg,
       aluodereg, carryinreg, multcarryinreg, carryinsselreg,
       pattern, mask),
  log(autoreset_pattern_detect),
  clock(clk),
  value(a, "bits:30"),
  value(b, "bits:18"),
  value(c, "bits:48"),
  value(opmode, "uint:7"),
  value(alumode, "uint:4"),
  value(carryinsel, "uint:3"),
  value(acin, "bits:30"),
  value(bcin, "bits:18"),
  value(pcin, "bits:48"),
  value({carryin, cea1, cea2, ceb1, ceb2, cec, cem, cep, cealumode,
         cectl, cemultcarryin, cecarryin, rsta, rstb, rstc, rstm,
         rstp, rstctrl, rstallcarryin, rstalumode, carrycascin,
         multsignin}, "log")
) {}

```

Output parameters:

acout - 30 bit A carry out

bcout - 18 bit B carry out

pcout - 48 bit P carry out
p - 48 bit P out
carryout -
carrycascout -
multsignout -
patterndetect -
patternbdetect -
overflow -
underflow -

Input parameters:

AttributesS -

loc - optional location.
a_input - select A input cascade.
b_input - select B input cascade.
use_mult -
use_simd -
sel_pattern -
sel_mask -
sel_rounding_mask -
autoreset_pattern_detect_optinv -
use_pattern_detect -
areg - 0, 1 or 2 for number of pipeline registers.
acascereg -
breg - 0, 1 or 2 for number of pipeline registers.
bcascereg -
creg - 0 or 1 for number of pipeline registers.
mreg - 0 or 1 for number of pipeline registers.
preg - 0 or 1 for number of pipeline registers.
opmodereg - 0 or 1 for number of pipeline registers.
alumodereg -
carryinreg - 0 or 1 for number of pipeline registers.
multcarryinreg - 0 or 1 for number of pipeline registers.
carryinselreg - 0 or 1 for number of pipeline registers.
pattern -
mask -
Input signals -
clk - clock input.
a - 18 bit a input.
b - 18 bit b input.
c - 48 bit c input.
opmode - 7 bit operation mode.
alumode - 7 bit operation mode.
carryinsel - select carry source.
acin - 30 bit B carry in.
bcin - 18 bit B carry in.
pcin - 48 bit P carry in.
carryin - carry input.
cea1 - A register clock enable.
cea2 - A register clock enable.
ceb1 - B register clock enable.
ceb2 - B register clock enable.
cec - C register clock enable.
cem - M register clock enable.
cep - P register clock enable.
cealumode -
cectrl -
cemultcarryin -

cecarryin - carry input register clock enable.
rsta - reset A register.
rstb - reset B register.
rstc - reset C register.
rstm - reset M register.
rstp - reset P register.
rstctrl -
rstallcarryin -
rstalumode -
carrycascin -
multsignin -

In-line target execution:

no

description:

This procedure creates a Virtex5 DSP48E block. The first 24 input parameters are attributes, the remaining inputs being input signals.

11.3.2.4 dsp48e1 - create a DSP48E1 block for Virtex6 and series 7 FPGAs

Library: xilinx/dsp48.3pl

Prototype:

```

global procedure
dsp48e1 (
  value(acout, "bits:30"),
  value(bcout, "bits:18"),
  value(pcout, "bits:48"),
  value(p, "bits:48"),
  value(carryout, "[4]log"),
  value({carrycascout, multsignout, patterndetect, patternbdetect,
         overflow, underflow}, "log")

  <-
  str(loc, a_input, b_input, use_mult, use_simd, sel_pattern, sel_mask,
      sel_rounding_mask, autoreset_pattern_detect_optinv, use_pattern_detect),
  int(areg, acascreg, breg, bcascreg, creg, mreg, preg, opmodereg,
      alumodereg, carryinreg, multcarryinreg, carryinsselreg,
      pattern, mask),
  log(autoreset_pattern_detect),
  clock(clk),
  value(a, "bits:30"),
  value(b, "bits:18"),
  value(c, "bits:48"),
  value(d, "bits:25"),
  value(opmode, "uint:7"),
  value(alumode, "uint:4"),
  value(carryinsel, "uint:3"),
  value(acin, "bits:30"),
  value(bcin, "bits:18"),
  value(pcin, "bits:48"),
  value({carryin, cea1, cea2, ceb1, ceb2, cec, cem, cep, cealumode,
         cectrl, cemultcarryin, cecarryin, rsta, rstb, rstc, rstm,
         rstp, rstctrl, rstallcarryin, rstalumode, carrycascin,
         multsignin}, "log")
) {}
  
```

Output parameters:

acout - 30 bit A carry out

bcout - 18 bit B carry out
pcout - 48 bit P carry out
p - 48 bit P out
carryout -
carrycascout -
multsignout -
patterndetect -
patternbdetect -
overflow -
underflow -

Input parameters:

AttributesS -

loc - optional location.
a_input - select A input cascade.
b_input - select B input cascade.
use_mult -
use_simd -
sel_pattern -
sel_mask -
sel_rounding_mask -
autoreset_pattern_detect_optinv -
use_pattern_detect -
areg - 0, 1 or 2 for number of pipeline registers.
acascreg -
breg - 0, 1 or 2 for number of pipeline registers.
bcascreg -
creg - 0 or 1 for number of pipeline registers.
mreg - 0 or 1 for number of pipeline registers.
preg - 0 or 1 for number of pipeline registers.
opmodereg - 0 or 1 for number of pipeline registers.
alumodereg -
carryinreg - 0 or 1 for number of pipeline registers.
multcarryinreg - 0 or 1 for number of pipeline registers.
carryinselreg - 0 or 1 for number of pipeline registers.
pattern -
mask -

Input signals -

clk - clock input.
a - 18 bit a input.
b - 18 bit b input.
c - 48 bit c input.
d - 25 bit c input.
opmode - 7 bit operation mode.
alumode - 7 bit operation mode.
carryinsel - select carry source.
acin - 30 bit B carry in.
bcin - 18 bit B carry in.
pcin - 48 bit P carry in.
carryin - carry input.
cea1 - A register clock enable.
cea2 - A register clock enable.
ceb1 - B register clock enable.
ceb2 - B register clock enable.
cec - C register clock enable.
cem - M register clock enable.
cep - P register clock enable.
cealumode -

cectrl -
cemultcarryin -
cecarryin - carry input register clock enable.
rsta - reset A register.
rstb - reset B register.
rstc - reset C register.
rstm - reset M register.
rstp - reset P register.
rstctrl -
rstallcarryin -
rstalumode -
carrycasc -
multsignin -

In-line target execution:

no

description:

This procedure creates a Virtex6 DSP48E1 block. The first 24 input parameters are attributes, the remaining inputs being input signals.

11.3.3 library functions

None.

11.4 Xilinx GTP library - xilinx/gtp_dual.3pl

This library creates GTP dual tiles.

This library must be incorporated by using a **module** statement -

```
module "xilinx/gtp_dual.3pl"
```

The struct **gtp_tile_params** describes the pins and some control parameters for a GTP tile (GTP channel pair).

```
struct(global.gtp_tile_params,
    loc,                                "str",
    {refclkp_pin, refclk_n_pin},        "str",
    {boost0, boost1},                  "log",
    {txdiffctrl0, txdiffctrl1},        "int",
    {txpreemphasis0, txpreemphasis1},  "int"
);
```

The following structs are created for GTP data input and output -

```
struct(global.gt_16,
    data,    "bits:16", // data
    k,       "log",     // true if top 8 bits of data form a K-character
    contin,  "log"      // this is a continuation word
);
```

```
struct(global.gt_16_e,
    data,    "bits:16" // data
    k,       "log",     // true if top 8 bits of data form a K-character
    first,   "log",     // previous word was an idle
    error,   "log"      // disparity error or not-in-table error
);
```

```
struct(global.gt_16_ee,
    data,    "bits:16", // data
    k,       "log",     // true if top 8 bits of data form a K-character
    first,   "log",     // previous word was an idlgtp\_duale
    disperr, "[2]log",  // disparity errors
    niterr,  "[2]log"   // not-in-table errors
);
```

The global structure **ds16** is also used and it is created in **stdlib.3pl**. It is

```
struct(global.ds16,
    data,    "bits:16", // data
    k,       "log"      // true if top 8 bits of data form a K-character
);
```

11.4.1 library modules

11.4.1.1 gtp_dual_16 - a 16-bit dual high-speed serial I/O channel tile

Library: xilinx/gtp_dual.3pl

Prototype:

```

global module
gtp_dual_16 (
  queue(out0),
  queue(out1),
  clock(usrc1k2),
  value({rxdiag0, rxdiag1}, "[10]log"),
  value({txdiag0, txdiag1}, "[2]log")
  <-
  var(tile, gtp_tile_params),
  float(freq, 156.25),
  int(run_length),
  queue(in0),
  queue(in1),
  value({txpowerdown0, txpowerdown1, rxpowerdown0, rxpowerdown1}, "uint:2", 0),
  value({gtpreset, rxreset0, rxreset1}, "log", false),
  value({loopback0, loopback1}, "uint:3", 0),
  value({diff0, diff1, pe0, pe1}, "uint:3", 0),
  log(los_fsm)
) {}

```

Output parameters:

out0 is the data output queue for GTP0.

out1 is the data output queue for GTP1.

usrc1k2 is the user interface clock.

rxdiag0 is an array of diagnostic signals for the GTP0 receiver.

rxdiag1 is an array of diagnostic signals for the GTP1 receiver. **txdiag0** is an array of diagnostic signals for the GTP0 transmitter.

txdiag1 is an array of diagnostic signals for the GTP1 transmitter.

Input parameters:

tile indicates which GTP_DUAL tile to use.

run_length is the maximum data run length.

in0 is the data input queue fro GTP0.

in1 is the data input queue fro GTP1.

txpowerdown0 control GTP0 transmitter powerdown.

txpowerdown1 control GTP1 transmitter powerdown.

rxpowerdown0 control GTP0 receiver powerdown.

rxpowerdown1 control GTP1 receiver powerdown.

gtpreset when true resets the entire GTP_DUAL tile.

rxreset0 when true resets the GTP0 receiver.

rxreset1 when true resets the GTP1 receiver.

loopback0 controls the GTP0 loopback mode.

loopback1 controls the GTP1 loopback mode.

diff0 overrides the GTP0 differential output voltage table entry.

diff1 overrides the GTP1 differential output voltage table entry.

pe0 overrides the GTP0 transmitter pre-emphasis table entry.

pe1 overrides the GTP1 transmitter pre-emphasis table entry.

los_fsm when true instantiates the loss-of-sync FSM

In-line target execution:

no

Description:

This module is both an input and an output interface to a GTP_dual GTP tile. The data width is 16 bit. A custom protocol is used. This calls **gtp_dual()** [subsubsection 11.4.2.1](#) p418, described below. The default external clock frequency is 156.25 MHz (serial rate 3.125Gbit/s) but lower clock frequencies can be used.

The particular GTP_DUAL tile is selected by the argument to input parameter **tile**. This parameter has the following type -

```

struct(global.gtp_tile_params,
    loc,                                "str",
    {refclkp_pin, refclk_n_pin},        "str",
    {boost0, boost1},                  "log",
    {txdiffctrl0, txdiffctrl1},        "int",
    {txpreemphasis0, txpreemphasis1},  "int"
);

```

where the fields are -

- loc - the tile location, e.g. "GTP_DUAL_X0Y4"
- refclkp_pin - the external reference clock positive pin
- refclk_n_pin - the external reference clock negative pin
- boost0 - GTP0 TX pre-emphasis boost
- boost1 - GTP1 TX pre-emphasis boost
- txdiffctrl0 - GTP0 TX differential Voltage
- txdiffctrl1 - GTP1 TX differential Voltage
- txpreemphasis0 - GTP0 TX pre-emphasis
- txpreemphasis1 - GTP1 TX pre-emphasis

boost0 and **boost1** are logical values which select one of two sets of levels of transmitter high frequency pre-emphasis (see **txpreemphasis0** and **txpreemphasis1** description below)

txdiffctrl0 and **txdiffctrl1** are 3 bit values which control the transmitter output voltage swing. These table values can be overridden by input parameters **diff0** and **diff1**. The values are -

3-bit value	transmitter differential swing (mV)
000	1100
001	1050
010	1000
011	900
100	800
101	600
110	400
111	0

txpreemphasis0 and **txpreemphasis1** are 3 bit values which control the transmitter high frequency pre-emphasis. These table values can be overridden by input parameters **pe0** and **pe1**. The values are -

3-bit value	transmitter pre-emphasis (%)	
	pre-emphasis boost off boost = false	pre-emphasis boost on boost = true
000	2	3
001	2	3
010	2.5	4
011	4.5	10.5
100	9.5	18.5
101	16	28
110	23	39
111	31	52

Input parameter **run_length** is an immediate integer which specifies the maximum contiguous data run for the transmitter. If data input is continuously available such that this length would be exceeded (and the data **contin** field is not true - see above) then an idle sequence is inserted in the output stream. When no data are available for transmission a continuous stream of idles is transmitted. These idles are used by the receiver to lock initially and to continue checking that lock is maintained. If receiver lock is lost then lock can only be regained at the next idle. In the custom protocol used in this interface the idle sequence is a single 16-bit word containing a comma byte. If no argument is supplied to parameter **run_length** then idles will only be inserted when the input queue is unavailable.

Input parameters **in0** and **in1** are the input data queues for the transmitters. If a transmitter is not used the associated argument is omitted. The type is either the struct **ds16** (defined in **stdlib.3pl**) or the struct **gt_16**. The most significant byte of the word may be a K-character, which is signified by setting

field **k** true. The byte must be one of the allowed K-characters (this is not checked!). For type **gt_16** the **contin** field should be true if this word is a continuation word, i.e. it is not the first word in a block. This ensures that despite the run length constraint a block will not be split by inserted idles. At the receiving end the first word following an idle can reset a block boundary.

Input parameters **txpowerdown0**, **txpowerdown1**, **rxpowerdown0** and **rxpowerdown1** control transmitter and receiver powerdown - see Xilinx "Virtex-5 FPGA RocketIO GTP Transceiver - User Guide" UG196.

Input parameter **gtpreset** when high resets the entire GTP_DUAL tile including the PLL.

Input parameters **rxreset0** and **rxreset1** when high reset the associated receiver.

Input parameters **loopback0** and **loopback1** control the loopback mode of each GTP - see Xilinx "Virtex-5 FPGA RocketIO GTP Transceiver - User Guide" UG196.

Output parameters **out0** and **out1** are the output data queues for the receivers. If the receiver is not used the argument is omitted. The type can be struct **ds16**, struct **gt_16_e** or struct **gt_16_ee** depending on the level of error detail required. If a receiver loses sync or byte alignment there is no data output. A word in which a disparity error or a not-in-table error occurs is written to the output queue but the error field is true. If the top byte of a word is a K-character the **k** field is true. If the **first** field is true then this word follows an idle, which information may be used to reset a block boundary. **Important note - the output queues must not block! The destination module must read them at a rate that precludes them ever becoming full, if necessary reading them to null. Generally the read clock domain must have an equal or higher frequency than the user interface clock ruckl, but this may be relaxed if it is known that the serial data source has a guaranteed maximum data rate that would allow a lower queue read clock domain.**

Output parameter **usrclk2** is the user interface clock (default 156.25MHz).

Output parameters **rxdiag0** and **rxdiag1** provide ten diagnostic outputs for the associated receiver. These are -

- **diag[0]** - receiver synchronisation FSM bit 1, if **los_fsm** true
- **diag[1]** - receiver synchronisation FSM bit 0, if **los_fsm** true
- **diag[2]** - is true when the upper byte of the current output word has a disparity error
- **diag[3]** - is true when the lower byte of the current output word has a disparity error
- **diag[4]** - is true when the upper byte of the current output word is a comma
- **diag[5]** - is true when the lower byte of the current output word is a comma
- **diag[6]** - **rxbytisaligned**, which is true when the idles are word boundary aligned
- **diag[7]** - **rxcommadet**, which is true when a positive or negative comma is being received
- **diag[8]** - ok, which is true when synchronisation is locked and the byte pair is aligned properly to a 16 bit word
- **diag[9]** - idle, which is true when the receiver is idle

diag[0] and **diag[1]** are both false if **los_fsm** is false since the loss-of-sync FSM will not have been instantiated. These outputs might be connected to status LEDs to indicate that the receiver is locked and functioning correctly. **diag[5]** will be true while the receiver is receiving idles but will be false when data are received.

Output parameters **txdiag0** and **txdiag1** provide two diagnostic outputs for the associated transmitter. These are -

- **diag[0]** - **is_k**, which is true when a data word with the K bit set is being transmitted
- **diag[1]** - idle, which is true when the receiver is idle

11.4.2 library procedures

11.4.2.1 gtp_dual - create a GTP dual high-speed serial I/O channel tile

Library: **xilinx/gtp_dual.3pl**

Prototype:

```

global procedure
gtp_dual (
  // outputs
  clock({refclkout, rxrecclk0, txoutclk0, rxrecclk1, txoutclk1}),
  value({
    drdy, phystatus0, phystatus1, pllclkdet, resetdone0, resetdone1,
    rxbyteisaligned0, rxbyteisaligned1, rxbyterealign0, rxbyterealign1,
    rxchanbondseq0, rxchanbondseq1, rxchanisaligned0, rxchanisaligned1,
    rxchanrealign0, rxchanrealign1, rxcommadet0, rxcommadet1, rxelecidle0,
    rxelecidle1, rxoversampleerr0, rxoversampleerr1, rxprbserr0, rxprbserr1,
    rxvalid0, rxvalid1},
    "log"
  ),
  value(rxconfigout, "bits:16"), // port do
  value({rxbufstatus0, rxbufstatus1}, "uint:3"),
  value({rxchbondo0, rxchbondo1}, "[3]log"),
  value({rxclkcorcnt0, rxclkcorcnt1}, "uint:3"),
  value({rxdata0, rxdata1}, "[2]bits:8"),
  value({rxdisperr0, rxdisperr1, rxnotintable0, rxnotintable1,
    rxlosssofsync0, rxlosssofsync1, rxrundisp0, rxrundisp1,
    rxchariscomma0, rxchariscomma1, rxcharisk0, rxcharisk1,
    txkerr0, txkerr1, txrundisp0, txrundisp1},
    "[2]log"),
  value({rxstatus0, rxstatus1}, "uint:3"),
  value({txbufstatus0, txbufstatus1}, "[2]log")

  <-

  // attributes
  str({loc, refclkp_pin, refclk_n_pin}),
  log({
    ac_cap_dis_0, ac_cap_dis_1, chan_bond_seq_2_use_0,
    chan_bond_seq_2_use_1, clk_cor_insert_idle_flag_0,
    clk_cor_insert_idle_flag_1, clk_cor_keep_idle_0, clk_cor_keep_idle_1,
    clk_cor_precedence_0, clk_cor_precedence_1, clk_cor_seq_2_use_0,
    clk_cor_seq_2_use_1, clk_correct_use_0, clk_correct_use_1,
    comma_double_0, comma_double_1, dec_mcomma_detect_0,
    dec_mcomma_detect_1, dec_pcomma_detect_0, dec_pcomma_detect_1,
    dec_valid_comma_only_0, dec_valid_comma_only_1, mcomma_detect_0,
    mcomma_detect_1, oversample_mode, pci_express_mode_0,
    pci_express_mode_1, pcomma_detect_0, pcomma_detect_1, rcv_term_gnd_0,
    rcv_term_gnd_1, rcv_term_mid_0, rcv_term_mid_1, rcv_term_vttrx_0,
    rcv_term_vttrx_1, rx_buffer_use_0, rx_buffer_use_1,
    rx_decode_seq_match_0, rx_decode_seq_match_1, rx_loss_of_sync_fsm_0,
    rx_loss_of_sync_fsm_1, termination_ovrd, tx_buffer_use_0,
    tx_buffer_use_1
  }),
  log(tx_diff_boost_0, true),
  log(tx_diff_boost_1, true),
  int({
    align_comma_word_0, align_comma_word_1, chan_bond_1_max_skew_0,
    chan_bond_1_max_skew_1, chan_bond_2_max_skew_0, chan_bond_2_max_skew_1,
    chan_bond_level_0, chan_bond_level_1, chan_bond_seq_1_1_0,
    chan_bond_seq_1_1_1, chan_bond_seq_1_2_0, chan_bond_seq_1_2_1,
    chan_bond_seq_1_3_0, chan_bond_seq_1_3_1, chan_bond_seq_1_4_0,
    chan_bond_seq_1_4_1, chan_bond_seq_1_enable_0, chan_bond_seq_1_enable_1,
    chan_bond_seq_2_1_0, chan_bond_seq_2_1_1, chan_bond_seq_2_2_0,
    chan_bond_seq_2_2_1, chan_bond_seq_2_3_0, chan_bond_seq_2_3_1,
    chan_bond_seq_2_4_0, chan_bond_seq_2_4_1, chan_bond_seq_2_enable_0,
    chan_bond_seq_2_enable_1, clk_cor_seq_1_1_0, clk_cor_seq_1_1_1,
    clk_cor_seq_1_2_0, clk_cor_seq_1_2_1, clk_cor_seq_1_3_0,
    clk_cor_seq_1_3_1, clk_cor_seq_1_4_0, clk_cor_seq_1_4_1,
    clk_cor_seq_1_enable_0, clk_cor_seq_1_enable_1, clk_cor_seq_2_1_0,

```

```

        clk_cor_seq_2_1_1, clk_cor_seq_2_2_0, clk_cor_seq_2_2_1,
        clk_cor_seq_2_3_0, clk_cor_seq_2_3_1, clk_cor_seq_2_4_0,
        clk_cor_seq_2_4_1, clk_cor_seq_2_enable_0, clk_cor_seq_2_enable_1,
        com_burst_val_0, com_burst_val_1, comma_10b_enable_0, comma_10b_enable_1,
        mcomma_10b_value_0, mcomma_10b_value_1, oobdetect_threshold_0,
        oobdetect_threshold_1, pcomma_10b_value_0, pcomma_10b_value_1,
        sata_burst_val_0,
        sata_burst_val_1, sata_idle_val_0, sata_idle_val_1, termination_ctrl
    }},
    int({ pcs_com_cfg, pma_cdr_scan_0, pma_cdr_scan_1, pma_rx_cfg_0,
        pma_rx_cfg_1, prbs_err_threshold_0, prbs_err_threshold_1,
        trans_time_from_p2_0, trans_time_from_p2_1, trans_time_non_p2_0,
        trans_time_non_p2_1, trans_time_to_p2_0, trans_time_to_p2_1}),
    int(chan_bond_seq_len_0, 1),
    int(chan_bond_seq_len_1, 1),
    int(clk25_divider, 1),
    int(clk_cor_adj_len_0, 1),
    int(clk_cor_adj_len_1, 1),
    int(clk_cor_det_len_0, 1),
    int(clk_cor_det_len_1, 1),
    int(clk_cor_max_lat_0, 3),
    int(clk_cor_max_lat_1, 3),
    int(clk_cor_min_lat_0, 3),
    int(clk_cor_min_lat_1, 3),
    int(clk_cor_repeat_wait_0, 0),
    int(clk_cor_repeat_wait_1, 0),
    int(oob_clk_divider, 1),
    int pll_divsel_fb, 1),
    int(pll_divsel_ref, 1),
    int(pll_rxdivsel_out_0, 1),
    int(pll_rxdivsel_out_1, 1),
    int(pll_txdivsel_comm_out, 1),
    int(pll_txdivsel_out_0, 1),
    int(pll_txdivsel_out_1, 1),
    int(rx_los_invalid_incr_0, 1),
    int(rx_los_invalid_incr_1, 1),
    int(rx_los_threshold_0, 4),
    int(rx_los_threshold_1, 4),
    int(sata_max_burst_0, 7),
    int(sata_max_burst_1, 7),
    int(sata_max_init_0, 22),
    int(sata_max_init_1, 22),
    int(sata_max_wake_0, 7),
    int(sata_max_wake_1, 7),
    int(sata_min_burst_0, 5),
    int(sata_min_burst_1, 5),
    int(sata_min_init_0, 12),
    int(sata_min_init_1, 12),
    int(sata_min_wake_0, 4),
    int(sata_min_wake_1, 14),
    str({ chan_bond_mode_0, chan_bond_mode_1, rx_slide_mode_0,
        rx_slide_mode_1, rx_status_fmt_0, rx_status_fmt_1, rx_xclk_sel_0,
        rx_xclk_sel_1  }},
    // str(pma_com_cfg),
    // inputs
    clock({
        txusrclk0, txusrclk20, rxusrclk0, rxusrclk20,
        txusrclk1, txusrclk21, rxusrclk1, rxusrclk21,
        clkkin, dclk
    }),
    value({den, dwe, gtpreset, intdatawidth, pllclkdeten, pllpowerdown,
        prbscntreset0, prbscntreset1, refclkpwrdbn, rxbufreset0, rxbufreset1,
        rxcdrreset0, rxcdrreset1, rxcommadetuse0, rxcommadetuse1,

```



```

        rxdatawidth0, rxdatawidth1, rxdec8b10buse0, rxdec8b10buse1,
        rxelecidlereset0, rxelecidlereset1, rxenchansync0, rxenchansync1,
        rxenelecidleresetb, rxeneqb0, rxeneqb1, rxenmcommaalign0,
        rxenmcommaalign1, rxenpcommaalign0, rxenpcommaalign1,
        rxensamplealign0, rxensamplealign1, rxpmasetphase0, rxpmasetphase1,
        rxpolarity0, rxpolarity1, rxreset0, rxreset1, rxslide0, rxslide1,
        txcomstart0, txcomstart1, txcomtype0, txcomtype1, txdatawidth0,
        txdatawidth1, txdetectrx0, txdetectrx1, txelecidle0, txelecidle1,
        txenc8b10buse0, txenc8b10buse1, txenpmaphasealign, txinhibit0,
        txinhibit1, txpmasetphase, txpolarity0, txpolarity1, txreset0, txreset1
    }, "log"
),
value(daddr, "uint:7"),
value(txconfigin, "bits:16"), // port di
value({rxchbondi0, rxchbondi1}, "[3]log"),
value({rxeqpole0, rxeqpole1}, "uint:4"),
value({txbypass8b10b0, txbypass8b10b1, txchardispmode0, txchardispmode1, txchardispval0, txchardispval1,
        txcharisk0, txcharisk1},
        "[2]log"),
value({rxeqmix0, rxeqmix1}, "uint:2"),
value({txdata0, txdata1}, "[2]bits:8"),
value({txenprbstst0, rxenprbstst0,
        txenprbstst1, rxenprbstst1},
        "uint:2"),
value({txpowerdown0, rxpowerdown0,
        txpowerdown1, rxpowerdown1},
        "uint:2"),
value({txpreemphasis0, txpreemphasis1, txdiffctrl0, txdiffctrl1,
        txbufdiffctrl0, txbufdiffctrl1, loopback0, loopback1},
        "uint:3")
) {

```

Output parameters:

See Xilinx documentation.

Input parameters:

See Xilinx documentation.

In-line target execution:

no

description:

This procedure creates a dual GTP tile. For a simpler interface see [gtp_dual_16 subsubsection 11.4.1.1 p415](#) above.

11.4.3 library functions

None.

11.4.4 library classes

None.

11.5 Xilinx Input/Output library for Virtex4 and later - xilinx/io.3pl

This library provides input and output delays and serialisers/deserialisers for Virtex5 and later.

This library must be incorporated by using a module statement, e.g. -

```
module "xilinx/io.3pl"
```

11.5.1 library modules

11.5.1.1 crc32 - 32 bit CRC

Library: xilinx/io.3pl

Prototype:

```
global module
crc32 (
    value(out, "bits:32")
    <-
    value(in),
    value(valid, "log"),
    clock(clk),
    value(reset, "log")
) {}
```

Output parameters:

out CRC output.

Input parameters:

in input data.

valid data valid.

clk clock.

reset reset to CRC_INIT initial value.

description:

This module creates an interface to a CRC32 element for Virtex5 and later FPGAs.

11.5.1.2 crc64 - 64 bit CRC

Library: xilinx/io.3pl

Prototype:

```
global module
crc64 (
    value(out, "bits:64")
    <-
    value(in),
    value(valid, "log"),
    clock(clk),
    value(reset, "log")
) {}
```

Output parameters:

out CRC output.

Input parameters:

in input data.

valid data valid.

clk clock.

reset reset to CRC_INIT initial value.

description:

This module creates an interface to a CRC64 element for Virtex5 and later FPGAs.

11.5.1.3 dcireset - reset DCI pins

Library: xilinx/io.3pl

Prototype:

```
global module
dcireset (value(locked, "log") <- value(rst, "log")) {}
```

Output parameters:

locked DCI pins calibrated and available.

Input parameters:

rst calibrate DCI impedences.

description:

This module creates an interface to the DCI reset hardware. When **rst** is asserted (high) the DCI impedances are calibrated. When calibration is complete **locked** will become true. DCI pins are unavailable during calibration.

11.5.1.4 sysmon - system monitor

Library: xilinx/io.3pl

Prototype:

```
global module
sysmon (
    value(dout, "bits:16"),
    value(drdy, "log"),
    value(alm, "[3]log"),
    value(ot, "log"),
    value(channel, "uint:5"),
    value({eoc, eos, busy, jtaglocked, jtagmodified, jtagbusy}, "log")
    <-
    value(din, "bits:16"),
    value(daddr, "uint:7"),
    clock(dclk),
    value({den, dwe, reset, convst, convstclk}, "log"),
    int({
        init_40, init_41, init_42,
        init_48, init_49, init_4a, init_4b,
        init_4c, init_4d, init_4e, init_4f,
        init_50, init_51, init_52, init_53,
        init_54, init_55, init_56, init_57 })
    ) {}
```

Output parameters:

do reconfiguration port output data.
drdy reconfiguration port data ready.
alm temperature sensor alarm.
ot over temperature alarm.
channel current ADC channel.
eoc end of conversion.
eos end of conversion sequence.
busy ADC busy.

Input parameters:

di reconfiguration port input data.
daddr reconfiguration port address.
dclk reconfiguration port clock.
den reconfiguration port enable.
dwe reconfiguration port write enable.
reset reset.
convst convert start.
convstclk convert start clock.
init_40 to **init_57**, initial configuration data.

description:

This module creates an interface to the system monitor. See for example Xilinx document "Virtex-5 FPGA System Monitor" UG192.

11.5.2 library procedures

11.5.2.1 iddr - input double data rate interface

Library: xilinx/io.3pl

Prototype:

```
global procedure
iddr (
    value(o1),
    value(o2)
    <-
    value(d),
    clock({c, cb}),
    str(mode),
    value({ce, r, s}, "log")
) {}
```

Output parameters:

o1 is the +ve clock edge data output.
o2 is the -ve clock edge data output.

Input parameters:

d is the DDR data input.
c is an optional primary clock input.
cb is an optional secondary clock input.
mode specifies the operating mode.
ce is an optional clock enable.
r is an optional synchronous reset.
s is an optional synchronous set.

In-line target execution:

no

description:

This procedure creates an input DDR (Double Data Rate) interface.

The argument to input parameter **d** may be an input mode variable or it may be the output from an interposed `idelay()` call.

c is an optional clock input. It is the primary or only clock.

cb is an optional clock input. It is a secondary clock.

If neither **c** nor **cb** are supplied, the current clock will be used to capture data. If **c** is supplied it will be used instead of the current clock. In either case both +ve and -ve edges of the clock are used to capture input data.

If **c** and **cb** are both supplied **c** will be used to capture +ve edge data and **cb** will be used to capture -ve edge data

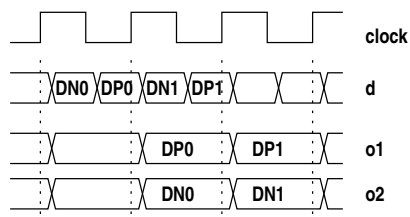
Output parameter **o1** is assigned input data clocked from the positive clock edges. Output parameter **o2** is assigned input data clocked from the negative clock edges.

ce is a clock enable and controls assignment to both outputs.

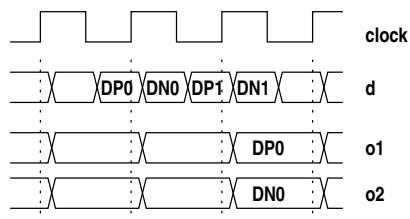
R can be used to clear the output registers and **s** to set them to all ones.

Parameter **mode** may have one of three string values -

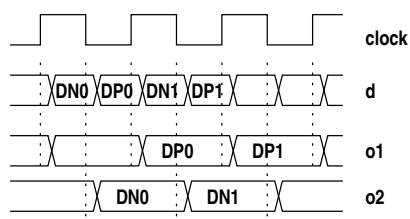
`mode == "nfirst"` - input data is assigned to an internal register on negative clock edges. Input data is assigned to **o1** on positive clock edges. Data in the internal register is assigned to **o2** on positive clock edges. The two outputs thus capture -ve:+ve pairs in that order and both outputs are clocked on the +ve clock edge.



`mode == "pfirst"` - this is the same as above but there is an additional pipeline register inserted before the positive edge register. The two outputs thus capture +ve:-ve pairs in that order and both outputs are clocked on the +ve clock edge. There is an extra cycle of latency in this mode.



`mode == "split"` - input data is assigned to **o1** on positive clock edges and to **o2** on negative clock edges. The two outputs are thus in different (complementary) clock domains.



No input parameters may contain queue reads.

Example code; input from a 2-channel DDR analog to digital converter - `io.3pl`

```

str(lvcmos33, "LVCMOS33");

// package pins
iopins(adc_clk_pin, lvcmos33, "str", "E17");
iopins(adc_pins, lvcmos33, "[8]str",
      {"F2", "G2", "H2", "J2", "H1", "E1", "G1", "E2"});

// ADC clock
clock(adc_clk_in, frequency=100.0e6, loc=adc_clk_pin, ibufg=true);
clock({global.adc_clk, adc_clk_raw}, frequency=100.0e6);
clkdcn(clk0=adc_clk_raw <- clk_in=adc_clk_in, clkfb=adc_clk);
clockbufg(adc_clk <- adc_clk_raw);
attributes(adc_clk, tnm="ADC_CLK");

// ADC interface
useclock(adc_clk);
value({adc0, adc1}, "uint:8");
input(adc_in, "uint:8", loc=adc_pins);
iddr(adc0, adc1 <- adc_in,, "nfirst");

```

Value variables **adc0** and **adc1** now 'point' to registers which are assigned ADC values on each positive edge of the clock, one from each channel.

11.5.2.2 idelay - input data delay

Library: xilinx/io.3pl

Prototype:

```

global procedure
idelay (
    value(o)
    <-
    input(d),
    int(delay),
    clock(refclk),
    value({ce, inc, rst}, "log"),
    log(isclock)
) {}

```

Output parameters:

o is the data output.

Input parameters:

d is the data input.

delay gives a delay count.

refclk is a reference clock to control the delay.

ce is an optional clock enable for changing the delay.

inc is an optional direction control for changing the delay.

rst is an optional reset for the delay.

islock should be true for a clock signal and false for a data signal.

In-line target execution:

no

description:

This procedure generates an input delay block.

The data input **d** is an input mode variable and must be a single bit.

The output parameter **o** is the delayed signal and is also a single bit.

ce, **inc** and **rst** inputs are used to change the delay value. These must be in the current clock domain.

If input parameters **delay** or **ce** have an argument then the input parameter **refclk** must be supplied with a clock whose frequency lies between 190.0MHz and 210.0MHz but is ideally 200MHz. If **idelay()** is called more than once when **variable** has an argument, then the same reference clock must be supplied to each call. Any calls to **idelay()** must also use the same reference clock.

If input parameter **ce** has no argument then the delay will be fixed. Input parameter **delay** gives the required delay. The delay is $400 + 78 * \text{delay}$ pS for a reference clock frequency of 200MHz. If **delay** has no argument the delay count will be zero (giving 400pS).

If input parameter **ce** has an argument then the delay will be variable. Input parameter **delay** gives the initial delay count (default 0). The delay can be changed by asserting **ce**. If **inc** is true the delay will be incremented, otherwise it will be decremented. If **rst** is asserted the delay will be changed back to the initial delay value. The clock domain of **ce**, **inc** and **rst** is the current clock domain.

Parameter **islock** describes the signal type which optimises the timing analyser for periodic or non-periodic signals.

No input parameters may contain queue reads.

Implementation note: this procedure will automatically create an IDELAYCTRL block on the first call to **idelay()** that has a non-default delay. The Xilinx software will duplicate this across all locations where required. Xilinx recommend explicitly creating an IDELAYCTRL for each I/O region. This seems like hard work requiring a detailed floor plan and it is not clear why they suggest doing so.

11.5.2.3 idelay - input/output data delay

Library: xilinx/io.3pl

Prototype:

```
global procedure
idelay (
    value(dataout)
    <-
    input(idatain),
    value(odatain),
    log(variable),
    int({idelay, odelay}),
    clock(refclk),
    value({t, ce, inc, rst}, "log")
) {}
```

Output parameters:

dataout is the data output.

Input parameters:

idatain is the data input from an IBUF (input mode variable).

odatain is the data input from OLOGIC.

variable specifies the type of input delay.

idelay gives the input delay count.

odelay gives the output delay count.

refclk is a reference clock to control the delays.

t is 3-state output control.

ce is an optional clock enable for changing the delay.

inc is an optional direction control for changing the delay.

rst is an optional reset for the delay.

In-line target execution:

no

description:

This procedure generates an input and output delay block.

idatain is an input mode variable and is the delay data source on input (**t** is true). It must be one bit.

odatain is the delay data source on output (**t** is false). This may be a CLB source (**value** or **static** mode variable) or the output of an ODDR block (output argument of `oddr()`). If this is a static mode variable it will be placed in an OLOGIC block. It must be a single bit.

t switches between input and output mode and must also control the output 3-state drive via an output mode variable with conditional assignment using **t**.

The output parameter **o** is the delayed signal. It can be used directly or connected to an IDDR block (input argument to `iddr()`) and must also be connected to an OBUFT (via an **output** mode variable).

If input parameter **variable** is not given an argument then the input delay is the default fixed delay necessary for zero hold time on registered input data when no DCM is used. Arguments **ce**, **inc** and **rst** must be omitted.

Input parameter `/bf refclk` must be supplied with a clock whose frequency lies between 190.0MHz and 210.0MHz but is ideally 200MHz. Multiple calls to `iodelay()` must use the same reference clock. Any calls to `iodelay()` where parameter **variable** has an argument must also use this same reference clock.

If input parameter **variable** has an argument whose value is false then the input delay will be fixed. Input parameter **idelay** gives the required delay. The delay is $400 + 78 * \text{delay}$ pS for a reference clock frequency of 200MHz. If **idelay** has no argument the delay count will be zero (giving 400pS).

If input parameter **variable** has an argument whose value is true then the input delay will be variable. Input parameter **idelay** gives the initial delay count (default 0). The delay can be changed by asserting **ce**. If **inc** is true the input delay will be incremented, otherwise it will be decremented. If **rst** is asserted the input delay will be changed back to the initial delay value. The clock domain of **ce**, **inc** and **rst** is the current clock domain.

The output delay is set by **odelay** and is $400 + 78 * \text{delay}$ pS for a reference clock frequency of 200MHz. If **odelay** has no argument the delay count will be zero (giving 400pS).

No input parameters may contain queue reads.

Implementation note: this procedure will automatically create an IDELAYCTRL block on the first call to `iodelay()` that has a non-default delay. The Xilinx software will duplicate this across all locations where required. Xilinx recommend explicitly creating an IDELAYCTRL for each I/O region. This seems like hard work requiring a detailed floor plan and it is not clear why they suggest doing so.

11.5.2.4 iserdes - input deserialiser

Library: xilinx/io.3pl

Prototype:

```
global module
iserdes (
    value(q),
    value({shiftout1, shiftout2}, "log")
    <-
    str(interface, "NETWORKING"),
    log(master, true), log(DDR, true),
    int(width),
    clock({clk, clkb, oclk, oclkb, clkdiv}), value({din, ddlyin}),
    value(bitslip, "log", false),
    value({ce1, ce2}, "log"),
    value({shiftin1, shiftin2, rst}, "log", false)
) {}
```


Output parameters:

see below.

Input parameters:

See below.

In-line target execution:

no

description:

A serial to parallel converter interface for the Xilinx ISERDES element. These may be used as a master-slave pair in DDR mode for output widths greater than 8.

q is the data output and must be of type "[n]bits:1" where 'n' is the output width. For a master ISERDES 'n' will be 2, 3, 4, 5, 6, 7 or 8. For a slave ISERDES 'n' will be 2 for a data width of 10, or 6 for a data width of 14.

shiftout1 and **shiftout2** are connections to the slave ISERDES for widths 10 and 14.

interface selects the interface type which can be "MEMORY", "MEMORY_DDR3", "MEMORY_QDR", "OVERSAMPLE" or "NETWORKING" (default).

master selects master mode if true, or slave mode otherwise. Default is true (master mode)

ddr selects DDR mode if true, SDR mode if false. Default is true (DDR mode).

width is the data width.

	networking	memory
sdc	2, 3, 4, 5, 6, 7, 8	not allowed
ddr	4, 6, 8, 10, 14(*)	4

(* ddr mode networking interface width 14 not allowed for XC5V ISERDES_NODELAY).

Where a master-slave pair are used for a width of 10 or 14 that width argument must be the full width for both modules of the pair even though the data outputs **q** are smaller, i.e. for a data width of 14 both ISERDES modules will have a width argument 14 even though the master has an 8-bit output and the slave a 6-bit output.

clk the main or only serial data input clock.

clkb second serial data input clock for MEMORY_QDR mode. For other modes this should be the inverted main clock.

oclk is the strobe clock for memory mode.

oclkb is the inverted **oclk**.

clkdiv is the frame clock input which must have a frequency which is serial clock frequency divided by the serial frame length.

din is the serial data input direct from an input pad. **ddlyin** is the serial data input from an IDELAY element. One and only one of these should be given an argument.

bitslip when asserted increases the deserialiser frame delay by one bit position.

ce1 is the clock enable for **clk**.

ce2 is the clock enable for **clkb**.

shiftin1, **shiftin2** are connections from a master ISERDES to a slave ISERDES for widths 10 and 14.

rst is a reset signal.

See the Xilinx documentation for details on the use of this element.

11.5.2.5 oddr - output double data rate interface

Library: xilinx/io.3pl

Prototype:

```
global procedure
oddr (
    value(o)
    <-
    value(i1),
    value(i2),
    value({ce, s, r}, "log")
) {}
```

Output parameters:

o is the DDR data output.

Input parameters:

i1 is the -ve edge data input.

i2 is the +ve edge data input.

ce is an optional clock enable.

r is an optional reset.

s is an optional set.

In-line target execution:

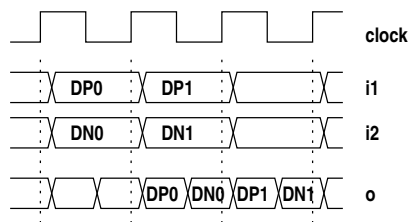
no

description:

This procedure creates an output DDR (Double Data Rate) interface.

The argument to output parameter **o** would usually also be passed as the third argument to an **output()** inbuilt procedure call to drive output buffers.

Input parameters **i1** and **i2** are registered on the current clock positive edge. The data from **i1** appears at the output **o** following a positive clock edge and the data from **i2** appears at the output **o** on the next negative clock edge.



The clock used for the DDR registers is the current clock.

ce is an optional clock enable.

r can be used to clear the output register and **s** to set it to all ones. These can be synchronous or asynchronous depending on attribute SRTYPE and are synchronous by default.

No input parameters may contain queue reads.

Example; DDR output, including a clock -

```
str(lvcmos33, "LVCMOS33");

// package pins
iopins(out_pins, lvcmos33, "[8]str",
    {"F2", "G2", "H2", "J2", "H1", "E1", "G1", "E2"});
iopins(dds_clk_pin, lvcmos33, "str", "E17");

static({out0, out1}, "uint:8");
```

```

out0 <- .....;
out1 <- .....;

value(dds, "uint:8");
value(dds_clk, "log");

odds(dds <- out0, out1);
odds(dds_clk <- false, true);

output(, , dds, loc=out_pins);
output(, , dds_clk, loc=dds_clk_pin);

```

Forwarding a clock via a DDR output is a scheme suggested by Xilinx (XAPP 853, p9) to maintain matching phase.

11.5.2.6 oserdes - output serialiser/deserialiser

Library: xilinx/io.3pl

NOT YET IMPLEMENTED!

11.5.3 library functions

None.

11.5.4 library classes

None.

11.6 Xilinx VirtexII Pro MGT library - xilinx/mgt.3pl

This library provide a simple 16 bit interface to a VirtexII Pro Multi-Gigabit Transceiver (referred to as an MGT or Rocket).

This library must be incorporated by using a module statement, e.g. -

```
module "xilinx/mgt.3pl"
```

The global structures **mgt_16**, **mgt_16_e** and **mgt_16_ee** are created. They are defined as follows -

```
struct(global.gt_16,
  data,  "bits:16",  // data
  k,     "log",      // true if top 8 bits of data form a K-character
  contin, "log"      // this is a continuation word
);
```

```
struct(global.gt_16_e,
  data,  "bits:16",  // data
  k,     "log",      // true if top 8 bits of data form a K-character
  first, "log",      // previous word was an idle
  error, "log"       // disparity error or not-in-table error
);
```

```
struct(global.gt_16_ee,
  data,  "bits:16",  // data
  k,     "log",      // true if top 8 bits of data form a K-character
  first, "log",      // previous word was an idle
  disperr, "[2]log", // disparity errors
  niterr, "[2]log"   // not-in-table errors
);
```

mgt_16 is used for input data. **mgt_16_e** and **mgt_16_ee** are used for output data, the former providing a simpler error indication than the latter.

The global structure **ds16** is also used and it is created in **stdlib.3pl**. It is

```
struct(global.ds16,
  data,  "bits:16",  // data
  k,     "log"       // true if top 8 bits of data form a K-character
);
```

11.6.1 library modules

11.6.1.1 gt_custom_16 - a 16-bit Rocket high-speed serial I/O channel

Library: xilinx/mgt.3pl

Prototype:

```
global module
  gt_custom_16 (
    queue(out),
    clock(ruclk),
    value(rxdiag, "[5]log"),
    value(txdiag, "[2]log")
  <-
  str({loc, txp_pin, txn_pin, rxp_pin, rxn_pin, clkp_pin, clkn_pin}),
  float(freq),          // MGT base frequency
```

```

        queue(in),
        int(run_length),
        value(reset, "log"),
        int(tx_diff_ctrl, 500),
        int(tx_preemphasis, 0),
        value(loopback, "uint:2", 0),
        log(los_fsm)
    ) {}

```

Output parameters:

out is the data output.

ruckl is the user interface clock.

rxdiag provides five diagnostic receiver signals.

txdiag provides two diagnostic transmitter signals.

Input parameters:

loc gives the MGT location coordinates.

txp_pin specifies the transmitter positive output pin.

txn_pin specifies the transmitter negative output pin.

rxp_pin specifies the receiver positive input pin.

rxn_pin specifies the receiver negative input pin.

clkp_pin specifies the clock input positive pin.

clkn_pin specifies the clock input negative pin.

freq is the parallel input and output data frequency.

in is the data input.

run_length is the maximum data run length.

reset when true resets the initialisation flag.

tx_diff_ctrl specifies the differential output voltage.

tx_preemphasis specifies the transmitter preemphasis.

loopback when true connects the MGT in loopback mode.

los_fsm when true connects enables the MGT loss-of-sync FSM.

Description:

This module provides a simple interface to procedure **mgt()** [subsubsection 11.6.2.1](#) p435, in this library. It creates both an input and an output interface to the VirtexII Pro MGT. The data width is 16 bit. A custom protocol is used.

The particular MGT is selected by the first 7 input arguments. The **loc** does not seem to be required but the transmitter pin, receiver pin and clock pin pairs must be provided. The following combinations apply to the XC2VP4 -

MGT location	TXP pin	TXN pin	RXP pin	RXN pin
GT_X0Y0	AB8	AB7	AB9	AB10
GT_X0Y1	A8	A7	A9	A10
GT_X1Y0	AB14	AB13	AB15	AB16
GT_X1Y1	A14	A13	A15	A16

and for the XC2VP40 -

MGT location	TXP pin	TXN pin	RXP pin	RXN pin
GT_X0Y0	ap32	ap33	ap31	ap30
GT_X0Y1	a32	a33	a31	a30
GT_X1Y0	ap28	ap29	ap27	ap26
GT_X1Y1	a28	a29	a27	a26
GT_X2Y0	ap20	ap21	ap19	ap18
GT_X2Y1	a20	a21	a19	a18
GT_X3Y0	ap16	ap17	ap15	ap14
GT_X3Y1	a16	a17	a15	a14
GT_X4Y0	ap8	ap9	ap7	ap6
GT_X4Y1	a8	a9	a7	a6
GT_X5Y0	ap4	ap5	ap3	ap2
GT_X5Y1	a4	a5	a3	a2

freq is an immediate floating point value which specifies the MGT base frequency in Hz. For 2.5GHz serial the base frequency is 125.0MHz (serial rate divided by 20).

Input parameter **in** is the input data queue for the transmitter. If the transmitter is not used the argument is omitted. The type is either the struct **ds16** (defined in `stdlib.3pl`) or the struct **gt_16**. The most significant byte of the word may be a K-character, which is signified by setting field **k** true. The byte must be one of the allowed K-characters (this is not checked!). For type **gt_16** the **contin** field should be true if this word is a continuation word, i.e. it is not the first word in a block. This ensures that despite the run length constraint (see Input parameter **run_length** below) a block will not be split by inserted idles. At the receiving end the first word following an idle can reset a block boundary.

Input parameter **run_length** is an immediate integer which specifies the maximum contiguous data run for the transmitter. If data input is continuously available such that this length would be exceeded (and the data **contin** field is not true - see above) then an idle sequence is inserted in the output stream. When no data are available for transmission a continuous stream of idles is transmitted. These idles are used by the receiver to lock initially and to continue checking that lock is maintained. If receiver lock is lost then lock can only be regained at the next idle. In the custom protocol used in this interface the idle sequence is a single 16-bit word containing a comma byte. If no argument is supplied to parameter **run_length** then idles will only be inserted when the input queue is unavailable.

Input parameter **reset** is used to clear the initialisation flag and execute initialisation process. This initialisation process expects to receive idles only for 6 seconds without error. If an error or a non-idle word is received in this time the received is reset and a further wait ensues. When 6 seconds of idles have been received the initialisation flag is set and the channel can now receive data. The initialisation process occurs immediately following configuration of the FGA. Clearly during this initialisation, and any others that are explicitly induced, the transmitter must be sending idles.

Input parameter **tx_diff_ctrl** specifies the differential output voltage and is 500 by default. Values are 400, 500, 600, 700 or 800 which is the differential voltage in mV. The peak-to-peak voltage is twice this figure.

Input parameter **tx_preamphasis** specifies the transmitter preemphasis and is 0 by default. values are 0 (10%), 1 (20%), 2 (25%) or 3 (33%).

Input parameters **tx_diff_ctrl** and **tx_preamphasis** control the transmitter output waveforms to correct for different cable types and lengths. Any changes to the default values are usually done empirically based on the measured error rate.

Input parameter **loopback** is an immediate integer which specifies the loopback mode. If the argument is omitted or is 0, operation is normal, i.e. loopback is not connected. Value 1 specifies internal parallel mode. Value 2 specifies internal serial mode.

Input parameter **los_fsm** controls the use of the MGT loss-of-sync finite state machine which detects when sync is held/lost. If the argument to this is false then **rxdiag[3]** will only indicate if channel initialisation was successful. If **los_fsm** is true then **rxdiag[3]** will be asserted when the receiver has been successfully initialised and remain asserted while sync is OK.

Output parameter **out** is the output data queue for the receiver. If the receiver is not used the argument is omitted. The type can be struct **ds16**, struct **gt_16_e** or struct **gt_16_ee** depending on the level of error detail required. The **k** field in each case is true if the most significant byte of the received word contains a K-character. For **gt_16_e** or **gt_16_ee** the **first** field is true then this word follows an idle, which information may be used to reset a block boundary. **Important note - the output queue must not block! The destination module must read it at a rate that precludes it ever becoming full, if necessary reading it to null. Generally the read clock domain must have an equal or higher frequency than the user interface clock rucclk, but this may be relaxed if it is known that the serial data source has a guaranteed maximum data rate that would allow a lower queue read clock domain.** For **gt_16_e** the **error** field is true if either byte exhibits a disparity error or not-in-table error. **gt_16_ee** provides each of these errors separately for each byte.

A word in which a disparity error or a not-in-table error occurs is written to the output queue but the error field is true. If the top byte of a word is a K-character the **k** field is true.

Output parameter **rucclk** is the user interface clock at the base frequency.

Output parameter **rxdiag** provides five diagnostic outputs for the receiver. These are -

- **diag[0]** - rxcommadet, which is true when a positive or negative comma is being received
- **diag[1]** - rxdisperr[0] or rxnotintable[0], which is true when a disparity error or a not-in-table error is flagged for the current upper byte of the 16-bit output
- **diag[2]** - rxdisperr[1] or rxnotintable[1], which is true when a disparity error or a not-in-table error is flagged for the current lower byte of the 16-bit output
- **diag[3]** - is true if the receiver is synchronised, as indicated by the receiver synchronisation FSM ⁴, and the initialisation flag is high
- **diag[4]** - is true if the receiver is receiving idles

These outputs might be connected to status LEDs to indicate that the receiver is locked and functioning correctly. **diag[0]** will be true while the receiver is receiving idles but will be false when data are received.

Output parameter **txdiag** provides two diagnostic outputs for the transmitter. These are -

- **diag[0]** - is true if the transmitter is sending idles
- **diag[1]** - is true if the transmitter is sending a word in which the top byte is a control character

11.6.2 library procedures

11.6.2.1 mgt - create an MGT high-speed serial I/O channel

Library: xilinx/mgt.3pl

Prototype:

```
global procedure
mgt (
    // outputs
    value(rxdata),
    value(rxchariscomma, "[log]"),
    value(rxcharisk, "[log]"),
    value(rxcommadet, "log"),
    value(rxlosssofsync, "uint:2"),
    value(rxnotintable, "[log]"),
    value(rxbufstatus, "[2]log"),
    value(rxcheckingcrc, "log"),
    value(rxclkcorcnt, "uint:3"),
    value(rxrcerr, "log"),
    value(rxdisperr, "[log]"),
    value(rxrealign, "log"),
    value(rxrecclk, "log"),
    value(rxrundisp, "[log]),
```

⁴if parameter los_fsm is true.

```

value(txbuferr, "log"),
value(txkerr, "[log]"),
value(txrundisp, "[log]"),
value(chbondo, "[4]log"),
value(chbonddone, "log")
<-
// attributes
str(loc),
str(txn_pin),
str(txp_pin),
str(rxn_pin),
str(rxp_pin),
log(aligned_comma_msb),
int(chan_bond_limit),
str(chan_bond_mode),
int(chan_bond_offset),
log(chan_bond_one_shot),
int(chan_bond_seq_1_1),
int(chan_bond_seq_1_2),
int(chan_bond_seq_1_3),
int(chan_bond_seq_1_4),
int(chan_bond_seq_2_1),
int(chan_bond_seq_2_2),
int(chan_bond_seq_2_3),
int(chan_bond_seq_2_4),
log(chan_bond_seq_2_use),
int(chan_bond_seq_len),
int(chan_bond_wait),
log(clk_cor_insert_idle_flag),
log(clk_cor_keep_idle),
int(clk_cor_repeat_wait),
int(clk_cor_seq_1_1),
int(clk_cor_seq_1_2),
int(clk_cor_seq_1_3),
int(clk_cor_seq_1_4),
int(clk_cor_seq_2_1),
int(clk_cor_seq_2_2),
int(clk_cor_seq_2_3),
int(clk_cor_seq_2_4),
log(clk_cor_seq_2_use),
int(clk_cor_seq_len),
log(clk_correct_use),
int(comma_10b_mask),
int(crc_end_of_pkt),
str(crc_format),
int(crc_start_of_pkt),
log(dec_mcomma_detect),
log(dec_pcomma_detect),
log(dec_valid_comma_only),
int(mcomma_10b_value),
log(mcomma_detect),
int(pcomma_10b_value),
log(pcomma_detect),
int(ref_clk_v_sel),
log(rx_crc_use),
int(rx_data_width),
log(rx_decode_use),
int(rx_los_invalid_incr),
int(rx_los_threshold),
log(rx_loss_of_sync_fsm),
log(serdes_10b),
int(termination_imp),
int(tx_crc_force_value),

```



```

    log(tx_crc_use),
    int(tx_data_width),
    int(tx_diff_ctrl),
    int(tx_preemphasis),
    // Inputs
    value(chbondi, "[4]log"),
    value(enchansync, "log"),
    value(enmcommaalign, "log"),
    value(enpcommaalign, "log"),
    value(loopback, "uint:2"),
    value(powerdown, "log"),
    value(refclk, "log"),
    value(refclk2, "log"),
    value(brefclk, "log"),
    value(brefclk2, "log"),
    value(refclksel, "log"),
    value(rxpolarity, "log"),
    value(rxreset, "log"),
    clock(rxusrclk),
    clock(rxusrclk2),
    value(txbypass8b10b, "[1]log"),
    value(txchardispmode, "[1]log"),
    value(txchardispval, "[1]log"),
    value(txcharisk, "[1]log"),
    value(txdata),
    value(txforcecercerr, "log"),
    value(txinhibit, "log"),
    value(txpolarity, "log"),
    value(txreset, "log"),
    clock(txusrclk),
    clock(txusrclk2)

) {}

```

Output parameters:

See Xilinx documentation.

Input parameters:

See Xilinx documentation.

In-line target execution:

no

description:

This procedure creates an MGT high speed serial I/O channel. For a simpler interface see **gt_custom_16** [subsubsection 11.6.1.1](#) p432 above.

11.6.3 library functions

None.

11.6.4 library classes

None.

11.7 Xilinx netlist postprocessing module - xilinx/postprocess.3pl

This module converts the EDIF netlist produced by 3PL to a configuration file. It has a single procedure.

This library must be incorporated by using a postmodule statement, e.g. -

```
postmodule ("xilinx/postprocess.3pl");
```

See [subsection 4.5.2](#) p104 and [section 19.3](#) p627.

More than one postprocessing module can be specified in which case an additional integer argument can be supplied to specify ordering. If omitted the order in which the modules have been added to the list will be the order of execution. Thus far only this module has been used.

11.7.1 library modules

None.

11.7.2 library procedures

11.7.2.1 postprocess - post-process the netlist file

Prototype:

```
global procedure
postprocess(
    str(rname),
    str(tool),
    float(minver, 1.0),
    float(maxver, 9999.0)
) {}
```

Output parameters:

None.

Input parameters:

	mode	type	
rname	immediate	str	optional file name for recipe file
tool	immed	str	"ise" or "vivado"
minver	immediate	float	minimum version number
maxver	immediate	float	maximum version number

In-line target execution:

no

description:

This procedure executes the Xilinx place-and-route tools to generate the configuration bit file from the EDIF netlist file produced by 3PL. To do this it executes the Xilinx tools on the host CPU via calls to `exec()`.

rname optional and is the name of a file containing a recipe of commands. If supplied it overrides the recipe file specified by directive **postProcessRecipe**. Recipe files **ise_cp1d.cmd**, **ise_fpga.cmd**, **ise_xflow.cmd** and **vivado.tcl** are available in the directory containing **postprocess.3pl**.

tool "ise" or "vivado" must be supplied and specifies the main place-and-route tool to be used.

If directives **XILINX_ISE** or **XILINX_VIVADO** are supplied (as appropriate to the platform) then this specifies the specific path of the place-and-route tool to execute. Otherwise the path specified by directive **XILINX** will be used along with **minver** and **maxver** to select the specific version of the ISE or Vivado software. The highest version $\geq$ minver and $\leq$ maxver will be used.

if directive **postProcessAppendFile** exists it specifies a file to append to the bit file, usually a .info file

if directive **postProcessCompress** exists and is true the bit file will be compressed.

If directive **postProcessFilter** exists it specifies that the standard output from the place-and-route tools should be filtered. Values may be -

- "error" - lines starting with "ERROR" are retained
- "critical" - the above plus lines starting with "CRITICAL" are retained
- "warning" - the above plus lines starting with "WARNING" are retained
- "info" - the above plus lines starting with "INFO" are retained

any other values being ignored. If this directive is not defined or has an invalid value then no filtering is applied.

11.7.3 library functions

None.

11.7.4 library classes

None.

11.8 Embedded PowerPC Interface Library - xilinx/ppc405lib.3pl

This library has never been used in earnest so its functionality has not been tested!

This library provides an instantiation of an embedded PowerPC.

This library must be incorporated by using a **module** statement -

```
module "xilinx/ppc405lib.3pl"
```

11.8.1 library modules

None.

11.8.2 library procedures

11.8.2.1 ppc405 - a PPC405

Library: xilinx/ppc405lib.3pl

Prototype:

```
global procedure
ppc405 (
    value({ C405CPMCORESLEEPREQ, C405CPMMSRCE, C405CPMMSREE, C405CPMTIMERIRQ,
            C405CPMTIMERRESETRREQ, C405DBGMSRWE, C405DBGSTOPACK,
            C405DBGWBCOMPLETE, C405DBGWBFULL, C405DCRREAD, C405DCRWRITE,
            C405JTGCAPTUREDR, C405JTGETEST, C405JTGPGMOUT, C405JTGSHIFTR,
            C405JTGTD0, C405JTGTD0EN, C405JTGUPDR, C405PLBDCUABORT,
            C405PLBDCUCACHEABLE, C405PLBDCUGUARDED, C405PLBDCUREQUEST,
            C405PLBDCURNW, C405PLBDCUSIZE2, C405PLBDCUUOATTR,
            C405PLBDCUWRITETHRU, C405PLBICUABORT, C405PLBICUUOATTR,
            C405RSTCHIPRESETRREQ, C405RSTCORERESETRREQ, C405RSTSYSRESETRREQ,
            C405TRCCYCLE, C405PLBICUREQUEST, C405TRCTRIGGEREVENTOUT,
            C405XXXMACHINECHECK, DSOCMBRAMEN, DSOCMBUSY, ISOCMBRAMEN,
            ISOCMBRAMEVENWRITEEN, ISOCMBRAMODDWRITEEN, C405PLBICUCACHEABLE}, "log"),
    value(C405DBGWBIAR, "bits:30"),
    value(C405DCRABUS, "bits:10"),
    value(C405DCRDBUSOUT, "bits:32"),
    value(C405PLBDCUABUS, "bits:32"),
    value(C405PLBDCUBE, "bits:8"),
    value(C405PLBDCUPRIORITY, "uint:2"),
    value(C405PLBDCUWRDBUS, "bits:64"),
    value(C405PLBICUABUS, "uint:30"),
    value(C405PLBICUPRIORITY, "uint:2"),
    value(C405PLBICUSIZE, "uint:2"),
    value(C405TRCEVENEXECUTIONSTATUS, "uint:2"),
    value(C405TRCODEEXECUTIONSTATUS, "uint:2"),
    value(C405TRCTRACESTATUS, "uint:4"),
    value(C405TRCTRIGGEREVENTTYPE, "uint:11"),
    value(DSOCMBRAMABUS, "uint:22"),
    value(DSOCMBRAMBYTEWRITE, "[4]log"),
    value(DSOCMBRAMWRDBUS, "bits:32"),
    value(ISOCMBRAMRDABUS, "uint:21"),
    value(ISOCMBRAMWRABUS, "uint:21"),
    value(ISOCMBRAMWRDBUS, "bits:32")
    <-
    int({XL0C, YL0C}),
```

```

clock({ BRAMDSOCCLK, BRAMISOCCLK, CPMC405CLOCK, PLBCLK}),
value({ CPMC405CORECLKINACTIVE, CPMC405CPUCLKEN, CPMC405JTAGCLKEN,
        CPMC405TIMERCLKEN, CPMC405TIMERTICK, DBG405DEBUGHALT,
        DBG405EXTBUSHOLDACK, DBG405UNCONDDEBUGEVENT, DCRC405ACK,
        EICC405CRITINPUTIRQ, EICC405EXTINPUTIRQ, JTGC405BNDSCANTDO,
        JTGC405TCK, JTGC405TDI, JTGC405TMS, JTGC405TRSTNEG,
        MCBCPUCLKEN, MCBJTAGEN, MCBTIMEREN, MCPPCRST,
        PLBC405DCUADDRACK, PLBC405DCUBUSY, PLBC405DCUERR,
        PLBC405DCURDDACK, PLBC405DCUFSIZE1, PLBC405DCUWRDACK,
        PLBC405ICUADDRACK, PLBC405ICUBUSY, PLBC405ICUERR,
        PLBC405ICURDDACK, PLBC405ICUFSIZE1, RSTC405RESETCHIP,
        RSTC405RESETCORE, RSTC405RESETSYS, TIEC405DETERMINISTICMULT,
        TIEC405DISOPERANDFWD, TIEC405MMUEN, TRCC405TRACEDISABLE,
        TRCC405TRIGGEREVENTIN}, "log"),
value(BRAMDSOCMRDDBUS, "bits:32"),
value(BRAMISOCMRDDBUS, "bits:64"),
value(DCRC405DBUSIN, "bits:32"),
value(DSARCVALUE, "uint:8"),
value(DSCNTLVALUE, "uint:8"),
value(ISARCVALUE, "uint:8"),
value(ISCNTLVALUE, "uint:8"),
value(PLBC405DCURDDBUS, "bits:64"),
value(PLBC405DCURDWDADDR, "uint:3"),
value(PLBC405ICURDDBUS, "bits:64"),
value(PLBC405ICURDWDADDR, "uint:3"),
value(TIEDSOCMDCRADDR, "uint:8"),
value(TIEISOCMDCRADDR, "uint:8")
) {}

```

Output parameters:

See Xilinx documentation.

Input parameters:

See Xilinx documentation.

description:

This procedure creates an instance of a PowerPC 405 core. For a simpler interface see procedure **ppcbr** [subsubsection 11.9.2.1](#) p442 above.

11.8.3 library functions

None.

11.8.4 library classes

None.

11.9 Embedded PowerPC Interface Library - xilinx/ppclib.3pl

This library provides a way of instantiating an embedded PowerPC. The library was under development but work on it has ceased. Its functionality has not been tested!

This library must be incorporated by using a **module** statement -

```
module "xilinx/ppclib.3pl"
```

11.9.1 library modules

None.

11.9.2 library procedures

11.9.2.1 ppcbr - a PPC405 with OCM

Library: xilinx/ppclib.3pl

Prototype:

```
global module
ppcbr (
    var({iramp, dramp0, dramp1, dramp2, dramp3}, "->")
    <-
    int({imsk, dmsk}),
    log({iramea, dramea}),
    value({cpuclocken, cpureset}, "log")
) {}
```

Output parameters:

iramp pointer to 64 bit instruction RAM

dramp0 pointer to data RAM byte 0 (most sig)

dramp1 pointer to data RAM byte 1

dramp2 pointer to data RAM byte 2

dramp3 pointer to data RAM byte 3 (least sig)

Input parameters:

imsk instruction memory size (bytes) in K - each 2K takes a block RAM - must be 4, 8, 16, 32, 64 or 128

dmsk data memory size (bytes) in K - each 2K takes a block RAM - must be 2, 4, 8, 16, 32 or 64

iramea true if instruction ram 2nd port required for external access

dramea true if data ram 2nd port required for external access

cpuclocken enable CPU core clock

cpureset reset the CPU - this should be raised for a clock cycle

description:

This procedure creates an interface to an embedded PowerPC 405. It uses on-chip block RAM for separate instruction and data memory. Both memories are located at address 0. The memory management is disabled. Pointers to the memories are available as output parameters to enable access via the second RAM port. This procedure calls **ppc405** [subsubsection 11.8.2.1](#) p440 above to create the PPC instance.

11.9.3 library functions

None.

11.9.4 library classes

None.

11.10 Zynq ARM processor interface Library - xilinx/zynq_axi.3pl

This library instantiates a Zynq CPU and provides interfaces to the two general purpose memory-mapped slave buses and the four high performance DMA master buses. It also connects the DDR RAM and general purpose I/O pins to the CPU.

This library must be incorporated by using a **source** statement -

```
source "xilinx/zynq_axi.3pl"
```

however xilinx/zynq_axi.3pl is already included in board header files -
boards/avnet/microzed_7010.3pl
boards/avnet/microzed_7020.3pl
boards/digilent/zybo.3pl

Code to create the interface to the CPU must be executed as a trailer module by -

```
trailermodule ("zynq_processor") {
    zynq_ps();
}
```

and this is also incorporated in the header files listed above.

Application code only needs to incorporate the board header file and create an instance of the CPU interface, e.g. -

```
module "boards/avnet/microzed_7010.3pl"

class(cpu, cpu_api());
```

The CPU_API class is documented in [section 11.11](#) p455 **FPGA to CPU interface Library - xilinx/cpu.3pl**. This library can also be used by FPGA boards which have a USB serial interface to a host computer.

Since cpu.3pl is included via a source statement its code shares the scope of this library, UART_CPU. Thus it can call the modules, functions and procedures therein.

cpu.3pl defines a class **CPU_API()** which provides the following class procedures for data transfer between FPGA and CPU -

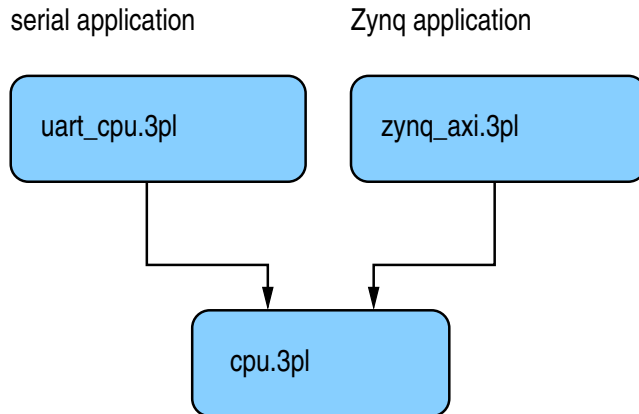
- read()** generic read [subsection 11.11.2.2](#) p459
- write()** generic write [subsection 11.11.2.3](#) p464
- read_memory()** memory read [subsection 11.11.2.4](#) p469
- read_queue()** queue read [subsection 11.11.2.5](#) p471
- read_value()** value read [subsection 11.11.2.6](#) p474
- write_memory()** memory write [subsection 11.11.2.8](#) p477
- write_queue()** queue write [subsection 11.11.2.9](#) p479
- write_static()** static write [subsection 11.11.2.10](#) p481
- dma_read()** to FPGA from CPU [subsection 11.11.1.2](#) p457
- dma_write()** from FPGA to CPU [subsection 11.11.1.3](#) p458
- signal_interrupt()** interrupt from a signal [subsection 11.11.1.1](#) p456
- interrupt()** interrupt procedure [subsection 11.11.2.1](#) p458

each of which can generate an instance of a CPU interface to a specific 3PL variable of appropriate mode. An internal class **CPU_CODEGEN** is also created but this is not visible to the application code

cpu.3pl is sourced by both **uart_cpu.3pl** and **zynq_axi.3pl** to provide CPU interfaces. The former employs a serial data link and the latter parallel busses.

When any of the procedures above are called a defining structure is created which is added to a list of CPU interfaces. At the end of immediate execution module **create_bus_interfaces()** is called. **create_bus_interfaces()** is defined in both **uart_cpu.3pl** and **zynq_axi.3pl** as it is specific to each. This

module traverses the interface list constructed above and for each interface calls a matching procedure in class **CPU_CODEGEN** to generate the logic for each instance.



Bus GP0 is used for reading 3PL expression values and queue mode variables and writing static and queue mode variables.

Bus GP1 is used to read and write 3PL memory mode variables.

For both GP buses burst transfers are used and burst lengths may be up to 8 32-bit words. Burst transfers are executed in the CPU interface code by load/store multiple commands which use a contiguous set of registers. A maximum of 8 of the 16 available registers has been imposed, hence the maximum burst size of 8 32-bit words.

An information file, design.info, is generated. This file lists the directives, AXI bus interface details and interrupt event parameters. The file is in JSON format. A python utility program, info2c, extracts data from this information file and generates a .h file for CPU programs in C. Similar utilities may be written for other languages.

info2c is a shell script invoking python. The first or only argument is the design name and info2c expects to find a file design.info. It will create a file design.h. If the optional second argument is provided then it is used as the output file name (".h" is appended).

The design.info data may also be stored in block RAM (rmemory) for read-time access by CPU code. This is enabled by default but may be avoided via a directive, i.e. **directive("excludeinfo", true);**, which can be executed at any time during 3PL immediate execution. The compression is done by calling inbuilt function **deflate()** [section 10.25](#) p279. The readback is initialised by writing to address 1 and 32-bit packed .info data is then read by successive 32-bit reads from address 1. The first read returns a 32 bit CRC-32 checksum for the remainder of the block, The second read returns the uncompressed data size in bytes and the third read returns the compressed data size in bytes. Reads beyond the data length will return zeros.

The data block size depends on the size of the compressed .info data and varies from a minimum of 1024 32 bit words (1 RAM block) to 32,768 words (32 RAM blocks).

These bus interfaces allocate successive addresses. These addresses are embedded in C functions generated by info2c and hence the programmer does not need to know the addresses (they can be determined from the .info file). Each interface is supplied a unique name in the 3PL procedure call and this name is used in the associated generated function names.

For 3PL int types info2c will generate matching C types int8_t, int16_t, int32_t or int64_t as appropriate to the size of the datum. The maximum primitive size is 64 bits.

For 3PL uint or bits types info2c will generate matching C types uint8_t, uint16_t, uint32_t or uint64_t as appropriate to the size of the datum. The maximum primitive size is 64 bits.

For 3PL log type info2c will generate matching type bool_t

For 3PL structs or enums info2c will generate a matching C bit field struct. The maximum size for a struct is 64 bits.

For a 3PL float type info2c will generate a C float or double type if the the type matches IEEE 754 format, otherwise it will generate a C bit field struct containing the exponent and mantissa as fields.

The 3PL data type may be an array of uint, bits or int of exactly 32 or 64 bits where the dimension is a maximum of 8 for 32 bit members or 4 for 64 bit members. The matching C type generated will be a struct containing the array.

11.10.1 library modules

11.10.1.1 cpu_write_dma_buffer - DMA transfer to CPU RAM

Library: xilinx/zynq_axi.3pl

Prototype:

```
global module
cpu_write_dma_buffer (
    str(name),           // interface name
    int(blockwords),     // block length in data words
    int(nblocks),        // number of blocks
    queue(in),           // data in
    log(use_wcount),     // use queue count rather than interrupt
    log(bigendian)       // transfer data as bigendian
) {}
```

Output parameters:

none

Input parameters:

name is a string identifier for the interface.

blockwords is the DMA block length in data words.

nblocks is the number of DMA blocks used.

in is a queue of data to be transferred.

use_wcount determines whether an interrupt or a count are to be used to determine when to read the 'done' queue.

This module creates a DMA buffer management interface allowing the FPGA to write directly to CPU RAM using DMA. For a simpler interface see **dma_write()** [subsection 11.10.1.4](#) p451. The parameter **name** is used when creating the interface functions for this module.

The interface will use the next unused HP bus of the four available. If all four HP buses have been used then a fatal error is invoked.

The interface has two queues accessible to the CPU, a 'free' queue and a 'done' queue.

The 'free' queue contains block numbers that are available to the DMA for writing. This queue is loaded initially by the CPU. The DMA interface removes a block number from the 'free' queue when it is to begin writing a block. If this queue is empty then DMA will wait until a block number is written to it by the CPU.

When the DMA has finished filling a block in CPU RAM then it writes the block number to a 'done' queue. The block numbers read from the 'done' queue can be used to compute the associated buffer address in mapped DMA memory.

If **use_wcount** has no argument or the argument value is FALSE then there is an interrupt associated with that queue on which the CPU can wait for completed blocks to be signalled. The interrupt is a level which can be used to control a loop reading one item from the queue on each iteration.

If **use_wcount** has a TRUE argument then the number of block numbers queued in the done queue can be determined and read in a count-controlled loop.

The CPU must return a block number read from the 'done' queue to the 'free' queue after it has completed processing the data in the block.

The status of block numbers in the range of blocks available for use is recorded in memory in this DMA interface. This status can be used when recovering from a CPU crash but not reloading the FPGA configuration. The status for a block can be read or written by the CPU and will be one of -

- **missing** - the block number is missing from the DMA interface. It presumably has been removed by the CPU and not yet returned.
- **free** - the block is free for a DMA transfer
- **done** - a DMA transfer has been completed for this block
- **busy** - the block is currently being written by a DMA transfer

The data input queue **in** must be of type -

```
struct(...,
    data,      ...user data type ...,
    term,      dma_block_term,
    tag,       tagtype
);
```

where

- **data** is the user data which can be any type
- **term** is a block termination enumerated type 'dma_block_term' which is
 - **'none'** (default) to add the datum to the current block
 - **'ignore'** to terminate the block, discarding the accompanying datum
 - **'first'** to start a new block with the datum
 - **'last'** to add the current datum to the block then terminate the block
- **tag** is a user-defined field which can be used to distinguish blocks of different content. The tag field of the last datum to be added to the current block is used as the block tag, all previous ones being ignored.

The block termination mechanism will not terminate an empty block.

Blocks will be terminated when full if not explicitly terminated by use of the termination field.

The input data type is rounded up in size to 8, 16, 32 or 64 bits, zero padded at the MSB end where necessary. Data are packed into 64-bit words for transmission across an HP bus to CPU memory. The packing is 8x8, 4x16, 2x32 or 1x64. Where a block is terminated before being filled any unfilled portion of the last 64-bit word will be zero. The remainder of the block in CPU memory remains unwritten.

On completion of a block transfer the structure that is read by the CPU from the 'done' queue is -

```
struct(bnctype,
    blocknum,    "uint:"+size(nblocks),
    words,       "uint:"+size(blockwords),
    tag,         tagtype
);
```

where -

- **blocknum** is the block number in the DMA buffer space
- **words** gives the number of words of packed user data in the block
- **tag** is a user-defined block tag

The Zynq ARM processor is little-endian and this is the default transfer order. If the argument to parameter **bigendian** is supplied and is true the data will be transferred in big-endian order instead. This might be used if the data is to be transferred unprocessed out of the Zynq to another processor which is big-endian.

The following C functions will be created from the .info file and placed into an include file by executing the python script info2c -

fpga_name_done_read()		read done queue block number and word count
fpga_name_done_avail_read()	1	read number of words in done queue
fpga_name_done_avail_thresh_write()		set interrupt threshold for done queue
fpga_name_done_enable()	2	enable done queue interrupt
fpga_name_done_disable()	2	disable done queue interrupt
fpga_name_done_wait()	2	wait on done queue interrupt
fpga_name_free_write()		write to free queue
fpga_name_status_read()		read block status memory

Notes:

1 - only if argument wcount is true

2 - only if argument wcount is omitted or false

in addition entries similar to the following will also be placed in the include file -

```
const unsigned fpga_name_blocks 4;
const unsigned fpga_name_block_size 160;
const unsigned fpga_name_block_words 80;
const unsigned fpga_name_port 0;
typedef uint32_t fpga_name_t;
```

In the following **fpga** is the pointer to the FPGA structure returned by **fpga_open()**.

fpga_name_done_enable(fpga)

Enable the interrupt on the block completion queue non-empty. This is only available if **use_wcount** is FALSE.

fpga_name_done_wait(fpga, NULL)

Wait on the interrupt on the block completion queue. Execution will continue, and the interrupt is disabled, when the queue is not empty. This is only available if **use_wcount** is FALSE. The second argument if not NULL is a timeout period (struct timeval).

fpga_name_done_avail_read(fpga)

Read the availability count on the block completion queue. This is only available if **use_wcount** is TRUE.

fpga_name_done_read(fpga)

Read the block completion queue. The function returns a structure similar to the following -

```
typedef union {
    struct {
        uint16_t blocknum: 3;
        uint16_t words: 6;
        uint16_t blocktype: 2;
        uint16_t : 5;
    };
    uint16_t __word;
} fpga_{\em name}_done_t;
```

where **blocknum** is the block number of a completed block in the DMA buffer, **words** is the number of words of type **fpga_name_t** written to that buffer (i.e. data words, not necessarily 64-bit words) and **blocktype** is the user's arbitrary block type code. The last unnamed field (5) is padding to make up 16 bits.

fpga_name_free_write(fpga, blocknum)

Write a block number to the free block queue for re-use by the DMA interface (**blocknum** is just an integer, not the struct shown above).

fpga_name_status_read(fpga, blocknum)

Read the status of a block from the DMA interface. For an interface named "xyz" the returned status would be a byte of type -

```
typedef enum {
    fpga_xyz_status_busy = 2,
    fpga_xyz_status_done = 3,
    fpga_xyz_status_free = 1,
    fpga_xyz_status_missing = 0,
} fpga_xyz_status_t;
```

Any block number that has status 'missing' can be written to the free queue to make it once again available for use.

const unsigned fpga_name_blocks

The number of DMA blocks for this interface.

const unsigned fpga_name_block_size

The block length in bytes for this interface, rounded up to a multiple of 8.

const unsigned fpga_name_block_words

The number of data words in this block.

const unsigned fpga_name_port

The HP bus used for this interface, 0 to 3.

typedef uint32_t fpga_name_t

The type of data word.

The CPU C code (using interrupt) might appear as follows, "DMA1" being the interface name -

```
fpga_t *      fpga = NULL;
fpga_DMA_done_t bn;

nrete( fpga = fpga_open(fpga_unit_default, fpga_flag_default) );
erete( fpga_configdata_load(fpga, prog_fpga_config_data, prog_fpga_config_size) );
erete( fpga_map (fpga) );

// Load the block numbers into the 'free' queue.
for (i=0 ; i<HP0_buffers ; i++)
    fpga_DMA1_free_write(fpga, i);

fpga_DMA1_done_enable(fpga);                // enable interrupt
for (;;) {
    erete( fpga_DMA1_done_wait(fpga, NULL) );
    // When there is a completed block number available
    // read it, process the data in the associated buffer,
    // then return the block number of the now free block.
    bn = fpga_DMA1_done_read(fpga);          // fetch block number
    p = fpga->dma.base + bn.blocknum * fpga_DMA1_block_size;
    for (int i=0 ; i<bn.words ; i++) {
        .
        . process data in DMA buffer block pointed to by p
        .
    }
    fpga_DMA1_free_write(fpga, bn.blocknum); // return block number
    fpga_DMA1_done_enable(fpga);            // re-enable interrupt
}
```

The code in this module executes in the bus clock domain (c100). The clock domain in use when this module is called is restored when the module is exited.

11.10.1.2 create_bus_interfaces - create the individual CPU interfaces

Library: xilinx/zynq_axi.3pl

Prototype:

```
module
create_bus_interfaces () {}
```

Output parameters:

none.

Input parameters:

none.

description:

This module creates the logic for each of the CPU interfaces. Each of these communicates via queues to logic within `zynq-ps()` [subsection 11.10.1.6](#) p452 below.

11.10.1.3 dma_read - DMA transfer from memory to FPGA

Library: xilinx/zynq_axi.3pl

Prototype:

```
global module
dma_read (
    queue(data, "bits:64")
    <-
    int(bus),
    queue(address, "uint:32"),
    queue(burstlen, "uint:4")
) {}
```

Output parameters:

Zynq **data** is the memory data read

Input parameters:

bus is the high performance master bus, 0 to 3

address is the physical memory address from which to read

burstlen is the burst length, which is the number of data words to be read

description:

DMA from RAM to FPGA using an AXI HP bus. This is a low-level module, providing only a simple DMA interface.

This module can also be called indirectly via library `cpu.3pl` as a procedure in class **CPU_API**. [section 11.11](#) p455 **FPGA to CPU interface Library - xilinx/cpu.3pl**.

A datum on each of queues 'address' and 'burstlen' will trigger the DMA transfer if queue 'data' is not full. The transfer will consume one datum from each of 'address' and 'burstlen' and will write the number of data to queue 'data' necessary to satisfy the burst length. changed by this procedure,

The code in this module executes in the bus clock domain (c100). The clock domain in use when this module is called is restored when the module is exited.

11.10.1.4 dma_write - DMA transfer from FPGA to memory

Library: xilinx/zynq_axi.3pl

Prototype:

```
global module
dma_write (
  <-
  int(bus),
  queue(data, "bits:64")
  queue(address, "uint:32"),
  queue(burstlen, "uint:4")
) {}
```

Output parameters:

data is the memory data read

Input parameters:

bus is the high performance master bus, 0 to 3

address is the physical memory address to write to

burstlen is the burst length, which is the number of data words to be written minus 1

description:

DMA from PL to RAM using an AXI HP bus. This is a low-level module, providing only a simple DMA interface. See `cpu_write_dma_buffer()` [subsubsection 11.10.1.2](#) p450 for a higher level interface supporting CPU buffer control. This is a low-level module, providing only a simple DMA interface.

This module can also be called indirectly via library `cpu.3pl` as a procedure in class **CPU_API**. [section 11.11](#) p455 **FPGA to CPU interface Library - xilinx/cpu.3pl**.

A datum on each of queues 'address' and 'burstlen' will trigger the DMA transfer if queue 'data' is not empty. The transfer will consume one datum from each of 'address' and 'burstlen' and will read the number of data from queue 'data' necessary to satisfy the burst length. The 'burstlen' is from 0 for one word to 15 for 16 words.

The code in this module executes in the bus clock domain (c100). The clock domain in use when this module is called is restored when the module is exited.

not empty.

11.10.1.5 sigint - interrupt the CPU on a signal

Library: xilinx/zynq_axi.3pl)

Prototype:

```
module
sigint(int(n), value(sig, "log")) {}
```

Output parameters:

none.

Input parameters:

n is the interrupt index, 0 to 16.

sig is a signal on whose rising edge the interrupt is triggered.

description:

When the signal **sig** is asserted the interrupt is triggered. The interrupt number, 0 to 15, is given by **index**.

11.10.1.6 zynq_ps - instantiate Zynq CPU

Library: xilinx/zynq_axi.3pl

Prototype:

```
module
zynq_ps () {}
```

Output parameters:

none

Input parameters:

none

In-line target execution:

no

description:

This instantiates the Zynq CPU and connects the DDR RAM and general purpose I/O pins to the CPU. It also defines all the CPU pins. After calling this module all the following procedures are available.

The code in this module executes in the bus clock domain (c100). The clock domain in use when this module is called is restored when the module is exited.

11.10.2 library procedures

11.10.2.1 add_interface - add an interface to the interface list

Library: xilinx/zynq_axi.3pl)

Prototype:

```
procedure
add_interface (var(v, "->")) {}
```

Output parameters:

none.

Input parameters:

v is an instance of class **busCommon**, defined in cpu.3pl.

description:

The instance of **busCommon** is added to the interface list. The list is processed at completion of immediate execution to create the whole CPU interface.

11.10.2.2 cpu_read - write to CPU

Library: xilinx/zynq_axi.3pl)

Prototype:

```
procedure
cpu_read (
    value(access, "log"),
    value(init, "log"),
    queue(out)
```



```

    <-
    int(bus),
    int(addr),
    value(data_in),
    value(contin, "log")
  ) {}

```

Output parameters:

access is an optional output parameter which is asserted when a CPU to FPGA write occurs.

init is an output parameter which is asserted in the case of a memory read to initiate the read.

out is an optional queue variable which will be written to when a transfer occurs.

Input parameters:

bus is not used (applies to Zynq).

address is a 6 bit interface address.

data_in input data.

contin is only used for a memory read and is asserted when the memory read has occurred and the transfer can begin.

description:

This procedure provides a low-level interface to transfer data from the FPGA to the CPU via the USB serial port. It is used by modules and procedures in library `cpu.3pl` and is not intended for use directly by application code.

The CPU initiates the transfer. The transfer may consist of multiple bytes.

11.10.2.3 cpu_write - write to CPU

Library: `xilinx/zynq_axi.3pl`

Prototype:

```

procedure
cpu_write (
    var(out),
    value(access, "log")
    <-
    int(bus),
    int(addr)
) {}

```

Output parameters:

var is a static or queue mode variable.

access is an optional output parameter which is asserted when a CPU to FPGA write occurs.

Input parameters:

bus is not used (applies to Zynq).

address is a 6 bit interface address.

description:

This procedure provides a low-level interface to transfer data from the CPU to the FPGA via the USB serial port. It is used by modules and procedures in library `cpu.3pl` and is not intended for use directly by application code.

The CPU initiates the transfer. The transfer may consist of multiple bytes.

11.10.3 library functions

11.10.3.1 cpu_api - CPU communications interface

Library: xilinx/zynq_axi.3pl)

Prototype:

```
global function  
cpu_api () {}
```

Return mode and type:

immediate, class

Input parameters:

none.

description:

This function returns an instance of class CPU_API

Library: xilinx/zynq_axi.3pl

Prototype:

```
global function cpu_api () {}
```

None.

11.10.4 library classes

None.

11.11 FPGA to CPU interface Library File - xilinx/cpu.3pl

This library provides a class for basic FPGA-CPU interface for platforms which provide a communications path between a CPU and FPGA. This includes the zynq FPGA and various FPGA prototype boards which have a USB port for configuration loading and serial data communications. These procedures allow the CPU to read and write static variables, queues and memories and the FPGA to interrupt the CPU.

This library contains alternative interfaces for reading and writing 3PL variables from the CPU. There are class procedures which are mode-specific -

```
CPU_API.read_value()
CPU_API.read_queue()
CPU_API.read_memory()
CPU_API.write_static()
CPU_API.write_queue()
CPU_API.write_memory()
```

and there are generic class procedures -

```
CPU_API.read()
CPU_API.write()
```

The former have the advantage that the mode of the variable being accessed is reflected in the procedure identifier and this may improve the readability the source code. On the other hand the latter are generic and shorter.

There are DMA class modules for reading and writing CPU memory from the FPGA -

```
CPU_API.dma_read()
CPU_API.dma_write()
```

and an interrupt class module and an interrupt class procedure -

```
CPU_API.sigint()
CPU_API.interrupt()
```

For example the following three examples are equivalent -

```
api.read_queue(entries, , acc <- q, "QR", true, 10 );
```

```
api.read_queue(entries, , acc <- q, "QR", wordcount=true, ilevel=10 );
```

```
api.read(count=entries, access=acc <- q, "QR", wordcount=true, ilevel=10);
```

A more detailed description of use of this class can be found in ([chapter 21](#) p645) for platforms with a USB serial link to the FPGA and ([chapter 20](#) p631) for Zynq platforms. These sections include both C and Python3 coding examples.

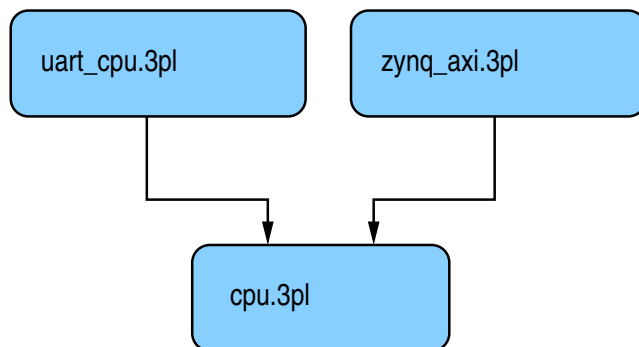
This library is incorporated by using a **source** statement -

```
source "xilinx/cpu.3pl"
```

however this library is already included via xilinx/uart_cpu.3pl ([section 11.12](#) p485) or xilinx/zynq_axi.3pl ([section 11.10](#) p444) and so should not be included directly by application programs.

serial application

Zynq application



xilinx/zynq_axi.3pl is included in -
 boards/avnet/microzed_7010.3pl
 boards/avnet/microzed_7020.3pl
 boards/digilent/zybo.3pl

xilinx/uart_cpu.3pl is included in -
 boards/avnet/aes_sp3a_eval400.3pl
 board/gadget_factory/papilio_one_100.3pl
 board/gadget_factory/papilio_one_250.3pl
 board/gadget_factory/papilio_one_500.3pl
 board/gadget_factory/papilio_pro_lx4.3pl
 board/gadget_factory/papilio_pro_lx9.3pl

The class for the FPGA interface is created by -

```
class(cpu, CPU_API());
```

cpu.3pl -

- defines class **busCommon** which is used to describe each interface.
- creates an instance of class **CPU_CODEGEN** whose procedures create the logic to implement each type of interface.
- creates an instance of class **CPU_API** whose modules provide the application program interface. Procedures within this class create instances of **busCommon** and add them to an interface list. **busCommon** contains all the information necessary to create a CPU interface. Class **CPU_API** in turn assigns module `create_bus_interfaces()` as a trailer module.

Module `create_bus_interfaces()` executes at the end of immediate execution (it is a trailer module). It creates the basic CPU interface logic by calling module **cpu_interface** and then traverses the interface list and creates code for each interface via procedures in class **CPU_CODEGEN**.

CPU_API, **CPU_CODEGEN**, **create_bus_interfaces** and **cpu_interface** are defined in both **UART_CPU.3pl** and **zynq_axi.3pl**, as the underlying interface logic is quite different for the two platforms.

11.11.1 class library modules

11.11.1.1 CPU_API.signal_interrupt - interrupt from a signal

Library: xilinx/cpu.3pl

Prototype:

```
module
  signal_interrupt (str(event), value(sig, "log")) {}
```

Output parameters:

none.

Input parameters:

event is the event identifier string.

sig is the signal causing the event.

In-line target execution:

no

description:

This procedure generates a module which will cause a CPU interrupt when its input signal **sig** is asserted. The argument to parameter **event** is an identifier with which to refer to the interrupt in the CPU interface software.

See also class procedure **interrupt()** [subsection 11.11.2.1](#) p458 which generates a CPU interrupt when executed in the FPGA.

11.11.1.2 CPU_API.dma_read - DMA transfer to FPGA from CPU

Library: xilinx/cpu.3pl

Prototype:

```
module
dma_read (
    queue(data, "bits:64")      // read data
    <-
    queue(address, "uint:32"),  // memory address
    queue(burstlen, "uint:4"),  // burst length - 1
    int(bus)                    // HP bus 0 to 3
) {}
```

Output parameters:

data is the queue to which data from the transfer is returned.

Input parameters:

address is a queue supplying the read address.

burstlen is a queue giving the burst length in the case of Zynq.

bus is the HP bus, 0 to 3, in the case of Zynq.

In-line target execution:

no

description:

This procedure generates a module which reads data from the CPU memory.

In the case of a platform with a serial interface only one input argument, **address**, is valid - **burstlen** and **bus** must not be given arguments. When an address becomes available on the **address** input queue and the output **data** queue is not full a DMA transfer (read) will take place. For a serial interface platform this is not a real DMA transfer but is carried out in software by a separate CPU thread.

For the Zynq When an address and a burstlength become available on the **address** and **burstlen** input queues and the output **data** queue is not full, a DMA transfer (read) will take place. The HP bus selected cannot be used for any other interfaces.

See also class module **CPU_API.dma_write()** [subsection 11.11.1.3](#) p458 which generates a DMA write module.

11.11.1.3 CPU_API.dma_write - DMA transfer from FPGA to CPU

Library: xilinx/cpu.3pl

Prototype:

```
module
  dma_write (
    queue(data, "bits:64"),    // write data
    queue(address, "uint:32"), // memory address
    queue(burstlen, "uint:4"), // burst length - 1
    int(bus)                   // HP bus 0 to 3
  ) {}
```

Output parameters:

None.

Input parameters:

data is a queue supplying data for the transfer.

address is a queue supplying the read address.

burstlen is a queue giving the burst length in the case of Zynq.

bus is the HP bus, 0 to 3, in the case of Zynq.

In-line target execution:

no

description:

This procedure generates a module which writes data to the CPU memory.

In the case of a platform with a serial interface only two input arguments, **data** and **address**, are valid - **burstlen** and **bus** must not be given arguments. When data and an address become available on the **data** and **address** input queues a DMA transfer (read) will take place. For a serial interface platform this is not a real DMA transfer but is carried out in software by a separate CPU thread.

For the Zynq When data an address and a burstlength become available on the **data**, **address** and **burstlen** input queues a DMA transfer (write) will take place. The HP bus selected cannot be used for any other interfaces.

See also class module **CPU_API.dma_read()** [subsection 11.11.1.2](#) p457 which generates a DMA read module.

11.11.2 class library procedures

11.11.2.1 CPU_API.interrupt - interrupt procedure

Library: xilinx/cpu.3pl

Prototype:

```
procedure
  interrupt (str(event)) {}
```

Output parameters:

none.

Input parameters:

event is the event identifier string.

In-line target execution:

yes

description:

This procedure generates an interrupt when executed in the FPGA. The argument to parameter **event** is an identifier with which to refer to the interrupt in the CPU interface software.

See also class module **signal_interrupt()** [subsection 11.11.1.1](#) p456 which creates a module which sources an interrupt from a signal.

11.11.2.2 CPU_API.read - a CPU-readable interface

Library: xilinx/cpu.3pl

Prototype:

```

procedure
  read (
    static(access, "log"),
    varargs
    <-
    str(name),
    var(in),
    varargs
  ) {}

```

Output parameters:

access is an optional static logical variable which is true for one clock cycle following a CPU read.
varargs optional output, parameters **out** and **count**.

Input parameters:

in is the value, queue, cmemory or rmemory mode variable to be read.

name is a string identifier for the interface.

varargs optional input parameters **wordcount**, **ilevel** and **port**.

In-line target execution:

no

description:

This procedure creates a module which provides a CPU-readable interface for the the CPU for 3PL variables of mode value, static, queue, cmemory or rmemory.

In the case of Zynq the interface uses AXI bus GP0. The size of the input data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the input data type must not exceed 1016 bits (127 bytes).

The module is created at the end of immediate execution from arguments queued by this procedure. It can be any type, primitive or compound.

It is a common wrapper for the procedures

CPU_API.read.value() [subsection 11.11.2.6](#) p474

CPU_API.read.queue() [subsection 11.11.2.5](#) p471

CPU_API.read.memory() [subsection 11.11.2.4](#) p469

Parameter **name** is a string identifier that can be used in the CPU code to refer to this interface.

Parameter **in** is the input variable or expression to be read by the CPU.

Optional input and output varargs parameters are -

	input varargs	output varargs
value static		
queue	wordcount ilevel	out count
cmemory rmemory	port	

Value or Static mode

Parameter **in** is the value-mode input variable or expression. Since it is value mode the argument may be a value mode variable, static mode variable or an expression containing these. An expression must not contain a queue read.

The module is created at the end of immediate execution from arguments queued by this procedure. It can be any type, primitive or compound. The optional output static logical variable **access** goes true for one clock cycle to indicate that a read has occurred.

This procedure can be called via the more cryptic and general procedure **cpu_api.read()** ?? p?? instead.

In the case of Zynq the interface uses AXI bus GP0. The size of the input data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the input data type must not exceed 1016 bits (127 bytes).

The optional output static logical variable **access** goes true for one clock cycle to indicate that a read has occurred.

The clock domain of output **access** need not be the same as that of **in**. If **access** has no clock domain it will be given the bus clock domain.

The parameters associated with this bus interface, including an allocated bus address, are written to the *design.info* file.

The utility info2c is used to process this file and create an include file *design.h* which contains a function definition to access this interface. for example the 3PL code

```
cpu_read_value("alpha", v);
```

will result in info2c producing a declaration like

```
uint32_t alpha_read (fpga_t * fpga) {}
```

When the CPU reads a value it is sampling that value, hence data may be lost if the sampling rate is lower than the rate at which data changes. The data sampling logic ensures that the data is a valid sample even when the source is changing at the time of the read. This is a primitive interface which depends on additional protocols being imposed by the application code. For example to guarantee data integrity a write to register the data in a static variable prior to reading could be used. For a more complex interface see **cpu_api.read.queue()** [subsubsection 11.11.2.5](#) p471.

On running info2c on the .json file the following C function will have been defined for interface *name* -

function	
fpga_name_read()	read a datum

Example:

If 3PL source file prog.3pl contains the following


```

class(cpu, cpu_api());

static(s, "uint:16");
.
.
cpu.read_value("S1", s);
.
.

```

the CPU code should contain something like

```

#include "prog_fpga.h"
.
.
fpga_t *fpga = NULL;
int d;
.
.
fpga_open(portName);
.
.
d = fpga_S1_read(fpga);
.

```

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure, The CPU code in Python3 should contain something like

```

import sys
sys.path.append("/Users/...../install/3pl")
import info2py
import prog_fpga as fpga

fpga_open()

fpga = info2py.info_from_file("prog") # load info from prog.info file

d = fpga.P1_read()

```

Queue mode

Parameter **in** is the queue-mode input variable.

The queue must be written somewhere in the program (queues without sources result in a fatal error on completion of immediate execution). When the module is created the argument queue must have an established write clock domain or a fatal error is generated.

If the input queue is read anywhere else in the program (i.e. other than in this module) then it must be read in the bus clock domain since that is the clock domain in which it is read in the module created by this procedure.

If optional parameter **wordcount** has an argument and it is true, the number of words in the queue buffer can be interrogated by the CPU. The input queue may have any of the allowed buffer sizes. If the queue is asynchronous, i.e. the read domain is not the bus clock domain, the count will not be exact but will be a guaranteed minimum number of readable data.

If optional parameter **wordcount** has an argument and it is false, the data returned from reading the number of words in the buffer is 0 or 1 instead of a count. A return of 1 means the queue buffer is not

empty, i.e. can be read. This option is provided to simplify the logic where a simple empty/non-empty status is required.

If optional parameter **wordcount** has no argument then there will no interface to directly read the number of words in the queue buffer, however an interrupt may be used instead.

Note that an empty queue is guaranteed to return all ones to the CPU when read. For most data types this single value will usually be invalid and hence in that case it can be used as an indication of an empty queue. This can be used instead of explicitly reading the available word count.

If an argument to optional parameter **ilevel** is provided then a level interrupt on queue buffer status is established as follows -

If there is no argument then no interrupt is provided.

If the argument is type "log" with value false then the interrupt will be asserted when the queue buffer is not empty.

If the argument is type "log" with value true then the interrupt will be asserted when the number of words in the buffer equals or exceeds a level which has been written to the interface from the CPU. The initial value of this level is 1. A level value of 0 should not be written to the interface as that will result in a continuous level interrupt regardless of queue state.

If the argument is type "uint" or "int" then the interrupt will be asserted when the number of words in the buffer equals or exceeds the fixed value specified by the integer argument.

The optional output parameter **count** provides a count of the number of data in the queue. The inbuilt function `queuewords()` can only be used in a module which reads the queue, which is usually not the case where `cpu_api.read_queue()` is the queue destination. **count** must have clock domain **bus_clk**. If it has no clock domain, `cpu_api.read_queue()` will set it to **bus_clk**. The width of **count** must be sufficient to hold the input queue depth.

If optional output parameter **out** has an argument, each datum read from the input queue **in** to the CPU local bus is also written to queue **out**. The type of **out** must be such that it can be assigned from **in**. **out** should not be allowed to block since it cannot stop data being popped from **in** to the CPU bus. While **out** is unavailable for writing, no attempt is made to write data popped from **in** to it.

If optional output parameter **access** has an argument, the static variable of type "log" is assigned true for one clock cycle every time a datum is popped from the input queue **in**. **access** may have any clock domain.

On running info2c on the .json file the following C functions will have been defined for interface *name* -

function	when	
<code>fpga_name.read()</code>	always	read a datum
<code>fpga_name.avail.read()</code>	wordcount has argument	read number of words
<code>fpga_name.avail.thresh.write()</code>	ilevel "log" true	set interrupt threshold
<code>fpga_name.enable()</code>	ilevel has argument	enable interrupt
<code>fpga_name.disable()</code>	ilevel has argument	disable interrupt
<code>fpga_name.wait()</code>	ilevel has argument	wait on interrupt

Example:

If 3PL source file `prog.3pl` contains the following

```
class(cpu, cpu_api());

queue(p, "uint:16");
.
.
cpu.read("P1", p, wordcount=false, ilevel=false);
.
.
```

the CPU code in C should contain something like

```
#include "prog_fpga.h"
.
.
fpga_t *fpga = NULL;
int d[...];
int i = 0;
.
.
fpga_open();
.
.
fpga_P1_event_wait(fpga);
while (fpga_P1_avail_read(fpga) != 0)
    d[i++] = fpga_P1_read(fpga);
.
```

The code will wait until the queue p goes from empty to non-empty when an interrupt will release the wait. A loop while the queue remains non-empty reads until the queue is empty. Note that since **wordcount** is given an argument but the value is false, the count will only have values 0/1 for empty/not-empty.

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

The CPU code in Python3 should contain something like

```
import sys
sys.path.append("/Users/...../install/3pl")
import info2py
from cpu_serial import *

fpga_open()

fpga = info2py.info_from_file("prog") # load info from prog.info file

d = []
fpga.P1_wait()
while (fpga.P1_avail_read() != 0)
    d.append = fpga.P1_read()
```

Note that in the unusual case where a queue variable is both read and written by the CPU and is not otherwise accessed in the FPGA the input and output clock domains of the queue will be undefined. To overcome this the clock domain can be set when the queue is created, e.g.

```
queue(q, "uint10", domain=null);
```

which will set both the input (write) and output (read) clock domains to the current clock domain - see [section 9.57](#) p234).

cmemory or rmemory mode

Parameter **in** is the cmemory-mode or rmemory-mode input variable.

If optional output parameter **access** has an argument, the static variable of type "log" is assigned true for one clock cycle every time the port of **in** is read. **access** may have any clock domain.

In the case of Zynq the interface uses AXI bus GP0. The size of the input data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the input data type must not exceed 1016 bits (127 bytes).

This interface can use any memory port and that port can also be used by other application code, but no arbitration of shared access via a port is provided. Coincident access will usually result in data corruption. If it is not feasible to dedicate a port to CPU access then the CPU application code must limit access to the memory to times when it is known that the FPGA logic will not also be accessing the memory port.

On running info2c on the .json file the following C function will have been defined for interface *name* -

function	
fpga_name_read()	read a datum from FPGA memory

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api());
rmemory(m, 2, "uint:8", "uint:16");
.
.
cpu.read_memory("MEM1", m, 0);
.
.
```

then for zynq the CPU code should contain something like

```
#include "prog.h"
.
.
fpga_t *fpga = NULL;
int d;
.
.
fpga_open();
.
.
d = fpga_MEM1_read(addr);
.
```

where addr is the memory address. The returned data is assigned to variable **d**.

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

11.11.2.3 CPU_API.write - a CPU-writable interface

Library: xilinx/cpu.3pl

Prototype:

```
procedure
write (
    var(out),
    varargs
    <-
    str(name),
    varargs
) {
```

Output parameters:

out is the target variable to be written to

varargs optional output parameter **access**

Input parameters:

name is a string identifier for the interface.

varargs optional input parameters **freecount**, **ilevel**, **port** and **reset**.

In-line target execution:

no

description:

This procedure provides a write interface for the the CPU for 3PL variables of mode static, queue, cmemory or rmemory.

In the case of Zynq the interface uses AXI bus GP0. The size of the input data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the input data type must not exceed 1016 bits (127 bytes).

The module is created at the end of immediate execution from arguments queued by this procedure. It can be any type, primitive or compound.

It is a common wrapper for the procedures

CPU_API.write_static() [subsubsection 11.11.2.10](#) p481

CPU_API.write_queue() [subsubsection 11.11.2.9](#) p479

CPU_API.write_memory() [subsubsection 11.11.2.8](#) p477

Parameter **name** is a string identifier that can be used in the CPU code to refer to this interface.

Parameter **out** is the output variable to be written to by the CPU.

The only optional output parameter is **access** which must have a static mode argument of type "log". If this optional parameter is given an argument then parameter **out** may be omitted. In that case the write serves only to generate a single cycle pulse to the **access** argument in the FPGA.

Optional input varargs parameters are -

	input varargs
static	
queue	freecount ilevel reset
cmemory rmemory	port

Static mode

Parameter **out** is the static-mode output variable. Its argument may be omitted if the optional parameter **access** is given an argument.

In the case of Zynq the interface uses AXI bus GP0. The size of the output data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the output data type is not limited.

Output parameter **out** is examined in order to determine its clock domain. If it has no clock domain it is given the bus clock domain.

On running info2c on the .json file the following C function will have been defined for interface *name* -

function	
fpga_name_write()	write a datum

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api());

static(s, "uint:16");
.
.
cpu.write_static(s <- "S2");
.
.
```

then for zynq the CPU code should contain something like

```
#include "prog.h"
.
.
int    d;
.
.
fpga_open();
.
.
fpga_S2_write(d);      // write
.
```

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

Queue mode

In the case of Zynq the interface uses AXI bus GP0. The size of the output data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the output data type is not limited.

The queue must be read somewhere in the program (queues without sinks result in a fatal error on completion of immediate execution). When the module is created the argument queue must have an established read clock domain or a fatal error is generated.

If the output queue is written anywhere else in the program (i.e. other than in this module) then it must be written in the bus clock domain (c100) since that is the clock domain in which it is written in this module.

If the argument to **freecount** is true the number of free entries in the queue buffer can be interrogated by the CPU. The output queue may have any of the allowed buffer sizes. If the queue is asynchronous, i.e. the write domain is not the bus clock domain (c100), the count will not be exact but will be a guaranteed minimum number of writable spaces.

If the argument to **freecount** is false the data returned from reading the number of free spaces in the buffer is 0 or 1 instead of a count. A return of 1 means the queue buffer is not full, i.e. can be written. This option is provided to simplify the logic where a simple full/non-full status is required.

If **freecount** has no argument then there will no interface to directly read the space remaining in the queue buffer, however an interrupt may be used instead.

Arguments to parameter **ilevel** control provision of a level interrupt on queue buffer status as follows -

If there is no argument then no interrupt is provided.

If the argument is type "log" with value false then the interrupt will be asserted when the queue buffer is not full.

If the argument is type "log" with value true then the interrupt will be asserted when the number of free spaces in the buffer equals or exceeds a level which has been written to the interface from the CPU. The initial value of this level is 1. A level value of 0 should not be written to the interface as that will result in a continuous level interrupt regardless of queue state.

If the argument is type "uint" or "int" then the interrupt will be asserted when the number of free spaces in the buffer equals or exceeds the fixed value specified by the integer argument. Again this should not be 0.

If parameter **reset** has an argument it provides a means of resetting (clearing) the output queue. The argument may have a different clock domain to that of the write side of the queue and hence this may be more convenient than directly resetting the queue where a change of clock domain may need to be explicitly provided.

Note that a write to a queue variable from **cpu_api.write_queue()** (or indeed from anywhere) can be detected within the FPGA by means of the inbuilt function **waswritten()** [section 10.169](#) p347. When using this function the clock domain must be the same as that of the queue write, i.e. **bus_clk**. Parameter **out** is the queue-mode output variable.

On running info2c on the .json file the following C functions will have been defined for interface *name* -

function	when	
<code>fpga_name_write()</code>	always	write a datum
<code>fpga_name_avail_read()</code>	freecount has argument	read number of spaces free
<code>fpga_name_avail_thresh_write()</code>	ilevel "log" true	set interrupt threshold
<code>fpga_name_enable()</code>	ilevel has argument	enable interrupt
<code>fpga_name_disable()</code>	ilevel has argument	disable interrupt
<code>fpga_name_wait()</code>	ilevel has argument	wait on interrupt

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api());
queue(p, "uint:16");
.
.
cpu.write_queue(p <- "P2",, true);
.
.
```

then for zynq the CPU code should contain something like

```
#include "prog_fpga.h"
.
.
fpga_t *fpga = NULL;
int d[...];
int i = 0;
int n = sizeof(d) / sizeof int;
.
.
fpga_open(portName, INTERRUPTS);
.
.
fpga_P2_avail_thresh_write(fpga, n); // set interrupt level
```

```

fpga_P2_wait(fpga);           // wait for interrupt
for (i=0 ; i<n ; i++) {      // transmit data block
    fpga_P2_write(fpga, d[i]); // write item
    fpga_P2_enable(fpga);      // enable interrupt
    .

```

or perhaps -

```

class(cpu, cpu_api());
queue(p, "uint:16");
.
.
cpu.write_queue(p <- "P2", false);
.
.

```

then for zynq the CPU code should contain something like

```

#include "prog_fpga.h"
.
.
fpga_t *fpga = NULL;
int      d[...];
int      i = 0;
int      n = sizeof(d) / sizeof int;
.
.
fpga_open(portName, INTERRUPTS);
.
.
for ( i=0 ; i<n ; i++) {      // transmit data block
    while (fpga_P2_aval(fpga) == 0) // poll until not full
        ;
    fpga_P2_write(fpga, d[i]);    // write item
}
.

```

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

The support CPU software for the UART CPU interface is still under development, but application code will be similar to the above.

cmemory or rmemory mode

Parameter **out** is the cmemory or rmemory mode output variable.

In the case of Zynq the interface uses AXI bus GP0. The size of the output data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the output data type is not limited.

This interface can use any memory port and that port can also be used by other application code, but no arbitration of shared access via a port is provided. Coincident access will usually result in data corruption. If it is not feasible to dedicate a port to CPU access then the CPU application code must limit access to the memory to times when it is known that the FPGA logic will not also be accessing the memory port.

On running info2c on the .json file the following C function will have been defined for interface *name* -

function	
fpga_name_write()	write a datum to FPGA memory

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api())
rmemory(m, 2, "uint:8", "uint:16");
.
.
cpu.write_memory(m <- "MEM2", 1);
.
.
```

then for zynq the CPU code should contain something like

```
#include "prog.h"
.
.
fpga_t *fpga = NULL;
int d;
.
.
fpga_open(portName, INTERRUPTS);
.
.
fpga_MEM2_write(fpga, addr, d);
.
```

where addr is the memory addressd the data to be written.

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

11.11.2.4 CPU_API.read_memory - a CPU-readable interface to a memory

Library: xilinx/cpu.3pl

Prototype:

```
global procedure
cpu_api.read_memory (
    static(access, "log")
    <-
    str(name),
    var(mem),
    int(port)
) {}
```

Output parameters:

access is an optional access indicator.

Input parameters:

name is a string identifier for the interface.

mem is the cmemory or rmemory variable.

port is memory port, 0 or 1, to be used for the read.

In-line target execution:

no

description:

This procedure creates a module which allows the CPU to read from a registered output memory (rmemory) or a CLB memory (cmemory) via the GP1 AXI bus.

This procedure can be called via the more cryptic and general procedure **CPU_API.read()** ?? p?? instead.

Parameter **in** is the cmemory-mode or rmemory-mode input variable.

If optional output parameter **access** has an argument, the static variable of type "log" is assigned true for one clock cycle every time the port of **in** is read. **access** may have any clock domain.

In the case of Zynq the interface uses AXI bus GP0. The size of the input data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the input data type must not exceed 1016 bits (127 bytes).

This interface can use any memory port and that port can also be used by other application code, but no arbitration of shared access via a port is provided. Coincident access will usually result in data corruption. If it is not feasible to dedicate a port to CPU access then the CPU application code must limit access to the memory to times when it is known that the FPGA logic will not also be accessing the memory port.

On running info2c on the .json file the following C function will have been defined for interface *name* -

function	
fpga_name_read()	read a datum from FPGA memory

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api());
rmemory(m, 2, "uint:8", "uint:16");
.
.
cpu.read_memory("MEM1", m, 0);
.
.
```

then for zynq the CPU code should contain something like

```
#include "prog.h"
.
.
fpga_t *fpga = NULL;
int d;
.
.
fpga_open();
.
.
d = fpga_MEM1_read(addr);
.
```

where addr is the memory address. The returned data is assigned to variable **d**.

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

See also -

CPU_API.read() ?? p??

CPU_API.write() ?? p??

CPU_API.read_value() [subsection 11.11.2.6](#) p474
CPU_API.write_static() [subsection 11.11.2.10](#) p481
CPU_API.read_queue() [subsection 11.11.2.5](#) p471
CPU_API.write_queue() [subsection 11.11.2.9](#) p479
CPU_API.write_memory() [subsection 11.11.2.8](#) p477
CPU_API.dma_write() [subsection 11.10.1.4](#) p451
CPU_API.dma_read() [subsection 11.10.1.3](#) p450

11.11.2.5 CPU_API.read_queue - a CPU read from a queue variable

Library: xilinx/cpu.3pl

Prototype:

```

global procedure
cpu_api.read_queue (
  static(access, "log"),
  static(count),
  queue(out)
  <-
  str(name),
  queue(in),
  log(wordcount),
  var(ilevel)
) {}

```

Output parameters:

access is an optional access indicator.

count is an optional queue occupancy count.

out is an optional output queue.

Input parameters:

name is a string identifier for the interface.

in is the input queue.

wordcount is an optional flag to enable the number of words available in the queue to be determined.

ilevel is an optional argument to allow the queue buffer level at which an interrupt occurs to be set.

In-line target execution:

no

description:

This procedure creates a module which is an interface to allow the CPU to read a queue variable. The module is created at the end of immediate execution from arguments queued by this procedure. The queue can be any type, primitive or compound. When the CPU reads from the queue a datum is removed from the queue buffer. If the CPU reads from the specified address when the queue is empty the read will return all ones .

This procedure can be called via the more cryptic and general procedure **CPU_API.read()** ?? p?? instead.

In the case of Zynq the interface uses AXI bus GP0. The size of the input data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the input data type must not exceed 1016 bits (127 bytes).

Parameter **in** is the queue-mode input variable.

The queue must be written somewhere in the program (queues without sources result in a fatal error on completion of immediate execution). When the module is created the argument queue must have an established write clock domain or a fatal error is generated.

If the input queue is read anywhere else in the program (i.e. other than in this module) then it must be read in the bus clock domain since that is the clock domain in which it is read in the module created by this procedure.

If optional parameter **wordcount** has an argument and it is true, the number of words in the queue buffer can be interrogated by the CPU. The input queue may have any of the allowed buffer sizes. If the queue is asynchronous, i.e. the read domain is not the bus clock domain, the count will not be exact but will be a guaranteed minimum number of readable data.

If optional parameter **wordcount** has an argument and it is false, the data returned from reading the number of words in the buffer is 0 or 1 instead of a count. A return of 1 means the queue buffer is not empty, i.e. can be read. This option is provided to simplify the logic where a simple empty/non-empty status is required.

If optional parameter **wordcount** has no argument then there will no interface to directly read the number of words in the queue buffer, however an interrupt may be used instead.

Note that an empty queue is guaranteed to return all ones to the CPU when read. For most data types this single value will usually be invalid and hence in that case it can be used as an indication of an empty queue. This can be used instead of explicitly reading the available word count.

If an argument to optional parameter **ilevel** is provided then a level interrupt on queue buffer status is established as follows -

If there is no argument then no interrupt is provided.

If the argument is type "log" with value false then the interrupt will be asserted when the queue buffer is not empty.

If the argument is type "log" with value true then the interrupt will be asserted when the number of words in the buffer equals or exceeds a level which has been written to the interface from the CPU. The initial value of this level is 1. A level value of 0 should not be written to the interface as that will result in a continuous level interrupt regardless of queue state.

If the argument is type "uint" or "int" then the interrupt will be asserted when the number of words in the buffer equals or exceeds the fixed value specified by the integer argument.

The optional output parameter **count** provides a count of the number of data in the queue. The inbuilt function `queuewords()` can only be used in a module which reads the queue, which is usually not the case where `cpu_api.read_queue()` is the queue destination. **count** must have clock domain **bus_clk**. If it has no clock domain, `cpu_api.read_queue()` will set it to **bus_clk**. The width of **count** must be sufficient to hold the input queue depth.

If optional output parameter **out** has an argument, each datum read from the input queue **in** to the CPU local bus is also written to queue **out**. The type of **out** must be such that it can be assigned from **in**. **out** should not be allowed to block since it cannot stop data being popped from **in** to the CPU bus. While **out** is unavailable for writing, no attempt is made to write data popped from **in** to it.

If optional output parameter **access** has an argument, the static variable of type "log" is assigned true for one clock cycle every time a datum is popped from the input queue **in**. **access** may have any clock domain.

On running `info2c` on the .json file the following C functions will have been defined for interface *name* -

function	when	
<code>fpga_name_read()</code>	always	read a datum
<code>fpga_name_avail_read()</code>	wordcount has argument	read number of words
<code>fpga_name_avail_thresh_write()</code>	ilevel "log" true	set interrupt threshold
<code>fpga_name_enable()</code>	ilevel has argument	enable interrupt
<code>fpga_name_disable()</code>	ilevel has argument	disable interrupt
<code>fpga_name_wait()</code>	ilevel has argument	wait on interrupt

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api());

queue(p, "uint:16");

.
.
cpu.read("P1", p, wordcount=false, ilevel=false);
.
.
```

the CPU code in C should contain something like

```
#include "prog_fpga.h"

.
.
fpga_t *fpga = NULL;
int d[...];
int i = 0;

.
.
fpga_open();

.
.
fpga_P1_event_wait(fpga);
while (fpga_P1_avail_read(fpga) != 0)
    d[i++] = fpga_P1_read(fpga);

.
```

The code will wait until the queue p goes from empty to non-empty when an interrupt will release the wait. A loop while the queue remains non-empty reads until the queue is empty. Note that since **wordcount** is given an argument but the value is false, the count will only have values 0/1 for empty/not-empty.

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

The CPU code in Python3 should contain something like

```
import sys
sys.path.append("/Users/...../install/3pl")
import info2py
from cpu_serial import *

fpga_open()

fpga = info2py.info_from_file("prog") # load info from prog.info file

d = []
fpga.P1_wait()
while (fpga.P1_avail_read() != 0)
    d.append = fpga.P1_read()
```

Note that in the unusual case where a queue variable is both read and written by the CPU and is not otherwise accessed in the FPGA the input and output clock domains of the queue will be undefined. To overcome this the clock domain can be set when the queue is created, e.g.

```
queue(q, "uint10", domain=null);
```

which will set both the input (write) and output (read) clock domains to the current clock domain - see [section 9.57](#) p234).

See also -

CPU_API.read() ?? p??
CPU_API.write() ?? p??
CPU_API.read_value() [subsection 11.11.2.6](#) p474
CPU_API.write_static() [subsection 11.11.2.10](#) p481
CPU_API.write_queue() [subsection 11.11.2.9](#) p479
CPU_API.read_memory() [subsection 11.11.2.4](#) p469
CPU_API.write_memory() [subsection 11.11.2.8](#) p477
CPU_API.dma.write() [subsection 11.10.1.4](#) p451
CPU_API.dma.read() [subsection 11.10.1.3](#) p450

11.11.2.6 CPU_API.read_value - CPU read from a value

Parameter **in** is the queue-mode input variable.

Library: xilinx/cpu.3pl

Prototype:

```
global procedure
cpu_api.read_value (
  static(access, "log")
  <-
  str(name),
  value(in)
) {}
```

Output parameters:

access is an optional static logical variable which is true for one clock cycle following a CPU read.

Input parameters:

name is a symbolic identifier string for the bus object.

in is a value to be read by the CPU. It takes its type from the argument. It must not contain queue reads. It may have any clock domain.

In-line target execution:

no

description:

This procedure creates a module which provides a CPU-readable interface to a value.

Parameter **name** is a string identifier that can be used in the CPU code to refer to this interface.

Parameter **in** is the value-mode input variable or expression. Since it is value mode the argument may be a value mode variable, static mode variable or an expression containing these. An expression must not contain a queue read.

The module is created at the end of immediate execution from arguments queued by this procedure. It can be any type, primitive or compound. The optional output static logical variable **access** goes true for one clock cycle to indicate that a read has occurred.

This procedure can be called via the more cryptic and general procedure **cpu_api.read()** ?? p?? instead.

In the case of Zynq the interface uses AXI bus GP0. The size of the input data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the input data type must not exceed 1016 bits (127 bytes).

The optional output static logical variable **access** goes true for one clock cycle to indicate that a read has occurred.

The clock domain of output **access** need not be the same as that of **in**. If **access** has no clock domain it will be given the bus clock domain.

The parameters associated with this bus interface, including an allocated bus address, are written to the *design.info* file.

The utility info2c is used to process this file and create an include file *design.h* which contains a function definition to access this interface. for example the 3PL code

```
cpu_read_value("alpha", v);
```

will result in info2c producing a declaration like

```
uint32_t alpha_read (fpga_t * fpga) {}
```

When the CPU reads a value it is sampling that value, hence data may be lost if the sampling rate is lower than the rate at which data changes. The data sampling logic ensures that the data is a valid sample even when the source is changing at the time of the read. This is a primitive interface which depends on additional protocols being imposed by the application code. For example to guarantee data integrity a write to register the data in a static variable prior to reading could be used. For a more complex interface see **cpu_api.read_queue()** [subsubsection 11.11.2.5](#) p471.

On running info2c on the .json file the following C function will have been defined for interface *name* -

function	
fpga_name_read()	read a datum

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api());

static(s, "uint:16");
.
.
cpu.read_value("S1", s);
.
.
```

the CPU code should contain something like

```
#include "prog_fpga.h"
.
.
fpga_t *fpga = NULL;
int d;
.
.
fpga_open(portName);
.
.
d = fpga_S1_read(fpga);
.
```

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure, The CPU code in Python3 should contain something like

```
import sys
sys.path.append("/Users/...../install/3pl")
```

```

import info2py
import prog_fpga as fpga

fpga_open()

fpga = info2py.info_from_file("prog")    # load info from prog.info file

d = fpga.P1_read()

```

See also -

CPU_API.read() ?? p??
CPU_API.write() ?? p??
CPU_API.write_static() [subsection 11.11.2.10](#) p481
CPU_API.read_queue() [subsection 11.11.2.5](#) p471
CPU_API.write_queue() [subsection 11.11.2.9](#) p479
CPU_API.read_memory() [subsection 11.11.2.4](#) p469
CPU_API.write_memory() [subsection 11.11.2.8](#) p477
CPU_API.dma_write() [subsection 11.10.1.4](#) p451
CPU_API.dma_read() [subsection 11.10.1.3](#) p450

11.11.2.7 CPU_API.receive_event - interrupt the CPU

Library: xilinx/cpu.3pl

Prototype:

```

global procedure
cpu_receive_event (value(sig, "log"), name) {}

```

Output parameters:

none

Input parameters:

sig is an interrupt signal

name is the interrupt event identifier string.

In-line target execution:

no

description:

This procedure connects a target variable of type log to the interrupt logic to drive the CPU interrupt on execution. The signal may be any clock domain and may be true for any period, the false to true transition causing the interrupt.

This procedure can be called more than once for the same event number, the signals from the instances being ORed. Parameter **in** is the queue-mode input variable. Because the signals are ORed a long signal may mask several shorter ones so the interrupt signal should be kept short.

On running info2c on the .json file the following C functions will have been defined for interface *name* -

function	
fpga_name.enable()	enable interrupt
fpga_name.disable()	disable interrupt
fpga_name.wait()	wait on interrupt

Example:

If 3PL source file prog.3pl contains the following


```

value(sig, "log");

cpu_receive_event("EV1", sig);
.
.
assert(sig <-);
.

```

then for zynq the CPU code should contain something like

```

#include "prog_fpga.h"
.
.
fpga_t *fpga = NULL;
.
.
fpga_open(portName, INTERRUPTS);
.
.
fpga_EV1_wait(fpga);
.

```

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

The support CPU software for the UART CPU interface is still under development, but application code will be similar to the above.

See also -

interrupt() Parameter **in** is the queue-mode input variable. [subsection 11.11.2.11](#) p483

11.11.2.8 CPU_API.write_memory - a CPU-writable interface to a memory

Library: xilinx/cpu.3pl

Prototype:

```

global procedure
cpu_write_memory (
    var(mem)
    <-
    str(name),
    int(port)
) {}

```

Output parameters:

mem is the cmemory or rmemory variable.

Input parameters:

name is a string identifier for the interface.

port is memory port, 0 or 1, to be used for the write. For cmemory port 1 is read-only.

In-line target execution:

no

description:

This procedure creates a module which allows the CPU to write to a registered output memory (rmemory) or a CLB memory (cmemory) via the AXI GP1 bus.

This procedure can be called via the more cryptic and general procedure **CPU_API.write()** ?? p?? instead.

Parameter **out** is the cmemory or rmemory mode output variable.

In the case of Zynq the interface uses AXI bus GP0. The size of the output data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the output data type is not limited.

This interface can use any memory port and that port can also be used by other application code, but no arbitration of shared access via a port is provided. Coincident access will usually result in data corruption. If it is not feasible to dedicate a port to CPU access then the CPU application code must limit access to the memory to times when it is known that the FPGA logic will not also be accessing the memory port.

On running info2c on the .json file the following C function will have been defined for interface *name* -

function	
fpga_name_write()	write a datum to FPGA memory

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api())
rmemory(m, 2, "uint:8", "uint:16");
.
.
cpu.write_memory(m <- "MEM2", 1);
.
.
```

then for zynq the CPU code should contain something like

```
#include "prog.h"
.
.
fpga_t *fpga = NULL;
int d;
.
.
fpga_open(portName, INTERRUPTS);
.
.
fpga_MEM2_write(fpga, addr, d);
.
```

where addr is the memory addressd the data to be written.

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

See also -

CPU_API.read() ?? p??

CPU_API.write() ?? p??

CPU_API.read.value() [subsection 11.11.2.6](#) p474

CPU_API.write_static() [subsection 11.11.2.10](#) p481

CPU_API.read.queue() [subsection 11.11.2.5](#) p471

CPU_API.write.queue() [subsection 11.11.2.9](#) p479

CPU_API.read.memory() [subsection 11.11.2.4](#) p469

CPU_API.dma.write() [subsection 11.10.1.4](#) p451

CPU_API.dma.read() [subsection 11.10.1.3](#) p450

11.11.2.9 CPU_API.write_queue - a CPU write to a queue variable

Library: xilinx/cpu.3pl

Prototype:

```
global procedure
cpu_write_queue (
    queue(out)
    <-
    str(name),
    log(freecount),
    var(ilevel),
    value(reset, "log")
) {}
```

Output parameters:

out is the queue written to from the CPU. It takes its type from the argument.

Input parameters:

name is a string identifier for the interface.

freecount is an optional flag to enable the number of free spaces available in the queue to be determined.

ilevel is an optional argument to allow the queue buffer level at which an interrupt occurs to be set.

reset is an optional queue reset.

In-line target execution:

no

description:

This procedure creates a module which writes to a queue variable from the CPU. The module is created at the end of immediate execution from arguments queued by this procedure. The queue can be any type, primitive or compound (including type null). When the CPU writes to the bus data address given by **daddr** a datum is appended to the queue buffer. If the CPU writes to the specified address when the queue is full the data is lost.

This procedure can be called via the more cryptic and general procedure **CPU_API.write()** ?? p?? instead.

In the case of Zynq the interface uses AXI bus GP0. The size of the output data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the output data type is not limited.

The queue must be read somewhere in the program (queues without sinks result in a fatal error on completion of immediate execution). When the module is created the argument queue must have an established read clock domain or a fatal error is generated.

If the output queue is written anywhere else in the program (i.e. other than in this module) then it must be written in the bus clock domain (c100) since that is the clock domain in which it is written in this module.

If the argument to **freecount** is true the number of free entries in the queue buffer can be interrogated by the CPU. The output queue may have any of the allowed buffer sizes. If the queue is asynchronous, i.e. the write domain is not the bus clock domain (c100), the count will not be exact but will be a guaranteed minimum number of writable spaces.

If the argument to **freecount** is false the data returned from reading the number of free spaces in the buffer is 0 or 1 instead of a count. A return of 1 means the queue buffer is not full, i.e. can be written. This option is provided to simplify the logic where a simple full/non-full status is required.

If **freecount** has no argument then there will no interface to directly read the space remaining in the queue buffer, however an interrupt may be used instead.

Arguments to parameter **ilevel** control provision of a level interrupt on queue buffer status as follows -

If there is no argument then no interrupt is provided.

If the argument is type "log" with value false then the interrupt will be asserted when the queue buffer is not full.

If the argument is type "log" with value true then the interrupt will be asserted when the number of free spaces in the buffer equals or exceeds a level which has been written to the interface from the CPU. The initial value of this level is 1. A level value of 0 should not be written to the interface as that will result in a continuous level interrupt regardless of queue state.

If the argument is type "uint" or "int" then the interrupt will be asserted when the number of free spaces in the buffer equals or exceeds the fixed value specified by the integer argument. Again this should not be 0.

If parameter **reset** has an argument it provides a means of resetting (clearing) the output queue. The argument may have a different clock domain to that of the write side of the queue and hence this may be more convenient than directly resetting the queue where a change of clock domain may need to be explicitly provided.

Note that a write to a queue variable from **cpu_api.write_queue()** (or indeed from anywhere) can be detected within the FPGA by means of the inbuilt function **waswritten()** [section 10.169](#) p347. When using this function the clock domain must be the same as that of the queue write, i.e. **bus_clk**. Parameter **out** is the queue-mode output variable.

On running info2c on the .json file the following C functions will have been defined for interface *name* -

function	when	
<code>fpga_name_write()</code>	always	write a datum
<code>fpga_name_avail_read()</code>	freecount has argument	read number of spaces free
<code>fpga_name_avail_thresh_write()</code>	ilevel "log" true	set interrupt threshold
<code>fpga_name_enable()</code>	ilevel has argument	enable interrupt
<code>fpga_name_disable()</code>	ilevel has argument	disable interrupt
<code>fpga_name_wait()</code>	ilevel has argument	wait on interrupt

Example:

If 3PL source file prog.3pl contains the following

```
class(cpu, cpu_api());
queue(p, "uint:16");
.
.
cpu.write_queue(p <- "P2",, true);
.
.
```

then for zynq the CPU code should contain something like

```
#include "prog_fpga.h"
.
.
fpga_t *fpga = NULL;
int d[...];
int i = 0;
int n = sizeof(d) / sizeof int;
.
.
fpga_open(portName, INTERRUPTS);
.
.
fpga_P2_avail_thresh_write(fpga, n); // set interrupt level
```

```

fpga_P2_wait(fpga);           // wait for interrupt
for (i=0 ; i<n ; i++) {      // transmit data block
    fpga_P2_write(fpga, d[i]); // write item
    fpga_P2_enable(fpga);      // enable interrupt
    .

```

or perhaps -

```

class(cpu, cpu_api());
queue(p, "uint:16");
.
.
cpu.write_queue(p <- "P2", false);
.
.

```

then for zynq the CPU code should contain something like

```

#include "prog_fpga.h"
.
.
fpga_t *fpga = NULL;
int     d[...];
int     i = 0;
int     n = sizeof(d) / sizeof int;
.
.
fpga_open(portName, INTERRUPTS);
.
.
for ( i=0 ; i<n ; i++) {      // transmit data block
    while (fpga_P2_availl(fpga) == 0) // poll until not full
        ;
    fpga_P2_write(fpga, d[i]);      // write item
}
.

```

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

The support CPU software for the UART CPU interface is still under development, but application code wil be similar to the above.

See also -

CPU_API.read() ?? p??
CPU_API.write() ?? p??
CPU_API.read_value() [subsubsection 11.11.2.6](#) p474
CPU_API.write_static() [subsubsection 11.11.2.10](#) p481
CPU_API.read_queue() [subsubsection 11.11.2.5](#) p471
CPU_API.read_memory() [subsubsection 11.11.2.4](#) p469
CPU_API.write_memory() [subsubsection 11.11.2.8](#) p477
CPU_API.dma_write() [subsubsection 11.10.1.4](#) p451
CPU_API.dma_read() [subsubsection 11.10.1.3](#) p450

11.11.2.10 CPU_API.write_static - a CPU-writable static variable

Library: xilinx/cpu.3pl

Prototype:

```

global procedure
cpu_write_static (
  static(out)
  static(access, "log")
  <-
  str(name)
) {}

```

Output parameters:

out is the static variable to be written to by the CPU. It takes its type from the argument. The argument may be omitted if only an access signal is required.

access is a static logical variable which is true for one clock cycle following a CPU write to. This argument is optional.

Input parameters:

name is a string identifier for the interface.

In-line target execution:

no

description:

This procedure creates a module which allows the CPU to write to a static variable. The module is created at the end of immediate execution from arguments queued by this procedure. The variable can be any type, primitive or compound.

The optional output parameter **access** must have a static mode argument of type "log". If this optional parameter is given an argument then an argument to parameter **out** may be omitted. In that case the write serves only to generate a single cycle pulse to the **access** argument in the FPGA.

Parameter **out** is the static-mode output variable. Its argument may be omitted if the optional parameter **access** is given an argument.

In the case of Zynq the interface uses AXI bus GP0. The size of the output data type must not exceed 256 bits as the CPU interface software limits burst transfers to 8 32-bit words.

Boards having a serial interface to the CPU use that interface. The size of the output data type is not limited.

Output parameter **out** is examined in order to determine its clock domain. If it has no clock domain it is given the bus clock domain.

On running info2c on the .json file the following C function will have been defined for interface *name* -

function	
fpga_name_write()	write a datum

Example:

If 3PL source file prog.3pl contains the following

```

class(cpu, cpu_api());

static(s, "uint:16");
.
.
cpu.write_static(s <- "S2");
.
.

```

then for zynq the CPU code should contain something like

```

#include "prog.h"
.
.
int    d;
.
.
fpga_open();
.
.
fpga_S2_write(d);      // write
.

```

This procedure creates no in-line target code so the current clock domain when this procedure is called is not relevant. The clock domain is not changed by this procedure,

See also -

CPU_API.read() ?? p??

CPU_API.write() ?? p??

CPU_API.read_value() [subsection 11.11.2.6](#) p474

CPU_API.read_queue() [subsection 11.11.2.5](#) p471

CPU_API.write_queue() [subsection 11.11.2.9](#) p479

CPU_API.read_memory() [subsection 11.11.2.4](#) p469

CPU_API.write_memory() [subsection 11.11.2.8](#) p477

CPU_API.dma_write() [subsection 11.10.1.4](#) p451

CPU_API.dma_read() [subsection 11.10.1.3](#) p450

11.11.2.11 CPU_API.interrupt - interrupt the CPU

Library: xilinx/cpu.3pl

Prototype:

```

global procedure
interrupt (var(v)) {}

```

Output parameters:

none

Input parameters:

v is an event interrupt event string identifier or the interrupt event number, 0 to 15.

In-line target execution:

yes

description:

This procedure causes an interrupt event. It executes in one cycle.

The in-line code for this procedure executes in the current clock domain which is unchanged on exiting the procedure.

On running info2c on the .json file the following C functions will have been defined for interface *name* -

function	
fpga_name.enable()	enable interrupt
fpga_name.disable()	disable interrupt
fpga_name.wait()	wait on interrupt

Example:

If 3PL source file prog.3pl contains the following

```
.  
.   
interrupt("EV2");  
.   
.
```

then
info2c prog
is run, the CPU code should contain something like

```
#include "prog_fpga.h"  
.   
.   
fpga_t *fpga = NULL;  
.   
.   
fpga_open(portName, INTERRUPTS);  
.   
.   
fpga_EV2_wait(fpga);  
.
```

See also
CPU_API.receive_event() [subsubsection 11.11.2.7](#) p476

11.11.3 class library functions

None.

11.12 UART CPU interface Library - xilinx/uart_cpu.3pl

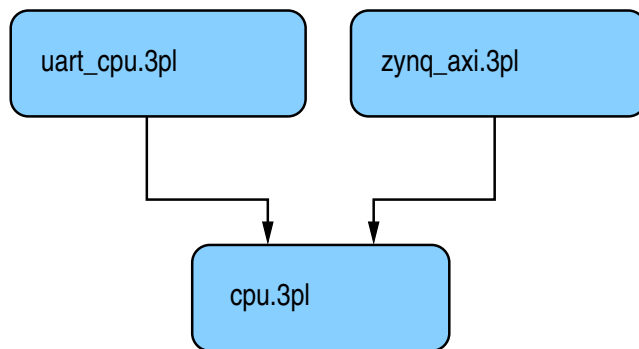
This library provides a CPU interface using a serial interface. This is by an FTDI USB port or similar.

This library must be incorporated by using a **module** statement -

```
module "xilinx/uart_cpu.3pl"
```

serial application

Zynq application



xilinx/uart_cpu.3pl is included in -
 boards/avnet/aes_sp3a_eval400.3pl
 board/gadget_factory/papilio_one_100.3pl
 board/gadget_factory/papilio_one_250.3pl
 board/gadget_factory/papilio_one_500.3pl
 board/gadget_factory/papilio_pro_lx4.3pl
 board/gadget_factory/papilio_pro_lx9.3pl

uart_cpu begins by -

```
source "xilinx/cpu.3pl"           // CPU API classes file
class(cpu, CPU_API());           // Create API class here as used below
global function cpu_api () {     // Pass the API class to application code
    return(cpu);
}
```

Since cpu.3pl is included via a source statement its code shares the scope of this library, UART_CPU. Thus it can call the modules, functions and procedures therein.

cpu.3pl defines a class **CPU_API()** which provides the following class procedures for data transfer between FPGA and CPU -

- read()** generic read [subsubsection 11.11.2.2](#) p459
- write()** generic write [subsubsection 11.11.2.3](#) p464
- read_memory()** memory read [subsubsection 11.11.2.4](#) p469
- read_queue()** queue read [subsubsection 11.11.2.5](#) p471
- read_value()** value read [subsubsection 11.11.2.6](#) p474
- write_memory()** memory write [subsubsection 11.11.2.8](#) p477
- write_queue()** queue write [subsubsection 11.11.2.9](#) p479
- write_static()** static write [subsubsection 11.11.2.10](#) p481
- dma_read()** to FPGA from CPU [subsubsection 11.11.1.2](#) p457
- dma_write()** from FPGA to CPU [subsubsection 11.11.1.3](#) p458
- signal_interrupt()** interrupt from a signal [subsubsection 11.11.1.1](#) p456
- interrupt()** interrupt procedure [subsubsection 11.11.2.1](#) p458

each of which can generate an instance of a CPU interface to a specific 3PL variable of appropriate mode. An internal class **CPU_CODEGEN** is also created but this is not visible to the application code

cpu.3pl is sourced by both **uart_cpu.3pl** and **zynq_axi.3pl** to provide CPU interfaces. The former employs a serial data link and the latter parallel busses.

When any of the procedures above are called a defining structure is created which is added to a list of CPU interfaces. At the end of immediate execution module **create_bus_interfaces()** is called. **create_bus_interfaces()** is defined in both **uart_cpu.3pl** and **zynq_axi.3pl** as it is specific to each. This module traverses the interface list constructed above and for each interface calls a matching procedure in class **CPU_CODEGEN** to generate the logic for each instance.

11.12.1 library modules

11.12.1.1 cpu_interface - create the CPU interface low-level logic

Library: xilinx/uart_cpu.3pl)

Prototype:

```
module
cpu_interface () {}
```

Output parameters:

none.

Input parameters:

none.

description:

This module is called by module **create_bus_interfaces** to generate the low-level interface logic to the CPU.

11.12.1.2 create_bus_interfaces - create the individual CPU interfaces

Library: xilinx/uart_cpu.3pl)

Prototype:

```
module
create_bus_interfaces () {}
```

Output parameters:

none.

Input parameters:

none.

description:

This module creates the logic for each of the CPU interfaces. Each of these communicates via logic created by module **cpu_interface** described above.

11.12.1.3 dma_read - DMA read to the FPGA from CPU RAM

Library: xilinx/uart_cpu.3pl)

Prototype:

```
module
dma_read (
    queue(data)
    <-
    queue(address, "uint:16")    // relative memory address
) {}
```

Output parameters:

data is the data output queue.

Input parameters:

address is a queue supplying the CPU RAM address relative to the DMA base address.

description:

This module performs a DMA read to the FPGA from CPU RAM. The transfer occurs when input queue **address** is not empty and the output queue **data** is not full.

11.12.1.4 dma_write - DMA write from the FPGA to CPU RAM

Library: xilinx/uart_cpu.3pl)

Prototype:

```
module
dma_write (
    queue(data),
    queue(address, "uint:16")    // relative memory address
) {}
```

Output parameters:

none.

Input parameters:

data is the data input queue.

address is a queue supplying the CPU RAM address relative to the DMA base address.

description:

This module performs a DMA write from the FPGA to CPU RAM. The transfer occurs when input queues **address** and **data** are not empty.

11.12.1.5 sigint - interrupt the CPU on a signal

Library: xilinx/uart_cpu.3pl)

Prototype:

```
module
  sigint(int(n), value(sig, "log")) {}
```

Output parameters:

none.

Input parameters:

n is the interrupt index, 0 to 16.

sig is a signal on whose rising edge the interrupt is triggered.

description:

When the signal **sig** is asserted the interrupt is triggered. The interrupt number, 0 to 15, is given by **index**.

11.12.2 library procedures

11.12.2.1 add_interface - add an interface to the interface list

Library: xilinx/uart_cpu.3pl)

Prototype:

```
procedure
  add_interface (var(v, "->")) {}
```

Output parameters:

none.

Input parameters:

v is an instance of class **busCommon**, defined in cpu.3pl.

description:

The instance of **busCommon** is added to the interface list. The list is processed at completion of immediate execution to create the whole CPU interface.

11.12.2.2 cpu_read - write to CPU

Library: xilinx/uart_cpu.3pl)

Prototype:

```
procedure
  cpu_read (
    value(access, "log"),
    value(init, "log"),
    queue(out)
  <-
    int(bus),
    int(addr),
    value(data_in),
    value(contin, "log")
  ) {}
```

Output parameters:

access is an optional output parameter which is asserted when a CPU to FPGA write occurs.

init is an output parameter which is asserted in the case of a memory read to initiate the read.

out is an optional queue variable which will be written to when a transfer occurs.

Input parameters:

bus is not used (applies to Zynq).

address is a 6 bit interface address.

data.in input data.

contin is only used for a memory read and is asserted when the memory read has occurred and the transfer can begin.

description:

This procedure provides a low-level interface to transfer data from the FPGA to the CPU via the USB serial port. It is used by modules and procedures in library `cpu.3pl` and is not intended for use directly by application code.

The CPU initiates the transfer. The transfer may consist of multiple bytes.

11.12.2.3 cpu_write - write to CPU

Library: `xilinx/uart_cpu.3pl`)

Prototype:

```
procedure
cpu_write (
    var(out),
    value(access, "log")
    <-
    int(bus),
    int(addr)
) {}
```

Output parameters:

var is a static or queue mode variable.

access is an optional output parameter which is asserted when a CPU to FPGA write occurs.

Input parameters:

bus is not used (applies to Zynq).

address is a 6 bit interface address.

description:

This procedure provides a low-level interface to transfer data from the CPU to the FPGA via the USB serial port. It is used by modules and procedures in library `cpu.3pl` and is not intended for use directly by application code.

The CPU initiates the transfer. The transfer may consist of multiple bytes.

11.12.3 library functions**11.12.3.1 cpu_api - CPU communications interface**

Library: `xilinx/uart_cpu.3pl`)

Prototype:

```
global function  
cpu_api () {}
```

Return mode and type:

immediate, class

Input parameters:

none.

description:

This function returns an instance of class CPU_API

11.13 The Standard Library - stdlib.3pl

This library contains modules, procedures and functions common to all platforms.

This library must be incorporated by using a **module** statement -

```
module "stdlib.3pl"
```

This library includes **multilib.3pl** and **divlib.3pl**. These are not documented separately.

11.13.1 global structs

Global structs **ds16**, **ds16e** and **ds32** are used for block protocols and for high speed serial channel data.

```
struct(global.ds16,
  data,  "bits:16", // data word
  k,     "log"      // data word contains K-character in MS byte
);
```

```
struct(global.ds16e,
  data,  "bits:16", // data word
  k,     "log",     // data word contains K-character in MS byte
  error, "log"      // data word is erroneous
);
```

```
struct(global.ds32,
  data,  "uint:32", // data word
  k,     "log"      // data word contains K-character in MS byte
);
```

11.13.2 library modules

11.13.2.1 array_to_queue - serialise an array to a queue

Library: stdlib.3pl

Prototype:

```
global module
array_to_queue (
  queue(pout),
  value(busy, "log")
  <-
  value(in),
  value(enable),
  value(go, "log")
);
```

Output parameters:

pout is the output queue variable

busy is true when the module cannot accept another input

Input parameters:

in is the input array.

enable is an optional argument controlling the number of items in **in** to process.
go is asserted to trigger the serialisation.

description:

Parameter **in** is an array. It may be value, static, queue, selectvalue or priority mode.

Parameter **enable** is a one dimensional array of "log" and may be value, static, or selectvalue mode. It must have the same dimension as the first or only dimension of parameter **in**.

When **go** is true the input array is copied to an internal structure which is then shifted out to the output queue **pout**. The **enable** array indicates which members of array **in** are to be copied and shifted, the lowest index false member of **enable** indicating that all members of array **in** from that index up are ignored. Thus array **enable** must have a number of contiguous entries which are true, starting at index 0, and that selects the number of contiguous entries in array **in** which will be copied and shifted to the output queues, again starting with index 0. **go** may contain queue reads.

The second input argument, for parameter **enable**, may be omitted in which case all members of **in** will be shifted and copied to the output queue.

The output queue must be the same type as the input array members. While the serialisation is in progress output value **busy** is true. If input **go** is true when **busy** returns to false another conversion will take place. If the input **in** is a queue then **go** may be held true.

The serialisation takes a number of clock cycles equal to the number of entries in the input array that are enabled plus one and hence the input data rate is limited to $\frac{1}{n+1}$ where n is the number of enabled entries in the input array.

11.13.2.2 crc_8_16 - 16 bit CRC from 8 bit data

Library: stdlib.3pl

Prototype:

```
global module
crc_8_16 (
    static(out, "uint:16")
    <-
    queue(in, "uint:8"),
    str(poly, "CRC16")
) {}
```

Output parameters:

out is the output static variable.

Input parameters:

in is the input queue.

poly is an optional string argument to select the polynomial used.

description:

This module accumulate a 16 bit CRC from 8 bit data. Parameter **in** is the data input queue, type "uint:8". The CRC is updated as each input datum is read. The optional parameter **poly** selects one of two polynomials to use. "CRC16" (default) selects $x^{16} + x^{15} + x^{13} + 1$. "CCITT" selects $x^{16} + x^{12} + x^5 + 1$. parameter **out** is a "uint:16" static variable in which the CRC value is accumulated Input data can be read at clock rate.

11.13.2.3 crc_16_16 - 16 bit CRC from 8 bit data

Library: stdlib.3pl

Prototype:


```

global module
crc_16_16 (
    static(out, "uint:16")
    <-
    queue(in, "uint:16"),
    str(poly, "CRC16")
) {}

```

Output parameters:

out is the output static variable.

Input parameters:

in is the input queue.

poly is an optional string argument to select the polynomial used.

description:

This module accumulate a 16 bit CRC from 16 bit data. Parameter **in** is the data input queue, type "uint:16". The CRC is updated as each input datum is read. The CRC is computed successively from each 8-bit half of each input datum but in one clock cycle. The optional parameter **poly** selects one of two polynomials to use. "CRC16" (default) selects $x^{16} + x^{15} + x^{13} + 1$. "CCITT" selects $x^{16} + x^{12} + x^5 + 1$. parameter **out** is a "uint:16" static variable in which the CRC value is accumulated Input data can be read at clock rate.

11.13.2.4 crcblock - block 32 bit data with a CRC

Library: stdlib.3pl

Prototype:

```

global module
crcblock (
    queue(out, ds16)
    <-
    queue(in, ds32),
    int(max_blk_len, 20),
    int({sob, eob}),
    log(fixed)
) {}

```

Output parameters:

out is the output queue, type ds16.

Input parameters:

in is the input queue, type ds32.

max_blk_len is the maximum variable block size or the fixed block size.

sob is the start-of-block K-character to be used.

eob is the end-of-block K-character to be used.

fixed is an optional logical argument to select fixed length blocks.

description:

This module encloses data in a block containing a cyclic redundancy check word. The 16 bit CRC polynomial is $x^{16} + x^{15} + x^{13} + 1$. Blocks can be fixed or variable length.

When input data become available the start-of-block word is written to the output queue. The start-of-block word has the most significant byte set to the start-of-block K character, the least significant byte to zero and the K flag is true. If 'fixed' is false (default), incoming data is read until either unavailable or max_block_len items have been read and the block is then terminated. If 'fixed' is true, incoming data is read until max_block_len items have been read and the block is then terminated. each 32 bit word is written to the output as a pair of 16 bit words. A 16 bit CRC is accumulated from each 16 bit data word. When the block is terminated the 16 bit CRC is written to the output followed by an end-of-block

word. The end-of-block word has the most significant byte set to the end-of-block K character, the least significant byte to zero and the K flag is true.

K flag	upper byte	lower byte	
1	sob	0	start-of-block
0	data word 1		first data word
0	data word 2		second data word
.	.		
.	.		
0	data word n		last data word
0	CRC		CRC
1	eob	0	end-of-block

11.13.2.5 crcunblock - unblock 32 bit data with a CRC

Library: stdlib.3pl

Prototype:

```
global module
crcunblock (
    queue(dout, ds32),
    static(eout)
    <-
    queue(in, ds16),
    int(max_blk_len, 20),
    int({sob, eob})
) {}
```

Output parameters:

out is the output queue, type ds32.

eout is an optional error counter static variable.

Input parameters:

in is the input queue, type ds16.

max_blk_len is the maximum variable block size or the fixed block size.

sob is the start-of-block K-character to be used.

eob is the end-of-block K-character to be used.

description:

This module unwraps data from a block containing a cyclic redundancy check word. The 16 bit CRC polynomial is $x^{16} + x^{15} + x^{13} + 1$. Blocks can be fixed or variable length. The start-of-block word, CRC word and end-of-block word are stripped off the input stream and data words are packed into 32 bit words (ds32 struct). A CRC is computed from the block data including the transmitted CRC word. If the resulting CRC is zero the block is written to the output queue. If the CRC is not zero the block is discarded.

When a block CRC error is encountered if the output parameter 'eout' has an argument the data length of the discarded block is added to the value of that variable. Any words received that do not match the block format cause 'eout' to be incremented. The size of the 'eout' counter is determined by the argument passed in the call to crcunblock().

If the block length exceeds the maximum block length the block is terminated and discarded and if the optional error counter output parameter is used, the block length is added to the error count.

Unpacked data is written to a dual-port circular buffer so that completed correct data words can be copied to the output data queue in parallel with unpacking of incoming blocks. If the output queue becomes unavailable for writing, data words will be discarded.

11.13.2.6 cread - read combinatorial output memory to a queue

Library: stdlib.3pl

Prototype:

```
global module
cread (queue(data) <- rmemory(m), int(port), queue(addr)) {}
```

Output parameters:

data is the queue variable to be written to from the memory read. It takes its type from the argument.

Input parameters:

m is the memory.

port is the memory port.

addr is the memory address.

description:

Read the combinatorial output memory when the input and output queues are available. The program logic must ensure that this memory port access is mutually exclusive to all other accesses to that port.

This module is just a wrapper around a single statement. The source is

```
global module
cread (queue(data) <- rmemory(m), int(port), queue(addr)) {
    data <<- cast(cmemoryread(m, port, addr), type(data));
}
```

It is provided to complete the set **cread()**, **cwrite()**, **rread()** and **rwrite()** where only **rread()** contains significant code.

See also -

cwrite() [subsubsection 11.13.2.7](#) p495,

rread() [subsubsection 11.13.2.15](#) p502,

rwrite() [subsubsection 11.13.2.16](#) p502.

11.13.2.7 cwrite - write combinatorial output memory from a queue

Library: stdlib.3pl

Prototype:

```
global module
cwrite (rmemory(m), int(port), queue({addr, data})) {}
```

Output parameters:

none

Input parameters:

m is the memory.

port is the memory port.

addr is the memory address.

data is the data to be written.

description:

Write to the combinatorial output memory when the input queues are available. The program logic must ensure that this memory port access is mutually exclusive to all other accesses to that port.

This module is just a wrapper around a single statement. The source is

```
global module
cwrite (rmemory(m), int(port), queue(addr), queue(data)) {
    cmemorywrite(m, port, addr, data);
}
```

It is provided to complete the set `cread()`, `cwrite()`, `rread()` and `rwrite()` where only `rread()` contains significant code.

See also -

`cread()` [subsection 11.13.2.6](#) p495,
`rread()` [subsection 11.13.2.15](#) p502,
`rwrite()` [subsection 11.13.2.16](#) p502.

11.13.2.8 demultiplex - demultiplex a source

demultiplex_l - demultiplex a source using port bit

demultiplex_e - demultiplex a source using port number

Library: `stdlib.3pl`

Prototypes:

```
global module
demultiplex.... (varargs <- queue(in)) {}
```

Output parameters:

varargs any number of output sinks, an array of pointers to sinks or a list of sinks.

Input parameters:

in is a single input queue.

description:

These modules are intended to demultiplex a single input queue to a number of output queues. Outputs need not be the same type but they must be assignable from the input type (or **body** field of the input - see below). The input need not contain queue reads in which case it will be read on every clock cycle.

These modules may also be called with a single output argument which is an array of pointers to queues. In this case they behave as though called with the queues as separate arguments. Similarly the single output argument may be of type "list", the list containing pointers to queues.

The output types must be a type assignable from the input data type. By 'assignable' is meant that the types need not be identical, e.g. the number of bits may differ in which case truncation or padding will occur or a "uint" may be assigned to an "int".

For **demultiplex()** the input can be any type. The input is copied to the outputs in turn. If an output is not available the next available output in cyclic order is selected. If no outputs are available the module waits until any output is available and this is written to.

For **demultiplex_l()** the input type is a wrapper struct containing the datum and an output port identifier. That identifier is an array of type "log" in which one member is **true** to identify the output port. The struct can be created by calling the library global function **multiplex_struct_l()** [subsection 11.13.4.17](#) p519.

For **demultiplex_e()** the input type is a wrapper struct containing the datum and an output port identifier. That identifier is a "uint" containing the port number (starting at 0). The struct can be created by calling the library global function **multiplex_struct_e()** [subsection 11.13.4.17](#) p519.

If the selected output queue is an unmatched parameter then the input is still read but the data is discarded. For **demultiplex_l()** and **demultiplex_e()** the directive "demultiplex_skip_null_outputs" can be set to true to ignore any unmatched output parameters, but note that if an input is directed to that output port the input will not be read.

These modules read and write in the current clock domain. Input and output queues may be asynchronous.

The module **demultiplex()** operates by means of a token placed on one output. When the input becomes available the first available output downstream of the token is written and the token is moved to the output following the one written. There are no clock cycles lost, as in a traditional token ring, in waiting for the token to move around the ring to an available output, thus the module is capable of forwarding an input datum on every clock cycle. The implementation uses the Xilinx carry chain to minimise the delay through the logic cascade of the token ring. If the carry chain were not used the logic delays would severely limit the clock rate at which this module could be used. Modules **demultiplex_l()** and **demultiplex_e()** do not need the token ring as the required output is explicitly specified by the input.

See also -

multiplex() [subsection 11.13.2.11](#) p498
multiplex_l() [subsection 11.13.2.11](#) p498
multiplex_e() [subsection 11.13.2.11](#) p498
multiplexp() [subsection 11.13.2.11](#) p498
multiplexp_l() [subsection 11.13.2.11](#) p498
multiplexp_e() [subsection 11.13.2.11](#) p498
multiplex_t() [subsection 11.13.2.11](#) p498
multiplex_t_l() [subsection 11.13.2.11](#) p498
multiplex_t_e() [subsection 11.13.2.11](#) p498
multiplex_struct_l() [subsection 11.13.4.17](#) p519
multiplex_struct_e() [subsection 11.13.4.17](#) p519.

11.13.2.9 divide - pipeline divide

Library: stdlib.3pl

Prototype:

```
global module
divide (queue(quot), queue(rem) <- value(num), value(denom)) {}
```

Output parameters:

quot is the quotient.

rem is the remainder.

Input parameters:

num is the dividend.

denom is the divisor.

This module is in divlib.3pl which is included in stdlib.3pl.

description:

This module is a pipeline divider. The numerator and denominator arguments may be int, uint, int:, uint:, fixed: or ufixed: where the trailing ":" indicates target types. For integer types the quotient and remainder outputs are integer, signed if either input is signed, and either or both the quotient and remainder outputs may be used. For one or both inputs fixed point the quotient output will be fixed point, signed if either input is signed, and the remainder output is not used.

If neither input is a queue or an expression containing a queue the divider will accept input on every clock cycle. Either input may be immediate mode but both inputs must not be immediate mode. The quotient and remainder will become available at the outputs after a latency which is determined by the numerator width. Data in transit through the module will continue to the outputs regardless of the availability of further input data.

The remainder queue output argument should be at least as wide as the denominator argument otherwise the resulting truncation may lose data (if the range is known to be constrained then this is acceptable).

For unsigned integer divide (both numerator and denominator arguments unsigned target types or non-negative immediate) the result width is the width of the numerator argument.

For signed integer divide (either or both numerator or denominator arguments int target types or negative immediate) the result width will be one bit wider than for unsigned divide. The extra bit is only required for the single case where both numerator and denominator are the maximum negative values (e.g. $-8 / -1$ gives $+8$, but -8 has a width of 4 bits and $+8$ as a signed int has a width of 5 bits). If this case is known not to arise a narrower output queue may be used.

For fixed point divide the result width is the sum of the numerator and denominator widths and the result binary point offset is the numerator offset + denominator width - denominator offset.

The number of pipeline stages is equal to the number of bits in the numerator plus one.

Divide by zero is not detected and gives an undefined result.

11.13.2.10 fifo - a fifo with static interface

Library: stdlib.3pl

Prototype:

```
global module
fifo (
    value(out), value({emptyout, fullout}, "log")
    <-
    value(in), value({read, write}, "log")
) {}
```

Output parameters:

out is the data output for a fifo read.

emptyout is true if the fifo is empty.

fullout is true if the fifo is full.

Input parameters:

in is the data input for a fifo write.

read causes a fifo read on a clock cycle when high.

write causes a fifo write on a clock cycle when high.

description:

Read or write a fifo. On each clock cycle **emptyout** and **fullout** indicate when reads and writes respectively cannot be executed. The program must ensure that these conditions are checked as the module does not do so! If **read** or **write** are high in a clock cycle then a read or write will be performed. These can be done on the same clock cycle.

11.13.2.11 multiplex - multiplex sources

multiplex_l - multiplex sources giving port bit

multiplex_e - multiplex sources giving port number

multiplexp - multiplex queues with priority

multiplexp_l - multiplex queues with priority giving port bit

multiplexp_e - multiplex queues with priority giving port number

multiplex_t - multiplex queues via a tree

multiplex_t_l - multiplex queues via a tree giving port bit

multiplex_t_e - multiplex queues via a tree giving port number

Library: stdlib.3pl

Prototype:

```
global module
multiplex.... (queue(out) <- varargs) {}
```

Output parameters:

out is a single output queue.

Input parameters:

varargs any number of input sources, an array of pointers to sources or a list of sources.

description:

These modules are intended to multiplex a number of input queues to a single output queue. Inputs need not be the same type but they must be assignable to the output type (or **body** field of the output queue - see below). By 'assignable' is meant that the types need not be identical, e.g. the number of bits may differ in which case truncation or padding will occur or a "uint" may be assigned to an "int".

When inputs become available simultaneously modules **multiplex()**, **multiplex.l()** and **multiplex.e()** use a pseudo-token ring to give equal treatment to all inputs. Note that although these modules were written to multiplex queues (or more correctly expressions containing queue reads), the input arguments could be other modes, say value or static. Where no queue reads are involved an expression is always available. In that case the inputs would be sampled and copied to the output queue continuously in a cyclic order. In principle the inputs could be a mixture of modes in which case they would be read cyclically but a queue would be skipped if not available for reading.

Modules **multiplexp()**, **multiplexp.l()** and **multiplexp.e()** handle simultaneous inputs by choosing the left-most available queue in the argument list. For these modules the inputs must contain queue reads.

Modules **multiplext()**, **multiplext.l()** and **multiplext.e()** handle simultaneous inputs by means of a balanced tree structure.

All nine modules may also be called with a single input argument which is an array of pointers to sources. In this case they behave as though called with the sources as separate arguments. Similarly the single input argument may be of type "list", the list containing pointers to sources.

In many applications it is necessary to be able to identify the origin of an output datum, i.e. the input argument to the module from which the datum was read. To facilitate this the module output queue type may be a struct which is a wrapper containing the output datum and an input port identifier. The output type for **multiplex()** and **multiplexp()** do not use a wrapper type, the output type being directly compatible with the input type(s).

The output type for **multiplex.l()**, **multiplexp.l()** and **multiplext.l()** is a wrapper with two field, **body** containing the datum and **port** which is an array of type "log" in which one member is **true** to identify the input port. The struct can be created by calling the library global function **multiplex_struct.l()** [subsubsection 11.13.4.17](#) p519.

The output type for **multiplex.e()**, **multiplexp.e()** and **multiplext.e()** is a wrapper with two fields, **body** containing the datum and **port** which is a "uint" containing the port number (starting at 0). The struct can be created by calling the library global function **multiplex_struct.e()** [subsubsection 11.13.4.17](#) p519.

In the case of **multiplex()**, **multiplexp()** and **multiplext()** the inputs are copied to the output queue when available. For **multiplex.l()**, **multiplexp.l()** and **multiplext.l()** the selected input is copied to the **body** field output queue when available and the associated port field has the member of the array indexed by the port number assigned true. For **multiplex.e()**, **multiplexp.e()** and **multiplext.e()** the same applies except that the port number is assigned to the "uint" **port** field.

The modules read and write in the current clock domain. Input and output queues may be asynchronous.

The modules **multiplex()**, **multiplex.l()** and **multiplex.e()** operate by means of a token placed on one input. When one or more inputs become available the first input downstream of the token is read and the token is moved to the input following the one read. There are no clock cycles lost, as in a traditional token ring, in waiting for the token to move around the ring to an available input, thus the module is capable of forwarding an input datum on every clock cycle. The implementation uses the

Xilinx carry chain to minimise the delay through the logic cascade of the token ring. Unfortunately the carry chain must be circular and the link back from the top of a linear carry chain to the bottom introduces a significant delay. The propagation time around the token ring is large enough to severely lower the clock rate that can be used as the number of inputs increases.

The modules **multiplex()**, **multiplex_l()** and **multiplex_e()** use a priority mode variable to select which input to read. For FPGAs priority mode variables are implemented using the Xilinx carry chain if the number of inputs exceeds the maximum number of LUT inputs. This is to minimise the delay through cascaded logic.

The modules **multiplext()**, **multiplext_l()** and **multiplext_e()** use a balanced tree with nodes of degree 2 to converge the inputs to the output. Each tree node arbitrates between its two inputs based on the number of leaves converging on each branch. The disadvantage of this scheme is that arbitration is based on the fixed configuration, not on the traffic through each branch. For example a four input multiplexer where input 4 is idle will arbitrate among the other three unfairly giving input 3 half the clock cycles, inputs 1 and 2 sharing the other half. The advantage of this multiplexer is that it will operate at a high clock rate since the arbitration is local to the nodes.

See also -

demultiplex() [subsection 11.13.2.8](#) p496

demultiplex_l() [subsection 11.13.2.8](#) p496

demultiplex_e() [subsection 11.13.2.8](#) p496

multiplex_struct_l() [subsection 11.13.4.17](#) p519

multiplex_struct_e() [subsection 11.13.4.17](#) p519.

11.13.2.12 multiply - pipelined multiply

Library: stdlib.3pl

Prototype:

```
global module
multiply (queue(prod) <- value(m1), value(m2)) {}
```

Output parameters:

prod is the product.

Input parameters:

m1 is the multiplicand.

m2 is the multiplier.

This module is in multlib.3pl which is included in stdlib.3pl.

description:

This module is a pipeline multiplier for signed or unsigned integers. The module is useful for Virtex, Virtex E and Spartan 2 FPGAs which do not contain hardware multipliers. Later Virtex and Spartan3 FPGAs containing hardware multipliers offer no reason to use this module.

If neither input is a queue or an expression containing a queue the multiplier will accept input on every clock cycle unless the output queue becomes unavailable. Either input may be a constant but both inputs must not be constant. If either input is signed (a target int or a -ve constant) the output queue must be signed (an int, not a uint).

The product will become available at the output after a latency which is determined by the input widths. Data in transit through the module will continue to the output regardless of the availability of further input data. The number of pipeline stages depends on the size and types of input arguments as follows -

unsigned	expr x expr	bits in smallest expr
signed	expr x expr	bits in smallest expr + 1
unsigned	expr x constant	1-bits in constant
signed	expr x constant	1-bits in constant + 1

where signed means that either input argument is signed. **expr** means a target variable or expression, possibly containing queues. **constant** means an immediate constant, variable or expression. Either input argument may be the constant.

The width of the output queue argument should be at least the sum of the widths of the two input arguments. If narrower the result will be truncated at the most significant end. If the range of the product is known to be constrained then truncation of the product is acceptable.

11.13.2.13 pfifo - a prewrite fifo with static interface

Library: stdlib.3pl

Prototype:

```
global module
pfifo (
    value(out),
    value({emptyout, fullout}, "log")
    <-
    value(in),
    value({read, prewrite, write}, "log"),
    value(reset, "log")
) {}
```

Output parameters:

out is the data output for a fifo read.

emptyout is true if the fifo is empty.

fullout is true if the fifo is full.

Input parameters:

in is the data input for a fifo write.

read causes a fifo read on a clock cycle when high.

prewrite notifies the fifo that a write will be done in the future, reserving a place. When no free places remain the full condition will be come true.

write causes a fifo write on a clock cycle when high. The empty condition takes account of data actually written, not notified writes (prewrites).

reset is an optional reset to clear the fifo.

description:

This fifo is intended to be used where data is to be passed through a fifo but there is a delay between execution of the data-producing operation and availability of that data. An example of this is external memory reads where the data only becomes available a number of cycles after the read command has been sent. The resulting data are then to be buffered in this fifo. To ensure that the memory reads can be pipelined it is necessary to notify the fifo that a write will occur later so that its test for space can take that into account. Once the memory read has been signaled it cannot be stopped so the fifo must have space available for the resulting datum some cycles later.

On each clock cycle **emptyout** and **fullout** indicate when reads and writes respectively cannot be executed. The program must ensure that these conditions are checked as the module does not do so! If **read**, **prewrite** or **write** are high in a clock cycle then a read, prewrite or write will be performed. Any combination of these can all occur on one clock cycle.

11.13.2.14 pdelay - delay pipeline data

Library: stdlib.3pl

Prototype:

```

global module
pdelay (
  queue(out)
  <-
  queue(in),
  int(delay),
  value(reset, "log")
) {}

```

Output parameters:

out is the data output.

Input parameters:

in is the data input.

delay is the number of data to be delayed.

reset is an optional reset.

description:

This module delays data by the number specified by argument **delay**. The module will only operate when both input and output queues are available. Initially *delay* input data will be consumed while outputting default values (0, false etc.). Following this the output datum will be that input *delay* data cycles ago. The optional reset can be used to return the module to its initial condition.

11.13.2.15 rread - read registered output memory to a queue

Library: stdlib.3pl

Prototype:

```

global module
rread (queue(data) <- rmemory(m), int(port), queue(addr)) {}

```

Output parameters:

data is the queue variable to be written to from the memory read. It takes its type from the argument.

Input parameters:

m is the memory.

port is the memory port.

addr is the memory address.

description:

Read the registered output memory when the input and output queues are available. There is a latency of 2 cycles between input and output.

The program logic must ensure that this memory port access is mutually exclusive to all other accesses to that port.

See also -

cread() [subsection 11.13.2.6](#) p495,

cwrite() [subsection 11.13.2.7](#) p495,

rwrite() [subsection 11.13.2.16](#) p502.

11.13.2.16 rwrite - write registered output memory from a queue

Library: stdlib.3pl

Prototype:

```

global module
rwrite (rram(m), int(port), queue({addr, data})) {}

```

Output parameters:

none

Input parameters:

m is the memory.

port is the memory port.

addr is the memory address.

data is the data to be written.

description:

Write to the registered output memory when the input queues are available. The program logic must ensure that this memory port access is mutually exclusive to all other accesses to that port.

The registered output of the port is also changed to the written value.

This module is just a wrapper around a single statement. The source is

```
global module
rwrite (rmemory(m), int(port), value(addr), value(data)) {
    rmemorywrite(m, port, addr, data);
}
```

It is provided to complete the set **cread()**, **cwrite()**, **rread()** and **rwrite()** where only **rread()** contains significant code.

See also -

cread() [subsection 11.13.2.6](#) p495,

cwrite() [subsection 11.13.2.7](#) p495,

rread() [subsection 11.13.2.15](#) p502.

11.13.2.17 sort - generate a parallel Batcher sort network

Library: stdlib.3pl

Prototype:

```
global module
sort (queue(out) <- queue(in)) {}
```

Output parameters:

out is the output queue.

Input parameters:

in is the input queue.

description:

This module generates a parallel sort network. The input and output arguments are queues which are 1-dimensional arrays of type uint or int. The output ordering depends on the The input and output array dimensions and types must be the same. The number of pipeline stages in this module depends on the dimensions of the queues. The sort network is generated using Batchers's algorithm. Note that the amount of logic used grows rapidly with the dimensionality of the queues!

11.13.2.18 sqrt - target fixed point square root

Library: stdlib.3pl

Prototype:

```
global module
sqrt (queue(out) <- queue(in)) {}
```

Output parameters:

out is the output queue.

Input parameters:

in is the input queue.

description:

This module performs a bitwise iterative square root on an unsigned integer or unsigned fixed point value. The input and output arguments must be queue mode variables. The input and output types must be both "uint:" or both "ufixed:". The integer or fractional widths of input and output need not match. For integers the output width will usually be half the input width. For fixed point the output type would usually have the point moved left with respect to the input. If a result value exceeds the integer or fractional width of the output type, left or right truncation will occur.

Where the operand is not an exact square the result is truncated at the least significant end, i.e. the result is the largest value whose square is not larger than the operand.

The module does not process data at clock rate; it processes an operand in a fixed time of $n/2+2$ clock cycles where 'n' is the width of the (internal) operand.

For an integer square root 'n' is the width of the input, plus 1 if the width is odd.

For a fixed point square root 'n' is calculated as follows -

```
int(n, size(in));      // total operand width - initially that of input
int(ifb, offset(in));  // input fraction part width
int(iib, n - ifb);     // input integer part width
int(ioffset);          // input left offset to form internal integer
int(ofb, offset(out)); // output fraction part width
int(xoffset);          // extra offset for more fractional precision

if ((ifb % 2) != 0) {   // if input fractional part width odd --
    n++;               // increment n to make even
    ioffset = 1;       // input must now be offset by 1 bit
}
xoffset = (2 * ofb) - (ifb + ioffset); // extra precision for output fractional part
if (xoffset > 0) {     // if extra width for output fractional precision --
    n += xoffset;      // add to n
    ioffset += xoffset; // add to input offset
}
if ((n % 2) != 0)      // if total width odd --
    n++;               // increment n to make even
```

This increments n for either integer or fraction part of odd width and also widens the the internal operand to account for extra precision in the output fractional part compared to the input.

11.13.2.19 stretch - generate a long pulse from a transition

Library: stdlib.3pl

Prototype:

```
global module
stretch (static(out, "log") <- value(in, "log"), int(n)) {}
```

Output parameters:

out is the output static variable.

Input parameters:

in is the input value.

n is the required length of the output pulse.

description:

When input argument **in** goes from false to true generate an output **out** which is true for the number of cycles specified by argument **n**.

11.13.2.20 timebase - generate decadic pulses

Library: stdlib.3pl

Prototype:

```
global module
timebase(
    static({tb_1us, tb_10us, tb_100us, tb_1ms, tb_10ms, tb_100ms, tb_1s}, "log")
    <-
) {}
```

Output parameters:

tb_1us is a $1\mu S$ interval pulse stream.

tb_10us is a $10\mu S$ interval pulse stream.

tb_100us is a $100\mu S$ interval pulse stream.

tb_1ms is a $1mS$ interval pulse stream.

tb_10ms is a $10mS$ interval pulse stream.

tb_100ms is a $100mS$ interval pulse stream.

tb_1s is a $1S$ interval pulse stream.

Input parameters:

None.

description:

This module generates pulse streams at decade intervals of $1\mu S$ up to $1S$. Each pulse is one cycle in the current clock domain. Each output pulse is aligned with the pulses from higher frequency outputs. The current clock frequency must be $\geq 2MHz$.

The $1\mu S$ pulses are derived from the current clock by integer division and hence if the current clock is not an integer multiple of $1MHz$ the interval will differ from $1\mu S$. Each of the other outputs is derived from the next higher frequency output by division by 10.

11.13.2.21 to_pulse - generate a pulse from a transition

Library: stdlib.3pl

Prototype:

```
global module
to_pulse (static(out, "log") <- value(in, "log")) {}
```

Output parameters:

out is the output static variable.

Input parameters:

in is the input value.

description:

Generate a pulse on **out** when the input argument **in** changes from false to true.

11.13.2.22 tws_recv - two wire serial receiver

Library: stdlib.3pl

Prototype:

```
tws_recv(
    queue(out)
    <-
    value(sck, "log"), value(sdo, "log"), varargs
) {}
```

Output parameters:

out is the output queue variable.

Input parameters:

sck is the input serial clock.

sdo is the input serial data.

varargs matches two optional arguments, **silence** and **settle**.

description:

This module receives a two-wire serial stream and produces a parallel output on a queue. Input **sck** is the input clock and input **sdo** is the serial data. The data is captured on the positive edges of the clock. The clock is low between words.

Optional argument **silence** (first argument matched by varargs) specifies the minimum time between serial words. If **sck** goes high before the specified interval has elapsed since the end of the current word, the current word will be discarded. The argument may be an immediate integer, in which case it specifies the number of clock cycles in the current clock domain, or it may be immediate floating point, which will be interpreted as a time in second. If the argument is missing or is zero no minimum time period between words will be enforced.

Optional argument **settle** (second argument matched by varargs) specifies the minimum time **sck** or **sdo** must remain unchanged following a transition for that transition to be recognised, i.e. it is a deglitch time period. The argument may be an immediate integer, in which case it specifies the number of clock cycles in the current clock domain, or it may be immediate floating point, which will be interpreted as a time in second. If the argument is missing or is zero no deglitching will be applied. Note that deglitching will introduce a data output latency.

The clock domain within which this module is called must be of a frequency high enough to properly sample **sck** and **sdo**, i.e. at least twice the frequency of **sck**.

See also library module `tw_ssend()` [subsubsection 11.13.2.23](#) p506.

11.13.2.23 tw_ssend - two wire serial transmitter

Library: `stdlib.3pl`

Prototype:

```
tw_ssend(
    static(sck, "log"), static(sdo, "log")
    <-
    queue(in), varargs
) {}
```

Output parameters:

sck is the output serial clock.

sdo is the output serial data.

Input parameters:

in is the input queue variable.

varargs matches two required arguments, **timebase** and **silence**.

description:

This module transmits a two-wire serial stream from a parallel input on a queue. Output **sck** is the serial clock and output **sdo** is the serial data. The data changes on the negative edges of the clock. **sck** is low between words.

Argument **timebase** (first argument matched by varargs) may be immediate, value or static mode.

If **timebase** is immediate mode it must be type "float", "int" or "uint". If type "float" it specifies the period of the output clock **sck** in second. If it is type "int" or "uint" it gives the period of **sck** as a multiple of the period of the current clock.

If **timebase** is value or static mode it must be a stream of pulses in the current clock domain, each pulse initiating the next **sck** transition during serial transmission. If **timebase** is continuously high then **sck** will be at half the current clock rate.

Argument **silence** specifies the minimum gap between transmitted words. The argument may be an immediate integer, in which case it specifies the number of clock cycles in the current clock domain, or it may be immediate floating point, which will be interpreted as a time in second. If the argument is zero no gap between successive words will be inserted.

The clock domain within which this module is called must be at least twice the frequency of **sck**.

See also library module **tw_s_recv()** [subsection 11.13.2.22](#) p505.

11.13.3 library procedures

11.13.3.1 assign - generalised assignment

Library: **stdlib.3pl**

Prototype:

```
global procedure
assign (var(out) <- varargs) {}
```

Output parameters:

out is the variable to be assigned.

Input parameters:

varargs expects a single argument which is the input value.

In-line target execution:

yes

description:

This procedure is provided for cases where assignment is to be made to a variable whose mode is not known or which varies from instance to instance.

The output variable may be of mode immediate, value, priority, clock, static, queue or selectvalue. The input may be any expression, immediate or target mode and in the latter case may contain queue reads.

11.13.3.2 cinclude - convert a C include file

Library: **stdlib.3pl**

Prototype:

```
global procedure
cinclude(str(filename)) {}
```

Output parameters:

none

Input parameters:

filename is the C include file name.

In-line target execution:

no

description:

This procedure reads a C include file and defines variables corresponding to each **#define** entry that

it finds. This enables a C include file to be shared between a CPU C program and an associated 3PL FPGA program where there is a CPU/FPGA interface and constants such as bus or register addresses must be the same.

cinclude() processes integer, string and floating point definitions. It also handles a **#define** with no value, creating a **log** type initialised to **true**.

cinclude() ignores C style comments (which must not extend beyond one line), C++ style comments and empty lines.

Lines beginning with **#define** must contain an identifier and an optional associated constant. Integer constants may be decimal (0x... or 0X...), octal (0...) or binary (0b... or 0B...). String constants must be enclosed quotes - "...".

#defines cannot be nested.

Each definition produces a corresponding global immediate **int**, **str**, **float** or **log** as appropriate whose initial value is the associated value from the definition. If there is no value a **log** type is created initialised to **true**.

For a C include file def.h as follows -

```
/* Include file */
/* has only single line C style comments */

#define VARA    3    // also ignores c++ style comments
#define VARB    0xf /* second */
#define VARC    "name"
#define VARD    5.67
#define ENABLE
```

The 3PL source

```
cinclude("def.h");
```

will read the file def.h and effectively execute the following in the global module -

```
int(VARA, 3);
int(VARB, 0xf);
str(VARC, "name");
float(VARD, 5.67);
log(ENABLE, true);
```

11.13.3.3 netlistcommentfromdirs - directives to netlist comment

Library: stdlib.3pl

Prototype:

```
global procedure
netlistcommentfromdirs () {}
```

Output parameters:

none.

Input parameters:

none.

In-line target execution:

no

description:

This procedure copies a listing of the current identifiers and values of all directives to the directive

"netlistcomment". Directive "netlistcomment" if defined is copied as comments to the output netlist and constraint files. If this procedure is called at the beginning of source code only directives defined by -D command line options and the directive "sourceversion" will be listed. If called at the end of execution then directives "part" and "family" will be listed along with the optional "author" directive and any other directives defined by the program.

11.13.3.4 queuearray - construct an array of queues

Library: stdlib.3pl

Prototype:

```
global procedure
queuearray(var(ptr) <- type(type)) {}
```

Output parameters:

ptr is an array of pointers to queues. Each queue has the type given by the input parameter.

Input parameters:

type is the type of the queues.

In-line target execution:

no

description:

Create an array of queues. Since 'queue' is a mode, not a type, there cannot be an array of them, hence an array of pointers to queues must be used instead (the queue type itself can be an array, but that is not what is wanted in this case).

The pointers to created queues are assigned to the output argument which must be a pointer or pointer array. The array may have more than one dimension. The queue type is given by the input argument which may be a type or a string equivalent. The created queues have the default depth of 2. If the output argument is a single pointer then a single new queue will be assigned to that pointer.

11.13.3.5 prelementIntPropertys - print a list of variable attributes

Library: stdlib.3pl

Prototype:

```
global procedure
prelementIntPropertys (str(lv, "[] (key, str, mode, str, value, str)")) {
```

Output parameters:

none

Input parameters:

A call to listattributes().

In-line target execution:

no

description:

This procedure prints a list of attributes for a variable. Each line list in order -

```
key mode value
```

the fields being separated by horizontal tabs. The format is not very sophisticated.

The argument must be a call to inbuilt function **listattributes()** [section 10.80](#) p303 which itself has one argument, the variable identifier.

11.13.3.6 printdirs - print a list of directives

Library: stdlib.3pl

Prototype:

```
global procedure
printdirs () {}
```

Output parameters:

none

Input parameters:

none.

In-line target execution:

no

description:

This procedure prints a list of directives. Each line list in order -

```
identifier type value
```

the fields being separated by horizontal tabs. The format is not very sophisticated.

11.13.3.7 printvars - print a list of variables

Library: stdlib.3pl

Prototype:

```
global procedure
printvars (var(lv, "[](scope, str, ident, str, var, ->)" )) {}
```

Output parameters:

none

Input parameters:

A call to listvars().

In-line target execution:

no

description:

This procedure prints a list of variables. Each line list in order -

```
scope identifier mode type
```

the fields being separated by horizontal tabs. The format is not very sophisticated.

The argument must be a call to inbuilt function **listvars()** [section 10.81](#) p303 which itself has an optional argument.

Argument "global" causes only global variables to be printed.

Argument "file" causes only file scope variables to be printed.

Argument "files" causes variables in all file scopes to be printed.

Argument "local" causes only local variables to be printed, i.e. only those declared within the current module, procedure or function, and not variables in nested control structures.

Argument "" selects the default, i.e. all block and local variables, the file scope variables and global scope variables will be printed.

See also inbuilt function **listvars()** [section 10.81](#) p303.

11.13.3.8 wait - wait for a time period

Library: stdlib.3pl

Prototype:

```
global procedure
wait (var(t)) {}
```

Output parameters:

none

Input parameters:

t is the delay in cycles or second.

In-line target execution:

yes

description:

This procedure waits until a time has elapsed. The input argument may be immediate, value or static mode. If immediate the wait period is determined on immediate execution. If a target mode the wait period is given by the value of the argument at the time the procedure is executed in the FPGA.

If the input argument is immediate mode, it must be type **uint**, **int** or **float**. If the argument is type **float** it is the number of second to wait. The delay will be rounded to the nearest integer number of clock cycles of the current clock. If the argument is an integer it is the number of clock cycles to wait.

If the input argument is value or static mode it must be type **uint** or **int** and is the number of clock cycles to wait. The value must not be zero - this will generate the longest wait consistent with the width of the argument variable.

11.13.3.9 wait_until - wait on a logical value

Library: stdlib.3pl

Prototype:

```
global procedure
wait_until (value(v, "log")) {}
```

Output parameters:

none

Input parameters:

v is the input value to wait on.

In-line target execution:

yes

description:

This procedure waits until the logical input argument 'v' is true.

11.13.3.10 wait_while - wait on a logical value

Library: stdlib.3pl

Prototype:

```
global procedure
wait_while (value(v, "log")) {}
```

Output parameters:

none

Input parameters:

v is the input value to wait on.

In-line target execution:

yes

description:

This procedure waits while the logical input argument '**v**' is true.

11.13.4 library functions

11.13.4.1 argstomap - create a map from key=value arguments

Library: stdlib.3pl

Prototype:

```
global function
argstomap (varargs) {}
```

Return mode and type:

immediate, map

Input parameters:

See description.

description:

This function takes a number of key=value arguments and from them builds a map.

11.13.4.2 asyncqueuespaces - space count for asynchronous queue

Library: stdlib.3pl

Prototype:

```
global function
asyncqueuespaces (queue(p)) {}
```

Return mode and type:

value, "uint:"

Input parameters:

p is an asynchronous queue variable.

description:

This function returns a target value which gives a guaranteed minimum number of free spaces in the asynchronous queue, rather than a precise count. The count is determined from status bits derived from the most significant two bits of the of the read and write gray coded addresses (see inbuilt function **queuewritestatus()** ([section 10.118](#) p321). This uses much less logic than an exact determination of the word count.

This function is intended for an external interface to the FPGA which writes a queue. The count can be used to avoid the extra transfers which would be required to determine non-full status prior to every write.

This must be called in the write clock domain.

11.13.4.3 asyncqueuewords - word count for asynchronous queue

Library: stdlib.3pl

Prototype:

```
global function
asyncqueuewords (queue(p)) {}
```

Return mode and type:

value, "uint:"

Input parameters:

p is an asynchronous queue variable.

description:

This function returns a target value which gives a guaranteed minimum number of words in the asynchronous queue, rather than a precise count. The count is determined from status bits derived from the most significant two bits of the of the read and write gray coded addresses (see inbuilt function **queuereadstatus**((section 10.115 p319)). This uses much less logic than an exact determination of the word count.

This function is intended for an external interface to the FPGA which reads a queue. The count can be used to avoid the extra transfers which would be required to determine non-empty status prior to every read.

This must be called in the read clock domain.

11.13.4.4 bin_to_gray - convert binary count to gray count

Library: stdlib.3pl

Prototype:

```
global function
bin_to_gray (value(b)) {}
```

Return mode and type:

immediate, ->

Input parameters:

b is the input binary count value.

description:

This function converts an input weighted binary value to

11.13.4.5 bitset - test if a bit is set

Library: stdlib.3pl

Prototype:

```
global function
bitset (value(in), int(bit)) {}
```

Input parameters:

in is the input value. It may be any target type.

bit is the index of the required bit.

return mode:

value

return type:

log

Description:

Returns true if a selected bit from the first (target) argument is set. The second argument is an immediate integer giving the bit index, 0 selecting the LSB. The first argument type need not be primitive, but it is better coding practice to select a bit from an array member or struct field rather than from the compound type.

See also `getbit()` [subsection 11.13.4.9](#) p516 which does the same but returns as type "bits:1".

11.13.4.6 deglitch - deglitch an input signal

Library: stdlib.3pl

Prototype:

```
global function
deglitch (value(in), varargs) {}
```

Return mode and type:

value, same type as input

Input parameters:

in is the input signal.

varargs expects a single argument (at most) giving the deglitch time constant.

The output of this function follows the input value, however the output will not follow an input change until the input has maintained its new value for a time specified by the second argument.

The first argument is the input signal or value. It may be any target type. It would normally be input mode or an expression containing an input mode variable.

Other modes are allowed for the input (except cmemory, rmemory or clock mode) and it may contain queue reads, but there would not seem to be any reason to use any of these.

The second argument may be missing, in which case the output is just connected to the input and there is no deglitching performed. The second argument may be immediate mode of type "int", "uint" or "float", or static mode of type "int:" or "uint:". If integer it specifies the deglitching time in terms of clock cycles of the current clock. If it is floating point the argument specifies an explicit time in second and the delay will be of the minimum number of clock cycles which encompass the delay time. The second argument will be ignored if it is immediate and the number of clock cycles specified, or the number derived from the floating point time, is 1 or 0 (or -ve). In that case the output is again simply connected to the input.

11.13.4.7 delay - fixed delay

Library: stdlib.3pl

Prototype:

```
global function
delay (value(in), int(len), value(step, "log"), log(ram)) {}
```

Return mode and type:

value, same type as input

Input parameters:

in is the input signal.

len is the length of the delay line.

step is an optional step control.

ram is an optional switch to force usage of RAM

This has up to 4 input arguments. Input argument **len** is immediate mode. **step** may be immediate or target mode as described below. The data input argument **in** must not be modes **immediate**, **queue**, **memory** or **clock** and value mode arguments must not contain queue reads.

The output of this function follows the input value delayed by **del** clock cycles.

The argument to **step** must be type "log" and may be as follows -

- no argument - the delay will advance on each clock cycle
- an immediate mode value (or constant value mode) which is **true** - the delay will advance on each clock cycle
- an immediate mode value (or constant value mode) which is **false** - the delay will advance on clock cycles in which the function is evaluated in the target code, e.g. on used in assignment
- a static, selectvalue or value mode value - the delay will advance on clock cycles when **step** is true.

For example -

```
when (1)
  a <- delay(b, 10, false);
```

will shift the current value of **b** into the delay line every time the assignment to **a** is made (**1** is true), the assigned value being popped off the end.

step false can only be used where **delay()** is evaluated in a statement executing within an execution thread. For example -

```
output(., delay(b, 10, false), loc=A_pins[0]);
```

would give an error (function used() called in illegal context).

Whereas -

```
when (1)
  a <- delay(b, 10);
```

will shift **b** into the delay line every clock cycle, popping off the value at the other end. When **1** is true the value popped off will also be assigned to **a**.

```
when (1)
  a <- delay(b, 10, true);
```

will behave the same way.

Where **step** is variable during target execution the shifting is controlled solely by step while the assignment, controlled by **1**, assigns the end value of the delay line to **a** regardless of whether a shift has occurred or not.

```
when (1)
  a <- delay(b, 10, step);
```

The data width is not limited. This function can operate as a sequence generator if the output is connected back to the input.

For the first **del** steps the output will be 0, false etc.

in may be any type. It must not contain queue reads (fatal error).

The return value is the same type as **in** and is the delayed input.

If the delay is small enough (see table below) this will call inbuilt module **del()** [section 8.1](#) p184 to implement the delay using CLB shift registers, otherwise rmemory (block RAM) will be used. In the latter case the length **len** is limited to the maximum allowed memory depth for the FPGA family -

	CLB SR maximum length	maximum RAM depth
Spartan-II, Spartan IIE, Virtex, Virtex-E	16	4096
Spartan-3, Virtex-II, Virtex-II Pro, Virtex 4,	16	16384
Spartan 6, Virtex 5, Virtex 6, series 7, Zynq	32	32768

If parameter **ram** is true block RAM will be used for the delay despite a small delay being specified.

See also library function **del()** [subsubsection 11.2.3.2](#) p396 in library xilinx/common which allows a variable-length delay line but does not offer the option of using block RAM rather than shift registers. That section also illustrates a way of using a queue input, getting around the no-queue limitation described above.

11.13.4.8 forcecast - force a prohibited cast

Library: stdlib.3pl

Prototype:

```
global function
forcecast (value(v), type(t)) {}
```

Input parameters:

v is the input value. It may be any target type.

t is the target type to which the input value is to be cast.

return mode:

value

return type:

given by 2nd argument

Description:

Returns the target first argument value cast to the target type given by the second argument.

This function can be used to perform an otherwise prohibited cast operation. It does this by casting in two steps via an intermediate "bits:" type. This should be used with caution!

11.13.4.9 getbit - get a bit

Library: stdlib.3pl

Prototype:

```
global function
getbit (value(in), int(bit)) {}
```

Input parameters:

in is the input value. It may be any target type.

bit is the index of the required bit.

return mode:

value

return type:

bits:1

Description:

Returns a selected bit from the first (target) argument. The second argument is an immediate integer giving the bit index, 0 selecting the LSB. The first argument type need not be primitive, but it is better coding practice to select a bit from an array member or struct field rather than from the compound type.

See also **bitset()** [subsubsection 11.13.4.5](#) p513 which does the same but returns as type "log".

11.13.4.10 **incrdecr** - dynamically increment or decrement

Library: stdlib.3pl

Prototype:

```
global function
incrdecr (value(in), value(incr, "log")) {}
```

Return mode and type:

value, type(in)

Input parameters:

in is the input value to increment or decrement. It may be types int or uint.

incr selects incrementing (true) or decrementing (false).

description:

Increment or decrement a value according to a logical input.

11.13.4.11 **isclear** - verify a run of clear bits

Library: stdlib.3pl

Prototype:

```
global function
isclear (value(in), int(bit), int(len, 1)) {}
```

Return mode and type:

value, log

Input parameters:

in is the value to be tested.

bit is the bit index from the least significant end.

len is the number of bits.

description:

This target function returns true if **len** bits starting at bit **bit** are all clear.

11.13.4.12 **isset** - verify a run of set bits

Library: stdlib.3pl

Prototype:

```
global function
isset (value(in), int(bit), int(len, 1)) {}
```

Return mode and type:

value, log

Input parameters:

in is the value to be tested.

bit is the bit index from the least significant end.

len is the number of bits.

description:

This target function returns true if **len** bits starting at bit **bit** are all set.

11.13.4.13 lfsr - linear feedback shift register

Library: stdlib.3pl

Prototype:

```
global function
lfsr (var(t), value(step, "log", true), int(init)) {}
```

Return mode and type:

value, uint

Input parameters:

t is the result type.

step is an optional control input to advance the output.

init is an optional initial value.

description:

This target function returns a pseudo-random number. The first argument specifies the type of the function value. It may be a type "str" or type "type". If the former it must be a valid type specification string. Any output type may be specified though it is difficult to imagine occasions where a compound type would be appropriate. The size of the type must be in the range 3 to 32 bits.

If the second argument is provided it controls the advance of the pseudo-random sequence from one to the next. If absent the sequence will advance on every clock cycle.

If the last parameter has an argument it provides an initial value for the register. This may have any value. If the value is larger than can be accommodated in the register it will be truncated at the most significant end. If the last parameter has no argument then the initial value of the register will be a random number derived from calls to inbuilt function random() for each bit.

For an output type of size n bits all 2^n values are produced (there is no lock-up state, or rather the 'lock-up' state is explicitly forced and exited).

11.13.4.14 lszeros - count least significant zero bits

Library: stdlib.3pl

Prototype:

```
global function
lszeros (value(in, "uint:")) {}
```

Return mode and type:

value, uint

Input parameters:

in is the operand.

description:

This target function returns the number of least significant zero bits in a non-zero unsigned integer. If the argument is zero the result is also zero. This function was written for use in floating point number normalisation (the mantissa is the argument) but may find other uses.

11.13.4.15 maxval - target maximum value

Library: stdlib.3pl

Prototype:

```
global function
maxval (varargs) {}
```

return mode:
immediate or value

return type:
type of arguments

Description:

Returns the maximum value of any number of immediate or target variables or expressions. The arguments must be type int or uint and must all be the same type. A mixture of immediate and target arguments is allowed, but in that case only a single immediate argument should be used as the function will not optimise the immediate arguments to a single immediate value.

11.13.4.16 minval - target maximum value

Library: stdlib.3pl

Prototype:

```
global function
minval (varargs) {}
```

return mode:
immediate or value

return type:
type of arguments

Description:

Returns the minimum value of any number of immediate or target variables or expressions. The arguments must be type int or uint and must all be the same type. A mixture of immediate and target arguments is allowed, but in that case only a single immediate argument should be used as the function will not optimise the immediate arguments to a single immediate value.

11.13.4.17 multiplex_struct_l - multiplex output type with log array multiplex_struct_e - multiplex output type with uint

Library: stdlib.3pl

Prototypes:

```
multiplex_struct_l (type(inttype), int(n)) {}

multiplex_struct_p (type(inttype), int(n)) {}
```

return mode:
immediate

return type:
a struct type

Description:

These functions are used in conjunction with library modules **multiplex()** [subsection 11.13.2.11](#) p498

multiplex_l() [subsection 11.13.2.11](#) p498

multiplex_e() [subsection 11.13.2.11](#) p498

multiplexp() [subsection 11.13.2.11](#) p498

multiplexp_l() [subsection 11.13.2.11](#) p498

multiplexp_e() [subsubsection 11.13.2.11](#) p498

multiplext() [subsubsection 11.13.2.11](#) p498

multiplext_l() [subsubsection 11.13.2.11](#) p498

multiplext_e() [subsubsection 11.13.2.11](#) p498

demultiplex() [subsubsection 11.13.2.8](#) p496

demultiplex_l() [subsubsection 11.13.2.8](#) p496

demultiplex_e() [subsubsection 11.13.2.8](#) p496 They return a struct which is a wrapper for the output data type. The wrapper allows for an associated identifier for the source port of a datum. The struct has two fields. Field **body** is the same type as the multiplexer inputs and contains a multiplexed datum. Field **port** contains an identifier for the source port from which the datum originated.

The first argument is the type of the inputs to module **multiplex()** or **multiplexp()** or the outputs of **demultiplex()**.

The second argument is the number of inputs to module **multiplex()** or **multiplexp()** or outputs to **demultiplex()**.

In the case of **multiplex_struct_l()** field **port** will be an array of type log whose dimension is equal to the number of inputs to the associated multiplexer or the outputs of the associated demultiplexer. This field has one member of the array true to identify the associated datum source port.

In the case of **multiplex_struct_e()** field **port** will be a "uint" large enough to contain the port number (starting at 0).

See also -

multiplex() [subsubsection 11.13.2.11](#) p498

multiplex_l() [subsubsection 11.13.2.11](#) p498

multiplex_e() [subsubsection 11.13.2.11](#) p498

multiplexp() [subsubsection 11.13.2.11](#) p498

multiplexp_l() [subsubsection 11.13.2.11](#) p498

multiplexp_e() [subsubsection 11.13.2.11](#) p498

multiplext() [subsubsection 11.13.2.11](#) p498

multiplext_l() [subsubsection 11.13.2.11](#) p498

multiplext_e() [subsubsection 11.13.2.11](#) p498

demultiplex() [subsubsection 11.13.2.8](#) p496

demultiplex_l() [subsubsection 11.13.2.8](#) p496

demultiplex_e() [subsubsection 11.13.2.8](#) p496

11.13.4.18 mux - multiplex inputs

Library: **stdlib.3pl**

Prototype:

```
global function
mux (value(s), varargs) {}
```

return mode:

value

return type:

type of data argument

Description:

Returns one of the data inputs. The first argument must be target mode of type **uint** and selects which of the following input arguments is to be returned as the value of the function. The remaining arguments are the data inputs.

If the data input expressions are primitive they must be the same type but their numeric widths may differ. The function value will be the width of the widest data input. If the data inputs are not primitive then they must all be exactly the same type, including primitive component widths.

11.13.4.19 ptr2expr - a pointer to an expression

Library: stdlib.3pl

Prototype:

```
global function
ptr2expr (varargs) {}
```

return mode:

value

return type:

type of argument

Description:

Returns a pointer to a value mode variable which has been assigned the target expression passed as the argument. The argument may also be a variable or constant. The argument cannot be mode clock or mode memory.

11.13.4.20 resync - resynchronise a pulse or level to another clock

Library: stdlib.3pl

Prototype:

```
global function
resync (value(in, "log")) {}
```

return mode:

value

return type:

log

Description:

in is an expression or a value, selectvalue, static or priority variable of type log. The argument must not contain queues.

When the input is clocked true in its clock domain the function returned value will go true at the next positive edge of the current clock and will then go false at the following edge. This process cannot repeat until the input has returned to false for at least one clock cycle in its clock domain after the output has gone false.

This function is used to produce a single cycle pulse in the current clock domain from a pulse or level in another clock domain.

11.13.5 select - a combinatorial selector

Library: stdlib.3pl

Prototype:

```
global function
select(s1, d1, s2, d2,...) {}
```

return mode:

value

return type:

type of data inputs

Description:

An inbuilt selector function using gate logic. This has any number of argument pairs.

The 1st input argument of each pair is a logical expression which selects the following data input expression if true.

The 2nd input argument of each pair is a data input signal. If the data input expressions are primitive they must be the same type but their numeric widths may differ. The function value will be the width of the widest data input. If the data inputs are not primitive then they must all be exactly the same type, including primitive component widths.

The function has a defined value when at most one select input is available and true and the associated data input is available, the value reflecting the selected data input. If more than one input is selected the value is undefined. If no inputs are selected, the value is zero or false depending on data type(s).

If none of the function inputs contain queues, the function value is always available. If any function inputs contain queues the function value is available when an input selector and its associated data input are available and the selector is true. Note that if some inputs contain queues but one input selector and its associated data do not, the function will be available when that selector is true. Note that the use of this function as a gate with a clear value when no selectors are true is only possible if the inputs contain no queues since any queues in any inputs will render the function value unavailable when no input is selected.

See also inbuilt function `selecta()` [section 10.129](#) p327 where the select and data inputs are arrays of values or pointers.

11.13.5.1 sprintf - format a string from variables

Library: stdlib.3pl

Prototype:

```
global function
sprintf (fmt, v1, v2, ...) {}
```

return mode:

immediate

return type:

str

Description:

This function is a wrapper for the inbuilt procedure `sprintf()` [section 9.72](#) p250. It returns the formatted string.

The format argument may be followed by a single argument which is an array of pointers, for example **inargs** in the calling module, procedure or function. In that case the pointers are resolved and provide the input variable list to be printed.

11.13.5.2 svn_version - get the current subversion version number

Library: stdlib.3pl

Prototype:

```
global function
svn_version () {}
```

Return mode and type:

immediate, "str"

Input parameters:

None.

description:

This function returns the current subversion version number as a string. It does this by running the command "svnversion" in a shell in the current directory and returning the resulting string from the shell standard output with the trailing newline removed.

11.13.5.3 to_pulse - generate a pulse from a transition

Library: stdlib.3pl

Prototype:

```
global function
to_pulse (value(in, "log")) {}
```

Return mode and type:

value, "log"

Input parameters:

in is the input value.

description:

Generate a pulse out when the input argument **in** changes from false to true.

11.13.5.4 truthtable - an expression from a truth table

Library: stdlib.3pl

Prototype:

```
global function
truthtable (
    var(table, "[64][6]str"),
    value({i1, i2, i3, i4, i5, i6}, "log")
) {}
```

Return mode and type:

value, "log"

Input parameters:

table is the input value. **i1** to **i6** are the input values.

description:

The first argument is the truth table as a 2-dimensional array of strings. This argument would usually be a 2-dimensional compound value, in which case its dimensions need not match the parameter dimensions as long as they do not exceed them. If the argument is a string array its dimensions must match those of the parameter, but the array would normally be initialised from a 2-dimensional compound value which again may have smaller dimensions. The string array dimensions are such as to allow a maximum of all 64 combinations of 6 variables.

Truth table entries must be "0" or "1" for explicit entries or "", "x" or "X" for 'don't care' entries. Compound values for string arrays may contain constant values of other types in which case these will be converted to strings. Thus a true entry may be given as "1", 1 or true, a false entry as "0", 0 or false and a 'don't care' entry may be simply omitted.

The 2nd and following arguments are the input variables to the function described by the truth table. One to six of these arguments may be provided, trailing arguments being omitted. The arguments provided must be contiguous from **i1**, i.e. there must be no omitted arguments except for trailing arguments. The number of arguments is used to determine how many columns of the truth table to process but no checks are made, e.g. extra columns are ignored and omitted columns will be treated as 'don't care' entries.

Example 1 - 4 input variables, compound value argument -

```
value({i, j, k, l, v}, "log");
v := truthtable(
    { {1, , 0, 1},
      {0, 0, 1, } },
      i, j, k, l
    );
```

Note that the simplest and neatest truth table scheme is to use integers 0 and 1 for explicit entries (these will be converted to strings "0" and "1") and omit 'don't care' entries. Note also that the input variables occur in the same order as the truth table columns and hence readability is aided by positioning them under the truth table when formatting the source code.

Example 2 - 3 input variables, string array argument -

```
value({i, j, k, v}, "log");
var(tt, "[64][6]str", { {1, , 0},
                       {0, 0, 1} });
v := truthtable(tt, i, j, k);
```

Here a string array of the correct dimensions has been supplied as the truth table argument but the array initialisation is the same compound value as before. Of course the truth table string array could be generated by an algorithm if that were required.

Example 3 - truth table function as a value variable initialiser -

```
value({i, j, k, l}, "log");
value({v, "log", truthtable( { {1, , 0, 1},
                              {0, 0, 1, } },
                              i, j, k, l
                              )});
```

11.13.6 library classes

None.

11.14 Capture a data stream - capture.3pl

This library contains a single function. That function captures a data stream when triggered.

This library must be incorporated by using a **source** statement, i.e.

```
source "capture.3pl"
```

11.14.1 library modules

11.14.1.1 capture - capture a data sequence

Library: capture.3pl

Prototype:

```
queue(out),
static({running, triggered}, "log")
<-
value(in),
int(length),
value(trigger_position, "uint:20", 0),
value({start, trigger, readout}, "empty")
) {}
```

Output parameters:

out is the output queue variable

running is true when data storage is running

triggered is true when a trigger has occurred

Input parameters:

in is the data stream input.

length is the length of captured data.

trigger_position controls the length of data captured after the trigger.

start is asserted to start data storing.

trigger is asserted to halt data storage after some number of samples.

readout is asserted to trigger stored data readout.

description:

The **capture()** module acts like a logic analyser. Following assertion of **start** input data **in** is stored continuously, the data buffer wrapping around. When **trigger** is asserted data will continue to be stored such that the data value that occurred coincident with the trigger is stored at the point in the stored data given by **trigger_position** (buffer wraparound is taken care of).

The length of captured data is given by immediate input parameter length. This must be a power of 2 and may have values -

256 to 4096 for Virtex and Virtex-E

512 to 16384 for Virtex2 and Virtex2 Pro

1024 to 32768 for Virtex5.

The input data may be any type.

Input parameter **start** starts data recording. If the argument does not contain queue reads it must be type "log" in which case the transition from false to true starts recording (it may stay true for some time as only the transition is recognised). If the argument does contain queue reads then it may be any type, including "null", and arrival of a datum starts recording (the type and value is not significant). If this

parameter is not supplied with an argument then data recording will start following global logic start and will continue following readout. **start** is ignored if already running or during readout.

Input parameter **trigger** finalises the data capture, data recording running on for some time if necessary to position the captured data around the trigger position. If the argument does not contain queue reads it must be type "log" in which case the transition from false to true finalises the data capture (it may stay true for some time as only the transition is recognised). If the argument does contain queue reads then it may be any type, including "null", and arrival of a datum finalises the data capture (the type and value is not significant). If the trigger position is not zero then data recording must have started an appropriate number of clock cycles prior to the trigger to have stored valid data. The trigger is ignored if data recording is not running, the trigger has already occurred or if readout is in progress.

Input parameter **readout** starts a readout of the stored data to the output queue. If the argument does not contain queue reads it must be type "log" in which case the transition from false to true starts the readout (it may stay true for some time as only the transition is recognised). If the argument does contain queue reads then it may be any type, including "null", and arrival of a datum starts the readout (the type and value is not significant). If the output queue blocks, readout will pause until write availability is restored. If this parameter is not supplied with an argument then readout will occur automatically following a trigger and subsequent completion of data capture. **readout** is ignored if data recording is running.

The input parameter **trigger_position** may have either an immediate or target argument. The trigger position is recognised each time a trigger occurs. The trigger position is the index into the output data block of the clock cycle in which the trigger occurred.

The output queue **out** should be the same type as **in** (it can differ if it can nevertheless be assigned from the type of **in**, e.g. it may be an integer type of different width).

Output parameter **running** is true when data recording is running, including after the trigger.

Output parameter **triggered** is true following the trigger input. It is false on completion of readout or when data recording is restarted if no readout is performed.

11.14.2 library procedures

None.

11.14.3 library functions

None.

11.14.4 library classes

None.

11.15 Conversion of 3PL struct and enum types to C types - ctypes.3pl

This library contains two functions. The first takes a 3PL struct or enumerated type and generates an equivalent C type as a string. The second function creates a string array of identifiers from an enumerated type.

This library must be incorporated by using a **module** statement, i.e.

```
module "ctypes.3pl"
```

11.15.1 library modules

None.

11.15.2 library procedures

None.

11.15.3 library functions

11.15.3.1 ctypes - converts 3PL types to C types

Library: ctypes.3pl

Prototype:

```
global function
ctypes (type(t), str(name), str(prefix, "")) {}
```

Return mode and type:

immediate, "str"

Input parameters:

t is the type to be converted.

name is the identifier to be used in the C type.

prefix is an optional string prefix to prepend to enum identifiers.

description:

This function takes a 3PL variable of type **type** and returns a string representing the equivalent C type. It is intended to allow non-primitive data to be passed between an FPGA and a CPU via a 32 bit port. The conversion allows types declared in 3PL to be converted to C types for inclusion and compilation.

The only types converted are **struct** and **enum**. A struct may contain enums but must not contain nested structs. Any enums contained within a struct must have been previously converted by this function. Enums where the ordinals have been specified explicitly and non-sequentially are converted correctly.

The second argument is the identifier to be given to the enum or struct. Since the type is passed by value from the argument there is no type identifier available, so one must be provided.

3PL structs are converted to C bit field structs with an additional field to pack the struct to 32 bits if necessary. If the total number of bits is greater than 32 an error exit occurs. The packed bit fields are in reverse order as required.

As an example, the following 3PL code -

```

type(grptype, "[16]log");
enum(src_type,
    TIME,
    PIXEL,
    CPU
);
enum(subj_type,
    NONE,
    LINEARISE,
    GAINTRIM,
    GROUP,
    PILEUP,
    THROTTLE,
    ENERGY,
    DACOLMAP,
    TIMER,
    TIMER_IM
);
struct(command,
    src,          src_type,
    subject,      subj_type,
    index,        "uint:4",
    groups,       grpype,
    pileup,       "log",
    discard,      "log"
);
file(tf, open("fpga_types.h", "w"));
writeln(tf, ctypes(src_type, "src_type", "SRC_"));
writeln(tf, ctypes(subj_type, "subj_type"));
writeln(tf, ctypes(command, "command"));
close(tf);

```

would create file fpga_types.h and write the following to it -

```

enum src_type {
    SRC_TIME = 0,
    SRC_PIXEL = 1,
    SRC_CPU = 2
};

enum subj_type {
    NONE = 0,
    LINEARISE = 1,
    GAINTRIM = 2,
    GROUP = 3,
    PILEUP = 4,
    THROTTLE = 5,
    ENERGY = 6,
    DACOLMAP = 7,
    TIMER = 8,
    TIMER_IM = 9
};

typedef union {
    struct {
        unsigned packing: 4;
        unsigned discard: 1;
        unsigned pileup: 1;
        unsigned groups: 16;
        unsigned index: 4;
        unsigned subject: 4;
        unsigned src: 2;
    } bits;
}

```

```
uint32_t word;
} command;
```

To allow structs packed into 32 bit words to be both transmitted and received as words and also accessed via struct fields it is best to declare a suitable union, e.g.

```
typedef union {
    struct command assign;
    uint32_t word;
} COMMAND;

COMMAND c;

c.assign.subject = ENERGY; /* assign using fields */
.
.
c.assign.src = CPU;

transmit_to_fpga(c.word); /* transmit as 32 bit word */
```

11.15.3.2 cdatas - generate a C name array for 3PL enum type

Library: ctypes.3pl

Prototype:

```
global function
cdatas (type(t), str(name)) {}
```

Return mode and type:

immediate, "str"

Input parameters:

t is the enum type.

name is the identifier to be used for the C string array.

description:

This function takes a 3PL variable of type **type** whose value is type **enum** and returns a string defining a C data object that matches the given type. The object has the C name given by **name**. Presently this function only operates on enum types, where the C object is a NULL-terminated string table formed from the enum values. The string array can be indexed by an enum ordinal to get the associated identifier string.

The following 3PL code -

```
enum(e, V1, V2, V3);
println(cdatas(e, "vname"));
```

would print -

```
static const char *vname[] __attribute__((unused)) = {
    "V1",
    "V2",
    "V3",
    NULL
};
```

11.15.4 library classes

None.

11.16 Floating Point Library - float.3pl

This library provides floating point modules, procedures and functions for all Xilinx FPGAs.

This library must be incorporated by using a **module** statement, i.e.

```
module "float.3pl"
```

This library was developed for a specific project, but it was soon determined that fixed point operations were far better suited and used far fewer logic resources. Hence a fixed point target type was incorporated fully into 3PL and this floating point library has never been used in earnest. It passed early testing but probably needs much more development.

11.16.1 library modules

11.16.1.1 fadd - floating add

Library: float.3pl

Prototype:

```
global module
fadd (queue(out) <- value(in1), value(in2)) {}
```

Output parameters:

out is the output queue variable.

Input parameters:

in1 is the first input value.

in1 is the second input value.

description:

This module adds the two input floating point values and assigns the sum to the output queue. The input values and the output queue must be type **float**. The input exponent and mantissa widths may differ from each other and from those of the output format. The module will assign data to the output queue when both inputs and the output are available.

The latency through this module is $5 + \lceil \log_2 m \rceil + r$ where m is the number of bits in the mantissa and r is 1 if either of the input formats differs to the output format and 0 otherwise.

Overflow and underflow are not handled.

11.16.1.2 fdiv - floating divide

Library: float.3pl

Prototype:

```
global module
fdiv (queue(out) <- value(in1), value(in2)) {}
```

Output parameters:

out is the output queue variable.

Input parameters:

in1 is the first input value.

in1 is the second input value.

description:

This module divides the first input floating point value by the second and assigns the quotient to the output queue. The input values and the output queue must be type **float**. The input exponent and mantissa widths may differ from each other and from those of the output format. The module will assign data to the output queue when both inputs and the output are available.

Overflow and underflow are not handled.

This module has not yet been written

11.16.1.3 fmul - floating multiply

Library: float.3pl

Prototype:

```
global module
fmul (queue(out) <- value(in1), value(in2)) {}
```

Output parameters:

out is the output queue variable.

Input parameters:

in1 is the first input value.

in1 is the second input value.

description:

This module multiplies the two input floating point values and assigns the product to the output queue. The input values and the output queue must be type **float**. The input exponent and mantissa widths may differ from each other and from those of the output format. The module will assign data to the output queue when both inputs and the output are available.

The latency through this module is $2 + \lceil \log_2 m \rceil + r$ where m is the number of bits in the mantissa and r is 1 if either of the input formats differs to the output format and 0 otherwise.

Overflow and underflow are not handled.

11.16.1.4 fneg - negate a floating point value

Library: float.3pl

Prototype:

```
global module
fneg (queue(out) <- value(in)) {}
```

Output parameters:

out is the output queue variable.

Input parameters:

in is the input value.

description:

This module negates the input floating point value and assigns the result to the output queue. The input value and the output queue must be type **float**. The input exponent and mantissa widths may differ from those of the output format. The module will assign data to the output queue when both input and output are available.

The latency through this module is 2 if the input format differs to that of the output and 1 otherwise.

This module simply changes the sign bit, including the

11.16.1.5 fsub - floating subtract

Library: float.3pl

Prototype:

```
global module
fsub (queue(out) <- value(in1), value(in2)) {}
```

Output parameters:

out is the output queue variable.

Input parameters:

in1 is the first input value.

in2 is the second input value.

description:

This module subtracts the second input floating point value from the first and assigns the difference to the output queue. The input values and the output queue must be type **float**. The input exponent and mantissa widths may differ from each other and from those of the output format. The module will assign data to the output queue when both inputs and the output are available.

The latency through this module is $5 + \lceil \log_2 m \rceil + r$ where m is the number of bits in the mantissa and r is 1 if either of the input formats differs to the output format and 0 otherwise.

Overflow and underflow are not handled.

11.16.1.6 itof - convert an integer to a floating point value

Library: float.3pl

Prototype:

```
global module
itof (queue(out) <- value(in)) {}
```

Output parameters:

out is the output queue variable.

Input parameters:

in is the input queue variable.

description:

This module converts the integer in the input argument to a floating point value which is assigned to the output queue. The input value must be type **int** or **uint** and the output queue must be type **float**. The module will assign data to the output queue when both input and output arguments are available.

The latency through this module is $1 + \lceil \log_2 m \rceil$ where m is the number of bits in the mantissa.

Overflow is not handled, but is very unlikely to occur in converting integer to floating point.

11.16.2 library procedures

None.

11.16.3 library functions

None.

11.16.4 library classes

None.

11.17 JSON type Library - jsontype.3pl

This library provides a conversion between a 3PL type and a JSON string.

This library must be incorporated by using a **module** statement, i.e.

```
module "jsontype.3pl"
```

11.17.1 library modules

none.

11.17.2 library procedures

None.

11.17.3 library functions

11.17.3.1 jsontype - generate a JSON string from a 3PL type

Library: jsontype.3pl

Prototype:

```
global function
jsontype (type(t), int(indent)) {}
```

Return mode and type:

immediate, "str"

Input parameters:

t is the 3PL type.

indent is the level of indenting of the formatted string, default 0.

description:

This function takes a 3PL type and generates a JSON description as a single returned string. The string contains leading indenting and newlines as required so that the resulting string is formatted to be easily human-readable.

11.17.4 library classes

None.

11.18 General UART Library File - uart.3pl

This library provides a general Universal Asynchronous Receiver/Transmitter which can be dynamically configured. For a simple UART see the next chapter, "Simple UART library file - uarts.3pl". These modules can only be used where the current clock is 100MHz, 50MHz or 16MHz.

This library must be incorporated by using a **module** statement -

```
module "uart.3pl"
```

11.18.1 global variables

A **value** mode variable is created for each baud rate. These are type **log** and oscillate at 16 times the baud rate. The variables are -

uart_brc16_500000	uart_brc16_460800	uart_brc16_307200
uart_brc16_230400	uart_brc16_153600	uart_brc16_115200
uart_brc16_76800	uart_brc16_57600	uart_brc16_38400
uart_brc16_19200	uart_brc16_9600	uart_brc16_4800
uart_brc16_2400	uart_brc16_1200	uart_brc16_600
uart_brc16_300		

Two **static** mode variables are created, one a millisecond clock, the other a microsecond clock. These are **uart_brc_us** and **uart_brc_ms** and are type **log**.

11.18.2 library modules

11.18.2.1 uart_tx - general UART transmitter

Library: uart.3pl

Prototype:

```
global module
uart_tx (
    value({txd, txbusy}, "log")
    value(brc, "log")
    <-
    queue(in, "bits:8"),
    value(brc16, "log"),
    value(txen, "log", true),
    value(cts, "log"),
    value(csize, "uint:2", 3),
    value(parenb, "log")
    value(parodd, "log", false)
    value(cstopb, "log")
) {}
```

Output parameters:

txd is the serial output.

brc is the baud rate clock.

Input parameters:

in is the 8-bit input.

br16 is the 16 times baud rate clock.

txen is transmit enable.

cts is 'clear to send'.

csize is 0 to 2 for 6, 7 or 8 bits.
parenb is parity enable.
parodd selects odd parity.
cstopb adds a second stop bit.

description:

A UART output section. **txd** is the transmit output, usually connected direct to an output pin.

in is the input data, always 8 bits.

Transmission occurs only if **txen** and **cts** (if connected) are asserted. If either go false the transmitter will stop at the end of the current character. **cts** usually connects to an output pin. **txbusy** indicates the transmitter is operating.

brc16 is a clock enable signal at 16 times the output baud rate. **csize** is the char size (0 = 5 bits, 1 = 6 bits, 2 = 7 bits, 3 = 8 bits). If PARENb is true, an even parity bit is added, odd if **parodd** is also true. If **cstopb** is true, a second stop bit is added. It is unsafe to change any of these while **txbusy** is high.

brc is the internal baud clock, 1/16 the rate of **brc16**, and is brought out in case anything wants it

The default is 8-bit characters, no parity, 1 stop bit.

Flow-control (**cts**) has not yet been tested. **uart_tx** cannot produce BREAK.

11.18.2.2 uart_rx - general UART receiver

Library: uart.3pl

Prototype:

```
global module
uart_rx (
    queue(out, "bits:8"), value({oferr, operr, overflow, rts}, "log")
<-
    value({rxd, brc16}, "log"),
    value(rxen, "log", true),
    value(csize, "uint:2", 3),
    value(parenb, "log")
    value(parodd, "log", true)
    value(cstopb, "log")
) {}
```

Output parameters:

out is the 8-bit output output.
oferr indicates a framing error.
operr indicates a parity error.
overflow indicates an output queue buffer overflow.
rts is 'ready to send'.

Input parameters:

rxd is the serial input.
br16 is the 16 times baud rate clock.
rxen is receive enable.
parenb is parity enable.
parodd selects odd parity.
cstopb expects a second stop bit.

description:

A UART receiver section. **rxd** is the receive input, usually connected direct to an input pin.

oferr is true if there was a framing error, and **operr** indicates a parity error (can only occur if **parenb** is true). **oferr** and **operr** are valid whenever out is available.

Optional input **rxen** if false will disable the receiver when the current character is complete. Output **rts** is true when the receiver will accept a character. It goes false when **rxen** is false or a received character was not acknowledged before the expected start of the next character.

brc16, **csize**, **parenb**, **parodd** and **cstopb** have the same meanings as in **uart_out()**.

Flow-control (**rts**) is supported but has not been tested. BREAK should be detect but is not.

11.18.3 library procedures

11.18.3.1 uart_brc16 - create baud rate clocks

Library: uart.3pl

Prototype:

```
global procedure
uart_brc16 () {
```

Output parameters:

None.

Input parameters:

None.

description:

This procedure should be called once in each UART installation. It generates the global 16 x baud rate clock signals from the given input clock, which must be the same clock given to the calls to **uart()**.

11.18.4 library functions

None.

11.18.5 library classes

None.

11.19 Simple UART Library File - uarts.3pl

This library provides a simple Universal Asynchronous Receiver/Transmitter whose configuration is set at compile time. For a UART which can be dynamically reconfigured see the previous chapter "General UART library file - uart.3pl". The number of data bits is determined from the size of the data input or output argument.

This library must be incorporated by using a **module** statement -

```
module "uart.3pl"
```

For a baud rate br and a current clock frequency f_c , $br < f_c/32$ unless $br = f_c/16$. Serial data sampling is controlled by a counter which runs at approximately 16 times the baud rate. The error on the rate of this counter results in a cumulative error which has most impact on the sampling of the last serial data bit. If the sampling tolerance exceeds 1/16 of the bit width of the last bit a warning message is printed. The calculated sampling tolerance includes both the error in the sampling counter and an allowance of plus or minus 100ppm in the clocks at either end.

11.19.1 library modules

11.19.1.1 subsecond_clocks - millisecond and microsecond clocks

Library: uarts.3pl

Prototype:

```
global module
subsecond_clocks (value({ms, us}, "log") <- ) {}
```

Output parameters:

ms is a millisecond clock.

us is a microsecond clock.

Input parameters:

none

description:

Outputs a millisecond clock and a microsecond clock.

11.19.1.2 uart_rx - simple UART receiver

Library: uarts.3pl

Prototype:

```
global module
uart_rx (
  queue(out), value({oferr, operr, overflow, rts}, "log")
  <-
  value(rxd, "log"),
  int(br),
  value(rxn, "log", true),
  log(parenb),
  log(parodd, true),
  log({cstopb, tq})
) {}
```

Output parameters:

out is the 8-bit output output.

oferr indicates a framing error.

operr indicates a parity error.

overflow indicates an output queue buffer overflow.

rts is 'ready to send'.

Input parameters:

rx is the serial input.

br is the baud rate.

rxen is receive enable.

parenb is parity enable.

parodd selects odd parity.

cstopb expects a second stop bit.

description:

A UART receiver section. **rx** is the receive input, usually connected direct to an input pin.

out is the output data. The size of the argument to this parameter determines the number of bits to be received.

oferr is true if there was a framing error, and **operr** indicates a parity error (can only occur if **parenb** is true). **oferr** and **operr** are valid whenever out is available.

Optional input **rxen** if false will disable the receiver when the current character is complete. Output **rts** is true when the receiver will accept a character. It goes false when **rxen** is false or a received character was not acknowledged before the expected start of the next character.

br is the baud rate.

If **parenb** is true, an even parity bit is added, odd if **parodd** is also true. If **cstopb** is true, a second stop bit is added.

tq if true shifts the data sampling point from the middle of the serial bit period to the 3/4 point. This is useful where an open collector driver is the serial data source as the -ve transition of the start bit and any data bits is fast but +ve transitions rely on the pullup resistor and may be slow. Thus the start bit starts the data sampling at the right time but later slow +ve transitions on data bits dictate later sampling.

Flow-control (**rts**) is supported but has not been tested. BREAK should be detect but is not.

11.19.1.3 uart_tx - simple UART transmitter

Library: uarts.3pl

Prototype:

```
global module
uart_tx (
    value({txd, txbusy}, "log"),
    value(brc, "log")
    <-
    value(in),
    int(br),
    value(txen, "log", true),
    value(cts, "log", true),
    log(parenb),
    log(parodd, false),
    log(cstopb)
) {}
```

Output parameters:

txd is the serial output.

brc is the baud rate clock.

Input parameters:

in is the input.

br is the baud rate.

txen is transmit enable.

cts is 'clear to send'.

csize is the number of data bits.

parenb is parity enable.

parodd selects odd parity.

cstopb adds a second stop bit.

description:

A UART output section. **txd** is the transmit output, usually connected direct to an output pin.

in is the input data. The size of **in** determines the number of bits transmitted. If the data input is a queue or an expression containing a queue, transmission will only occur when the input is available.

Transmission occurs only if **txen** and **cts** (if connected) are asserted. If either go false the transmitter will stop at the end of the current character. **cts** usually connects to an output pin. **txbusy** indicates the transmitter is operating.

br is the baud rate.

If **parenb** is true, an even parity bit is added, odd if **parodd** is also true. If **cstopb** is true, a second stop bit is added.

brc is the internal baud clock and is brought out in case anything wants it.

The default is 8-bit characters, no parity, 1 stop bit.

Flow-control (**cts**) has not yet been tested. **uart_tx** cannot produce BREAK.

11.19.2 library procedures

None.

11.19.3 library functions

None.

11.19.4 library classes

None.

11.20 External QDR II memory Library - qdr2_ram.3pl

This library provides read and write interfaces for quad data rate (QDR II) synchronous static RAM.

This library must be incorporated by using a **module** statement -

```
module "qdr2_ram.3pl"
```

The RAM address and data dimensions are inferred from the dimensions of the address and data FPGA pin arguments supplied to the class call.

Multiple read and write ports are allowed. Addresses and data are transferred using queue mode variables and simultaneous requests are handled by arbitration based on fair access or priority - see below.

The interfaces to a single external RAM chip are incorporated in an instance of a class. Reads and writes are provided by modules in that class.

11.20.1 library modules

None.

11.20.2 library procedures

None.

11.20.3 library functions

None.

11.20.4 library classes

11.20.4.1 class qdr2_ram

Prototype:

```
global class
qdr2_ram (
    var(addr\_pins, "[str]"),
    var(data\_pins, "[str]"),
    var(q\_pins, "[str]"),
    str(\_r\_pin),
    str(\_w\_pin),
    str(\_bw0\_pin),
    str(\_bw1\_pin),
    str(\_doff\_pin),
    str(cq\_pin),
    str(\_cq\_pin),
    str(k\_pin),
    str(\_k\_pin),
    clock(clk_0),           // clock domain for the interface
    clock(clk_90),         // clk_0 delayed 90 degrees
    clock(del_clk),        // nominally 200MHz clock for IDELAY timing
    str(slew, "slow"),     // output pad slew rate
    int(drive, 2),         // output pad drive current - 2mA
```

```

    str(tnm, "QDRPAD") // pad timing group name
    int( qdel, 0),      // Q input delay
    int( cqdel, 12),    // CQ input delay
    int(_cqdel, 12),    // _CQ input delay
    str(cqbufloc),      // CQ input buffer location - probably not necessary
    str(_cqbufloc)      // _CQ input buffer location - probably not necessary
) {}

```

Output parameters:

none.

Input parameters:

addr_pins are the RAM address FPGA output pins

data_pins are the RAM write data FPGA output pins

q_pins are the RAM read data FPGA input pins

_r_pin is **_w_pin** is **_bw0_pin** is **_bw1_pin** is **_doff_pin** is the

cq_pin is **_cq_pin** is **k_pin** is **_k_pin** is **clk_0** is the clock domain of the RAM interface

clk_90 is the clock domain of the RAM interface

del_clk is the clock domain of the RAM interface

slew is the output pad slew rate

drive is the output pad drive current in mA

tnm is a timing group name applied to all pads connected to this interface.

qdel is the default input delay for the Q input

cqdel is the default input delay for the CQ input

_cqdel is the default input delay for the _CQ input

cqbufloc is an optional location for the CQ input buffer - probably not necessary. **_cqbufloc** is an optional location for the _CQ input buffer - probably not necessary.

It is presumed that the external RAM chip has a clock source which is also input to the FPGA and that argument **clk_0** is the FPGA version of that clock. **clk_0** is the clock domain in which the interface logic to the QDR II RAM is to operate. If this argument is omitted the current clock domain is assumed.

Argument **clk_90** is a clock delayed 90 degrees from **clk_0** (or from the current clock if **clk_0** is omitted).

Argument **del_clk** must be an approximately 200MHz clock to control IDELAY element input signal delays. The **clk_0**, **clk_90** and **del_clk** arguments must be provided.

Optional arguments **slew** ("slow" or "fast" and **drive** (2, 4, 6, 8, 12, 16 or 24 mA) specify the slew rate and drive current for FPGA output pads.

Optional argument **tnm** specifies a timing group name for the pads of the interface (default "QDRPAD"),

Optional arguments **qdel**, **cqdel** and **_cqdel** specify input delays for the Q, CQ and _CQ inputs from the QDR RAMs. These can be adjusted from the default values to centre the clocks in the data 'eyes' if need be - see class procedures **incr_qdel()**, **decr_qdel()**, **reset_qdel()**, **incr_cqdel()**, **decr_cqdel()**, **reset_cqdel()**, **incr__cqdel()**, **decr__cqdel()**, **reset__cqdel()** below.

It is assumed that there is no data skew and hence all data inputs have the same delay.

Optional arguments **scqbufloc** and **_cqbufloc** are location strings for the CQ and _CQ BUFIO input buffers but are probably not necessary.

There is no provision for digitally controlled impedance in this interface. If this is necessary it must be provided separately.

It may prove advantageous to provide locations constraints for clock BUFIOs but this is probably not necessary and has not.

The following is an example of creation of a class instance -

```

class(qdr2);
qdr = qdr2_ram(
    addr_pins=ext_ram_a_pins[0],

```

```

        data_pins=ext_ram_d_pins[0],
        q_pins=ext_ram_q_pins[0],
        _r_pin=_ext_ram_r_pins[0],
        _w_pin=_ext_ram_w_pins[0],
        _bw0_pin=_ext_ram_bw0_pins[0],
        _bw1_pin=_ext_ram_bw1_pins[0],
        _doff_pin=_ext_ram_doff_pins[0],
        cq_pin=ext_ram_cq_pins[0],
        _cq_pin=_ext_ram_cq_pins[0],
        k_pin=ext_ram_k_pins[0],
        _k_pin=_ext_ram_k_pins[0],
        clk_0_pin=ext_ram_clk_0,
        clk_90_pin=ext_ram_clk_90,
        del_clk=c200
    );

```

11.20.4.2 class module read - read from external QDR2 memory class module readp - read from external QDR2 memory with priority

Library: xilinx/hymod/qdr2_ram.3pl

Prototypes:

```

global module
read (queue(data) <- queue(address))) {}

```

```

global module
readp (queue(data) <- int(pri), queue(address)) {}

```

Output parameters:

data is the queue receiving the data read from the memory.

Input parameters:

pri is the priority of the read request.

addr is the address to be read.

description:

When the output and input queues are available an external memory read will be executed, subject to access arbitration. The address input queue and the data output queue may be any type, but a crude pad/truncate cast to the address and data primitive sizes will be performed.

The internal code of this module executes in clock domain **clk_0** (but the call to the module returns with the current clock domain restored). If the clock domains in which input queue *addr* is written or output queue *data* is read are not the c166 clock domain then these queues will be asynchronous, i.e. their read and write clock domains will be different. Any other source to queue *data* must be in the c166 clock domain.

For **read()** access arbitration among competing read requests is based on a fair strategy. Read requests are funnelled to a single queue via a binary tree of queues. At each node conflicting requests are resolved in favour of one of the two inputs by a scheme which alternates priority between the inputs based on the number of tree leaves on each side.

For **readp()** access is subject to access priority, priority order being given by input parameter **pri**. Lower numbers are higher priority. The priorities do not need to form a contiguous set but the same priority must not be used on more than one call to **readp()**.

For one bank of external RAM, calls to **read()** and **write()** cannot be mixed with calls to **readp()** and **writep()**.

11.20.4.3 class module write - write to external QDR2 memory class module writep - write to external QDR2 memory with priority

Library: xilinx/hymod/qdr2_ram.3pl

Prototype:

```
global module
write (queue(addrout) <- varargs) {}
```

Prototype:

```
global module
writep (queue(addrout) <- int(pri), varargs) {}
```

Output parameters:

addrout is optional and is the address of the datum that has been written.

Input parameters:

pri is the priority of the write request.

varargs expects either two arguments, the address and the data, or a single argument which is a struct containing both address and data.

description:

The two forms allow the address and data to be given in a single queue or as two separate queues or expressions.

For the first form the address and data are contained in a struct with fields **address** and **data**.

For the second form the address and data are separate arguments which may be any type, but a crude pad/truncate cast to the address and data primitive sizes will be performed. At least one of the input arguments must contain a queue read.

When the input queue or queues is/are available an external memory write will be executed, subject to access arbitration.

If the output parameter is passed a queue variable argument the write address is written to the queue when the memory write occurs. This can be used to verify that the write has occurred and at what address. This might be used for example where the external memory is being used as a FIFO and the read address must not pass the write address. The output argument may be a queue of type "null" if write notification is required but the address value is not of interest.

The internal code of this module executes in clock domain **clk_0** (but the call to the module returns with the current clock domain restored). If the clock domains in which input queues *addr* and *data* are written are not the external memory clock domain then these queues will be asynchronous, i.e. their read and write clock domains will be different. Any other sinks to these queues must be in the c166 clock domain.

For **write()** access arbitration among competing write requests is based on a fair strategy. Access requests are funnelled to a single queue via a binary tree of queues. At each node conflicting requests are resolved in favour of one of the two inputs by a scheme which alternates priority between the inputs based on the number of tree leaves on each side.

For **writep()** access is subject to access priority, priority order being given by input parameter **pri**. Lower numbers are higher priority. The priorities do not need to form a contiguous set but the same priority must not be used on more than one call **writep()**.

For one bank of external RAM, calls to **read()** and **write()** cannot be mixed with calls to **readp()** and **writep()**.

11.20.4.4 **class procedure incr_qdel - increment Q input delay** **class procedure decr_qdel - decrement Q input delay** **class procedure reset_qdel - reset Q input delay**

Library: xilinx/hymod/qdr2_ram.3pl

Prototype:

<code>procedure incr_qdel () {}</code>
<code>procedure decr_qdel () {}</code>
<code>procedure reset_qdel () {}</code>

Output parameters:

none.

Input parameters:

none.

description:

Increment, decrement or reset the input delay of the **Q** signal from the RAM. This procedure is provided to allow the Q signal delay to be altered during execution of test code to position data signals within the clock eye.

11.20.4.5 **class procedure incr_cqdel - increment CQ input delay** **class procedure decr_cqdel - decrement CQ input delay** **class procedure reset_cqdel - reset CQ input delay**

Library: xilinx/hymod/qdr2_ram.3pl

Prototype:

<code>procedure incr_cqdel () {}</code>
<code>procedure decr_cqdel () {}</code>
<code>procedure reset_cqdel () {}</code>

Output parameters:

none.

Input parameters:

none.

description:

Increment, decrement or reset the input delay of the **CQ** signal from the RAM. This procedure is provided to allow the CQ signal delay to be altered during execution of test code to position data signals within the clock eye.

11.20.4.6 **class procedure incr__cqdel - increment _CQ input delay** **class procedure decr__cqdel - decrement _CQ input delay** **class procedure reset__cqdel - reset _CQ input delay**

Library: xilinx/hymod/qdr2_ram.3pl

Prototype:

<code>procedure incr__cqdel () {}</code>
<code>procedure decr__cqdel () {}</code>
<code>procedure reset__cqdel () {}</code>

Output parameters:

none.

Input parameters:

none.

description:

Increment, decrement or reset the input delay of the **_CQ** signal from the RAM. This procedure is provided to allow the **_CQ** signal delay to be altered during execution of test code to position data signals within the clock eye.

11.21 External ZBT RAM interface - zbt_ram.3pl

This library provides read and write interfaces for zero bus turnaround (ZBT) synchronous static RAM.

This library must be incorporated by using a **module** statement -

```
module "zbt_ram.3pl"
```

The RAM address and data dimensions are inferred from the dimensions of the address and data FPGA pin arguments supplied to the class call.

Multiple read and write ports are allowed. Addresses and data are transferred using queue mode variables and simultaneous requests are handled by arbitration based on fair access or priority - see below.

The interfaces to a single external RAM chip are incorporated in an instance of a class. Reads and writes are provided by modules in that class.

11.21.1 library modules

none.

11.21.2 library procedures

None.

11.21.3 library functions

None.

11.21.4 library classes

11.21.4.1 class zbt_ram

Prototype:

```
global class
zbt_ram (
    var(addr_pins, "[]str"),
    var(data_pins, "[]str"),
    str(_we_pin),
    str(_wea_pin),
    str(_web_pin),
    str(_wec_pin),
    str(_wed_pin),
    str(_cke_pin),
    clock(clk),
    str(slew, "fast"),
    int(drive, 12),    // 12mA
    str(tnm, "ZBTPAD")
) {}
```

Output parameters:

none.

Input parameters:

addr_pins are the RAM address FPGA output pins

data_pins are the RAM data FPGA bi-directional pins

_we_pin is the RAM write enable FPGA output pin

_wea_pin is the RAM byte A write enable FPGA output pin

_web_pin is the RAM byte B write enable FPGA output pin

_wec_pin is the RAM byte C write enable FPGA output pin

_wed_pin is the RAM byte D write enable FPGA output pin

_cke_pin is the RAM clock enable FPGA output pin

clk is the clock domain of the RAM interface

slew is the output pad slew rate ("fast", "slow")

drive is the output pad current drive (2 to 24 mA)

tnm is the timing group name for the pads

It is presumed that the external RAM chip has a clock source which is also input to the FPGA and that **clk** is the FPGA version of that clock. If that argument is missing the current clock domain is used for the interface logic.

The output pad slew rate and drive current are optional arguments which if omitted have the default values shown.

The timing group name **tnm** is applied to all input and output pads connected to this interface.

The following is an example of creation of a class instance -

```
class(zbt);
zbt = zbt_ram(
    addr_pins=ext_ram_a_pins[0],
    data_pins=ext_ram_d_pins[0],
    _we_pin=_ext_ram_we_pins[0],
    _wea_pin=_ext_ram_wea_pins[0],
    _web_pin=_ext_ram_web_pins[0],
    _wec_pin=_ext_ram_wec_pins[0],
    _wed_pin=_ext_ram_wed_pins[0],
    _cke_pin=_ext_ram_cke_pins[0],
    clk=c166
);
```

11.21.4.2 class module read - read from external ZBT memory class module readp - read from external ZBT memory with priority

Library: zbt_ram.3pl

Prototypes:

```
module
read (queue(data) <- queue(address)) {}
```

```
module
readp (queue(data) <- int(pri), queue(address)) {}
```

Output parameters:

data is the queue receiving the data read from the memory.

Input parameters:

pri is the priority of the read request.

addr is the address to be read.

description:

When the output and input queues are available an external memory read will be executed, subject to access arbitration. The address input queue and the data output queue may be any type, but a crude pad/truncate cast to the address and data primitive sizes will be performed.

The internal code of this module executes in clock domain **clk** (but the call to the module returns with the current clock domain restored). If the clock domains in which input queue *addr* is written or output queue *data* is read are not the external memory clock domain then these queues will be asynchronous, i.e. their read and write clock domains will be different.

For **read()** access arbitration among competing read and write requests is based on a fair strategy (but see the final paragraph below). Access requests are funnelled to a single queue via a binary tree of queues. At each node conflicting requests are resolved in favour of one of the two inputs by a scheme which alternates priority between the inputs based on the number of tree leaves on each side.

For **readp()** access is subject to access priority, priority order being given by input parameter **pri**. Lower numbers are higher priority. The priorities do not need to form a contiguous set but the same priority must not be used on more than one call to **readp()** or **writep()**.

Calls to **read()** and **write()** cannot be mixed with calls to **readp()** and **writep()**.

11.21.4.3 class module write - write to external ZBT memory class module writep - write to external ZBT memory with priority

Library: **zbt_ram.3pl**

Prototype:

```
global module
write (queue(addrout) <- varargs) {}
```

Prototype:

```
global module
writep (queue(addrout) <- int(pri), varargs) {}
```

Output parameters:

addrout is optional and is the address of the datum that has been written.

Input parameters:

pri is the priority of the write request.

varargs expects either two arguments, the address and the data, or a single argument which is a struct containing both address and data.

description:

The two forms allow the address and data to be given in a single queue or as two separate queues or expressions.

For the first form the address and data are contained in a struct with fields **address** and **data** that is defined in the class -

```
struct(in_struct,
      address,   addr_t,
      data,      data_t,
      write,     "log"
);
```

(the **write** field is assigned internally and can be ignored.)

For the second form the address and data are separate arguments which may be any type, but a crude pad/truncate cast to the address and data primitive sizes will be performed. At least one of the input arguments must contain a queue read.

When the input queue or queues is/are available an external memory write will be executed, subject to access arbitration.

If the output parameter is passed a queue variable argument the write address is written to the queue when the memory write occurs. This can be used to verify that the write has occurred and at what address. This might be used for example where the external memory is being used as a FIFO and the read address must not pass the write address. The output argument may be a queue of type "null" if write notification is required but the address value is not of interest.

The internal code of this module executes in clock domain **clk** (but the call to the module returns with the current clock domain restored). If the clock domains in which input queues *addr* and *data* are written are not the external memory clock domain then these queues will be asynchronous, i.e. their read and write clock domains will be different. Any other sinks to these queues must be in the external memory clock domain.

For **write()** access arbitration among competing read and write requests is based on a fair strategy (but see the final paragraph below). Access requests are funnelled to a single queue via a binary tree of queues. At each node conflicting requests are resolved in favour of one of the two inputs by a scheme which alternates priority between the inputs based on the number of tree leaves on each side.

For **writep()** access is subject to access priority, priority order being given by input parameter **pri**. Lower numbers are higher priority. The priorities do not need to form a contiguous set but the same priority must not be used on more than one call to **readp()** or **writep()**.

Calls to **read()** and **write()** cannot be mixed with calls to **readp()** and **writep()**.

11.22 SPI Library File - spi.3pl

SPI is a simple, synchronous serial protocol used to control peripheral chips through a four-wire bus, at typical speeds of 1Mbit/s. SPI is more or less defined in Section 8 of the Motorola MC68HC11 Reference Manual. A subset of SPI is known as Microwire.

This implementation provides separate modules for SPI master and slave. Each module can receive and transmit, although in many cases it will only do one or the other. The SPI master is the initiator of all transfers. It generates the synchronous clock (**SCK**) and chip select (**_nSS**) signals, and drives the master-out-slave-in (**MOSI**) data line. The slave drives the master-in-slave-out (**MISO**) data line.

Often there will be one master and one slave. However, single-master, multi-slave situations are common. In this case, the master drives multiple chip-select signals (**_nSS**), one per slave, and **SCK**, **MOSI** and **MISO** are common. Slaves must in these cases use tri-state or open-collector logic to drive MISO. Multi-master operation is possible, but requires separate arbitration arrangements between masters, usually performed at a software level.

Peripheral chip manufacturers often alter aspects of the interface, for example, inverting signals, using wired-or or tri-state logic, omitting **_nSS**, using a common wire for **MOSI/MISO**, and so on. This implementation should still be useful in these cases, using some additional glue logic to account for the differences.

This library must be incorporated by using a **module** statement -

```
module "spi.3pl"
```

11.22.1 library modules

11.22.1.1 spi_master - SPI master transmitter/receiver

Library: spi.3pl

Prototype:

```
global module spi_master(
    static(SCK, "log"),
    static(MOSI, "log"),
    static(_NSS, "log"),
    queue(rx)
<-
    queue(tx),
    value(MISO, "log", false),
    value(timebase, "log", true),
    log(chpa, false)
) {}
```

Output parameters:

SCK is the output data clock.

MOSI is the master data output to a slave.

_NSS is a slave chip select.

rx is the data output queue for the input data from MISO.

Input parameters:

tx is the data input queue for the output data to MOSI.

MISO is the master data input from a slave.

timebase is the .

chpa controls the clock sense.

description:

SPI master transmitter/receiver. Data written to the tx queue is serialised and clocked out to the **MOSI**

(master-out slave-in) output in synchronism with the generated **SCK** output. Simultaneously, input data on **MISO** is sampled, deserialised and written to the **rx** queue. **_NSS** goes low for the duration of the transmission. **SCK**, **MOSI**, **MISO** and **_NSS** are normally routed to FPGA pins for connection to SPI peripherals.

The data word size is the larger of that of the rx and tx queues. Generally these will be the same. Where the receive size is larger, transmit words will be zero-padded at trailing end. Where the transmit size is larger, the only the trailing receive bits will be returned.

Timing is regulated by the **timebase** signal. Normally this is a stream of single-clock pulses at twice the SPI baud rate; SPI peripheral chips normally run at 1Mbaud or lower. If the timebase signal is tied true, the baud rate will be one quarter the prevailing clock rate, and if the timebase parameter is omitted, it will be one half. Note that SPI slaves may not work reliably at high speeds.

Input **chpa** is an immediate input which controls the phase of **SCK**. When false (the default), data will be sampled on the rising edge of **SCK**. When true, the falling edge is used. (Note this is not the same as inverting **SCK**).

The master may be configured transmitter-only by omitting the **rx** queue and **MISO** input. This is a common situation. Similarly, the master may be configured receiver-only by making the type of the **tx** queue null. In this case (empty) writes to the **tx** queue initiate receives.

If, following a receive, the **rx** queue is not available, the received data word is silently discarded.

11.22.1.2 spi_slave - SPI slave transmitter/receiver

Library: spi3pl

Prototype:

```
global module spi_slave(
    value(MISO, "log"),
    queue(rx),
    static(SOSI, "log")
<-
    value(tx),
    value(SCK, "log", false),
    value(MOSI, "log", false),
    value(_NSS, "log", true),
    log(chpa, false)
) {}
```

Output parameters:

MISO is the data output to the master.

rx is the output queue for data received from the master.

SOSI is an optional output to a daisy-chained slave.

Input parameters:

tx is the data input for transmission to the master.

SCK is the input clock from the master.

MOSI is the data input from the master.

_NSS is the slave chip select from the master.

chpa controls the clock sense.

description:

SPI slave transmitter/receiver. When the connected SPI master initiates a transfer, the value of input **tx** is sampled, serialised and clocked out to the **MISO** (master-out slave-in) output in synchronism with the received **SCK** input. Simultaneously, input data on **MOSI** is sampled, deserialised and written to the **rx** queue. **_NSS** must be low throughout the operation.

Input **chpa** controls the phase of the expected **SCK**. The meaning is as described in module spi_master() (not implemented yet).

The data word length is determined by the connected SPI master, although generally the slave will match. However, the word length may be dynamic, especially in a multi-master system.

The slave may be configured as a transmitter only by omitting the **rx** queue and **MOSI** input. Similarly, the slave may be configured as a receiver only by omitting the **tx** parameter.

If, during a receive, the **rx** queue is not available (i.e., full), the received data word is silently discarded.

In the unlikely event slave daisy-chaining is desired, the **SOSI** output can be connected to the **MOSI** input of subsequent slaves. **SOSI** is the output of the receive shift-register, and so is the same as the slave's **MOSI** input but delayed by the bit-width of the receiver. **This feature is untested.**

11.22.2 library procedures

None.

11.22.3 library functions

None.

11.22.4 library classes

None.

11.23

AVNET Spartan3A Evaluation Board Header File - boards/avnet/aes_sp3a_eval400.3pl

This library must be incorporated by using a **module** statement -

```
module "boards/avnet/aes_sp3a_eval400.3pl"
```

A 16MHz clock is available as **c16**. The clock source is a Maxim silicon oscillator hence the frequency is not accurate and displays short term variation.

The header code sets the current clock as **c16**.

The following pins are defined (as strings for single pins and arrays of strings otherwise) -

name	pins	
c12_pin		12MHz clock input - seems to be 3MHz!
c16_pin		16MHz clock input
c32_pin		32MHz clock input - doesn't seem to work!
j6_pins	4	4 pin diligent header J6
j7_pins	4	4 pin diligent header J7
j4_r0_pins	17	row of 2 x 20 pin connector J4
j4_r1_pins	18	row of 2 x 20 pin connector J4
flash_ce_pin		flash CE
flash_oe_pin		flash OE
flash_we_pin		flash WE
flash_ry_by_pin		flash ready/busy
flash_word_pin		flash 8/16 bit mode selector
flash_reset_pin		flash reset
flash_addr_8	22	flash 8-bit mode address
flash_data_8	8	flash 8-bit mode data
flash_addr_16	21	flash 16-bit mode address
flash_data_16	16	flash 16-bit mode data
cap_button_pins	4	touch buttons
led_pins	4	LEDs

11.23.1 library modules

None.

11.23.2 library procedures

None.

11.23.3 library functions

None.

11.23.4 library classes

None.

11.24 AVNET Microzed Evaluation Board Header File - boards/avnet/microzed_70x0.3pl

These library must be incorporated by using a **module** statements -

```
module "boards/avnet/microzed_7010.3pl"
```

or

```
module "boards/avnet/microzed_7020.3pl"
```

The following is included -

```
source "boards/avnet/microzed_base.3pl"
```

which in turn includes -

```
module "xilinx/xc7.3pl"
source "xilinx/zynq_axi.3pl"
```

Module zynq_ps() is called as a trailer module (defined in xilinx/zynq_axi.3pl). It instantiates the Zynq CPU and connects up the DDR RAM and some general purpose I/O pins. It also derives a 100MHz clock from the CPU which is the bus clock and is available as **c100**.

The following are defined as I/O pins with standard "LVCMOS33" -

Test pins tp6_pin to tp20_pin("str").

The 50 pins on each of connector 1 and connector 2, con1_pins and con2_pins ("[50]str").

The I/O pins on each connector are grouped with GND and VCC pins between groups. For convenience the pin groups are also defined separately as con1_3_16_pins, con1_19_38_pins etc., each being an array of "str".

11.24.1 library modules

None.

11.24.2 library procedures

None.

11.24.3 library functions

None.

11.24.4 library classes

None.

11.25 Digilent Zybo Evaluation Board Header File - boards/digilent/zybo.3pl

These library must be incorporated by using a **module** statement -

```
module "boards/digilent/zybo.3pl"
```

The following are included -

```
module "xilinx/xc7.3pl"  
source "xilinx/zynq_axi.3pl"
```

Module zynq_ps() is called as a trailer module (defined in xilinx/zynq_axi.3pl). It instantiates the Zynq CPU and connects up the DDR RAM and some general purpose I/O pins. It also derives a 100MHz clock from the CPU which is the bus clock and is available as **c100**.

See header file for I/O pins.

11.25.1 library modules

None.

11.25.2 library procedures

None.

11.25.3 library functions

None.

11.25.4 library classes

None.

11.26 Xilinx ML5050 Virtex-5 Demonstration Board Header File - boards/xilinx/ml505.3pl

This library must be incorporated by using a **module** statement -

```
module "boards/xilinx/ml505.3pl"
```

This header file incorporates "xilinx/xc5v.3pl", "xilinx/common.3pl" and "stdlib.3pl".

This header file contains some useful pin definitions and device interfaces but is far from complete. What is there can be used. Thus far the initial code in the header file is not yet worth documenting!

11.26.1 library modules

None.

11.26.2 library procedures

None.

11.26.3 library functions

None.

11.26.4 library classes

None.

11.27 memec Board Header Files - boards/memec/v4fx12lc.3pl boards/memec/v4fx12mm.3pl

Description yet to be written.

11.28 papilio Spartan 3E Board Header Files - boards/gadgetfactory/papilio_250.3pl boards/gadgetfactory/papilio_500.3pl

These libraries must be incorporated by using a **module** statement -

```
module "boards/gadgetfactory/papilio_250.3pl"
```

or

```
module "boards/gadgetfactory/papilio_500.3pl"
```

This header file incorporates "xilinx/xc3s.3pl", "xilinx/common.3pl" and "stdlib.3pl".

These header files define the pins and create the 32MHz clock, c32.

If the following lines begin the source file

```
float(global.user_clock_ratio, 2.0);  
module "demo/.3pl"
```

clock **uclk** will also be established, in this case at half the **c32** frequency. Values allowed are 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 9.0, 10.0, 11.0, 12.0, 13.0, 14.0, 15.0 and 16.0.

The header code in both cases leaves the current clock as **c32** for the beginning of application code.

11.28.1 library modules

None.

11.28.2 library procedures

None.

11.28.3 library functions

None.

11.28.4 library classes

None.

Chapter 12 - Miscellaneous tools

The 3PL source directory **sources/tools** contains miscellaneous scripts and programs related to configuration file generation or CPU to FPGA interfaces. The **make** files for each of these installs them in **~/install/3pl**. In most cases the installation simply copies the files.

The following utility programs should be executed on the host computer -

```
bin2c
bin2py
info2c
info2py
edif2bit
run_ISE
run_Vivado
```

bin2c and **bin2py** can be run on the Zynq CPU however they require the .bit file as input. Similarly **info2c** and **info2py** can be run on the Zynq CPU but they require the .info file as input.

edif2bit, run_ISE and run_Vivado execute Xilinx software tools and hence cannot execute on the Zynq CPU.

The remaining tools are used by CPU code on the Zynq and hence should be copied to the Zynq CPU, probably again to **~/install/3pl** there -

```
cpu_serial.c
cpu_serial.h
cpu_serial.py
fpga_py_extension.c
fpga.c
fpga.h
fpga.py
mk_setup
setup.py
```

12.1 bin2c/

Type: Python3 script

Source: **sources/tools/bin2c**

Target: any

Execute: on host computer or on Zynq CPU

Convert a binary compressed FPGA configuration bit file to a C include file containing an initialised byte array as C source code. By this means the configuration can be included in the application program and can be loaded by the CPU application code without the need to read an external file during execution.

Execute by -

bin2c *design_name*

which will read *design_name.bit.gz* and produce a C include file *design_name.include.h*.

This is currently only applicable to the Zynq platform. In principle it could be used for platforms whose configuration can be loaded via a serial link from CPU to FPGA, or some other channel, but a procedure to do that would have to be created for each specific platform.

The following application code for design name *prog* illustrates the inclusion of the configuration data and the configuration loading procedure -

```
.
.

#include "prog_cinclude.h"
.
.

int main (int argc, const char *argv[]) {

    fpga_open();

    fpga_configdata_load(prog_fpga_config_data, prog_fpga_config_size);

    .
    .
}
```

The source for `fpga_configdata_load` is in `sources/tools/fpga` and also `~/install/3pl/src/fpga.c` and the object file is `~/install/3pl/lib/fpga.o`.

Variables `prog_fpga_config_data` and `prog_fpga_config_size` are defined in the included file `prog_cinclude.h`.

12.2 bin2py/

Type: Python3 script

Source: `sources/tools/bin2py`

Target: any

Execute: on host computer or on Zynq CPU

Convert a binary compressed or uncompressed FPGA configuration bit file to a Python3 source file containing an initialised byte array as Python3 source code to be imported. By this means the configuration can be embedded in the application program and can be loaded by the CPU application code without the need to read an external file during execution.

Execute by -

bin2py *design_name*

which will read *design_name.bit* or *design_name.bit.gz* and produce a Python3 file *design_name.config.py*.

The following application code for design name *prog* illustrates the inclusion of the configuration data and the configuration loading procedure -

```
import sys
import os
sys.path.append(os.environ['HOME'] + "/install/3pl")
from fpga import *
import prog_fpga as fpga
import prog_config as config

FPGA_OPEN()
```

```
FPGA_CONFIGDATA_LOAD(config.data, len(config.data))
.
.
```

In the above **from fpga import *** defines low level functions for communication between the CPU and the FPGA.

import prog_fpga as fpga imports a definition of the application level FPGA interface functions and has been created by executing **info2py**. The same effect could be obtained by importing **info2py.py**.

import prog_config as config imports a Python3 source version of the FPGA configuration file (preferably compressed) and is created by **edif2bit**.

12.3 info2c/

Type: Python3 script

Source: **sources/tools/info2c**

Target: Zynq or any with serial interface

Execute: on host computer or on Zynq CPU

Generate CPU/FPGA zynq interface functions for C from the 3PL information file (design.info). This generates an include file **design_name_include.h**.

12.4 info2py/

Type: Python3 script Source: **sources/tools/info2py**

Target: Zynq or any with serial interface

Execute: on host computer or on Zynq CPU

Generate CPU/FPGA interface functions for Python3 from the 3PL information file (design.info).

This can be executed as a main program (~/install/3pl/bin/info2py) and will generate a Python3 source file of CPU/FPGA interface functions that can be imported into the CPU application program.

Alternatively this file (~/install/3pl/include/info2py.py) can be imported into the application program in which case it will create the interface functions internally and will compile them into a module.

info2py is a copy of into2py.py included so that it can be invoked as a main program without the file name extension, e.g. **info2py prog** rather than **python3 infotopy.py prog**.

This source will import **serial/cpu_serial/cpu_serial.py** for FPGA platforms which have a serial interface, or **zynq/fpga/fpga.py** in the case of Zynq.

To execute **info2py** and get an importable file -

info2py design_name

which will read **design_name.info** and create a Python3 source file **design_name_fpga.py** which can be imported by the Python3 application program. The imported file defines all the FPGA interface functions.

If info2py.py is imported into the application program the FPGA interface JSON description can be read from the .info file or from a version stored in block RAM in the FPGA. The following code illustrates both of these cases with one of them commented out -

```
import sys
import os
sys.path.append(os.environ['HOME'] + "/install/3pl")
from fpga import *
import info2py
```

```

FPGA_OPEN()

# From .info file
#fpga = info2py.info_from_file("prog")

# From FPGA
fpga = info2py.info_from_fpga()
.
.

```

12.5 edif2bit/

Type: shell/TCL script

Source: **sources/tools/edif2bit**

Target: any

Execute: on a Linux host computer

Generate an FPGA configuration bit file from the EDIF netlist file. This invokes the Xilinx ISE or Vivado tools as appropriate and hence must be executed on a host containing these tools.

Execute by -

edif2bit *design_name*

which will read *design_name.edif* and either *design_name.ncf* (ISE) or *design_name.xdc* (Vivado) and produce the configuration file *design_name.bit*.

optional command line arguments include -tools ("ise" or "vivado") to manually select place-and-route software, -compress to run gzip on the bit file and -info to append the .info file and .rpt file to the configuration file. This last option appends meta-data to the configuration file to document its production.

The bit file can be incorporated into the CPU application program by using bin2c or bin2py and then loaded into the FPGA during program execution as described in previous sections.

Alternatively the FPGA configuration can be loaded from the bit file by the application program. The bit file may be compressed, i.e. "prog.bit" or "prog.bit.gz".

The following example is in C -

```

#include <stdio.h>
#include <stdint.h>
#include "fpga.h"
#include "prog.h"

int main (int argc, const char *argv[]) {
    int i;

    fpga_open();
    fpga_configfile_load("prog.bit");
    .
    .

```

Or in Python3 -

```

import sys
import os
sys.path.append(os.environ['HOME'] + "/install/3p1")
from fpga import *
import prog_fpga as fpga
import time
import signal
import prog_config as config

```



```
FPGA_OPEN()
FPGA_CONFIGFILE_LOAD("prog.bit.gz"))
.
.
```

12.6 run_ISE

Type: shell script

Source: **sources/tools/run_ISE**

Target: any

Execute: on a Linux host computer

Generate an FPGA configuration bit file from the EDIF netlist file using Xilinx ISE tools. This is a very much more primitive and direct script than edif2bit above.

Execute by -

run_ISE *design_name*

12.7 run_vivado

Type: shell script

Source: **sources/tools/run_Vivado**

Target: any

Execute: on a Linux host computer

Generate an FPGA configuration bit file from the EDIF netlist file using Xilinx Vivado tools. This is a very much more primitive and direct script than edif2bit above.

Execute by -

run_Vivado *design_name*

12.8 serial/

This directory contains interface code for FPGA platforms which have a serial interface between the CPU and the FPGA. An example of this type of platform is the Gadget Factory Papilio board.

These procedures can be called directly but for convenience are usually called through the included/imported intervening procedures generated by info2c or info2py

See also - [chapter 21](#) p645.

12.8.1 cpu_serial.py

Type: Python3

Source: **sources/tools/serial/cpu_serial.py**

Target: platforms with serial interface to CPU

Execute: on a Linux or OSX host computer

As well as the open and close procedures listed below this file defines necessary low-level CPU/FPGA interface communications functions. These functions are called from the interface code in the Python3 file generated by **info2py** and imported into the application code.

fpga_open(port_name)

Open the serial interface to the FPGA. **port_name** is a string specifying the serial device to the FPGA, e.g. `"/dev/tty.usbmodem14101"`.

fpga_close()

Close the serial interface to the FPGA.

DMA_buffer(b)

Set a pointer to the DMA buffer, eg.

```
dmabuffer = bytearray(4096)
DMA_buffer(dmabuffer)
```

12.8.2 cpu_serial.c/, cpu_serial.h

Type: C

Source: `sources/tools/serial/cpu_serial.c`, `sources/tools/serial/cpu_serial.h`

Target: platforms with serial interface to CPU

Execute: on a Linux or OSX host computer

As well as the open and close procedures listed below this file defines necessary low-level CPU/FPGA interface communications functions. These functions are called from the interface code in the C include file generated by **info2c**.

fpga_open(port_name)

Open the serial interface to the FPGA. **port_name** is a string specifying the serial device to the FPGA, e.g. `"/dev/tty.usbmodem14101"`.

fpga_close()

Close the serial interface to the FPGA.

12.9 zynq/

This directory contains CPU/FPGA interface functions for the Zynq FPGA platform.

These procedures can be called directly but for convenience are usually called through the included/imported intervening procedures generated by `info2c` or `info2py`

See also - [chapter 20](#) p631.

For this interface to operate the Zynq Linux device tree must provide entries for `axi_mgp0`, `axi_mgp1` and `axi_hpbuff` and must also provide interrupt events 0 to 15. An example is shown in [Appendix H](#) p753.

12.9.1 fpga.c, fpga.h

Type: C source

Source: `sources/tools/zynq/fpga/fpga.c`, `sources/tools/fpga/fpga.h`

Target: Zynq

Execute: on Zynq CPU

fpga_open ()

This maps register space (AXI mgp0), memory space (AXI mgp1), DMA space (AXI HP0, HP1, HP2 and HP3) and events 0 to 15.

fpga_close ()

This unmaps the previously mapped spaces.

fpga_event_wait(uint32_t event_index)

Wait for an event. This enables the event specifies by the event index **ev** and then does a blocking read on /dev/uio_x where x is the event unit number. When the read occurs the event is now disabled.

fpga_event_control(uint32_t event_index, uint32_t c)

Enable or disable the event specified by event index **ev**. This is done by writing 1 or 0 respectively to /dev/uio_x where x is the event unit number.

uio_event_fd(uint32_t event_index)

Return the file descriptor for /dev/uio_x where x is the event unit number.

uio_event_count(uint32_t event_index)

Return the number of events since the previous wait. If greater than 1 then a number of events have occurred before the wait has executed its blocking read.

uio_dma_phys_address()

Return the physical address of the mapped DMA space.

fpga_configfile_load(char *filename)

Load an FPGA configuration from the specified bit file.

fpga_configdata_load(uint8_t *data, size_t len)

Load an FPGA configuration from an array of char.

data is the array of char.

len is the dimension of the array.

FPGA_REG_READ(addr, type, dw)

Read an FPGA value or queue mode variable.

addr is the address of the interface.

type is the type of the returned value.

dw is the number of 32-bit words to be transferred.

FPGA_REG_WRITE(addr, val, dw)

Write to an FPGA static or queue mode variable.

addr is the address of the interface.

val is the value to be written.

dw is the number of 32-bit words to be transferred.

FPGA_MEM_READ(addr, loc, type, dw)

Read an FPGA cmemory or rmemory mode variable.

addr is the address of the interface.

loc is the address within the memory.

type is the type of the returned value.

dw is the number of 32-bit words transferred.

FPGA_MEM_WRITE(addr, loc, val, dwv)

Write to an FPGA cmemory or rmemory mode variable.

addr is the address of the interface.

loc is the address within the memory.

val is the value to be written.

dww is the number of 32-bit words transfered.

12.9.2 fpga.py

Type: Python3 source

Source: **`sources/tools/zynq/fpga/fpga.py`**

Target: Zynq

Execute: on Zynq CPU

Python3 low-level CPU/FPGA interface communications. functions. This imports low-level communication functions from **`fpga_read_write.c`** which has been added to the Python3 library by **`sources/tools/zynq/lib_py_ext/setup.py`** below as Python3 module **`fpga.py`**.

fpga_open ()

This calls **`fpga.c.open()`** in **`fpga_read_write.c`** (below) to map register space (AXI mgp0), memory space (AXI mgp1), DMA space (AXI HP0, HP1, HP2 and HP3) and events 0 to 15.

It then saves the current value of each event device `/sys/class/uio/uiox/event` where x is 0 to 15. This value increments on each event occurrence since open-up and the initial value prior to execution is needed to determine how many events have occurred between interrupts (blocking reads of the event device `/dev/uiox`).

fpga_close ()

This calls **`fpga.c.close()`** in **`fpga_read_write.c`** (below) to unmap the previously mapped spaces.

comms_write (mem, address, memaddress, dba)

Write to the FPGA.

mem is 0 for a register or queue, 1 for cmemory or rmemory.

address is the address of the interface.

memaddress is the address within a memory if **mem** is 1.

dba is a byte array containing the data. The size of the byte array indicates the number of bytes to write.

comms_read (mem, address, memaddress, bytes)

Read from the FPGA.

mem is 0 for a value or queue, 1 for cmemory or rmemory.

address is the address of the interface.

memaddress is the address within a memory if **mem** is 1.

bytes is the number of bytes to read.

The data is returned as a byte array.

event_wait (ev)

Wait for an event. This enables the event specifies by the event index **ev** and then does a blocking read on `/dev/uiox` where x is the event index. When the read occurs the event is now disabled.

event_enable (ev)

Enable the event specified by event index **ev**. This is done by writing 1 to `/dev/uiox` where x is the event index.

event_disable (ev)

Disable the event specified by event index **ev**. This is done by writing 0 to `/dev/uiox` where x is the event index.

dma_phys_address ()

This function returns the physical address of the DMA mapped space.

dma_get_word (addr)

This function gets one 32-bit word from the DMA mapped space. **addr** is an address offset into the DMA address space.

dma_put_word (addr, data)

This function puts one 32-bit word into the DMA mapped space. **addr** is an address offset into the DMA address space and **data** is the data word.

12.9.3 fpga_py_extension.c

Type: C source

Source: `sources/tools/zynq/fpga_py_extension.c`

Target: Zynq

Execute: on Zynq CPU

Python3 C extension code. This provides links from the Python3 interface functions into functions in **fpga.c**. It is added to the Python3 library as module **fpga.read_write** by executing **setup.py** (below).

py_fpga_open()

Called as `fpga.open()` in python.

Calls `fpga.open()` in `fpga.c`. This maps register space (AXI mcp0), memory space (AXI mcp1), DMA space (AXI HP0, HP1, HP2 and HP3) and events 0 to 15.

py_fpga_close()

Called as `fpga.close()` in python.

Calls `fpga.close()` in `fpga.c`. This unmaps the previously mapped spaces.

py_fpga_configfile_load(fname)

Called as `fpga_configfile.load()` in python.

Calls `fpga_configfile.load()` in `fpga.c`.

fname is the file name of the configuration bit file. This will have a `.bit` file name extension or `.bit.gz` if compressed,

py_fpga_configdata.load(data, len)

Called as `fpga_configdata.load(data, len)` in python.

data is a byte array containing the configuration bit pattern.

len is the dimension of the array.

The configuration byte array is contained in an imported module created by **bin2py** (which itself is a Python3 script).

py_fpga_burst_read(is_mem, address, n_words)

Called as `fpga_burst_read(mem, adress, words)` in python.

Read an FPGA value, queue, cmemory or rmemory mode variable.

is_mem is 1 for cmemory or rmemory mode and 1 otherwise.

address is the address of the interface.

n_words is the number of 32-bit words to be transfered.

py_fpga_burst_write(is_mem, address, n_words, data)

Called as `fpga_burst_write(mem, address, words, data)` in python.

Write to an FPGA static, queue, cmemory or rmemory mode variable.

is_mem is 1 for cmemory or rmemory mode and 1 otherwise.

address is the address of the interface.

n_words is the number of 32-bit words to be transfered.

data is the data, a byte array.

py_uio_event_wait(event_no)

Called by `uio_event_wait(ev)` in python.

Calls `fpga_event_wait()` in `fpga.c`.

event_no is the event index, 0 to 15.

Wait for an event. This enables the event specifies by the event index **ev** and then does a blocking read on `/dev/uioX` where `x` is the event unit number. When the read occurs the event is now disabled.

py_uio_event_fd(event_no)

Called by `uio_event_fd(ev)` in python.

event_no is the event index, 0 to 15.

Return the file descriptor for `/dev/uioX` where `x` is the event unit number.

py_uio_event_unit(event_no)

Called by `uio_event_unit(ev)` in python.

event_no is the event index, 0 to 15.

Return the event number `x` for `/dev/uioX` where `x` is the event unit number.

py_uio_event_count(event_no)

Called by `uio_event_count(ev)` in python.

Return the number of events since the previous wait. If greater than 1 then a number of events have occurred before the wait has executed its blocking read.

py_uio_event_control(event_no, c)

Called by `uio_event_control(ev, c)` in python.

Calls `fpga_event_control()` in `fpga.c`.

Enable or disable the event specified by event index **ev**. This is done by writing 1 or 0 respectively to `/dev/uioX` where `x` is the event unit number.

py_uio_dma_get_word(address)

Called by `uio_dma_get_word(addr)` in python.

Returns a 32-bit word from the DMA space with offset **address**.

py_uio_dma_put_word(address, data)

Called by `uio_dma_put_word(addr, data)` in python.
Copies a 32-bit word **data** from the DMA space with offset **address**.

py_uio_dma_phys_address()

Called by `uio_dma_phys_address()` in python.
Returns the physical address of the DMA space.

12.9.4 setup.py

Type: Python3 source
Source: **sources/tools/zynq/setup.py**
Target: Zynq
Execute: on Zynq CPU

A Python3 source file to add **fpga_read_write.c** above as a C extension to the Python3 library. **mk_setup** is a shell script to execute **setup.py** which will add module **fpga_read_write** to the Python3 library.

12.9.5 mk_setup

Type: shell command file
Source: **sources/tools/zynq/mk_setup**
Target: Zynq
Execute: on Zynq CPU

This command file adds the C extension described above to the Python3 library. It also compiles `fpga.c` to `fpga.o`. Execute by typing **mk_setup** to the shell.

Chapter 13 - Code structure

A program usually consists of modules interconnected by queues. A module code body contains one or more code elements each of which is initiated repeatedly, i.e. is contained in an infinite loop.

A program file contains optional global scope and file scope structure and variable creation calls and executable code intermixed with module, procedure and function declarations. It is preferable to have top level code first followed by the module, procedure and function declarations.

Execution begins with the initial file module which in turn calls other modules, procedures and functions.

Execution of statements in **3PL** is controlled by two mechanisms -

- execution sequence
- queue availability

Statements in a module (including the main file module) which are not nested within control structures, such as conditional statements or loops, will be referred to as **top level** statements.

All top level statements are executed in an implied infinite loop. Statements within control structures are executed under the control of those structures. Statements may additionally wait upon the availability of queues.

Top level simple assignment statements occurring immediately within a module will execute whenever referenced queues are available. If no queues are referenced, the statements will execute on every clock cycle.

An assignment statement enclosed in a **when** which itself is immediately inside a module will be controlled primarily by the **when** which will execute every clock cycle. The **when** body may wait on queues when the **when** is true, otherwise it will execute at once.

A great deal of code can be written without using explicit serial and parallel constructs, relying on queue availability alone to control execution.

A simple program might be -

```
cmemory(m, 2, 8, 8);
queue(a, "uint:8");
queue(b, "uint:8");
queue(addr, "uint:8");
queue(d, "bits:8");

[addr <<- a + b;]
[d <<- cmemoryread(d <- m, 0, addr);]
```

The two target statements within the initial file module are themselves modules. The first is an inline module and the single assignment statement within will execute when its queues are available. The second is also an inline module and again will wait on the availability of its queues.

If a module has more than one statement, e.g. -

```
[
    y <<- a + b;
    z <<- c - d;
]
```

each statement in the module would execute when its queues were available. We could have simply made these independent modules -

```
[y <<- a + b;]
[z <<- c - d;]
```

The first form would be used had we needed to share local static variables.

There are three structures giving parallelism within a module which we must understand.

The first has been used above, simply enclosing statements in inline module brackets

```
y <<- a + b;
z <<- c - d;
```

Each statement in the module will execute independently and repeatedly whenever it can. If a statement contains only static variables it will execute on every clock cycle (such statements would normally be contained within a **when** or other such conditional statement to control their execution). If a statement contains queue variables it will execute when these are available (both input on the RHS and output on the LHS).

If we enclose the statements in a **par** block

```
[
    par {
        y <<- a + b;
        z <<- c - d;
    }
]
```

each statement within the block will execute exactly once as soon as it can. The parallel block will then have completed execution and since it is immediately enclosed within a module the parallel block will execute repeatedly. Note that the statements within the parallel block are paired in execution but will not necessarily execute at the same time.

If we wish to ensure that parallel statements execute simultaneously we must enclose them in a **sync** block

```
[
    sync {
        y <<- a + b;
        z <<- c - d;
    }
]
```

which will execute both statements in the block when all queue variables in the block are available, i.e. they are guaranteed to execute simultaneously and in a single clock cycle. Note that if a **sync** block contains a sequential block or an iterative statement the **sync** simply controls entry into the block after which it behaves like a **par** block, hence it would not control any further reads or writes of those queues.

A situation to be wary of with queue dependencies occurs when queue fields are being assigned. For example

```
[
    v.a <<- y;
    v.b <<- z;
]
```

If **y** and **z** are not available simultaneously then the assignment will occur separately on each becoming available, the other field of **v** being assigned zero. In that case they must be locked together by

```
[
  sync {
    v.a <<- y;
    v.b <<- z;
  }
]
```

We can use parallel blocks to control static assignments, e.g.

```
[
  static(s, "uint:8");

  par {
    a <<- b + c;
    s++;
  }
]
```

where the static assignment **s++** will only occur as often as the queue assignment, and the availability of input and output queues will control both. For actual simultaneous execution we would again write

```
[
  static(s, "uint:8");

  sync {
    a <<- b + c;
    s++;
  }
]
```

In the previous examples we might have been using **s** to count queue data processed. If we wished to count total cycles to determine the proportion of active cycles we could write -

```
[
  static({s, t}, "uint:8");

  par {
    a <<- b + c;
    s++;
  }
  t++;
]
```

The static assignment to **t** will occur every clock cycle since it is immediately within a module.

For queues which are structures or arrays we can simply assign them, e.g.

```
struct {t1,
  d, "uint:8",
  l, "log"
};
queue({a, b}, t1);

module
m () {
  a <<- b;
}
```

will assign the entire structure. We can mix different queue types as long as each component assignment type matches, e.g.

```

struct {t1,
      d,  "uint:8",
      l,  "log"
};
queue(a, t1);
queue(pixel, "uint:8");
queue(j, "log");

module
m () {
  sync {
    a.d <<- pixel;
    a.l <<- j;
  }
}

```

but note that we have used the **sync** {...} parallel block to ensure that the two fields are assigned simultaneously. If we do not ensure simultaneity then two separate data will be written, one with the **l** field false, the other with the **d** field zero. There is no conflict assigning to output queue **a** as the assignments are to separate fields of the structure.

If we want to examine two inputs but only advance one of them -

```

queue({o, i1, i2}, "uint:8");

[
  sync {
    when (?i1 > ?i2)
      o <<- i1;
    else
      o <<- i2;
  }
]

```

The input queues **i1** and **i2** are examined without reading (the ? operator) and then the larger is read and assigned to the output queue. The *when* will not wait for queues **i1** and **i2** to become available in the test expression because the queues are examined, not read, but we have enclosed the *when* in a *sync* block and there are reads of queues **i1** and **i2** within that block so the *sync* will wait until queues **i1**, **i2** and **o** are available before executing the *when*.

We could shorten the code, using a ? : operator, to

```

queue({o, i1, i2}, "uint:8");

[o <<- (?i1 > ?i2) ? i1 : i2;]

```

Note that we now have a single assignment statement and that statement will not execute until all queues read or written (but not examined) within it are available.

If we wanted to do the same but advance both inputs each time we could do this by

```

queue({o, i1, i2}, "uint:8");

[o <<- (i1 > i2) ? i1 : i2;]

```

since both inputs are read regardless of which is selected for assignment to the output queue.

If we wished to take inputs when they become available and channel them to one output queue we might write -

```

// BAD CODE!
queue({o, i1, i2}, "uint:8");

```

```
[
  when (<i1)
    o <<- i1;
  when (<i2)
    o <<- i2;
]
```

but this is both pointless and incorrect. Firstly the availability tests are not necessary as the assignments will not execute unless the input queues are available anyway. Secondly if both input queues are available the assignments would conflict. We could check both explicitly by -

```
queue({o, i1, i2}, "uint:8");

[
  when (<i1 && !<i2)
    o <<- i1;
  when (!<i1 && <i2)
    o <<- i2;
  when (<i1 && <i2)
    o <<- i1;
]
```

Here there are no conflicts as the three conditions are mutually exclusive (we do not need mutual exclusion like this if there is no assignment conflict). The above code will favour the first input when both are available. To give both inputs an even chance

```
queue({o, i1, i2}, "uint:8");
static(first, "log");

[
  when (<i1 && !<i2)
    o <<- i1;
  when (!<i1 && <i2)
    o <<- i2;
  when (<i1 && <i2) par {
    when (first)
      o <<- i1;
    else
      o <<- i2;
    first <- !first;
  }
]
```

The inner *when* and the static assignment will be done in parallel in one cycle. It is preferable to not nest *whens* as they are not optimised and the nesting increases the length of logic paths. We could instead write -

```
queue({o, i1, i2}, "uint:8");
static(first, "log");

[
  when (<i1 && !<i2)
    o <<- i1;
  when (!<i1 && <i2)
    o <<- i2;
  when (<i1 && <i2 && first) par {
    o <<- i1;
    first <- false;
  }
  when (<i1 && <i2 && !first) par {
    o <<- i2;
    first <- true;
  }
]
```

```
]
```

The inner *when* could be contracted into a single assignment using the ternary operator `? :` on the RHS.

```
queue({o, i1, i2}, "uint:8");
static(first, "log");

[
  if (<i1 && !<i2)
    o <<- i1;
  if (!<i1 && <i2)
    o <<- i2;
  if (<i1 && <i2) par {
    o <<- first ? i1 : i2;
    first <- !first;
  }
]
```

Value variables serve two purposes - they can represent a combinatorial expression that is to be used in more than one place without duplicating the logic, and they can also be used simply to refer to other variables (the same thing really - an expression consisting of a single variable).

The first usage is quite straightforward, for example

```
queue({q1, q2, q3, q4}, "uint:8");

[
  q3 <<- q1 + q2;
  q4 <<- q1 + q2;
]
```

would duplicate the logic performing the addition, whereas

```
queue({a, b, c, d}, "uint:8");

[
  value(v, "uint:8");

  v := q1 + q2;
  q3 <<- v;
  q4 <<- v;
]
```

would build only one instance of the addition. Note that where a value expression contains queues those queues will be read and acknowledged as though they appeared separately in the statements in which the value variable is used, even though any operators within the expression will only be instantiated once. This of course applies equally to queues appearing in array subscript expressions within a value expression.

The second usage of a value variable is as a reference or pointer. For example we may wish to devise a module each invocation of which reports to a central module. We can do this as follows

```
queue({q1, q2, ... }, "uint:8");
value(p_report, "[10]uint:8");
int(reports);

.
.
m(q1);
.
.
m(q2);
.
```

```

.
report();

module
m (queue(in, "uint:8")) {
    p_report[reports++] := in;
}

module
report () {
    int(i);

    for (i=0 ; i<reports ; i++)
        ..... p_report[i];
}

```

Each time module **m** is invoked it establishes a reference in value array **p_report** to the module input queue. At the end of the module **report** is invoked which iterates through the value array **p_report** and assigns the referenced queues to something. We might like to do something like the following

```

when (<p_report[0])
    p_out <<- p_report[0];
else when (<p_report[1])
    p_out <<- p_report[1];
else when (<p_report[2])
    p_out <<- p_report[2];
.
etc
.

```

which will read from one of the queues in a priority order. We could generalise the above program by means of recursion -

```

queue({result, q1, q2, q3}, "uint:8");
var(v, value, "[3]uint:8");
.
.
v[0] := q1;
.
.
v[1] := q2;
.
.
v[2] := q3;
.
.
[pri(result <- v, 0, 3);]

procedure
pri (queue(o) <- value(i), int({index, size})) {
    when (<i[index])
        o <<- i[index];
    else
        if (index < (size - 1))
            pri(o <- i, index+1, size);
}

```

All of the above code examples using nested *whens* will generate cascaded logic which can quickly lead to control logic paths which are too long for signals to propagate within a clock cycle.

The use of immediate code to control target code generation can be illustrated by the following examples. If we wish to create a delay line we could write the following -

```

static({y, z}, "uint:8");

.
.
z <- del(y, 5);
.
.

function
delay (value(in, "uint:8"), int(n)) {
  var(d, static, "[10]uint:8");
  [
    d[0] <- in;
    for (int(i,1) ; i<n ; i++)
      d[i] <- d[i-1];
  ]
  return(d[n-1]);
}

```

We have had to include the code for the delay line itself in a module, the function just returning a value which is the last static variable in the delay line. Thus the function itself does not contain any in-line target code (functions cannot), the module being an independent executing entity.

The *for* loop is immediate and executes *n* times. Each time it iterates it generates one target assignment statement. Since we are at the top level of a module the target assignment statement before the loop and each of the target assignment statements generated by the loop will each be wrapped in their own infinite loops to run completely independently in parallel. Had we used a loop like this below the top level of a module it would have had to be enclosed in a *seq*, *par* or *sync*. The function above, as called, is equivalent to having written -

```

function
delay (value(in, "uint:8"), int(n)) {
  var(d, static, "[10]uint:8");
  [
    d[0] <- in;
    d[1] <- d[0];
    d[2] <- d[1];
    d[3] <- d[2];
    d[4] <- d[3];
  ]
  return(d[4]);
}

```

There are two annoying limitations of the above code: the data type is fixed and the static array in the function has a fixed dimension. The following version removes both these limitations -

```

static({y, z}, "uint:8");

.
.
z <- del(y, 5);
.
.

function
delay (value(in), int(n)) {
  var(d, static, "["+n+"]"+type(in));
  [
    d[0] <- in;
    for (int(i,1) ; i<n ; i++)
      d[i] <- d[i-1];
  ]
  return(d[n-1]);
}

```



```
}

```

Parameter *in* is now created without a data type, which it will take from the matching argument. It now uses that data type by getting the string version by means of *type()* and prepending the required dimension in brackets from parameter *n*. The data type of the function return will be the same as *d*, which is the same as the input *in*.

If an immediate loop is to contain more than one target statement then these must be bracketed, e.g.

```
for (i=0 ; i<3 ; i++) {
    a[i] <- b[i] + i;
    c[i] <- d[i] - i;
}
```

is equivalent to writing

```
a[0] <- b[0] + 0;
c[0] <- d[0] - 0;
a[1] <- b[1] + 1;
c[1] <- d[1] - 1;
a[2] <- b[2] + 2;
c[2] <- d[2] - 2;
```

If we were to use such a loop inside a procedure we would have to enclose the resulting target statements in a *seq*, *par* or *sync* because the body of a procedure can only contain one top level target statement -

```
procedure
p () {
    par {
        for (i=0 ; i<3 ; i++) {
            a[i] <- b[i] + i;
            c[i] <- d[i] - i;
        }
    }
}
```

in this case a *par* block.

An immediate loop that contains target code in its body simply replicates the target code and it is the context in which the immediate loop occurs that decides whether the resultant replicated statements are to be executed sequentially, in parallel or in individual infinite loops. For example array variables are not allowed in expressions, so if we wish to do array arithmetic we must code it explicitly. For example to add two 2D arrays we could write -

```
static({a, b, c}, "[3][4]uint:8");

par {
    for (int(i) ; i<3 ; i++)
        for (int(j) ; j<4 ; j++)
            c[i][j] <- a[i][j] + b[i][j];
}
```

If this code is at the top level in a module then we may not want the *par*. However, note that this will execute on every clock cycle so we may want some execution control using a *when*. If the variables were queues rather than statics then their execution would be controlled by queue availability.

Chapter 14 - Coding Recommendations

14.1 minimising variable assignments

Assignments to a static or queue variable should be kept to a minimum. It is preferable to use different variables where possible rather than re-use the same variable. The more assignments a variable has the more delay is incurred in its input logic and hence the more likely it is that the logic will not meet timing constraints. In general flip-flops are in excess in FPGAs and it is better to use many variables with few assignments each rather than a few variables with many assignments each.

14.2 reset() better than assignment

It is better to use the `reset()` inbuilt procedure rather than re-assigning the initial value (usually zero) to a static variable. The reset does not require the input multiplexing logic used by assignments, has a separate input to the flip-flops, has precedence over assignment in a clash should that prove useful and multiple resets in the code are simply ORed which is much simpler than assignment multiplexing. Note: the compiler now recognises when an assignment of a constant to a static variable returns it to its initial state and generates a reset instead of an assignment. In view of the priority of a reset over an assignment should they clash it may still enhance readability to use `reset()` explicitly.

14.3 queue subsets of structs

Where a processing pipeline has more fields in the queue structs as one progresses along the pipeline it is better to use one struct type containing all the necessary fields for every intermediate queue, rather than giving each queue its own struct, each a superset of the previous. This simplifies the source code while not affecting the logic resources used as queue (and static) elements which are not referenced are removed during netlist optimisation.

14.4 queue writes exclusivity

Writes to a queue must be mutually exclusive (simultaneous writes to component array members or fields is OK). In order to control writes to a queue it is usually best to restrict writing to any one queue to a single module.

14.5 nesting of target control structures

Nesting of target control structures (`when`, `branch`, `seq while`, `par while`, `sync while`, `do seq while`, `do par while`, `do sync while`) should be minimised. For example it is better to use independent wider

whens than to nest them. For example if the following is at the top level (not nested in any other control structures) -

```
when (l1) par {
  when (l2)
    a <- b;
  c <- d;
}
```

it would be better written as -

```
when (l1 & l2)
  a <- b;
when (l1)
  c <- d;
```

Here the two **when** statements are executed independently in parallel since they are at the top level. If the example was not at the top level then instead we could code it as -

```
par {
  when (l1 & l2)
    a <- b;
  when (l1)
    c <- d;
}
```

In these sorts of transformations care must be taken to ensure that the two forms are equivalent in execution since statements in a **par** do not necessarily execute the same as statements in separate top-level infinite loops. for example -

```
par {
  seq {
    a <- x;
    b <- y;
  }
  c <- z;
}
```

executes differently to top-level -

```
seq {
  a <- x;
  b <- y;
}
c <- z;
```

since the **seq** will take 2 clock cycles to execute and thus assignment to **c** will occur every 2 clock cycles in the first example but every clock cycle in the second.

As another example -

```
sync {
  when (l1)
    a.f1 <- 2;
  else when (l2 && l3)
    a.f1 <- 3;
  else when (l2 && !l3)
    a.f1 <- 4;
  else
    a.f1 <- 0;
  a.f2 <- b;
}
```

Note that a **sync** is needed here to ensure that the queue is available to write before testing **l1**, **l2** etc as these variables could be changed by other code running in parallel while execution here is waiting for queue availability. This code is valid but will have long cascaded control logic. We could code this instead as -

```

sync {
  when (l1 && !l2))
    a.f1 <- 2;
  when (!l1 && l2 && l3)
    a.f1 <- 3;
  when (!l1 && l2 && !l3)
    a.f1 <- 4;
  a.f2 <- b;
}

```

as here the **when** statements execute in parallel but with a shallower depth of expressions to test. Note that we have been able to eliminate the zero assignment case because a queue assignment will occur because of **a.f2 <- b;** and hence if none of the **whens** are true, field **f1** will be assigned zero by default, so the **else** is not needed.

The following example illustrates ways of recoding to overcome timing constraint problems. An input queue carries a struct with two numeric fields **f1** and **f2** and a logical field **coord**. If **coord** is true the input queue is to be copied to the output queue. If **coord** is false **f1** is to be written to the output queue a number of times where the count is given by **f2**. This can be neatly coded as -

```

struct(s,
  f1,      "uint:12",
  f2,      "uint:19",
  coord,   "log"
);
queue({in, out}, s);
static(e, "uint:12");
static(count, "uint:19", 1);
when (in.coord)
  out <- in;
else seq {
  sync {
    count <- in.f2;
    e <- in.f1;
    out.f1 <- in.f1;
  }
  par while (count != 1) {
    out.f1 <- e;
    count--;
  }
}
}

```

This version is succinct but it did not meet a 10ns timing constraint by 5ns. The main problem is a path from static variable **count** through the **while()** test back through the containing when and then through to one of the assignments to **out**. We can eliminate the **while()** by realising that if we move to the top level of the module we have an implied infinite loop. A **when()** at the top level therefore behaves like a **while()** lower down. With this in mind we can recode the example as follows -

```

sync {
  when (count == 1) {
    when (in.coord)
      out <- in;
    else par {
      count <- in.f2;
      e <- in.f1;
      out.f1 <- in.f1;
    }
  }
}

```

```

}
when (count != 1) par {
  out.f1 <<- e;
  count--;
}

```

This code is perhaps less clear, but it is simpler than the previous version and familiarity with the implied infinite loops of 3PL reduces its apparent obscurity. This code still failed to meet a 10ns timing constraint by 1ns. The 19 bit comparison introduces a delay of 3ns or so and we could eliminate this by doing the comparison in parallel in the previous iteration and storing the result in a static variable as follows -

```

static(counteq1, "log", true);
sync {
  when (counteq1) {
    when (in.coord)
      out <<- in;
    else par {
      count <- in.f2;
      counteq1 <- (in.f2 == 1);
      e <- in.f1;
      out.f1 <<- in.f1;
    }
  }
}
when (!counteq1) par {
  out.f1 <<- e;
  count--;
  counteq1 <- (count == 2);
}

```

This version easily met the 10ns timing constraint but at the price of more obscure coding.

14.6 uncontrolled input sources

It is often the case that input data is from a source that has no flow control, for example a video camera. In that case pipeline signalling cannot extend back past the input module. This is a problem if any module in the processing pipeline fails to read an input queue in the cycle it becomes available. Buffering does not help as any pipeline delays accumulate. It is preferable to use a processing pipeline with a higher clock frequency than the input data. The pipelines are connected by means of an asynchronous queue variable which provides a FIFO between modules whose clock frequencies may differ.

14.7 bit access

There is usually no reason to access individual bits in a word. If a word represents flags then it should be an array of **log**. If there are packed fields then it should be a **struct**. If data is being handled on a word basis but a single bit is to be examined at one point in the code one method is to cast the word into an array of single bits and then reference the required bit directly, e.g.

```

static(a, "uint:8");
value(b, "[8]bits:1");

b := cast(a, "[8]bits:1");

.
... <- .. b[3] ..; // the 4th bit of 'a'
.

```

The alternative is to use masking and shifting in order to access bit fields. If the mask word and shift count are constant then the two forms will generate equivalent code and are efficient. There are functions in `stdlib.3pl` called `getbit()`, `isset()` and `isclear()` which already do this and can be used for single bit access. Note that there is no way to assign a single bit of a multibit primitive - the whole datum must be assigned with the selected bit changed. If single bits are to be assigned the type should be declared as an array of type **log** or of **bits:1** and later cast to the primitive multibit type required.

14.8 pipeline resets

Many library modules have optional reset inputs, which are usually value mode. These reset inputs are usually intended for recovery from error conditions. It must be recognised that the use of these inputs will usually destroy the current state of the pipeline. Modules along a pipeline may be buffering zero, one or more data and the exact position of successive data in the pipeline is usually not known. If an asynchronous reset is to be used it is usually necessary to reset the entire pipeline to an initial (empty) state and restart it from the input end.

14.9 pipeline conflicts

There are two pipeline topological features which have the potential to cause trouble. The first is alternate queue paths between a source and destination module. The second is queue loops.

14.9.1 alternate queue paths

When there are two or more queue paths from a source module to a destination module there is the potential for the queues to interact. In the simple case where the destination module waits on both inputs and then reads both, problems arise if the latency through the paths differs. Data in the shorter path will reach the destination module first and that path will then block. The point of the block will move back until all the queue buffers in the path are full. The longer path will in the meantime be flowing but not blocking. When the short path fills and blocks the source the longer path will cease to read any more data from the source and so will start to empty from the input end. Eventually data in the longer path will reach the destination module at which point the destination will start to read both queues. After a while the short path will unblock the source so that data flows again into both paths, but soon the data gap in the long path will reach the destination which will block again. This is an oversimplified view and in practice the queue interactions can lead to a complicated disjointed data flow. The data will always be correct and in step and the pipeline will not deadlock, but data processing will be sporadic and the throughput will be less than it could be.

The simple solution is to put a buffer (queue) in the shorter path. This will take up the excess data and keep the source module from blocking, thus maintaining the flow to the longer path. The minimum buffer size should be the difference in latency between the two paths. A queue variable whose buffer size has been increased by setting attribute **depth()** or **mindepth()** can be used for this purpose.

This phenomenon has not been analysed. It can probably be modelled using Petri nets.

14.9.2 queue loops

A closed queue loop should only be used after careful consideration as it can easily lead to deadlock. It is possible to set up a loop if either reading of the inputs is conditional at the point at which the loop feeds back, or else if modules in the loop are initialised so that the pipeline starts already in the steady state condition.

Chapter 15 - Code Examples

Many of these examples have not been compiled, and those that have may not have been executed, hence there may be coding or algorithmic errors.

15.1 queue switching

If we want a pipeline to diverge into one of two or more branches we might write a module to do this as follows -

```
module
diverge (queue({out0, out1}) <- value(select, "log"), queue(in)) {
  when (select)
    out0 <<- in;
  else
    out1 <<- in;
}
```

This module has two output queues and one input queue. It continuously performs one of two assignments depending on the value of logical variable **select**. This code however has behaviour that is not ideal. If for example queue **in** is not available, execution will wait on the selected assignment statement until **in** is available, even though **select** might change in the meantime. We could rewrite the code as -

```
module
diverge (queue({out0, out1}) <- value(select, "log"), queue(in)) {
  sync when (select) {
    out0 <<- in;
  }
  sync when (!select) {
    out1 <<- in;
  }
}
```

Each of the two **sync when** statements will execute on every clock cycle. One of these will terminate because the test expression is false and the other will either execute the code block because the input and output queues are both available or else will also terminate.

Similarly if we wish to converge two queues we could use -

```
module
converge (queue(out) <- value(select, "log"), queue({in0, in1})) {
  sync when (select) {
    out <<- in0;
  }
  sync when (!select) {
```

```

        out <- in1;
    }
}

```

Alternatively we could select inputs based on queue availability -

```

module
converge (queue(out) <- queue({in0, in1})) {
    out <- in0;           // always when both available
    sync when (!in0)     // when out and in1 are available but not in0
        out <- in1;
}

```

(which favours **in0** over **in1**). A **sync when()** will only execute the body when all queues in the test expression or the body can be read or written (as appropriate) and when the test expression is true. When these conditions are not met it does not block!

15.2 a two-cycle module

This module adds pairs of input data, outputting one datum per input pair.

```

queue({q1, q2}, "uint:8");

.
.
[
    static(temp, "uint:8");
    static(second, "log");

    when (second) par {
        q2 <- temp + q1;
        second <- false;
    } else par {
        temp <- q1;
        second <- true;
    }
]
.
.

```

The **when** will execute repeatedly since it is contained immediately within a module, i.e. on completion it will begin execution again on the next clock cycle. Since the test expression is a static variable and thus is always available the **when** will pass execution to one of the code bodies in the current cycle. Each code body is a parallel block. The selected block will execute its enclosed statements as soon as possible and will complete execution when both have completed. In each block the first assignment statement may wait on queue availability but the second assignment will execute at once.

If **second** is false, the second code body will begin execution. This is a parallel block so both contained statements may execute. The second has no queue dependencies so will execute (it toggles the value of the test variable). The first has a queue dependency so may wait (it saves the datum from the input queue). When both have executed, control will pass back to the **when** which will complete then be executed again.

If **second** is TRUE, the first code body will begin execution. It is similar, but performs the sum and writes the result to the output queue. Note that the sum will generate a 9-bit result which will be truncated (since it is unsigned) on assignment to 8 bits.

Pairs of input data are summed and the result is the output, thus the output queue runs at half the speed of the input queue. Logical static variable **second** is inverted each time.

We could force the two assignment statements in each block to execute simultaneously by changing the **par** blocks to **sync** blocks.

```

queue({q1, q2}, "uint:8");

.
.
[
  static(temp, "uint:8");
  static(second, "log");

  when (second) sync {
    q2 <- temp + q1;
    second <- false;
  } else sync {
    temp <- q1;
    second <- true;
  }
]
.
.

```

In this case all statements in a block wait until all queues referenced anywhere in the block are available at which time the statements all execute simultaneously. In this example that is not necessary and does not change the result. We could also put a **sync** block around the **when** instead -

```

queue({q1, q2}, "uint:8");

.
.
[
  static(temp, "uint:8");
  static(second, "log");

  sync{
    when (second) par {
      q2 <- temp + q1;
      second <- false;
    } else par {
      temp <- q1;
      second <- true;
    }
  }
]
.
.

```

which is subtly different, though it leads to a similar result. In this case execution will hang at the **sync** until both queues are available at which it will enter the **when** and the true or the false block in the **when**, which will execute both statements in the **par** simultaneously since we are guaranteed there will be no queue waits. This differs slightly from the previous cases if **second** is false and **q2** is not available as the **sync** will wait for **q2** to become available, but this is not necessary since **q2** is not being written to.

We could move the assignment of the logical static variable **second** out of the **when**, but now must enclose the code in a parallel block -

```

queue({q1, q2}, "uint:8");

.
.
[
  static(temp, "uint:8");
  static(second, "log");

  par {

```

```

        when (second)
            q2 <<- temp + q1;
        else
            temp <- q1;
            second <- !second;
    }
]
.
.

```

or could use a **sync** block instead of **par**. Without the containing parallel block the two components (the **when** and the static assignment) would execute independently. Thus logical static variable **second** would toggle continuously regardless of any queue waits.

But after all that we could code this example using a sequential block! -

```

queue({q1, q2}, "uint:8");

.
.
[
    static(temp, "uint:8");

    seq {
        temp <- q1;
        q2 <<- temp + q1;
    }
]
.
.

```

15.3 loops

3PL provides target loop control structures, but these are not always the best way to implement loops. Take for example a module which is to produce a repeating sequence of integers from 0 to 9. We might write it as -

```

module
sequence (queue(out, "uint:4") <- ) {
    static(i, "uint:4");

    seq {
        i <- 0;
        sync while (i < 10) {
            out <<- i;
            i++;
        }
    }
}

```

This code will execute as required, but it wastes one cycle at every start of the sequence initialising **i** to 0 and another cycle on the final **while** test at which point the loop exits. We could test **i** inside the loop and when it reaches 9 reset it to 0, but then we would no longer need the **while**. We could instead code the module using implicit infinite loops as -

```

module
sequence (queue(out, "uint:4") <- ) {
    static(i, "uint:4");

    sync {

```

```

        out <<- i;
        when (i == 9)
            reset(i); // better than i <- 0
        else
            i++;
    }
}

```

This generates the sequence with no lost cycles. This will enter the **sync** block repeatedly whenever the output queue **out** is available. Each time it assigns the next sequence value to the output queue and in parallel either increments or zeros the index. It will always execute the output queue assignment in one cycle because the **sync** has ensured that it is available. Only one write to the queue is done and then the **sync** repeats as there are no loops within the **sync**.

For nested loops we could again use **whiles** -

```

module
sequence (queue(out, "uint:16") <- ) {
    static(x, "uint:4");
    static(y, "uint:4");

    seq {
        reset(y);
        seq while (y < 10) {
            reset(x);
            sync while (x < 10) {
                out <<- x | (y << 8);
                x++;
            }
            y++;
        }
    }
}

```

but here we waste more cycles sequentially initialising or incrementing loop variables. The above code can probably be improved but at the expense of readability. We could again change this to use implicit infinite loops -

```

module
sequence (queue(out, "uint:16") <- ) {
    static(x, "uint:4");
    static(y, "uint:4");

    sync {
        out <<- x | (y << 8);
        when (x == 9) par {
            reset(x);
            when (y == 9)
                reset(y)
            else
                y++;
        } else
            x++;
    }
}

```

This generates the combined sequence from two nested sequences with no lost cycles. This enters the **sync** block repeatedly whenever the output queue is available. Each time through it assigns the combined sequence value to the output queue and in parallel computes the next **x** and **y** values. A disadvantage of this code is that it nests **when** statements, which while logically correct can lead to long combinatorial logic paths which may not fit within timing constraints. The compiler cannot transform this to optimise the code. The conditional statements could be separated -

```

module
sequence (queue(out, "uint:16") <- ) {
    static(x, "uint:4");
    static(y, "uint:4");

    sync {
        out <<- x | (y << 8);
        when (x == 9) par {
            reset(x);
            when (y == 9)
                reset(y)
            else
                y++;
        } else
            x++;
        when (x == 9 && y == 9)
            reset(y);
        when (x == 9 && y != 9)
            y++;
    }
}

```

Now we have traded logic depth for logic width but in the process have duplicated tests. It is now debatable if this is good or bad as duplicated logic can be faster by avoiding fanout routing problems. For this reason the compiler does not attempt to eliminate this code duplication, leaving it up to the programmer to do so explicitly. If we feel we should eliminate duplication we could rewrite the above as

```

module
sequence (queue(out, "uint:16") <- ) {
    static({x, y}, "uint:4");
    value(xlimit, "log", x == 9);
    value(ylimit, "log", y == 9);

    sync {
        out <<- x | (y << 8);
        when (xlimit)
            reset(x);
        else
            x++;
        when (xlimit && ylimit)
            reset(y);
        when (xlimit && !ylimit)
            y++;
    }
}

```

Note that the local variables x, y, xlimit and ylimit could have been created inside the sync block, rather than in the scope of the module itself, since they are only used within that block.

15.4 a data-dependent module

A module to sum inputs until a zero datum, at which point the sum is output.

```

queue(q1, "uint:8");
queue(q2, "uint:16");

.
.
[
    static(sum, "uint:16");

```

```

    when (q1 == 0) sync {
      q2 <<- sum;
      sum <- 0;    // could use reset(sum);
    } else
      sum <- sum + q1;
  ]
  .
  .

```

Here the **when** test expression will only evaluate when queue **q1** is available. If the input queue value is zero, the sum will be assigned to the output, otherwise the input will be added to the sum. Input queue **q1** was read in the test expression so the datum will always be removed. If it was used in the last statement, it will be read in the same clock cycle as in the **when** test so this will constitute a single read of the queue and the removal of one datum. The assignment to queue **q2** may wait on availability. Because of this we must delay execution of the entire parallel block until all queues are ready (in this case just **q2**) so that the two assignments are simultaneous since we are changing **sum** in the same parallel block in which we are using its value.

Multiple simultaneous reads of the same input queue in an execution cycle is normal practice. Examines and reads can also be used on a queue simultaneously. If one or more reads are executed on an input queue in an execution cycle the queue datum will be removed.

15.5 a module with extra cycles

Input data are copied to the output, but if the MSB is set, a zero is also output. This example generates another execution cycle independent of input queue availability.

```

queue({q1, q2}, "uint:8");

.
.
[
  static(extra, "log");

  when (extra) par {
    q2 <<- 0;
    extra <- false;
  } else sync {
    q2 <<- q1;
    when ((q1 & 0x80) != 0)
      extra <- true;
  }
]
.
.

```

The **when** will execute repeatedly inside the implied infinite loop (the statement is top-level). The test expression involves no queues so will flow on immediately passing execution to one of the code bodies. Normally **extra** is false so the second code body will be executed. It is a parallel block which will assign the input queue value to the output queue. This assignment may wait on either queue. The second statement tests the MSB of the input queue value (it may also wait on that queue) and if that bit is set, it assigns TRUE to the static logical variable **extra**. When both statements have executed the parallel block finishes, passing control back to the outer **when** which also finishes. The outer **when** now executes again in its implied infinite loop, but if **extra** is true it will select the first code body. This first parallel block assigns zero to the output queue and returns **extra** to false. This code has nested **when** statements so would be better coded as -

```

queue({q1, q2}, "uint:8");

```

```

.
.
[
    static(extra, "log");

    when (extra) par {
        q2 <<- 0;
        extra <- false;
    } else sync {
        q2 <<- q1;
        extra <- (q1 & 0x80) != 0;
    }
]
.
.

```

Both this code and the one above read queue **q1** in two places. Because these take place simultaneously (forced to do so by the **sync**) they both pop the same datum.

We could also code this example using a sequential block -

```

queue({q1, q2}, "uint:8");

.
.
[
    sync {
        seq { // bad - always takes 2 cycles!
            q2 <<- ?q1;
            when ((q1 & 0x80) != 0)
                q2 <<- 0;
        }
    }
]
.
.

```

The **sync** block is immediately inside the module structure so again will execute repetitively. By putting a **sync** around the **seq** we have ensured that both queues are available before we enter the **seq**. The first sequential statement assigns the value of the input queue to the output queue, but examines the input rather than reading it. This leaves the datum in place. The second sequential statement reads the input queue and if the MSB is set it writes a zero to the output queue. However, this code is not good as it always takes a minimum of two cycles since the **when** will take one cycle even when the test is false.

15.6 interfacing external data

The following code illustrates an interface to an external data source. The data is clocked but is not present on every clock cycle. A separate clock enable input is asserted when data is present on the next clock edge.

```

useclock(c100); // 100MHz default platform clock

float(dclk_freq, 10.0e6); // data clock frequency, 10MHz
clock(dclk_in, frequency=dclk_freq, loc=A_pins[8]); // data clock input
clock({dclk, dclk_raw}, frequency=dclk_freq);
clkdcn(dclk_raw <- dclk_in, clkfb=dclk); // DCM for data clock
clockbufg(dclk <- dclk_raw); // data clock // clock buffer

queue(d, "uint:8");

dataInput(d <-);

```



```

process1(..... <- d);
.
.
.

module
dataInput (queue(out, "uint:8") <- ) {
    useclock(dclk); // use the 10MHz data clock

    input(data_in, "uint:8", loc=A_pins[0..7]);
    input(enable_in, "log", loc=A_pins[20]);
    static(data, "uint:8");
    static(enable, "log");

    data <- data_in;
    enable <- enable_in;

    when (enable)
        out <<- data;

    useclock(); // revert to the default 100MHz clock
}

```

The first section interfaces the external 10MHz data clock through a DCM and a BUFG. **dclk_in** is the input from the PAD and associated IBUF. This is connected to the input of a DCM. The DCM output is **dclk_raw** which is buffered through a BUFG to get the internal distributed data clock **dclk**.

Queue **d** is the data source for further processing. **process1** is a module or procedure which contains a pipeline to process the data and is not defined here. Queue **d** is asynchronous and is written sporadically using the 10MHz **dclk** in module **dataInput()** and read using **c100**, the default processing clock.

The module **dataInput** has a single parameter, the output queue **out**. This queue delivers valid data from the data input when they become available. We assume that the 100MHz pipeline **process1()** will keep up with the maximum input data rate of 10MHz.

The module **dataInput** changes the current clock domain to **dclk**. The data and enable inputs are synchronous with this clock domain and the data is written to queue **out** in this clock domain.

The data and enable inputs are clocked on every cycle of **dclk** into static variables **data** and **enable**. Now synchronised to **dclk** within the FPGA when enable is asserted the data is written to the module output queue **out** (whose matching argument is queue **d**). The current clock domain is then reverted to **c100** on exit from the module.

Suppose that we wish to reverse the data bits. There are three ways we could do that -

- We could simply reverse the order of the data input pins.
- We could just reverse subscripts.
- We could assign the bits from **data** to **out** singly (but in parallel of course) in reverse order

The first is obvious -

```
input(data_in, "uint:8", loc=A_pins[7..0]);
```

The second can be done by library function **reverse()** -

```
out <<- reverse(data);
```

The third requires casting the data to and from an array -

```

type(t, "[8]bits:1");
value(v1,, cast(data, t));
value(v2, t);

for (int(i) ; i<8 ; i++)
    v2[i] := v1[7-i];
out <<- cast(v2, type(out));

```

This could also be done by using library function `getbit()` -

```
type(t, "[8]bits:1");
value(v2, t);

for (int(i) ; i<8 ; i++)
  v2[7-i] := getbit(data, i);
out <<- cast(v2, type(out));
```

If the enable input is such that it remains asserted for multiple clock cycles to delineate a data frame, we may want to include a flag on the first datum in each frame.

```
struct (dtype,
  data, "uint:8", // data
  first, "log" // is this the first frame datum?
);

module
dataInput (queue(out, dtype) <- ) {
  useclock(dclk); // use the 10MHz data clock

  input(data_in, "uint:8", loc=A_pins[0..7]);
  input(enable_in, "log", loc=A_pins[20]);
  static(data, "uint:8");
  static(enable, "log");

  data <- data_in;
  enable <- enable_in;

  static(first, "log", true);

  when (enable) {
    out.data <<- data;
    out.first <<- first;
    first <- false;
  }
  when (!enable)
    first <- true;

  useclock(); // revert to the default 100MHz clock
}
```

This code is a little slack in that it assumes that the queue **out** will always be available for writing. This assumption is reasonable in that there is no flow control of the external data source so data would be lost if the output of module **dataInput()** did not keep up with the data input stream since the write to **out** would hang until it became available. Meanwhile the next datum might have overwritten the previous. Enclosing the code in a **sync{}** would not help as that simply moves the write availability check from the individual assignment statements to an enclosing block where it would still wait.

15.7 a 1-dimensional pixel window

To create a sliding 3 pixel window from a data stream we might consider creating the structure shown in [Figure 15.1](#) p599. This is not the best way to implement a window operator.

Firstly module **m4** would have to be written so that is first accepted **q1** only, then **q1** with **q2** and finally all three queue inputs. If **m4** waits on its 3 inputs it will read the same datum from all 3 and it will only output every 3 cycles, the pipeline being delayed 2 cycles in every 3. **m2** and **m3** must hold exactly one datum each, otherwise the window will not contain contiguous data. We could get around the startup problem by giving queues **q2** and **p3** initial data.

Secondly these modules as shown would only be unit delays for which the queue overheads would be inefficient.

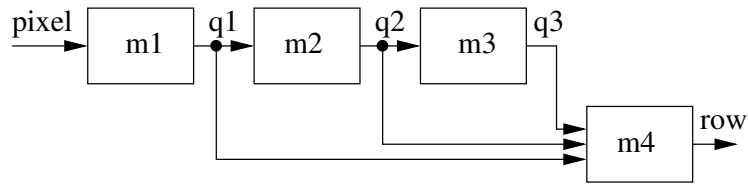


Figure 15.1: A 3 pixel window

It is far better to implement the window delays as static variables within a module.

```

module
win3 (queue(out, "[3]uint:8") <- queue(in, "uint:8")) {
    static(del, "[2]uint:8");

    sync {
        del[0] <- in;
        del[1] <- del[0];
        out[0] <<- in;
        out[1] <<- del[0];
        out[2] <<- del[1];
    }
}

```

Note that we needed a **sync** block to bind the execution of the queue and static assignments to the availability of queues **in** and **out**. With a **par** the two static assignments would have executed straight away regardless of the availability of the queues.

We could parameterise this in a number of ways. Firstly we could make the window width a parameter -

```

module
win1D (queue(out, "["+width+"]uint:8") <- queue(in, "uint:8"), int(width)) {
    static(del, "["+(width-1)+"uint:8");

    sync {
        del[0] <- in;
        for (int(i, 1) ; i<(width-1) ; i++)
            del[i] <- del[i-1];
        out[0] <<- in;
        for (int(i, 1) ; i<width ; i++)
            out[i] <<- del[i-1];
    }
}

```

The above code could be written more simply as -

```

module
win1D (queue(out, "["+width+"]uint:8") <- queue(in, "uint:8"), int(width)) {
    static(del, "["+(width-1)+"uint:8");

    sync {
        del[0] <- in;
        del[1..width-1] <- del[0..width-2];
        out[0] <<- in;
        out[1..width-1] <<- del[0..width-2];
    }
}

```

Because input parameters are created and matched with arguments before output parameters we can use an immediate input parameter value as a dimension for an output parameter. An alternative would have been to simply omit any type for the output parameter and let the output argument (which must after all be correct) supply the parameter dimension.

```

module
win3 (queue(out) <- queue(in, "uint:8"), int(width)) {
    .
    .
    .
}

```

We could now make the window module independent of data type, picking up the input and output types from the arguments -

```

module
win1D (queue(out) <- queue(in), int(width)) {
    static(del, "["+(width-1)+"]"+type(in));

    sync {
        del[0] <- in;
        del[1..width-1] <- del[0..width-2];
        out[0] <- in;
        out[1..width-1] <- del[0..width-2];
    }
}

```

We could simplify this further by making the static the same type as the output queue, even though we need an array one smaller -

```

module
win1D (queue(out) <- queue(in), int(width)) {
    static(del, type(out));

    .
    .
    .
}

```

Here we rely on code optimisation to remove the unused array member of static variable **del**.

The previous examples do not recognise frame boundaries. In practice we might have a frame input signal which is asserted during contiguous pixels in a frame and not asserted during frame gaps. To carry this signal along with the pixel data a struct would be used for the input stream -

```

struct(inpix,
    data, "uint:8",
    frame, "log"
);

```

where **frame** is true for valid pixels and false for data between frames. For the data output there would no longer be a gap in the data stream so we could perhaps flag the last window in a frame by using another struct -

```

struct(pixwin1,
    data, "["+winwidth+"]uint:8",
    last, "log"
);

```

We have the choice of filling off-edge pixels in the 3-pixel window with a default value (0), adding another field to struct **pixwin1** or else restricting the scan width to exclude off-edge pixels. The following example takes the second alternative. It is assumed that the gap between frames is wider than the window.

```

int(winwidth, 3);
queue(q1, inpix);
queue(q2, pixwin1);

```

```

win1D(q2 <- q1, winwidth);
.
.
.

module
win1D (queue(out, pixwin1) <- queue(in, inpix), int(width)) {
  if (width%2 == 0)
    texit("win1D - window width is not odd");
  else
    int(centre, width / 2);
  static(del, "["+width+"]inpix");

  sync {
    del[0] <- in;
    del[1..width-1] <- del[0..width-2];
    when (del[centre].frame) par {
      for (int(i) ; i<width ; i++)
        out.data[i] <<- del[i].data;
      out.last <<- !del[centre-1].frame;
    }
  }
}

```

Note that the **sync** requires that both the input and output queues be available before the input queue will be read even when the window centre is off-edge. This is unlikely to be of concern.

15.8 a 2-dimensional pixel window

We can extend on the previous example to make a 2D window.

Since we are processing a 2D raster we can envisage an input signal that is asserted when a pixel is within a line and another that is asserted when a pixel is within a frame (frame now has a different meaning to that in the previous example). The input pixels will now have two fields in addition to the data -

```

struct(inpix,
  data,  "uint:8",  // pixel
  line,  "log",     // pixel within line
  frame, "log"      // pixel within frame
);

```

Here field **line** will be true when a pixel is within a horizontal line and field **frame** will be true when a pixel is within a frame. During line gaps **line** will be false but **frame** will remain true. Thus a pixel is valid when **line** and **frame** are both true.

```

struct(inpix,
  data,  "uint:8",  // pixel
  line,  "log",     // pixel within line
  frame, "log"      // pixel within frame
);

```

For a 2D array specification we can use a string -

```

int(winwidth, 3);
str(pixarray, "["+winwidth+"] ["+winwidth+"]");

```

which we can then insert in types -

```

type(pixwin2D, pixarray+"inpix");
struct(outpix,
  data,  pixarray+"uint:8",

```

```

    lastl, "log",
    lastf, "log"
);

```

The output struct contains the 2D pixel array **data**, a flag indicating that the central pixel is the last pixel in a line **lastl** and flag indicating the last line in a frame, **lastf**. **lastf** will be true for every pixel window in the last line.

```

module
win2D (queue(out, outpix) <- queue(in, inpix), int(linelen)) {
    int(ww, dimension(out.data, 0)); // window width
    if (ww%2 == 0)
        texit("win2D - window width is not odd");
    else
        int(centre, ww / 2); // centre pixel of window
        static(del, pixwin2D); // window - 2D array of pixels

    sync {
        del[0][0] <- in;
        del[1..ww-1][0] <- delay(del[0..ww-2][ww-1], linelen-ww-1,, true);
        del[0..ww-1][1..ww-1] <- del[0..ww-1][0..ww-2];
        when (del[centre][centre].line) sync {
            for (int(i) ; i<ww ; i++)
                for (int(j) ; j<ww ; j++)
                    out.data[i][j] <<- del[i][j].data;
            out.lastl <<- !del[centre-1][centre-1].line;
            out.lastf <<- !del[centre-1][centre-1].frame;
        }
    }
}

```

The .. construction indicates a subscript range. The code above connects up the window elements such that each row shifts one left and the the input for each row is the leftmost pixel of the previous row delayed by the line length - the widow length - 1. If expanded explicitly the above would look like -

```

int(ed, linelen-ww-1); // extra delay - line length - window length - 1

del[0][0] <- in; // input to 1st line

del[1][0] <- delay(del[0][2], ed); // 2nd line from end of 1st
del[2][0] <- delay(del[1][2], ed); // 3rd line from end of 2nd

del[0][1] <- del[0][0]; // shift 1st line left
del[0][2] <- del[0][1]; // " " " "
del[1][1] <- del[1][0]; // shift 2nd line left
del[1][2] <- del[1][1]; // " " " "
del[2][1] <- del[2][0]; // shift 3rd line left
del[2][2] <- del[2][1]; // " " " "

```

The library function **delay()** implements a fixed delay. The 1st argument is the input. The 2nd argument is the delay line length. The 3rd argument, if true, specifies that block RAM is to be used to contain the data rather than flip-flops or CLB shift registers. This should be specified where too many shift registers would be required otherwise.

15.9 pixel window operations

To carry out window operations on an image it is necessary to stream the image in some useful pixel order. This is not a limitation of 3PL, which can access arrays in parallel, but rather a limitation of the hardware on which 3PL target code executes. Because of the size of image arrays current FPGA-based processing boards can only store images in conventional RAM. To access the image it must be read

through a port, usually one pixel at a time. To create a small window which scans through the image the image stream is delayed through successive line delay elements and pixel delay elements to form a parallel pixel window. Thus an $n \times n$ window will require $n-1$ line delays and $n \times (n-1)$ pixel delays. A pixel delay is simply a register but a line delay requires a small FPGA block RAM (rmemory). These blocks constitute the main limitation to the number and size of window operations we can perform in an FPGA. Of course if we can carry out a number of parallel operations on the same window we avoid proliferation of these delay elements.

Using the above pixel window code we can perform the usual image processing array operations, for example a Sobel operator could be coded as -

```
int(iwidth, 1024);
queue({pixel, sob}, "uint:8");
queue(window, "[3][3]uint:8");

.
win3x3(window <- pixel, iwidth);
sobel(sob <- window);
.

module
sobel (queue(out) <- queue(in)) {
    value({s1, s2, s3, s4}, "int:10");
    value({m1, m2}, "int:11");

    s1 := in[0][0] + 2 * in[1][0] + in[2][0];
    s2 := in[0][2] + 2 * in[1][2] + in[2][2];
    s3 := in[0][0] + 2 * in[0][1] + in[0][2];
    s4 := in[2][0] + 2 * in[2][1] + in[2][2];
    m1 := s2 - s1;
    m2 := s4 - s3;
    out <<- (m1 > 0 ? m1 : -m1) + (m2 > 0 ? m2 : -m2);
}
```

Further window operations would require more stages in the pipeline of calls to module **pixwin()** and window operator modules.

Note that the above sobel module is correct but it generates large combinatorial expressions whose logic depth is likely to prove a problem at high clock rates. We could recode it to use internal pipelining within the module -

```
module
sobel (queue(out) <- queue(in)) {
    queue({s1, s2, s3, s4}, "int:10");
    queue({m1, m2}, "int:11");

    sync {
        s1 <<- in[0][0] + 2 * in[1][0] + in[2][0];
        s2 <<- in[0][2] + 2 * in[1][2] + in[2][2];
        s3 <<- in[0][0] + 2 * in[0][1] + in[0][2];
        s4 <<- in[2][0] + 2 * in[2][1] + in[2][2];
    }
    m1 <<- s2 - s1;
    m2 <<- s4 - s3;
    out <<- (m1 > 0 ? m1 : -m1) + (m2 > 0 ? m2 : -m2);
}
```

Note that there is a **sync** around the four queue assignment statements that read queue **in**. This is not necessary for synchronisation since the internal queues **s1** to **s4** can only become unavailable simultaneously due to the symmetry of the code and hence these four assignments will execute simultaneously as they have a common input queue. However, the use of the **sync** means that queue availability is only tested once after which the four assignments execute without further queue testing. If the **sync** were omitted then each assignment would individually test queue availability before executing. This would

result in duplicated logic, but it is not obvious if this would be give larger or smaller signal delays (code duplication can give smaller signal delays by keeping signal lines short and local).

15.10 traversing a memory

The following module traverses a memory from an address supplied as an argument (a queue), summing the contents until a zero entry is encountered, at which time the sum is output. We need to write the code differently depending on whether we use combinatorial output memory or registered output memory.

For combinatorial output memory the read occurs within the current clock cycle so the data can be used directly. For this reason the combinatorial output memory read is a function rather than a procedure.

```
cmemory(mem, 1, "uint:8", "uint:8");

module
sum (queue(out, "uint:16") <- queue(inaddr, "uint:8")) {
  static(searching, "log");
  static(addr, "uint:8");
  static(sum, "uint:16");
  value(mrv, "uint:8", cmemoryread(mem, 0, addr));

  when (searching)
    when ((mrv == 0)) sync {
      reset(searching, sum);
      out <<- sum;
    } else par {
      sum <- sum + mrv;
      addr++;
    }
  else sync {
    sum <- cmemoryread(mem, 0, inaddr);
    addr <- inaddr + 1;
    searching <- true;
  }
}
```

Initially **searching** is false (by default) so the **else** of the outer **when** will be executed. This is a **sync** which will wait for the starting address queue **inaddr** to be available. When **inaddr** is available the memory location whose address in **in** will be read and assigned to the sum **sum** and logical static **searching** will be assigned **true**. The search address **addr** will be assigned **in + 1** ready to continue the scan.

The **sync** starting the outer **else** block is not strictly necessary as there is only one queue to be waited on and both dependent statements will hence execute at the same time. Also it does not matter if the assignment to **searching** occurs in a separate clock cycle to the queue reads. The use of **sync** there however will save checking the availability of **inaddr** on two deparate statements.

Note that for combinatorial output memory the output always reflects the current address input so despite the function name there is no explicit read.

The outer **when** will repeat in its implied infinite loop and this time will execute the true body. We want to do the memory read and then decide what action to take. By assigning the memory read function to a value variable, **mrv** (memory read value), we can use the result in two places, the inner **when** test expression and then optionally adding the value to the sum. It is important that the memory read and the address increment occur in the same clock cycle in the **par** block, but there is no problem there as no queues are involved so there will be no queue waits.

When **searching** is true and **mrv** is zero the sum can be written to queue **out**. Since **sum** is zeroed (along with **searching**) in the same parallel block as it is written to queue **out** the block must be a **sync** otherwise the sum will be zeroed whie waiting for the output queue to be available should it not already be so.

Note that the memory scan proceeds even if the output queue is not available right up until it is time to write the result to the output queue at which point there might be a wait on output queue availability.

The nested when statements in the code above could be eliminated by detecting the three distinct states

-

```
cmemory(mem, 1, "uint:8", "uint:8");

module
sum (queue(out, "uint:8") <- queue(inaddr, "uint:8")) {
  static(searching, "log");
  static(addr, "uint:8");
  static(sum, "uint:16");
  value(mrv, "uint:8", cmemoryread(mem, 0, addr));
  value(ieq0, "log", mrv == 0);

  sync {
    when (searching && ieq0) sync {
      reset(searching, sum);
      out <<- sum;
    }
    when (searching && !ieq0) par {
      sum <- sum + mrv;
      addr++;
    }
    when (!searching) sync {
      sum <- cmemoryread(mem, 0, inaddr);
      addr <- in + 1;
      searching <- true;
    }
  }
}
```

The sync will execute in an implied infinite loop as usual. The three **when** statements will execute simultaneously within the **sync** but the conditional expressions are mutually exclusive so that only one of them will execute its code block. The surrounding **sync** locks them together, otherwise if one were waiting on a queue wait another may find that its conditional expression has changed and then it might begin execution at the same time. The **sync** ensures that only one can execute at a time, given that their conditional expressions are mutually exclusive at the time of entry to the **sync**. We could also achieve the same by using individual **sync** statements -

```
cmemory(mem, 1, "uint:8", "uint:8");

module
sum (queue(out, "uint:8") <- queue(inaddr, "uint:8")) {
  static(searching, "log");
  static(addr, "uint:8");
  static(sum, "uint:16");
  value(mrv, "uint:8", cmemoryread(mem, 0, addr));
  value(ieq0, "log", mrv == 0);

  when (searching && ieq0) sync {
    reset(searching);
    out <<- sum;
  }
  when (searching && !ieq0) sync {
    sum <- sum + mrv;
    addr++;
  }
  when (!searching) sync {
    sum <- cmemoryread(mem, 0, inaddr);
    addr <- inaddr + 1;
    searching <- true;
  }
}
```

```
}

```

For registered output memory the output becomes available in the cycle following a read, the read being explicit.

```
rmemory(mem, 1, "uint:8", "uint:8");

module
sum (queue(out, "uint:8") <- queue(inaddr, "uint:8")) {
  static(searching, "log");
  static(addr, "uint:8");
  static(sum, "uint:16");
  value(mrv, "uint:8", rmemoryout(mem, 0));
  value(ieq0, "log", mrv == 0);

  when (searching && ieq0) sync {
    reset(searching);
    out <<- sum;
  }
  when (searching && !ieq0) sync {
    sum <- sum + mrv;
    rmemoryread(mem, 0, addr);
    addr++;
  }
  when (!searching) sync {
    rmemoryread(mem, 0, inaddr);
    reset(sum);
    addr <- inaddr + 1;
    searching <- true;
  }
}
```

This executes much as the previous example but will execute one extra memory read, which will not be used. The variable **mr_v** now represents the value of the memory output register holding the result of a memory read on a previous cycle.

On the first memory read the result is not available until the next clock cycle. On the following clock cycle the previous read result is added to **sum**, another memory read is executed and the address is incremented, all in parallel.

We could code the solution differently as follows

```
rmemory(mem, 1, "uint:8", "uint:8");

module
sum (queue(out, "uint:8") <- queue(inaddr, "uint:8")) {
  static(addr, "uint:8");
  static(sum, "uint:16");
  value(mrv, "uint:8", rmemoryout(mem, 0));

  seq {
    sync {
      rmemoryread(mem, 0, inaddr);
      addr <- inaddr + 1;
      sum <- 0;
    }
    while (mrv != 0) par {
      rmemoryread(mem, 0, addr);
      sum <- sum + mrv;
      addr++;
    }
    out <<- sum;
  }
}
```

This is an endlessly repeating sequence of three parts. The first reads the memory location given by the input queue **inaddr**. This will wait until the input queue is available. Following this a **while** loop continues until it encounters a zero entry, summing the values as it goes. The final part writes out the sum to the output queue (possibly waiting on its availability). This solution is more readable.

The three examples above might be shortened by overlapping the output assignment with the next address input.

15.11 searching a linked list in memory

This illustrates linked list traversal using static variables. The memory contains items structured as link plus data. The queue function is passed an initial memory address. It follows links in the memory until it finds a zero address. At that point it outputs the associated data item.

```
str(type, "uint:8");
struct(item,
  data, type,
  addr, type
);
rmemory(mem, 1, type, item);

module
find (queue(out, type) <- queue(inaddr, type)) {
  static(searching, "log");
  value(mrv, item, rmemoryout(mem, 0));

  if (!searching) par {
    rmemoryread(mem, 0, inaddr);
    searching <- true;
  } else par {
    if (mrv.addr == 0) par {
      reset(searching);
      out <<- mrv.data;
    } else
      rmemoryread(mem, 0, mrv.addr);
  }
}
```

As in the previous section we could write instead -

```
str(type, "uint:8");
struct(item,
  data, type,
  addr, type
);
rmemory(mem, 1, type, item);

module
find (queue(out, type) <- queue(in, type)) {
  static(i, item);

  seq {
    rmemoryread(mem, 0, in);
    while (i.addr != 0)
      rmemoryread(mem, 0, i.addr);
    out <<- i.data;
  }
}
```

15.12 function generation using look-up tables

Arbitrary function generation can be performed by means of a look-up table. For example to generate a stream of sin values to a queue -

```
var(mi, "[256]uint");
for (int(i) ; i<256 ; i++)
    mi[i] = 2048 + floor(2047.0 * sin(2 * i * math_pi / 256.0));

rmemory(m, 1, "uint:8", "uint:12", mi);

static(addr, "uint:8");
queue(sinstream, "uint:12", depth=16);

sync {
    rmemoryread(m, 0, addr);
    sinstream <<- rmemoryout(m, 0);
    addr++;
}
```

The sin values are generated by immediate code and used as an initialisation array for the block RAM (rmemory). The 12 bit values are offset by 2048 so that they are positive. The rmemory has an 8 bit address to hold 256 values and the stored values are 12 bit unsigned.

The following code takes the sin stream and drives an external MCP4822 DAC to produce an analog sin wave. The data stream to the DAC is much slower than the sin generation which just waits, generating values when the queue **sinstream** is not full.

```
module "spi.3pl"

useclock(c32); // 32MHz clock domain

// sin generation code inserted here

static({_cs, _ldac}, "log", true);
static({sdi, sck}, "log");
struct(s,
    val,          "uint:12",
    active,       "log",
    gain_low,     "log",
    dummy,        "log",
    select,       "uint:1"
);
queue(data, s);

output(, _cs, loc=C_pins[0]); // DAC chip select
output(, sdi, loc=C_pins[1]); // DAC serial data in
output(, sck, loc=C_pins[2]); // DAC serial clock
output(, _ldac, loc=C_pins[3]); // DAC load value

seq {
    sync {
        data.select <<- 0; // select A output
        data.gain_low <<- true; // X1 gain
        data.active <<- true; // active
        data.val <<- sinstream; // DAC value
    }
    _ldac <- false;
    _ldac <- true;
    wait(781.241e-6); // delay for 5Hz 256 sample/cycle sin
}

value(spi_step, "log"); // SPI serial transmission step - 4MHz 32 / 8)
```

```

seq {
  assert(spi_step <- ); // 1 cycle
  nop(7);               // wait 7 cycles
}

spi_master(sck, sdi, _cs <- data,, spi_step);

```

This can be expanded to multi-phase by interleaved reads of the sin table since the table reads are very much faster in this case than the data transmitted to the DAC. Multiple DACs can be driven in parallel.

15.13 signal processing

It is not usually appropriate to use queue mode variables for signal processing pipelines. Queue mode variables are queues whereas what is required is a chain of registers. In a processing pipeline where queue mode variables are used each datum moves along the pipeline at clock rate regardless of other data. If input data is unavailable the pipeline empties from the input end. If the output is blocked then data fill the pipeline and become queued. A signal processing pipeline is implemented using a chain of registers where data is available on every clock cycle and the output is never blocked. Each register in the pipeline contains one datum. The data move in lock step every clock cycle.

A signal processing schematic is shown in [Figure 15.2](#) p609.

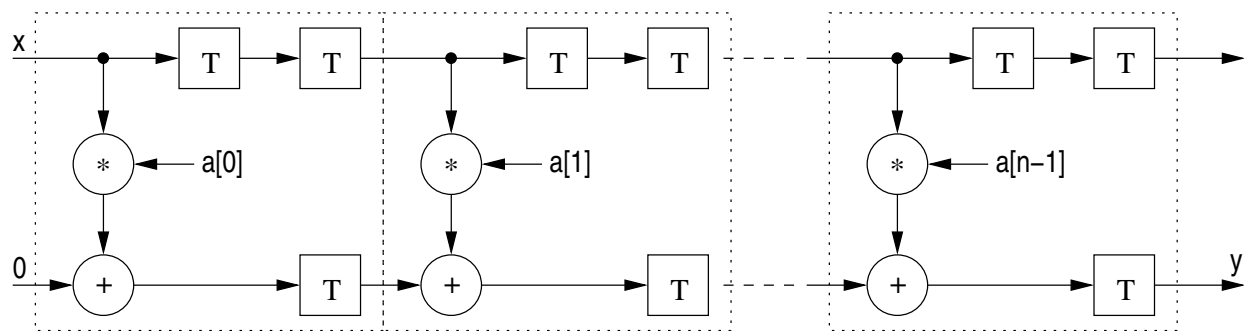


Figure 15.2: Simple signal processing example

The code for a single stage might be -

```

module
section (
  static({out0, out1}, "fixed:12.6")
  <-
  value({in0, in1}, "fixed:12.6"), float(a)
) {
  static(s, "fixed:12.6");

  s <- in0;
  out0 <- s;
  out1 <- a * in0 + in1;
}

```

For simplicity and generality the input and output parameter types can be omitted, in which case the types of the arguments will be used -

```

module
section (static({out0, out1}) <- value({in0, in1}), float(a)) {
  static(s, type(in0));

  s <- in0;
  out0 <- s;
}

```

```

    out1 <- a * in0 + in1;
}

```

The complete filter could be constructed by -

```

int(n, 50);
var(x_pins, "["+n+"]string", {"A2", "B4", .....}); // FPGA input pins
var(y_pins, "["+n+"]string", {.....}); // FPGA output pins
var(a, "["+n+"]float", {3.1, 2.7, .....}); // coefficients
input(x, "fixed:12.6", loc=x_pins);
value(v1, "fixed:12.6", i1);
value(v2, "fixed:12.6", 0);

for (int(i) ; i<n ; i++) {
    section(v1, v2 <- v1, v2, a[i]);
}

output(, v2, loc=y_pins);

```

Note the use of value variables **v1** and **v2** as both input and output arguments to the module **section**. On each call the prior value of each variable is used and on each return these variables point to the new output parameter static variables. Thus **v1** and **v2** act as links between successive calls to construct the processing pipeline chain.

A less regular example is shown in [Figure 15.3](#) p610. This example is an IIR, or recursive, filter from a

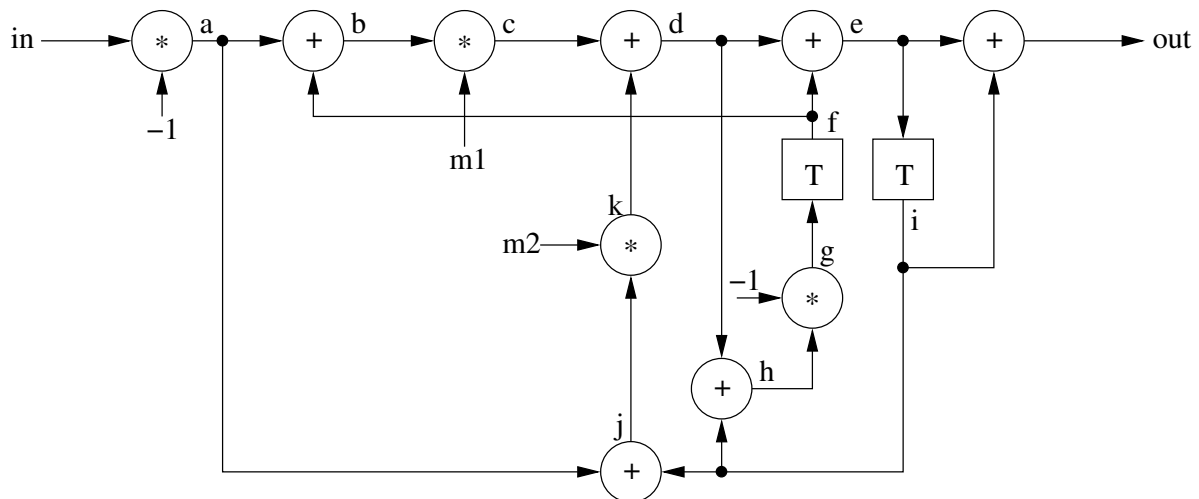


Figure 15.3: Complicated signal processing example

text on digital filters. The code for this is -

```

module filter (queue(in, "uint:16") <- queue(out, "uint:16")) {
    static({f, i}, "uint:16");
    value({a, b, c, d, e, g, h, j, k}, "uint:16");
    int(m1, ...);
    int(m2, ...);

    sync {
        f <- g;
        i <- e;

        a := -in;
        b := a + f;
        c := b * m1;
        d := c + k;
        e := d + f;
    }
}

```

```

        g := -h;
        h := i + d;
        j := a + i;
        k := j * m2;

        out <<- e + i;
    }
}

```

This could be shortened to -

```

module filter (queue(in, "uint:16") <- queue(out, "uint:16")) {
    static({f, i}, "uint:16");
    value({d, e, g}, "uint:16");
    int(m1, ...);
    int(m2, ...);

    sync {
        f <- g;
        i <- e;

        d := (f - in) * m1 + (i - in) * m2;
        e := d + f;
        g := d - i;

        out <<- e + i;
    }
}

```

15.14 A batcher parallel sorting network

A batcher sort network is generated by an algorithm, which must be written as immediate code. The following module creates a batcher sorting network for any number of inputs greater than 2. The input and output queues must be arrays of the same size and type **uint** or **int**.

```

module
sort (queue(out) <- queue(in)) {
    int(inputs, dimension(in));
    int(stages, get_stages(inputs));
    int(sbits);      // bits to hold 'size'
    int(size2);      // power of 2 containing window size
    int(i);
    int({p, q, r}); // Batcher parameters
    int(d); // Batcher parameter - distance between exchange pairs
    int(wi);      // window member index
    int(si);      // network stage index
    var(ignore, "["+inputs+"]log"); // flags for skipping 2nd of exchange pair
    var(net, "["+(stages+1)+"]->");
    net[0] = &in;
    queuearray(net[1..stages-1] <- type(in));
    net[stages] = &out;

    for (i=inputs-1 ; i!=0 ; i=i>>1)
        sbits++;
    size2 = 1 << (sbits - 1);

    for (p=size2 ; p>0 ; p=p>>1) {
        r = 0;
        d = p;
        for (q=size2 ; q!=(p/2) ; q=q>>1) {
            [

```

```

        for (wi=0 ; wi<inputs ; wi++)
        if (ignore[wi])          // already connected
            ignore[wi] = false;
        else if ((wi < (inputs-d)) && ((wi & p) == r)) { // exchange
            cond_exchange ( net[si+1]->[wi], net[si+1]->[wi+d]
                <-
                net[si]->[wi], net[si]->[wi+d] );
            ignore[wi+d] = true; // ignore later
        } else                    // straight through
            net[si+1]->[wi] <- net[si]->[wi];
    ]
    d = q - p;
    r = p;
    si++;
}
}
}

```

Notice that the input and output queues are the only arguments passed to the module, all necessary parameters being derived from the characteristics of the input queue. The intermediate **for** loop with loop variable **q** generates an inline module on each iteration, each module handling one parallel stage of the sort. The stages are pipelined together. The inner loop generates conditional exchange assignments or direct assignments.

The code needs an array of queues for the multi-stage network and since we cannot implement that directly a pointer array is used. The first and last pointers in the array are to the input and output queues. The intermediate pointers are to queues created by the library procedure **queuearray()**.

The immediate code which generates the target code uses Batcher's algorithm which will not be described here. The number of stages needed for the sort is determined by the following immediate function -

```

function
get_stages (int(size)) {
    int({sbits, stages});

    for (int(i, size-1) ; i!=0 ; i=i>>1)
        sbits++;
    for (stages=0 ; sbits>0 ; sbits--)
        stages = stages + sbits;
    return(stages);
}

```

Conditional exchanges are implemented by the following procedure -

```

procedure
cond_exchange (queue(o0), queue(o1) <- queue(i0), queue(i1)) {
    value(gt01, "log");

    gt01 := i0 > i1;
    par {
        o0 <- gt01 ? i1 : i0;
        o1 <- gt01 ? i0 : i1;
    }
}

```


Chapter 16 - Specific Limitations

16.1 multiply, divide and remainder operators

16.1.1 Virtex-E

In the Virtex-E there are no hardware multiply or divide functions. The multiply, divide and remainder operators carry out these operations using cascaded adders or subtractors. Since these are expression operators they cannot be pipelined. The result is that these operations have a large data delay, typically 30ns to 40ns.

If the operation can be allowed to take place over several clock cycles between static variables then the timing constraints can be changed to suit, e.g.

```
static(a, "uint:8"); attribute(a, {tnm="SRC"});
static(b, "uint:8"); attribute(b, {tnm="SRC"});
static(c, "uint:16"); attribute(c, {tnm="DEST"});
maxdelay("SRC", "DEST", 40.0e-9); // 40ns path constraint

c <- a * b;    // slow operation!
```

If the operation must be at clock rate then pipelined 3PL library modules should be used for multiply, divide and remainder.

16.1.2 Virtex-II or Virtex-II Pro

In the Virtex-II or Virtex-II Pro there are hardware multipliers but no hardware dividers.

The multiply operator uses one hardware multiplier for operands up to 18 bits if signed, 17 bits if unsigned. If one operand is between 19 and 35 bits signed, or 18 to 34 bits unsigned, then two hardware multipliers plus an adder are used. If both operands are between 19 and 35 bits signed, or 18 to 34 bits unsigned, then four hardware multipliers plus two adders are used.

No attempt has been made to share hardware multipliers between products of small operands. Hardware multipliers are a limited resource and care must be taken not to exceed the number available in the particular FPGA being used.

There are no hardware dividers and divide and remainder operators carry out these operations using cascaded adders and subtractors. Since these are expression operators they cannot be pipelined. The result is that these operations have a large data delay, typically 30ns to 40ns. For approaches to this see the previous subsection above (Virtex-E).

16.1.3 Virtex-5

In the Virtex-5 there are hardware multipliers but no hardware dividers. The multipliers are contained in the DSP48E blocks.

The multiply operator uses one hardware multiplier for operands up to 25 bits by 18 bits signed. Unsigned operands have a 0 sign bit appended at the most significant end and thus become one bit larger.

Selectvalue, static and queue mode variables which are assigned arithmetic or logical expressions can employ DSP blocks implicitly by setting the DSP attribute. In the case of static variables this will also use the DSP block output register rather than using flip-flops (but not if the variable is initialised to a value other than zero since the DSP block output register cannot be initialised). Selectvalue, static and queue mode variables can have the ALU attribute (and not the DSP attribute) in which case additions and subtractions in assigned expressions will be combined in a common ALU implemented in CLBs.

To take advantage of the detailed architecture of the DSP blocks they can be instantiated directly by calling **DSP48E()** [subsection 11.3.2.3](#) p410 in library xc5v.3pl. Alternatively there are some useful modules providing a simpler interface in that library -

adder/subtractor - **dspaddsub** [subsection 11.3.1.1](#) p401

counter - **dspcounter** [subsection 11.3.1.2](#) p402

logical functions - **dsplog** [subsection 11.3.1.3](#) p403

multiplier - **dspmuladd** [subsection 11.3.1.4](#) p404

multiplier adder - **dspmuladd** [subsection 11.3.1.5](#) p406

multiplier subtracter - **dspmulsub** [subsection 11.3.1.6](#) p406

subtracter multiplier - **dspsubmul** [subsection 11.3.1.7](#) p406

If these modules are used with no DSP48E registers then the blocks can be used as a combinatorial elements in simple static and queue variable assignments. If DSP48E internal pipeline registers are used then the pipeline delays must be taken into account. This complicates the programming style considerably and is a tradeoff for the performance and resource usage advantages gained.

16.1.4 Virtex-6

At the moment the support is identical to that for virtex-5 - see above.

16.2 registered output memory

Registered output memory in Xilinx FPGAs is block RAM. These are a limited resource.

16.2.1 XCV300E

Block RAMs in the XCV3 have 4,096 bits which can be configured as -

data width	address width	depth
1	12	4096
2	11	2048
4	10	1024
8	9	512
16	8	256

i.e. anywhere from 8 bit address to 12 bit address.

The 3PL compiler will select blocks of appropriate size and configuration to provide depths of 256 to 4096, using as many blocks as required to match the data width.

16.2.2 XCV3, XC2VP

Block RAMs in the XCV3 have 8,432 bits which can be configured as -

data width	address width	depth
1	14	16384
2	13	8192
4	12	4096
9	11	2048
18	10	1024
36	9	512

i.e. anywhere from 9 bit address to 14 bit address (the unusual data widths result from the use of extra 'parity' bits at the wider configurations).

The 3PL compiler will select blocks of appropriate size and configuration to provide depths of 512 to 16384, using as many blocks as required to match the data width.

16.2.3 Virtex-5, Virtex-6

Block RAM elements are grouped in pairs, each block containing 18Kbits. The pair can be configured as a single 36Kbit block. 18K blocks can be configured as -

data width	address width	depth
1	14	16384
2	13	8192
4	12	4096
9	11	2048
18	10	1024

36K blocks can be configured as -

data width	address width	depth
1	15	32768
2	14	16384
4	13	8192
9	12	4096
18	11	2048
36	10	1024

The 3PL compiler will select blocks of appropriate size and configuration to provide depths of 1024 to 32768, using as many blocks as required to match the data width.

Block RAM is also used to implement deep queues. For queues whose depth is 512, 1024, 2048, 4096 or 8192 a FIFO primitive element is used. This uses a block RAM with inbuilt FIFO control logic, saving considerable external CLB logic and probably giving improved performance (but note that the clock to output time of nearly 2nS still persists). The hardware FIFO is not used in the case of asynchronous queues on which the `queureadstatus()` and `queuewritestatus()` have been used.

16.3 selectvalue variables

Selectvalue variables implement buses using CLB logic unless explicitly directed to use 3-state buffers driving partitionable 'long lines'. This is done by setting attribute "threestate" on individual selectvalue variables. These are only available on older FPGAs such as Virtex-E, Virtex-II and Virtex-II Pro.

16.4 high speed serial

16.4.1 Virtex-II Pro

MGT high speed serial ports are interfaced using library `gt_custom_16()` [subsection 11.6.1.1](#) p432 in library `xilinx/mgt.3pl`.

16.4.2 Virtex-5, Virtex-6

GTP high speed serial ports are interfaced using library module `gtp_dual_16()` [subsection 11.4.1.1](#) p415 in library `xilinx/gtp_dual.3pl`.

GTX high speed serial ports are not yet supported.

Chapter 17 - Using CPLDs

The Xilinx Coolrunner and XC9500 CPLDs are supported. The variable modes **selectvalue**, **queue**, **cmemory** and **rmemory** cannot be used.

For **static** mode variables the **continuous** attribute is true by default. This means that top level static assignments will not have an initiated execution thread containing an infinite loop but instead the register clock enable will be continuously asserted.

For **output** mode variables the **direct** attribute is true by default. This means that conditional (3-state) output will have the enable driven directly from the conditional test expression rather than from an initiated execution thread.

Target multiplication, division, remainder and variable left or right shift are not implemented for CPLDs.

The following 3PL program illustrates XC9500 usage -

```
part("XC9536PC44-5");

module "xilinx/xc95.3pl";

var(inpins, "[]str", {"P11", "P12", "P13", "P14", "P18", "P19", "P20"});
str(outpin, "P22");
var(countpins, "[]str", {"P34", "P35", "P36"});
str(clkpin, "P5");

clock(clk, frequency=1.0e6, loc=clkpin);

useclock(clk);

input(in, "[7]log", loc=inpins);

static(o, "log");
static(count, "uint:3", 5);
output(, count, loc=countpins);

count++;

seq {
  o <- in[0] && in[1] && in[2] && in[3] && in[4] && in[5] && in[6];
  o <- false;
}

output(, o, loc=outpin);
```

For Coolrunner CPLDs the I/O standard can be specified -

```
part("XC2C32APC44-6");

module "xilinx/xc2c.3pl";

var(inpins, "[]str", {"P25", "P26", "P27", "P28", "P29", "P33", "P34"});
var(outpin, "str", "P35");
var(clkpin, "str", "P5");

clock(clk, frequency=1.0e6, loc=clkpin);

useclock(clk);

input(in, "[7]log", loc=inpins);

static(o, "log");

seq {
  o <- in[0] && in[1] && in[2] && in[3] && in[4] && in[5] && in[6];
  o <- false;
}

output(, o, loc=outpin);
```

The 3PL compiler will run the appropriate Xilinx CPLD fitting software, if post-processing is not disabled. The final product is a JEDEC file with a **.jed** file name extension.

Chapter 18 - Setting up an FPGA board

The Gadget Factory papilio board provides a good example of the way 3PL is set up for a specific board.

The following is a very simple 3PL application program for the papilio -

```
module "boards/gadget_factory/papilio_one_500.3pl"

static(t, "log");

seq {
    wait(0.5);
    t <- !t;
}

output(., t, loc=A_pins[0]);
```

This trivial program outputs a 1Hz square wave on connector A pin 0 (P18). The only requirement in setting up the platform is the first statement which imports a file module specific to the papilio one 500. A 32MHz clock is established by default and is the current clock domain.

The imported file looks as follows -

```
nosource

// 3pl board support for GadgetFactory Papilio One 500k board

part("XC3S500EVQ100-4");

module "xilinx/xc3s.3pl"

// all IO is 3.3V on this board
str(lvcmos33, "LVCMOS33");

// clock generator 32MHz
iopins(c32_pin, "str", "P89", frequency=32.0e6, ibufg=true, iostandard=lvcmos33);

// 48 I/O pins across A (wing 1A), B (wing 1B), C (wing 2A)
iopins(global.A_pins, "[16]str", {
    "P18", "P23", "P26", "P33", "P35", "P40", "P53", "P57",
    "P60", "P62", "P65", "P67", "P70", "P79", "P84", "P86"
}, iostandard=lvcmos33
);
iopins(global.B_pins, "[16]str", {
.
.
.
.
```

```

iopins(global.flash_cs_pin, "str", "P24", iostandard=lvcmos33);
iopins(global.flash_clk_pin, "str", "P50", iostandard=lvcmos33);
iopins(global.flash_mosi_pin, "str", "P27", iostandard=lvcmos33);
iopins(global.flash_miso_pin, "str", "P44", iostandard=lvcmos33);

// FTDI USB channel B serial receive and transmit
// Labelling is reversed to schematic!
// 'tx' should be an FPGA output, 'rx' an input
iopins(global.serial_rxd_pin, "str", "P88", iostandard=lvcmos33);
iopins(global.serial_txd_pin, "str", "P90", iostandard=lvcmos33);

// create 32MHz clock
clock(c32_in, loc=c32_pin);
simpleclock(global.c32, c32_in);
useclock(c32);

directive("comms_baud_rate", 115200); // define default serial device baud rate
module "xilinx/uart_cpu.3pl"

trailermodule ("create_uart_bus_interface") {
    create_bus_interfaces();
}

```

Some of the pin definitions have been omitted ... to shorten the listing.

This file module specifies parameters for this particular board.

The call to inbuilt procedure part() specifies the Xilinx FPGA part number.

The next statement imports the file module xilinx/xc3s.3pl, described later. That file module specifies parameters appropriate for the particular FPGA.

The calls to iopins() build a map of all input and output FPGA pins and attributes associated with them, such as the FPGA pin name and the I/O standard. Additional attributes can be specified when the pins are passed to an input(), output() or clock() inbuilt procedure.

The first call to iopins() specifies the clock input pin (P89), the clock frequency, (32MHz), specifies that an IBUFG should be used and that the I/O standard is LVCMOS.

A call to clock() creates a clock mode variable, c32_in. The call to simpleclock() in xilinx/clocks.3pl creates a global clock mode variable c32, a clock manager (DCM) and a clock buffer. c32 is then made the current clock.

Module xilinx/uart_cpu() is imported to provide a CPU interface to the FPGA using the serial link. That interface is set up at the end by a trailer module which calls create_bus_interfaces()

Finally a trailer module is created to call create_bus_interfaces() after the source code exits to build the logic required to flesh out the CPU/FPGA interface. Calls to the interface routines are made throughout the source code and this logic is required at the end to tie it all together.

The file module xilinx/xc3s.3pl, while specific to the Spartan 3S, also imports libraries relevant to all Xilinx FPGAs.

```

// Include file for Xilinx Spartan 3 (XC3S) elements
nosource

presetdirective("postProcessTool", "ise"); // ISE will be used

source "xilinx/common.3pl" // xilinx common library
module "xilinx/postprocess.3pl" // code to run ISE or Vivado
module "stdlib.3pl" // standard 3PL generic library
module "xilinx/clocks.3pl" // xilinx clock managers
module "xilinx/io.3pl" // xilinx I/O facilities

```



```

// Specify allowed special I/O and clock elements valid for
// this device.
// Map "elementMap" is in common.3pl.
elementMap["BUFG"] = true;
elementMap["BUFGMUX"] = true;
elementMap["BUFGCTRL"] = true;
elementMap["DCM"] = true;

leadermodule ("attributes", 100) {
    attributeprocedure(processPinAttributes);
}

trailermodule ("constraints", 100) {
    netlistcommentfromdirs();
    generateConstraints();
}

postmodule ("Xilinx postprocessing") {
    postprocess("xilinx/ise_fpga.cmd", directive("postProcessTool"), 9.1);
}

```

This sets a directive to select the place-and-route tools to be used ("ise" or "vivado") and imports several appropriate libraries that might be needed.

A leader module, which is executed prior to other source code, calls inbuilt procedure `attributeprocedure()` to notify 3PL that the library procedure `processPinAttributes()` in `xilinx/common.3pl` is to be called whenever an `input()` `output()` or `clock()` inbuilt procedure is called with I/O pins. This procedure is called for each pin and processes associated attributes and constraints for the pin.

A trailer module, which is executed after the source program exits, calls two procedures, `netlistcommentfromdirs()` in `stdlib.3pl` and `generateConstraints()` in `xilinx/common.3pl`. The first of these simply copies all directives and their values as comments to the netlist (EDIF) file. The second generates constraints from information gathered during immediate execution.

Finally a post-processing module is created which calls `postprocess()` in `xilinx/postprocess.3pl`. (Note that post processing by the Xilinx tools can be suppressed by the `-s` option when running 3pl. This might be done if the tools are to be run manually or perhaps on another computer. For example if 3pl is executed on Apple OSX the Xilinx tools have to be executed on a Linux or Windows emulator such as Virtualbox or else run on another computer since Xilinx do not support OSX).

Chapter 19 - Running the Compiler

The 3PL compiler is written in Java. It can be run on any computing platform that supports Java 1.6 or later. The 3PL compiler was developed and used in UNIX and MACOS environments, but can be run in a Windows environment as well.

The 3PL compiler and libraries are generally installed in a system directory such as **/opt/3pl**. Each version (comprising both compiler and matching library files) is installed in a subdirectory such as **/opt/3pl/11.4.3**, so multiple versions can exist side-by-side. A symbolic link **/opt/3pl/current** points to the most recent version. The compiler executable would then be **/opt/3pl/current/bin/3pl**.

The compiler generates a Xilinx EDIF netlist file, some associated ISE and Vivado constraint files and three optional report files. It can then run ISE or Vivado to produce a configuration file that can be executed on the FPGA.

Error messages during parsing or immediate execution display variable identifiers in a number of different formats depending on the specific error. In many instances the identifier is displayed literally. Where more information would seem to be useful the identifier is displayed as *proc/var* for example where *var* was created within procedure *proc*. The same format applies to variables created within modules or functions. Note however that variables created within a structure nested within a module, procedure or function are in a deeper scope which has no name so the identifier will just be displayed literally. In the case of file variables the display format is *file:var* and for global variables *glob:var*. Where arguments are passed to module, procedure or function parameters a parameter identifier will be displayed as a chain of variables separated by *->* showing the linkage from the parameter back through one or more calls, for example *p1/v->m2/a->glob:x* where the first variable in the chain is the parameter in question.

19.1 Environment variables

The following environment variables should be defined -

PATH should include the location of the 3PL compiler, typically **/opt/3pl/current/bin**. However, you can select a specific version if you wish. The latter is typically done in Makefiles or build scripts.

XILINX should be set to the installation directory of the Xilinx post-processing tools, typically **/opt/Xilinx** or similar, if 3PL is to run these tools. This directory would have been selected when the Xilinx tools (either ISE or Vivado) were installed. The standard Xilinx installation structure under this directory is assumed. It can be overridden by a 3PL directive of the same name.

The following environment variables generally *should not* be defined by the user, but may be required in unusual situations -

THREEPL_JAVA. It is assumed the Java virtual machine is run by executing the command **java**. If not, or if you wish to use a specific Java, the command to execute it can be overridden by defining this environment variable.

THREEPL_JAVA_OPTS. This is by default **-Xmx800M** which sets the Java machine maximum heap size to 800Mbyte.

THREEPL_INCLUDE_DIRS. This is the location of the 3PL libraries, and is automatically generated relative to the execution path of the compiler executable. This ensures the correct version of the libraries is used. However it can be overridden using this environment variable.

XILINX_ISE. Specifies a precise version of Xilinx ISE to use, for example `/opt/Xilinx/14.7i`. It can be overridden by a 3PL directive of the same name. This overrides the **XILINX** environment variable, if defined. Normally the 3PL post-processing code will choose the newest version appropriate for the device.

XILINX_VIVADO. Specifies a precise version of Xilinx Vivado to use, for example `/opt/Xilinx/Vivado/2019.1`. It can be overridden by a 3PL directive of the same name. This overrides the **XILINX** environment variable, if defined. Normally the 3PL post-processing code will choose the newest version appropriate for the device.

19.2 Generating EDIF

The 3PL compiler is run by -

```
3pl {-Idirectories} {-dhlpst} name
```

where the source file is "name.3pl" (the file name extension need not be given in the call to the compiler, though it is accepted). The file name argument may be a path containing "/"s. There must be only one source file argument.

The *design name* is derived from the source file name argument by removing the trailing ".3pl" if present and removing the preceding file path up to the rightmost "/" if any.

The design name may be set by a command line option **-N name**. It may also be changed by calling inbuilt procedure **designname()** where the string argument supplies a new design name. The command line option overrides the inbuilt procedure call which in turn prevails over the default derivation from the file name.

The design name is also changed if inbuilt procedure **export()** is called to create a logic core EDIF file.

The design name should comprise only alphanumeric characters or "_" (underscore). Any other characters are silently converted to "_". This is to ensure the design name matches the output file names and also observes EDIF identifier name restrictions.

The compiler will create a Xilinx EDIF netlist file called "name.edn" and either a Xilinx ISE constraint file, "name.ncf", or a Vivado constraint file, "name.xdc", where "name" is the design name.

The **-I** option specifies a ":"-separated list of directories to be searched for files to be included by **source** and **module** statements. There may be more than one **-I** and they may precede or follow the source file name. The "-I" may be separated from its list by spaces, i.e. the list is the following argument. Directories are searched in the order - current directory, **-I** option lists in order, then finally the 3PL library directory. Therefore, source/module files may be preempted by including alternative versions earlier in the directory search path.

To write the three output files to a directory other than the current directory a **-o** option can be used. The directory name can be appended to the **-o** or can be the following argument. The directory will be created if it does not exist.

The call options are -

- -c - compile only, do not execute
- -d - display the interactive netlist schematic window after outputting an EDIF netlist
- -D d=s - define a directive d whose value is string s
- -f - force compilation even if sources have not been changed
- -F - error messages use full path names rather than file names
- -G - use gates rather than LUTs to help optimisation
- -h - print a help message

- -i - intermediate code only
- -I d1:d2:... - list of directories to search for libraries
- -l - annotate printed TDE list with source file locations
- -n - print a list of netlist net identifiers and their equivalent signal names to a file 'name'.net
- -N name - change design name to 'name'
- -o dir - write file to directory 'dir'
- -p - print the TDE list prior to optimisation
- -R - just print token generator output - testing only
- -r - generate a verbose report to file 'name'.rpt
- -s - skip post-processing
- -T - print the TDE list (including signal definitions)
- -t - print the TDE list (excluding signal definitions)
- -V - print the compiler version information (with -v, prints more)
- -x - suppress netlist code optimisation - testing only
- -X - development mode - testing only
- -Y - print clock domain resolutions (compiler diagnostic)

3PL will by default check the times of last modification of all referenced source and library files and the compiler make date. If all of these are older than the output netlist file and report file the compiler will exit without compiling and executing. This can be overridden by using the -f option to force compilation and execution.

At the end of netlist generation 3PL will execute any post-processing modules. If this is not desired the -s option will cause this to be skipped.

The -r option causes a detailed report to be produced in file *design_name.rpt*.

The -t option will result in a listing of the intermediate code in file *design_name.icl*. The -l, -p and -T options modify the intermediate code listing and are only used for compiler testing.

The -c option terminates 3PL after parsing the source file.

The -i option terminates 3PL after immediate execution.

The -R, -x, -X and -Y options are only used for compiler testing.

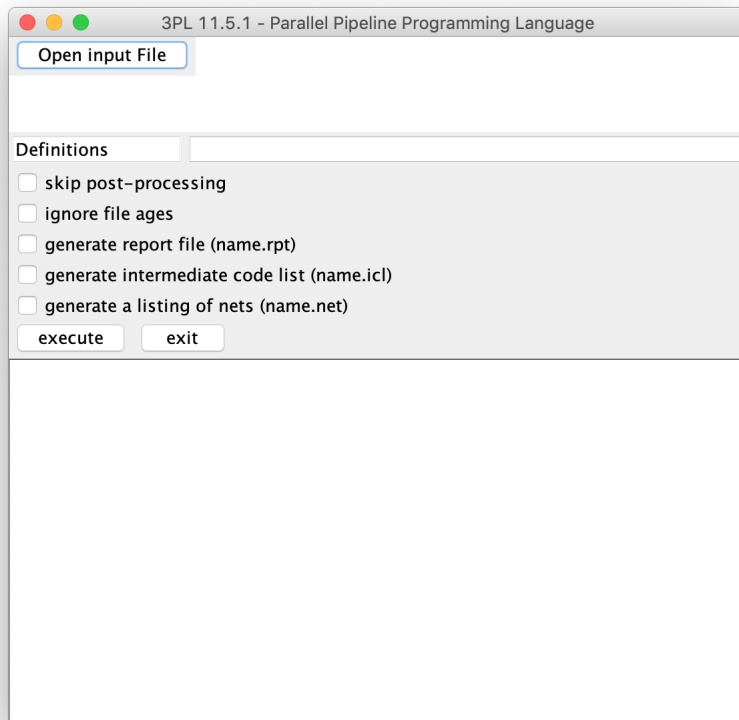
The -D command defines a directive. The directive may be just a key, in which case a directive of type "log" with that key and value **true** will be created. If key=value is given then the directive will be given the specified value and its type will be as appropriate for the value ("log", "int" or "float"). Space between the -D and the key is optional.

The -G option replaces many LUTs normally produced by the code generated with nested gates. The logic remains the same but the use of nested gates instead of explicit LUTs may change the way the vendor place-and-route software optimises and packs the logic which in turn may change the ease, or lack thereof, with which the timing constraints can be met. Thus when timing constraints fail to be met trying the -G option may improve the situation. The same effect can be achieved by setting directive **gatesNotLUTs** true prior to completion of immediate execution.

-I, -o, -D and -N may each be, but need not be, separated from its argument by spaces.

These options also set associated directives and for most of them the directives can also be set during immediate execution - see [section 3.10](#) p75.

If 3PL is called with no command line arguments, it pops up a GUI to open the file.



The GUI has text fields to specify the source file and any directives to define, buttons to set some options and a button to start (parsing, immediate execution and then code generation). The output appears in a scrollable text window. A file can be repeatedly compiled and executed by a single button press. Run this way it is still convenient to run via a script so that the library directory is specified and the Java memory heap maximum size is set. Directive definitions are space-separated and may be *name=value* or simply *name*. The value may be an integer, floating point or the strings "true" or "false". If none of those then it will be assumed to be a string value, any enclosing double quotes being removed. Where just a directive name is given it will be type "log" (boolean) with value TRUE. If the directives string is erroneous it will change colour to red.

Unfortunately the 3PL compiler terminates immediately on most parsing errors. This means that you can only fix one error at a time. Making the compiler continue following a parsing error is something the author has not yet worked out how to accomplish with the compiler compiler that was used.

The error messages have been designed to be as clear and informative as possible. However, the following parse error is an exception -

```
Parse error: file 'prog.3pl' line 2
Encountered "#m#prog.3pl#2#\n" in column 1.
Was expecting one of:
    "break" ...
    "calling." ...
    "case" ...
    .
    .
    .
```

This has resulted from unmatched braces around a module definition in a file included via a **module** or **source** statement!

An immediate execution error causes termination, which is reasonable.

If a post-processing module is provided then conditional code within it will carry out place-and-route and any other such tasks. This is described in the next section.

19.3 Generating Xilinx FPGA and CPLD configurations

See also [section 11.7](#) p438

3PL post-processing code will by default run the Xilinx ISE or Vivado tools to transform the EDIF and constraint files generated by 3PL into configuration files that may be downloaded into the device for execution. To prevent this either the `-s` option or the `"skipPostProc"` directive must be set true. This might be done where 3PL is executed on a different host to that of the place-and-route tools or where the user prefers to run the place-and-route separately for some reason.

The post-processing code automatically selects the Xilinx tool set and version based on the FPGA or CPLD part. Xilinx Vivado is used for 7-series FPGAs, and ISE for all other FPGAs and CPLDs.

The code uses the newest installed version of ISE or Vivado, found under the installation directory specified by the **XILINX** environment variable or 3PL directive, that supports the part. This is because Xilinx dropped support for older parts as newer versions of ISE were released, and older versions of the software did not support later parts.

This version can be overridden if necessary using either the **XILINX_ISE** environment variable or 3PL directive, or the **XILINX_VIVADO** environment variable or 3PL directive, as appropriate. The need to do so should be rare.

Actual post-processing uses a 'recipe' file. For ISE the recipe file is a simple list of ISE commands to run. The default recipe for FPGAs is the file **xilinx/ise_fpga.cmd** -

```
ngdbuild @designName@
map -detail -w -timing -ol high -pr b @designName@
par -w -ol high @designName@ @designName@.ncd
bitgen -m -w -g LCK_cycle:4 -g GTS_cycle:1 @designName@
```

Items in the recipe file enclosed with @ characters are replaced with the 3PL directive of the same name. Generally only the "designName" and "part" directives are useful. The ISE commands are executed in order, until one of them returns an error status or the script is completed.

For Vivado the recipe file is a Tcl script run in the Vivado Tcl interpreter in non-project mode. The same @-substitution is performed on the recipe first. The default recipe is **xilinx/vivado.tcl** -

```
# read the EDIF netlist file into Vivado
read_edif @designName@.edn

# read the constraints file into Vivado (if there is one)
if {[file exists @designName@.xdc]} {
    read_xdc -ref @designName@ @designName@.xdc
}

# place and route the design
link_design -top @designName@ -part @part@
opt_design
power_opt_design
place_design
phys_opt_design
route_design
phys_opt_design
write_checkpoint -force @designName@.dcp
```

```
# emit the reports
report_timing_summary -file @designName@.tim
report_io -file @designName@.pad
report_utilization -file @designName@.xlsx
report_clock_utilization -file @designName@.clku

# write bitstream
write_bitstream -force @designName@.bit
```

Note that a checkpoint file is created so that it can be read into Vivado in GUI mode and the design layout can be examined.

Both ISE and Vivado are run in the 3PL compiler output directory, as specified by the -o option. This means that the recipe files can expect their input netlist and constraint files in the 'current' directory, and the many output files from the ISE and Vivado programs are also written to the output directory.

The default ISE and Vivado recipe files should be adequate for most 3PL programs, but if necessary they can be overridden by defining the 3PL directive "postProcessRecipe". This should be the filename of the recipe, not the recipe itself, and is searched for on the same include directory path as the 3PL **source** and **module** statements. It would typically be located within the user's source code.

The voluminous output from the ISE and Vivado programs is filtered so that only warning and error messages are printed. This may be changed using the 3PL directive "postProcessFilter", which should have string values "error", "critical", "warning", "info", or "all". The 3PL compiler -v option (verbose) has the same effect as "all".

If the 3PL directive "postProcessAppendFile" is defined and the file it names is readable, it is appended to the FPGA configuration file with a separator string. Board support modules use this to attach a text report or information file to the configuration. This can be examined to determine the provenance of the configuration. The extra data has been found not to upset FPGA configuration loaders, and it seems to be ignored.

If the 3PL boolean directive "postProcessCompress" is defined and true the FPGA configuration file is compressed using gzip.

19.4 The report file

The report file is produced when the -r option is provided on calling 3PL. It has the following sections -

Source files

Source files are listed starting with the primary source file in the 3PL call and then files included by **module** or **source** statements the order they are read. The list entries are indented to reflect the nesting depth of each included file.

Each line has three components, a label, a directory path and the 3PL source file in that directory.

The label is "SRC" for a file included by a **source** statement and is "FMODn", where 'n' is 1, 2, 3 etc. for a file included by a **module** statement.

Signal names are constructed using the file path as a prefix. This generates quite long signal identifiers. To reduce this the "FMODn" labels are used in place of directories. Thus unique signal names are still ensured but are shortened. When reading the intermediate code list the path string of a source file can be determined by consulting the source file list and matching the label.

Leader modules main module and trailer modules

Each leader module is listed as it is executed using the identifying string supplied in the source code. The ordering number is also listed.

The main module is then executed

Finally each trailer module is listed as it is executed, again using the identifying string supplied in the source code. Again the ordering number is also listed.

In any of these there may be warnings printed or there may be an exception thrown for a code error in the module.

Logic for variables

The logic for target variables is created. Variables which are unused are listed and code for them is not generated. There may also be interspersed warnings about possible assignment conflicts. These can not usually be resolved by 3PL and it is up to the programmer to verify that these assignments are mutually exclusive by the logic of the program.

Queue variables with multiple sinks

Queue variables which have more than one destination (reader) are listed as a convenience to the programmer. The rules for multiple queue sinks are complicated and this list is a reminder to the programmer to verify that modules which pop entries from queues are correctly structured. A coding error here can be quite difficult to resolve.

Clock variables

Clock variables, as with all variables, are usually passed as inputs or outputs to modules or procedures, in each case acquiring a new identifier. These multiple identifiers are all equivalent as the named signals are connected. 3PL attempts to extract the most meaningful of these for each distinct clock and use that identifier in all occurrences of the clock signal.

These primary identifiers are listed in a section titled "clocks driving logic".

Following this all other clocks are listed. These are intermediate clocks between input buffers, clock managers and clock buffers and do not drive clock inputs of logic elements.

Directives

This section lists all directives at completion of immediate execution.

Intermediate code optimisation

This section lists a few details such as optimisation stages and the number of passes in each. This is probably mainly of interest to the 3PL maintainer.

A detailed list of clocks appears again here, in this case showing which are unused and which are connected to other clock signals and are aliases. These latter are often identifiers of clock variables which are module or procedure parameters.

Generating EDIF code

This section reports statistics on netlist optimisation processes and passes. It also lists the number of elementary FPGA elements of each type generated. Finally CPU and elapsed time for parsing, immediate execution etc. are listed along with memory usage.

19.5 Netlist display and debugging

When calling 3PL and the **-n** option is given, a list of netlist file net identifiers will be written to file *design.name.net*. Each identifier is listed at the beginning of a line and is followed by a colon and then a list of equivalent net identifiers. If the **-o** option is also supplied then the file will appear in the specified directory. The list of equivalent net identifiers can be used to identify source program variable identifiers associated with signal identifiers appearing in the netlist file or in any vendor place-and-route report files.

19.6 The intermediate code list file

The intermediate code produced by the 3PL compiler following immediate execution can be listed in file *design_name.icl* by using option `-t` when calling 3PL. If option `-pt` is given instead the intermediate code prior to optimisation will also be listed. The listing will not include signal declarations (SIG elements - see below) but these can be included by using `-T` and `-pT`.

The intermediate code is described in [Appendix E](#) p689.

Examination of the intermediate code can be very useful in verifying what 3PL has generated and in attempting to determine signal paths which do not meet timing constraints. The listing has been formatted to be concise but human-readable.

19.7 Netlist display and debugging

The compiler can also display the generated netlist as a schematic if the `-d` option is given. A window will appear which has a text area on the bottom right in which to click and enter a signal, pin or block identifier. The type of identifier is selected by clicking on buttons "signal", "pin" or "block".

For signals a signal identifier, or part of an identifier, is entered followed by a return. If the string entered is an identifier and it is not a substring of any other identifier the element which is the source of the signal will be displayed in the window. If the string entered is a substring of multiple identifiers then a sorted list of those identifiers will appear in a scrollable pop-up list. Left-clicking an identifier in the list will display its source element and hide the list. If the string is also a complete identifier as well as a substring of other identifiers then that identifier will appear at the head of the list. Note that the signal identifiers are those present in the netlist file and these will often have been translated from the intermediate code identifiers by replacing dots and brackets by underscores (these characters are not accepted in netlist identifiers).

For pins the FPGA pad designator must be entered. The block attached to the pin will be displayed.

For blocks the unique block identifier must be provided.

An element in the schematic window may have any input wire left-clicked to display its source element(s) or any output wire left-clicked to display its destination element(s). A left click in the body of an element will expand all its inputs (GND and VCC inputs are not expanded). A middle click on an input will remove the displayed source element for that input and a middle click in an element will remove all displayed input source elements. If an element is right-clicked its properties will be displayed in a pop-up text window. These properties include RAM and flip-flop initialisation, LUT initialisation and associated equation and any UCF or NCF file strings.

Note: the schematic display software was written to aid compiler development and debugging. The display sometimes malfunctions as it has not been completely debugged. Because the display tool is probably of little if any use to normal 3PL users its debugging has a low priority.

Chapter 20 - Zynq Communication Between the CPU and the FPGA

The following assumes the use of a Linux kernel on the Zynq platform and a Linux, Unix or OSX operating system on a host computer for execution of some of the support programs (and optional CPU code cross-compilation if a cross-compile is available).

Note that the version of Linux on the Zynq CPU for which this software was developed provided device /dev/xdevcfg, to which the configuration file could be copied. Versions of Petalinux have now dispensed with this device and instead provide an FPGA manager to perform the configuration loading. This manual and the associated software tools presume the former arrangement and have not been updated to accomodate the FPGA manager.

This interface is very similar to the serial CPU interface described in the previous chapter ([chapter 21](#) p645). Much of the code is common to both.

The board file xilinx/zynq_axi.3pl includes source xilinx/cpu.3pl. Procedure create_bus_interfaces() in xilinx/zynq_axi.3pl executes at the end of the main code to create the bus interface logic.

GP bus 0 is used for reading 3PL expressions (i.e. value mode variables) or writing 3PL static or queue mode variables. GP bus 1 is used to read or write 3PL memory variables. Burst transfers of up to 8 32-bit words are supported. The four HP buses are used to implement 64-bit wide DMA transfers between the FPGA and CPU. Thus far the default 100MHz clock has been used for all buses. The CPU to FPGA interfaces are described in [section 11.10](#) p444.

For this interface to operate the Zynq Linux device tree must provide entries for axi_mgp0, axi_mgp1 and axi_hpbuff and must also provide interrupt events 0 to 15. An example is shown in [Appendix H](#) p753.

The following source code, prog.3pl, is an example of a simple 3PL program using CPU interfaces -

```
module "boards/avnet/microzed_7010.3pl"

class(cpu, cpu_api()); // CPU API class

static(a, "int:25");
static(b, "int:18");

cpu.write(a <- "A"); // write to {\bf a} from CPU
cpu.write(b <- "B"); // write to {\bf a} from CPU

static(p, "int:48");

p <- a * b;

cpu.read(p, "P"); // read from {\bf p} to CPU

cpu.receive_event("INT", p == 1); // interrupt when p == 1
```

Static variables **a** and **b** can be written to by the CPU. The product is assigned to **p** (in every clock cycle), which can be read by the CPU. The strings "A", "B" and "P" are names with which to refer to the interfaces within CPU code.

Note that the interrupt **INT** occurs when the expression **p == 1** goes from false to true.

When the program is compiled and executed by 3pl [chapter 19](#) p623) the following files will be produced

-

- **prog.edif** - netlist file
- **prog.xdc** - Vivado constraints file
- **prog.bit** - configuration bit file
- **prog.info** - CPU/FPGA interface information (JSON) file
- **prog.rpt** - report file
- **prog.net** - optional list of net labels

The bit file is produced by a default post-processing module and can be suppressed by specifying the **-s** option (skip) to 3PL. In that case the place-and-route can be performed by shell script `edif2bit` (`sources/fpga_platforms/edif2bit/edif2bit`) or by the simple bash script `run_vivado` (`sources/fpga_platforms/run_vivado`).

The configuration file may be compressed using `gzip`

The configuration file can be loaded into the FPGA manually by `cat >/dev/xdevcfg`. Alternatively The configuration file or compressed configuration file can be converted into a byte array in CPU application code - see below.

The CPU/FPGA interface information file `prog.info` is in JSON format and describes the CPU/FPGA interface.

```
{
  "version": {
    "major": 1,
    "minor": 0
  },
  "directives": {
    "ALU": "false",
    "FIFO": "false",
    "OS": "mac os x",
    "QUEUEREG": "false",
    "arch": "x86_64",
    "compileOnly": "false",
    "compilerMakeDate": "2022-04-04 14:45:06 +1000",
    "continuous": "false",
    "currentDirectory": "/Users/dun202/snark/src/test/zynq_int_test",
    "date": "2022-07-05 10:26:31 +1000",
    "designName": "prog",
    "family": "XC7",
    "filePath": "false",
    "forceExec": "true",
    "gatesNotLuts": "false",
    "intTruncWarning": "false",
    "intermediateOnly": "false",
    "listNets": "true",
    "listTDEs": "true",
    "listTDEsSel": "0",
    "locations": "false",
    "netFile": "/Users/dun202/snark/src/test/zynq_int_test/prog.net",
    "netlistDisplay": "false",
    "netlistFile": "/Users/dun202/snark/src/test/zynq_int_test/prog.edn",
    "optimiseConnect": "true",
    "outputDirectory": "/Users/dun202/snark/src/test/zynq_int_test/",
    "parentDirectory": "/Users/dun202/snark/src/test/zynq_int_test/",
```

```

    "part": "xc7z010clg400-1",
    "postProcessCompress": "true",
    "postProcessTool": "vivado",
    "reportFile": "/Users/dun202/snark/src/test/zynq_int_test/prog.rpt",
    "rptToFile": "true",
    "sigList": "false",
    "skipPostProc": "true",
    "sourceFile": "prog.3pl",
    "srlAddrWidth": "5",
    "unassOut": "fatal",
    "version": "11.7.0M (devel svn 10011:10036M, dun202)"
  },
  "spaces": {
    "reg": {
      "bus": "axi_mgp0",
      "read_addr_bits": 2,
      "write_addr_bits": 2,
      "interfaces": {
        "infoword": {
          "interface": "read_value",
          "write": false,
          "data_addr": 1,
          "data_bits": 32,
          "data_type": {
            "type": "uint",
            "bits": 32
          }
        },
        "inforeset": {
          "interface": "write_static",
          "write": true,
          "data_addr": 1,
          "data_bits": 0
        }
      },
      "A": {
        "interface": "write_static",
        "write": true,
        "data_addr": 2,
        "data_bits": 25,
        "data_type": {
          "type": "int",
          "bits": 25
        }
      },
      "B": {
        "interface": "write_static",
        "write": true,
        "data_addr": 3,
        "data_bits": 18,
        "data_type": {
          "type": "int",
          "bits": 18
        }
      },
      "P": {
        "interface": "read_value",
        "write": false,
        "data_addr": 2,

```

```

        "data_bits": 48,
        "data_type": {
            "type": "int",
            "bits": 48
        }
    }
}
},
"events": {
    "INT": {
        "intr": 0
    }
}
}

```

This file contains a description of the CPU bus interfaces including data width, data type and allocated bus addresses.

A list of the interface functions is also written to the report file.

The design.info data may also be stored in block RAM (rmemory) for access by CPU code. This is enabled by default but may be avoided via a directive, i.e. **directive("excludeinfo", true);**, which can be executed at any time during 3PL immediate execution. The compression is done by calling inbuilt function **deflate()** [section 10.25](#) p279. The readback is initialised by writing to address 1 and 32-bit packed .info data is then read by successive 32-bit reads from address 1. The first read returns a 32 bit CRC-32 checksum for the remainder of the block, The second read returns the uncompressed data size in bytes and the third read returns the compressed data size in bytes. Reads beyond the data length will return zeros.

The data block size depends on the size of the compressed .info data and varies from a minimum of 1024 32 bit words (1 RAM block) to 32,768 words (32 RAM blocks).

For C applications the .info file must be converted to a C include file by utility **info2c**. That include file defines functions and procedures to read and write to specified FPGA variables.

For Python3 applications the .info file cab be converted by **info2py** to a Python3 file that can be imported into the python application. Alternatively **info2py** can itself be imported into the python application and it can then read th .info file or read out the embedded version in the FPGA.

20.1 C interface

The configuration bit file can be manually loaded into the FPGA by `cat prog.bit >/dev/xdevcfg`.

The configuration data can instead be incorporated into the CPU application program by using the utility **bin2c** (sources/fpga_platforms/bin2c). This will create an include file **prog_cinclude.h**. Although the uncompressed configuration file prog.bit can be used it is clearly preferable to use the compressed configuration file prog.bit.gz. The configuration data can then be loaded into the FPGA by the application program by calling **fpga_configdata_load()** as illustrated later.

Alternatively the application program can load the FPGA configuration from the configuration file prog.bit or prog.bit.gz by **fpga_configfile_load("prog.bit")** or **fpga_configfile_load("prog.bit.gz")**.

When the **prog.info** file is processed by the python script **info2c** (sources/fpga_platforms/zynq/info2c/info2c) by **info2c prog** a C include file, **prog_fpga.h**, results.

```

// event "INT"
static __inline__ void
fpga_INT_enable() {

```

```

        fpga_event_control(0, 1);
    }
    static __inline__ void
    fpga_INT_disable() {
        fpga_event_control(0, 0);
    }
    static __inline__ void
    fpga_INT_wait() {
        fpga_event_wait(0);
    }
    static __inline__ int
    fpga_INT_count() {
        return(uio_event_count(0));
    }

    // interface "A" in space "reg" (bus "axi_mgp0")
    typedef int32_t fpga_A_t;
    const unsigned fpga_A_addr = 4;
    static __inline__ void
    fpga_A_write(fpga_A_t val) {
        FPGA_REG_WRITE(fpga_A_addr, val, 1);
    }

    // interface "P" in space "reg" (bus "axi_mgp0")
    typedef int64_t fpga_P_t;
    const unsigned fpga_P_addr = 2;
    static __inline__ fpga_P_t
    fpga_P_read() {
        FPGA_REG_READ(fpga_P_addr, fpga_P_t, 2);
    }

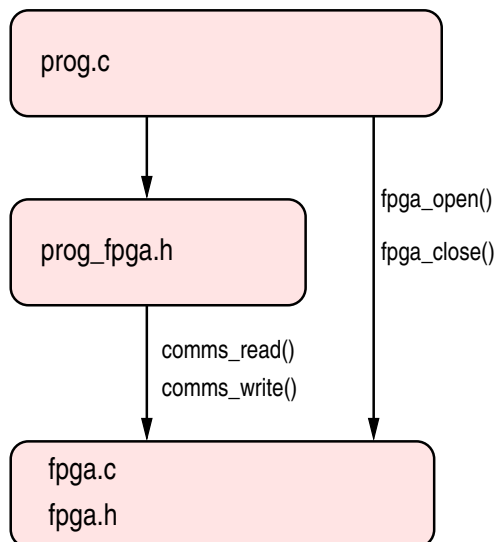
    // interface "B" in space "reg" (bus "axi_mgp0")
    typedef int32_t fpga_B_t;
    const unsigned fpga_B_addr = 6;
    static __inline__ void
    fpga_B_write(fpga_B_t val) {
        FPGA_REG_WRITE(fpga_B_addr, val, 1);
    }

    // space and event bitfield for fpga_map() 2nd argument
    #define fpga_map_needed \
        fpga_map_none \
        | fpga_space_reg \
        | fpga_INT_event

```

At the moment reads and writes of single variables will defines types **int8_t** or **uint8_t** up to **int_64_t** or **uint64_t** for data widths 8, 16, 32 or 64. A struct up to 64 bits in size will be mapped to a bit field. To transfer bursts larger than 2 (64 bits) subroutines **FPGA_REG_READ()** and **FPGA_REG_WRITE()** can be called directly for bursts up to 8 32-bit words. In that case the bus address must be determined from the .info file and passed explicitly.

This file, **prog_fpga.h**, defines functions which specifically read or write interface objects in the FPGA. These funtions in turn call functions in the communications layer defined by **cpu_serial.c**, **cpu_serial.h**. Some function such as **open_cpu_serial()** and **close_cpu_serial()** are call directly skipping **prog_fpga.h**.



The following C program includes the above file. Note that in the case of a Python3 program the interface information can be read from the FPGA and compiled as a module. If the interface information has been incorporated into block RAM in the FPGA. This cannot be done with a C interface and thus directive "excludeinfo" should be set true to avoid wasting these logic and memory resources;

```

#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <pthread.h>
#include "/root/install/3pl/fpga.h"

int64_t    p;

void*      INT_handler ();
pthread_t  thread_start(
    void *   (*func)(void*),
    void *   arg,
    char *   name,
    int      priority
);

#include    "prog_cinclude.h"

#include    "prog_fpga.h"

int
main (
    int      argc,
    char**   argv
) {
    pthread_t  INT_thread;
    int        pri_min = 1;

    // Open the CPU interface.
    fpga_open();

    // Load the configuration.
    //fpga_configdata_load(prog_fpga_config_data, prog_fpga_config_size);
    fpga_configfile_load("prog.bit");

    // Start a thread for the interrupt
    INT_thread = thread_start(INT_handler, NULL, "INT_handler", pri_min);

    fpga_A_write(123);
  
```



```

    fpga_B_write(1);
    p = fpga_P_read();
    printf("read %lld (should be 123)\n", p);
    fpga_A_write(1);    // will generate an interrupt
    sleep(2);           // wait for the interrupt
    fpga_B_write(2);
    fpga_B_write(1);    // will generate another interrupt

    // Terminate the interrupt thread.
    pthread_cancel(INT_thread);

    // Close the CPU interface.
    fpga_close();

    exit(0);
}

void*
INT_handler () {
    for(;;) {
        fpga_INT_enable();
        fpga_INT_wait();
        printf("interrupt INT, p = %lld\n", p);
        printf("number of events %d\n", fpga_INT_count());
    }
}

// Start a thread.
pthread_t
thread_start(
    void * (*func)(void*),
    void * arg,
    char * name,
    int priority
) {
    int status;
    struct sched_param sp;
    pthread_t tid;

    if ((status = pthread_create(&tid, NULL, func, arg)) != 0) {
        printf("thread_start(): pthread_create() failed thread '%s' (err = %s)\n",
            name, strerror(status));
        exit(1);
    }

    sp.sched_priority = priority;

    status = pthread_setschedparam(tid, SCHED_RR, &sp);
    if (status != 0) {
        printf("thread_start(): pthread_setschedparam() failed for thread '%s' (err = %s)\n",
            name, strerror(status));
        exit(1);
    }
    return(tid);
}

```

Include file "fpga.h" defines the basic FPGA communication interface functions.

Include file "prog_fpga.h" is created by python script 'info2c' and contains function definitions for each FPGA read, write or interrupt (event). These in turn use the functions in fpga.h above.

fpga.open() sets up the ARM memory mapping for the AXI buses and devices associated with interrupts

(events).

fpga_configdata_load() loads the FPGA configuration. In this example the compressed FPGA configuration is compiled into the CPU program and is included from the file **prog_cinclude.h**. That file has been created by running program **bin2c** which takes the configuration file **prog.bit** compresses it and then converts it to an array of hexadecimal constants. The FPGA configuration can alternatively be loaded from a file by the CPU if desired by calling **fpga_configfile_load()** instead.

Bus interfaces are used by calling the appropriate subroutines defined in **prog_fpga.h**, in this case **fpga_A_write()**, **fpga_B_write()** and **fpga_P_read()**, the bus addresses having been subsumed into the subroutine definitions. These functions are defined in 'prog_fpga.h'.

The C program must be linked with object file 'fpga.o' which can be compiled from source file 'fpga.c' located in the 3PL source directory 'sources/fpga_platforms/zynq/fpga'.

20.2 Python interface

As described previously the configuration file can be loaded into the FPGA manually by **cat prog.bit >/dev/xdevcfg**. Alternatively the configuration file **prog.bit** or compressed configuration file **prog.bit.gz** can be converted into a byte array in source code.

The configuration data can be incorporated into the CPU application program by using the utility **bin2py** (sources/fpga_platforms/bin2py). This will create a source file **prog_config.py** which can be imported into the application program. The configuration data can then be loaded into the FPGA by the application program by calling **fpga_configdata_load()** -

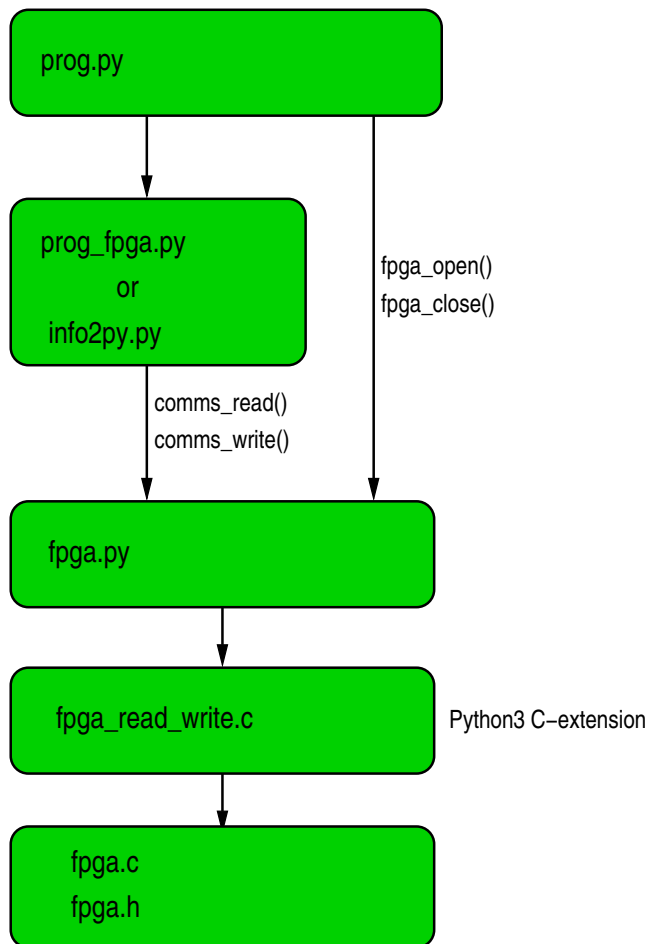
```
fpga_configdata_load(config.data, len(config.data))
```

The configuration can also be loaded from the uncompressed or compressed bit file during execution of the application program by calling **fpga_configfile_load()**-

```
fpga_configfile_load("prog.bit.gz"))
```

or

```
fpga_configfile_load("prog.bit"))
```



For a Python interface to the FPGA the modules **info2py.py** (in directory `fpga_platforms/info2py`) and functions in **fpga.py** (in directory `fpga_platforms/zynq/fpga`) must be imported.

Functions in **fpga.py** provide the code to open the interface to the FPGA, read and write to the FPGA and generate interrupts if required.

`info2py.py` either reads the `.info` file (call `info_from_file(file)`) or downloads the compressed info data in the FPGA (call `info_from_fpga()`) and generates an internal module containing functions to access the FPGA register or memory interfaces.

The 3PL module `zynq_axi3pl` will always produce a `.info` file. The compressed version loaded into block RAM in the FPGA configuration will be included by default but can be omitted by creating the directive `"excludeinfo"`.

The identifiers of the interface functions are listed in the report file, along with the data type. Each function is contained in the CPU-API class **fpga**.

The previous 3PL source code can be used again as an example.

The following Python source code would provide a CPU interface to the FPGA. The configuration file `prog.bit` is loaded into the FPGA by the application code by .

The CPU/FPGA interface functions are defined in imported module **prog_fpga.py** which has been created by **info2py prog**.

fpga_configfile_load("prog.bit"). This line could be omitted if the configuration was manually preloaded by executing `cat prog.bit >/dev/xdevcfg`.

```

import sys
import os
sys.path.append(os.environ['HOME'] + "/install/3pl")
from fpga import *

```

```

import prog_fpga as fpga
import time
import signal
import threading

fpga_configfile_load("prog.bit")

fpga_open()

def INT_handler():
    while True:
        fpga.INT_wait()
        print("event INT")
        print("number of events " + str(fpga.INT_count()))

ev_INT = threading.Thread(target=INT_handler, args=(), daemon=True)
ev_INT.start()

a = 123
b = 1

fpga.A_write(a)
fpga.B_write(b)
print(str(fpga.P_read()) + "\t(should be 123)")
a = 1
fpga.A_write(a);    # will generate an interrupt
a = 2
fpga.A_write(a);
a = 1
fpga.A_write(a);    # will generate an interrupt

time.sleep(1.0)     # wait for the interrupts

fpga_close()

os.kill(os.getpid(), signal.SIGTERM)    # use this to kill all threads!

```

As an alternative the configuration could be stored within the application code as an initialised byte array. Executing **bin2py prog** will take prog.bit.gz and produce a Python3 module **prog_config.py** which can be imported to define the CPU/FPGA interface.

```

import sys
import os
sys.path.append(os.environ['HOME'] + "/install/3p1")
from fpga import *
import prog_fpga as fpga
import time
import signal
import threading
import prog_config as config

fpga_configdata_load(config.data, len(config.data))

fpga_open()

.
.
.

```

Instead of the CPU/FPGA interface definitions being imported from file **prog_fpga.py** as above they can be read from **prog.info** during execution using module **info2py.py**.

```

.
.
import info2py
.
.
fpga_open()

fpga = info2py.info_from_file("prog")    # load info from prog.info file
.
.

```

Another possibility is to import the CPU/FPGA interface definitions from the interface description embedded in the FPGA, assuming this has not been omitted by the directive "excludeinfo" true -

```

.
.
import info2py
.
.
fpga_open()

fpga = info2py.info_from_fpga()          # load info from FPGA
.
.

```

The additional Python3 package path "/Users/.../install" depends on the layout of the file system, but the path component must contain "fpga.py" and "info2py".

The Python3 library module **fpga.py** must have been added by executing shell script **mk_setup** in **/install/3pl/** on the Zynq CPU. This script runs setup.py which creates and adds the library module fpga_py_extension.c. Both **setup.py** and **mk_setup** are installed in **/install/3pl/setup.py** and the sources are in **sources/tools/zynq**

The Python CPU interface to the FPGA can read or write variables of any type to any degree of nesting, but there is a maximum variable size of 256 bits (ldm/stm using 8 registers for a burst transfer). The following type conversions between 3PL and Python will occur -

FPGA	Python
log	logical
bits	integer
uint	integer
int	integer
ufixed	IEEE754 64bit float
fixed	IEEE754 64bit float
float	IEEE754 64bit float
struct	dictionary
array	list

There are no run-time tests for data integrity, for example overflows.

3PL arbitrary ufixed and fixed integer and fraction sizes are converted in or out of IEEE 64 bit float in Python. Similarly 3PL arbitrary floating point biased exponent and mantissa sizes are again converted in or out of IEEE 64 bit floating point.

A struct in 3PL is converted in or out of a dictionary in Python, the field name being the key.

3PL multi-dimensional arrays are nested lists in Python.

20.3 An aside

As a matter of interest the following C function illustrates retrieval of the information string from the FPGA. The string length is returned as the function value and a pointer to the information string is

assigned to the pointer argument. It is not necessary to use this function normally as the .info file can be converted to a C include file by using Python script 'info2c' as explained previously.

```
int fetch_info (unsigned char ** info) {
    int          i;
    int          n;
    int          e;
    unsigned long * buf;
    unsigned char * ubuf;
    unsigned long * bufp;
    unsigned long ulen;

    fpga_reg_write(1, 0);
    n = fpga_reg_read(1);      // number of bytes of compressed data
    ulen = n * 10;            // enough room for uncompressed data (a guess)

    buf = malloc(n+3);        // enough room for compressed data, 32-bit packed
    ubuf = malloc(ulen);
    bufp = buf;
    for (i=0 ; i<n ; i+=4)
        *bufp++ = fpga_reg_read(1);

    e = uncompress(ubuf, &ulen, ((Byte *)buf), n);
    if (e != Z_OK) {
        printf("info uncompress error %d\n", e);
        return(-1);
    }

    free(buf);
    *info = realloc(ubuf, ulen); // eliminate unused trailing buffer space
    return(ulen);
}
```

But alas in C this interface information cannot be used, as it can in Python3.

In the case of Python3, the following code would retrieve the JSON info from the FPGA data as a string

```
#-----
# Read the compressed JSON structure from the FPGA
# and expand it. Return a code module containing the interface
# functions.
#
def info_from_fpga ():
    # reset the json interface
    fpga_rw.comms_write(0, 2, 0, None)

    # get the checksum
    rba = fpga_rw.comms_read(0, 1, 0, 4)
    checksum = int.from_bytes(rba, byteorder=bo, signed=False)

    # get the uncompressed byte count
    rba = fpga_rw.comms_read(0, 1, 0, 4)
    ucl = int.from_bytes(rba, byteorder=bo, signed=False)

    # get the compressed byte count
    rba = fpga_rw.comms_read(0, 1, 0, 4)
    cl = int.from_bytes(rba, byteorder=bo, signed=False)

    # read the compressed json info string
    words = cl // 4 + 1
    rba = bytearray()
    for i in range(words):
        rba += fpga_rw.comms_read(0, 1, 0, 4)
```

```
# decompress the json string
infostring = zlib.decompress(rba[2:-4], -15)

return(str(infostring, 'utf-8'))
```


Chapter 21 - Serial Communication Between the CPU and the FPGA

FPGA demonstration boards such as the Gadget Factory Papilio provide a serial interface to the FPGA for loading the configuration into the FPGA. These interfaces also provide serial communication between the host CPU and the FPGA. The 3PL library provides an interface which allows FPGA static, value, queue and memory mode variables to be written and read from the CPU and the FPGA to interrupt the CPU. For FPGA boards that do not have a serial interface inexpensive USB to serial adapters are available.

The FPGA code uses procedures `uart_rx()` and `uart_tx()` in library `uarts.3pl`.

This interface is very similar to the Zynq/CPU interface described in the next chapter ([chapter 20](#) p631). Much of the code is common to both. The DMA interface is of course missing here however it is intended to add a pseudo-DMA interface.

The CPU to FPGA interfaces are described in [section 11.12](#) p485.

The board file `papilio_one.500.3pl` for example includes module `xilinx/uart_cpu.3pl` which includes source `xilinx/cpu.3pl` and `uarts.3pl`. Trailing procedure `create_bus_interfaces()` in `xilinx/uart_cpu.3pl` creates the bus interface logic following completion of the main code.

The board header, e.g. `board/gadget_factory/papilio_one.500.3pl`, will have defined a directive to specify the serial link baud rate, directive("comms_baud_rate", 115200);.

The following source code, `prog.3pl`, is an example of a simple 3PL program using CPU interfaces -

```
module "boards/gadget_factory/papilio_one_500.3pl"

class(cpu, cpu_api());

static(a, "int:25");
static(b, "int:18");

cpu.write(a <- "A");
cpu.write(b <- "B");

static(p, "int:48");

p <- a * b;

cpu.read(p, "P");

cpu.receive_event(p == 1, "INT"); // interrupt when p == 1
```

This example has two static mode integer variables, **a** and **b**, that can be written to by the CPU. The product is written on every clock cycle to static mode integer variable **p**, which can be read by the CPU. When **p** is assigned 1 a CPU interrupt is generated

The papilio header file includes module xilinx/uart_cpu.3pl which in turn includes xilinx/cpu.3pl.

The strings "A", "B" and "P" are names with which to refer to the interfaces in CPU interface code.

When the program is compiled and executed by 3pl the file (see [chapter 19](#) p623) file **prog.info** (amongst others) will be produced.

```
{
  "version": {
    "major": 1,
    "minor": 0
  },
  "directives": {
    "ALU": "false",
    "FIFO": "false",
    "OS": "mac os x",
    "QUEUEREG": "false",
    "arch": "x86_64",
    "comms_baud_rate": "53475",
    "compileOnly": "false",
    "compilerMakeDate": "2022-08-24 14:24:48 +1000",
    "continuous": "false",
    "currentDirectory": "/Users/dun202/src/mine/3PL/test/11.6.0/manual_example_serial",
    "date": "2022-09-23 16:02:55 +1000",
    "designName": "prog",
    "family": "XC3S",
    "filePath": "false",
    "forceExec": "true",
    "gatesNotLuts": "false",
    "intTruncWarning": "false",
    "intermediateOnly": "false",
    "listNets": "true",
    "listTDEs": "true",
    "listTDEsSel": "0",
    "locations": "false",
    "netFile": "/Users/dun202/src/mine/3PL/test/11.6.0/manual_example_serial/prog.net",
    "netlistDisplay": "false",
    "netlistFile": "/Users/dun202/src/mine/3PL/test/11.6.0/manual_example_serial/prog.edi",
    "optimiseConnect": "true",
    "outputDirectory": "/Users/dun202/src/mine/3PL/test/11.6.0/manual_example_serial/",
    "parentDirectory": "/Users/dun202/src/mine/3PL/test/11.6.0/manual_example_serial/",
    "part": "xc3s400aft256-4",
    "postProcessTool": "ise",
    "reportFile": "/Users/dun202/src/mine/3PL/test/11.6.0/manual_example_serial/prog.rpt",
    "rptToFile": "true",
    "sigList": "false",
    "skipPostProc": "true",
    "sourceFile": "prog.3pl",
    "srlAddrWidth": "4",
    "tlimit": "504",
    "uart_cpu": "true",
    "unassOut": "fatal",
    "version": "11.7.0M (devel svn 10083:10129M, dun202)"
  },
  "spaces": {
    "reg": {
      "bus": "serial",
      "read_addr_bits": 6,
      "write_addr_bits": 6,

```

```

    "interfaces": {
      "inword": {
        "interface": "read_value",
        "write": false,
        "data_addr": 1,
        "data_bits": 32,
        "data_type": {
          "type": "uint",
          "bits": 32
        }
      },
      "inforeset": {
        "interface": "write_static",
        "write": true,
        "data_addr": 1,
        "data_bits": 0
      },
      "A": {
        "interface": "write_static",
        "write": true,
        "data_addr": 2,
        "data_bits": 25,
        "data_type": {
          "type": "int",
          "bits": 25
        }
      },
      "B": {
        "interface": "write_static",
        "write": true,
        "data_addr": 3,
        "data_bits": 18,
        "data_type": {
          "type": "int",
          "bits": 18
        }
      },
      "P": {
        "interface": "read_value",
        "write": false,
        "data_addr": 2,
        "data_bits": 48,
        "data_type": {
          "type": "int",
          "bits": 48
        }
      }
    },
    "events": {
      "INT": {
        "intr": 0
      }
    }
  }
}

```

This file contains a description of the CPU bus interfaces including data width, data type and allocated bus address.

A list of the interface functions is written to the report file.

The design.info data may also be stored in block RAM (rmemory) for access by CPU code. This is enabled by default but may be avoided via a directive, i.e. **directive("excludeinfo", true);**, which can be executed at any time during 3PL immediate execution. The embedded .info is only useful for Python application code as there is no easy way to incorporate the interface description into an executing C program.

The compression is done by calling inbuilt function **deflate()** [section 10.25](#) p279. The readback is initialised by writing to address 1 and 32-bit packed .info data is then read by successive 32-bit reads from address 1. The first read returns a 32 bit CRC-32 checksum for the remainder of the block, The second read returns the uncompressed data size in bytes and the third read returns the compressed data size in bytes. Reads beyond the data length will return zeros.

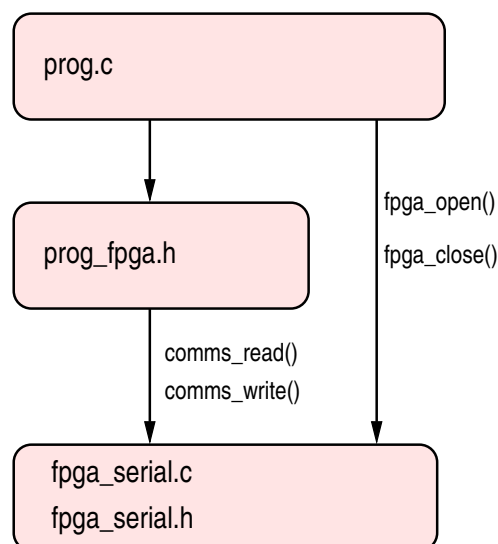
The data block size depends on the size of the compressed .info data and varies from a minimum of 1024 32 bit words (1 RAM block) to 32,768 words (32 RAM blocks).

21.1 C interface

The python script info2c (in directory tools/serial/info2c) reads the .info file and produces a C .h include file, e.g.

```
info2c prog
```

reads file **prog.info** and produces **prog_fpga.h**. This file provides a wrapper for each FPGA interface. These in turn call functions in **fpga.c**. The application must also directly call **fpga_open()** and **fpga_close()** in fpga.c.



For the .info file listed previously the created include file **prog_fpga.h** contains -

```
// interface "P" in space "reg"
typedef int64_t fpga_P_t;
const unsigned fpga_P_addr = 2;
static fpga_P_t
fpga_P_read() {
    fpga_P_t v;
    comms_read(0, fpga_P_addr, 0, 6, 1, 2, (char *)&v);
    if ((v & 0x800000000000) != 0)
        v |= 0xffff000000000000;
```

```

    return(v);
}

// interface "B" in space "reg"
typedef int32_t fpga_B_t;
const unsigned fpga_B_addr = 3;
static void
fpga_B_write(fpga_B_t val) {
    comms_write(0, fpga_B_addr, 0, 3, (char *)&val);
}

// interface "A" in space "reg"
typedef int32_t fpga_A_t;
const unsigned fpga_A_addr = 2;
static void
fpga_A_write(fpga_A_t val) {
    comms_write(0, fpga_A_addr, 0, 4, (char *)&val);
}

// event "INT"
void
fpga_INT_wait() {
    sem_wait(sem[0]);
}

const unsigned fpga_interrupts = 1;

```

Note that for each interface function the string "fpga_" is prepended to the function identifier.

At the moment reads and writes of single variables will defines types **int8_t** or **uint8_t** up to **int_64_t** or **uint64_t** for data widths 8, 16, 32 or 64. A struct up to 64 bits in size will be mapped to a bit field.

The following C program includes the above file -

```

#include <stdio.h>
#include "cpu_serial.h"

void*      INT_handler ();

#include    "prog_fpga.h"

int
main (
    int      argc,
    char**   argv
) {
    pthread_t  INT_thread;
    int        pri_min = 1;

    // Establish the CPU interface.
    fpga_open("/dev/tty.usbmodem14201");

    // Start a thread for the interrupt
    INT_thread = thread_start(INT_handler, "INT_handler", pri_min);

    fpga_A_write(123);
    fpga_B_write(1);
    printf("read %lld (should be 123)\n", fpga_P_read());
    fpga_A_write(1);    // will generate an interrupt
    sleep(1);           // wait for the interrupt

    // Close the interrupt thread.
    pthread_cancel(INT_thread);
}

```

```

        // Close the CPU interface.
        fpga_close();

        exit(0);
    }

    void*
    INT_handler () {
        for(;;) {
            fpga_INT_wait();
            printf("interrupt INT\n");
        }
    }
}

```

The CPU code uses library **cpu_serial.c** (see appendix F) and hence the application program must be linked to **cpu_serial.o**.

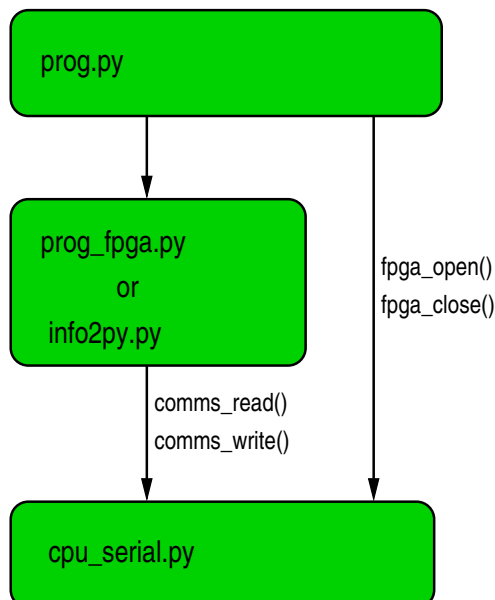
Procedure **findserialdev()** is called to find the file path for the serial device to which the FPGA board is connected. In this case the code is for Apple OSX and device names will be different in Linux or Unix.

fpga_open() opens the serial device, starts a thread to handle the communications and establishes any semaphores required for interrupts. Writes to the FPGA do not supply any acknowledgement. Data returned from the FPGA included requested reads and asynchronous 'interrupts'. Details of the interface are described in [section 11.11](#) p455.

fpga_close() terminates the thread, removes any semaphores and closes the serial port.

Bus interfaces are used by calling the appropriate subroutines defined in **prog.h**, in this case **fpga_A_write()**, **fpga_B_write()** and **fpga_P_read()**, the bus addresses having been subsumed into the subroutine definitions.

21.2 Python interface



The 3PL module **uart_cpu.3pl** will always produce a **.info** file. A compressed version loaded into block RAM in the FPGA configuration will also be included by default but can be omitted by creating the directive "excludeinfo".

For a Python interface to the FPGA the module **cpu_serial.py** (in directory **fpga_platforms/serial/uart_cpu** must be imported.)

Optionally **info2py.py** (in directory `fpga_platforms/info2py`) can be imported to convert the `.info` file to an internal compiled Python3 module. If **info2py.py** is imported it either reads the `.info` file (call `info_from_file(file)`) or downloads the compressed info data in the FPGA (call `info_from_fpga()`) and generates an internal module containing functions to access the FPGA register or memory interfaces.

As an alternative the Python3 script **info2py** can be executed to create a Python3 module **prog_fpga.py** which is then imported into the application program. A version of this file for the present example is listed below -

```
import os
import math
import string
import textwrap
import argparse
import json
import sys
import time
sys.path.append(os.environ['HOME'] + '/install/3pl')
import cpu_serial as fpga_rw

def B_write(wdata):
    val = wdata
    if val < 0:
        wba = val.to_bytes(3, byteorder='little', signed=True)
    else:
        wba = val.to_bytes(3, byteorder='little')
    fpga_rw.comms_write(0, 3, 0, wba)

def P_read():
    rba = fpga_rw.comms_read(0, 2, 0, 6)
    val = int.from_bytes(rba, byteorder='little', signed=False)
    fval = val & 0xfffffffffff
    if (fval & 0x800000000000) != 0:
        fval = -((fval ^ 0xfffffffffff) + 1)
    return(fval)

def A_write(wdata):
    val = wdata
    if val < 0:
        wba = val.to_bytes(4, byteorder='little', signed=True)
    else:
        wba = val.to_bytes(4, byteorder='little')
    fpga_rw.comms_write(0, 2, 0, wba)

# event "INT"
def INT_wait():
    fpga_rw.semaphores[0].acquire()

def INT_count():
    return(fpga_rw.event_count(0))
```

`cpu_serial.py` provides the code to open the serial interface to the FPGA, read and write to the FPGA and generate interrupts if required.

The identifiers of the interface functions are listed in the report file, along with the data type. Each function is contained in the CPU_API class **fpga**.

Python source code equivalent to the previous C interface might be -

```
import sys
sys.path.append("/Users/...../install/3pl")
import info2py
from cpu_serial import *
import time
```

```

import signal
import threading

fpga_open("/dev/tty.usbmodem14101")

fpga = info2py.info_from_fpga()          # load info from FPGA

def INT_handler():
    while True:
        fpga.INT_wait()
        print("event INT")

ev_INT = threading.Thread(target=INT_handler, args=(), daemon=True)
ev_INT.start()

a = 123
b = 1

fpga.A_write(a)
fpga.B_write(b)
print(str(fpga.P_read()) + "\t(should be 123)")
a = 1
fpga.A_write(a);    # will generate an interrupt

time.sleep(1.0)     # wait for the interrupt

fpga_close()

os.kill(os.getpid(), signal.SIGTERM)    # use this to kill all threads!

```

In the code above the application program loads the interface information from the version embedded in the FPGA in block RAM. With the following modification the interface information will be loaded from the prog.info file instead.

```

import sys
.
.
.

fpga_open("/dev/tty.usbmodem14101")

fpga = info2py.info_from_file("prog")    # load info from prog.info file
.
.
.

```

Alternatively the following imports a version of the interface function definitions created by Python3 script **info2py**.

```

import sys
.
.
.
import prog_fpga as fpga    # import interface info from Python3 file

fpga_open("/dev/tty.usbmodem14101")
.
.
.
.

```

The Python serial interface to the FPGA can read or write variables of any type to any degree of nesting, but there is a maximum variable size of 1016 bits (the 7-bit byte count in the communication protocol

limits the data length to 127 bytes). The following type conversions between 3PL and Python will occur

FPGA	Python
log	logical
bits	integer
uint	integer
int	integer
ufixed	IEEE754 64bit float
fixed	IEEE754 64bit float
float	IEEE754 64bit float
struct	dictionary
array	list

There are no run-time tests for data integrity, for example overflows.

3PL arbitrary ufixed and fixed integer and fraction sizes are converted in or out of IEEE 64 bit floatin point in Python. Similarly 3PL arbitrary floating point biased exponent and mantissa sizes are again converted in or out of IEEE 64 bit floatin point.

A struct in 3PL is converted in or out of a dictionary in Python, the field name being the key.

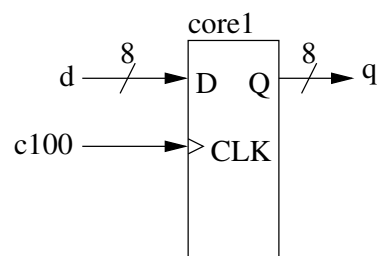
3PL multi-dimensional arrays are nested lists in Python.

Chapter 22 - Using EDIF Cores

By 'core' is meant an EDIF netlist file which implements a logic function but is not itself a complete logic design.

Note: in the case of the Zynq platform the EDIF file must be a core, the bi-directional ports being connected to resources external to the FPGA. The CPU interface is incorporated as a large cell called "PS7".

A Xilinx core EDIF netlist file contains an interface description which lists the input and output (and 3-state) connections to the core logic. As an example the following core was generated using Xilinx coregen -



This is a simple 8-bit register. The core is called "core2" and has the following inputs and outputs -

port	width	direction
D	8	input
CLK	1	input
Q	8	output

Note that the use of upper case identifiers for ports seems to be a convention with Xilinx but is not required by 3PL.

NGC files are generated from **coregen** via **xst** and are linked in by **ngdbuild**.

Note that the core may have an associated constraints file and this can be added to a constraints file generated by 3PL by means of library procedures **append_ncf_file()** [subsection 11.2.2.4](#) p378.

22.1 linking in EDIF or NGC cores

The following 3PL hymod program illustrates how this core can be used -

```

module "csiro/hymod/io1.3pl"

  useclock(c100);

  var(ipins, "[str", {.....});
  
```

```

var(opins, "[]str", {.....});
static(s, "uint:8");
output(., s, {loc=opins});
input(d, "uint:8", {loc=ipins});
value(q, "uint:8");

import(Q=q <- "core1", CLK=c100, D=d);
attributes(q, domain=c100);

s <- q;

```

Input ports can be driven by expressions or variables of mode input, value, static, priority or clock.

Variables connected to the output ports of the core can only be clock, value or selectvalue (for 3-state) mode. Output variables to **import()** will not have a clock domain. It may be set explicitly by **domain()** as in the example above, otherwise variables with no clock domain will not be checked against the current clock when evaluated. Note that this does not invalidate timing constraints which are set up according to which clocks state devices are actually clocked by in the final netlist.

Variables can be arrays or structs.

The number of bits in each argument must exactly match the associated core port width.

Where a clock is sourced in the core and is connected to the main program via *import()* the clock variable must be created in the main program, e.g.

```

clock(c, frequency=10e6, tnm="C");
useclock(c);

value({d, q}, "uint:8");
static(s, "uint:8");

import(Q=q, CLK=c <- "core2", D=d);

s <- q;

```

All input ports must be connected but connections to unwanted output ports may be omitted.

To place and route the first example above the procedure is no different to usual, e.g.

```
ngdbuild main
```

Ngdbuild will expect to find a file "core1.edn" which it will link in to main.edn.

22.2 creating EDIF cores

To create a core EDIF netlist file the inbuilt procedure *export()* is called. This generates the interface which enables the core to be linked to other EDIF netlist files. As an example -

```

source "xilinx/common.3pl" // to get declaration of part()
part("....."); // so core references correct Xilinx element library

clock(c);

useclock(c);

static(a, "log");
static(b, "uint:8");
value(v1, "log");
value(v2, "uint:8");

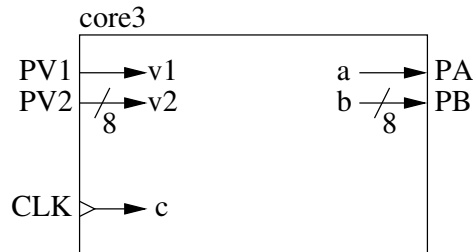
```

```

a <- v1;
b <- v2;

export(PA=a, PB=b <- CLK=c, PV1=v1, PV2=v2);

```



In this case *PA* and *PB* are output ports from this core and these are driven by static variables *a* and *b*. The core takes inputs from ports *CLK*, *PV1* and *PV2*. The clock variable *c* is brought in to the core from outside.

Since the clock *c* has an external source we do not have to supply a DLL or DCM but we still have to create the clock variable *c* that is to be connected to the external clock source. The call *useclock()* need not precede the *export()* call.

The core name must have been supplied using the -e option to the 3PL call. This name will also be used for the netlist and constraint output files.

Chapter 23 - Accessing Specific FPGA Hardware Elements

FPGA hardware elements can be created directly from 3PL source code. This can be done for the usual common elements that would normally be utilised during code generation or for little used or new elements that are not directly used by the code generator.

Note that state elements, such as flip-flops and RAMs, created this way are outside the control of the normal execution thread mechanism.

There are three main reasons for using these directly created elements -

- Special elements such as Virtex4 I/O serial/parallel converters cannot be directly incorporated into the semantics of the language.
- We may wish to use low-level components whose inputs are not connected to a normal 3PL execution thread, for example a flip-flop whose clock enable or reset inputs are connected directly to some source.
- We may wish to use absolute or relative placement of components, something which is very limited in 3PL proper.
- The compiler may produce combinatorial logic which is sub-optimal and which in a critical case needs to be replaced by hand-coded logic.

In such cases we could wrap the created elements in a 3PL module, procedure or function to provide a normal 3PL code interface to hide these specialised constructs.

As an example library `io.3pl` creates elements that are common to many FPGAs. These include IDDDR, ODDR, IDELAY, IODELAY, IDELAYCTRL, ISERDES, SYSMON and some DSP functions. These are all created by the mechanism described below.

Where an element is required which is not included in a library the following inbuilt procedures can be used to create it -

`element()` [section 9.18](#) p201,
`elementend()` [section 9.20](#) p203,
`elementproperty()` [section 9.23](#) p205,
`elementinput()` [section 9.21](#) p203,
`elementclockinput()` [section 9.19](#) p202
`elementoutput()` [section 9.22](#) p204
`elementtsoutput()` [section 9.24](#) p206.

The inbuilt procedure `elementproperty()` only takes strings as value arguments so a number of library procedures are available in `xilinx/common.3pl` to format other types. These are -

`elementLogProperty()` [subsubsection 11.2.2.11](#) p383,
`elementIntProperty()` [subsubsection 11.2.2.10](#) p382,

elementBinProperty() [subsubsection 11.2.2.6](#) p379,
elementHexProperty() [subsubsection 11.2.2.9](#) p381,
elementFloatProperty() [subsubsection 11.2.2.8](#) p381,
elementStrProperty() [subsubsection 11.2.2.12](#) p384

As an example if we wished to directly use a CLB shift register instead of using it implicitly with **del()**, we could create it by -

```
string(id, "srl123");
element("SRL16E", id);
```

where "SRL16E" is the name of the element in the Xilinx library and "srl123" is an optional identifier for the element. We would now connect up each of its ports -

```
value({d, ce, q}, "log");
static(a, "uint:4");
d := .....;
a <- .....;

elementinput(id, "D", d);
elementinput(id, "A", a, 1);
elementclockinput(id, "CLK", currentclock());
elementinput(id, "CE", ce);
elementoutput(q <- id, "Q");
elementend(id);
.... <- q;
.
.
```

We could have omitted the **id** argument in all the calls above in which case the currently open element would have been correctly implied.

The call to **element()** creates the element which is retained until the next call to **element()** as the current element. Calls to **elementinput()** or **elementoutput()** with no **id** apply to the current element. When an id is explicitly used then multiple elements can be created using interleaved calls to the above procedures, though there is probably little advantage in doing that. The arguments to **elementinput()** are -

- the optional element id (string)
- the port name (string)
- the signal variable
- an array format indicator (integer 0 for single signal, 1 for an array where the pins are specified individually, e.g. (**port A0 (direction INPUT)**) or 2 for an EDIF array format such as (**port (array (rename A "a[0:11]") 12) (direction INPUT)**)). Unfortunately the required format seems to depend on whether ISE or Vivado are used and also on the specific element.

Input signal variables can be the constants true or false, any value, selectvalue, static or priority variable or an expression. The signal cannot be from a queue read or an expression containing a queue read but can be from a queue variable if the examine operator ? is used.

This process is subject to the following observations -

1. The variables or constants are not checked against the port type for validity, e.g. if an output port was connected to a constant or a static (i.e. the output of the static) this would not be detected until code generation time when a fatal error resulting from multiple net sources would occur.
2. Errors may not be detected until the Xilinx place-and-route software is run at which point an obscure error message would be generated.

3. Elements generated this way cannot be handled by the simulator (however the simulator is obsolete and not currently available).

In the example above it would have been sensible to construct a module or procedure so that the code could be reused, for example -

```
module
srl16 (value(out) <- value(in), value(ce, "log"), value(addr, "uint:4")) {
  element("SRL16E");
  elementinput("D", in);
  elementinput("A", addr, 1);
  elementclockinput("CLK", currentclock());
  elementinput("CE", ce);
  elementoutput(out <- , "Q");
  elementend();
}
```

It makes no difference if this is a module or procedure as there is no in-line code and no queue reads or writes and the calling syntax and argument passing for these are identical.

To implement a similar delay for say an 8-bit integer -

```
module
srl16_8 (value(out, "int:8") <- value(in, "int:8"), value(ce, "log"), value(addr, "uint:4")) {
  value(o, "[8]bits:1");
  for (int(i) ; i<8 ; i++)
    srl16(o[i] <- getbit(in, i), ce, addr);
  out := cast(o, "int:8");
}
```

To access single bits of the input the library function `getbit()` (`stdlib.3pl`) has been used. There is no equivalent for assigning bits to the output so a local array of single bits has been defined and then this was cast to the output type.

The following code is a simplified example of the use of such a delay. To avoid complicating the example the inputs are assumed to be synchronous with the current clock.

```
iopins(io_pins, "[20]str", {"AB12", "AC13", .....});

input(din, "int:8", loc=io_pins[0..7], readdomain=null);
input(dvalid, "log", loc=io_pins[8], readdomain=null);
static(sum, "int:8");
value(del, "int:8");
value(step, "log");

srl16_8(del <- din, step, 5);

when (dvalid) par {
  sum <- din + del;
  assert(step <-);
}

output(, sum, loc=io_pins[9..16]);
```

In the above example the clock enable is sourced from an instance of the `assert()` [section 9.5](#) p190 inbuilt procedure. This procedure asserts the output when the execution stream in which it is contained executes.

Alternatively the SRL16E CE could have been driven directly from variable `dvalid`.

```
.
.
srl16_8(del <- din, dvalid, 5);
```

```
when (dvalid)
  sum <- din + del;
.
```

Chapter 24 - Constraints, Properties and Attributes

These three terms are used somewhat interchangeably by Xilinx. Some constraints/properties/attributes are generated automatically by 3PL. These are -

1. input and output port locations
2. CLB memory initialisation
3. block memory initialisation

These are written to the EDIF file as properties on instances of elements.

All other constraints are generated by 3PL code, in the case of Xilinx in library `xilinx/common.3pl`. The constraints are written to the NCF file in the case of ISE or an XDC file for Vivado.

Input and output connections have location constraints generated if a **loc** attribute has been set for the input or output mode variable. Attributes **ibufdelay**, **ifddelay**, **tnm**, **pullup**, **pulldown**, **differential**, **diff_term**, **iostandard**, **ibufg**, **slew**, **drive**, if set, also result in constraint file entries. Inbuilt procedures **input()** [section 9.39](#) p218, **output()** [section 9.47](#) p224 and **clock()** [section 9.10](#) p194 do not directly generate these constraints; for convenience and flexibility they are handled in 3PL software.

While attributes can be passed directly to inbuilt procedures **input()**, **output()** and **clock()**, they may also be associated with the input pins. This is done by using the library procedure **iopins()** [subsubsection 11.2.2.19](#) p388 to create variables which contain strings naming the FPGA I/O pins (locations). **iopins()** also creates a global map **pinAttributes** whose entry keys are I/O pin names, the associated values being attribute maps whose keys are attribute identifiers and values the attribute values.

A 3PL user-defined procedure to extract these pin-associated attributes can be automatically called whenever an input, output or clock mode variable is created. The Xilinx 3PL libraries contain such a procedure, **processPinAttributes()** [subsubsection 11.2.2.24](#) p393. This procedure reads the pin locations (loc attribute) from the variable and then reads the attribute map for each pin, adding these attributes to the attribute map for the variable. Note that these pin-related attributes will be overridden by attributes passed as arguments to **input()**, **output()** or **clock()**. **processPinAttributes()** also ensures that attributes are converted from single value to per-pin arrays where appropriate.

The above procedure must be registered with 3PL so that it can be called by the system when an input, output or clock mode variable is created. This is done by calling inbuilt procedure **attributeprocedure()** [section 9.6](#) p191 with a pointer to the procedure as argument. Clearly this should be done before any of the above I/O procedures are called, preferably by placing a call to **attributeprocedure()** early in a leader module.

While netlist generation attributes are handled by **input()**, **output()** or **clock()** following completion of immediate execution, constraint file attributes must be handled by 3PL software. This is done by library procedure **generateConstraints()** [subsubsection 11.2.2.18](#) p388. This procedure should be called late in a trailer module to ensure that no more calls to **input()**, **output()** or **clock()** will be executed. This procedure calls inbuilt function **varlist()** [section 10.167](#) p346 which returns a list of target mode variables.

The resulting list is traversed and the constraint-related attributes of input, output and clock mode variables are extracted and constraints are written to the constraints file.

An example of constraints that might need to be generated explicitly are input and output delays where combinatorial logic paths are included - see [section 7.4](#) p167.

The clock timing group identifier can be explicitly set using a clock attribute (tnm), otherwise it will be constructed from the clock identifier. Regardless of whether a frequency or period attribute has been set a period constraint will be generated. If no frequency or period attribute has been set a warning message will be produced to the effect that there will be no timing constraints for that clock.

CLB RAM and block RAM initialisation entries are added to the EDIF file. If no initialisation is supplied to CMEMORY or RMEMORY mode variables the initialisation entries will be 0.

The library **common.3pl** contains procedures for adding constraints to the NCF and XDC file.

The following raw low level procedures can be used for general constraint strings or constraint strings applying to netlists, but for most constraints there are more convenient ways to apply them -

addncfline() [subsection 11.2.2.2](#) p377
netaddncf() [subsection 11.2.2.21](#) p392
elementaddncf() [subsection 11.2.2.5](#) p378
addxdcline() [subsection 11.2.2.3](#) p377

I/O constraints such as slew rate, drive current or input delays are normally passed to inbuilt procedures **input()** [section 9.39](#) p218 and **output()** [section 9.47](#) p224 as attributes from which the constraint file entries are automatically generated. If I/O logic is being built from the bottom up using inbuilt procedure **element()** and associated procedures then I/O constraints can be applied by library procedure **IOconstraint()**. [subsection 11.2.2.1](#) p376.

Direct creation of elements is usually done in library procedures, for example in `xilinx/clocks.3pl`, but this can also be done in any application code using inbuilt procedure **element()** [section 9.18](#) p201. Where a vendor library element has been explicitly created by 3PL code properties can be added by using inbuilt procedure

```
elementproperty(id, property, value);
```

where all three arguments are strings. There are more convenient library procedures to call this indirectly -

elementLogProperty() [subsection 11.2.2.11](#) p383,
elementIntProperty() [subsection 11.2.2.10](#) p382,
elementBinProperty() [subsection 11.2.2.6](#) p379,
elementHexProperty() [subsection 11.2.2.9](#) p381,
elementFloatProperty() [subsection 11.2.2.8](#) p381,
elementStrProperty() [subsection 11.2.2.12](#) p384.

The main netlist constraints are

- I/O pad location
- absolute element location constraints
- relative element location constraints
- time constraints

Absolute location constraints are not inherently supported by 3PL except for I/O pads and their associated buffers and I/O registers (static mode variables). Single objects such as global clock buffers or clock managers can be given absolute locations by calling inbuilt procedure **elementproperty()** with a "LOC" property, or via library procedure **elementStrProperty()**. registers could be given location constraints by reading back block ID attributes from the static mode variable and applying location constraints to these.

Relative location constraints are not explicitly supported at the moment but they could be generated by application code using constraint file procedures described above.

3PL takes high-level statements and synthesises logic to implement these statements using basic logic elements. Since the logic synthesis process is not accessible to the programmer there is no way to supply location information pertinent to synthesised logic elements. In extreme cases logic can be constructed

using inbuilt procedure **element()** and location constraints can be attached to these, however this is very low level logic construction.

Clock managers (dll, dcm, pll, clkmmcm) are created by library procedures which can be passed a location argument.

Global clock buffers can be created by using library procedure **clockbufg()** in **xilinx/clocks.3pl** (all procedures in this library have optional location arguments). This library procedure is an example of the use of procedures **element()**, **elementinput()** and **elementoutput()** and its code is an example of the use of these procedures -

```
global procedure
clockbufg (clock(out) <- clock(in), str(loc), log(invert)) {
    elementCheck("BUFG", "clockbufg");
    element("BUFG");
    if (invert) {
        value(_in, "log", !in);
        elementinput("I", _in);
    } else
        elementinput("I", in);
    elementoutput(out <- , "O");

    attributes(out,
        frequency=frequency(in),
        period=period(in),
        dutycycle=dutycycle(in)
    );
    if (matched(loc)) elementStrProperty("LOC", loc);
    elementend();
}
```

The first input argument is omitted in each call to **elementinput()** and **elementoutput()** since we have not bothered using an element identifier.

Input arguments to parameters **loc** and **invert** are optional. The procedure calls **element()** to create an element called **BUFG** (a Xilinx global clock buffer). If **invert** is false (which it is by default if not passed an argument) then input argument **in**, a clock signal, is connected to pin **I** port type 0 (input). If **invert** is true an inverted signal **_in** is created (value mode variables of type **log** can be used with clock mode variables) which inverts **in** and is connected to the buffer. Likewise parameter **out** is connected to pin **O** port type 1 (output).

The next statement applies an absolute location constraint to the clock buffer if parameter **loc** has been supplied an argument. For example the argument might be the string "GCLKBUF2" in which case library procedure **elementStrProperty()** is called with arguments "LOC" and "GCLKBUF2". This library procedure simply makes the inbuilt procedure call **special(0, "LOC=GCLKBUF2")** which prepends "INST " and the element identifier to the location string and writes it to the NCF file.

The last three statement copies the frequency, period and duty cycle of the input parameter clock variable to the output parameter clock variable.

Location constraints can be applied to elements explicitly instantiated by using inbuilt procedures **element()**. When using **element()** the example above shows the adding of a location constraint by using library procedure **elementStrProperty()**.

As a further example suppose using ISE we wish to register an input signal from pad "A1" in a flip-flop at location "SLICE_X12Y10" (I don't believe it is possible to specify the X or Y flip-flop in the slice!).

```
input(in, "log", loc="A1");
value(out, "log");
element("fdre");
elementclockinput("C", c100); // connect clock
elementinput("D", in);        // connect data in
elementoutput(out <- , "Q");    // connect data out
```

```
elementaddncf("LOC=SLICE_X12Y10");
elementend();
```

The value mode variable **out** can now be used in 3PL as usual. The flip-flop will be clocked on every clock cycle, but a clock enable input could be connected.

To do the same as above using Vivado the location constraint would have to be generated differently, for which an explicit element identifier would be needed -

```
input(in, "log", loc="A1");
value(out, "log");
element("fdre", "dff123");
elementclockinput("C", c100); // connect clock
elementinput("D", in);        // connect data in
elementoutput(out <- , "Q");   // connect data out
elementend();
addxdcline("set_property LOC SLICE_X12Y10 [get_cells dff123]");
```

In the above example if we did not have the location constraint for the flip-flop we could have written it in 3PL as -

```
input(in, "log", loc="A1");
static(s, "log");
useclock(c100);
s <- in;
```

However, there is one important distinction - the high-level version will not start executing until it gets a start signal (this may be from an external source or may be automatic after configuration loading) as it is executing in a thread within an infinite loop initiated formally within 3PL. The previous low-level example will start executing immediately the clock is available.

Time constraints are already handled well in 3PL. These are usually set up in platform header files so that application programs can ignore timing constraints.

When clock variables are created they have a frequency or period specified. The clock period is used to automatically provide a timing constraint for all logic driven by that clock (inbuilt procedure **useclock()** sets the current clock domain for all following logic). However, there are some situations in which timing constraints are not automatically generated -

- timing constraints between I/O pads and clocked elements
- timing constraints between clock domains
- situations where a timing constraint can be longer than that of the period of the clock in use

Timing constraints between I/O pads and state elements are illustrated in [section 7.4](#) p167.

Timing constraints between clock domains is only meaningful if the clocks are related in frequency and have a known phase relationship. 3PL normally forbids data from one clock domain being simply assigned in another, but it provides three mechanisms to avoid this. Firstly queues can be read and written in different clock domains. The logic implementing this uses asynchronous FIFOs and there need be no relationship between the two clock domains. Secondly the inbuilt function **sample()** allows data in one clock domain to be sampled in another. Again there need be no relationship between the two clock domains. Thirdly the inbuilt function **accept()** can be used to force simple assignment to be accepted by the compiler. This can only be used safely under certain circumstances when the clock domains are related and have a known relative phase. In this case a timing constraint must be used between state elements in the two clock domains. A common example is where one clock is a multiple of the other and they are in phase, i.e. the positive edge of the slower clock coincides with a positive edge of the faster clock. For example if we have two clocks **c100** and **c50** which are 100MHz and 50MHz respectively and are in phase we can assign from one domain into the other -

```
maxdelay("C100", "C50", 10.0e-9);
static({v1, v2}, type);
useclock(c100);
v1 <- .....; // as side effect sets the clock domain of v1 to c100
useclock(c50);
v2 <- accept(v1); // assigning v1 in domain c50 - accept() prevents fatal error
```

When a clock variable is created it will be given a Xilinx timing group identifier which by default is the variable identifier converted to upper case. During place and route all state elements clocked by a clock are entered in a set associated with the clock timing group identifier. Data paths between state elements in the same timing group are automatically constrained to be no greater than the clock period minus setup time. Data paths between state elements in different timing groups are not constrained by default. The example above applies a timing constraint along data paths from state elements in timing group **C100** to state elements in timing group **C50**. The constraint is 10.0ns and for reporting purposes the constraint will be given the identifier "TS_C100_C50".

The application of arbitrary timing constraints between registers, for example where a register might be assigned every second clock cycle, is probably not a good idea. This is not in the spirit of the language and would involve code which was difficult to verify and maintain. It can be done by reading the block identifier attributes of the registers concerned and applying a timing constraint between them.

Appendix A - File system

classes/
 3pl.jar
 3PL Java archive file
 threep1
 3PL Java class files compiled from sources/threep1 below.
doc/
 api
 3PL Java API
 impl
 Latex and fig sources for the 3PL implementation description
 overview
 3PL overview
 ref
 Latex and fig sources for the 3PL reference manual
sources/
 tools/
 Miscellaneous scripts and programs for configuration file generation or CPU to - see [12](#)
 FPGA interfaces
 bin2c/
 Convert a binary compressed or uncompressed FPGA configuration bit file to a C include file containing an initialised byte array as C source code. By this means the configuration can be included in the application program and can be loaded by the CPU application code without the need to read an external file during execution.
 bin2py/
 Convert a binary compressed or uncompressed FPGA configuration bit file to a Python3 source file containing an initialised byte array as Python3 source code. By this means the configuration can be included in the application program and can be loaded by the CPU application code without the need to read an external file during execution.
 edif2bit/
 Generate an FPGA configuration bit file from the EDIF netlist file. This invokes Xilinx ISE or Vivado tools as appropriate.
 info2c/
 Generate CPU/FPGA interface functions for C from the 3PL information file (design.info). This is applicable to the Zynq AXI bus interface or to platforms with a serial link from the CPU to the FPGA. This generates a C include file **design_fpga.h**
 info2py/

Generate CPU/FPGA interface functions for Python3 from the 3PL information file (design.info). This can be executed as a main program and will generate a Python3 source file of CPU/FPGA interface functions that can be imported into the CPU application program. Alternatively this file can be imported into the application program in which case it will create the interface functions internally and will compile them into a module.

info2py is a copy of into2py.py included so that it can be invoked as a main program without the file name extension, e.g. **info2py prog** rather than **python3 infotopy.py prog**.

This source will import serial/cpu_serial/cpu_serial.py for FPGA platforms which have a serial interface, or zynq/fpga/fpga.py in the case of Zynq.

run_ise/

Generate an FPGA configuration bit file from the EDIF netlist file using Xilinx ISE tools. This is a very much more primitive and direct script than edif2bit above.

run_vivado/

Generate an FPGA configuration bit file from the EDIF netlist file using Xilinx Vivado tools. This is a very much more primitive and direct script than edif2bit above.

serial/

This directory contains interface code for FPGA platforms which use a serial interface to both load the FPGA configuration and to transfer data between the CPU and the FPGA. An example of this type of platform is the Gadget Factory Papilio board.

cpu_serial/

Python3 low-level CPU/FPGA interface communications functions.

uart_cpu/

C low-level CPU/FPGA interface communications functions.

zynq/

fpga/

fpga.c, fpga.h

C low-level CPU/FPGA interface communications functions.

fpga.py

Python3 low-level CPU/FPGA interface communications functions. This imports low-level communication functions from lib_py_ext/fpga_read_write.c which has been added to the Python3 library by lib_py_ext/setup.py below.

lib_py_ext/

fpga_py_extension.c

C extension functions for the zynq Python3 CPU/FPGA interface

setup.py

A Python3 source file to add fpga_read_write.c above as a C extension to the Python3 library.

include/

3PL library source files to be included in 3PL programs by module "..." or source "..." statements. These are described in [chapter 11](#) p351.

scripts/

Scripts used by the makefile when building 3PL.

threepl/

Source code for the 3PL compiler and interpreter.

codegen/

Java classes used to construct the intermediate code list.

exceptions/

Compiler and intermediate code execution exceptions.

exec/

Java classes instantiated during parsing and used during immediate execution.

funcs/

3PL inbuilt functions.

mods/

3PL inbuilt modules.

nodes/

Nodes in the code tree produced by the 3PL parser.

parser/

The 3PL parser.

procs/

3PL inbuilt procedures.

simulator/

A target code simulator that has atrophied with time and neglect. It is unlikely to be still functional and has not been included in 3PL for quite a few years. It has not been deleted in case someone wants to try to resurrect it. This would be difficult due to lack of documentation!

ThreePL.java

The 3PL main program.

Appendix B - regular expressions

The following summary of regular expressions is taken directly from the Java documentation -

Characters

```
x    The character x
\\   The backslash character
\On  The character with octal value On  (0 <= n <= 7)
\Onn The character with octal value Onn (0 <= n <= 7)
\Omnn The character with octal value Omnn (0 <= m <= 3, 0 <= n <= 7)
\xhh The character with hexadecimal value 0xhh
\uhhhh The character with hexadecimal value 0xhhhh
\t   The tab character ('\u0009')
\n   The newline (line feed) character ('\u000A')
\r   The carriage-return character ('\u000D')
\f   The form-feed character ('\u000C')
\a   The alert (bell) character ('\u0007')
\e   The escape character ('\u001B')
\cx  The control character corresponding to x
```

Character classes

```
[abc]  a, b, or c (simple class)
[^abc] Any character except a, b, or c (negation)
[a-zA-Z] a through z or A through Z, inclusive (range)
[a-d[m-p]] a through d, or m through p:
[a-dm-p] (union) [a-z&&[def]] d, e, or f (intersection)
[a-z&&[^bc]] a through z, except for b and c:
[ad-z] (subtraction)
[a-z&&[^m-p]] a through z, and not m through p:
[a-lq-z] (subtraction)
```

Predefined character classes

```
.    Any character (may or may not match line terminators)
\d   A digit: [0-9]
\D   A non-digit: [^0-9]
\s   A whitespace character: [ \t\n\x0B\f\r]
\S   A non-whitespace character: [^\s]
\w   A word character: [a-zA-Z_0-9]
```

\W A non-word character: [^\w]

Boundary matchers

^ The beginning of a line

\$ The end of a line

\b A word boundary

\B A non-word boundary

\A The beginning of the input

\G The end of the previous match

\Z The end of the input but for the final terminator, if any

\z The end of the input

Greedy quantifiers

X? X, once or not at all

X* X, zero or more times

X+ X, one or more times

X{n} X, exactly n times

X{n,} X, at least n times

X{n,m} X, at least n but not more than m times

Reluctant quantifiers

X?? X, once or not at all

X*? X, zero or more times

X+? X, one or more times

X{n}? X, exactly n times

X{n,}? X, at least n times

X{n,m}? X, at least n but not more than m times

Possessive quantifiers

X?+ X, once or not at all

X*+ X, zero or more times

X++ X, one or more times

X{n}+ X, exactly n times

X{n,}+ X, at least n times

X{n,m}+ X, at least n but not more than m times

Logical operators

XY X followed by Y

X|Y Either X or Y

(X) X, as a capturing group

Back references

\n Whatever the nth capturing group matched

Quotation

\ Nothing, but quotes the following character

\Q Nothing, but quotes all characters until \E

\E Nothing, but ends quoting started by \Q

Special constructs (non-capturing)

(? :X) X, as a non-capturing group
 (?idmsux-idmsux) Nothing, but turns match flags `i d m s u x` on - off
 (?idmsux-idmsux:X) X, as a non-capturing group with the given flags `i d m s u x` on - off
 (?=X) X, via zero-width positive lookahead
 (?!X) X, via zero-width negative lookahead
 (?<=X) X, via zero-width positive lookbehind
 (?<!X) X, via zero-width negative lookbehind
 (?>X) X, as an independent, non-capturing group
 Backslashes, escapes, and quoting

The backslash character (`'\'`) serves to introduce escaped constructs, as defined in the table above, as well as to quote characters that otherwise would be interpreted as unescaped constructs. Thus the expression `\\` matches a single backslash and `\{` matches a left brace.

It is an error to use a backslash prior to any alphabetic character that does not denote an escaped construct; these are reserved for future extensions to the regular-expression language. A backslash may be used prior to a non-alphabetic character regardless of whether that character is part of an unescaped construct.

Backslashes within string literals in Java source code are interpreted as required by the Java Language Specification as either Unicode escapes or other character escapes. It is therefore necessary to double backslashes in string literals that represent regular expressions to protect them from interpretation by the Java bytecode compiler. The string literal `"\b"`, for example, matches a single backspace character when interpreted as a regular expression, while `"\\b"` matches a word boundary. The string literal `"\hello\"` is illegal and leads to a compile-time error; in order to match the string `(hello)` the string literal `"\\(hello\\)"` must be used.

Groups and capturing

Capturing groups are numbered by counting their opening parentheses from left to right. In the expression `((A)(B(C)))`, for example, there are four such groups:

```

1      ((A)(B(C)))
2      (A)
3      (B(C))
4      (C)

```

Group zero always stands for the entire expression.

Capturing groups are so named because, during a match, each subsequence of the input sequence that matches such a group is saved. The captured subsequence may be used later in the expression, via a back reference, and may also be retrieved from the matcher once the match operation is complete.

The captured input associated with a group is always the subsequence that the group most recently matched. If a group is evaluated a second time because of quantification then its previously-captured value, if any, will be retained if the second evaluation fails. Matching the string `"aba"` against the expression `(a(b)?)+`, for example, leaves group two set to `"b"`. All captured input is discarded at the beginning of each match.

Groups beginning with (?: are pure, non-capturing groups that do not capture text and do not count towards the group total.

Appendix C - Syntax

This syntax is a little out-of-date. Classes have been introduced since this was written. It will be updated when time permits.

Tokens are enclosed in angle brackets - <...>. Optional elements are indicated by ? and | separates alternatives.

File module

```
Filemodule := <FMODSTART> ( ( Statement ) | Filemodule )* ( <FMODFINISH> | <EOF> )
```

Procedure call or module call

```
ProcModCall := DereferencedVar <LPAREN> ArgList <RPAREN>
```

Function call

```
FuncCall := DereferencedVar <LPAREN> ArgList <RPAREN>
```

A module, procedure or function call argument list

```
ArgList := ( ( Keymatch | Expr | CompVal ) )?
          ( ( <COMMA> | <LPOINT> ) ( ( Keymatch | Expr | CompVal ) )? ) *
Keymatch := <IDENTIFIER> <EQ> ( Expr | CompVal )
```

Module definition

```
ModDef := ( <GLOBAL> )? <MODULE> <IDENTIFIER> <LPAREN> ParamList <RPAREN> Block
```

Procedure definition

```
ProcDef := ( <GLOBAL> )? <PROCEDURE> <IDENTIFIER> <LPAREN> ParamList <RPAREN> Block
```

Function definition

```
FuncDef := ( <GLOBAL> )? <FUNCTION> <IDENTIFIER> <LPAREN> ParamList <RPAREN> Block
```

Module, procedure or function output or input parameters

```
ParamList := (
              ( ProcModCall | <VARARGS> )
              ( ( <COMMA> | <LPOINT> ) ( ProcModCall | <VARARGS> )? ) *
            )?
```

General code block

```
Block := <LBRACE> ( Statement ) * <RBRACE>
```

Statement

```
Statement := (
              ModDef |
              ProcDef |
```

```

FuncDef |
ProcModCall <SEMICOL> |
InlineModBlock |
ImAssignStat <SEMICOL> |
TargAssignStat <SEMICOL> |
NullQueueAssignStat <SEMICOL> |
SeqWhileStat |
SeqDoWhileStat |
SeqBlock |
ParWhileStat |
ParDoWhileStat |
ParBlock |
SyncWhileStat |
SyncDoWhileStat |
SyncBlock |
WhenStat |
IncrDecrStat <SEMICOL> |
ReturnStat |
BreakStat |
ContinStat |
IfStat |
WhileStat |
DoWhileStat |
ForStat |
SwitchStat |
CaseStat |
DefaultStat |
PetriNetStat |
<SEMICOL>
)

Immediate assignment (=)
  ImAssignStat := DereferencedVar <EQ> ( Expr | CompVal )

Target assignment (:=, |-, <- or <<-)
  TargAssignStat := DereferencedVar
    ( <COLEQ> | <BUSPOINT> | <LPOINT> | <LLPOINT> )
    ( Expr | CompVal )

A null queue assignment (pop a queue)
  NullQueueAssignStat := <NULL> <LLPOINT> DereferencedVar

An exception
  unless := <UNLESS> <LPAREN> Expr <RPAREN> ( ( Statement | Block ) )?

Par block
  ParBlock := <PAR> <LBRACE> ( Statement )* <RBRACE> ( unless )?

Seq block
  SeqBlock := <SEQ> <LBRACE> ( Statement )* <RBRACE> ( unless )?

Sync block
  SyncBlock := <SYNC> <LBRACE> ( Statement )* <RBRACE> ( unless )?

Inline module block
  InlineModBlock := <LBRACK> ( Statement )* <RBRACK>

Variable increment or decrement statement (e.g. v++;)

```

```

IncrDecrStat := FlaggedVar

Function or procedure return statement
ReturnStat := <RETURN> ( <LPAREN> Expr <RPAREN> )? <SEMICOL>

Break statement
BreakStat := <BREAK> <SEMICOL>

Continue statement
ContinStat := <CONTINUE> <SEMICOL>

If statement
IfStat := <IF> <LPAREN> Expr <RPAREN> ( Statement | Block )
        ( <ELSE> ( Statement | Block ) )?

While statement
WhileStat := <WHILE> <LPAREN> Expr <RPAREN> ( Statement | Block )

Do while statement
DoWhileStat := <DO> Block <WHILE> <LPAREN> Expr <RPAREN> <SEMICOL>

For statement
ForStat := <FOR> <LPAREN> ( ProcModCall | ImAssignStat )? <SEMICOL>
        ( Expr )? <SEMICOL>
        ( ImAssignStat | IncrDecrStat )? <RPAREN>
        ( Statement | Block )

Switch statement
SwitchStat := <SWITCH> <LPAREN> Expr <RPAREN> Block

Case statement
CaseStat := <CASE> ( Expr | CompVal ) <COLON>

Default statement
DefaultStat := <DEF> <COLON>

Petri net
PetriNetStat := <PETRINET> ( <LPAREN> Expr <RPAREN> )?
        <LBRACE> <IDENTIFIER> IdentList <SEMICOL>
        ( IdentList ( <RPOINT> | <RRPOINT> ) IdentList
        ( <WHEN> <LPAREN> Expr <RPAREN> )? ( Block )? <SEMICOL> )+
        <RBRACE>

Non-empty identifier list
IdentList := <IDENTIFIER> ( <COMMA> <IDENTIFIER> )*

When statement
WhenStat := <WHEN> <LPAREN> Expr <RPAREN> ( Statement | Block )
        ( <ELSE> ( Statement | Block ) )? ( unless )?

Seq while statement
SeqWhileStat := <SEQ> <WHILE> <LPAREN> Expr <RPAREN>
        <LBRACE> ( Statement )* <RBRACE> ( unless )?

Par while statement
ParWhileStat := <PAR> <WHILE> <LPAREN> Expr <RPAREN>
        <LBRACE> ( Statement )* <RBRACE> ( unless )?

```

Sync while statement

```
SyncWhileStat := <SYNC> <WHILE> <LPAREN> Expr <RPAREN>
               <LBRACE> ( Statement ) * <RBRACE> ( unless )?
```

Seq do while statement

```
SeqDoWhileStat := <SEQ> <DO> <LBRACE> ( Statement ) * <RBRACE>
                 <WHILE> <LPAREN> Expr <RPAREN> ( unless )?
```

Par do while statement

```
ParDoWhileStat := <PAR> <DO> <LBRACE> ( Statement ) * <RBRACE>
                 <WHILE> <LPAREN> Expr <RPAREN> ( unless )?
```

Sync do while statement

```
SyncDoWhileStat := <SYNC> <DO> <LBRACE> ( Statement ) * <RBRACE>
                  <WHILE> <LPAREN> Expr <RPAREN> ( unless )?
```

Expression (operator ? :)

```
Expr := Expr1 ( <QUEST> Expr1 <COLON> Expr1 )?
```

Expression (operator ||)

```
Expr1 := Expr2 ( <OROR> Expr2 )*
```

Expression (operator ^^)

```
Expr2 := Expr3 ( <UPUP> Expr3 )*
```

Expression (operator &&)

```
Expr3 := Expr4 ( <ANDAND> Expr4 )*
```

Expression (operator |)

```
Expr4 := Expr5 ( <OR> Expr5 )*
```

Expression (operator ^)

```
Expr5 := Expr6 ( <UP> Expr6 )*
```

Expression (operator &)

```
Expr6 := Expr8 ( <AND> Expr8 )*
```

Expression (operators == != <lt > <lt= >=)

```
Expr8 := Expr9 ( ( <EQEQ> | <NE> | <LANGE> | <RANGE> | <LE> | <GE> ) Expr9 )*
```

Expression (operators <lt<lt >gt>gt)

```
Expr9 := Expr10 ( ( <LS> | <RS> ) Expr10 )*
```

Expression (operators + -)

```
Expr10 := Expr11 ( ( <PLUS> | <MINUS> ) Expr11 )*
```

Expression (operators * / %)

```
Expr11 := Expr12 ( ( <STAR> | <SLASH> | <PERCENT> ) Expr12 )*
```

Expression (unary operators ~ ! -)

```
Expr12 := ( <TILDE> Expr13 | <EXCL> Expr13 | <MINUS> Expr13 | Expr13 )
```

Expression (unary operators ? <)

```
Expr13 := ( <QUEST> Expr14 | <LANGE> Expr14 | Expr14 )
```

Expression (constants, function call, parenthesised expression)

```
Expr14 := (
```

```

    <INTEGER_LITERAL> |
    <FLOATING_POINT_LITERAL> |
    <STRING_LITERAL> |
    <LOG_LITERAL> |
    <NULL> |
    FuncCall |
    Expr15 |
    <LPAREN> Expr <RPAREN>
)

```

Expression (unary operator >) grammar production

```
Expr15 := ( <RANGLE> DereferencedVar | DereferencedVar )
```

Compound value

```

CompVal := <LBRACE>
          ( ( Expr | CompVal ) )? ( <COMMA> ( ( Expr | CompVal ) )? ) *
          <RBRACE>
DereferencedVar := FlaggedVar ( ( <RPOINT> | <RRPOINT> ) ( Subscript | Field ) * ) *

```

Pre or post increment or decrement or indirect variable.

++, --, * and & are not handled as normal operator nodes.

The VarNode to which they apply has flags set

to indicate ++v, --v, v++, v-- , *v or &v .

These are handled when the variable is accessed.

```
FlaggedVar := ( <INCR> | <DECR> | <AND> )? Variable ( <INCR> | <DECR> )?
```

A variable (possibly with subscripts or field names)

```

Variable := ( <GLOBALDOT> | <FILEDOT> | <CALLINGDOT> | <LOCALDOT> )?
            <IDENTIFIER> ( Subscript | Field ) *

```

Subscript(s)

```
Subscript := <LBRACK> ( ( Expr ( <DOTDOT> Expr )? )? ) <RBRACK>
```

Struct field (on a variable)

```
Field := <DOT> <IDENTIFIER>
```

Tokens are -

<BREAK>	"break"
<CALLINGDOT>	"calling."
<CASE>	"case"
<CONTINUE>	"continue"
<DEF>	"default"
<DO>	"do"
<UNLESS>	"unless"
<FILEDOT>	"file."
<ELSE>	"else"
<FOR>	"for"
<FUNCTION>	"function"
<GLOBAL>	"global"
<GLOBALDOT>	"global."
<IF>	"if"
<LOCALDOT>	"local."
<MODULE>	"module"
<NULL>	"null"
<PAR>	"par"
<PETRINET>	"petrinet"

<PROCEDURE>	"procedure"
<RETURN>	"return"
<SEQ>	"seq"
<SWITCH>	"switch"
<SYNC>	"sync"
<VARARGS>	"varargs"
<WHEN>	"when"
<WHILE>	"while"
<EQ>	"="
<RRPOINT>	"->>"
<RPOINT>	"->"
<COLEQ>	":="
<BUSPOINT>	" -"
<LLPOINT>	"<<-"
<LPOINT>	"<-"
<EQEQ>	"=="
<NE>	"!="
<LE>	"<="
<GE>	">="
<PLUS>	"+"
<MINUS>	"_"
<STAR>	"*"
<SLASH>	"/"
<PERCENT>	"%"
<OROR>	" "
<UPUP>	"^^"
<ANDAND>	"&&"
<OR>	" "
<UP>	"^"
<AND>	"&"
<EXCL>	"!"
<TILDE>	"~"
<QUEST>	"?"
<LS>	"<<"
<RS>	">>"
<INCR>	"++"
<DECR>	"--"
<DOTDOT>	".."
<LBRACE>	"{"
<RBRACE>	"}"
<LBRACK>	"["
<RBRACK>	"]"
<LPAREN>	"("
<RPAREN>	")"
<LANGLE>	"<"
<RANGLE>	">"
<COMMA>	","
<SEMICOL>	";"
<COLON>	":"
<DOT>	"."
<INTEGER_LITERAL>	<DECIMAL_LITERAL> <HEX_LITERAL> <BIN_LITERAL> <OCTAL_LITERAL>
<DECIMAL_LITERAL>	["1"-"9"] (["0"-"9"])*
<HEX_LITERAL>	"0" ["x", "X"] (["0"-"9", "a"-"f", "A"-"F"])+
<BIN_LITERAL>	"0" ["b", "B"] (["0"-"9", "a"-"f", "A"-"F"])+
<OCTAL_LITERAL>	"0" (["0"-"7"])*>
<FLOATING_POINT_LITERAL>	(["0"-"9"])+ "." (["0"-"9"])+ (<EXPONENT>)?

	"." ([0-"9"])+ (<EXPONENT>)?
	([0-"9"])+ "." <EXPONENT>
	([0-"9"])+ (<EXPONENT>)?>
<EXPONENT>	["e","E"] (["+", "-"])? ([0-"9"])+
<STRING_LITERAL>	"\" (~ ["\"", "\\\"", "\n", "\r"]
	"\\\" (["n", "t", "b", "r", "f", "\\\"", "\'", "\""]
	[0-"7"] ([0-"7"])?
	[0-"3"] [0-"7"] [0-"7"]))* "\"
<LOG_LITERAL>	"true" "false"
<IDENTIFIER:>	<LETTER> (<LETTER> <DIGIT>)*
<LETTER>	["A"-"Z", "a"-"z", "_"]
<DIGIT>	[0-"9"]
<FMODSTART>	a token inserted by the file reader at the start of each source file
<FMODFINISH>	a token inserted by the file reader at the end of each source file

Appendix D - Compiler Versions

The current version is 11.6.

The principal enhancement from version 3.n to version 4.0 was the introduction of an extra scope based on source files. The **include** statement has now been removed, along with **globalvars** and **privatevars**; **module** and **source** statements have been added in their place. **nomodule** and **nosource** statements have been added to ensure that files are included using the correct mechanism. The compiler is not backward compatible in this respect though usually changes required to existing sources are quite small and local.

Version 4.1 had a new optional clause, the **unless** construct, appended to most target control structures to allow premature termination when an exception expression is true. This also includes an optional code block to be executed when the exception occurs.

Version 4.2 contains list ([subsection 3.7.4](#) p61) and map ([subsection 3.7.5](#) p63) types.

Version 5.0 introduces variable **attributes** which replace the arbitrary string of arguments previously required by inbuilt procedures **input()**, **externout()** and **clock()**. There is a new mode **output** and procedure **externout()** has been deleted. Procedure **attributes()** allows attributes to be applied to variables and function **attributes()** provides a means of examining attributes. Variable attributes are used for -

- applying constraints to generated code elements, particularly input and output buffers and pads
- controlling some aspects of code generation such as queue buffer depths and code optimisation

There are now two memory modes, **cmemory** for combinatorial output memory and **rmemory** for registered output memory, where previously there was a single mode (**memory**) used by both types of memory. The inbuilt procedures **cmemory()** and **rmemory()** are still used as before to create combinatorial output and registered output memories respectively

Version 5.0 also now takes note of the order of a subscript range. Previous versions sorted the range limits to smaller..larger but version 5.0 uses the order specified, creating a descending order if required.

Inbuilt procedures **externout()** and **bufferize()** have been deleted, inbuilt procedure **fatal()** has been renamed **exit()** and an inbuilt procedure **texit()** (same as **trace()** followed by **exit()**) has been added.

Variables of type "type" and type "str" can now be cross-assigned or initialised (no cast is needed). Variables of type "type" cannot otherwise be used where type "str" is accepted in which case the type can be converted by inbuilt function **typestring()**, but this restriction may be lifted in the future.

Version 6.0 requires Java 1.6 where previous versions used Java 1.5.

Version 6.1 will handle Xilinx Coolrunner and XC95 CPLDs with a subset of modes and operations. Queue, cmemory, rmemory and selectvalue modes cannot be used. Target multiplication, division, remainder and variable shift operations are not implemented.

Version 6.2 has undergone an internal code reorganisation and has a few obscure but important bug fixes.

Version 6.3 has deprecated the following procedures -

deprecated	replacement
rram()	rmemory()
rramread()	rmemoryread()
rramwrite()	rmemorywrite()
cram()	cmemory()
cramwrite()	cmemorywrite()

and the following functions -

deprecated	replacement
rramout()	rmemoryout()
cramread()	cmemoryread()

This has been done to use identifiers more appropriate to the variable mode and at the same time to reverse the address type and data type arguments to the variable creation procedures to match the order of address and data arguments to the associated write procedures. The deprecated procedures and functions remain in the implementation.

Version 6.4 supports Virtex-5, including gigabit transceivers.

Version 6.5 contains a branch statement, the target equivalent of a switch statement. It also has new inbuilt functions `timestring()` and `elapsedtime()`.

Version 6.6 now allows pin location attributes to be any type as long as they only contain "str" primitives. Previously these were restricted explicitly to types "str" and "[]str" but now may be nested arrays and structs as long as all primitives are "str". This version also contains the added procedures and functions -

<code>split()</code>	split a string into substrings
<code>getfield()</code>	get the nth field from a struct
<code>numfields()</code>	get the number of fields in a struct
<code>isstruct()</code>	determine if a variable type is a struct
<code>shell()</code>	execute a command in a shell
<code>exec()</code>	execute a command
<code>env()</code>	return a map of environment variables

Version 6.7 now implements target fixed-point types and arithmetic operations. This version also changed floating point implementation from using a struct for representation to a proper type with inbuilt functions to extract the sign, biased exponent and mantissa. Functions added are -

<code>Offset()</code>	the offset of a fixed point value
<code>BiasedExponent()</code>	the biased exponent of a floating point value
<code>BiasedExponentWidth()</code>	the biased exponent width of a floating point value
<code>FloatIsNegative()</code>	the sign of a floating point value
<code>Mantissa()</code>	the mantissa of a floating point value
<code>MantissaWidth()</code>	the mantissa width of a floating point value

Version 6.8 included the `iswritten()` inbuilt function.

Version 7.0 has conditional compilation. It also has had the `round()` inbuilt function extended to provide rounding from fixed or ufixed to int or uint using round-to-even.

Version 7.1 has substantially improved code optimisation. It also extends the 'continuous' attribute to queue mode variables.

Version 7.2 has a new target construct, 'sync when ()'.

Version 7.3 allows target mode values in a compound value.

Version 7.4 has improved implementation of combinatorial output memory. This is more efficient in some cases and also gives some more choices in the smaller queue buffer depths than previously.

Version 7.5 has the following new features -

- static assignments `+=`, `-<=` and `*<=` to add to, subtract from or multiply the LHS variable
- immediate floating point values can be added to, subtracted from or assigned to target fixed and ufixed types

- static variables can have an "ALU" attribute which will use a single adder/subtractor with an input selector to combine additions, subtractions, increments, decrements and assignments
- an option to generate nested logic gates rather than expressions in LUTs for many low-level constructs to change the way the vendor place-and-route software optimises the logic

This version also adds a flip-flop to the output of every SRL16 shift register to improve timing.

Version 7.6 has undergone some improvement in netlist optimisation. It also now handles the Virtex-5 DSP48E block multiplier and provides for specific instantiation of the DSP48E. In the case of Virtex-2 multipliers or Virtex-5 DSP48Es where these do not use the associated output register and their outputs drive flip-flops (static registers) with identical clock enables and resets, then the flip-flops are delayed and the element registers are used instead. A Virtex5 static variable with DSP attribute set will also use the same DSP48E block for ALU logic.

Version 8.0 has changed the way lists and maps are used. Lists and maps are now accessed using inbuilt procedures and functions. Also in this version functions can have the pointer dereference unary operator `->` operator appended where the function returns a pointer. In addition in that case the function call can occur on the left side of an assignment statement.

Version 8.1 has a reworked parser that now allows a function call with a pointer dereference followed by increment or decrement operator to be a statement. It also allows subscripts and fields on function call returns.

Version 8.2 now supports Virtex-5 and Virtex-6. It has been upgraded to use the DSP blocks in Virtex-5 and Virtex-6 under control of DSP attributes on selectvalue, static and queue mode variables. For all FPGAs it also allows addition and subtraction operations to be combined in a single ALU for selectvalue, static and queue mode variables when directed by an ALU attribute. Explicit support for IDELAY, ODELAY, IDDR, ODDR and DSP48E has been added to the library xc5v.3pl (CRC16, CRC32 and SYSMON were already included).

Version 8.3 contains the waitpri() construct. This is primarily intended for arbitration of conflicting writes to queue variables but can also be used to arbitrate between memory accesses and other statements. Call to priority() to create a priority mode variable no longer needs to be given a type in which case it will be given type "[20]log" by default. The inbuilt function isread() was added. This function is true for one clock cycle when its queue argument is read.

Version 8.4 allows queues of depth 0, i.e. unbuffered queues. The principal use of these is to synchronise two parallel execution threads, optionally passing a datum in the process. The use of these is of necessity somewhat restricted.

Version 8.5, when called with no source file command line argument, pops up a GUI to open the file. The GUI has buttons to set some options and a button to start (parsing, immediate execution and then code generation). The output appears in a scrollable text window. A file can be repeatedly compiled and executed by a single button press. Version 8.5 will run on Linux (or any Unix), MAC OSX and Windows.

Version 8.6 uses CLB logic for selectvalue mode variables by default, whereas previously 3-state buffers were used. An attribute is used to force 3-state implementation if desired. An additional attribute is now available to determine the output level when no inputs are selected. Inbuilt procedures and functions tselect() and tselecta() have been removed.

Version 8.7 allows selectvalue mode variables to have their default values, when no input is selected, to be specified. The environment variable **THREEPL_CONTINUOUS** is no longer recognised. There is a new directive, "output directory", produced by the compiler. The optional report file is now placed in the designated output directory instead of the parent directory.

Version 8.8 has added inbuilt functions memoryaddresstype() and memorydatatype().

Version 8.9 now allows compound values to contain target expressions. Inbuilt procedure float() will now accept an integer initialisation argument.

Version 8.10 dispenses with the unworkable conditional compilation scheme and substitutes conditional file inclusion via additional forms of 'module' and 'source' statements.

Version 8.11 no longer supports automatic conversion of "null" queues to "log" values. A queue of type "null" can be explicitly cast to a value of type "log" where required.

Version 8.12 removes some restrictions on the use of unbuffered queues.

Version 8.13 is functionally identical to version 8.12. The version number has been advanced because the source code has been extensively modified to incorporate generic Java typing and eliminate casts where possible. Also the parser has been separated from the main program, which was not the case previously.

Version 8.14 removes the restriction on cmemory and rmemory variables being passed as arguments to varargs parameters.

Version 9.0 has removed clock managers and high speed serial devices from within 3PL and implemented them instead as 3PL library modules. This is a significant change and is not backward compatible.

Version 9.1 checks for input variables with no clock domain that occur in target conditional statement test expressions.

Version 10.0 introduces a major change. Most constraints and attributes have been moved from the netlist constraints file (NCF) to the EDIF netlist file where they now appear as element properties. Remaining constraints (principally timing) are still written to the NCF but are also translated into a Vivado XDC TCL script file. This means that there are no changes required to the source file or to 3PL run options or directives when changing between ISE and Vivado. Where constraints are generated in the 3PL source, mostly in library files, there are now calls to generate both NCF and XDC constraints separately. Library procedures INPUT(), OUTPUT() and CLOCK() have been introduced to properly handle constraints. These in turn call inbuilt procedures input(), output() and clock().

Version 10.1 has the 'ignoremodules' option added to queue attributes.

Version 10.2 allows map indices or keys to be applied in brackets in the same way as array subscripts. It also allows subscripts to be used to access lists, though list insertion still requires the use of an inbuilt procedure.

Version 10.3 has changed inbuilt procedures input(), output() and clock() so that where possible I/O attributes (properties) normally contained in a constraint file are now handled in 3PL software, not through Java code as a side effect of netlist generation. Most of the attributes that these procedures accept are arrays, one member per pin, as this allows attributes to differ from pin to pin even within one variable.

Version 11.0 changed all occurrences of 'pipe' to 'queue' in 3PL and the libraries. It was felt that naming a variable mode 'pipe' and then explaining that it was actually a queue was confusing and unnecessary. Also inbuilt procedures scanf, fscanf, sscanf, printf, fprintf and sprintf have been added.

Version 11.1 allows static variables to be initialised via an "init" attribute. Inbuilt function strtotype() has been added.

Version 11.2 no longer has inbuilt modules FD and REGISTER. There are numerous other minor code changes which should be backward compatible with version 11.1.

Version 11.3 now allows many inbuilt modules and procedures and a few inbuilt functions to be called with parameter=argument pairs as well as matching arguments to parameters solely by position. Input mode and output mode variables no longer have clock domains. Period attributes have been removed, the reciprocal of frequency being used instead.

Version 11.4 provides conditional compilation and execution based on time of last modification of all included source files, the compiler build time and the output netlist file. It also provides a 3PL post-processing section where 3PL code can be executed after netlist generation is complete. This code can invoke place-and-route tools and any other post-processing operations which can again be conditional on the time of last modification of related files.

Version 11.5 has had a minor change to command line options (-v no longer available). It has had a major restructuring of intermediate code signal structures. The report file format has been restructured and improved and the optional intermediate code list is now in a separate file.

Version 11.6 has a change to the **unless** control structure. It now requires an exception code block in braces even if empty. The call to inbuilt procedure shell() has had its output argument order changed.

Version 11.7 has classes but is otherwise backward compatible with version 11.6.

Appendix E - Intermediate Code

The intermediate code produced by the 3PL compiler following immediate execution can be listed in file *design_name.icl* by using option -t when calling 3PL. If option -pt is given instead the intermediate code prior to optimisation will also be listed. The listing will not include signal declarations (SIG elements - see below) but these can be included by using -T and -pT.

The intermediate code consists largely of elements which can be expected to be available in any FPGA. This set of elements is enough to implement any 3PL program. Signals and signal arrays are arguments to these elements.

The intermediate code list has undergone significant optimisation. Some additional optimisation occurs during low-level code generation (converting the intermediate code list into the output EDIF netlist). Additional logic minimisation occurs during later place-and-route by the FPGA vendor software.

The intermediate code elements are -

- AND - AND 2 or more signals
- CONNECT - connect a signal or signal array to another or to a constant
- CRAM - a combinatorial-output RAM
- DEL - delay a signal one or more clock cycles
- DFF - a D-type flip-flop
- DIVERGE - handle queue divergence
- DOWHILE - a 'do while' statement
- ELEMENT - create a netlist element
- EXECP - delay execution on queue availability
- IBUF - an FPGA pad input buffer
- ILOOP - control an infinite loop
- INV - invert a signal
- OBUF - an FPGA pad output buffer
- OPERATOR - apply an expression operator to operands
- OR - OR 2 or more signals
- PORT - an external signal connection to/from another netlist
- PRIORITY - a priority encoder
- QUEUEBUFFER - a queue variable (buffer)
- REG - a register
- RESYNC - resynchronise a pulse or edge to another clock
- RRAM - a registered-output RAM
- SAMPLE - sample a value in another clock domain
- SELECT - select one of several inputs
- SIG - declare a signal or signal array
- SPECIAL - special functions for specific FPGA families
- START - start an execution stream
- TSELECT - a 3-state driver
- WAIT - wait for completion of parallel execution streams
- WHEN - a 'when' statement
- WHILE - a 'while' statement
- XOR - XOR 2 or more signals

E.1 Signals

E.1.1 variables

Global variables generate signals whose names begin with "glob.". Thus the output of a static global variable *a* would be a signal named *glob.a*.

file module variables generate signals whose names begin with the file name, e.g. "prog.a".

Modules, procedures and functions are nested within one of the file modules. A static variable *a* created within module *m* in file module prog would be signal "prog.m_1.a" where an integer has been appended to distinguish variables created during different calls to the module. Further nested calls append more such strings to the identifier.

Variables created within the scope of control structure blocks will have strings such as "if", "while" and "seq" appended to their signal identifiers.

Signal array references have a subscript range appended, even if the entire variable is accessed, e.g. "glob.a[0..7]". For multiple ranges comma-separated ranges are shown, e.g. "glob.a[0..7,16..23]"

E.1.1.1 static variables

A global static variable *s* of size 8 bits would have the following associated signal arrays -

- "glob.s[8]" - the data output signal array
- "glob.s.D[8]" - the data input signal array
- "glob.s.CE[8]" - the clock enable signal array
- "glob.s.RES[8]" - the set/reset signal array

The value of the variable is available via the data output signal. Assignments are made by driving the data input signal and asserting the clock enable signals to the selected bits. The set/reset signal array is used to return some portion of the variable to its initial state.

E.1.1.2 queue variables

A global queue variable *p* of size 8 bits would have the following associated signal arrays -

- "glob.p[8]" - the data output signal array
- "glob.p.D[8]" - the data input signal array
- "glob.p.NE" - output availability (not empty)
- "glob.p.POP" - pop a datum
- "glob.p.PUSH" - push a datum
- "glob.p.NF" - input availability (not full)
- "glob.p.CNT[4]" - buffer occupancy count
- "glob.p.SPC[4]" - buffer space count
- "glob.p.RSTAT[5]" - buffer status in read clock domain
- "glob.p.WSTAT[5]" - buffer status in write clock domain

These signals are further described in the section on queue buffers below.

E.1.1.3 value variables

Value variables are effectively pointers and can be assigned multiple times. To ensure that each assigned value is distinguished a unique integer is appended. A global value variable *v* of 8 bits might thus appear variously as "glob.v_1", "glob.v_2" etc. At the completion of immediate program execution the signal array "glob.v" would be equated with the last value the variable was assigned. This is an attempt to give the unappended variable identifier a sensible value during simulation (simulation is now disabled).

E.1.1.4 priority variables

A global priority variable p of size 8 bits would have the following associated signal arrays -

- "glob.p[8]" - the data output signal array
- "glob.p.D[8]" - the data input signal array
- "glob.p.PE" - the 'else' signal

E.1.1.5 memory variables

Memory variables have separate signal array identifiers for each port, e.g. global combinatorial-output memory m with two ports would have port output signal arrays "glob.m_0" and "glob.m_1".

E.1.2 expressions and execution control signals

The values resulting from expressions are given signal array identifiers of the form "E9".

Various other control signals are generated with prefixes "A", "AND", "CD", "CONST", "EO", "F", "FB", "FD", "FDCP", "FDRS", "FF", "FS", "I", "IN", "INT", "INV", "LOG", "MADDR", "MDATA", "P", "PE", "PNTBUSY", "Q", "RADDR_", "ROPAV", "S", "SAMPLEOUT", "SB", "SC", "SD", "SS", "start", "TF", "TS", "UINT" and "WADDR_", integers being appended in each case to ensure identifiers are unique. There is no particular significance in these identifiers, the initial strings having been chosen to reflect the context in an attempt to improve readability of the intermediate code.

E.2 Intermediate code elements

E.2.1 AND

E.2.1.1 purpose

An AND gate for control signals.

E.2.1.2 arguments

Parameters	none
Inputs	input signal input signal . .
Outputs	AND signal

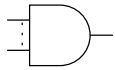
E.2.1.3 description

AND two or more input signals, the result appearing on the output.

E.2.1.4 dump format

S11 = A12 & A14 & S8

E.2.1.5 diagrammatic form



E.2.2 CONNECT

E.2.2.1 purpose

Connect signals or signal arrays.

E.2.2.2 arguments

Parameters	optional boolean for cast sign extension
Inputs	signal, signal array or constant
Outputs	signal or signal array

E.2.2.3 description

This connects signals or signal arrays.

If there is a single boolean parameter the operation is a cast. A true value indicates sign extension will be used if the output is wider than the input. A false value indicates zero padding will be used if the output is wider than the input. If the output is narrower than the input then truncation will occur and the parameter is not relevant. The parameter may be omitted or null.

If there is no cast parameter and the input and output arrays are the same size then they will be simply connected.

If there is no cast parameter and the input is a single bit then it will be connected to all outputs ('expand').

If there is a cast parameter then the inputs will be connected to the outputs with possible size adjustment. If the output is smaller than the input then truncation will occur. If the output is larger than the input then the cast parameter will determine if sign extension (true 'cast - sign_extend') or zero padding (false 'cast - pad') is used.

If the input is a constant TDEVar then the output signal array is connected to a pattern of GND and VCC to implement a 64-bit constant. The constant (which may be signed) will be truncated to fit the signal array. This case may include a sign extension parameter, but the parameter is not relevant since the constant is guaranteed to fit the output width, including the sign.

E.2.2.4 dump format

```
S11 = TS15
E1[0..7] = glob.a[8..15]
E2[0..2] = VCC expand
E3[0..7] = 17
E4[0..15] = glob.v[0..11] cast - pad
E5[0..15] = glob.v[0..11] cast - sign_extend
```


E.2.3 CRAM

E.2.3.1 purpose

Combinatorial-output RAM.

E.2.3.2 arguments

Parameters	Integer number of ports Integer data width in bits Integer address width in bits String[] optional initialisation String - "tnm" attribute key String - data for "tnm" attribute
Inputs	clock signal, port0 address array, port0 data.in array, or null, port0 write signal, or null, port0 clock signal, port1 address array, port1 data.in array, or null, port1 write signal, or null, port1 ..etc..
Outputs	data.out array, or null, port0 data.out array, or null, port1

E.2.3.3 description

This is a RAM with a clocked write but a combinatorial read. At all times the output for a port reflects the memory contents at the location indicated by the current address input.

The 2nd parameter gives the data width and the address width gives the memory depth (number of words). All port addresses must be the same width and all port data inputs and outputs must be the same width. The initialisation string array has one entry for each initial value. Each entry is a hexadecimal string. The initialisation array may be smaller than the depth of the RAM but not larger. The initialisation parameter may be omitted in which case the RAM is explicitly initialised to all 0. Ports which are not written have null arguments for the data in and the write enable. Ports which are not read have null arguments for the data out.

For Xilinx FPGAs Combinatorial-output RAM is CLB RAM. The address width is limited to a maximum of 9 bits (a depth of 512 words). The data width is not limited.

There is one optional attribute which is a timing group name.

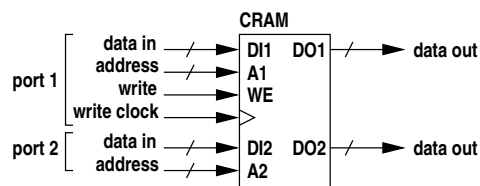
E.2.3.4 dump format

CRAM - 2 ports {	
OUT0	/Users/dun202/lib/3pl/3pl_lib/csiro/hymod/io2.trailermodule_0.ext_ram_trailer_0
OUT1	/Users/dun202/lib/3pl/3pl_lib/csiro/hymod/io2.trailermodule_0.ext_ram_trailer_0
<-	
CLK0	glob.ext_ram_clk
ADDR0	MADDR36129[0..3]
DATA0	MDATA36130[0..35]
WE0 S36137	
CLK1	null

ADDR1	MADDR36152[0..3]
DATA1	-
WE1	-
}	

The presence of an initialisation argument is flagged but the values are not listed.

E.2.3.5 diagrammatic form



E.2.4 DEL

E.2.4.1 purpose

To create a delay line to delay a signal one or more clock cycles. The input and output may be any width. The delay may be variable.

E.2.4.2 arguments

Parameters	Integer number of clock cycles to delay if constant, else 0 String array initial words in line as array of hexadecimal strings
Inputs:	clock signal input signal array delay signal array number of clock cycles to delay if variable, else null enable signal delay steps if high - null for every cycle reset or null
Outputs	output signal

E.2.4.3 description

If the reset is present then a fixed delay line will be created using flip-flops, in which case the variable length input must be null.

If the variable length input is not null the delay line will use CLB shift registers and the reset input must be null. The delay length is limited to 16 or 32 depending on the FPGA.

If the variable delay is used, this value selects the point in the delay line from which the output is taken. 0 selects the output of the 1st delay element and 15 the output of the 16th delay element. Note that changing this value will result in data being skipped (decreasing) or repeated (increasing).

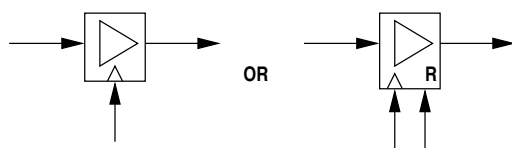
If there is initialisation data the number of words will be equal to the delay line length. If there is no initialisation data these parameters will be omitted.

The optional reset input returns the logic to its initial state.

E.2.4.4 dump format

DEL E69[0..3] <- p.in[0..3] CLK glob.c166 delay 18 initialised
--

E.2.4.5 diagrammatic form



E.2.5 DFF

E.2.5.1 purpose

A D-type flip-flop.

E.2.5.2 arguments

Parameters	Boolean optional initial value integer type of flip-flop
Inputs	data_in signal, or null clock signal, or null clock enable signal, or null reset, set, clear or preset signal, or null
Outputs	data_out signal

E.2.5.3 description

This provides a basic D-type flip-flop. Unused arguments may be null and unused trailing arguments may be omitted. The 2nd argument specifies the type of flip-flop and the function of the 4th input signal-

0 - FDRE - R (synchronous reset)

1 - FDSE - S (synchronous set)

2 - FDCE - CLR (asynchronous clear)

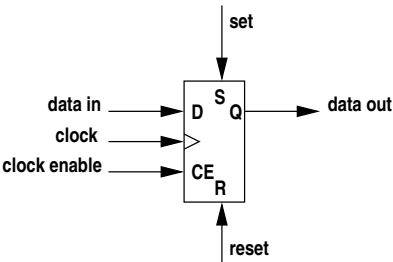
3 - FDPE - PRE (asynchronous set)

Any of the 1st (data in), 3rd (clock enable) or 4th (reset/set/clear/preset) signal inputs which are unconnectd will be connected to GND.

E.2.5.4 dump format

```
DFF {
  OUT      FDRS622
  <-
  D        -
  C        glob.ruclk
  CE       -
  R        OR621
  S        S601
}
```

E.2.5.5 diagrammatic form



E.2.6 DIVERGE

E.2.6.1 purpose

A controller for divergent queue signals. Where a queue is read within more than one module it is required that each module execute a read before the datum is popped. Until all destination modules have executed a read the queue must appear unavailable for reading to those modules that have already executed a read. This is accomplished by inserting this DIVERGE element between the queue .NE and .POP signals and the .NE and .POP signals seen by the queue reads in each destination module.

E.2.6.2 arguments

Parameters	none
Inputs	clock signal reset signal queue not empty signal module queue pop signal module queue pop signal . . .
Outputs	queue pop signal module queue not empty signal module queue not empty signal . . .

E.2.6.3 description

This splits the source queue .NE (read availability) and .POP (acknowledge) signals into a pair per destination module. The 1st input is the output (read) clock signal for the queue.

The 2nd input is the .NE (read availability) signal from the queue. The 1st output signal is the .POP (acknowledge) signal to the queue. The 2nd input is a reset which is asserted if the source queue is reset. The 3rd input is the .POP signal from the 1st module. The 2nd output is the .NE signal to the 1st module. Further input and output signals to other modules may follow. If there is only one module the .NE and .POP signals should be just connected through. If all destination acknowledge signals go high simultaneously the acknowledge signal to the source also goes high, otherwise it goes high when the last destination acknowledge signal goes high. The destination acknowledge signal is high for one clock cycle. When each destination acknowledge signal goes high the matching destination availability signal will go low at the next clock cycle unless all destination modules have also acknowledged in which case destination availability signals will be the be the same as the source availability signal.

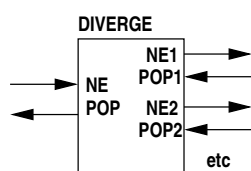
E.2.6.4 dump format

```

DIVERGE {
  in:
    c100
    S11
    glob.p.NE
    S21
    S30
  out:
    glob.p.POP
    S15
    S22
    S31
}

```

E.2.6.5 diagrammatic form



E.2.7 DOWHILE

E.2.7.1 purpose

A *do while* statement.

E.2.7.2 arguments

Parameters	none
Inputs	start signal test signal block finish delayed signal optional reset signal or null clock signal optional reset signal or null
Outputs	start signal for the body DO WHILE finish signal

E.2.7.3 description

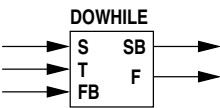
The start input signal starts the first iteration of the loop by propagating straight through to the body start output signal. The block finish delayed input signal comes from the block finish signal, possibly via an EXECp if the test signal comes from an expression containing queue reads. The block finish delayed input signal is then routed either through to the body start output signal (test succeeds) or via a one cycle delay to the finish output signal (test fails).

The optional reset signal clears internal state in the WHILE.

E.2.7.4 dump format

```
DO WHILE {
    START_B      S23
    FINISH       F35
    <-
    START        S24
    TEST         E31[0]
    CONTIN       F33
    CLK          glob.c100
    RESET        s34
}
```

E.2.7.5 diagrammatic form



E.2.8 ELEMENT

E.2.8.1 purpose

Create an instance of a netlist element.

E.2.8.2 arguments

Parameters	String the element name String netlist block name or null String port name — 1st port integer port type — integer port width — integer port array format — . — 2nd port .
Inputs	input 1
Inputs	input 2
.	
.	
Outputs	output 1
Outputs	output 2
.	
.	

E.2.8.3 description

This is used to explicitly create a component element. It can be used to access library components which are not available implicitly in normal code or to access external elements such as cores. It is useful for accessing hardware elements which are peculiar to a particular FPGA rather than being generic. Examples of this are Xilinx global clock buffers as these are not generic elements normally generated implicitly by the compiler.

The first parameter is the element name. The optional second parameter is a netlist identifier for this instance of the element. If null an arbitrary identifier of the form "b123" will be generated.

The following parameters are quadruples giving the port name, port type (0 input, 1 output, 2 3-state output, 4 clock input), port width in bits and port array format, 0 for non-array ports, 1 for pins as A1, A2, A3 etc. and 2 for EDIF pin arrays. For each quadruple there will be an associated signal in the input or output list in the order of occurrence in the parameter list.

E.2.8.4 dump format

```
ELEMENT DCM block dcm1
  PIN CLKIN I glob.ruclk
  PIN CLKFB I prog.cfb
  PIN CLK0 0 prog.cfb_raw
  PIN CLKDV 0 prog.pclk_raw
```

E.2.9 EXECP

E.2.9.1 purpose

An execution wait for queue availability and/or priority grant.

E.2.9.2 arguments

Parameters	none
Inputs	clock signal start signal buffered queue read/write availability signal unbuffered queue write availability signal unbuffered queue read availability signal priority output signal or null optional reset or null
Outputs	delayed start signal priority input signal or null unbuffered queue write pending signal or null unbuffered queue read pending signal or null

E.2.9.3 description

This element delays an execution signal if it is necessary to wait for queue availability, for priority or both. The buffered queue availability input is high if all associated buffered queues to be read or written are available. The unbuffered queue availability inputs are similar but the write and read availabilities are on separate inputs.

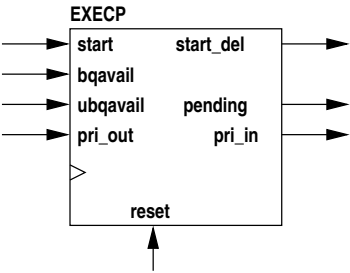
If the queue availability inputs are high and the priority encoder passes the signal, the execution signal is passed through without delay. If the queue availability inputs are not all high or priority has been pre-empted, the execution signal is stored until the queue availabilities are high and priority is granted at which time a single cycle output pulse is generated. An input execution signal must not occur when an output is pending however it may occur on the cycle following the output pulse. The queue availability inputs may be several queue availabilities ANDed together. The reset input, if not null, is used to clear any pending execution (waiting on queue availability or priority). The pending signals are used to notify unbuffered queues that a write or read is pending awaiting unbuffered queue availability. It is asserted when the start signal is asserted or the registered start signal is asserted and all buffered queues are available. The output need not have an argument (null) if unbuffered queues are not involved.

E.2.9.4 dump format

The dump format depends on which inputs and outputs are used.

```
EXECVP {
    START_DEL SD250
    PRI_IN -
    WPENDING -
    RPENDING -
    <-
    CLK glob.ruclk
    START_IN S262
    BQAV kandinski.link_0.if_3372.xc2vp_mgt16_0.q1.NF
    UBQWAV -
    UBQRAV -
    PRI_OUT -
    RESET -
}
```

E.2.9.5 diagrammatic form



E.2.10 IBUF

E.2.10.1 purpose

A buffered external input signal.

E.2.10.2 arguments

Parameters	String - optional element identifier String - input package pin String - optional input package pin for differential input Boolean - ibufg flag Boolean - differential flag
Inputs	input signal from pad optional differential input signal from pad or null
Outputs	output signal optional differential output signal or null

E.2.10.3 description

This buffers an external input signal. The first parameter specifies the ID given to the IBUF element. The second parameter gives the primary FPGA pad designation. The third parameter is an optional

second input pad for differential input or null. The fourth parameter is true for an IBUFG or IBUFGDS. The last parameter is true if the input is differential (IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT).

There is a primary input signal followed by a second input signal if there is differential input to the IBUF (IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT) or null (IBUF, IBUFG).

There is a primary output signal followed by a second output signal if there is differential output from the IBUF (IBUFDS_DIFF_OUT) or null (IBUF, IBUFG, IBUFDS, IBUFGDS).

There must be a **loc** entry to specify the package pin. For differential inputs two consecutive pins are used, positive input followed by negative input.

E.2.10.4 dump format

```
IBUF    INPUT327[0] <- PORT_P88 loc=P88, IBUFG
IBUF    INPUT430[0] <- PORT_P91, PORT_P92 loc=P91,P92 IBUFDS
```

E.2.11 ILOOP

E.2.11.1 purpose

An infinite loop.

E.2.11.2 arguments

Parameters	none
Inputs	start signal block finish signal
Outputs	block start signal

E.2.11.3 description

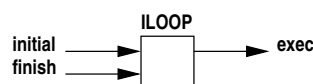
The start input signal is the start signal for all infinite loops in this clock domain. The block finish signal is high when the last statement in the code block executes. This element simply ORs the two inputs.

Infinite loops containing a single statement that always executes in one clock cycle are replaced by a D-flip-flop which is set by the clock domain start signal and stays set thereafter since the contained statement is to execute on every clock cycle.

E.2.11.4 dump format

```
ILOOP    S12 <- glob.c100.start F14
```

E.2.11.5 diagrammatic form



E.2.12 INV

E.2.12.1 purpose

An single signal inverter.

E.2.12.2 arguments

Parameters	none
Inputs	input signal
Outputs	output signal

E.2.12.3 description

The input and output are single signals, the output being the input inverted.

E.2.12.4 dump format

```
E17[1] = inv(glob.a[1])
```

E.2.12.5 diagrammatic form



E.2.13 OBUF

E.2.13.1 purpose

A buffered external output signal.

E.2.13.2 arguments

Parameters	String - optional array of element identifiers String - output package pin String - optional output package pin for differential output boolean - differential flag
Inputs	input signal optional 3-state enable signal (+ve for high-Z) or null
Outputs	output signal to pad optional differential output signal to pad or null

E.2.13.3 description

This buffers an external output signal. The first parameter specifies the ID given to the OBUF element. The second parameter gives the primary FPGA pad designation. The third parameter is an optional second output pad for differential output or null. The fourth parameter flags a differential output.

The first input signal is the data input. The second input signal is an optional 3-state enable which must be low to drive data onto the output pad (OBUFT, OBUFTDS). If this argument is null output drive to the pad is continuous (OBUF, OBUFDS).

There is a primary output signal followed by a second output signal if there is differential output from the OBUF (OBUFDS, OBUFTDS) or null (OBUF, OBUFT).

There must be a **loc** entry to specify the package pin. For differential outputs two consecutive pins are used, positive output followed by negative output.

E.2.13.4 dump format

OBUF	PORT_P90 <- OUTPUTBIT213 loc=P90
OBUF	PORT_P92 <- OUTPUTBIT412 _EN=T31 loc=P92

E.2.14 OPERATOR

E.2.14.1 purpose

An arithmetic or logical operator.

E.2.14.2 arguments

The general form is

Parameters	enumerated type operator code
Parameters	enumerated type 1st operand type
Parameters	enumerated type 2nd operand type if binary or ternary
Parameters	enumerated type 3rd operand type if ternary
Parameters	enumerated type result type
Inputs	1st operand array 2nd operand array if binary or ternary 3rd operand array if ternary
Outputs	output array

The unused parameters and inputs are missing, not null.

E.2.14.3 description

Operator codes are -

TDEOp.INV	unary logical inversion
TDEOp.NEG	unary arithmetic negation
TDEOp.ADD	binary integer addition
TDEOp.SUB	binary integer subtraction
TDEOp.ADDSUB	binary integer addition/subtraction
TDEOp.MUL	binary integer multiplication
TDEOp.DIV	binary integer division
TDEOp.REM	binary integer remainder
TDEOp.DIVREM	binary integer division/remainder
TDEOp.EQ	binary equality test
TDEOp.NE	binary inequality test
TDEOp.LT	binary less-than test
TDEOp.LE	binary less-than-or-equal-to test
TDEOp.GT	binary greater-than test
TDEOp.GE	binary greater-than-or-equal-to test
TDEOp.AND	binary logical and
TDEOp.OR	binary logical or
TDEOp.XOR	binary logical exclusive or
TDEOp.LSH	binary left shift
TDEOp.RSH	binary right shift
TDEOp.COND	ternary conditional (?:)

Arithmetic operations are either Ptype.UINT (unsigned integer) or Ptype.INT (signed integer). These may be mixed. If either operand is signed the result will be signed. The width of the result is that necessary to retain all significant bits. Operands to equality and inequality relational operators may be Ptype.UINT, Ptype.INT, Ptype.BITS or Ptype.LOG. Operands to the other four relational operators are Ptype.UINT or Ptype.INT. The first operand to the shift operators may be Ptype.UINT, Ptype.INT or Ptype.BITS. The second operand (shift count) must be Ptype.UINT. The shift count may be constant or variable. The ternary conditional operator first operator must be Ptype.LOG and the other two operands must be matching types (may be mixed Ptype.UINT and Ptype.INT) and types Ptype.UINT, Ptype.INT, Ptype.BITS or Ptype.LOG.

The integer addition/subtraction operator provides conditional addition or subtraction by means of two additional input signals. The first selects addition (high) or subtraction (low). The second gates the second operand, high for add/subtract and low to inhibit add/subtract (output is first operand). This operation is available as an inbuilt function, not an operator.

Parameters	TDEOp.ADDSUB
Parameters	enumerated type 1st operand type
Parameters	enumerated type 2nd operand type
Parameters	enumerated type result type
Inputs	1st operand array 2nd operand array add/subtract control (high for add) pass 2nd operand control (2nd operand 0 if this is low)
Outputs	output array

The integer division/remainder operator produces the quotient and remainder in a single division. For the division and remainder operators the quotient and remainder are both produced and one of these is discarded. This operation is available as an inbuilt procedure, not an operator.

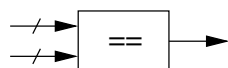
E.2.14.4 arguments

Parameters	TDEOp.DIVREM
Parameters	enumerated type 1st operand type
Parameters	enumerated type 2nd operand type
Parameters	enumerated type quotient type
Parameters	enumerated type remainder type
Inputs	1st operand array 2nd operand array
Outputs	quotient output array
Outputs	remainder output array

E.2.14.5 dump format

<pre> OP E3[0..8] = E2[0..7] + glob.a[8..15] OP E15[0] = !E14[0] OP E20[0..7] = E4[0] ? blog.a[0..7] : E3[0..7] </pre>
--

E.2.14.6 diagrammatic form



Example of equality relational operator shown.

E.2.15 OR

E.2.15.1 purpose

An OR gate for control signals.

E.2.15.2 arguments

Parameters	none
Inputs	input signal input signal . .
Outputs	AND signal

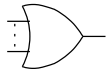
E.2.15.3 description

OR two or more input signals, the result appearing on the output.

E.2.15.4 dump format

<pre> S11 = A12 A14 S8 </pre>

E.2.15.5 diagrammatic form



E.2.16 PORT

E.2.16.1 purpose

Flags a signal which is external to the current source code but is not external to the FPGA (i.e. not a pad connection). These signals are either connections to an imported core or are exported from the current source code which is itself a core.

E.2.16.2 arguments

Parameters	Integer type of signal string port name
Inputs	signal or signal array
Outputs	none

E.2.16.3 description

The first parameter is an integer type code for the port type which is 0 for input, 1 for output or 2 for 3-state output. The second parameter is the port name.

E.2.16.4 dump format

```

PORT {
  par:
    Str    0
    Int    ADDRESS
  in:
    Var    glob.addr[0..18]
}
```

E.2.17 PRIORITY

E.2.17.1 purpose

A priority encoder.

E.2.17.2 arguments

Parameters	none
Inputs	input signal 0 input signal 1 . .
Outputs	output signal 0 output signal 1 . .

E.2.17.3 description

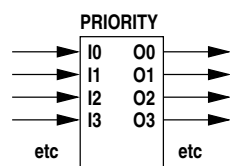
When one or more inputs are high the highest priority matching output signal will be high and all other outputs low. The inputs are ordered in decreasing priority (the 1st is highest).

The number of outputs is equal to the number of inputs or the number of inputs + 1. In the latter case the last output is the 'else' case, i.e. no inputs high.

E.2.17.4 dump format

```
PRIORITY {
  OUT  PRI_OUT40234[0,1,2,3,4]
  ELSE kandinski.processing_0.timer_0.mutex_var_0.pri.PE[0]
  <-
  IN    tPRI_IN40233[0,1,2,3,4]
}
```

E.2.17.5 diagrammatic form



E.2.18 QUEUEBUFFER

E.2.18.1 purpose

A queue buffer.

E.2.18.2 arguments

Parameters	Integer depth Boolean use hardware FIFO is possible, or null String initial value (hexadecimal string) or null
Inputs	input clock signal, or null if same as output clock output clock signal input data array push datum signal (.PUSH) pop datum signal (.POP) reset signal (optional)
Outputs	output data array read available signal (.NE - not empty) write available signal (.NF - not full) ready availability signal for unbuffered queue synchronous buffer occupancy count or null synchronous buffer space count or null asynchronous buffer write status asynchronous buffer read status reset signal (optional)

E.2.18.3 description

A normal queue buffer has a depth of 2. If the depth is ≥ 2 , an initial value is not allowed. If a separate input clock is supplied, depths start at 16 and are various powers of 2 depending on the size ranges of the block RAMs in the particular FPGA. If the input data array and output data array arguments are null then this is a null data queue buffer and the buffer depth is a power of 2 greater than 3.

The 2nd output signal is the read availability signal. This is high if the buffered queue is not empty. For an unbuffered queue it is high if the queue is waiting on a write, i.e. it is connected to the PUSH input.

The 3rd output signal is the write availability signal. This is high if the buffered queue is not full. For an unbuffered queue it is high if the queue is waiting on a read, i.e. it is connected to the POP input.

The 4th output is null for a buffered queue. For an unbuffered queue it is the AND of the push (write) and pop (read) inputs (which is also the AND of the read availability (NE) and write availability (NF) signals).

If the 5th output is not null this gives the number of words in a synchronous queue buffer synchronised to the queue clock (read and write).

If the 6th output is not null this gives the number of free spaces in a synchronous queue buffer synchronised to the queue clock (read and write).

If the 7th output is not null this gives the buffer status for an asynchronous queue buffer synchronised to the write (input) clock.

If the 8th output is not null this gives the buffer status for an asynchronous queue buffer synchronised to the read (output) clock.

If there is an 9th output this is a reset signal. This signal goes high for one clock cycle when the queuebuffer is reset via the reset() inbuilt procedure. For an asynchronous queuebuffer this signal is synchronised to the read (output) clock and is timed to follow the reset of the input (write) side of the queuebuffer.

The asynchronous queue buffer status is a 5-bit word whose bits are -

bit 0	not full
bit 1	empty to 1/4 full + 1
bit 2	2 words to 1/2 full + 1
bit 3	1/4 full + 1 to 3/4 full + 1
bit 4	1/2 full + 1 to full
bit 5	3/4 full + 1 to full

The bits are mutually exclusive and one of them is always set.

Synchronous data queue buffers have a default depth of 2. Synchronous null queue buffers have a default depth of 16. Asynchronous data queue buffers have a default depth of 16. Asynchronous null queue buffers have a default depth of 16. Null queue buffers may have a depth which is 4 or greater and which is a power of 2. Data queue buffers may have the following depths -

- Spartan2, Virtex - 16, 256, 512, 1024, 2048 or 4096
- Spartan3, Virtex2, Virtex2Pro, Virtex4 - 16, 512, 1024, 2048, 4096, 8192 or 16384
- Virtex-5 - 64, 1024, 2048, 4096, 8192, 16384 or 32768

and for synchronous buffers depth 2 is also allowed and is the default.

For queues of depth zero the signals have rather different meanings -

- **PUSH** - unbuffered queue write is pending from an EXECP
- **NE** - connected to **PUSH**
- **POP** - unbuffered queue read is pending from an EXECP
- **NF** - connected to **POP**

outputs **NE** and **NF** simply reflect the **PUSH** and **POP** inputs. These are ANDed together to form an unbuffered queue availability signal (ready) back to the relevant EXECPs to synchronise the write/read pair.

E.2.18.4 dump format

```

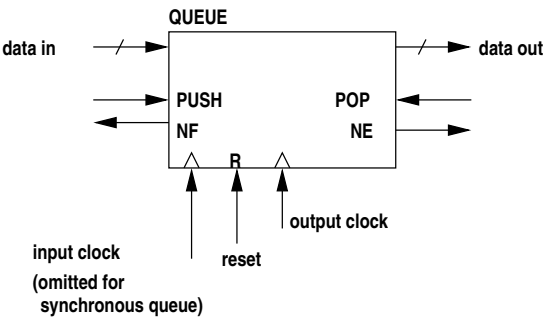
QUEUEBUFFER depth 2 {
    OUT      kandinski.dps[0..11,12..21,22..30]
    NE       kandinski.dps.NE
    NF       kandinski.dps.NF
    CNT      -
    SPC      -
    WSTAT    -
    RSTAT    -
    <-
    CLKIN    -
    CLKOUT   glob.ruclk
    DATA    kandinski.dps.D[0..11,12..21,22..30]
    PUSH     S5669
    POP      SD5715
    RESET    GND
}

```

The minimum size of the buffer is two and this is implemented as two registers. **NF** stands for "Not Full", meaning that at least one free register is available for writing. **PUSH** is ""push data onto queue". **NE** stands for "Not Empty, meaning one or more registers contains data which can be read. **POP** acknowledges a read, popping the word from the buffer (queue).

Where a queue depth of zero is specified the "queue" is unbuffered. In that case this QUEUEBUFFER element simply connects **OUT** to **DATA**, **NE** to **PUSH** and **NF** to **POP** and there is no included logic. **PUSH** is now a write pending signal and **POP** a read pending signal. **NE** and **NF** are ANDed to form the unbuffered queue availability signal to both the read and write EXECP elements involved.

E.2.18.5 diagrammatic form



Buffer word and space counts and status outputs not shown.

E.2.19 REG

E.2.19.1 purpose

A register (static variable).

E.2.19.2 arguments

Parameters	String[] - block ids String - initial value boolean - IOB flag boolean - DSP flag
Inputs	clock signal (may be null if .D null) data signal array (.D - may be null) clock enable signal or signal array (.CE) set/reset signal or signal array (.RES)
Outputs	data signal array

The block IDs are the identifiers for the DFF elements comprising the register, from LSB to MSB. The IOB flag indicates that the DFFs within I/O blocks should be used. This applies when the register input is connected to input buffers or its output is connected to output buffers. The DSP flag indicates that a DSP block may be used to implement the register instead of using CLB flip-flops.

The either or both the clock enable and reset inputs may be single signals in which case they will be connected across all bits.

E.2.19.3 description

When the clock enable is high data will be clocked in on the next clock edge. For a primitive type all clock enable inputs are tied together as are all reset inputs. For a compound type clock enables and resets are grouped in words corresponding to primitives. The resets return the state to the initial value (0 by default). If the clock is null, there are no inputs to this register so generate a constant instead.

E.2.19.4 dump format

REG
OUT glob.v[0..7]

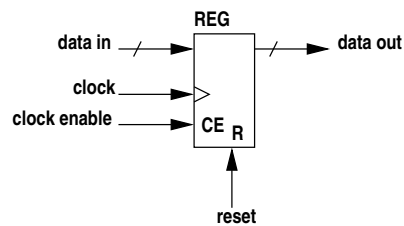
```

<-
CLK  glob.c100
DATA glob.v.D[0..7]
CE   glob.v.CE[0..7]
R    glob.v.RES[0..7]
{0x0}{tnm=reg1}

```

The last line shows the initial value followed by any flags.

E.2.19.5 diagrammatic form



This only shows one word of a multiword register. Although they share a common identifier, the words of a register of compound type (array or struct) are independent. The data input and output will be labelled with the associated subscript range.

E.2.20 RESYNC

E.2.20.1 purpose

A pulse or level resynchroniser.

E.2.20.2 arguments

Parameters	none
Inputs	input signal input clock signal output clock signal
Outputs	output signal

E.2.20.3 description

The input may be a pulse or a level. When the input rises the rise is captured by the next input clock edge and an output pulse of one period duration is generated on the next output clock edge.

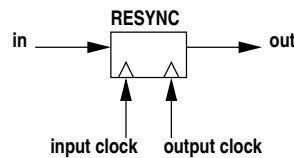
E.2.20.4 dump format

```

RESYNC {
  in:
    glob.a[0]
    glob.c100
    glob.bus_clk
  out:
    E41[0]
}

```

E.2.20.5 diagrammatic form



E.2.21 RRAM

E.2.21.1 purpose

Registered-output RAM.

E.2.21.2 arguments

Parameters	Integer number of ports Integer data width in bits Integer address width in bits String[] optional initialisation String - attribute key String - data for attribute key ... other attribute pairs
Inputs	clock signal, port0 address array, port0 data.in array, or null, port0 read signal, or null, port0 write signal, or null, port0 clock signal, port1 address array, port1 data.in array, or null, port1 read signal, or null, port1 write signal, or null, port1 ..etc..
Outputs	data.out array, or null, port0 data.out array, or null, port1

E.2.21.3 description

This is a RAM with clocked read and write. The output for a port reflects the memory contents following a read clock cycle.

The 2nd parameter gives the data width and the address width gives the memory depth (number of words). All port addresses must be the same width and all port data inputs and outputs must be the same width. The initialisation string array has one entry for each initial value. Each entry is a hexadecimal string. The initialisation array may be smaller than the depth of the RAM but not larger. The initialisation parameter may be omitted in which case the RAM is explicitly initialised to all 0. Ports which are not written have null arguments for the data in and the write enable. Ports which are not read have null arguments for the data out.

For Xilinx FPGAs registered-output RAM is block RAM. The address width is limited to a maximum of 12 bits (a depth of 4096 words) for Spartan 2, Virtex and Virtex E and 14 bits (a depth of 16384 words) for Spartan 3, Virtex 2 and Virtex 2 PRO. The data width is not limited.

Attributes are -

loc	block RAM location
tnm	timing group name
writemode	write mode
writemodea	port A write mode
writemodeb	port B write mode
continuous	continuous enable

There is one **loc** attribute per block. Block RAM

formats are selected by address width, then the number of blocks is determined by the total data width divided by the block data width. Block locations are in order from the least significant data bits. The other attributes are applied to each individual RAMB4_S, RAMB4_S_S, RAMB16_S or RAMB16_S_S prepended in each case with "INST name " where 'name' is the generated netlist identifier for the RAM component.

E.2.21.4 dump format

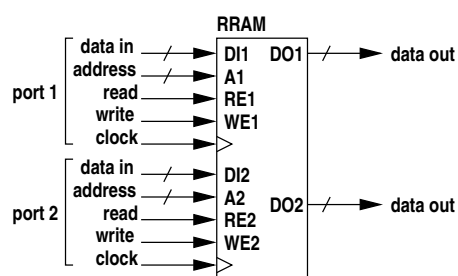
```

RRAM - 2 ports {
    OUT0      kandinski.link_0.if_3372.xc2vp_mgt16_0.crcunblock_0.buf_0[0..31,32]
    OUT1      kandinski.link_0.if_3372.xc2vp_mgt16_0.crcunblock_0.buf_1[0..31,32]
    <-
    CLK0      glob.ruclk
    ADDR0     MADDR1031[0..5]
    DATA0    MDATA1032[0..31,32]
    RE0 -
    WE0 S1057
    CLK1      glob.ruclk
    ADDR1     S38113[0..5]
    DATA1    -
    RE1 S38114
    WE1 -
        initialised
}

```

The presence of an initialisation argument is flagged but the values are not listed.

E.2.21.5 diagrammatic form



E.2.22 SAMPLE

E.2.22.1 purpose

Sample data from a different clock domain.

E.2.22.2 arguments

Parameters	Boolean input clock freq i , output clock freq
Inputs	input data array input clock signal output clock signal
Outputs	output data array

E.2.22.3 description

The input is sampled into the output clock domain. However, sampled data will not be corrupted depending on the relative clock rates data may be missed or multiply sampled. The input may be a different size to the output, zero padding or truncation being applied as required.

E.2.22.4 dump format

```

SAMPLE {
    params:
        true
    in:
        glob.a[0..7]
        glob.c100
        glob.bus_clk
    out:
        E41[0..7]
}
```

E.2.23 SELECT

E.2.23.1 purpose

Select one of several inputs.

E.2.23.2 arguments

Parameters	String - unselected output value boolean - ALU flag boolean - DSP flag
Inputs	select signal signal array select signal signal array . . .
Outputs	signal array

E.2.23.3 description

The inputs are in pairs. The first of each pair is an input signal which selects the second of the pair (a signal array) to be connected to the output signal array. If no select signals are high the output is explicitly zero. The input arrays may be smaller than the output array.

The data inputs may be narrower than the output in which case they will be padded or sign extended as appropriate.

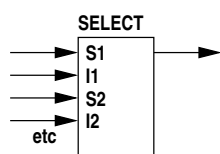
SELECT TDEs with only a single select/data input pair will be optimised away (data input connected to output) unless the output is a queue variable which is not primitive. In that special case the SELECT TDE functions as a gate to ensure that a field or array member of a queue that is not explicitly assigned when the queue as a whole is assigned is assigned zero/false.

E.2.23.4 dump format

```

SELECT {
    par:
    in:
        S11
        glob.a[0..7]
        S15
        glob.b[24..31]
    out:
        glob.c[0..7]
}
```

E.2.23.5 diagrammatic form



E.2.24 SIG

E.2.24.1 purpose

Declare a signal or signal array.

E.2.24.2 arguments

Parameters	none
Inputs	signal or signal array signal or signal array . .
Outputs	none

E.2.24.3 description

There may be any number of inputs. Each input may be a single signal or an array. Declarations are created which will be dimensioned if arrays.

This element is only used by the simulator, not the code generator. It is not generated unless the -s or -T options are passed to the compiler.

E.2.24.4 dump format

	SIG	glob.a
	SIG	E1[1]
	SIG	S11
	SIG	mod1_1.a[8]

E.2.25 SPECIAL

E.2.25.1 purpose

Provides special functions, perhaps vendor specific.

E.2.25.2 arguments

Parameters	Integer type of special function
Inputs	...
Outputs	...

E.2.25.3 description

The first parameter is an integer type code which selects the particular special function. At the moment there are only two special functions.

add line to the NCF or the XDC file

Parameters	Integer 0 or 1 String a format string
Inputs:	signal optional signal . .
Outputs:	none

The format string is written to the ISE NCF file or the Vivado XDC file. The format string may contain %n or %N conversions at which points netlist identifiers are to be inserted. The netlist identifiers are obtained from the input signals which are matched to the conversions in order. There will usually be at most one of these though the format allows any number. %n is replaced by the identifier of the next input signal argument. %N is similar to the above but changes the identifier to upper case and removes any leading "GLOB_" string This is used for generating clock timing group names from the clock signal identifier.

E.2.26 START

E.2.26.1 purpose

Start an execution stream.

E.2.26.2 arguments

Parameters	none
Inputs	start level signal (may be VCC) domain clock signal
Outputs	start pulse signal

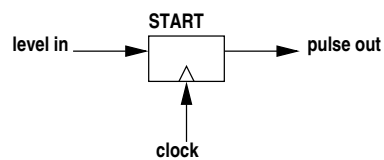
E.2.26.3 description

The start input may be a pulse or a level. When the start signal rises the rise is captured by the next clock edge and an output pulse of one period duration is generated on the next clock edge. The logic is locked after this pulse so that any further start signal rises are ignored. The start level signal may be VCC if execution is to start immediately after configuration loading, otherwise it must be derived combinatorially from external inputs.

E.2.26.4 dump format

```
START {
  OUT  glob.uclk.start
  <-
  IN   VCC
  COUT glob.uclk
}
```

E.2.26.5 diagrammatic form



E.2.27 TSELECT

E.2.27.1 purpose

A 3-state driver.

E.2.27.2 arguments

Parameters	none
Inputs	select signal signal array
Outputs	signal array

E.2.27.3 description

The first of each input is a source signal array to be conditionally connected to the output signal array. The second input is an enable signal. When the enable is high the output is driven by the input. When the enable is low the output is not driven (is high impedance). The input array and output array must be the same size.

E.2.27.4 dump format

```

TSELECT {
    par:
    in:
        glob.a[0..7]
        S11
    out:
        glob.c[0..7]
}

```

E.2.28 WAIT

E.2.28.1 purpose

A parallel block completion wait.

E.2.28.2 arguments

Parameters	none
Inputs	clock signal reset signal or null input signal . . .
Outputs	output signal

E.2.28.3 description

There are four or more inputs. The first input signal is the clock signal for the clock domain. The second input signal is a reset signal or is null if there is no reset. The third and following are finish signals. If all input signals go high simultaneously the output signal also goes high, otherwise the output signal goes high when the last input signal goes high. The output signal is high for one clock cycle. The reset input clears any pending latched completion signals.

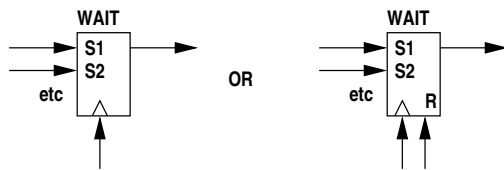
E.2.28.4 dump format

```

WAIT {
    in:
        c100
        null
        S11
        S13
    out:
        S15
}

```

E.2.28.5 diagrammatic form



E.2.29 WHEN

E.2.29.1 purpose

A *when* statement.

E.2.29.2 arguments

Parameters	none
Inputs	start signal test signal true block finish signal or null false block finish signal or null
Outputs	true block start signal false block start signal WHEN finish signal or null

E.2.29.3 description

A start signal will be routed to either the true block start output or the false block start output according to whether the test input signal is high or low. The finish signals for the true and false blocks are returned as inputs and the OR of these emerges as the third output, signalling the end of the statement.

A missing true or false block will have the block start signal connected back to the associated block finish signal via a 1-cycle delay (TDEType.DEL). If both the true block and the false block require a single DEL for their finish signals these will be optimised to a separate common DEL from the start signal to the finish signal. In that case -

- true block finish signal
- false block finish signal
- WHEN finish signal

arguments to the WHEN will all be null.

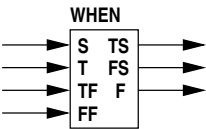
E.2.29.4 dump format

```

WHEN {
  T_START  S307
  F_START  FS270
  FINISH   F268
  <-
  START    S267
  TEST     E274[0]
  T_FINISH F281
  F_FINISH FF312
}

```

E.2.29.5 diagrammatic form



E.2.30 WHILE

E.2.30.1 purpose

A *while* statement.

E.2.30.2 arguments

Parameters	none
Inputs	start signal test signal continuation signal from the body clock signal optional reset signal or null
Outputs	start signal for the body WHILE finish signal

E.2.30.3 description

If the test input signal is high when start input signal goes high, the latter is routed to the body start output signal to execute the contained code block. If the test input signal is low when start input signal goes high, the latter is delayed by one clock cycle and then appears on the finish output signal (i.e. a *while* takes a minimum of one clock cycle). When the body code block finishes the last execution signal comes back to the continuation input signal. If the test input signal is high the continuation input signal is routed to the body start output signal to again execute the contained code block. If the test input signal is low the continuation input signal is routed through to the finish output signal, again with one clock cycle delay.

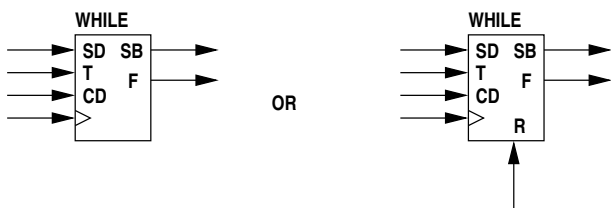
The start signal and the continuation signal may come from EXEC_P elements which wait on queue availability in the test value.

The optional reset signal clears internal state in the WHILE.

E.2.30.4 dump format

WHILE {	
START_B	S505
FINISH	S516
<-	
START	S502
TEST	E504[0]
CONTIN	F507
C	glob.ruclk
RESET	null
}	

E.2.30.5 diagrammatic form



E.2.31 XOR

E.2.31.1 purpose

An XOR gate for control signals.

E.2.31.2 arguments

Parameters	none
Inputs	input signal input signal . .
Outputs	AND signal

E.2.31.3 description

Exclusive OR two or more input signals, the result appearing on the output.

E.2.31.4 dump format

$$S11 = A12 \wedge A14 \wedge S8$$

E.2.31.5 diagrammatic form



E.3 Intermediate code optimisation

The intermediate code list is optimised by repeated applications of the following -

- Combine any **EXEC**s with a common start and identical avail signal.
- Combine any parallel **DEL**s with a common start driving a common **WAIT**.
- Eliminate any **DEL** in parallel with one or more **EXEC**s with a common start driving a common **WAIT**.
- Eliminate any **WAIT** that has a single input.
- Eliminate any duplicate **SELECT** inputs.

- Find any AND or OR gates which have duplicated inputs and remove the duplicates. If this results in a single input remaining, eliminate the gate and connect the input to the output.
- Find any **INVs** with GND or VCC input and eliminate them, connecting the output to the inverse of the constant input.
- Find any nested **AND**, **OR** or **XOR** gates and consolidate them.
- Find any control (not source code) **AND** or **OR** gates whose outputs are not connected and eliminate them.
- Find any **DELs** whose outputs are not connected and eliminate them.
- Find any **WHENs** which have a single **DEL** in both the true block and the false block. Remove these and instead connect a single **DEL** from the start input to both the true block finish input and the false block finish input. This will result in an OR with identical inputs which will be eliminated by another optimisation iteration.
- Eliminate any **ILOOPs** which have the same output start signal and input finish signal. These infinite loops are empty, probably because they contain only value equivalence statements.
- Find any **ILOOPs** which have a single **DEL** from the output start signal to the input finish signal. Replace the pair with a DFF whose SET input is driven by the start signal.
- Find multiple **DFFs** driven by a common **START** and eliminate all but one.
- Find any branch statements in which all branches, including the default, have a single **DEL**. Remove all the **DELs** and the final associated OR and replace with a single **DEL** from the start signal.

These optimisations are repeatedly applied to the list until a pass makes no change.

E.4 Intermediate code examples

Note that in the following examples the intermediate code dumps and the associated diagrams only show sections of the logic that illustrate the points under discussion - they are not complete listings or diagrams.

In many cases clock inputs are not shown. The examples are constructed to use clock **c100** and this can be assumed unless shown otherwise.

In all the examples the signal **glob.c100.start** appears. This is a single cycle pulse, synchronised to the **c100** 100MHz clock, which is distributed to all execution threads to initiate them. In practice a recent change to the compiler has inserted a single cycle delay between this signal and every thread that it initiates but this is not shown in the examples. The purpose of the extra cycle is simply to avoid the path delays associated with distributing **glob.c100.start** which can have a very high fanout. The inserted flip-flop in the path to each thread will be placed close to the logic that it initiates resulting in a much shorter path into the thread. **glob.c100.start** will still have a high fanout but there is no combinatorial logic between its source and the myriad destination flip-flops.

E.4.1 top level sequential block

The code -

```
module "csiro/hymod/io1.3pl"

useclock(c100);

static({s1, s2}, "uint:8");

seq {
    s1 <- s2;
    s2 <- s1;
}
```

contains one execution thread. That thread will be an infinite loop containing two executable statements. The assignments do not wait on queues. The intermediate code would look like -

```
DEL F14 <- (1) glob.c100 S13
DEL F17 <- (1) glob.c100 F14
ILOOP      S13 <- glob.c100.start F17
```

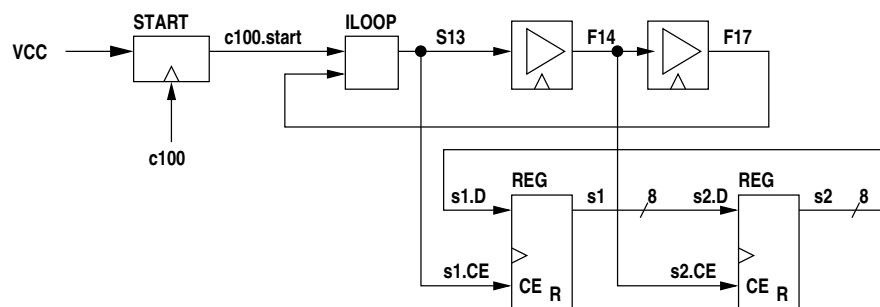
```

START      glob.c100.start <- START=VCC, CLK=glob.c100
glob.s1.CE[0..7] = S13
glob.s1.D[0..7] = glob.s2[0..7]
glob.s1.RES[0..7] = GND
REG
  OUT glob.s1[0..7]
  <-
  CLK glob.c100
  DATA glob.s1.D[0..7]
  CE glob.s1.CE[0..7]
  R glob.s1.RES[0..7]
  {0x00}
glob.s2.CE[0..7] = F14
glob.s2.D[0..7] = glob.s1[0..7]
glob.s2.RES[0..7] = GND
REG
  OUT glob.s2[0..7]
  <-
  CLK glob.c100
  DATA glob.s2.D[0..7]
  CE glob.s2.CE[0..7]
  R glob.s2.RES[0..7]
  {0x00}

```

The multiple versions of value variable **start** (created in header code `csiro/hymod/io1.3pl`) are a result of the internal intricacies of the compiler. The last value the variable contains is assigned to the unappended identifier **glob.start** in an attempt to provide a useful value in simulation (no longer enabled).

Diagrammatically this is -



The thread is started from signal **VCC** so element **START** generates a single pulse using clock **c100**. This is injected into **ILOOP** to start the infinite loop. The execution signal for the body of the infinite loop is **S13**. The loop body is a **seq** block containing two statements. The first is executed by **S13** and a finish signal is generated one clock cycle later (**F14**). This finish signal is the start signal for the next statement. A finish signal is generated one clock cycle later for the second statement (**F17**). This second finish signal marks the completion of the loop body code so it is fed back to the **ILOOP** to generate the next iteration. There is no delay in **ILOOP** (it is simply an OR gate) so the loop executes every two clock cycles.

The two execute signals, **S13** and **F14** control the assignments to the two static variables **s1** and **s2** via their CE inputs.

E.4.2 top level static assignment

If we simplify the above code so that the infinite loop contains only a single assignment -

```

static({s1, s2}, "uint:8");

s1 <- s2;

```

then the intermediate code is -

```

START      glob.c100.start <- START=VCC, CLK=glob.c100

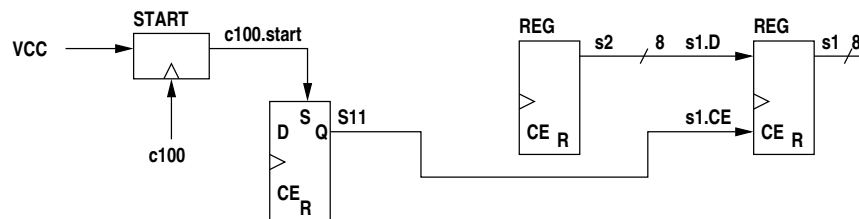
```

```

glob.s1.CE[0..7] = S11
glob.s1.D[0..7] = glob.s2[0..7]
glob.s1.RES[0..7] = GND
REG
  OUT  glob.s1[0..7]
  <-
  CLK  glob.c100
  DATA glob.s1.D[0..7]
  CE   glob.s1.CE[0..7]
  R    glob.s1.RES[0..7]
  {0x00}
glob.s2.RES[0..7] = GND
REG
  OUT  glob.s2[0..7]
  <-
  CLK  null
  DATA null
  CE   null
  R    glob.s2.RES[0..7]
  {0x00} make const!
DFF S11 <- C=glob.c100, S=glob.c100.start

```

which diagrammatically is -



Since the **ILOOP** contains a single statement which unconditionally takes one clock cycle, i.e. it executes on every clock cycle, the loop can be eliminated and instead the destination register clock enable can be asserted when execution starts and remain asserted from then on. This is done by a D-flip-flop which is set by the start signal. The output of the flip-flop provides an execute signal for every clock cycle.

If the assignment is conditional -

```

static({s1, s2}, "uint:8");

when (s1 != s2)
  s1 <- s2;

```

then the intermediate code becomes -

```

E8[0] = glob.s1[0..7] != glob.s2[0..7]      (uint, uint, log)
WHEN {
  START    SD7
  TEST     E8[0]
  T_FINISH null
  F_FINISH null
  T_START  S17
  F_START  FS6
  FINISH   null
}
glob._sdreset_1[0] = VCC
glob._sdreset_2[0] = VCC
START glob.c100.start <- START=VCC, CIN=glob.c100, COUT=glob.c100
glob._sdreset[0] = glob._sdreset_2[0]
glob.s1.CE[0..7] = S17
glob.s1.D[0..7] = glob.s2[0..7]
glob.s1.RES[0..7] = GND
REG
  OUT  glob.s1[0..7]
  <-
  CLK  glob.c100
  DATA glob.s1.D[0..7]
  CE   glob.s1.CE[0..7]
  R    glob.s1.RES[0..7]

```

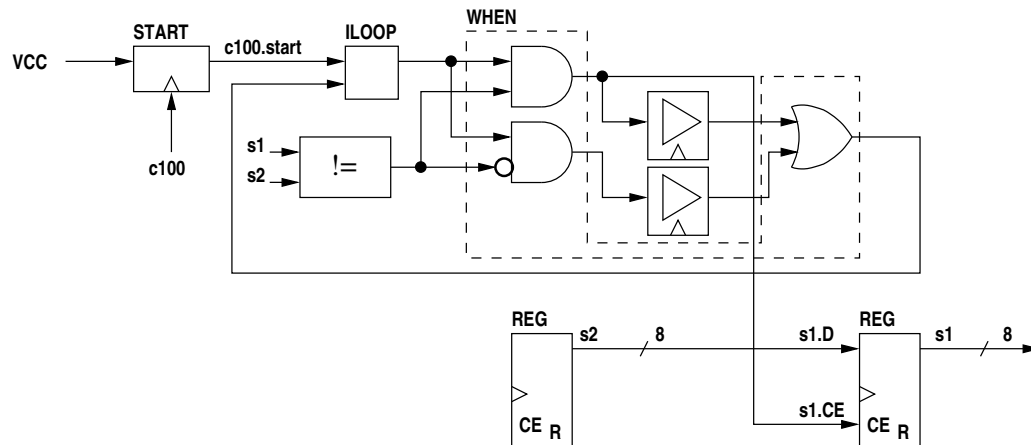


```

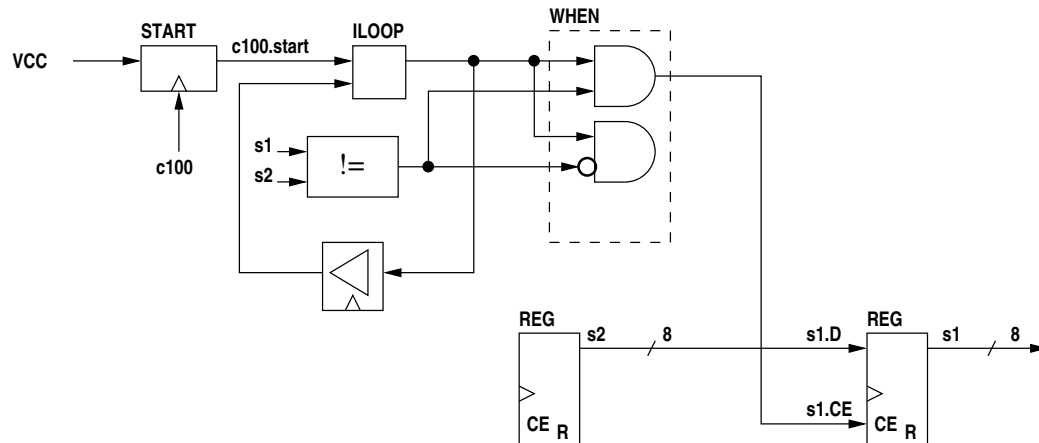
{0x00}
glob.s2.RES[0..7] = GND
REG
OUT  glob.s2[0..7]
<-
CLK  null
DATA null
CE   null
R    glob.s2.RES[0..7]
{0x00} make const!
DFF SD7 <- C=glob.c100, S=glob.c100.start

```

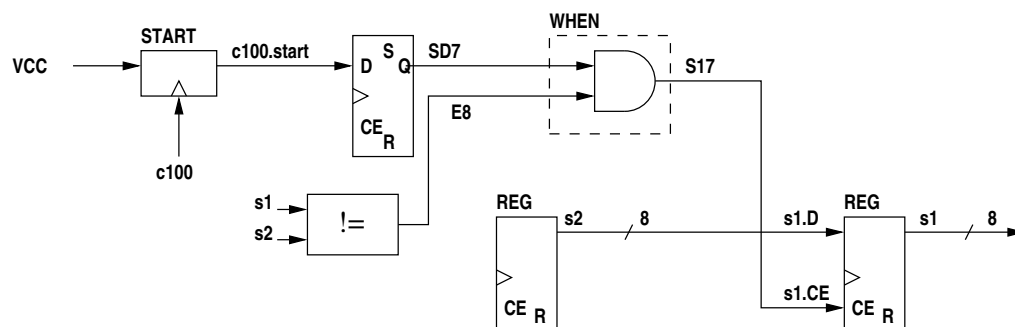
Now the unoptimised logic for the above code would be -



Since both the true and false bodies of the **WHEN** execute in a single cycle (the unused false statement still requires a single cycle delay) this will be optimised to a single shared delay -



The false side of the **WHEN** is now unused so its AND gate will be omitted. The **ILOOP** with a single delay feedback will be replaced with a set-once flip-flop -



Note that, like the simple assignment statement in the previous example, this logic has no execution thread any more. It is simply a register whose clock enable is asserted in clock cycles where the expression controlling the **WHEN** is true.

If the **WHEN** false code block contained a single cycle statement rather than being omitted then the second AND gate would be retained but otherwise the logic would be the same.

These examples show that top level assignments to static variables, even when conditional, have minimal logic overheads.

E.4.3 top level static assignment with queue reads

If the RHS of the assignment contains one or more queue reads then we must use an **EXEC**. For example -

```
static(s, "uint:8");
queue({q1, q2}, "uint:8");

s <- q1 + q2;
```

the intermediate code is -

```
START      glob.c100.start <- START=VCC, CLK=glob.c100
E14[0..8] = glob.q1[0..7] + glob.q2[0..7]    (uint, uint, uint)
S15[0..7] = cast(false){E14[0..8]}
AV16 = AV18[0] AND AV19[0]
ACK20[0] = S43
ACK22[0] = S43
EXEC      S43 <- C=glob.c100, EXEC=S11, AV16
DEL F13 <- (1) glob.c100 S43
ILOOP      S11 <- glob.c100.start F13
glob.s.CE[0..7] = S43
glob.s.D[0..7] = S15[0..7]
glob.s.RES[0..7] = GND
REG
    OUT  glob.s[0..7]
    <-
    CLK  glob.c100
    DATA glob.s.D[0..7]
    CE   glob.s.CE[0..7]
    R    glob.s.RES[0..7]
    {0x00}
glob.q1.RES = GND
QUEUEBUFFER (2)
    OUT  glob.q1[0..7]
    NE   glob.q1.NE[0]
    NF   glob.q1.NF[0]
    CNT  null
    SPC  null
    WSTAT null
    RSTAT null
    <-
    CLKOUT glob.c100
    DATA  glob.q1.D[0..7]
    PUSH   glob.q1.PUSH
    POP    glob.q1.POP
    R      glob.q1.RES
glob.q1.POP = ACK22[0]
AV19[0] = glob.q1.NE[0]
glob.q2.RES = GND
QUEUEBUFFER (2)
    OUT  glob.q2[0..7]
    NE   glob.q2.NE[0]
    NF   glob.q2.NF[0]
    CNT  null
    SPC  null
    WSTAT null
    RSTAT null
    <-
```

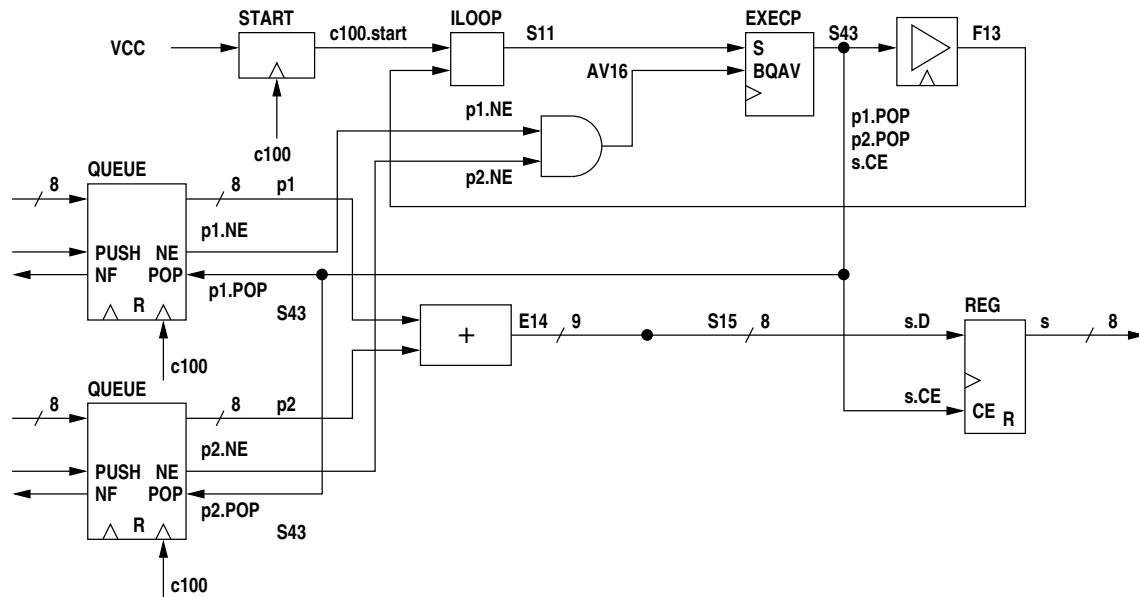
```

CLKOUT glob.c100
DATA   glob.q2.D[0..7]
PUSH   glob.q2.PUSH
POP     glob.q2.POP
R       glob.q2.RES
glob.q2.POP = ACK20[0]
AV18[0] = glob.q2.NE[0]

```

glob.q1.PUSH, **glob.q1.NF**, **glob.q2.PUSH** and **glob.q2.NF** are not connected here as they are used for assignments to the queues. Note that the code above would strictly give compiler errors since the queues have no sources (are not written).

The following shows the logic diagrammatically -



At the start **c100.start** is a single cycle pulse which will be passed through **ILOOP** to **S11**. If **AV16** is high (queues **q1** and **q2** are available for reading) then the pulse will in turn be passed as **S43**, which executes the assignment to static **s**. If **AV16** is low the input pulse **S11** is stored until **AV16** goes high when **S43** is finally driven by a single cycle pulse. The assignment executes in one cycle so a finish signal **F13** is generated by a one cycle delay from **S43**. The finish signal connects back to the **ILOOP** to cause the next iteration. If **AV16** remains high the assignment to **s** will occur on every clock cycle, **S11**, **S43**, and **F13** remaining high. **S43** executes the assignment by raising the clock enable to static register **s** (**s.CE**). At the same time it acknowledges queues **q1** and **q2** since **q1.POP** and **q2.POP** are connected to **S43**. If either queue becomes unavailable for reading (empty) then **q1.NE** or **q2.NE** will go low and hence **AV16** will go low, causing the **EXEC** to wait.

Note that the input to static register **s** is 8 bits wide but the sum of the outputs of queues **q1** and **q2** is 9 bits wide. A `cast()` element from **E14** to **S15** will truncate the sum, ignoring the MSB.

E.4.4 top level queue assignment

If we change the above code to a queue assignment -

```

static(s, "uint:8");
queue({q1, q2, q3}, "uint:8");

q3 <<- q1 + q2;

```

the intermediate code is -

```

E14[0..8] = glob.q1[0..7] + glob.q2[0..7]    (uint, uint, uint)
S15[0..7] = cast(false){E14[0..8]}

```

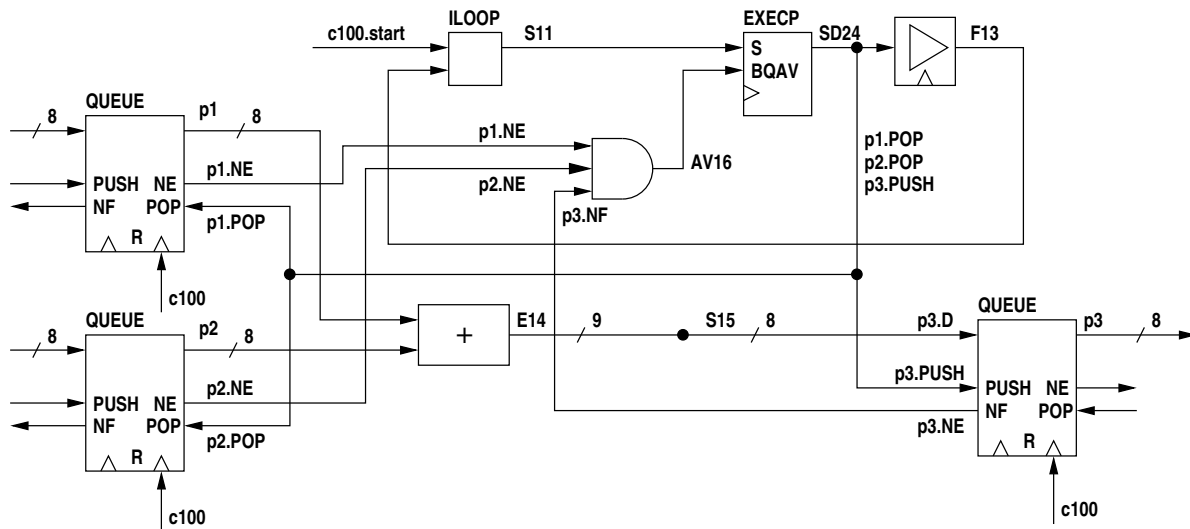
```

AV18[0] = glob.q2.NE[0]
AV19[0] = glob.q1.NE[0]
AV16 = glob.q3.NF[0] AND AV18[0] AND AV19[0]
EXEC    SD24 <- C=glob.c100, EXEC=S11, AV16
glob.q3.PUSH = SD24
ACK20[0] = SD24
glob.q2.POP = ACK20[0]
ACK22[0] = SD24
glob.q1.POP = ACK22[0]
DEL F13 <- (1) glob.c100 SD24
ILOOP    S11 <- glob.c100.start F13
glob.q1.RES = GND
QUEUEBUFFER (2)
    OUT    glob.q1[0..7]
    NE     glob.q1.NE[0]
    NF     glob.q1.NF[0]
    CNT    null
    SPC    null
    WSTAT  null
    RSTAT  null
    <-
    CLKOUT glob.c100
    DATA  glob.q1.D[0..7]
    PUSH   glob.q1.PUSH
    POP    glob.q1.POP
    R      glob.q1.RES

glob.q2.RES = GND
QUEUEBUFFER (2)
    OUT    glob.q2[0..7]
    NE     glob.q2.NE[0]
    NF     glob.q2.NF[0]
    CNT    null
    SPC    null
    WSTAT  null
    RSTAT  null
    <-
    CLKOUT glob.c100
    DATA  glob.q2.D[0..7]
    PUSH   glob.q2.PUSH
    POP    glob.q2.POP
    R      glob.q2.RES

glob.q3.D[0..7] = S15[0..7]
glob.q3.RES = GND
QUEUEBUFFER (2)
    OUT    glob.q3[0..7]
    NE     glob.q3.NE[0]
    NF     glob.q3.NF[0]
    CNT    null
    SPC    null
    WSTAT  null
    RSTAT  null
    <-
    CLKOUT glob.c100
    DATA  glob.q3.D[0..7]
    PUSH   glob.q3.PUSH
    POP    glob.q3.POP
    R      glob.q3.RES

```



The assignment statement execution signal is now **SD24**. The **EXEC** now waits on **q3.NF**, **q1.NE** and **q2.NE**. The assignment statement execution signal now drives **q3.PUSH** to write to queue **q3**.

E.4.5 multiple assignments to a variable

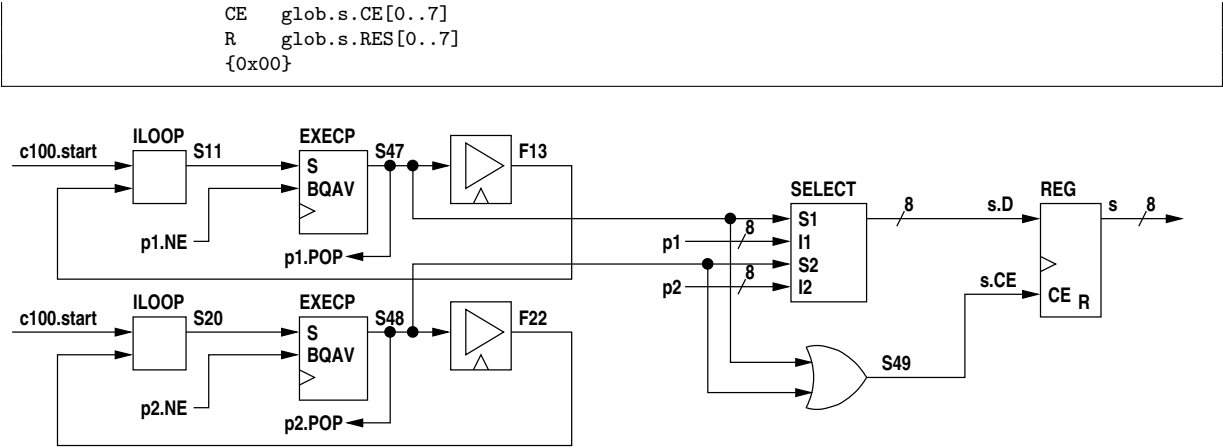
The following code has two threads, each an assignment to the same static variable **s**. We assume here for the code to make sense that queues **q1** and **q2** are never available for reading simultaneously.

```
static(s, "uint:8");
queue({q1, q2}, "uint:8");

s <- q1;
s <- q2;
```

the intermediate code is -

```
AV16[0] = glob.q1.NE[0]
AV14 = AV16[0]
EXEC    S47 <- C=glob.c100, EXEC=S11, AV14
ACK17[0] = S47
glob.q1.POP = ACK17[0]
DEL F13 <- (1) glob.c100 S47
ILOOP   S11 <- glob.c100.start F13
AV25[0] = glob.q2.NE[0]
AV23 = AV25[0]
EXEC    S48 <- C=glob.c100, EXEC=S20, AV23
ACK26[0] = S48
glob.q2.POP = ACK26[0]
DEL F22 <- (1) glob.c100 S48
ILOOP   S20 <- glob.c100.start F22
S49 = S48 OR S47
glob.s.CE[0..7] = S49
SELECT {
  in:
    Var S47
    Var glob.q1[0..7]
    Var S48
    Var glob.q2[0..7]
  out:
    Var glob.s.D[0..7]
}
glob.s.RES[0..7] = GND
REG
OUT glob.s[0..7]
<-
CLK glob.c100
DATA glob.s.D[0..7]
```



Each thread is an **ILOOP** with an **EXECP** to wait for queue availability and a 1-cycle delay. If the first assignment statement executes **A18** goes high. If the second assignment statement executes **S21** goes high. These are ORed to drive the clock enable of static variable **s** to carry out the assignment. The input data source is selected by the **SELECT** block with **A18** or **S21** selecting the appropriate data source (clearly it is a programming error if the assignments are not mutually exclusive).

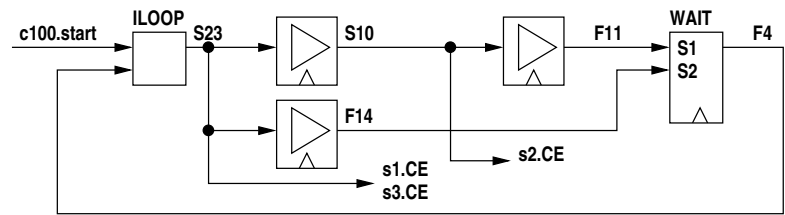
E.4.6 general parallel block

A PAR block is implemented by using the same start signal for all statements in the block and waiting for all their finish signals using a WAIT element. For example the code -

```
static({s1, s2, s3}, "uint:8");  
  
par {  
  seq {  
    s1 <- 3;  
    s2 <- 4;  
  }  
  s3 <- 5;  
}
```

generates the intermediate code -

```
DEL S10 <- (1) glob.c100 S23  
DEL F11 <- (1) glob.c100 S10  
DEL F14 <- (1) glob.c100 S23  
WAIT {  
  in:  
    Var glob.c100  
    Var F11  
    Var F14  
  out:  
    Var F4  
}  
ILOOP      S23 <- glob.c100.start F4
```



Execution signal **S23** executes the assignments to **s1** and **s2** and Execution signal **S10** executes the assignment to **s3**.

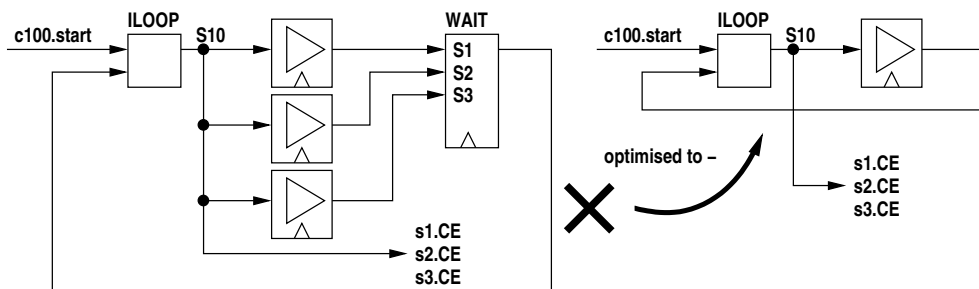
E.4.7 minimal parallel block

If the statements in a PAR block all take one clock cycle and have no queue reads the compiler will eliminate parallel **DEL** elements. For example the code -

```
static({s1, s2, s3}, "uint:8");

par {
  s1 <- 3;
  s2 <- 4;
  s3 <- 5;
}
```

does not generate parallel delays, e.g. -



where **S10** executes the three assignments.

E.4.8 general when statement

A **WHEN** element directs an execution signal out through one of two outputs, the **TRUE** output or the **FALSE** output. These output signals execute the associated code block. The finish signals from the two code blocks must be Ored (they are mutually exclusive) to get the finish signal for the **WHEN**. This **OR** is contained in the **WHEN** element. A missing code block is replaced by a single cycle delay, hence the minimum time to execute a **WHEN** is one clock cycle. Note that if this minimum delay were not enforced we would have a combinatorial loop in this and other similar cases.

In the following example -

- the test contains queue reads
- the FALSE block is missing

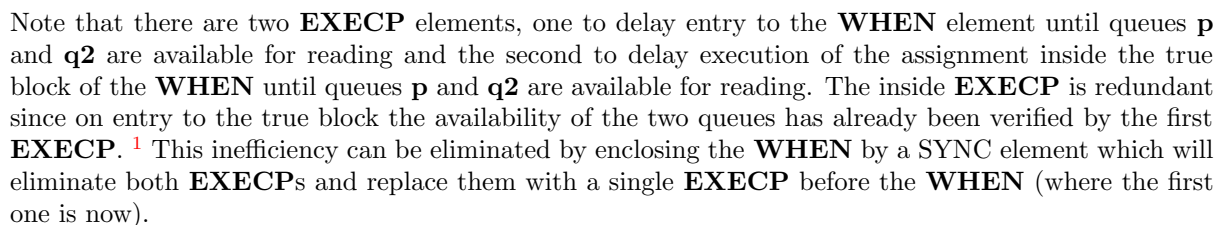
```
static(s, "uint:8");
queue({q1, q2}, "uint:8");
value(v, "uint:8", q1 + q2);

when (v == 0)
  s <- v;
```

the intermediate code is -

```
E3[0..8] = glob.q1[0..7] + glob.q2[0..7]    (uint, uint, uint)
S4[0..7] = cast(false){E3[0..8]}
E10[0] = S4[0..7] == 0    (uint, uint, log)
AV17[0] = glob.q1.NE[0]
AV16[0] = glob.q2.NE[0]
AV14 = AV16[0] AND AV17[0]
ACK18[0] = A19
ACK20[0] = A19
EXECVP    A19 <- C=glob.c100, EXEC=S11, AV14
DEL F13 <- (1) glob.c100 A19
DEL FF23 <- (1) glob.c100 FS8
ACK24[0] = A25
ACK26[0] = A25
EXECVP    A25 <- C=glob.c100, EXEC=S5, AV14
WHEN {
```

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E.4.9 minimal when statement

```
static({s1, s2}, "uint:8");
static(l, "log");

seq {
  nop();
  when (l) par{
    s1 <- 3;
    s2 <- 4;
  }
}
```

¹The compiler should omit queue availability testing inside the **WHEN** for queues already tested on entry to the **WHEN**, but when expressions containing queue reads are complicated this turns out to be surprisingly difficult.


```

    }
}

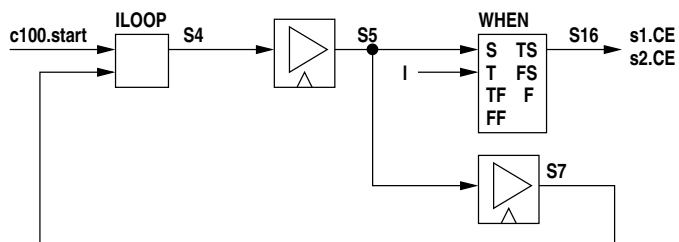
```

results in the intermediate code -

```

DEL S5 <- (1) glob.c100 S4
DEL F7 <- (1) glob.c100 S5
WHEN {
    START          S5
    TEST           glob.l[0]
    T_FINISH      null
    F_FINISH      null
    T_START        S16
    F_START        FS9
    FINISH         null
}
ILOOP          S4 <- glob.c100.start F7

```



Notes -

- the false start signal **FS9** goes nowhere
- the `nop()` generates the first **DEL** and has been included simply to stop the logic collapsing even further to a single D-flip-flop set by **c100.start** - see below
- **S16** is the execution signal for both assignments in the true code block of the **WHEN**

To expand on the comment about the `nop()`, if we simplify the code to -

```

static({s1, s2}, "uint:8");
static(1, "log");

when (1) par{
    s1 <- 3;
    s2 <- 4;
}

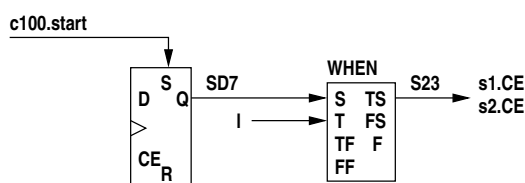
```

the intermediate code becomes -

```

WHEN {
    START          SD7
    TEST           glob.l[0]
    T_FINISH      null
    F_FINISH      null
    T_START        S23
    F_START        FS6
    FINISH         null
}
DFF SD7 <- C=glob.c100, S=glob.c100.start

```



The **ILOOP** and any **DELs** have disappeared. **SD7** goes high after **c100.start** and remains high thereafter. **s23** goes high whenever the **WHEN** test is true and causes the two assignments to occur.

E.4.10 simple target while statement

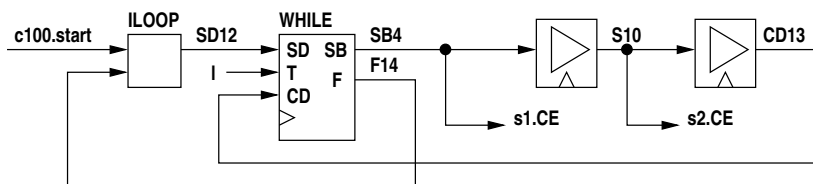
The following example is a target while statement where the loop test does not contain queue reads -

```
static({s1, s2}, "uint:8");
static(1, "log");

seq while (1) {
  s1 <- 5;
  s2 <- 6;
}
```

the intermediate code is -

```
DEL S10 <- (1) glob.c100 SB4
DEL CD13 <- (1) glob.c100 S10
WHILE {
  START_B      SB4
  FINISH       F14
  <-
  START        SD12
  TEST         glob.l[0]
  CONTIN       CD13
  C            glob.c100
}
ILOOP         SD12 <- glob.c100.start F14
```



The **WHILE** element has two entry points, **START_DEL** is the initial entry point and **CONTIN_DEL** is the entry point for further iterations. **START_B** is the start signal for the contained code block and **FINISH** is the finish signal from the **WHILE**.

The signal **SB4** executes the first assignment statement and **S10** executes the second. The loop takes two clock cycles per iteration with no extra overheads, but if on initial entry the test is false the **WHILE** has an internal delay of one clock cycle between the start input (**SD12** into F) and the finish signal (**F14** at F). This ensures that the **WHILE** takes a minimum of one clock cycle.

E.4.11 target while statement with queue waits

If the **WHILE** test expression contains queue reads an **EXECP** appears in both the **START_DEL** and **CONTIN_DEL** inputs. If the loop body contains queue reads then **EXECP** elements will appear there as well. The following example has both.

```
static(s, "uint:8");
queue({q1, q2, q3, q4, q5, q6}, "uint:8");
static(1, "log");

par while (q1 || q2) {
  s <- q3;
  q4 <- q5 + q6;
}
```

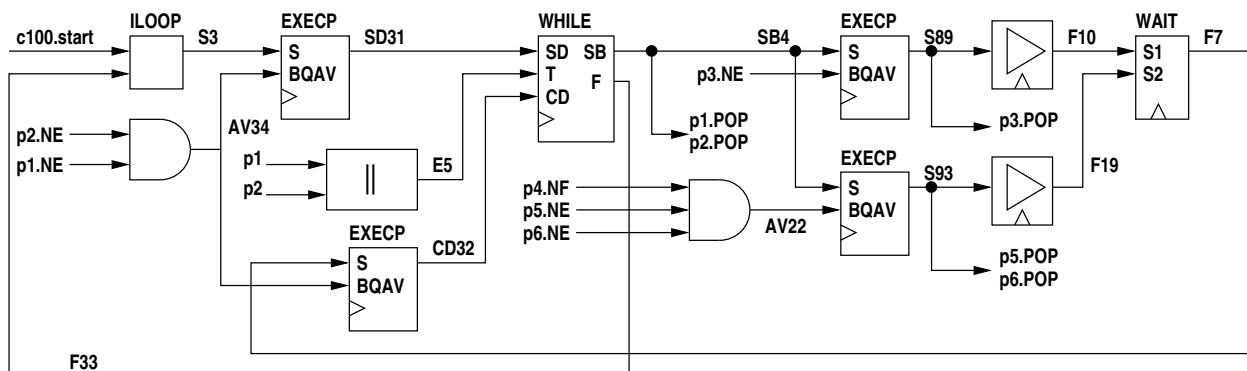
The intermediate code is -

```
E5[0] = glob.q1[0] OR glob.q2[0]
AV13[0] = glob.q3.NE[0]
AV11 = AV13[0]
EXECP      S89 <- C=glob.c100, EXEC=SB4, AV11
```

```

ACK14[0] = S89
glob.q3.POP = ACK14[0]
DEL F10 <- (1) glob.c100 S89
E20[0..8] = glob.q5[0..7] + glob.q6[0..7]    (uint, uint, uint)
S21[0..7] = cast(false){E20[0..8]}
AV24[0] = glob.q5.NE[0]
AV25[0] = glob.q6.NE[0]
AV22 = glob.q4.NF[0] AND AV24[0] AND AV25[0]
EXECVP      S93 <- C=glob.c100, EXEC=SB4, AV22
ACK26[0] = S93
ACK28[0] = S93
glob.q5.POP = ACK26[0]
glob.q6.POP = ACK28[0]
DEL F19 <- (1) glob.c100 S93
WAIT {
  in:
    Var glob.c100
    Var F10
    Var F19
  out:
    Var F7
}
WHILE {
  START_B      SB4
  FINISH       F33
  <-
  START        SD31
  TEST         E5[0]
  CONTIN       CD32
  C            glob.c100
}
ACK38[0] = SB4
ACK40[0] = SB4
glob.q2.POP = ACK38[0]
glob.q1.POP = ACK40[0]
AV37[0] = glob.q1.NE[0]
AV36[0] = glob.q2.NE[0]
AV34 = AV36[0] AND AV37[0]
EXECVP      SD31 <- C=glob.c100, EXEC=S3, AV34
EXECVP      CD32 <- C=glob.c100, EXEC=F7, AV34
ILOOP      S3 <- glob.c100.start F33

```



The variables have not been shown. **S89** is the execution signal for the static assignment and **S93** is the execution signal for the queue assignment.

E.4.12 simple target do-while statement

The following code illustrates a target **DO-WHILE** loop with no queue dependencies -

```

static({s1, s2}, "uint:8");
static(1, "log");

par do {
  s1 <- 4;

```

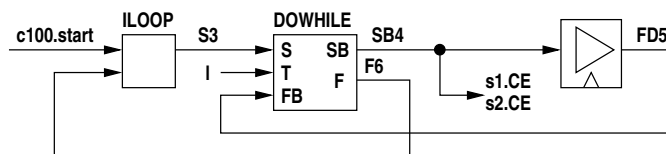
```
    s2 <- 5;
} while (1);
```

The intermediate code is -

```

DEL FD5 <- (1) glob.c100 SB4
DO WHILE {
    START_B SB4
    FINISH F6
    <-
    START S3
    TEST glob.l[0]
    FINISH_B_DEL FD5
}
ILOOP S3 <- glob.c100.start F6

```



Note that since the two parallel assignment statements take a single cycle a single delay element is used to time them and single signal **SB4** executes both. No parallel **WAIT** is necessary.

E.4.13 target do-while statement with queue waits

If we take the above example and introduce queue dependencies in both the test expression and one of the body statements -

```
static(s, "uint:8");
queue(q1, "log");
queue({q2, q3}, "uint:8");

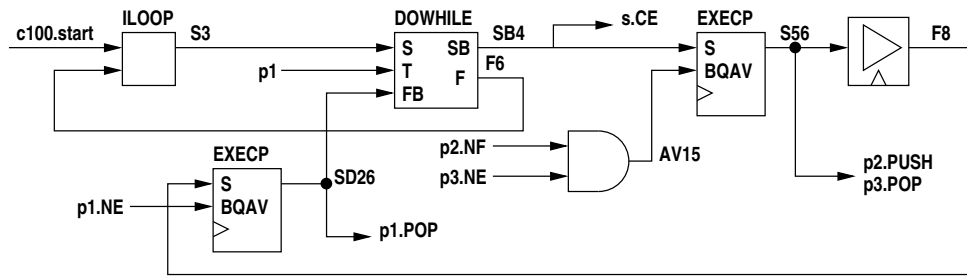
par do {
  s <- 4;
  q2 <<- q3;
} while (q1);
```

the intermediate code is -

```

AV17[0] = glob.q3.NE[0]
AV15 = glob.q2.NF[0] AND AV17[0]
EXEC      S56 <- C=glob.c100, EXEC=SB4, AV15
ACK18[0] = S56
glob.q3.POP = ACK18[0]
glob.q2.PUSH = S56
DEL F8 <- (1) glob.c100 S56
AV23[0] = glob.q1.NE[0]
AV21 = AV23[0]
EXEC      SD26 <- C=glob.c100, EXEC=F8, AV21
DO WHILE {
    START_B          SB4
    FINISH            F6
    <-
    START              S3
    TEST               glob.q1[0]
    FINISH_B_DEL      SD26
}
ACK24[0] = SD26
glob.q1.POP = ACK24[0]
glob.s.CE[0..7] = SB4
ILOOP      S3 <- glob.c100.start F6

```



The parallel body of the loop has one assignment with queue dependencies and one without. The latter takes a fixed single cycle so it is executed using the block execute signal **SB4** whereas the queue assignment has an **EXECP** before being executed by **S56**. Since the queue assignment will take at least one cycle the delay for the static assignment and the parallel **WAIT** have been omitted by the compiler.

Note that the **FINISH_BLOCK** input to the **DO_WHILE** is via an **EXECP** since the block iteration test contains a queue read.

E.4.14 multiple queue reads within a module

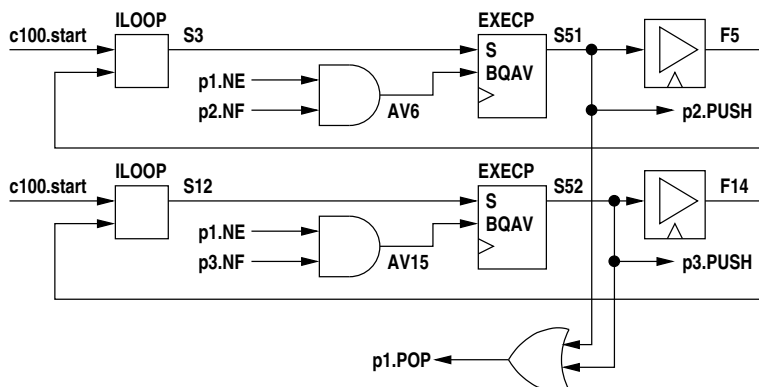
Multiple queue reads read the same datum if simultaneous or separate data otherwise. The queue acknowledges are ORed. The following code illustrates this -

```
queue({q1, q2, q3}, "uint:8");

q2 <<- q1;
q3 <<- q1;
```

The intermediate code is -

```
AV8[0] = glob.q1.NE[0]
AV6 = glob.q2.NF[0] AND AV8[0]
EXECP    S51 <- C=glob.c100, EXEC=S3, AV6
ACK9[0] = S51
DEL F5 <- (1) glob.c100 S51
ILOOP    S3 <- glob.c100.start F5
AV15 = glob.q3.NF[0] AND AV8[0]
EXECP    S52 <- C=glob.c100, EXEC=S12, AV15
ACK17[0] = S52
DEL F14 <- (1) glob.c100 S52
ILOOP    S12 <- glob.c100.start F14
glob.q1.POP = ACK17[0] OR ACK9[0]
glob.q2.D[0..7] = glob.q1[0..7]
glob.q2.PUSH = S51
glob.q3.D[0..7] = glob.q1[0..7]
glob.q3.PUSH = S52
```



E.4.15 multiple queue reads in different modules

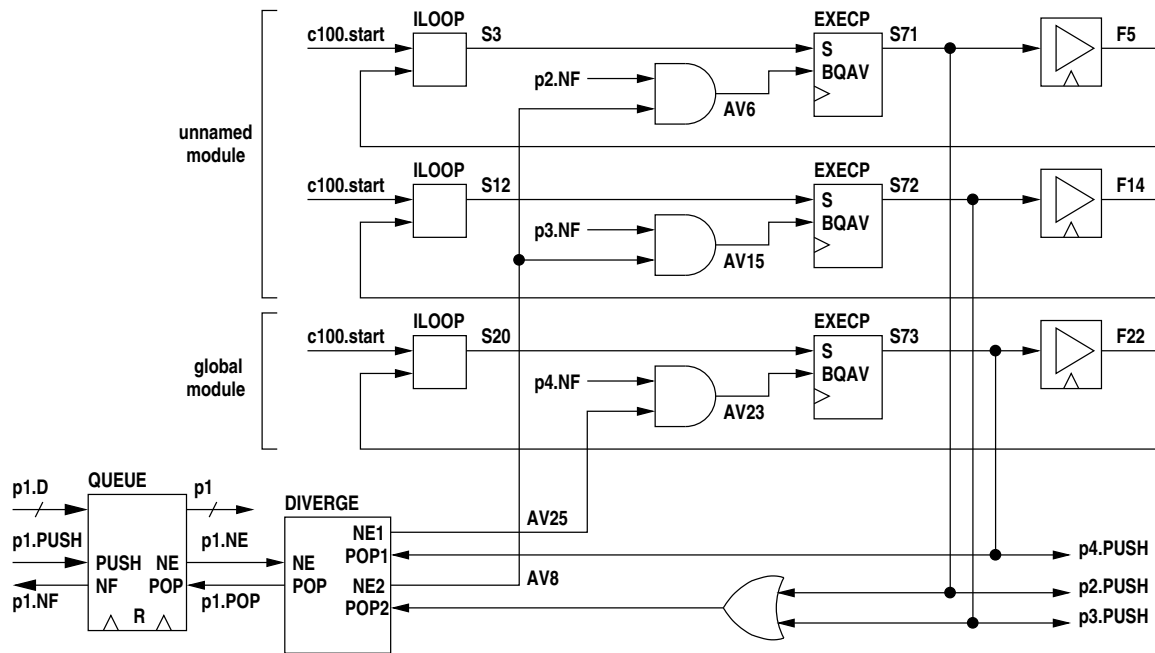
Where multiple queue reads are in different modules each module must execute a read before the queue is acknowledged, this occurring simultaneously with the last read. Multiple reads within a module are again ORed.

```
queue({q1, q2, q3, q4}, "uint:8");

[
  q2 <<- q1;
  q3 <<- q1;
]
q4 <<- q1;
```

Note that the first two assignments are in an inline module and the third assignment is in the file module. The intermediate code is -

```
AV6 = glob.q2.NF[0] AND AV8[0]
ACK9[0] = S71
EXECVP      S71 <- C=glob.c100, EXEC=S3, AV6
DEL F5 <- (1) glob.c100 S71
ILOOP      S3 <- glob.c100.start F5
AV15 = glob.q3.NF[0] AND AV8[0]
ACK17[0] = S72
EXECVP      S72 <- C=glob.c100, EXEC=S12, AV15
DEL F14 <- (1) glob.c100 S72
ILOOP      S12 <- glob.c100.start F14
AV23 = glob.q4.NF[0] AND AV25[0]
ACK26[0] = S73
EXECVP      S73 <- C=glob.c100, EXEC=S20, AV23
DEL F22 <- (1) glob.c100 S73
ILOOP      S20 <- glob.c100.start F22
S69 = ACK26[0]
S70 = ACK17[0] OR ACK9[0]
DIVERGE {
  in:
    Var glob.c100
    Var glob.q1.RES
    Var glob.q1.NE[0]
    Var S69
    Var S70
  out:
    Var glob.q1.POP
    Var AV25[0]
    Var AV8[0]
}
glob.q1.RES = GND
glob.q2.D[0..7] = glob.q1[0..7]
glob.q2.PUSH = S71
glob.q3.D[0..7] = glob.q1[0..7]
glob.q3.PUSH = S72
glob.q4.D[0..7] = glob.q1[0..7]
glob.q4.PUSH = S73
```



E.4.16 multiple queue writes

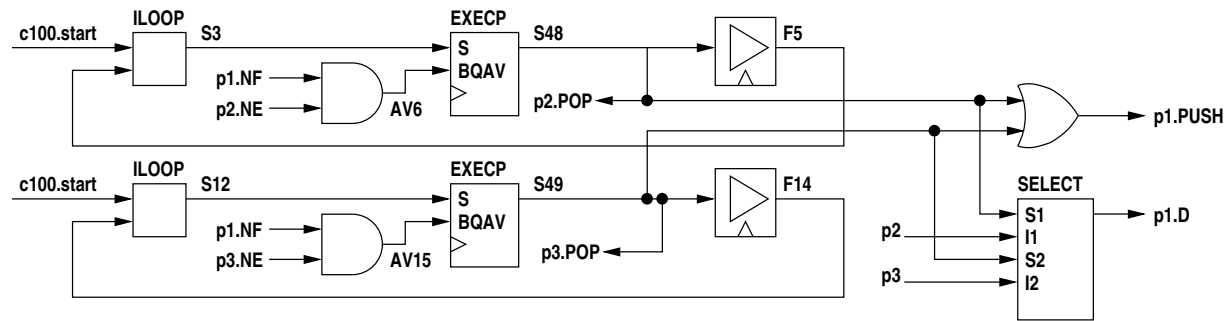
Multiple writes to queues are treated independently regardless of which module they are in. The writes are simply **OR**ed and the code must be constructed such that conflicts do not occur (there are no compile-time or run-time checks).

```
queue({q1, q2, q3}, "uint:8");
```

```
q1 <<- q2;
q1 <<- q3;
```

The intermediate code is -

```
AV8[0] = glob.q2.NE[0]
AV6 = glob.q1.NF[0] AND AV8[0]
EXECVP S48 <- C=glob.c100, EXEC=S3, AV6
ACK9[0] = S48
glob.q2.POP = ACK9[0]
DEL F5 <- (1) glob.c100 S48
ILOOP S3 <- glob.c100.start F5
AV17[0] = glob.q3.NE[0]
AV15 = glob.q1.NF[0] AND AV17[0]
EXECVP S49 <- C=glob.c100, EXEC=S12, AV15
ACK18[0] = S49
glob.q3.POP = ACK18[0]
DEL F14 <- (1) glob.c100 S49
ILOOP S12 <- glob.c100.start F14
S50 = S49 OR S48
glob.q1.PUSH = S50
SELECT {
  in:
    Var S48
    Var glob.q2[0..7]
    Var S49
    Var glob.q3[0..7]
  out:
    Var glob.q1.D[0..7]
}
```



E.4.17 "unless" constructs

The **unless** construct is implemented using ordinary intermediate code elements. The code follows one of two patterns -

- the exception expression contains no queue reads
- the exception expression does contain queue reads

If an exception code block is present then that is spliced into the execution logic. A flip-flop is added to the block which is subject to the **unless** construct to register whether execution is taking place within the subject block. If the subject block does not have any executing statements within it then the exception is ignored.

If the exception expression contains no queue reads, a **WHEN** is used to detect the exception. The value of the exception expression is **ANDed** with the subject execution flip-flop to form the test input to the **WHEN**. The output of the **WHEN** resets all **DEL**, **WAIT**, **EXECP** and **WHILE** elements in the subject killing all execution. A second **WHEN** then loops through a **DEL** waiting for the exception expression to go false. The false output of the second **WHEN** then drives the finish of an enclosing **ILOOP** driving the first **WHEN**.

If the exception expression contains queue reads, the first **WHEN** is preceded by an **EXECP** and the second **WHEN** is not needed as a queue read is a single event, not a duration.

In either case the finish signal is **ORed** with the finish signal of the subject block so that it appears to complete.

An exception code block is spliced in after the **WHEN** and prior to the finish signal.

The following example, **ee1.3pl**, illustrates the case where the exception expression contains no queue reads -

```

module "csiro/hymod/io1.3pl"

useclock(c100);

static({e, 1}, "log");
static({v1, v2, v3}, "uint:8");

when (1) {
  seq {
    v1 <- 3;
    par {
      v2 <- 5;
      v3 <- 7;
    }
  }
} unless (e) {
  seq {
    v1 <- 0;
    v2 <- 0;
  }
}

```

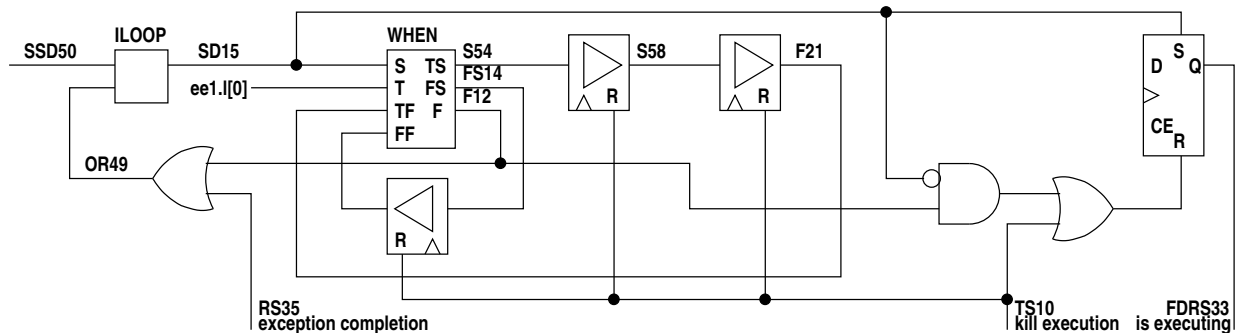
The intermediate code for the control logic for the **WHEN** is -


```

ILOOP SD15 <- SSD50 OR49
WHEN {
    START SD15
    TEST ee1.l[0]
    T_FINISH F21
    F_FINISH FF28
    T_START S54
    F_START FS14
    FINISH F12
}
DEL S58 <- (1) glob.c100 S54 TS10
DEL F21 <- (1) glob.c100 S58 TS10
DEL FF28 <- (1) glob.c100 FS14 TS10
OR49 = F12 OR RS35
INV30 = inv(SD15)
AND31 = F12 AND INV30
OR32 = AND31 OR TS10
DFF FDRS33 <- C=glob.c100, R=OR32, S=SD15
.
.
ee1.v1.CE[0..7] = S54
.
.
ee1.v2.CE[0..7] = S58
.
.
ee1.v3.CE[0..7] = S58

```

In diagrammatic form this is -



Note that there is no parallel wait - the optimiser has noted that the two assignments in parallel take exactly one clock cycle and has replaced two **DELS** and a **WAIT** with one **DEL** (**S58** to **F21**). The first **DEL** in the **SEQ**, **S54** to **S58** is for the first assignment, hence **v1.CE** is driven by **S54**. The two parallel assignments, **v2.CE** and **v3.CE**, are driven by **S58**. **F21** completes the true block of the **WHEN** and feeds back to the true finish input to emerge as the finish output for the **WHEN**, **F12**.

If there were no **UNLESS** construct the flip-flop and three gates would be omitted, **F12** connecting back to the **ILOOP**.

The OR gate between **F12** and the **ILOOP** is to allow an execution pulse to be injected as an alternative finish signal for the **WHEN** after execution of the **WHEN** has been killed. **RS35** is the new execution completion pulse.

Execution of the **WHEN** is killed by resetting all control logic state within it, in this case just the three **DELS**. The kill signal is **TS10**.

The exception logic needs to know if the **WHEN** is actually executing (it may be nested inside other constructs and not actually be executing at the moment - in this example it is always executing since it is at the top level, hence the **ILOOP**). The flip-flop is set when the **WHEN** begins execution and is reset when the **WHEN** finishes or execution is killed by **TS10**. If the false branch of the **WHEN** is taken there is a single **DEL** in the **ILOOP** and hence the **WHEN** executes on every cycle. The **AND** gate is there to inhibit resetting of the flip-flop if the **S** input is asserted.

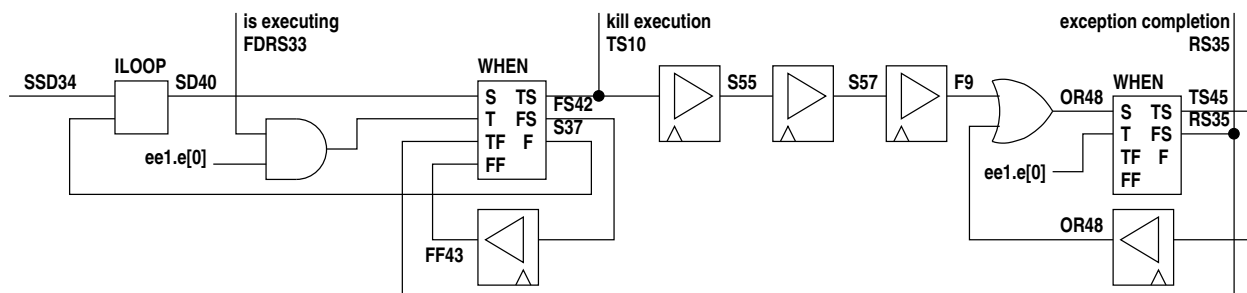
The intermediate code for the exception control logic is -

```

ILOOP SD40 <- SSD34 S37
AND41[0] = ee1.e[0] AND FDRS33
WHEN {
    START SD40
    TEST AND41[0]
    T_FINISH RS35
    F_FINISH FF43
    T_START TS10
    F_START FS42
    FINISH S37
}
DEL FF43 <- (1) glob.c100 FS42
DEL S55 <- (1) glob.c100 TS10
DEL S57 <- (1) glob.c100 S55
DEL F9 <- (1) glob.c100 S57
OR48 = F9 OR S47
WHEN {
    START OR48
    TEST ee1.e[0]
    T_FINISH null
    F_FINISH null
    T_START TS45
    F_START RS35
    FINISH null
}
DEL S47 <- (1) glob.c100 TS45
.
.
ee1.v1.RES[0..7] = S55
.
.
ee1.v2.RES[0..7] = S57

```

In diagrammatic form this is -



This logic runs independently in its own **ILOOP**. The first **WHEN** waits for the exception expression (**ee1.l[0]**) to be asserted, but only if the associated subject construct is executing (**FDRS33**). When this is not true, the execution emerges on the false output, **FS42**, through a **DEL** to the false finish **FF43** and then emerges from the when finish output as **S37** and feeds back to the **ILOOP** for the next clock cycle. Thus the **WHEN** tests every clock cycle. When the exception occurs the true output of the **WHEN** is asserted. This is **TS10** which kills all execution in the associated subject code. Execution proceeds through three **DELs** executing the **SEQ** in the exception code block. If there was no exception code block there would be a single **DEL** here. Progressive stages, **S55** and **S57**, reset variables **v1** and **v2** in the **SEQ** block. At the completion of the exception code block a second **WHEN** waits for the exception expression to go false, the true branch output looping back via a **DEL** and the OR gate. When **ee1.e[0]** goes false, the execution signal emerges on the true branch of the **WHEN**, **RS35**, which completes execution of the killed subject block. It also feeds back to the true finish input of the first when, emerging as the finish signal, **S37**, feeding back to the enclosing **ILOOP** to wait for the next exception.

If the exception expression contains queue reads a single **WHEN** combined with an **EXECP** is used, for example **ee2.3pl** -

```

module "csiroy/hymod/io1.3pl"

useclock(c100);

```

```

static(1, "log");
queue(e, "log");
static({v1, v2, v3}, "uint:8");

when (1) {
  seq {
    v1 <- 3;
    par {
      v2 <- 5;
      v3 <- 7;
    }
  }
} unless (e) {
  seq {
    v1 <- 0;
    v2 <- 0;
  }
}

```

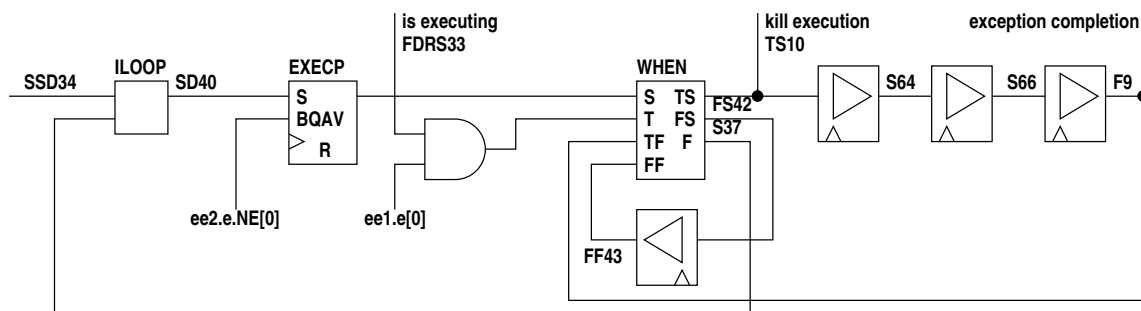
The intermediate code for the exception control logic containing a queue read is -

```

ILOOP S36 <- SSD34 S37
AV46[0] = ee2.e.NE[0]
AV44 = AV46[0]
EXECP SD47 <- C=glob.c100, EXEC=S36, AV44, null
AND41[0] = ee2.e[0] AND FDRS33
WHEN {
  START SD47
  TEST AND41[0]
  T_FINISH F9
  F_FINISH FF43
  T_START TS10
  F_START FS42
  FINISH S37
}
DEL S64 <- (1) glob.c100 TS10
DEL S66 <- (1) glob.c100 S64
DEL F9 <- (1) glob.c100 S66
DEL FF43 <- (1) glob.c100 FS42
.
.
ee2.v1.RES[0..7] = S64
.
.
ee2.v2.RES[0..7] = S66

```

In diagrammatic form this is -



The exception logic now waits for queue **e** to be not empty. Execution is then passed to the **WHEN**. If the value is false, the **WHEN** loops back to the **ILOOP** via a **DEL** and **FF43**, emerging from the **WHEN** on the finish output **S37**.

If the exception value is true subject execution is killed and the execution signal goes through three **DELs** as before. On emerging (**F9**) it completes execution of the subject block and loops back to the true completion input of the **WHEN**, emerging as the **WHEN** finish signal **S37**, completing the **ILOOP**.

Appendix F - cpu_serial.c, cpu_serial.h

The following code provides a serial interface to an FPGA. The FPGA code uses modules `uart_rx()` and `uart_tx()` in library `uarts.3pl` for serial transmission via the host computer port.

The .info file for the design contains the read and write addresses, data types and bit counts and the interrupt numbers. This is all transferred to the associated .h file by python script `uart_info2c` for inclusion in the CPU C source code.

Each command from the CPU to the FPGA begins with the following byte

com	address
2	6

The two-bit command is -

- 00 read value, read queue
- 10 write static, write queue
- 01 read memory
- 11 write memory

The 6-bit address allows access 64 writable and 64 readable locations in the FPGA.

For a write to a static or queue variable the format is -

10	address	data	-----
2	6	8	

where the command is followed by zero or more data bytes (a zero-length write has no data but the write can act as a trigger or event). There is no response from the FPGA.

For a read from a value or a queue the format is -

00	address
2	6

read command

0	data byte count	data	-----
1	7	8	

returned data

where the returned data consists of a data byte count and one or more data bytes.

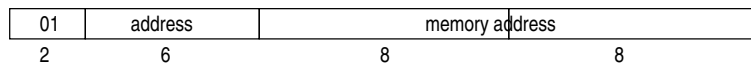
For a write to memory -

11	address	memory address	data	-----
2	6	8	8	

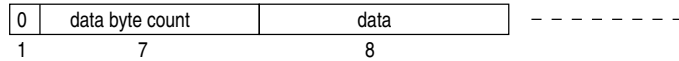
where the command is followed by 16-bit memory address and one or more data bytes. There is no response from the FPGA.

For a read from memory -

command



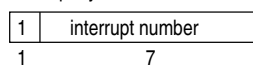
returned data



where the command is followed by 16-bit memory address. A data byte count and one or more data bytes are returned from the FPGA.

Interrupts are initiated by the FPGA unlike the previous transfers which are all initiated by the CPU. For an interrupt the FPGA sends a single byte to the CPU. The byte has the most significant bit set and the remaining 7 bits indicates the interrupt number.

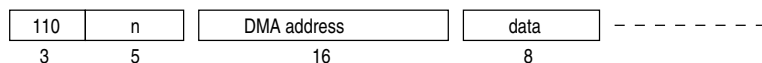
interrupt byte from FPGA



An interrupt byte will not be sent during the transmission of a CPU command and any associated return data or during a DMA transfer (see below).

DMA requests by the FPGA code are of course carried out by software in the CPU (there being no hardware connection for real DMA) and are initiated by the FPGA.

For DMA write from FPGA to CPU -

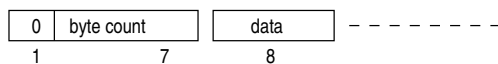


DMA write command

For DMA read to FPGA from CPU -



DMA read command



returned data

Commands and associated data are written to port file descriptor `fd`. Data received from the FPGA is read from `fd` in the input handler which runs in its own thread. A received byte with the top bit set is an interrupt, the remaining bits giving the interrupt number (semaphore index). A `sem_post()` is issued to the semaphore with that index. A received byte with the top bit clear is a byte count for data to be received from the FPGA. The following bytes indicated by the count are read from `fd` and written to pipe `pipefd[1]`. A command expecting received data waits on `pipefd[0]` and reads from it (a command knows in advance how many bytes to expect as this is contained in the `.info` file).

`sem_wait()`

```
#include "cpu_serial.h"

#define BSIZE 1024

sem_t** sem;
char** sem_name;
char buffer[BSIZE];
int fd;
struct termios tio;
static int pipefd[2];
pthread_t tideh;
int ninterrupts;
```

```

int      open_port (char* pname);
void     close_port ();
void     write_ (
        char*   ptr,
        int     bytes
    );
int      read_ (
        char*   ptr,
        int     bytes
    );
void*    input_handler ();

void
open_cpu_serial(char* pname, int interrupts) {
    ninterrupts = interrupts;
    fd = open_port(pname);
    pipe(pipefd);
    if (ninterrupts != 0) {
        sem = (sem_t**)malloc(ninterrupts * sizeof(sem_t*));
        sem_name = (char**)malloc(ninterrupts * sizeof(char*));
        for (int i=0 ; i<ninterrupts ; i++) {
            sem_name[i] = (char*)malloc(6);
            sprintf(sem_name[i], "/e%d", i);
            sem_unlink(sem_name[i]);    // remove if already exists!
            if ((sem[i] = sem_open(sem_name[i], O_CREAT, 0600, 0)) == SEM_FAILED) {
                perror("sem_open");
                exit(1);
            }
        }
    }
    tideh = thread_start(input_handler, "input_handler", 1);
}

void
close_cpu_serial() {
    pthread_cancel(tideh);
    for (int i=0 ; i<ninterrupts ; i++) {
        sem_close(sem[i]);
        sem_unlink(sem_name[i]);
    }
    close_port();
    close(pipefd[0]);
    close(pipefd[1]);
}

void
comms_write (
    int      mem,          // memory write 1, static or queue write 0
    uint8_t  address,      // FPGA register address
    uint16_t memaddress,   // memory address if required
    int      bytes,        // number of data bytes
    char*    ptr           // pointer to data
) {
    char      com = 0x80 | (mem << 6) | address;
    int       i;
    int       n = bytes;

    write_(&com, 1); // send command byte
    if (mem != 0)
        write_((char*)&memaddress, 2); // send memory address
    while (n > 0) {
        i = write(fd, ptr, n);
        n -= i;
        ptr += i;
    }
}

void
write_ (
    char*    ptr,
    int      bytes
) {
    int i;

```

```

    if (write(fd, ptr, bytes) < 0) {
        fprintf(stderr, "data write error\n");
        exit(1);
    }
}

void
comms_read (
    int          mem,          // memory read 1, value or queue read 0
    uint8_t      address,      // FPGA register address
    uint16_t     memaddress,   // memory address if required
    int          bytes,        // number of data bytes to be received
    int          sig,          // 1 if signed, 0 if unsigned
    int          fill,         // number of top bytes to fill
    char*        ptr           // pointer to data destination
) {
    char    com = (mem << 6) | address; // set address and memory flag in command
    int     n = bytes; // byte counter initialisation
    char    snap;      // most significant byte
    char    pad = 0;    // padding byte - 0 or 0xff
    int     i;
    // send command byte
    write_(&com, 1);

    // send memory address if is a memory read
    if (mem != 0)
        write_((char*)&memaddress, 2);

    // receive data and write to destination address
    while (n > 0) {
        i = read_(ptr, n);
        n -= i;
        ptr += i;
    }
    snap = *(ptr - 1);

    // if whole result type has been filled, return
    if (fill == 0)
        return;

    // fill remainder of result with 0x00 (unsigned or +ve) or 0xff (signed and -ve)
    if ((sig != 0) && ((snap & 0x80) != 0))
        pad = 0xff; // 0xff if signed and -ve
    while (fill-- != 0)
        *ptr++ = pad; // fill most significant bytes
}

int
read_ (
    char*    ptr,
    int      bytes
) {
    int r;
    r = read(pipefd[0], ptr, bytes);
    if (r < 0) {
        fprintf(stderr, "data read error\n");
        exit(1);
    }
    return(r);
}

void*
input_handler () {
    int    r;
    char   c;
    int    n;
    int    bytes;
    char*   ptr;

    for (;;) {
        r = read(fd, &c, 1);
        if (r < 0) {
            printf("uart read error\n");
            exit(1);
        }
    }
}

```

```

    }
    n = c & 0x7f;    // byte count or interrupt ID
    if (c & 0x80) {
        // is an interrupt
        if (n >= 20)
            printf("interrupt %d illegal (max 20) - ignored\n", n);
        else
            sem_post(sem[n]);
    } else {
        // is a command return -
        // read the data
        bytes = n;
        ptr = buffer;
        while (bytes > 0) {
            r = read(fd, ptr, bytes);
            bytes -= r;
            ptr += r;
        }
        // write the data back to the command requesting it
        bytes = n;
        ptr = buffer;
        while (bytes > 0) {
            r = write(pipefd[1], ptr, bytes);
            bytes -= r;
            ptr += r;
        }
    }
}

}

int
open_port (char* pname) {
    int fd;
    struct termios ntio;
    int flags;

    if (strlen(pname) == 0) {
        fprintf(stderr, "could not find USB serial device");
        exit(1);
    }

    fd = open(pname, O_RDWR | O_NOCTTY | O_NONBLOCK);
    if (fd < 0) {
        fprintf(stderr, "Cannot open USB serial device\n");
        exit(1);
    }
    printf("port %s open\n", pname);

    // Make the file descriptor blocking.
    flags = fcntl(fd, F_GETFL);
    flags &= ~O_NONBLOCK;
    fcntl(fd, F_SETFL, flags);

    tcgetattr(fd, &ntio); // get current port settings
    bzero(&ntio, sizeof(ntio)); // clear struct for new port settings
    ntio.c_cflag = CS8 | CREAD | CSTOPB | CLOCAL;
    ntio.c_iflag = IGNPAR;
    ntio.c_oflag = 0; // OCANON
    ntio.c_lflag = 0; // ICANON
    ntio.c_cc[VMIN] = 1; // blocking read until 1 character arrives
    ntio.c_cc[VSTART] = _POSIX_VDISABLE;
    ntio.c_cc[VSTOP] = _POSIX_VDISABLE;
    ntio.c_cc[VINTR] = _POSIX_VDISABLE;
    ntio.c_cc[VQUIT] = _POSIX_VDISABLE;
    ntio.c_cc[VSUSP] = _POSIX_VDISABLE;
    cfsetispeed(&ntio, B115200);
    cfsetospeed(&ntio, B115200);

    tcflush(fd, TCIFLUSH);
    tcsetattr(fd, TCSANOW, &ntio);

    return(fd);
}

```



```

void
close_port () {
    // Restore terminal settings
    tcsetattr(fd, TCSANOW, &tio);
    close(fd);
}

// Start a thread.
pthread_t
thread_start(
    void * (*func)(void*),
    char * name,
    int priority
) {
    int status;
    struct sched_param sp;
    pthread_t tid;

    if ((status = pthread_create(&tid, NULL, func, NULL)) != 0) {
        printf("thread_start(): pthread_create() failed thread '%s' (err = %s)\n",
            name, strerror(status));
        exit(1);
    }

    sp.sched_priority = priority;

    status = pthread_setschedparam(tid, SCHED_RR, &sp);
    if (status != 0) {
        printf("thread_start(): pthread_setschedparam() failed for thread '%s' (err = %s)\n",
            name, strerror(status));
        exit(1);
    }
    return(tid);
}

```

Appendix G - cpu_serial.py

The Python3 version of this code functions the same way as the C version above -

```
import os
import fcntl
import termios
import queue
import threading

tio = None
infile = None
fd = None
q = queue.Queue()

# Write data to FPGA.
# mem      0 if static or queue write, 1 if memory write
# address   FPGA interface address
# memaddress memory address if mem == 1
# dba       byte array of data to send
#
def comms_write (mem, address, memaddress, dba):
    ba = bytearray()
    ba += bytes([0x80 | (mem << 6) | address])          # command
    if mem != 0:
        ba += memaddress.to_bytes(2, byteorder='little')# append memory address
    if dba != None:
        ba += dba                                       # append data
    os.write(fd, ba)                                    # write to FPGA

# Read data from FPGA.
# mem      0 if value or queue read, 1 if memory read
# address   FPGA interface address
# memaddress memory address if mem == 1
# bytes     data length - NOT USED HERE (required by Zynq)!
#
def comms_read (mem, address, memaddress, bytecount):
    global q
    wba = bytearray()
    wba += bytes([(mem << 6) | address])                # command - address and memory flag
    if mem != 0:
        wba += memaddress.to_bytes(2, byteorder='little')# append memory address
    os.write(fd, wba)                                    # write to FPGA
    return(q.get())                                     # read response from FPGA

# Close the serial port to the FPGA.
def close_port ():
    # Restore terminal settings
    tcsetattr(fd, TCSANOW, tio)
    close(fd)

# Handle input from FPGA via serial port.
def input_handler ():
    global fd
```

```

global q
while True:
    try:
        r = os.read(fd, 1)
    except OSError:
        print("UART read error")
        exit(1)
    c = r[0]
    n = c & 0x7f    # Note: max of 127 bytes
    if c & 0x80:
        # is an interrupt
        if n >= 20:
            print("interrupt " + str(n) + " illegal (max 19)")
            exit(1)
        else:
            semaphores[n].release()
    else:
        # is a command return - read the data
        buff = bytearray()
        while n > 0:
            buff += bytes(os.read(fd, n))
            n -= len(buff)
        q.put(buff)

# Open the serial port to the FPGA.
def open_cpu_serial (pname, ninterrupts):
    global fd
    global tio
    global semaphores

    if len(pname) == 0:
        print("could not find USB serial device" + pname)
        exit(1)
    try:
        # Open initially as non-blocking so it does not wait trying to read
        # as it is opened.
        fd = os.open(pname, os.O_RDWR | os.O_NOCTTY | os.O_NONBLOCK)
    except OSError:
        print("Cannot open USB serial device")
        exit(1)

    # Having opened the file and continued, now make the file
    # descriptor blocking.
    flags = fcntl.fcntl(fd, fcntl.F_GETFL)
    flags &= ~os.O_NONBLOCK
    fcntl.fcntl(fd, fcntl.F_SETFL, flags)

    tio = termios.tcgetattr(fd) # get current port settings
    ntio = termios.tcgetattr(fd)

    ntio[0] = termios.IGNPAR          # iflags
    ntio[1] = 0                      # oflags - OCANON
    ntio[2] = termios.CS8 | termios.CREAD | termios.CSTOPB | termios.CLOCAL # cflags
    ntio[3] = 0                      # lflags - ICANON
    ntio[4] = termios.B115200         # ispeed
    ntio[5] = termios.B115200         # ospeed
    ntio[6][termios.VMIN] = 1         # blocking read until 1 character arrives
    ntio[6][termios.VSTART] = 0
    ntio[6][termios.VSTOP] = 0
    ntio[6][termios.VINTR] = 0
    ntio[6][termios.VQUIT] = 0
    ntio[6][termios.VSUSP] = 0

    termios.tcflush(fd, termios.TCIFLUSH)
    termios.tcsetattr(fd, termios.TCSANOW, ntio)

    # Create and start the input handler thread.
    t = threading.Thread(target=input_handler, args=())
    t.start()

    global semaphores
    semaphores = []
    for i in range(ninterrupts):
        semaphores.append(threading.Semaphore(0))

```

```
def close_cpu_serial ():
    close_port()

def open_port (pname):
    if pname == "":
        print("Cannot find USB serial device " + pname)
        exit(1)
    fd = open(pname, 'r')

def close_port ():
    # Restore terminal settings
    termios.tcsetattr(fd, termios.TCSANOW, tio)
    #os.close(fd) HANGS WAITING ON THREAD!
```

Appendix H - Zynq device tree

The following partial device tree source matches the Zynq CPU/FPGA interfaces described in [chapter 20](#) p631 and [chapter 12](#) p561.

```
reserved-memory {
    #address-cells = <0x1>;
    #size-cells = <0x1>;
    ranges;

    hpbuf_reserved {
        compatible = "xlnx,uio", "generic-uio";
        size = <0x1000000>;
        alignment = <0x1000000>;
        linux,phandle = <0x8>;
        phandle = <0x8>;
    };
};

amba_pl {
    #address-cells = <0x1>;
    #size-cells = <0x1>;
    compatible = "simple-bus";
    ranges;

    axi_mgp0@40000000 {
        compatible = "xlnx,axi_mgp0", "xlnx,uio", "generic-uio";
        reg = <0x40000000 0x1000000>;
        xlnx,s00-axi-addr-width = <0xa>;
        xlnx,s00-axi-data-width = <0x20>;
    };

    axi_mgp1@80000000 {
        compatible = "xlnx,axi_mgp1", "xlnx,uio", "generic-uio";
        reg = <0x80000000 0x1000000>;
        xlnx,s01-axi-addr-width = <0xa>;
        xlnx,s01-axi-data-width = <0x20>;
    };

    axi_hpbuf@0 {
        compatible = "xlnx,axi_hpbuf", "xlnx,uio", "generic-uio";
        memory-region = <0x8>;
    };
};
```

```

event0 {
    compatible = "xlnx,uio", "generic-uio";
    interrupts = <0x0 0x1d 0x1>;
    interrupt-names = "int61";
    interrupt-parent = <0x3>;
};

event1 {
    compatible = "xlnx,uio", "generic-uio";
    interrupts = <0x0 0x1e 0x1>;
    interrupt-names = "int62";
    interrupt-parent = <0x3>;
};

event2 {
    compatible = "xlnx,uio", "generic-uio";
    interrupts = <0x0 0x1f 0x1>;
    interrupt-names = "int63";
    interrupt-parent = <0x3>;
};

event3 {
    compatible = "xlnx,uio", "generic-uio";
    interrupts = <0x0 0x20 0x1>;
    interrupt-names = "int64";
    interrupt-parent = <0x3>;
};

event4 {
    compatible = "xlnx,uio", "generic-uio";
    interrupts = <0x0 0x21 0x1>;
    interrupt-names = "int65";
    interrupt-parent = <0x3>;
};

event5 {
    compatible = "xlnx,uio", "generic-uio";
    interrupts = <0x0 0x22 0x1>;
    interrupt-names = "int66";
    interrupt-parent = <0x3>;
};

event6 {
    compatible = "xlnx,uio", "generic-uio";
    interrupts = <0x0 0x23 0x1>;
    interrupt-names = "int67";
    interrupt-parent = <0x3>;
};

event7 {
    compatible = "xlnx,uio", "generic-uio";
    interrupts = <0x0 0x24 0x1>;
    interrupt-names = "int68";
    interrupt-parent = <0x3>;
};

event8 {
    compatible = "xlnx,uio", "generic-uio";

```

```

        interrupts = <0x0 0x34 0x1>;
        interrupt-names = "int84";
        interrupt-parent = <0x3>;
    };

    event9 {
        compatible = "xlnx,uio", "generic-uio";
        interrupts = <0x0 0x35 0x1>;
        interrupt-names = "int85";
        interrupt-parent = <0x3>;
    };

    event10 {
        compatible = "xlnx,uio", "generic-uio";
        interrupts = <0x0 0x36 0x1>;
        interrupt-names = "int86";
        interrupt-parent = <0x3>;
    };

    event11 {
        compatible = "xlnx,uio", "generic-uio";
        interrupts = <0x0 0x37 0x1>;
        interrupt-names = "int87";
        interrupt-parent = <0x3>;
    };

    event12 {
        compatible = "xlnx,uio", "generic-uio";
        interrupts = <0x0 0x38 0x1>;
        interrupt-names = "int88";
        interrupt-parent = <0x3>;
    };

    event13 {
        compatible = "xlnx,uio", "generic-uio";
        interrupts = <0x0 0x39 0x1>;
        interrupt-names = "int89";
        interrupt-parent = <0x3>;
    };

    event14 {
        compatible = "xlnx,uio", "generic-uio";
        interrupts = <0x0 0x3a 0x1>;
        interrupt-names = "int90";
        interrupt-parent = <0x3>;
    };

    event15 {
        compatible = "xlnx,uio", "generic-uio";
        interrupts = <0x0 0x3b 0x1>;
        interrupt-names = "int91";
        interrupt-parent = <0x3>;
    };
};

```

Appendix I - Alphabetic List of all Inbuilt and Library Modules, Procedures and Functions

Abbreviations -

I M - inbuilt module

I P - inbuilt procedure

I F - inbuilt function

L M - library module

L P - library procedure

L F - library function

L C - library class

L CM - library class module

L CP - library class procedure

L CF - library class function

abs	I F	absolute value function	10.1 p265
accept	I F	ignore clock domain differences	10.2 p265
acos	I F	trigonometric arc cosine function	10.3 p266
addmodeattribute	I P	add attribute to the mode attribute map	9.1 p188
addncfline	L P	add line to NCF file	11.2.2.2 p377
addsub	I F	conditionally add or subtract	10.3.1 p267
addxdcline	L P	add line to XDC file	11.2.2.3 p377
append	I P	append a list entry	9.2 p188
appendall	I P	append entries from another list	9.3 p189
append_ncf_file	L P	append a file to the NCF file	11.2.2.4 p378
argadjust	I P	adjust arguments for arithmetic operations	9.4 p189
arglist	I F	pass lists, arrays or maps of arguments	10.4 p268
argstomap	L F	create a map from key=value arguments	11.13.4.1 p512
array_to_queue	L M	serialise an array to a queue	11.13.2.1 p491
arraytostr	I F	convert an integer array to a string	10.5 p269
arraytype	I F	array type of a variable or expression	10.6 p269
asin	I F	trigonometric arc sine function	10.7 p270
assert	I P	assert a value mode variable for a clock cycle	9.5 p190
assign	L P	generalised assignment	11.13.3.1 p507
asyncqueuespaces	L F	space count for asynchronous queue	11.13.4.2 p512
asyncqueuewords	L F	word count for asynchronous queue	11.13.4.3 p513
atan	I F	single argument trigonometric arc tangent function	10.8 p270
atan2	I F	two argument trigonometric arc tangent function	10.9 p270
attribute	I F	get an attribute from a variable	10.10 p271
attributeprocedure	I P	register a procedure to handle attributes	9.6 p191
attributes	I F	get the attributes of a variable	10.11 p271
attributes	I P	assign attributes to a variable	9.7 p192
biasedexponent	I F	target float biased exponent	10.12 p272
biasedexponentwidth	I F	width of target float biased exponent	10.13 p272
bin_to_gray	L F	convert binary count to gray count	11.13.4.4 p513
bitset	L F	test if a bit is set	11.13.4.5 p513
bitsfor	I F	number of bits to represent an integer - DEPRECATED!	10.14 p273
capture	L M	capture a data sequence	11.14.1.1 p525
cast	I F	cast a value to a new type or type width	10.15 p273
cdatas	L F	generate C name array for 3PL enum type	11.15.3.2 p529
ceil	I F	round float to nearest integer towards $+\infty$	10.16 p274
change	I P	change a list entry	9.8 p193
cinclude	L P	convert a C include file	11.13.3.2 p507
class	I P	create a variable of type "class"	9.9 p193
clkdcn	L P	create a DCM digital clock manager	11.1.2.1 p357
clkdlle	L P	create an extended Spartan2, Virtex or VirtexE DLL	11.1.2.3 p361
clkdlhlf	L P	create a high frequency Spartan2, Virtex or VirtexE DLL	11.1.2.4 p363
clkdll	L P	create a DLL delay-locked loop clock manager	11.1.2.2 p360
clkmmcm_adv	L P	create an advanced MMCM multi-mode clock manager	11.1.2.6 p365
clkmmcm_base	L P	create a basic MMCM multi-mode clock manager	11.1.2.7 p368
clock	I P	create one or more clock variable	9.10 p194
clockused	L F	determine if a clock variable has been used	11.2.3.1 p395
clockbufg	L P	a BUFG clock buffer	11.1.2.9 p371
clockbufio	L P	a local clock buffer	11.1.2.10 p372
clockfromexpr	I P	drive a clock from an expression	9.11 p196
clockmuxbuffer	L P	a clock mux/buffer	11.1.2.11 p372
clockmuxbufferdisable	L P	a clock mux/buffer with disable	11.1.2.12 p373
clock_params	L P	derive clock synthesis parameters	11.1.2.13 p373
clock_params	L F	derive clock synthesis parameters	11.1.3.1 p375
close	I P	close a file	9.12 p196
cmemory	I P	create a combinatorial output memory	9.13 p197
cmemoryread	I F	read a combinatorial output memory	10.17 p275
cmemorywrite	I P	write to a combinatorial output memory	9.14 p198
componentpointer	I F	get a pointer to a static variable component	10.18 p276

componenttype	I F	get the type of a variable component	10.19 p277
copy	I F	copy a list or map	10.20 p277
cos	I F	trigonometric cosine function	10.21 p277
cpu_api	L C	a CPU/FPGA interface class (UART, ZYNQ)	11.11 p455
cpu_api.dma_read	L CM	DMA transfer from memory to FPGA (ZYNQ)	11.10.1.3 p450
cpu_api.dma_write	L CM	DMA transfer from FPGA to memory (ZYNQ)	11.10.1.4 p451
cpu_api.read	L CP	a CPU-readable interface (UART, ZYNQ)	11.11.2.2 p459
cpu_api.read_memory	L CP	a CPU-readable interface to a memory (UART, ZYNQ)	11.11.2.4 p469
cpu_api.read_queue	L CP	CPU read from a queue variable (UART, ZYNQ)	11.11.2.5 p471
cpu_api.read_value	L CP	CPU read from a value (UART, ZYNQ)	11.11.2.6 p474
cpu_api.receive_event	L CP	interrupt the CPU (UART, ZYNQ)	11.11.2.7 p476
cpu_api.write	L CP	a CPU-writable interface (UART, ZYNQ)	11.11.2.3 p464
cpu_api.write_dma_buffer	L CM	DMA transfer to CPU RAM (ZYNQ)	11.10.1.2 p450
cpu_api.write_memory	L CP	a CPU-writable interface to a memory (UART, ZYNQ)	11.11.2.8 p477
cpu_api.write_queue	L CP	a CPU write to a queue variable (UART, ZYNQ)	11.11.2.9 p479
cpu_api.api.interrupt	L CP	interrupt the CPU (UART, ZYNQ)	11.11.2.11 p483
crc32	L M	Virtex5 and later 32 bit CRC	11.5.1.1 p422
crc64	L M	Virtex5 and later 64 bit CRC	11.5.1.2 p422
crc_8_16	L M	16 bit CRC from 8 bit data	11.13.2.2 p492
crc_16_16	L M	16 bit CRC from 16 bit data	11.13.2.3 p492
crcblock	L M	block 32 bit data with a CRC	11.13.2.4 p493
crcunblock	L M	unblock 32 bit data with a CRC	11.13.2.5 p494
cread	L M	read combinatorial output memory to a queue	11.13.2.6 p495
cspan	I F	head string selected by characters not in a set	10.22 p278
ctypes	L F	converts 3PL types to C types	11.15.3.1 p527
currentclock	I F	get the current clock variable	10.23 p278
currentelement	I F	get the current element identifier	10.24 p279
cwrite	L M	write combinatorial output memory from queue	11.13.2.7 p495
dcireset	L M	reset DCI pins	11.5.1.3 p423
deglitch	L F	deglitch an input signal	11.13.4.6 p514
deflate	I F	compress a string	10.25 p279
delay	L F	fixed delay	11.13.4.7 p514
del	I M	variable length delay line	8.1 p184
del	L F	variable length delay line	11.2.3.2 p396
demultiplex	L M	demultiplex a source	11.13.2.8 p496
demultiplex_e	L M	demultiplex a source using port number	11.13.2.8 p496
demultiplex_l	L M	demultiplex a source using port bit	11.13.2.8 p496
designname	I P	change the design name	9.15 p199
dimension	I F	dimension - get a dimension of an array or size of a map or list	10.26 p279
dimensions	I F	get the number of dimensions of an array	10.27 p280
directive	I P	enter a directive	9.16 p200
directive	I F	get the value of a directive	10.28 p280
directives	I F	get the global directives map	10.29 p281
direxists	I F	determine if a directive exists	10.30 p281
divide	L M	pipelined integer division-remainder	11.13.2.9 p497
divrem	I P	combined quotient and remainder	9.17 p200
domain	I F	get the clock domain for a variable	10.31 p282
dsp48	L P	create a DSP48 block for Virtex4 FPGAs	11.3.2.1 p408
dsp48a	L P	create a DSP48 block for Spartan6 FPGAs	11.3.2.2 p409
dsp48e	L P	create a DSP48E block for Virtex5 FPGAs	11.3.2.3 p410
dsp48e1	L P	create a DSP48E1 block for Virtex6 FPGAs	11.3.2.4 p412
dspaddsub	L M	DSP add/subtract	11.3.1.1 p401
dspcounter	L M	DSP counter	11.3.1.2 p402
dsplog	L M	DSP logical operation	11.3.1.3 p403
dspmul	L M	DSP multiply	11.3.1.4 p404
dspmuladd	L M	DSP multiply and add	11.3.1.5 p406
dspmulsub	L M	DSP multiply and subtract	11.3.1.6 p406
dspsubmul	L M	DSP subtract and multiply	11.3.1.7 p406
dutycycle	I F	get the duty cycle of a clock variable	10.32 p282

elapsedtime	I F	get the elapsed time	10.33 p283
element	I P	add an element to the EDIF netlist	9.18 p201
elementaddncf	L P	add constraint to NCF file for an element	11.2.2.5 p378
elementBinProperty	L P	add a property of type "int" to an element in binary format	11.2.2.6 p379
elementcheck	I P	check that a Xilinx element is allowed in the current device family	11.2.2.7 p380
elementcheck	I F	check that a Xilinx element is allowed in the current device family	11.2.3.3 p398
elementclockinput	I P	connect a clock input signal to an EDIF netlist element	9.19 p202
elementend	I P	complete construction of an EDIF netlist element	9.20 p203
elementFloatProperty	L P	add a property of type "float" to an element	11.2.2.8 p381
elementHexProperty	L P	add a property of type "int" to an element in hexadecimal format	11.2.2.9 p381
elementinput	I P	connect an input signal to an EDIF netlist element	9.21 p203
elementIntProperty	L P	add a property of type "int" to an element	11.2.2.10 p382
elementLogProperty	L P	add a property of type "log" to an element	11.2.2.11 p383
elementoutput	I P	connect an output signal to an EDIF netlist element	9.22 p204
elementproperty	I P	set an element property	9.23 p205
elementStrProperty	L P	add a property of type "str" to an element	11.2.2.12 p384
elementtsoutput	I P	connect a 3-state output signal to an EDIF netlist element	9.24 p206
emptyvalue	I P	clear a value-mode variable	9.25 p207
endswith	I F	a string is the tail of another	10.34 p283
entries	I F	get the number of entries in a list or map - DEPRECATED	10.35 p283
enum	I P	create an enumerated type	9.26 p207
enumv	I P	create an enumerated type	9.27 p208
env	I F	get the environment variables	10.36 p284
exec	I P	execute a command	9.28 p209
exists	I F	determine if an identifier is recognised - DEPRECATED!	10.37 p284
exit	I P	terminate execution	9.29 p210
exp	I F	e to a power	10.38 p285
family	I F	string identifying platform	10.39 p285
export	I P	export a 'core'	9.30 p211
external	I P	declare a signal as an EDIF port	9.31 p211
fadd	L M	floating add	11.16.1.1 p530
fdiv	L M	floating divide	11.16.1.2 p530
fdre	L P	Create an FDRE flip-flop	11.2.2.13 p384
fdrse	L P	Create an FDRSE flip-flop	11.2.2.14 p385
fdse	L P	Create an FDSE flip-flop	11.2.2.15 p386
fdce	L P	Create an FDCE flip-flop	11.2.2.16 p386
fdpe	L P	Create an FDPE flip-flop	11.2.2.17 p387
fifo	L M	a fifo with static interface	11.13.2.10 p498
file	I P	create file variables	9.32 p212
filecanread	I F	check if a file is readable	10.40 p285
filecanwrite	I F	check if a file is writable	10.41 p286
filedelete	I F	delete a file	10.42 p286
fileexists	I F	check if a file exists	10.43 p286
filegetparent	I F	get the path to the file	10.44 p287
fileisdir	I F	check if a file is a directory	10.45 p287
fileisfile	I F	check if a file is a file	10.46 p287
filelastmod	I F	get the time of last modification	10.47 p288
filelength	I F	get the file length in bytes	10.48 p288
filelist	I F	get a list of files and directories	10.49 p289
filelistdirs	I F	get a list of directories	10.50 p289
filelistfiles	I F	get a list of files	10.51 p289
filemkdir	I F	create a directory	10.52 p290
filerename	I F	rename a file or directory	10.53 p290
filesetreadonly	I F	set a file or directory read-only	10.54 p291
find	I F	find a substring using a regular expression	10.55 p291
first	I F	get the first value of an enum	10.56 p292
float	I P	create immediate floating point variables	9.33 p213
floatisnegative	I F	sign of target float	10.57 p292
floor	I F	round float to nearest integer towards $-\infty$	10.58 p292

fmul	L M	floating multiply	11.16.1.3 p531
fneg	L M	negate a floating point value	11.16.1.4 p531
format	I F	format values into a string	10.59 p293
forcecast	L F	force a prohibited cast	11.13.4.8 p516
fprintf	I P	print a formatted line to an output file	9.34 p213
frequency	I F	get the frequency of a clock variable	10.60 p293
fscanf	I P	scan a line from a file into variables	9.35 p214
fsub	L M	floating subtract	11.16.1.5 p532
generateConstraints	I F	generate constraint file entries from I/O mode variable attributes	11.2.2.18 p388
get	I F	get a list or map entry	10.61 p294
getbit	L F	get a bit	11.13.4.9 p516
getfield	I F	get the nth field from a struct	10.62 p294
getiostandard	L F	get the I/O standard for a pin or pins	11.2.3.4 p399
gt_custom_16	L M	a 16-bit Rocket high-speed serial I/O channel	11.6.1.1 p432
gtp_dual	L M	create a GTP dual high-speed serial I/O channel tile	11.4.2.1 p418
gtp_dual_16	L M	a 16-bit dual high-speed serial I/O channel tile	11.4.1.1 p415
hadeof	I F	had EOF on the last file read	10.63 p295
hasfield	I F	check if a struct has a specific field	10.64 p295
hasnext	I F	is there a next value of an enum	10.65 p296
hasqueureads	I F	does an argument contain queue reads	10.66 p296
hasprev	I F	is there a previous value of an enum	10.67 p297
head	I F	get the head of a string	10.68 p297
iddr	L P	input double data rate interface	11.5.2.1 p424
idelay	L P	input data delay	11.5.2.2 p426
ident	L P	extract an identifier from an argument	9.37 p216
identifier	I F	get the identifier of a variable	10.69 p297
import	I P	import a 'core'	9.36 p215
imtotargettype	I F	convert an immediate type to a target type	10.70 p298
includedir	I P	add directory for include path	9.38 p218
includedir	I F	get the directories searched for module and source files	10.71 p299
incrdecr	L F	dynamically increment or decrement	11.13.4.10 p517
input	I P	create an input mode variable	9.39 p218
int	I P	create one or more immediate integer variables	9.40 p221
inttohexstring	I F	get a hex string representing an integer	10.72 p299
IOconstraint	L P	add an I/O constraint to the constraints file	11.2.2.1 p376
iodelay	L P	input/output data delay	11.5.2.3 p427
iopins	L P	create FPGA pin specifications	11.2.2.19 p388
isclear	L F	verify a run of clear bits	11.13.4.11 p517
iserdes	L P	input deserialiser	11.5.2.4 p428
isnull	I F	determine if null	10.73 p300
isread	I F	determine when a queue variable is read	10.74 p300
issigned	I F	determine if a type is signed	10.75 p300
isset	L F	verify a run of set bits	11.13.4.12 p517
iswritten	I F	determine when a static or queue variable is written	10.78 p302
isstruct	I F	determine if a variable type is a struct	10.76 p301
istargconst	I F	check if a value mode variable is a constant	10.77 p301
itof	L M	convert an integer to a floating point value	11.16.1.6 p532
jsontype	L F	generate a JSON string from a 3PL type	11.17.3.1 p534
last	I F	get the last value of an enum	10.79 p302
lfsr	L F	linear feedback shift register	11.13.4.13 p518
listattributes	I F	get a list of variable attributes	10.80 p303
liststacks	I P	list the current state of the body and scope stacks	9.41 p221
listvars	I F	get a list of variables	10.81 p303
location	I F	get the current source location	10.82 p304
log	I P	create one or more immediate logical variables	9.42 p222
log	I F	natural logarithm	10.83 p305
lowskew	L P	request low skew secondary clock routing	11.1.2.14 p374
lszeros	L F	count least significant zero bits	11.13.4.14 p518

simpleclock	L P	create a clock of specified frequency	11.1.2.5 p364
mantissa	I F	target float mantissa	10.84 p305
mantissawidth	I F	width of target float mantissa	10.85 p305
match	I F	find substrings using a regular expression	10.86 p306
matched	I F	determine if a parameter has an argument	10.87 p306
max	I F	maximum of integers	10.88 p307
maxdelay	L P	specify max delay between timing groups	11.2.2.20 p391
maxval	L F	target maximum value	11.13.4.15 p518
maxvalue	I F	maximum value of a target type	10.89 p307
memoryaddresstype	I F	get the address type of a memory variabl	10.90 p308
memorydatatype	I F	get the data type of a memory variable	10.91 p308
min	I F	minimum of integers	10.92 p309
minval	L F	target minimum value	11.13.4.16 p519
minvalue	I F	minimum value of a target type	10.93 p309
mod	I F	modulus of integer or float	10.94 p310
mode	I F	the mode of a variable	10.95 p310
msg	I P	print a message	9.43 p222
multiplex	L M	multiplex sources	11.13.2.11 p498
multiplex_e	L M	multiplex sources giving port number	11.13.2.11 p498
multiplex_l	L M	multiplex sources giving port bit	11.13.2.11 p498
multiplexp	L M	multiplex queues with priority	11.13.2.11 p498
multiplexp_e	L M	multiplex queues with priority giving port number	11.13.2.11 p498
multiplexp_l	L M	multiplex queues with priority giving port bit	11.13.2.11 p498
multiplex_struct_e	L F	multiplex output type with uint	11.13.4.17 p519
multiplex_struct_l	L F	multiplex output type with log array	11.13.4.17 p519
multiply	L M	pipelined multiply	11.13.2.12 p500
mux	L F	multiplex inputs	11.13.4.18 p520
negclock	I P	create a negative clock	9.45 p223
netaddncf	L P	add constraint to NCF file for a net	11.2.2.21 p392
netlistcommentfromdirs	I P	directives to nelist comment	11.13.3.3 p508
next	I F	get the next value of an enum	10.96 p310
nop	I P	a no-operation (delay)	9.44 p223
nullarg	I F	omit an unmatched parameter	10.97 p311
numfields	I F	get the number of field in a struct	10.98 p312
numinargs	I F	number of input arguments	10.99 p312
numoutargs	I F	number of output arguments	10.100 p312
numwords	I F	number of words in a variable, expression or type	10.101 p313
oddr	L P	output double data rate interface	11.5.2.5 p430
offset	I F	point offset of a fixed point value	10.102 p313
ord	I F	get the ordinal of an enum or a character	10.103 p314
open	I P	open a file	9.46 p223
output	I P	create an output mode variable	9.47 p224
part	L P	specify the FPGA	9.48 p228
pdelay	L M	delay pipeline data	11.13.2.14 p501
period	I F	get the period of a clock variable	10.104 p314
period_div	L P	timing group for divided clock	11.2.2.22 p392
period_mul	L P	timing group for divided clock	11.2.2.23 p393
pfifo	L M	a prewrite fifo with static interface	11.13.2.13 p501
pll_base	L M	create a series-7 basic PLL	11.1.2.8 p369
portdomain	I F	get the port clock domain for a memory variable	10.105 p315
postprocess	I F	post-process the netlist file	11.7.2.1 p438
pow	I F	raise to a power	10.106 p315
ppc405	L P	a power PC 405 core	11.8.2.1 p440
ppcbr	L P	a PPC405 with OCM	11.9.2.1 p442
presetdirective	I P	enter a directive during parsing	9.50 p229
prev	I F	get the previous value of an enum	10.107 p315
prevclock	I F	get the previous clock variable	10.108 p316
primetypestring	I F	type of a primitive variable as a string	10.110 p317

printattributes	L P	print a list of variable attributes	11.13.3.5 p509
printdirs	L P	print a list of directives	11.13.3.6 p510
printf	I P	print a formatted line to stdout	9.49 p228
println	I P	print a line	9.51 p230
printvars	L P	print a list of variables	11.13.3.7 p510
prielse	I F	else case value for priority variable	10.109 p316
priority	I P	create a priority variable	9.52 p230
processPinAttributes	L P	process pin attributes	11.2.2.24 p393
ptr2expr	L F	a pointer to an expression	11.13.4.19 p521
pullup	L P	connect pullups to a bus	11.2.2.25 p394
prunlock	I P	unlock priority execution	9.53 p232
pulse	I P	generate a pulse - DEPRECATED!	9.54 p232
put	I P	put a map entry	9.55 p233
putall	I P	put entries from another map	9.56 p233
qdr2_ram	L C	external QDR2 RAM class	11.20.4.1 p541
qdr2_ram.read	L CM	read from external QDR2 RAM	11.20.4.2 p543
qdr2_ram.readp	L CM	read from external QDR2 RAM with priority	11.20.4.2 p543
qdr2_ram.write	L CM	write to external QDR2 RAM	11.20.4.3 p544
qdr2_ram.writep	L CM	write to external QDR2 RAM with priority	11.20.4.3 p544
qdr2_ram.incr_qdel	L CP	increment Q input delay	11.20.4.4 p545
qdr2_ram.decr_qdel	L CP	decrement Q input delay	11.20.4.4 p545
qdr2_ram.reset_qdel	L CP	reset Q input delay	11.20.4.4 p545
qdr2_ram.incr_cqdel	L CP	increment CQ input delay	11.20.4.5 p545
qdr2_ram.decr_cqdel	L CP	decrement CQ input delay	11.20.4.5 p545
qdr2_ram.reset_cqdel	L CP	reset CQ input delay	11.20.4.5 p545
qdr2_ram.incr__cqdel	L CP	increment _CQ input delay	11.20.4.6 p546
qdr2_ram.decr__cqdel	L CP	decrement _CQ input delay	11.20.4.6 p546
qdr2_ram.reset__cqdel	L CP	reset _CQ input delay	11.20.4.6 p546
queue	I P	create one or more queue variables	9.57 p234
queuearray	L P	construct an array of queues	11.13.3.4 p509
queuedepth	I F	get the depth of a queue	10.111 p317
queueisempty	I F	determine if a queue is empty	10.112 p318
queuemaxdepth	I F	get the maximum allowed depth of a queue	10.113 p318
queuemindepth	I F	get the minimum allowed depth of an asynchronous queue	10.114 p319
queureadstatus	I F	get the read status of an asynchronous queue	10.115 p319
queuespaces	I F	get the number of free spaces in a queue	10.116 p320
queewords	I F	get the number of entries in a queue	10.117 p320
queewritestatus	I F	get the write status of an asynchronous queue	10.118 p321
random	I F	a random number	10.119 p322
readbyte	I F	read a byte from an input file	10.121 p323
readdomain	I F	get the read clock domain for a variable	10.120 p322
readln	I F	read a line from an input file	10.122 p323
readbyte	I F	read a byte from an input file	10.121 p323
ref	I P	create one or more variables	9.58 p237
remove	I P	remove a list entry	9.59 p238
removedirective	I P	remove a directive	9.60 p238
replace	I F	replace left substring	10.123 p324
replaceall	I F	replace all substrings	10.124 p324
reset	I P	reset a static or queue variable	9.61 p238
resetoutput	I F	get the reset output of a queue	10.125 p325
resync	I M	resynchronise a level	8.2 p185
resync	L F	resynchronise a pulse or level to another clock	11.13.4.20 p521
reverse	L F	reverse a value	11.2.3.5 p399
rmemory	I P	create a registered output memory	9.62 p239
rmemoryout	I F	get value of a reg output memory	10.127 p326
rmemoryread	I P	read reg output memory	9.63 p242
rmemorywrite	I P	write to reg output memory	9.64 p242

round	I F	round float or fixed to nearest integer	10.126 p325
rpt	I P	print to the report file	9.65 p243
rread	L M	read reg output memory to a queue	11.13.2.15 p502
rwrite	L M	write reg output memory from a queue	11.13.2.16 p502
sample	L F	sample an expression	10.128 p326
sscanf	I P	scan a string into variables	9.68 p246
scanf	I P	scan a line from stdin into variables	9.67 p245
select	L F	a combinatorial selector	11.13.5 p521
selecta	I F	a combinatorial selector from arrays	10.129 p327
selectvalue	I P	create one or more selectvalue variables	9.66 p244
sequence	L M	generate a sequence	11.2.1.1 p376
sequence	L F	generate a sequence	11.2.3.6 p399
setdirective	I P	enter a directive - DEPRECATED!	9.69 p247
setiostandard	L P	set the I/O voltage standard for a pin or pins	11.2.2.26 p395
shell	I P	execute a command in a shell	9.70 p248
signum	I F	sign function	10.130 p328
sin	I F	trigonometric sine function	10.131 p328
size	I F	get the size in bits	10.132 p328
sort	L M	generate a parallel Batcher sort network	11.13.2.17 p503
span	I F	head string selected by characters in a set	10.133 p329
special	I P	vendor-specific special functions	9.71 p249
spi_master	L M	SPI master transmitter/receiver	11.22.1.1 p551
spi_slave	L M	SPI slave transmitter/receiver	11.22.1.2 p552
split	I F	split a string into substrings	10.134 p330
sprintf	I P	copy a formatted line to a string variable	9.72 p250
sprintf	L F	return a formatted line from variables	11.13.5.1 p522
sqrt	I F	immediate square root	10.135 p330
sqrt	L M	target integer square root	11.13.2.18 p503
stacktrace	I F	get the current stack trace	10.136 p330
start	I P	set the start condition for execution threads	9.73 p251
startswith	I F	a string is the head of another	10.137 p331
statemachine	I P	create a state machine - DEPRECATED!	9.74 p252
static	I P	create one or more static variables	9.75 p253
stop	I P	stop a target execution stream	9.76 p256
str	I P	create one or more immediate string variables	9.77 p256
strtoarray	I F	convert a string to an array of integers	10.138 p331
strtofloat	I F	convert a string to a float	10.142 p333
strtoint	I F	convert a string to an integer	10.143 p333
strtotype	I F	convert a string to a type	10.144 p333
stretch	L M	generate a long pulse from a transition	11.13.2.19 p504
strlen	I F	find the length of a string	10.139 p332
strpos	I F	find leftmost surefpbstring position	10.140 p332
strrpos	I F	find rightmost substring position	10.141 p332
struct	I P	create a struct	9.78 p257
subsecond_clocks	L M	millisecond and microsecond clocks	11.19.1.1 p538
substitute	I F	substitute characters	10.145 p334
substring	I F	get a substring	10.146 p334
svn_version	L F	get the current subversion version number	11.13.5.2 p522
sysmon	L M	Virtex-5 system monitor	11.5.1.4 p423
tail	I F	get the tail of a string	10.147 p335
tan	I F	trigonometric tangent function	10.148 p335
targconsttoimmed	I F	convert a value mode constant to immediate	10.149 p336
targtoimtype	I F	convert a target type to an immediate type	10.150 p336
texit	I P	print stack trace and terminate execution	9.79 p257
time	I F	get the date and time	10.151 p336
timestring	I F	get the date and time as a string	10.152 p337

timebase	L M	decadic interval pulses	11.13.2.20 p505
toarray	I F	convert to a one-dimensional array	10.153 p338
todegree	I F	convert radian to degree	10.154 p339
tolower	I F	convert string to lower case	10.155 p339
toradian	I F	convert degree to radian	10.156 p340
tostring	I F	string value	10.157 p340
toupper	I F	convert string to upper case	10.158 p340
to_pulse	L F	generate a pulse from a transition	11.13.5.3 p523
to_pulse	L M	generate a pulse from a transition	11.13.2.21 p505
trace	I P	print a stack trace	9.80 p258
trunc	I F	truncate float or fixed to nearest integer towards zero	10.159 p341
truthtable	L F	an expression from a truth table	11.13.5.4 p523
twsend	L M	two wire serial receiver	11.13.2.22 p505
twsend	L M	two wire serial transmitter	11.13.2.23 p506
type	I P	create a type variable	9.81 p258
type	I F	type of a variable or expression	10.160 p341
typeisqual	I F	compare two types for equality	10.161 p342
typestring	I F	type of a variable as a string	10.162 p343
typeval	I F	value of a variable of type "type"	10.163 p343
typevalstring	I F	value of a variable of type "type" as a string	10.164 p344
uart_brc16	L P	create baud rate clocks	11.18.3.1 p537
uart_rx	L M	general UART receiver	11.18.2.2 p536
uart_rx	L M	simple UART receiver	11.19.1.2 p538
uart_tx	L M	general UART transmitter	11.18.2.1 p535
uart_tx	L M	simple UART transmitter	11.19.1.3 p539
useclock	I P	set the current clock	9.82 p259
used	I F	target value which is "true" when a function executes	10.165 p345
value	I P	create one or more value variables	9.83 p259
var	I P	create one or more variables	9.84 p261
varexists	I F	determine if a variable exists	10.166 p346
varlist	I F	return a list of target mode variables	10.167 p346
wait	L P	wait for a time period	11.13.3.8 p511
wait_until	L P	wait on a logical value	11.13.3.9 p511
wait_while	L P	wait on a logical value	11.13.3.10 p511
wasassigned	L P	determine if variable has been assigned	10.168 p347
waswritten	I F	determine when a static or queue variable has been written	10.169 p347
width	I F	get the width of a primitive type - DEPRECATED!	10.170 p348
words	I F	get the size in words	10.171 p348
writebyte	I P	write a byte to a file	9.85 p263
writedomain	I F	get the write clock domain for a variable	10.172 p349
writeln	I P	write a line to a file	9.86 p264
zbt_ram	L C	external ZBT RAM class	11.21.4.1 p547
zbt_ram.read	L CM	read from external ZBT RAM	?? p??
zbt_ram.readp	L CM	read from external ZBT RAM with priority	?? p??
zbt_ram.write	L CM	write to external ZBT RAM	?? p??
zbt_ram.writep	L CM	write to external ZBT RAM with priority	?? p??
zynq-ps	I M	instantiate ZYNQ CPU	11.10.1.6 p452

Appendix J - List of all Inbuilt and Library Modules, Procedures and Functions Arranged by Group

Abbreviations -

I M - inbuilt module

I P - inbuilt procedure

I F - inbuilt function

L M - library module

L P - library procedure

L F - library function L CM - library class module

L CP - library class procedure

L CF - library class function

J.1 variable creation

addmodeattribute	I P	add attribute to the mode attribute map	9.1 p188
attributes	I P	assign attributes to a variable	9.7 p192
attributeprocedure	I P	register a procedure to handle attributes	9.6 p191
class	I P	create a variable of type "class"	9.9 p193
clock	I P	create one or more clock variable	9.10 p194
enum	I P	create an enumerated type	9.26 p207
enumv	I P	create an enumerated type	9.27 p208
file	I P	create file variables	9.32 p212
float	I P	create immediate floating point variables	9.33 p213
identifier	I F	get the identifier of a variable	10.69 p297
ident	I P	extract an identifier from an argument	9.37 p216
input	I P	create an input mode variable	9.39 p218
int	I P	create one or more immediate integer variables	9.40 p221
iopins	L P	create FPGA pin specifications	11.2.2.19 p388
log	I P	create one or more immediate logical variables	9.42 p222
simpleclock	L P	create a clock of specified frequency	11.1.2.5 p364
open	I P	open a file	9.46 p223
output	I P	create an output mode variable	9.47 p224
queue	I P	create one or more queue variables	9.57 p234
queuearray	L P	construct an array of queues	11.13.3.4 p509
prielse	I F	else case value for priority variable	10.109 p316
priority	I P	create a priority variable	9.52 p230
processPinAttributes	L P	process pin attributes	11.2.2.24 p393
ref	I P	create one or more variables	9.58 p237
selectvalue	I P	create one or more selectvalue variables	9.66 p244
static	I P	create one or more static variables	9.75 p253
str	I P	create one or more immediate string variables	9.77 p256
struct	I P	create a struct	9.78 p257
type	I P	create a type variable	9.81 p258
value	I P	create one or more value variables	9.83 p259
var	I P	create one or more variables	9.84 p261

J.2 clocks

clkdcn	L P	create a DCM digital clock manager	11.1.2.1 p357
clkdlle	L P	create an extended Spartan2, Virtex or VirtexE DLL	11.1.2.3 p361
clkdllhf	L P	create a high frequency Spartan2, Virtex or VirtexE DLL	11.1.2.4 p363
clkdll	L P	create a DLL delay-locked loop clock manager	11.1.2.2 p360
clkmmcm_adv	L P	create an advanced MMCM multi-mode clock manager	11.1.2.6 p365
clkmmcm_base	L P	create a basic MMCM multi-mode clock manager	11.1.2.7 p368
clockbufg	L P	a BUFG clock buffer	11.1.2.9 p371
clockbufio	L P	a local clock buffer	11.1.2.10 p372
clockfromexpr	I P	drive a clock from an expression	9.11 p196
clockmuxbuffer	L P	a clock mux/buffer	11.1.2.11 p372
clockmuxbufferdisable	L P	a clock mux/buffer with disable	11.1.2.12 p373
clockmuxbuffer	L P	a clock mux/buffer	11.1.2.11 p372
clockused	L F	determine if a clock variable has been used	11.2.3.1 p395
clock_params	L P	derive clock synthesis parameters	11.1.2.13 p373
clock_params	L F	derive clock synthesis parameters	11.1.3.1 p375
currentclock	I F	get the current clock variable	10.23 p278
frequency	I F	get the frequency of a clock variable	10.60 p293
lowskew	L P	request low skew secondary clock routing	11.1.2.14 p374
simpleclock	L P	create a clock of specified frequency	11.1.2.5 p364
maxdelay	L P	specify max delay between timing groups	11.2.2.20 p391
negclock	I P	create a negative clock	9.45 p223
period	I F	get the period of a clock variable	10.104 p314
period.div	L P	timing group for divided clock	11.2.2.22 p392
period.mul	L P	timing group for divided clock	11.2.2.23 p393
pll_base	L M	create a series-7 basic PLL	11.1.2.8 p369
prevclock	I F	get the previous clock variable	10.108 p316
useclock	I P	set the current clock	9.82 p259

J.3 arithmetic

acos	I F	trigonometric arc cosine function	10.3 p266
abs	I F	absolute value function	10.1 p265
addsub	I F	conditionally add or subtract	10.3.1 p267
argadjust	I P	adjust arguments for arithmetic operations	9.4 p189
asin	I F	trigonometric arc sine function	10.7 p270
atan	I F	single argument trigonometric arc tangent function	10.8 p270
atan2	I F	two argument trigonometric arc tangent function	10.9 p270
biasedexponent	I F	target float biased exponent	10.12 p272
bitset	L F	test if a bit is set	11.13.4.5 p513
ceil	I F	round float to nearest integer towards $+\infty$	10.16 p274
cos	I F	trigonometric cosine function	10.21 p277
divide	L M	pipelined integer division-remainder	11.13.2.9 p497
divrem	I P	combined quotient and remainder	9.17 p200
dsp48	L P	create a DSP48 block for Virtex4 FPGAs	11.3.2.1 p408
dsp48a	L P	create a DSP48 block for Spartan6 FPGAs	11.3.2.2 p409
dsp48e	L P	create a DSP48E block for Virtex5 FPGAs	11.3.2.3 p410
dsp48e1	L P	create a DSP48E1 block for Virtex6 FPGAs	11.3.2.4 p412
dspaddsub	L M	DSP add/subtract	11.3.1.1 p401
dspcounter	L M	DSP counter	11.3.1.2 p402
dsplog	L M	DSP logical operation	11.3.1.3 p403
dspmul	L M	DSP multiply	11.3.1.4 p404
dspmuladd	L M	DSP multiply and add	11.3.1.5 p406
dspmulsub	L M	DSP multiply and subtract	11.3.1.6 p406
dspsubmul	L M	DSP subtract and multiply	11.3.1.7 p406
exp	I F	e to a power	10.38 p285
fadd	L M	floating add	11.16.1.1 p530
fdiv	L M	floating divide	11.16.1.2 p530
floatisnegative	I F	sign of target float	10.57 p292
floor	I F	round float to nearest integer towards $-\infty$	10.58 p292
fmul	L M	floating multiply	11.16.1.3 p531
fneg	L M	negate a floating point value	11.16.1.4 p531
fsub	L M	floating subtract	11.16.1.5 p532
getbit	L F	get a bit	11.13.4.9 p516
incrdecr	L F	dynamically increment or decrement	11.13.4.10 p517
isclear	L F	verify a run of clear bits	11.13.4.11 p517
isset	L F	verify a run of set bits	11.13.4.12 p517
lfsr	L F	linear feedback shift register	11.13.4.13 p518
itof	L M	convert an integer to a floating point value	11.16.1.6 p532
log	I F	natural logarithm	10.83 p305
mantissa	I F	target float mantissa	10.84 p305
mod	I F	modulus of integer or float	10.94 p310
max	I F	maximum of integers	10.88 p307
min	I F	minimum of integers	10.92 p309
maxval	L F	target maximum value	11.13.4.15 p518
minval	L F	target minimum value	11.13.4.16 p519
multiply	L M	pipelined multiply	11.13.2.12 p500
pow	I F	raise to a power	10.106 p315
random	I F	a random number	10.119 p322
round	I F	round float or fixed to nearest integer	10.126 p325
signum	I F	sign function	10.130 p328
sin	I F	trigonometric sine function	10.131 p328
sort	L M	generate a parallel Batcher sort network	11.13.2.17 p503
sqrt	I F	immediate square root	10.135 p330
sqrt	L M	target integer square root	11.13.2.18 p503
tan	I F	trigonometric tangent function	10.148 p335
todegree	I F	convert radian to degree	10.154 p339
toradian	I F	convert degree to radian	10.156 p340
trunc	I F	truncate float or fixed to nearest integer towards zero	10.159 p341

J.4 string

arraytostr	I F	convert an integer array to a string	10.5 p269
cspan	I F	head string selected by characters not in a set	10.22 p278
deflate	I F	compress a string	10.25 p279
endswith	I F	a string is the tail of another	10.34 p283
find	I F	find a substring using a regular expression	10.55 p291
format	I F	format values into a string	10.59 p293
head	I F	get the head of a string	10.68 p297
inttohexstring	I F	get a hex string representing an integer	10.72 p299
json_type	L F	generate a JSON string from a 3PL type	11.17.3.1 p534
match	I F	find substrings using a regular expression	10.86 p306
replace	I F	replace left substring	10.123 p324
replaceall	I F	replace all substrings	10.124 p324
sscanf	I P	scan a string into variables	9.68 p246
span	I F	head string selected by characters in a set	10.133 p329
split	I F	split a string into substrings	10.134 p330
sprintf	I P	copy a formatted line to a string variable	9.72 p250
sprintf	L F	return a formatted line from variables	11.13.5.1 p522
startswith	I F	a string is the head of another	10.137 p331
strtoarray	I F	convert a string to an array of integers	10.138 p331
strtofloat	I F	convert a string to a float	10.142 p333
strtoint	I F	convert a string to an integer	10.143 p333
strtotype	I F	convert a string to a type	10.144 p333
strlen	I F	find the length of a string	10.139 p332
strpos	I F	find leftmost substring position	10.140 p332
strrpos	I F	find rightmost substring position	10.141 p332
substitute	I F	substitute characters	10.145 p334
substring	I F	get a substring	10.146 p334
tail	I F	get the tail of a string	10.147 p335
tolower	I F	convert string to lower case	10.155 p339
tostring	I F	string value	10.157 p340
toupper	I F	convert string to upper case	10.158 p340

J.5 FPGA memory

cmemory	I P	create a combinatorial output memory	9.13 p197
cmemoryread	I F	read a combinatorial output memory	10.17 p275
cmemorywrite	I P	write to a combinatorial output memory	9.14 p198
cread	L M	read combinatorial output memory to a queue	11.13.2.6 p495
cwrite	L M	write combinatorial output memory from queue	11.13.2.7 p495
rmemory	I P	create a registered output memory	9.62 p239
rmemoryout	I F	get value of a reg output memory	10.127 p326
rmemoryread	I P	read reg output memory	9.63 p242
rmemorywrite	I P	write to reg output memory	9.64 p242
rread	L M	read reg output memory to a queue	11.13.2.15 p502
rwrite	L M	write reg output memory from a queue	11.13.2.16 p502

J.6 external memory

qdr2_ram	L C	QDR2 RAM class	11.20.4.1 p541
qdr2_ram.read	L CM	read from QDR2 RAM	11.20.4.2 p543
qdr2_ram.readp	L CM	read from QDR2 RAM with priority	11.20.4.2 p543
qdr2_ram.write	L CM	write to QDR2 RAM	11.20.4.3 p544
qdr2_ram.writep	L CM	write to QDR2 RAM with priority	11.20.4.3 p544
qdr2_ram.incr_qdel	L CP	increment Q input delay	11.20.4.4 p545
qdr2_ram.decr_qdel	L CP	decrement Q input delay	11.20.4.4 p545
qdr2_ram.reset_qdel	L CP	reset Q input delay	11.20.4.4 p545
qdr2_ram.incr_cqdel	L CP	increment CQ input delay	11.20.4.5 p545
qdr2_ram.decr_cqdel	L CP	decrement CQ input delay	11.20.4.5 p545
qdr2_ram.reset_cqdel	L CP	reset CQ input delay	11.20.4.5 p545
qdr2_ram.incr__cqdel	L CP	increment _CQ input delay	11.20.4.6 p546
qdr2_ram.decr__cqdel	L CP	decrement _CQ input delay	11.20.4.6 p546
qdr2_ram.reset__cqdel	L CP	reset _CQ input delay	11.20.4.6 p546
zbt_ram	L C	ZBT RAM class	11.21.4.1 p547
zbt_ram.read	L CM	read from ZBT RAM	?? p??
zbt_ram.readp	L CM	read from ZBT RAM with priority	?? p??
zbt_ram.write	L CM	write to ZBT RAM	?? p??
zbt_ram.writep	L CM	write to ZBT RAM with priority	?? p??

J.7 file

close	I P	close a file	9.12 p196
file	I P	create file variables	9.32 p212
filecanread	I F	check if a file is readable	10.40 p285
filecanwrite	I F	check if a file is writable	10.41 p286
filedelete	I F	delete a file	10.42 p286
fileexists	I F	check if a file exists	10.43 p286
filegetparent	I F	get the path to the file	10.44 p287
fileisdir	I F	check if a file is a directory	10.45 p287
fileisfile	I F	check if a file is a file	10.46 p287
filelastmod	I F	get the time of last modification	10.47 p288
filelength	I F	get the file length in bytes	10.48 p288
filelist	I F	get a list of files and directories	10.49 p289
filelistdirs	I F	get a list of directories	10.50 p289
filelistfiles	I F	get a list of files	10.51 p289
filemkdir	I F	create a directory	10.52 p290
filerename	I F	rename a file or directory	10.53 p290
filesetreadonly	I F	set a file or directory read-only	10.54 p291
fprintf	I P	print a formatted line to an output file	9.34 p213
fscanf	I P	scan a line from a file into variables	9.35 p214
hadeof	I F	had EOF on the last file read	10.63 p295
open	I P	open a file	9.46 p223
printf	I P	print a formatted line to stdout	9.49 p228
readbyte	I F	read a byte from an input file	10.121 p323
readln	I F	read a line from an input file	10.122 p323
scanf	I P	scan a line from stdin into variables	9.67 p245
writebyte	I P	write a byte to a file	9.85 p263
writeln	I P	write a line to a file	9.86 p264

J.8 CPU interface

cpu_api	L C	a CPU/FPGA interface class (UART, ZYNQ)	11.11 p455
cpu_api.dma_read	L CM	DMA transfer from memory to FPGA (ZYNQ)	11.10.1.3 p450
cpu_api.dma_write	L CM	DMA transfer from FPGA to memory (ZYNQ)	11.10.1.4 p451
cpu_api.read	L CP	a CPU-readable interface (UART, ZYNQ)	11.11.2.2 p459
cpu_api.read_memory	L CP	a CPU-readable interface to a memory (UART, ZYNQ)	11.11.2.4 p469
cpu_api.read_queue	L CP	a CPU read from a queue variable (UART, ZYNQ)	11.11.2.5 p471
cpu_api.read_value	L CP	CPU read from a value (UART, ZYNQ)	11.11.2.6 p474
cpu_api.receive_event	L CP	interrupt the CPU (UART, ZYNQ)	11.11.2.7 p476
cpu_api.write	L CP	a CPU-writable interface (UART, ZYNQ)	11.11.2.3 p464
cpu_api.write_dma_buffer	L CM	DMA transfer to CPU RAM (ZYNQ)	11.10.1.2 p450
cpu_api.write_memory	L CP	a CPU-writable interface to a memory (UART, ZYNQ)	11.11.2.8 p477
cpu_api.write_queue	L CP	a CPU write to a queue variable (UART, ZYNQ)	11.11.2.9 p479
cpu_api.write_static	LS CP	a CPU-writable static variable (UART, ZYNQ)	11.11.2.10 p481
cpu_api.interrupt	L CP	interrupt the CPU (UART, ZYNQ)	11.11.2.11 p483
dma_read	I CM	DMA transfer from memory to FPGA (ZYNQ)	11.10.1.3 p450
dma_write	I CM	DMA transfer from FPGA to memory (ZYNQ)	11.10.1.4 p451

J.9 serial I/O

crc_8_16	L M	16 bit CRC from 8 bit data	11.13.2.2 p492
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